

Appendix H

SCORE Placeholder Sheet for IFW Content

Application Number: 18059958

Document Date: 11/29/2022

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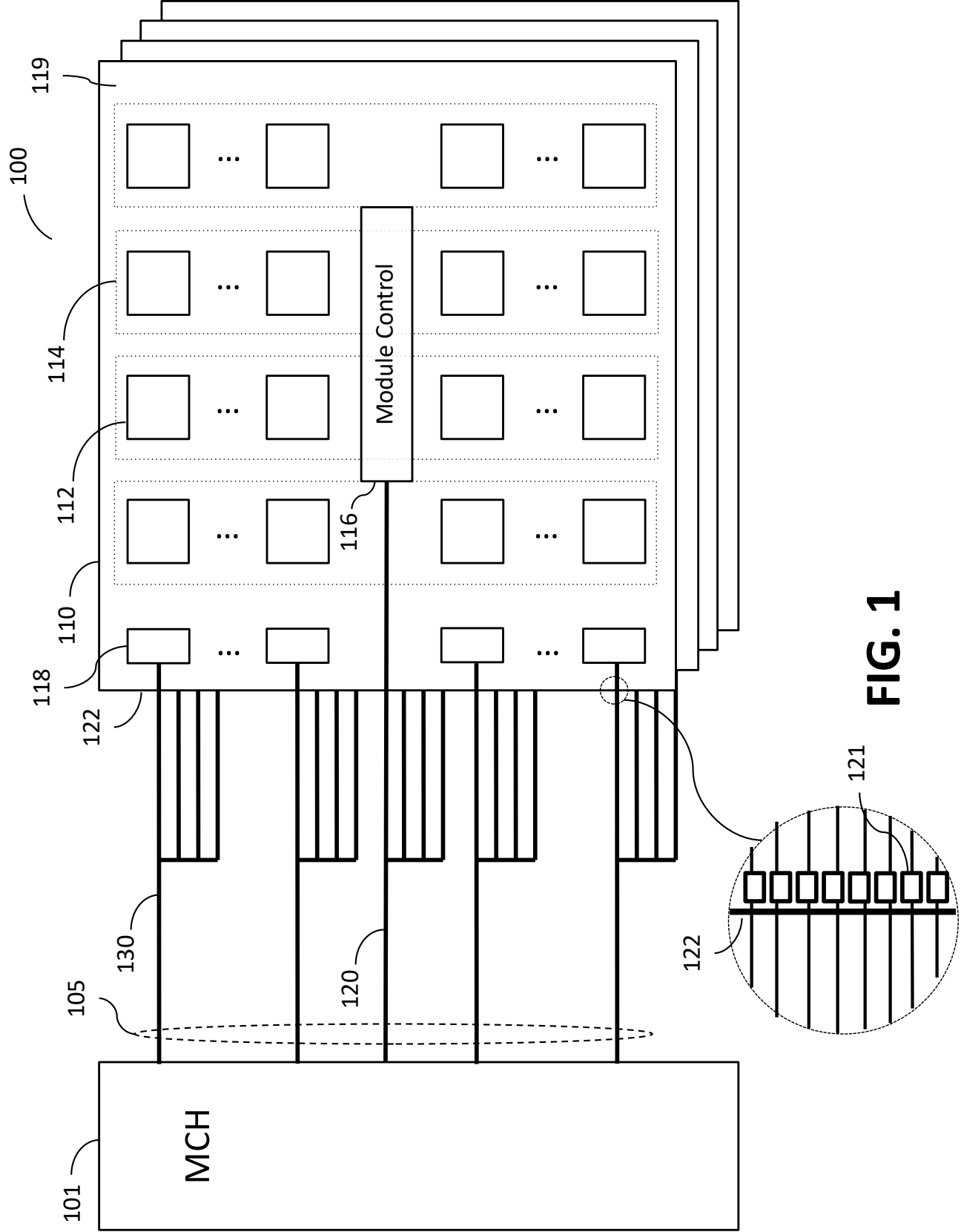


FIG. 1

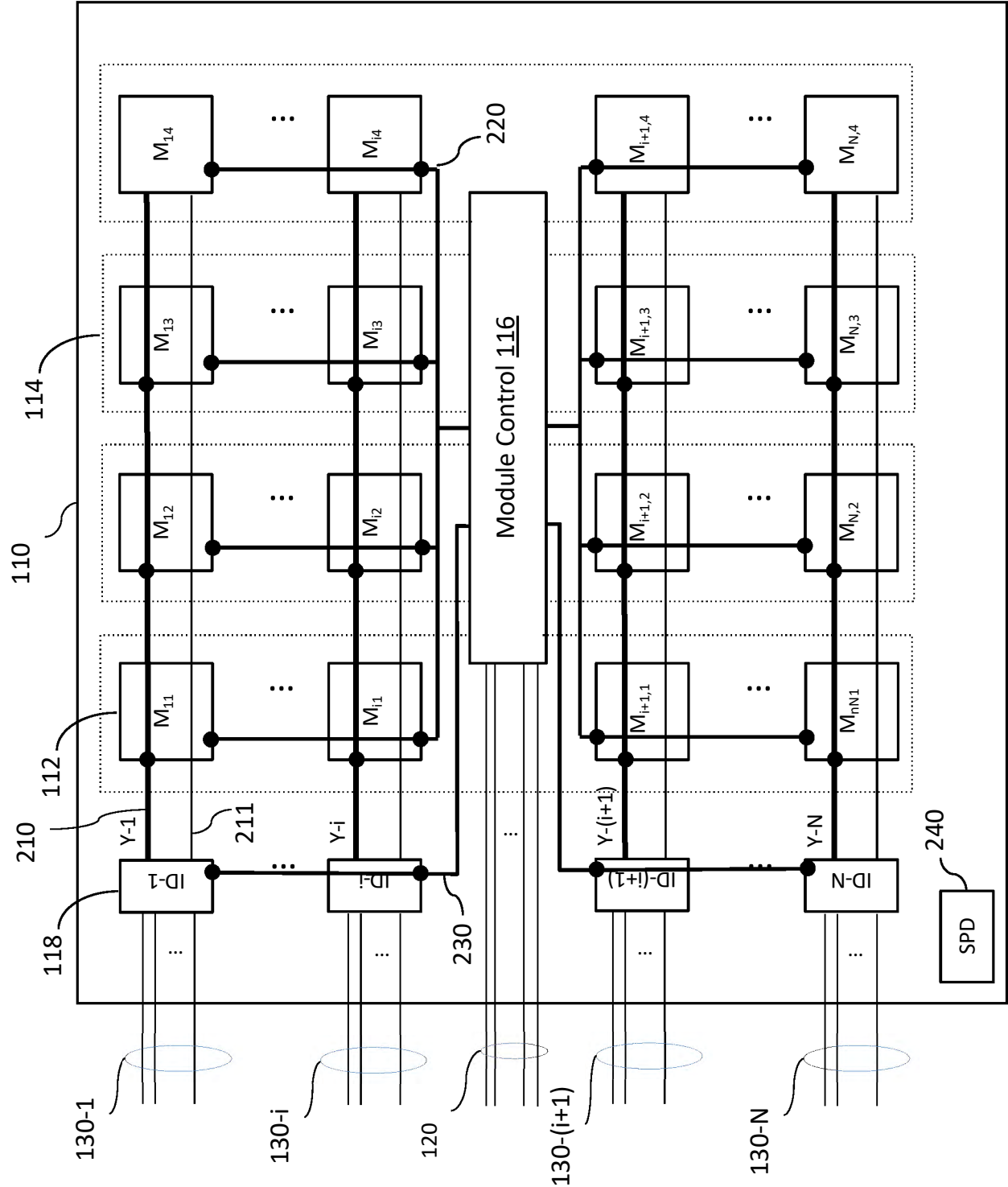


FIG. 2A

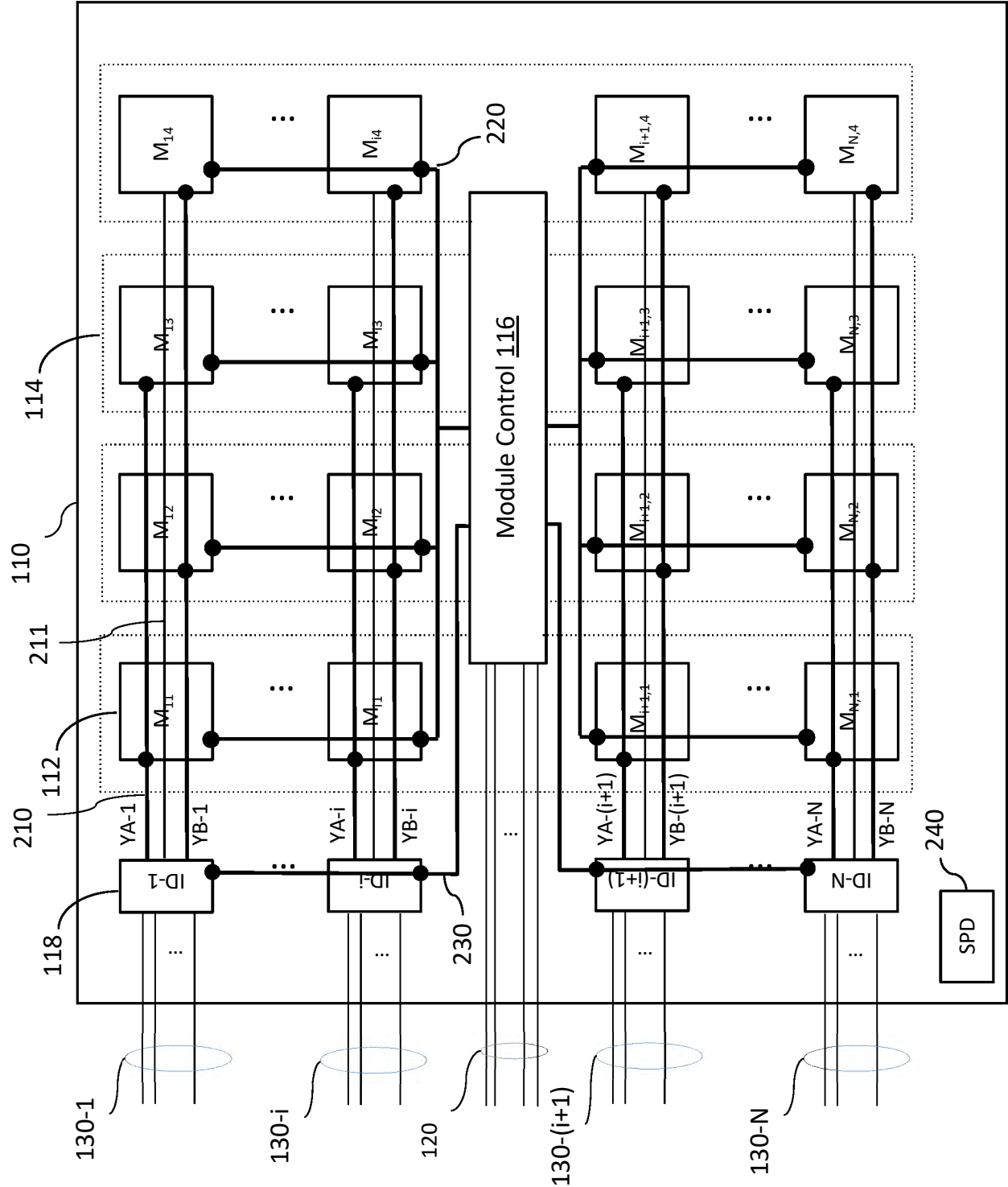


FIG. 2B

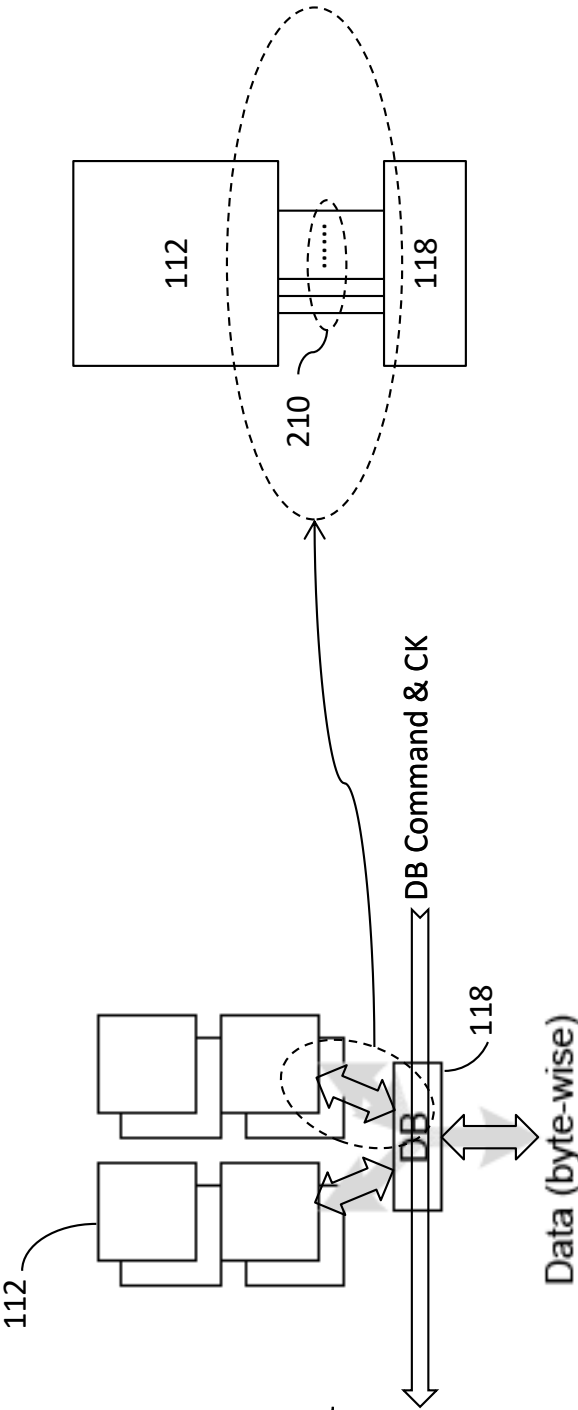
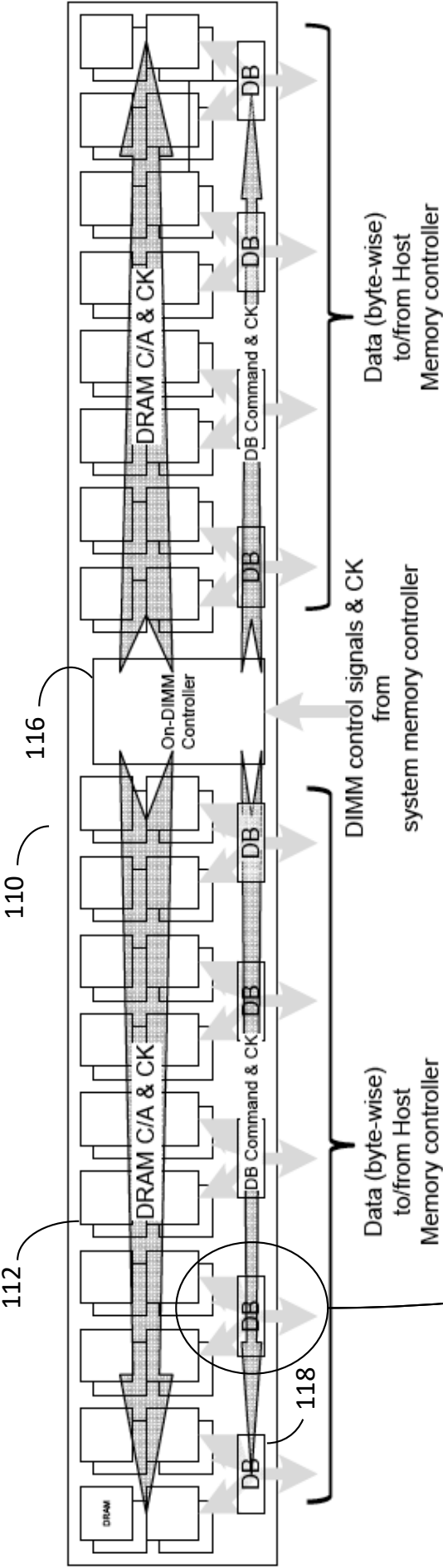


FIG. 2D

FIG. 2C

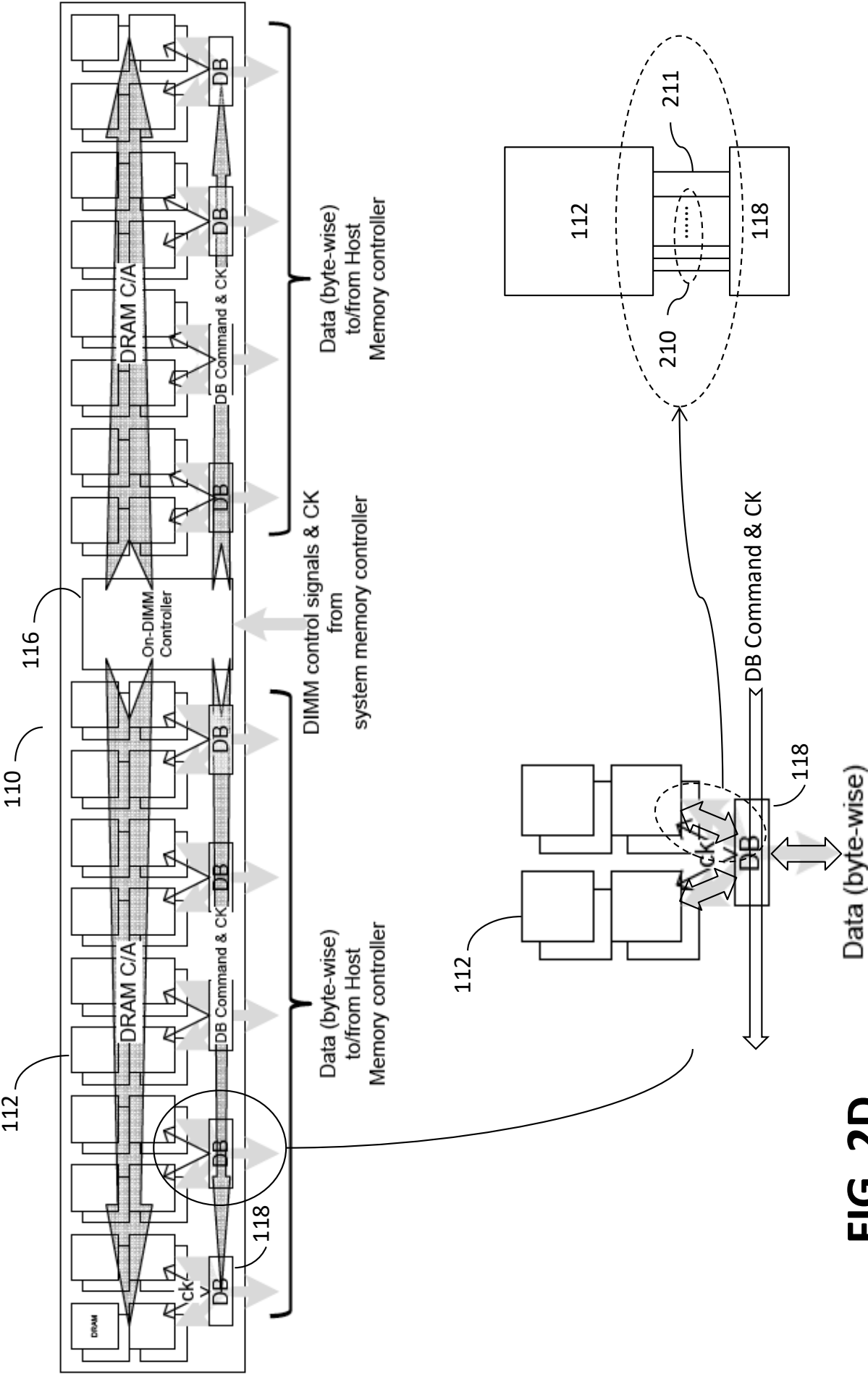


FIG. 2D

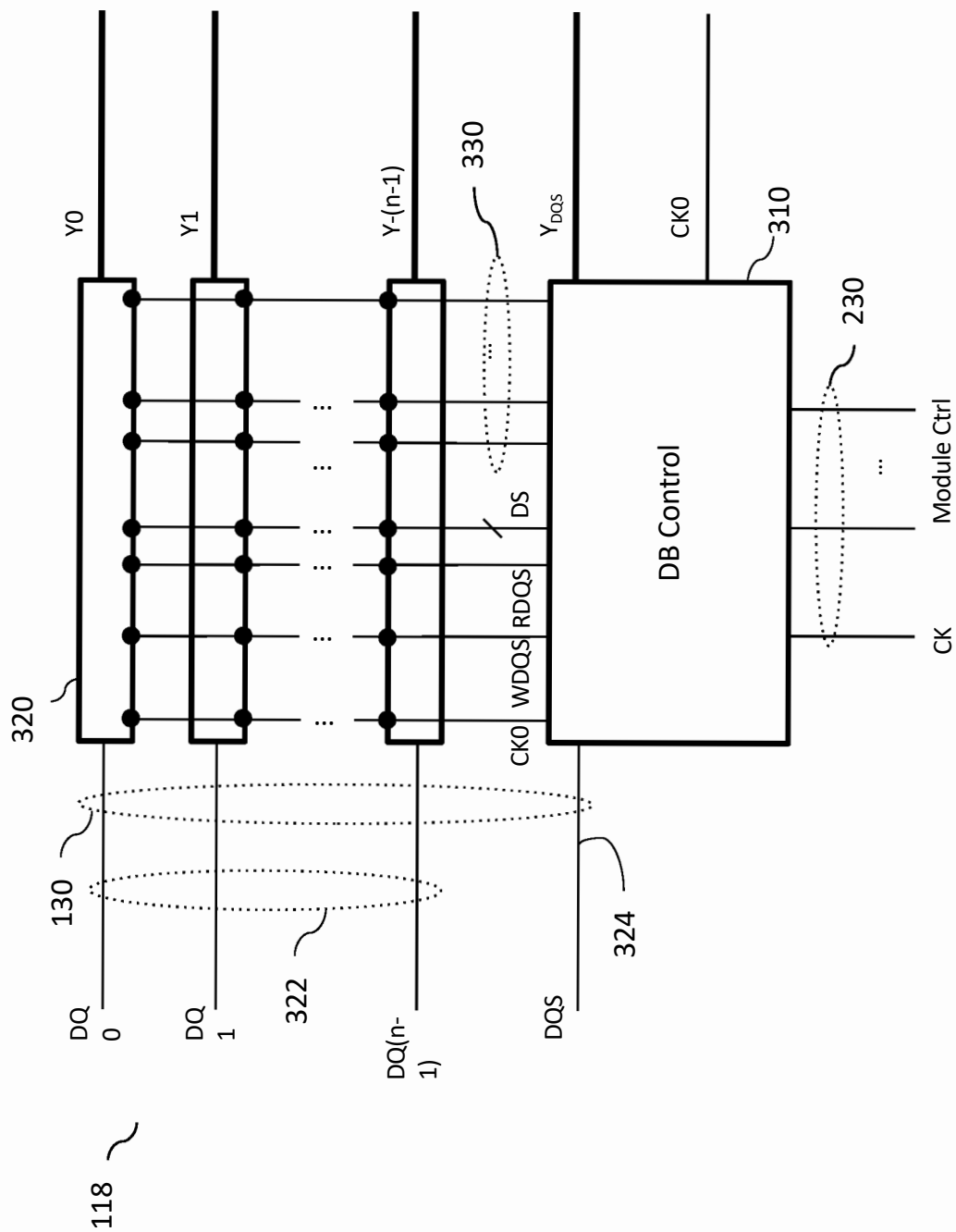


FIG. 3

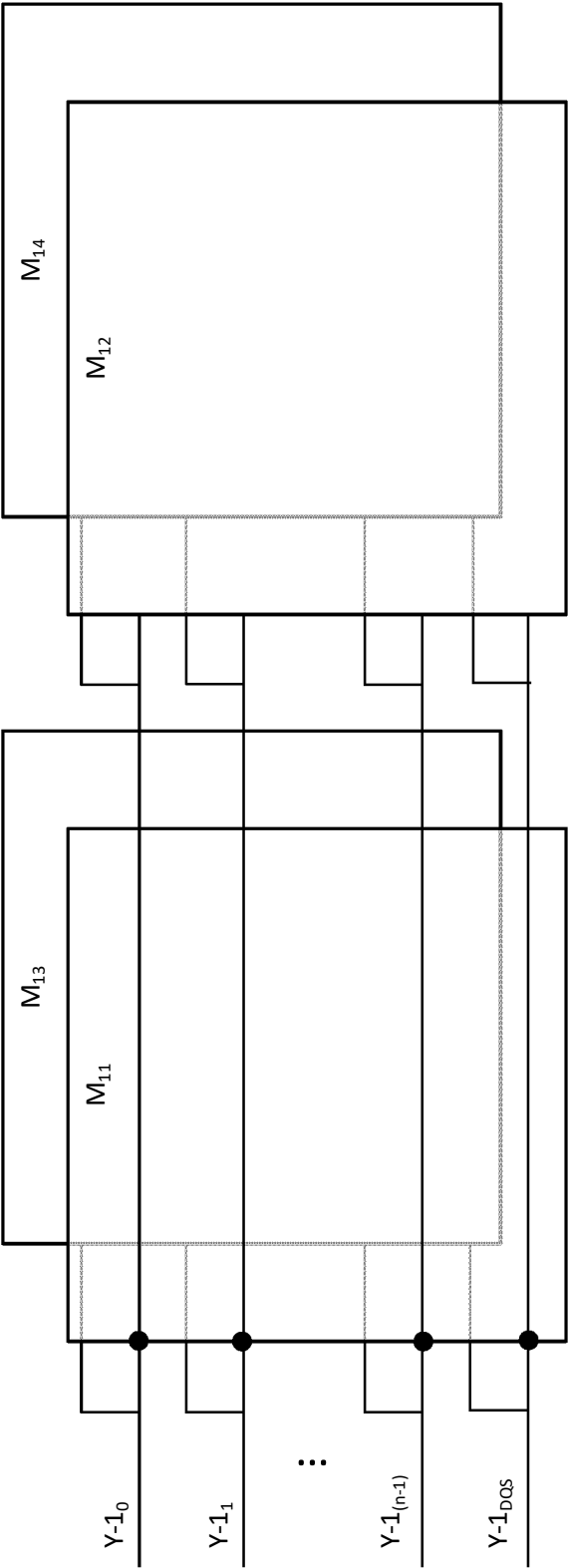


FIG. 4A

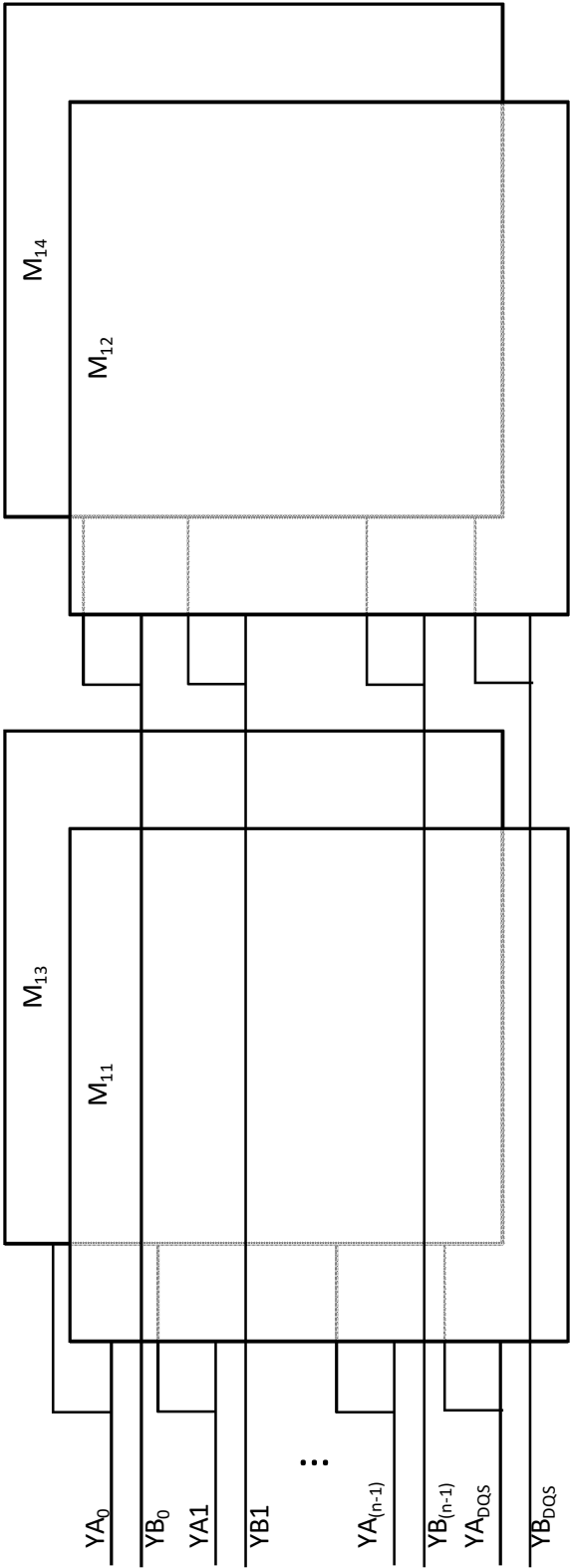


FIG. 4B

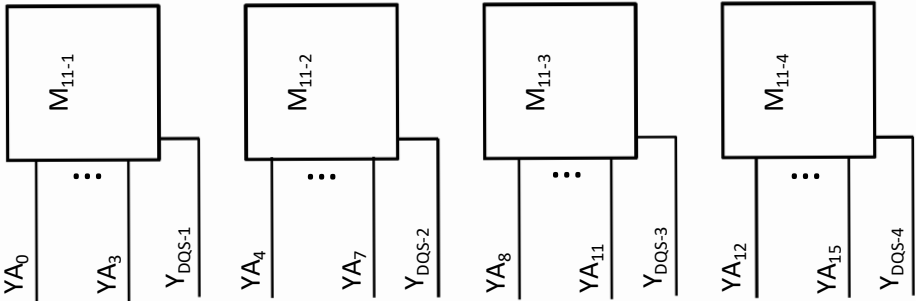


FIG. 5B

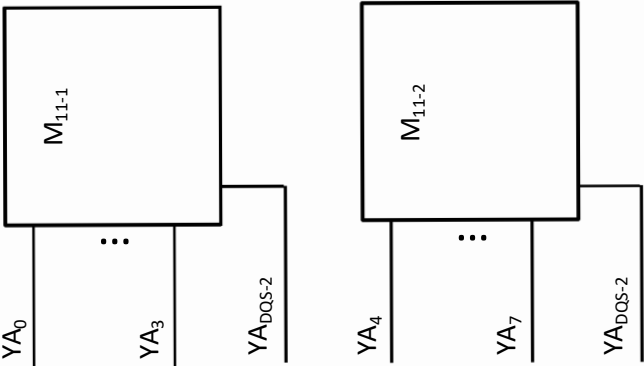


FIG. 5A

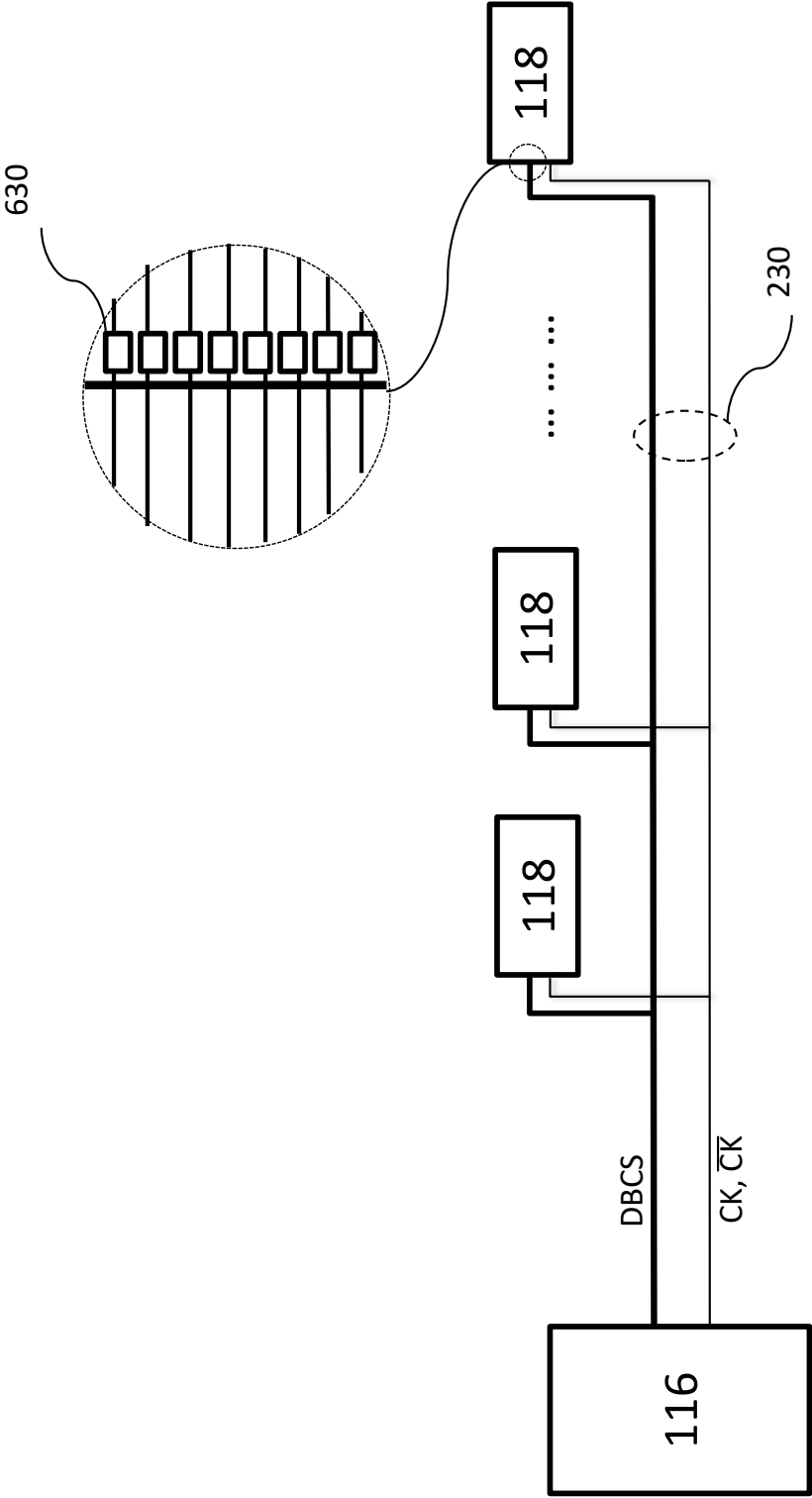


FIG. 6A

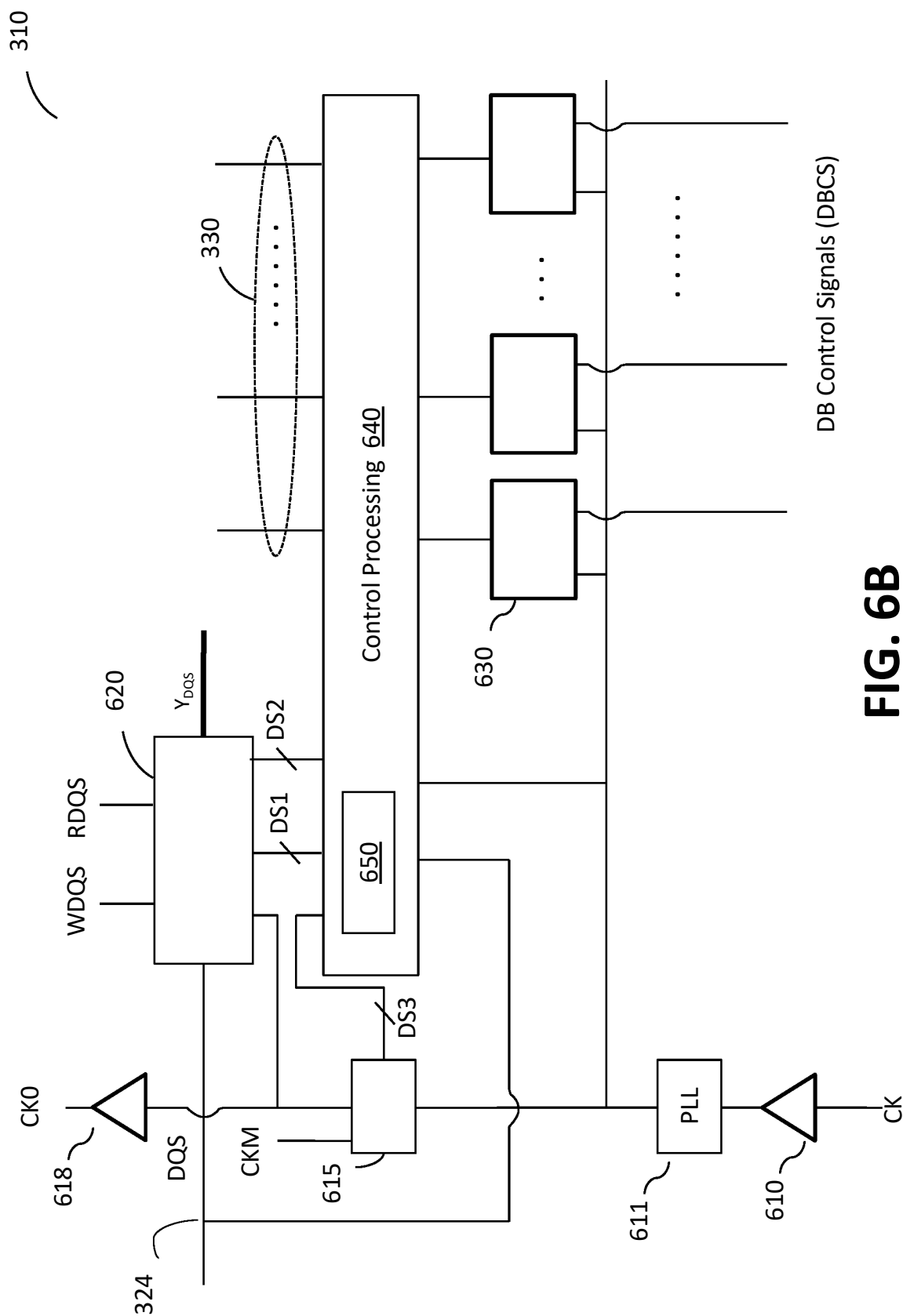


FIG. 6B

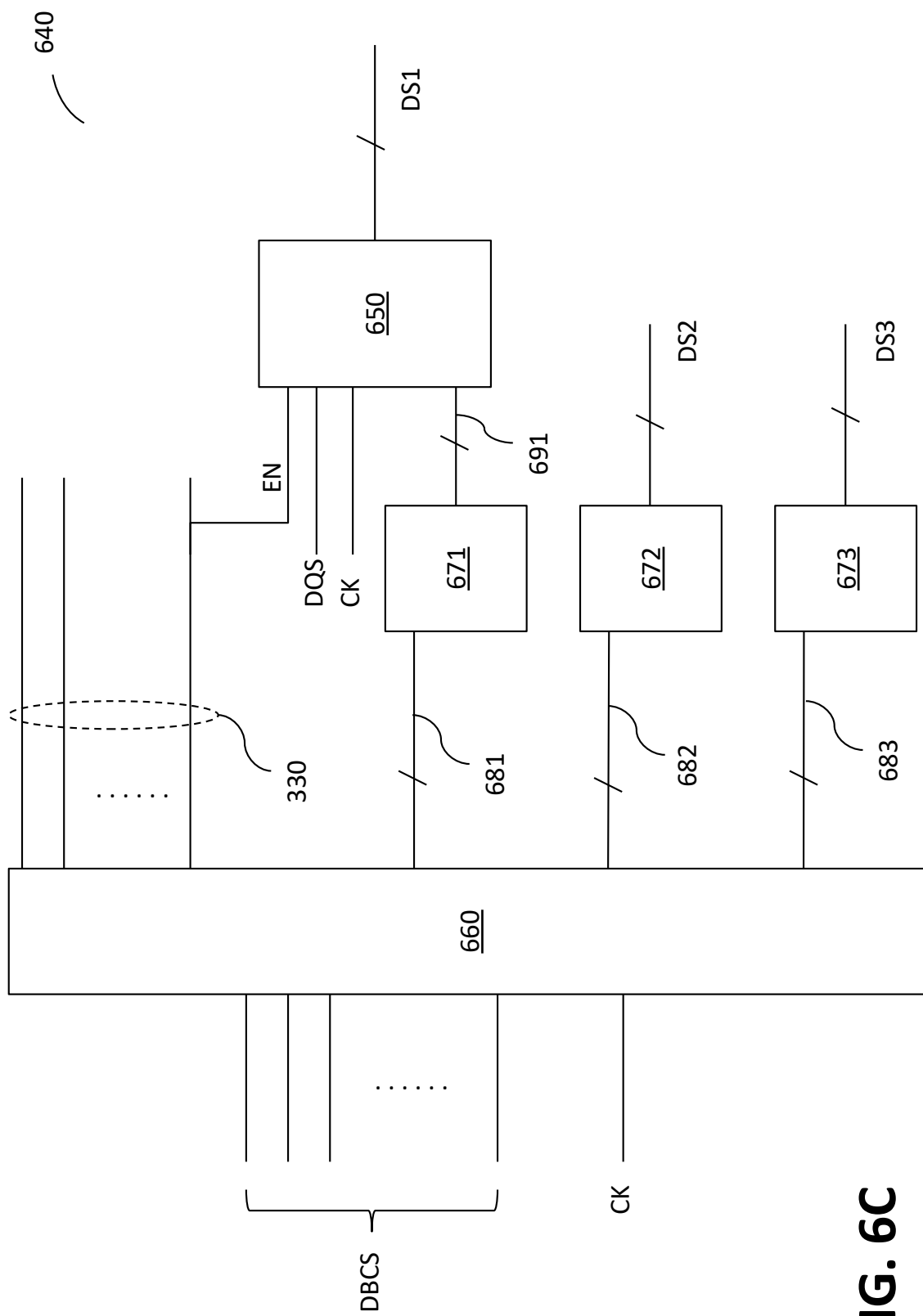


FIG. 6C

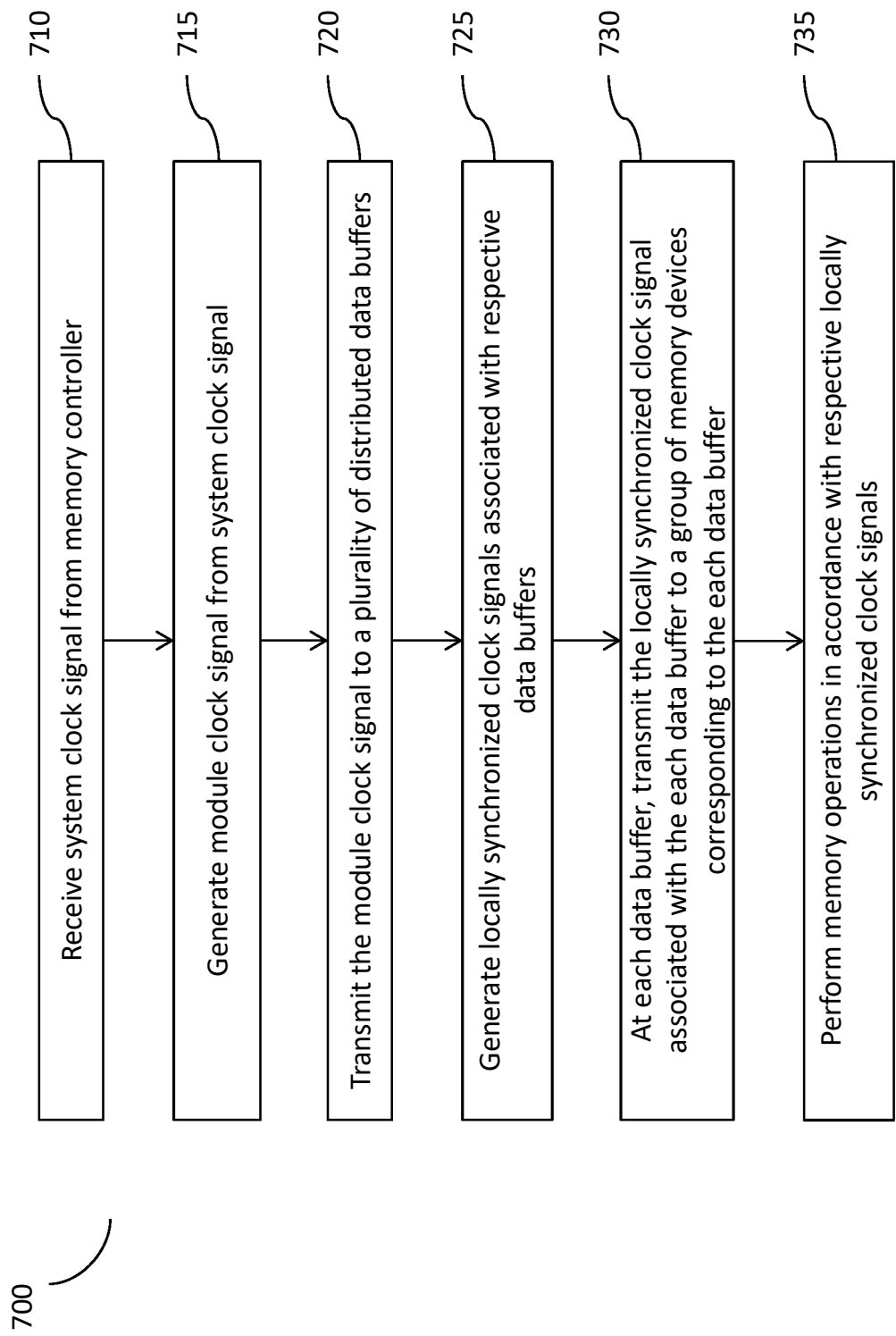


FIG. 7A

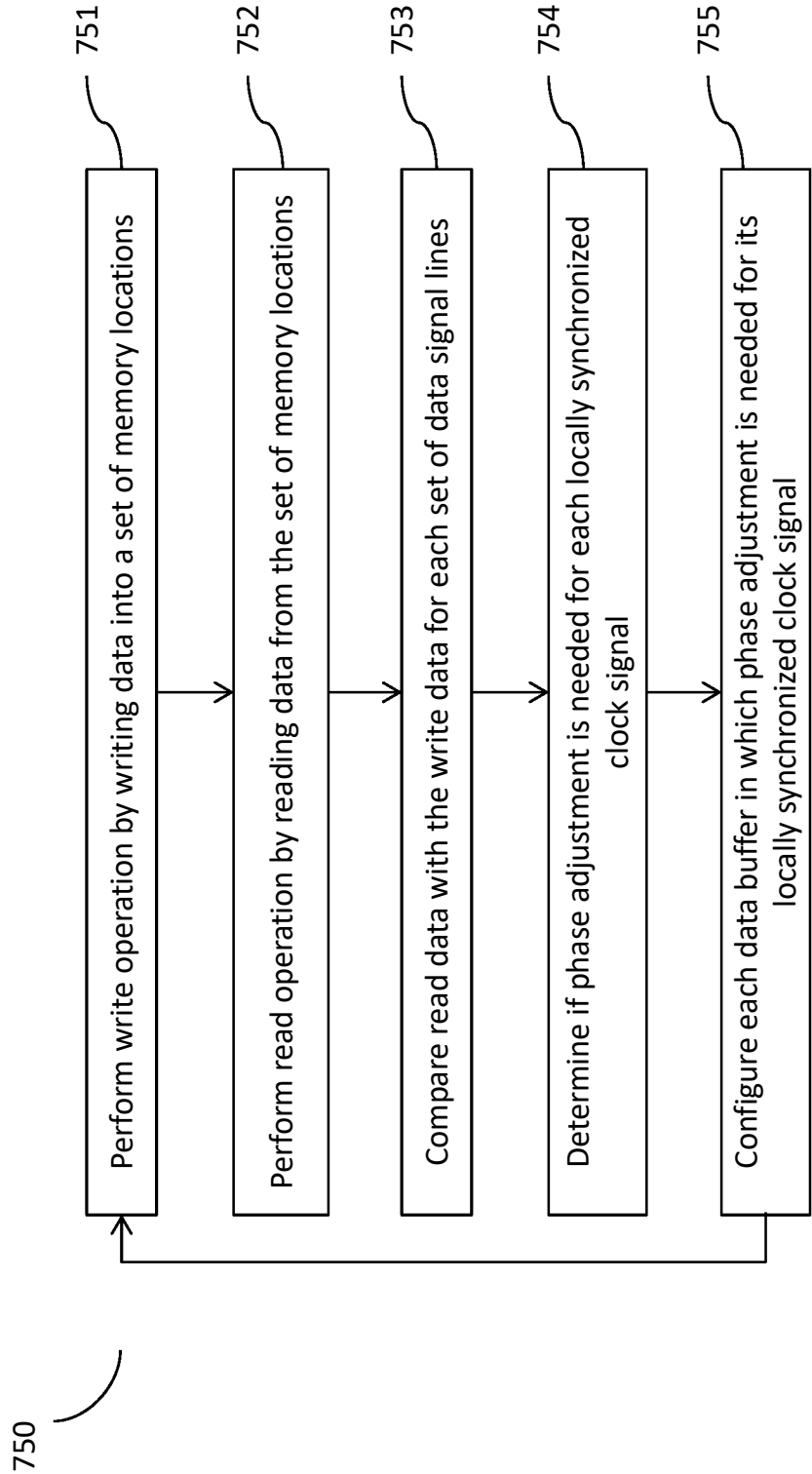


FIG. 7B

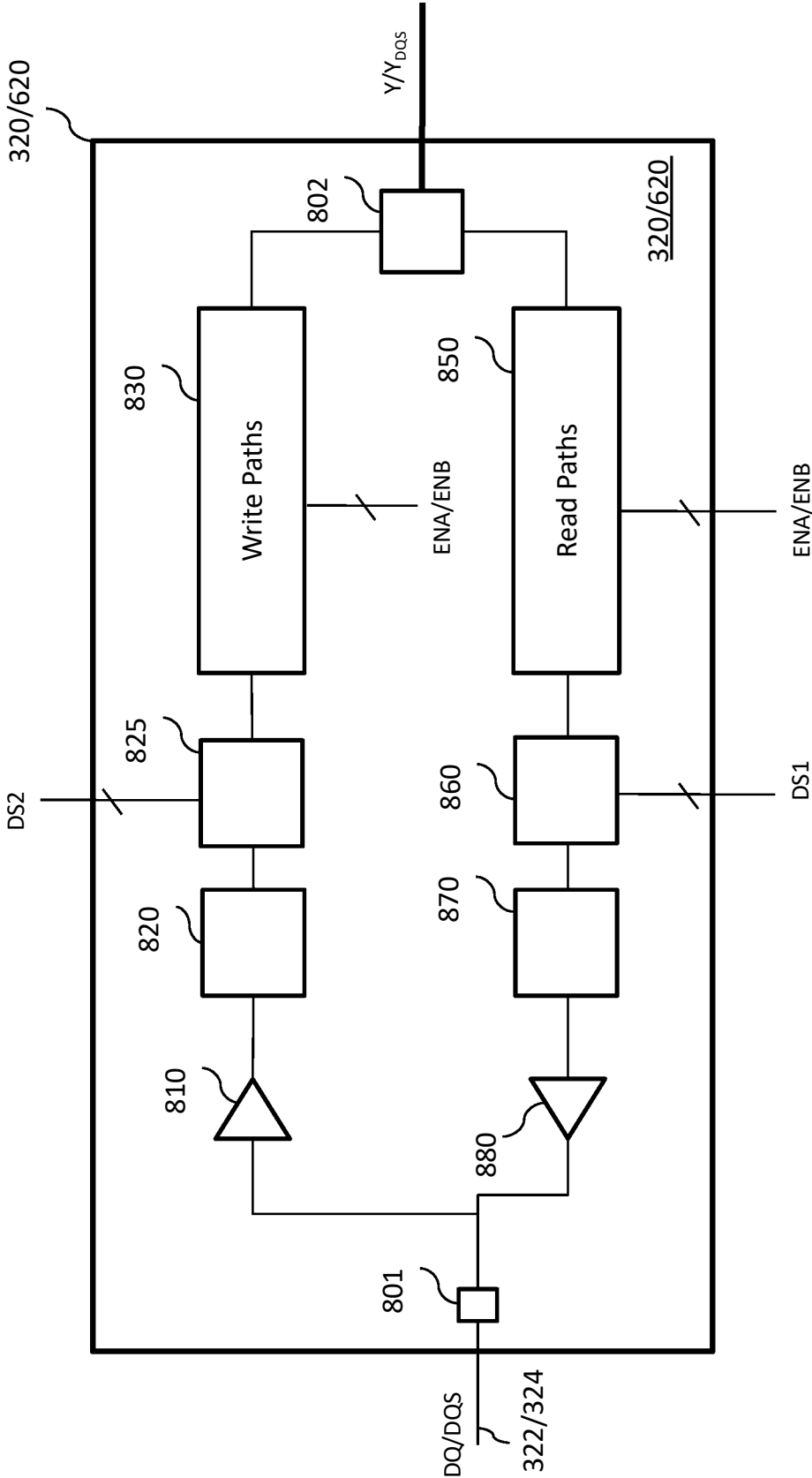


FIG. 8

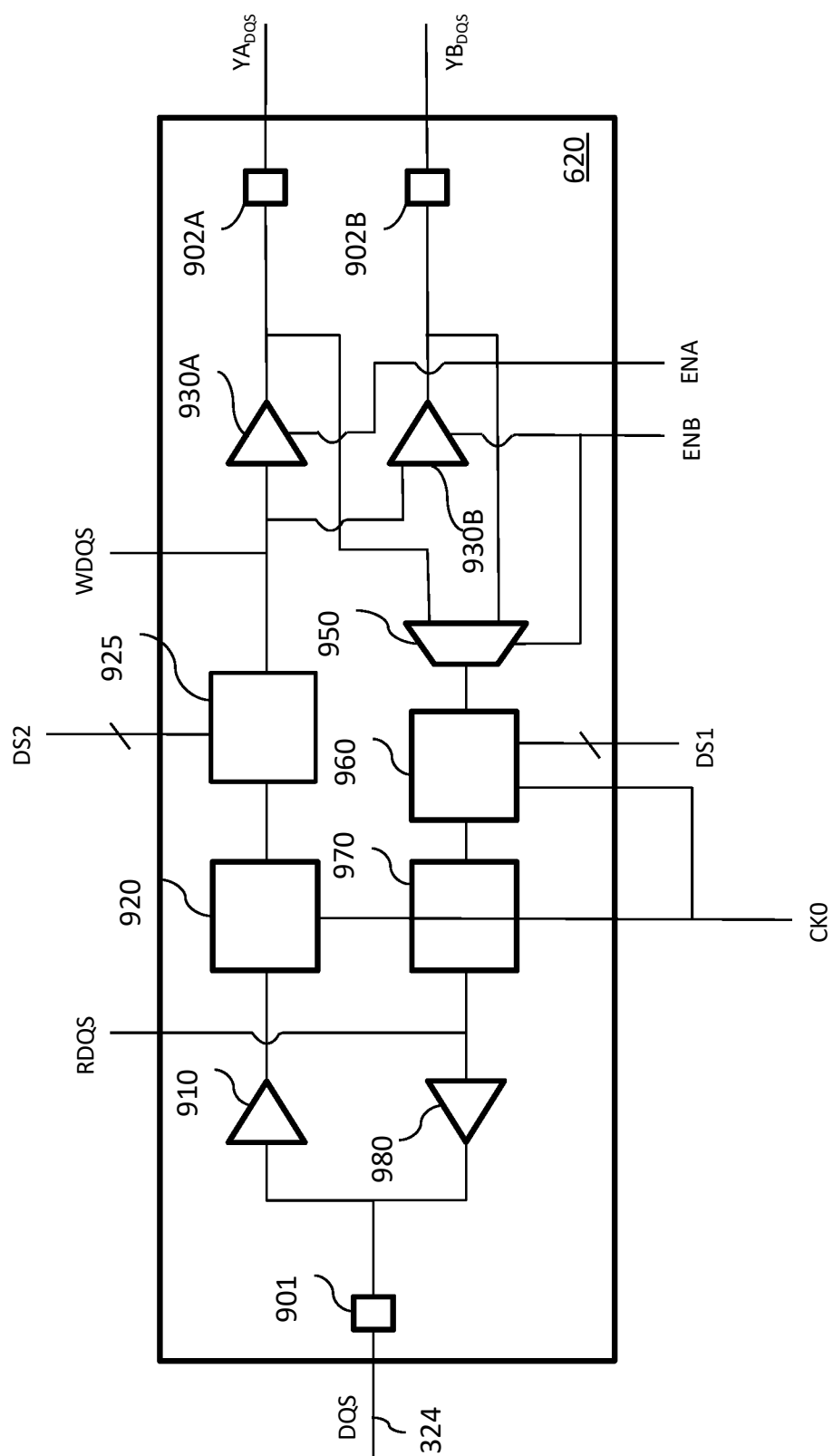


FIG. 9

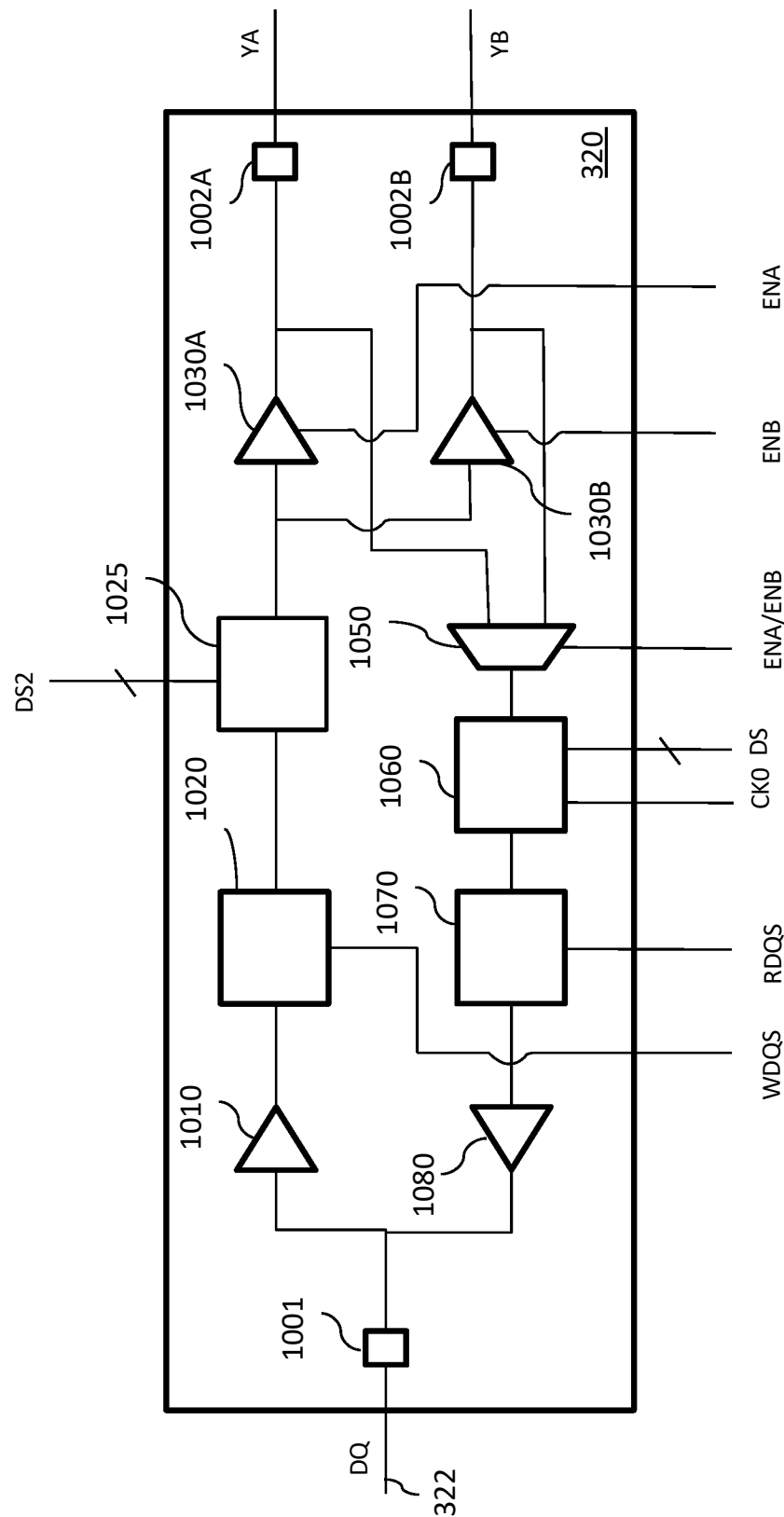


FIG. 10

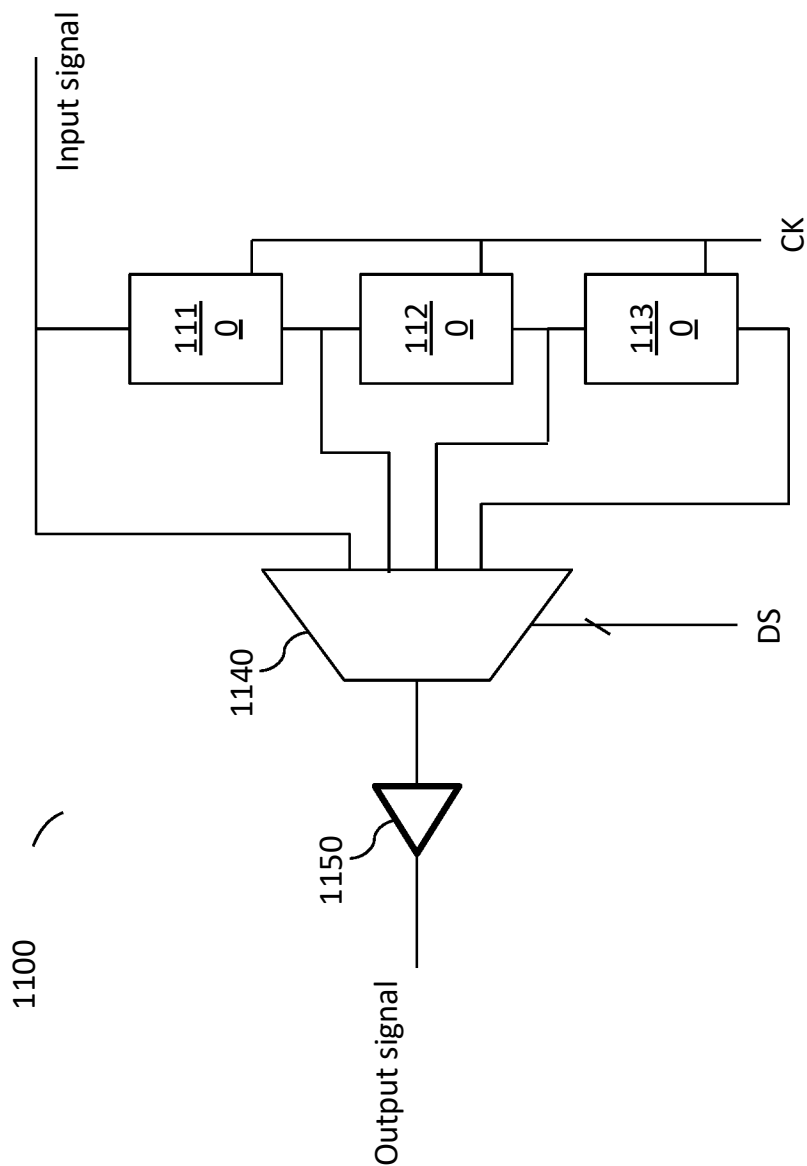


FIG. 11

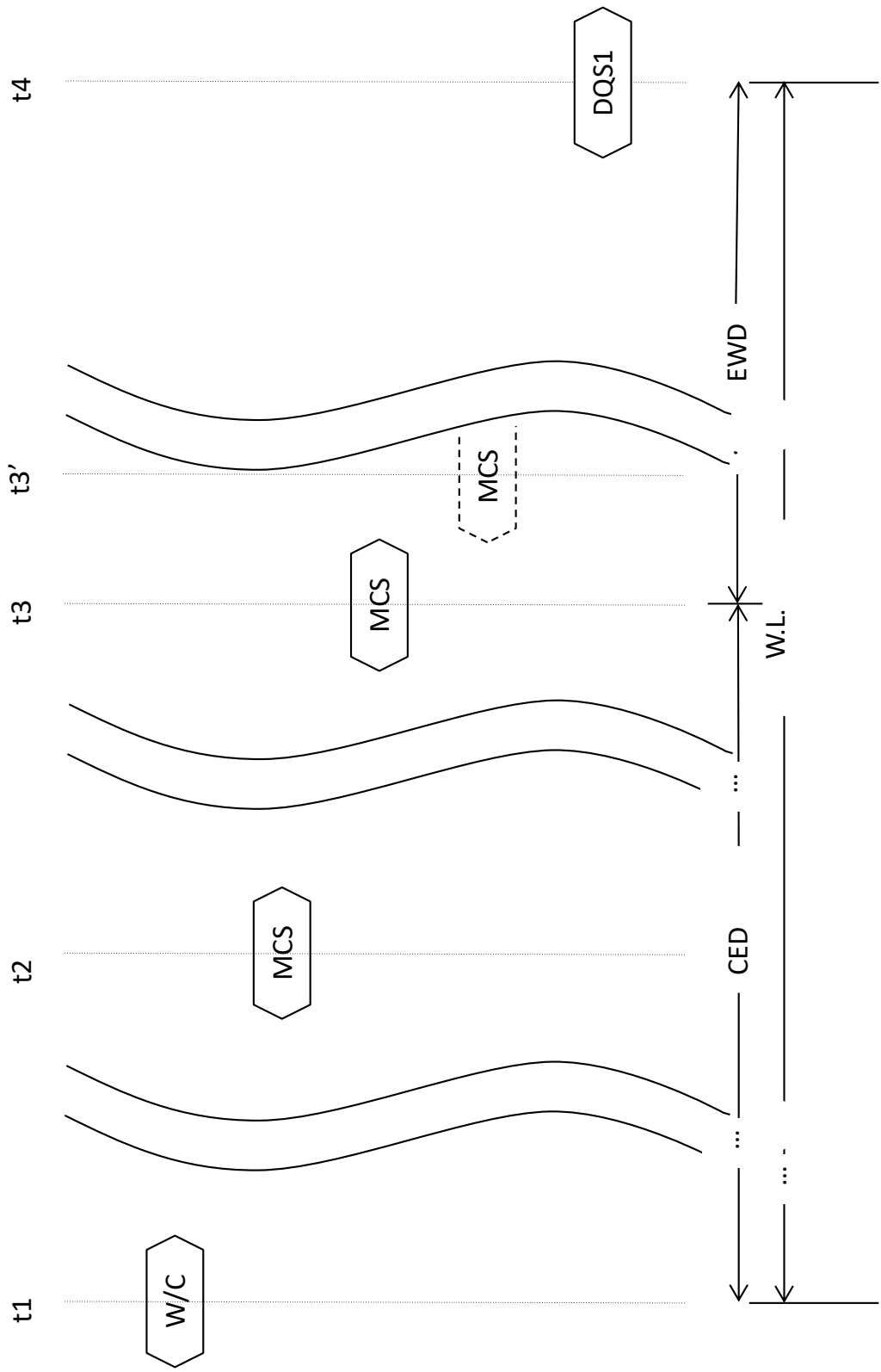


FIG. 12A

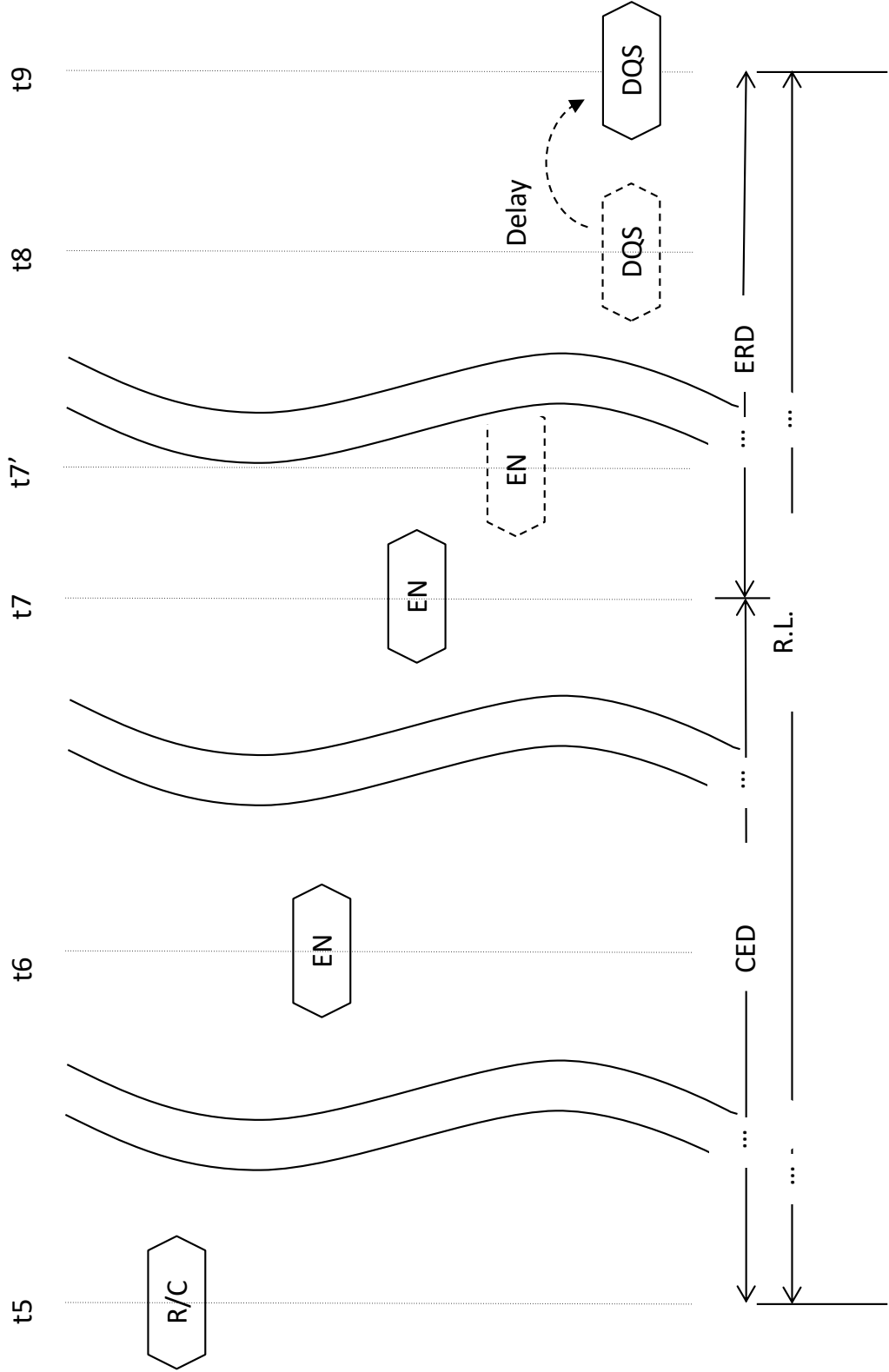


FIG. 12B

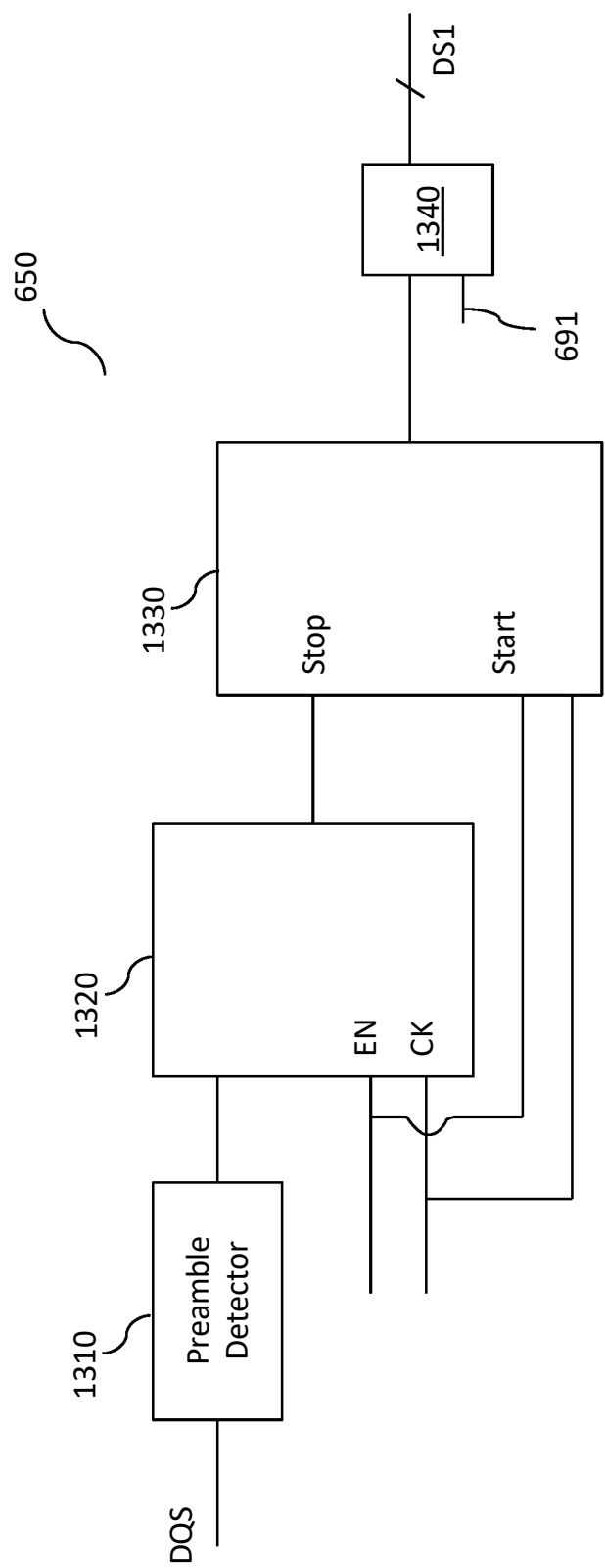


FIG. 13

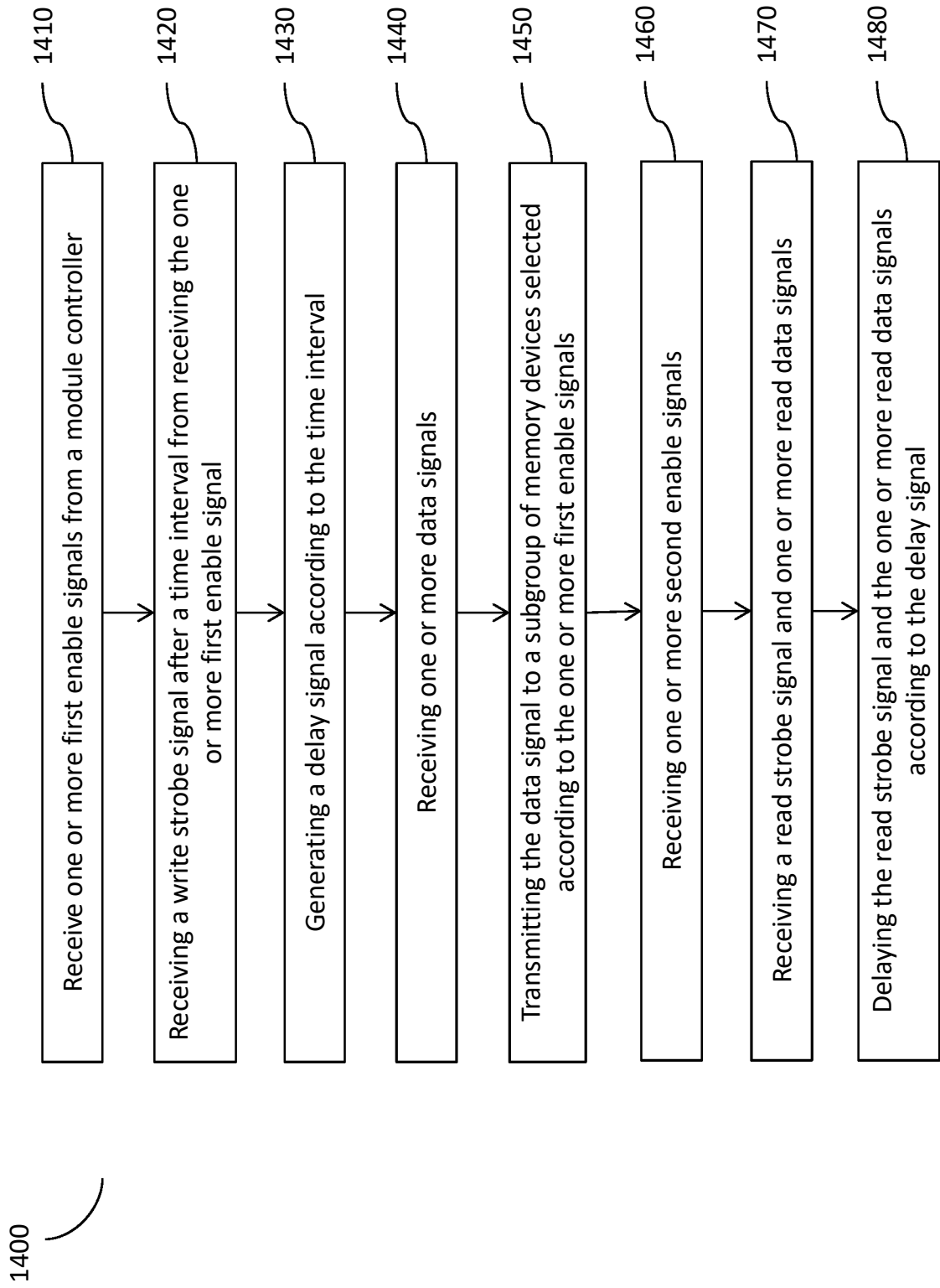


FIG. 14

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application a continuation of U.S. Patent Application Ser. No. 17/141,978, filed on January 5, 2021, entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 16/432,700, filed on Jun. 5, 2019 (now U.S. Pat. No. 10,884,923), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 14/445,035, filed on Jul. 28, 2014 (now U.S. Pat. No. 10,324,841), entitled “Memory Module with Local Synchronization,” which claims priority to U.S. Provisional Patent Application No. 61/859,215, filed July 27, 2013, each of which is incorporated herein by reference in its entirety.

FIELD

[0002] The disclosure herein is related generally to memory modules, and more particularly to memory modules having distributed data buffers.

BACKGROUND

[0003] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a

memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

[0004] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0005] The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0006] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0007] For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules. The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0008] In some conventional memory systems, the memory controllers include leveling

mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0010] FIGS. 2A-2D are each a diagrams illustrating interactions among components in a a memory module according to certain embodiments.

[0011] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0012] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0013] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0014] FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0016] FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

[0017] FIG. 7A is a flowchart illustrating a method of operation for a memory module according

to certain embodiments.

[0018] FIG. 7B is a flowchart illustrating a method of training locally synchronized clock signals according to certain embodiments.

[0019] FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0020] FIG. 9 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0021] FIG. 10 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0022] FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0023] FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0024] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0025] FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

[0026] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g.,

read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

[0027] In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

[0028] FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and

one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

[0029] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0030] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

[0031] Furthermore, the memory devices 112 may include any type of memory devices. For

example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory (SDRAM) devices in compliance with, for example, the Double Data Rate Type 3 (DDR3) or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0032] As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0033] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0034] Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Patent No. 7,532,537, entitled “Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation,” issued on May 12, 2009, the entirety of which is incorporated herein by reference.

[0035] Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers 118, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer control signals are used to convey configuration or module self-test control information from the module control device 116 to the data buffers 118. In one embodiment, the data buffer control signals are transmitted to the data buffers 118 via respective data buffer control signal lines 230. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers 118 via the data buffer control signal lines and decoded/processed at the data buffers.

[0036] Module control device 116 transmits the module C/A signals to the memory devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers 118 via data buffer control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the data buffers 118 share a same set of data buffer control signal lines 230.

[0037] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices $M_{11}, \dots, M_{i1}, M_{i+1,1}, \dots, M_{N1}$, a second rank includes memory devices $M_{12}, \dots, M_{i2}, M_{i+1,2}, \dots, M_{N2}$, and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11}, M_{12}, M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0038] In certain embodiments, the data buffers 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers 118 is associated with the first group of memory devices M_{11}, M_{12}, M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory devices, data buffers ID- i among the data buffers 118 is associated with the i^{th} group of memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be associated with a same group of memory devices, or vice versa.

[0039] In one embodiment, each group or sets of memory devices are coupled to the associated data buffer 118 via a set of module data/strobe lines 210 and one or more clock signal lines 211. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0040] In one embodiment, the data buffers 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each data buffer 118 is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the data buffer control signals. Further, each data buffers 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0041] In one embodiment, the data buffers 118 are distributed across the memory module 110 or the module board 119 in positions corresponding to edge connectors 121 coupled to respective groups of system data/strobe signal lines 130. For example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

[0042] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0043] In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0044] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system 100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the memory devices 112.

[0045] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the

control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

[0046] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

[0047] For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of interface timing training may be required because the system memory controller 101 has very limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0048] Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very significant portion

of a data period. As a result, the clock skew issues cannot simply be addressed by pipelining the data paths.

[0049] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer 118 in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit 116 and the group of locally synchronized components.

[0050] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices 112, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller 101, and/or programmable phase adjustment for one or more control signal from the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data buffers,

as disclosed in commonly-owned U.S. Patent 8,001,434, issued August 16, 2011, entitled “Memory Board with Self-Testing Capability,” which is incorporated herein by reference in its entirety.

[0051] In certain embodiments, operations of the data buffers 118 are controlled by the data buffer control signals from the module control circuit 116, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers 118 to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers 118 are distributed across the memory module, the data buffer control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. Patent Application No. 13/952,599, filed July 27, 2013, entitled “Memory Module with Distributed Data Buffers and Method Of Operation,” the entirety of which is incorporated herein by reference.

[0052] Because the data buffers 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signals lines 230 from the module control device 116 to the farthest positioned data buffers 118, such as data buffer ID-1 and data buffer ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of data

buffer control signals may reach different data buffers 118 at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

[0053] With the data buffers 118 receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the data buffers 118. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used to adjust the timing of the transmission of read data to the MCH 101 during a subsequent read operation, as explained in more detail below.

[0054] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0055] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n

data (DQ) signal lines 322 each for transmitting one of a set of bits $DQ_0, DQ_1, \dots, DQ_{n-1}$, in a data signal and at least one strobe (DQS) signal line 324 for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line Y_{DQS} . When the subsets of memory devices are coupled to the associated data buffer 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

[0056] Each data buffer 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, YA_1, \dots, YA_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each data buffer 118 further includes an DB Control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS} .

[0057] The DB Control circuit 310 also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines 230, and outputs datapath control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit 310 also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective

DQ signal line 322 with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

[0058] In certain embodiments, the DB Control circuit 310 also provides one or more delay signals DS, which is used by the DQ routing circuits 320 to control the timing of read and/or write data output by the data buffer 118, as explained in further detail below. In certain embodiments, the DB Control circuit 310 regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer 118 and the respective group of memory devices form a group of locally synchronized devices on the memory module 110.

[0059] The memory devices 112 in each group of locally synchronized devices are coupled to the data buffer 118 in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices 112 in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer 118 via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M_{11} , M_{12} , M_{13} , and M_{14} in the first group of memory devices can be coupled to the data buffer ID-1 via a same set of module data lines $Y-1_0$, $Y-1_1$, ..., $Y-1_{n-1}$ and module strobe line $Y-1_{DQS}$. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0060] Alternatively, as shown in FIG. 4B, memory devices M_{11} and M_{13} , which form a subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines $YA-1_0$, $YA-1_1$, ..., $YA-1_n$ and module strobe line $YA-1_{DQS}$ and memory devices M_{12} and M_{14} , which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines $YB-1_0$, $YB-1_1$, ..., $YB-1_n$ and module strobe line $YB-1_{DQS}$. Memory

devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board 119. Memory devices coupled to the same data buffers may be placed side-by-side, on opposite sides of the module boards 119, or stacked over each other, and/or over the associated data buffer.

[0061] Multiple memory devices having a data width that is less than a data width of the data buffers 118 may be used in place of one of the memory devices 112, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M_{11-1} and M_{11-2} may be used in place of the memory device M_{11} . Each of the two memory devices M_{11-1} and M_{11-2} has a data width of 4, and together they act like a memory device M_{11} of a data width of 8. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory circuit M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}.

[0062] In another embodiment, as shown in FIG. 5B, four memory devices M_{11-1} to M_{11-4} may be used as the memory device M_{11} . Each of the four memory devices M_{11-1} to M_{11-4} has a data width of 4, and together they act like a memory device M_{11} of a data width of 16. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory device M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}, and so on.

[0063] FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits 630 on each data buffer. In certain embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit 116 uses the DBCS to configure the programmable features in the data buffers 118

based on configuration information received from the MCH 101; in the test mode, the DBCS is used to control the self-test processes run by the data buffers; in the normal operation mode, the DBCS includes the one or more enable signals and one or more ODT signals.

[0064] FIG. 6B illustrates the DB Control circuit 310 in an data buffer 118. As shown, the DB Control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, a phase-locked loop circuit (PLL) 611 that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit 615 for adjusting the phase of the locally synchronized clock CK0, and a clock driver 618 that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to a corresponding system DQS signal line 324 and on another side to corresponding module DQS signal lines YADQS and YBDQS. The DB Control circuit 310 further includes receiver circuits 630 to receive respective ones of the DBCS.

[0065] The DB Control circuit 310 further includes a control processing circuit 640 that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more data path enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the control processing circuit 640, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

[0066] The strobe routing circuit 620 also buffers strobe signals received from either the MCH

101 or the memory devices 112, and outputs either a write strobe WDQS or read strobe RDQS, respectively, to the DQ routing circuits 320. In one embodiment, the DB Control circuit 310 further includes a dynamic delay control circuit 650 that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit 320 and the strobe routing circuit 620.

[0067] In certain embodiments, as shown in FIG. 6C, the control processing circuit 640 includes control logic 660 that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals 330. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals 681 to programmable registers 671 that provides static control of the timing of read data/strobe signals, configuration signals 682 to programmable registers 671 that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals 683 to programmable registers 673 that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

[0068] The programmable registers 671 is coupled via signal lines 691 to the dynamic delay control circuit 650, which also takes as input a DQS signal, clock signal CK, and one of the datapath enable signals EN, and output the delay signal DS1. The dynamic delay control circuit 650 is described in further detail below.

[0069] FIG. 7A is a flowchart illustrating a method 700 performed by the memory module 110 according to certain embodiments. As shown in FIG. 7A, the method includes receiving 710 clock signal together with system C/A signals from the MCH 101, and generating 715 module clock signal CK. The module clock signal CK is transmitted 720 to a plurality of distributed data buffers 118, which generates 725 locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits 730 its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform 735 memory operations in accordance with the respective locally synchronized clock

signals.

[0070] Since the module control circuit 116 generates the DBCS based on signals from the MCH 101, the MCH 101 can train the data buffers 118 to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit 116 generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH 101. As shown in FIG. 7B, a method 750 for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing 751 data to a set of memory locations in the memory devices 112, which is followed by performing a read operation by reading 752 data from the set of memory locations. The read data is compared 753 with the write data to determine 754 if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit 116 agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit 116 do not agree with the corresponding write data bits. The disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers 118 further away from the module control circuit 116 being excessively delayed during transmission, and proper phase adjustment can be programmed 755 into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

[0071] FIG. 8 illustrates a DQ or DQS routing circuit 320 or 620 according to an embodiment. As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 801 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 802 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS}, or YA/YA_{DQS} and YB/YB_{DQS}. The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 810 that buffers write data/strobe, a write data/strobe receiver 820 that samples the write data/strobe, and a write delay circuit 825

controlled by the delay signal DS2. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 830 that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

[0072] The DQS routing circuit further includes a plurality of read paths 850 that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit 860 by an amount controlled by the delay signal DS1, and sampled by a sampler circuit 870. The sampled read data/strobe is transmitted by transmitter 880 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 801.

[0073] FIG. 9 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 901 that is coupled to a corresponding DQS signal line 324, a second DQS pin 902A that is coupled to a corresponding module DQS line Y_{ADQS} , a third DQS pin 902B that is coupled to a corresponding module DQS line Y_{BDQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first DQS pin 901 and the second DQS pin 902A and a second write strobe path coupled between the first DQS pin 901 and the third DQS pin 902B. The first write strobe path includes a write strobe buffer 910 that buffers a write strobe, a write strobe receiver 920 that samples the write strobe according to the buffered module signal CK0 and a write delay circuit 925 controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 930A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{ADQS} . The second write strobe path includes the write strobe buffer 910, the write strobe receiver 920, the write delay circuit 825 controlled by the delay signal DS2, and a second write strobe transmitter 930B that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{BDQS} . The first and second write strobe transmitters, 930A and 930B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0074] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 901 and a selected one of the second and third DQS pins 902A and 902B. In the read strobe path, a select circuit 950 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 902A or a read strobe signal received via DQS pin 902B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 960 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 970 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 980 onto the corresponding strobe signal line 324 via the first DQS pin 901.

[0075] FIG. 10 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1001 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1002A that is coupled to a corresponding module DQ line Y_{ADQ} , a third DQ pin 1002B that is coupled to a corresponding module DQ line Y_{BDQ} . The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1001 and the second DQ pin 1002A and a second write data path coupled between the first DQ pin 1001 and the third DQ pin 1002B. The first write data path includes a write data buffer 1010, a write data receiver 1020 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, a write delay circuit 1025 controlled by the delay signal DS2, and a first write data transmitter 1030A that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{ADQ} . The second write data path includes the write data buffer 1010, the write data receiver 1020, the write delay circuit 825 controlled by the delay signal DS2, and a second write data transmitter 1030B that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{BDQ} . The first and second write data transmitters, 1130A and 1130B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0076] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1001 and a selected one of the second and third DQ pins 1002A and 1002B. In the read data path, a select circuit 1050 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1002A or a read data signal received via DQ pin 1002B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1060 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1070 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1080 onto the corresponding data signal line 130 via the first DQ pin 1001.

[0077] FIG. 11 illustrate a delay circuit 1100 that can be used as the delay circuit 825/860/925/960/1025/1060 according to an embodiment. As shown, the delay circuit 1100 includes a plurality of delay stages, such as delay stages 1110, 1120, and 1130, each delaying an input signal by a predetermined amount. The delay circuit 1100 further includes a select circuit 1140 (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

[0078] As stated above, the MCH 101 can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the interface between the data buffers and the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers 118 having programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

[0079] FIG. 12A is a timing diagram for a write operation according to one embodiment. As

shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by a data buffer 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle in which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the data buffer 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer 118 since both these signals are received by the data buffer. Based on such determination, the data buffer 118 can have knowledge of the time interval between t3 and t1, referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer 118 to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

[0080] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t5, the module control circuit 116 outputs one or more enable signals EN at time t6 in response to the read commands. The one or more enable signals are received by an data buffer 118 at time t7, which outputs at time t8 read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle which t3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers 118.

[0081] With knowledge of the time interval between t7 and t5, which should be about the same as the time interval between t3 and t1, i.e., the command-to-enable delay CED, in certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the

one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t9 by the data buffer to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t9 and t5 being consistent with a read latency R.L. associated with the system 100.

[0082] The time interval between t4 and t3, i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the DB Control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the dynamic delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers 671 in circuit 1340, which output the delay signal DS1.

[0083] Thus, as shown in FIG. 14, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1400. In the method, during a write operation, one or more data buffer control signals are received by an data buffer 118 from a module control circuit or module controller 116 (1410). The module controller 116 generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH 101. The one or more data buffer control signals are used to control the data buffer 118. For example, the one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer 118 from the MCH 101 (1420). In one embodiment, upon receiving the one or more first

enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0084] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the data buffer 118. The time interval CED can be used by the data buffer 118 to properly time the transmission of read data to the MCH 101.

[0085] As shown in FIG. 14, a delay signal DS is generated according to the time interval EWD (1430). Concurrent to receiving the write strobe signal DQS, the data buffer 118 also receives a set of write data signals DQ (1440). The received write data signals are transmitted to the subgroup of memory devices (1450), which are selected from the group of memory devices coupled to the data buffer 118 by the one or more first enable signals.

[0086] During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer 114 from the module controller 116 (1460). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the data buffer 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of memory devices (1470). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.

WE CLAIM:

1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors formed along an edge thereof for connecting to respective ones of the C/A signal lines and the plurality of sets of data/strobe signal lines;

memory devices mounted on the PCB and organized in one or more ranks and in a plurality of groups, a respective group of the memory devices including at least one respective memory device in each of the one or more ranks and configured to communicate data/strobe signals with the memory controller via a respective set of the plurality of sets of data/strobe signal lines;

circuits mounted on the PCB and configurable to:

receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines;

generate a module clock signal and module C/A signals in response to the system clock and input C/A signals;

generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals having respective phase relationships with the module clock signal; and

output the module C/A signals to the memory devices, wherein the module C/A signals including one or more chip select signals corresponding, respectively, to the one or more ranks; and

output the plurality of local clock signals to respective groups of the memory devices, wherein a respective local clock signal having a respective programmable phase relationship with the module clock signal is output to a corresponding group of the

memory devices, the corresponding group of the memory devices including at least one corresponding memory device that performs a memory read or write operation by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines in accordance with the respective local clock signal and in response to module C/A signals received by the at least one corresponding memory device.

2. The memory module of claim 1, wherein the circuits include logic and configuration registers programmable by the logic to control the respective programmable phase relationships.

3. The memory module of claim 2, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform a set of operations and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

ABSTRACT OF THE DISCLOSURE

A memory module is operable in a computer system having a memory controller and a system bus and comprises memory devices organized in one or more ranks and in a plurality of groups, and circuits configurable to receive from the memory controller a system clock and input control and address (C/A) signals, generate a module clock signal and module C/A signals in response to the system clock and input C/A signals, generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, and output the plurality of local clock signals to respective groups of the memory devices. A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices that includes at least one corresponding memory device in each of the one or more ranks.

UTILITY PATENT APPLICATION TRANSMITTAL Only for new nonprovisional applications under 37 CFR § 1.53(b))		Attorney Docket No. NL073-20US03	
		First Inventor Hyun Lee	
		Title <i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	
		Electronically filed	November 29, 2022
APPLICATION ELEMENTS See MPEP Chapter 600 concerning utility patent application contents.		Address to: Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450	
1. ☐ Fee Transmittal Form (with duplicate for fee processing) 2. ☐ Applicant claims Small Entity status, see 37 C.F.R. § 1.27 3. ☐ Applicant certifies Micro Entity status, see 37 C.F.R. § 1.29 4. ☒ Specification [Total Pages 31] 5. ☒ Drawing(s) (35 USC § 113) [Total Sheets 22] 6. ☒ Oath and Declaration [Total Pages 2] a. ☐ Newly executed (original or copy) b. ☒ Copy from a prior application (37 CFR § 1.63(d)) 7. ☒ Application Data Sheet, see 37 C.F.R. § 1.76 8. ☐ CD-Rom or CD-R in duplicate, large table or Computer Program (Appendix) ☐ Landscape Table on CD 9. ☐ Nucleotide and/or Amino Acid Sequence Submission (if applicable, all necessary) a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statement verifying identity of above copies		ACCOMPANYING APPLICATION PARTS	
		10. ☐ Assignment Papers (cover sheet & document(s)) Name of Assignee: Netlist, Inc. 11. ☒ 37 CFR § 3.73(c) Statement ☒ Power of Attorney 12. ☐ English Translation Document (if applicable) 13. ☐ Information Disclosure Statement and PTO-1449 ☐ Copies of citations attached 14. ☐ Preliminary Amendment 15. ☐ Return Receipt Postcard (MPEP 503) 16. ☐ Certified Copy of Priority Document(s) (if foreign priority is claimed) 17. ☐ Non-Publication Request under 35 U.S.C. § 122 (b)(2)(B)(i) 18. ☐ Other:	
Note: (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 MUST be included in an Application Data Sheet (ADS). (2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).			
19. CORRESPONDENCE ADDRESS: Customer Number 79141			
Signature		/Jamie J. Zheng/	
		Date	November 29, 2022
Name (Print/Type)		Jamie J. Zheng	
		Registration No. (Attorney/Agent)	51,167

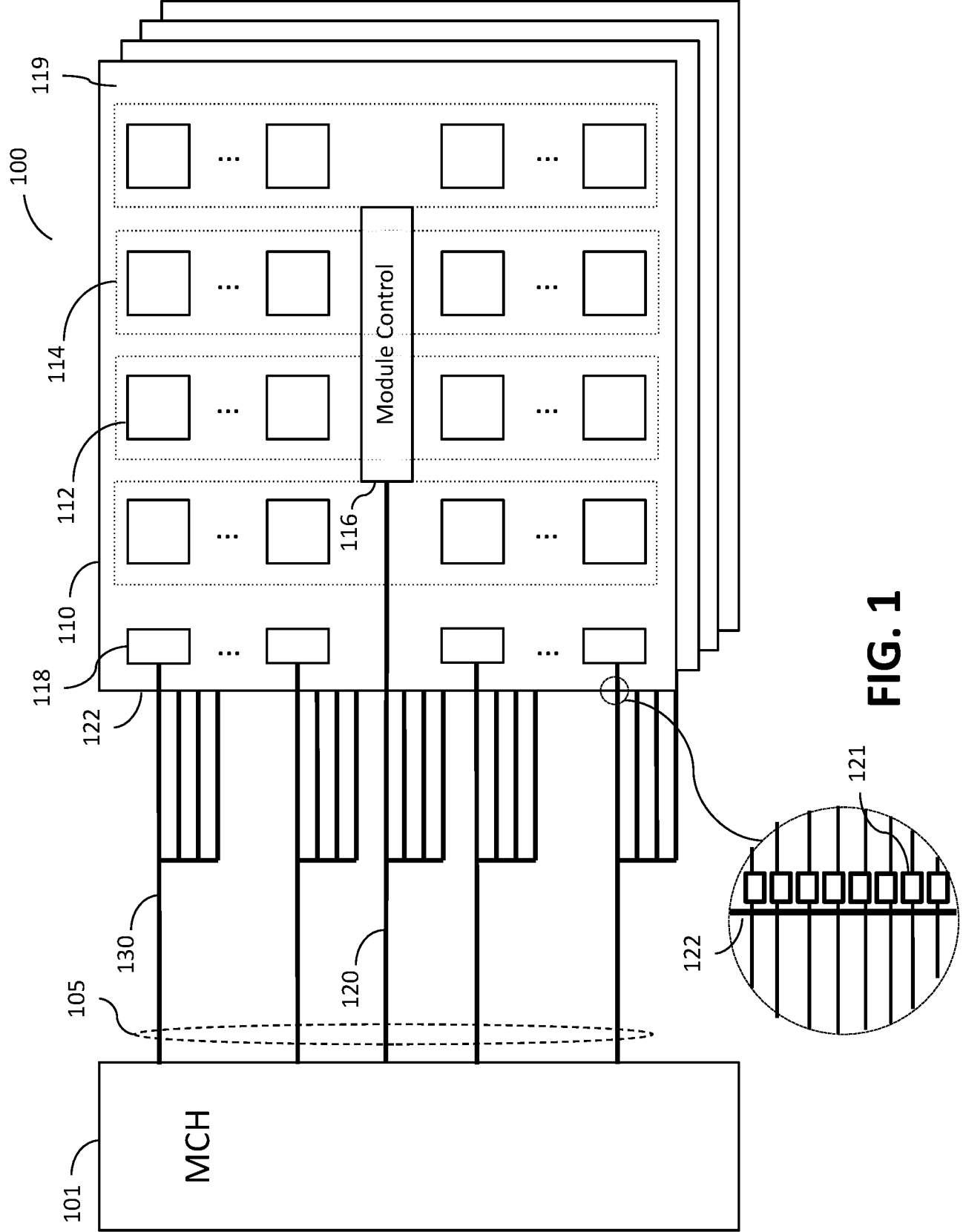


FIG. 1

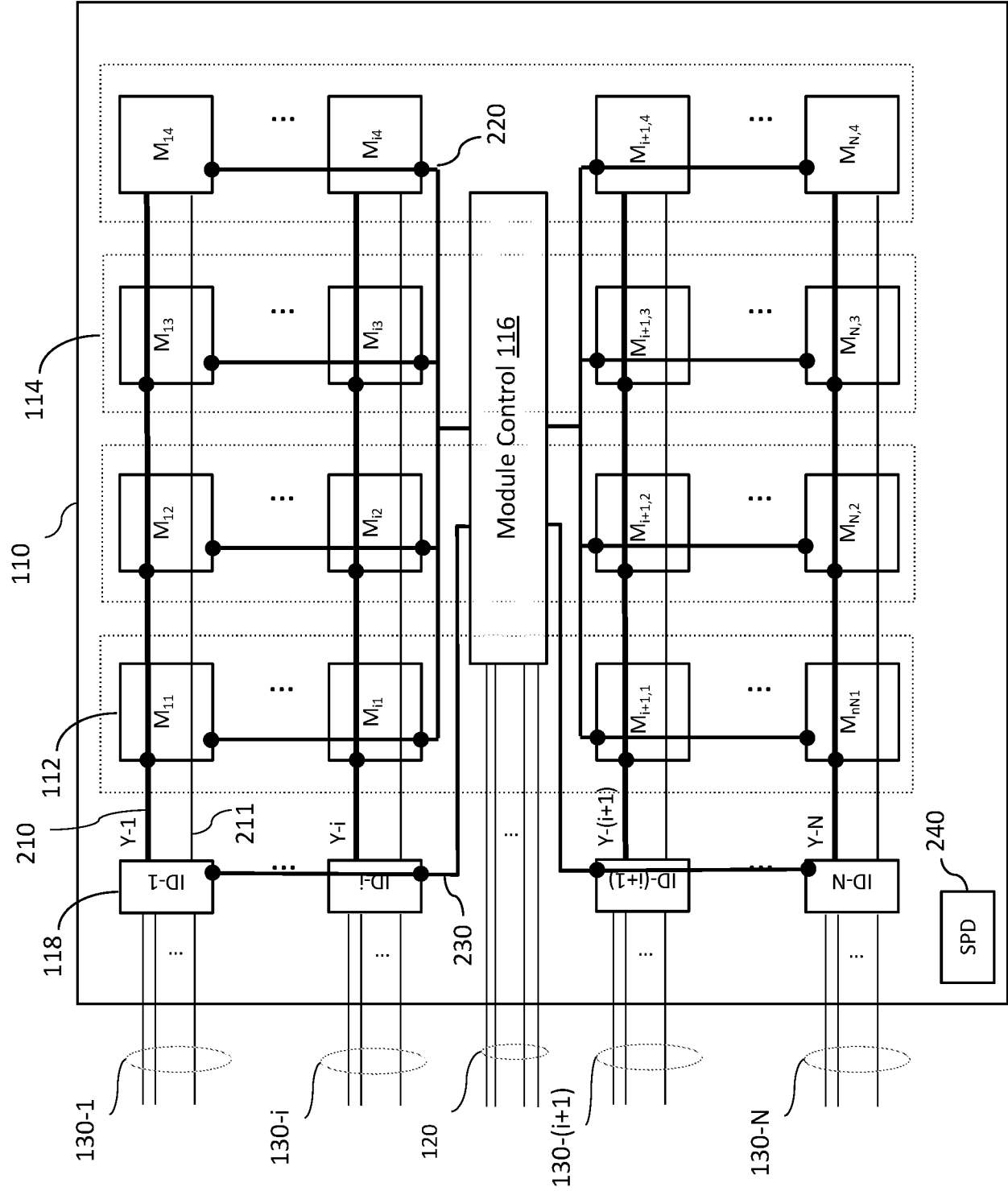


FIG. 2A

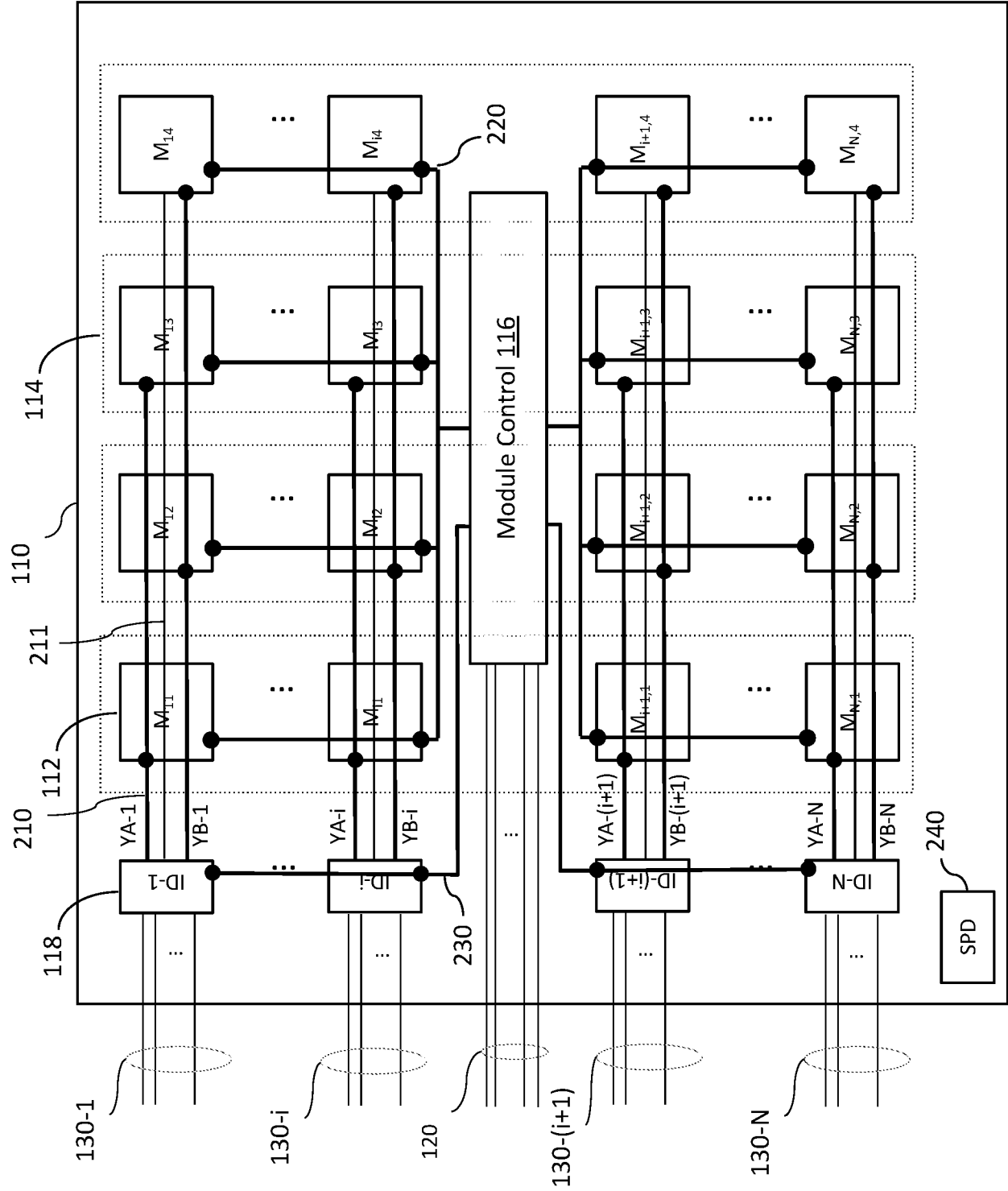


FIG. 2B

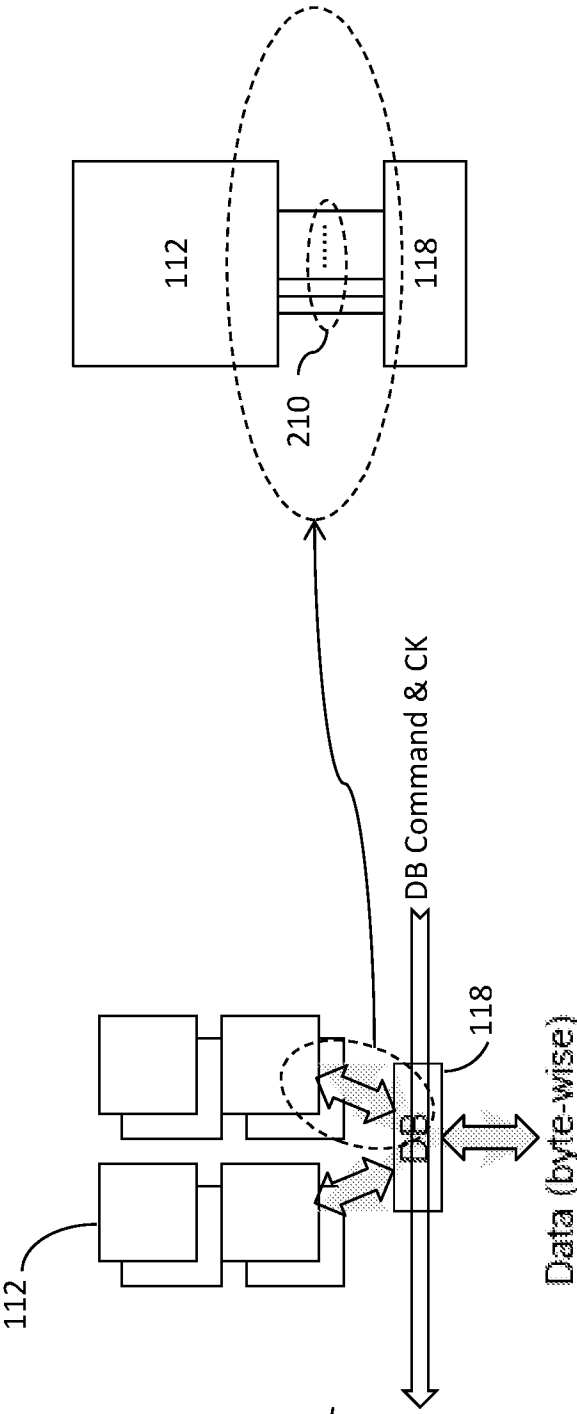
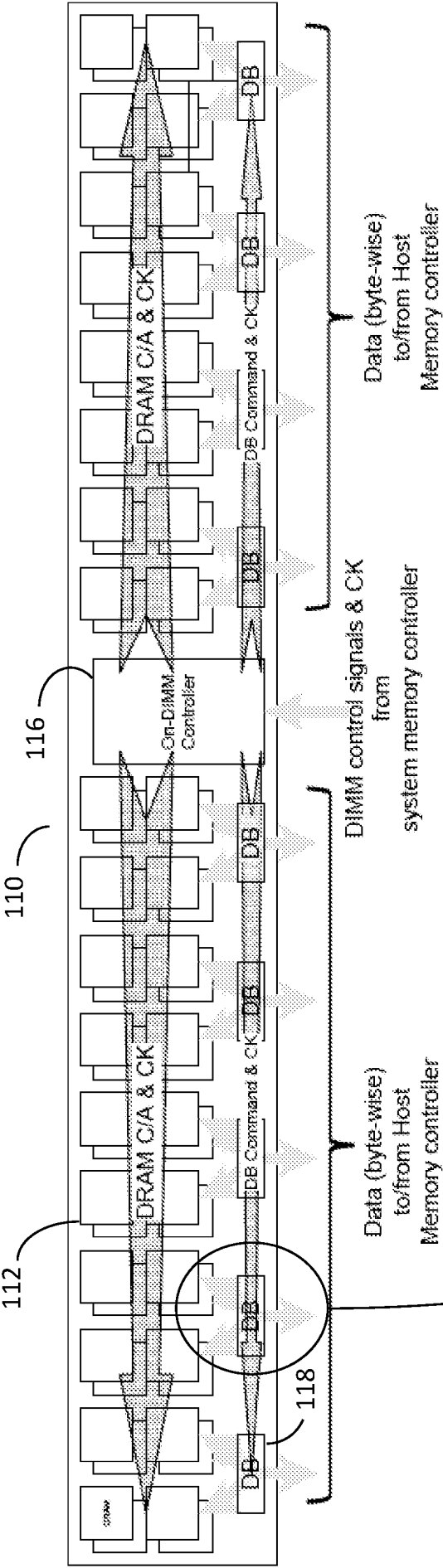


FIG. 2C

FIG. 2D

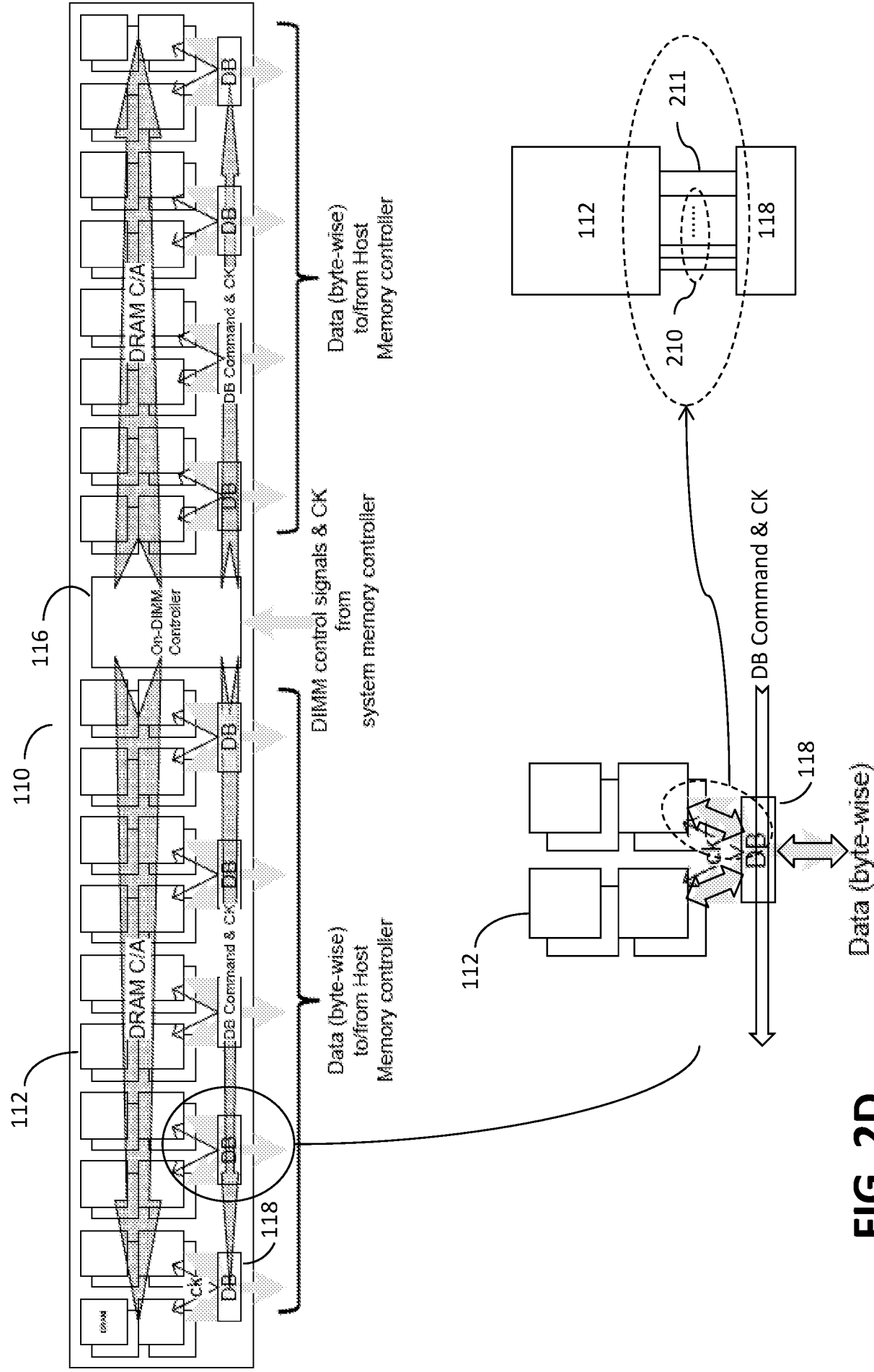
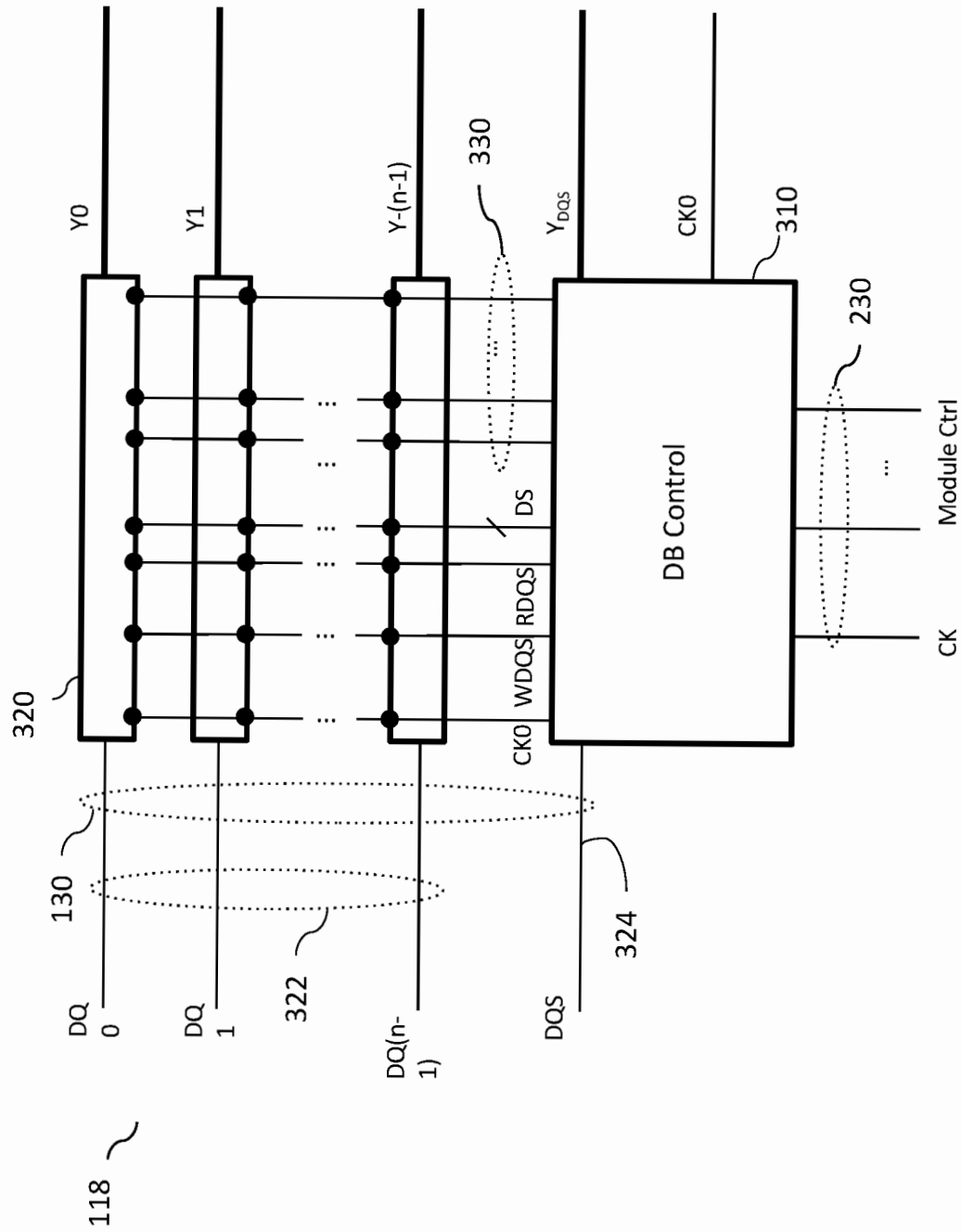


FIG. 2D

**FIG. 3**

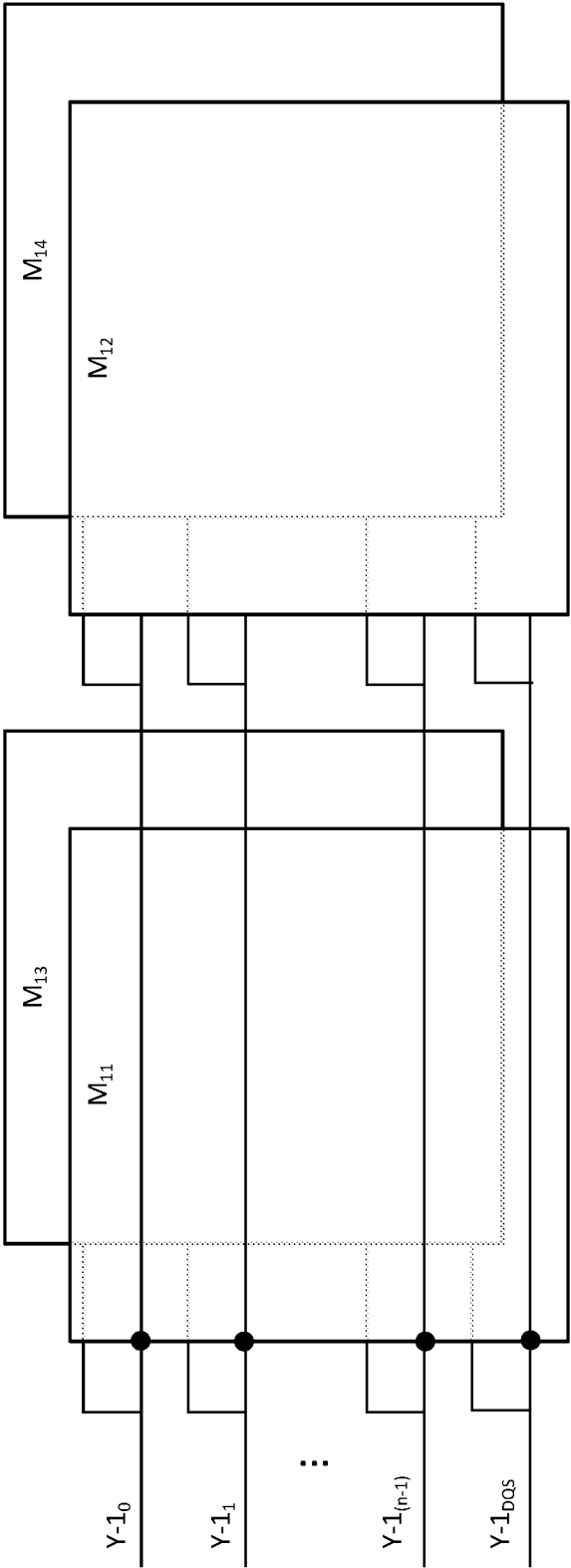


FIG. 4A

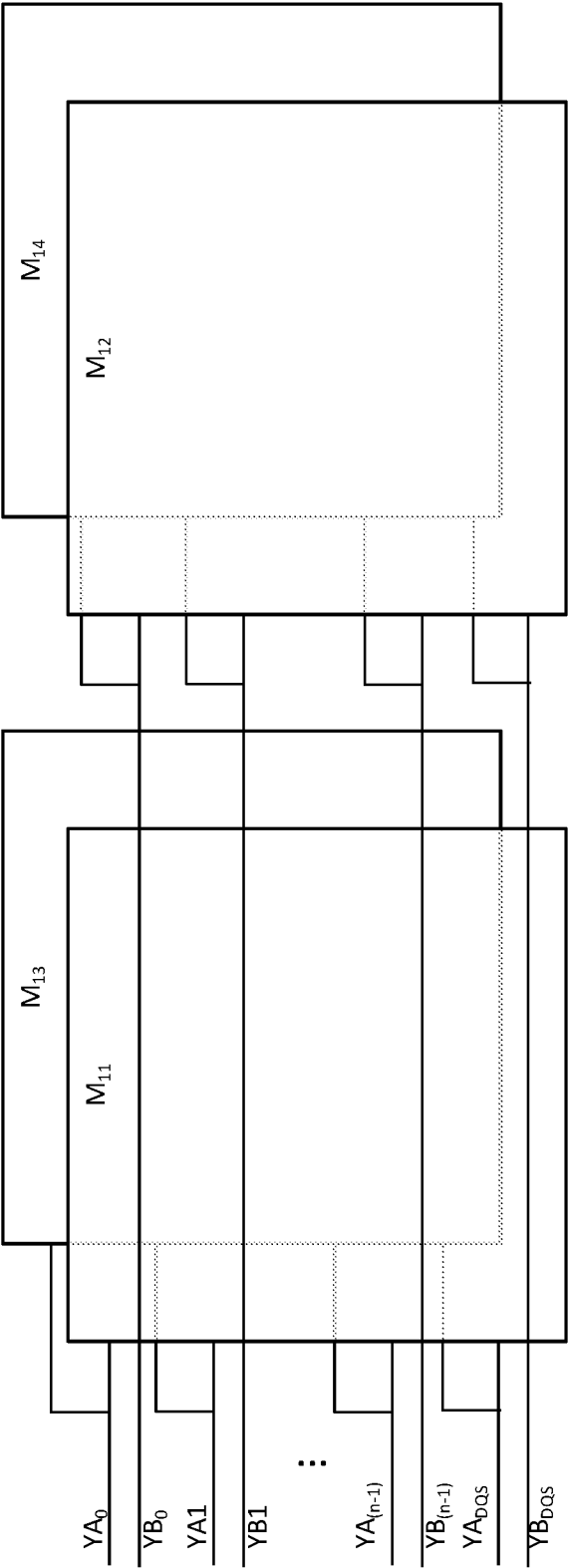


FIG. 4B

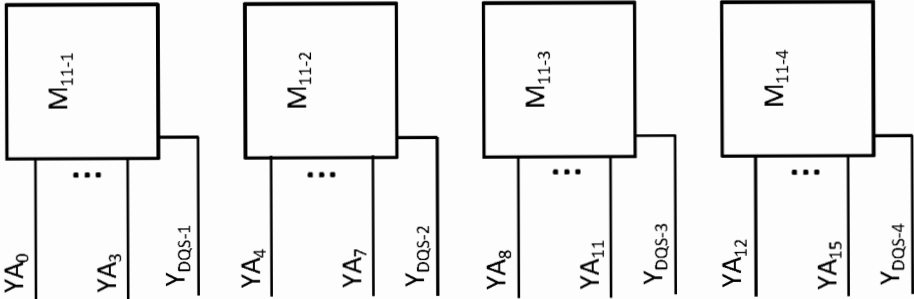


FIG. 5B

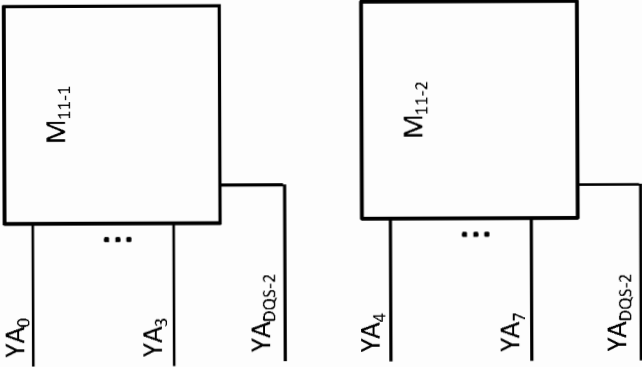


FIG. 5A

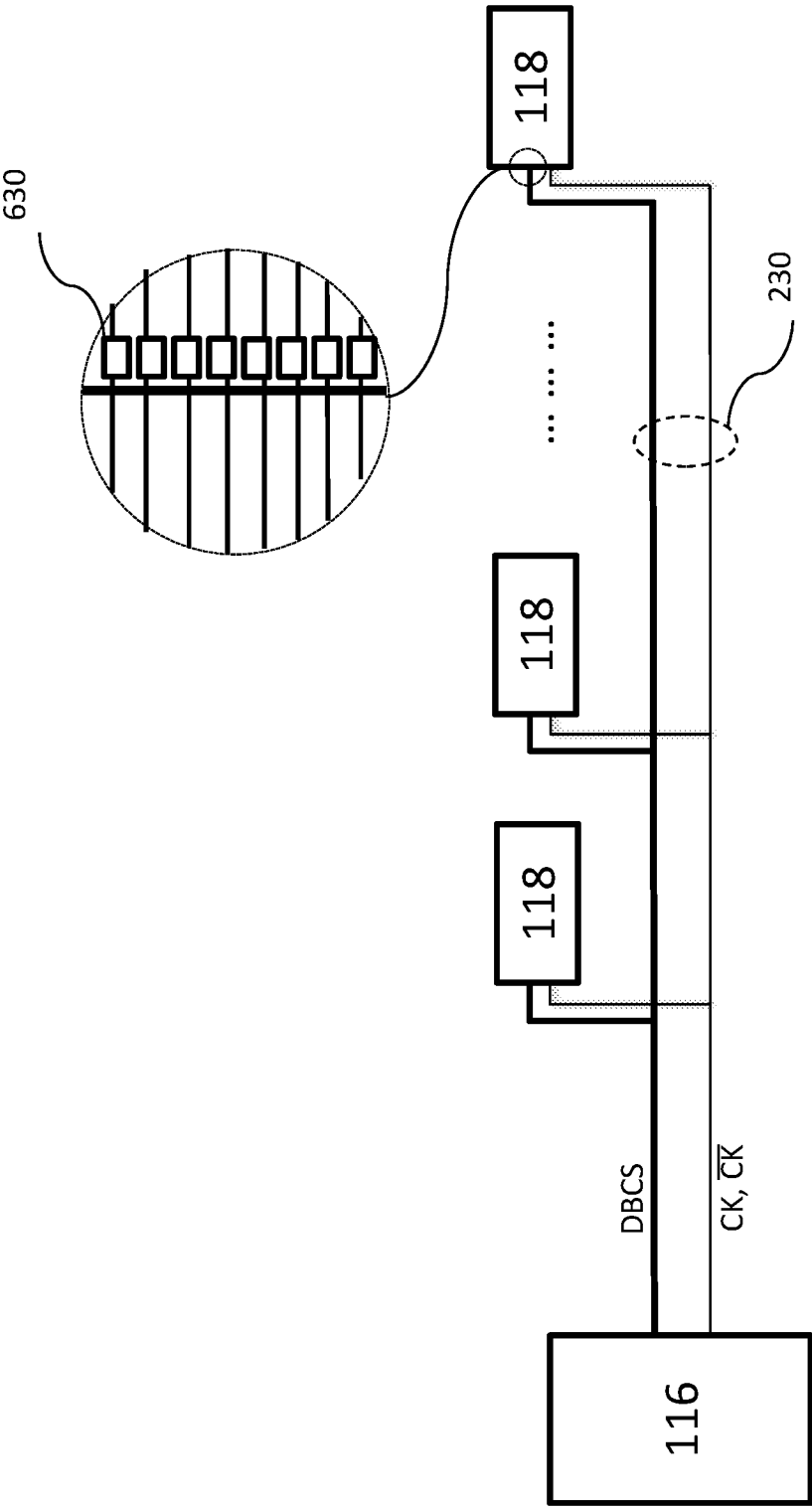
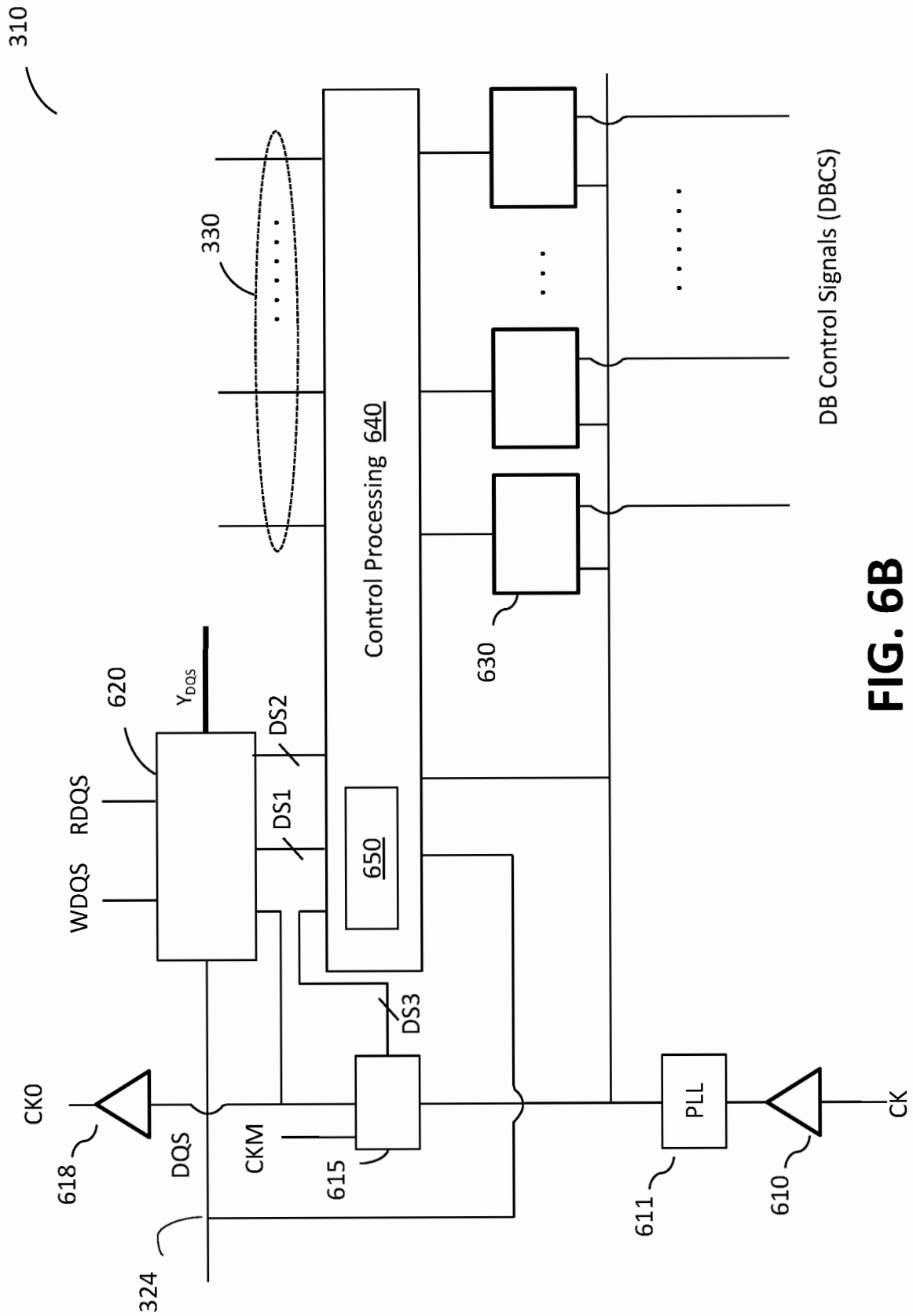


FIG. 6A

**FIG. 6B**

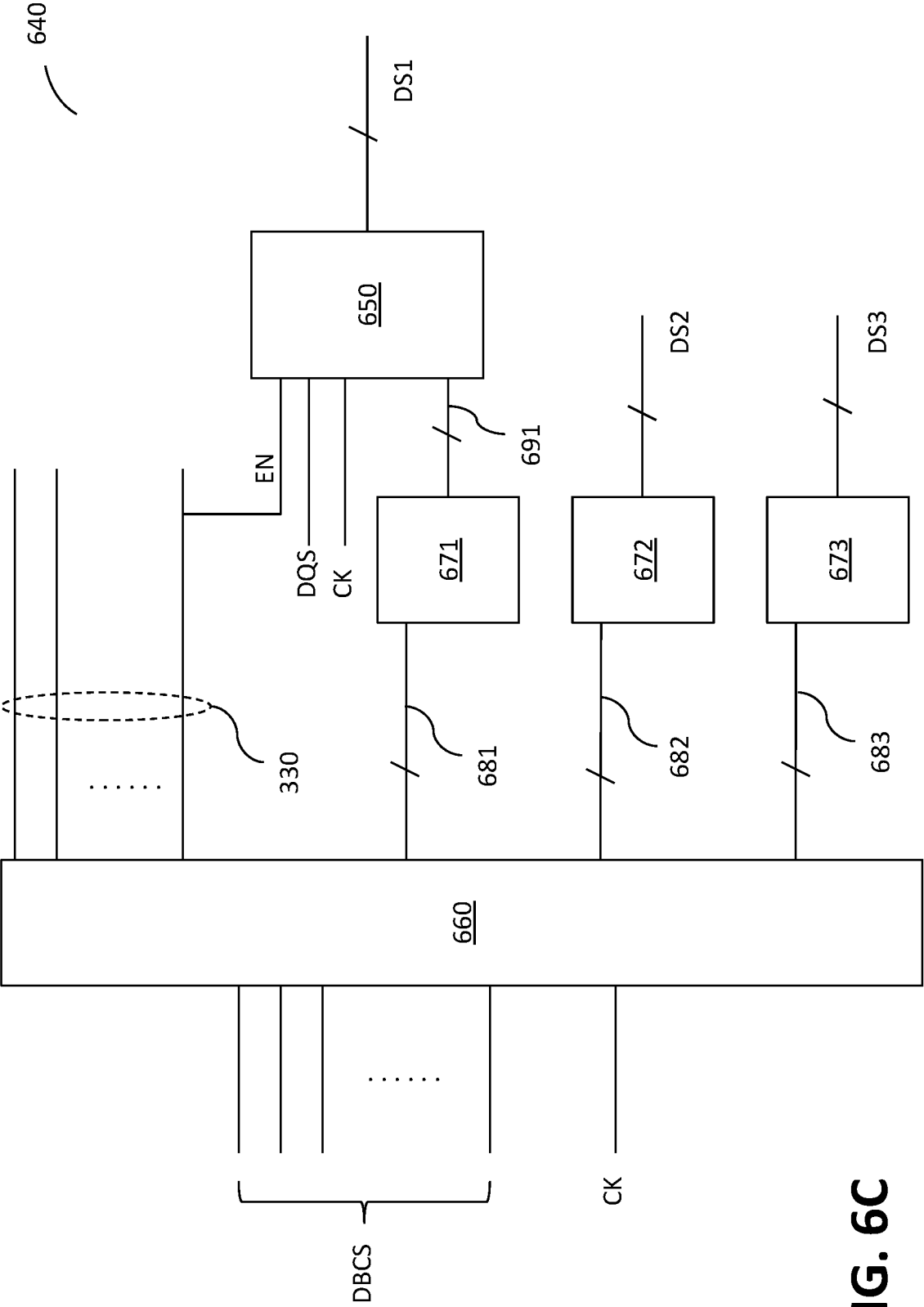


FIG. 6C

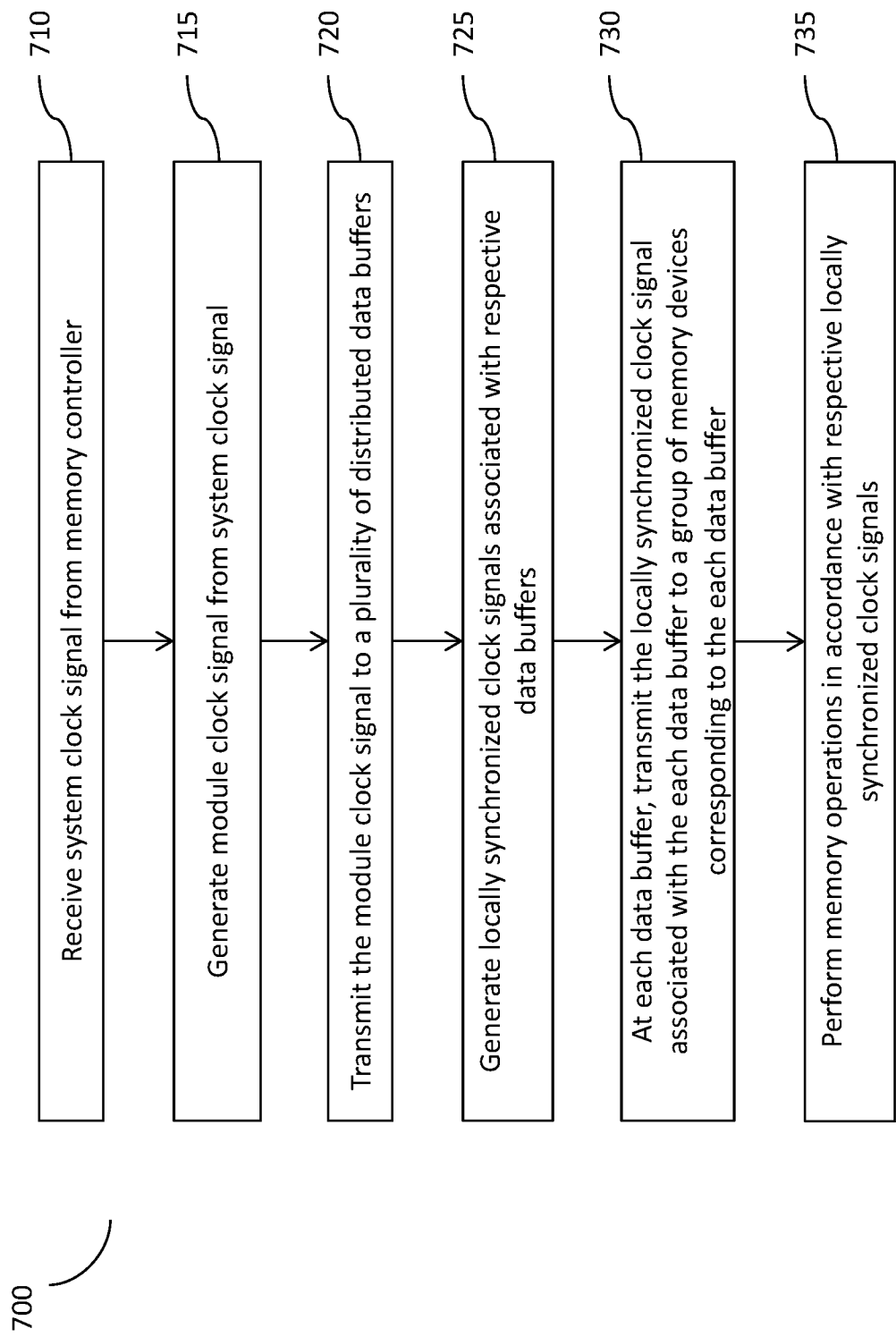


FIG. 7A

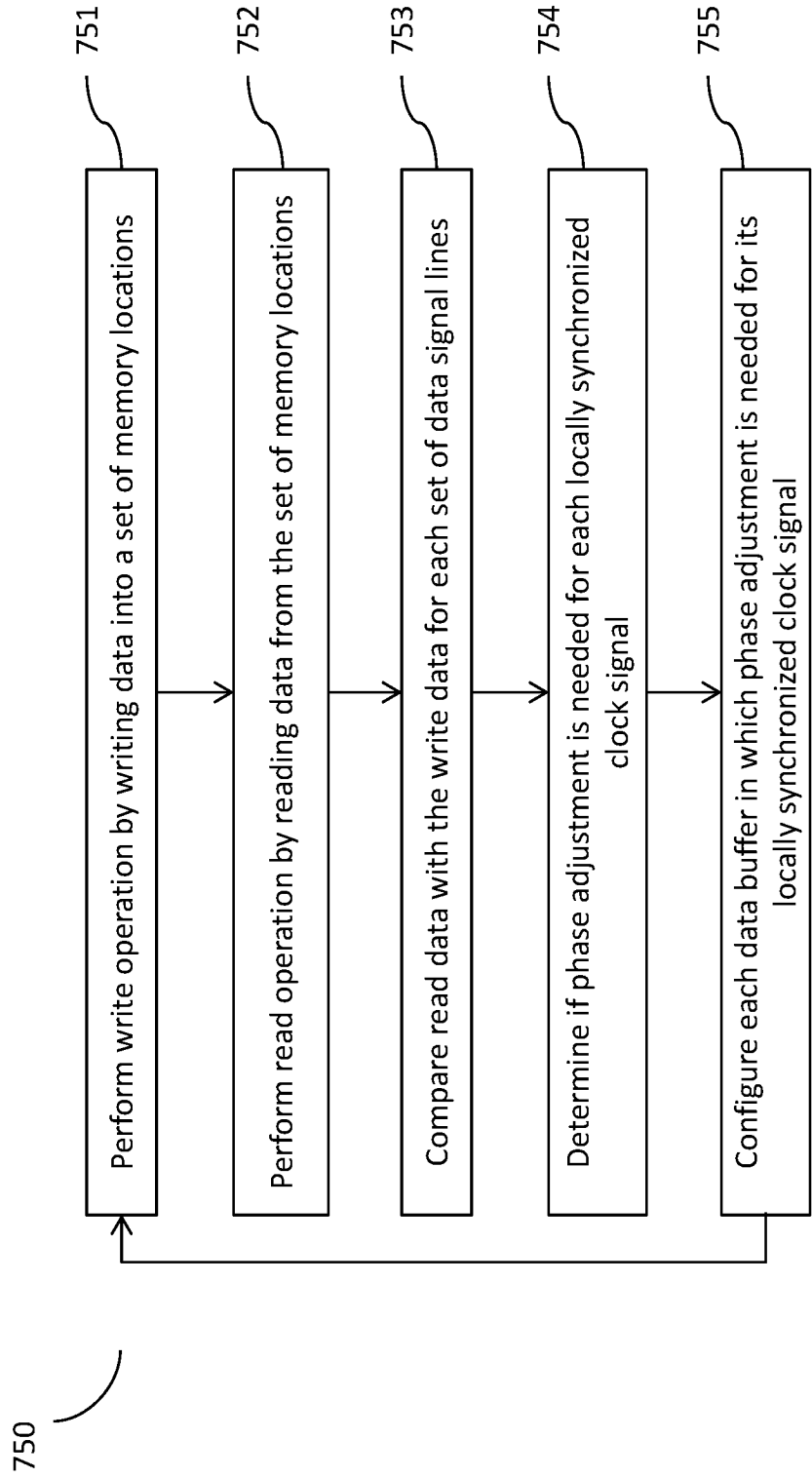


FIG. 7B

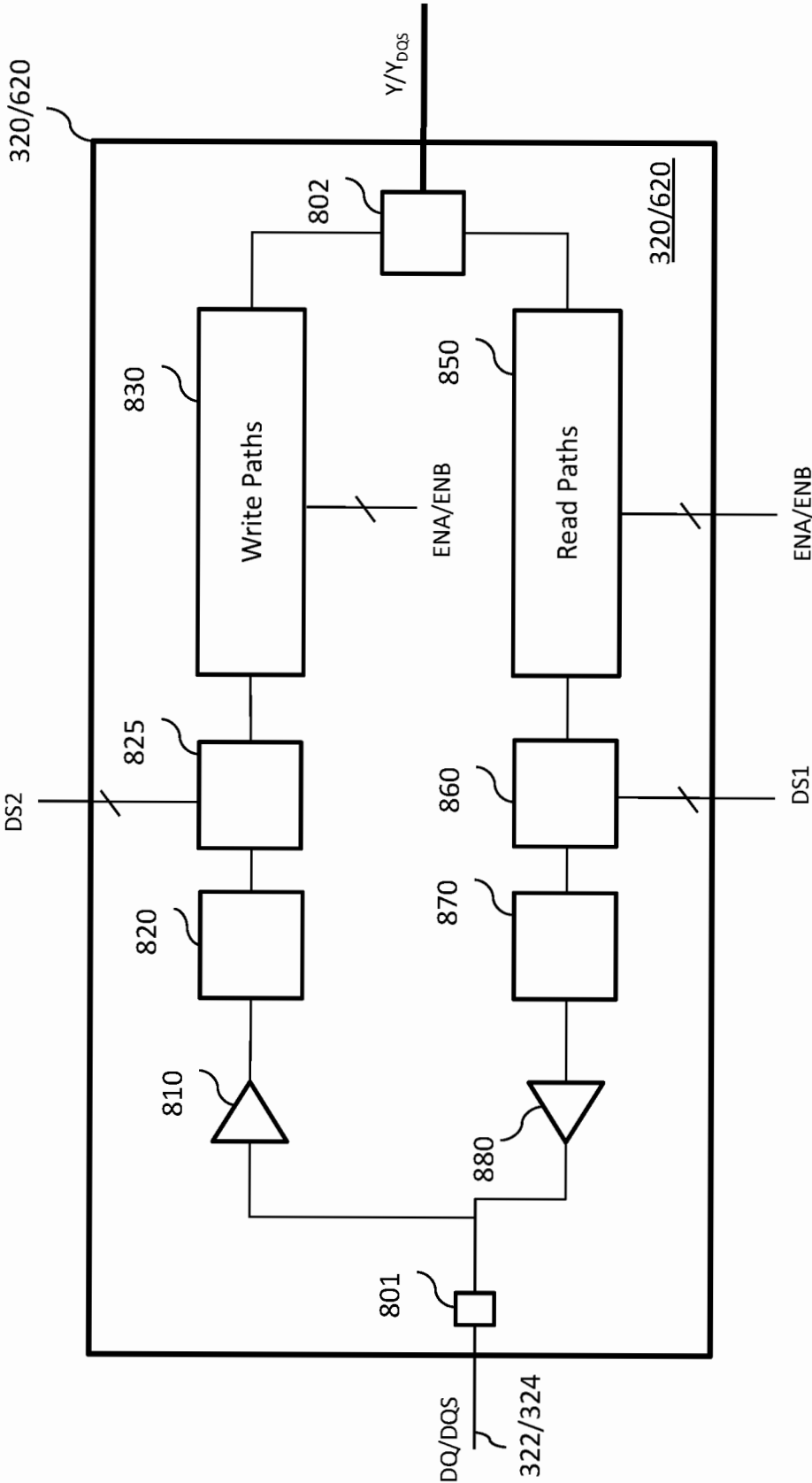


FIG. 8

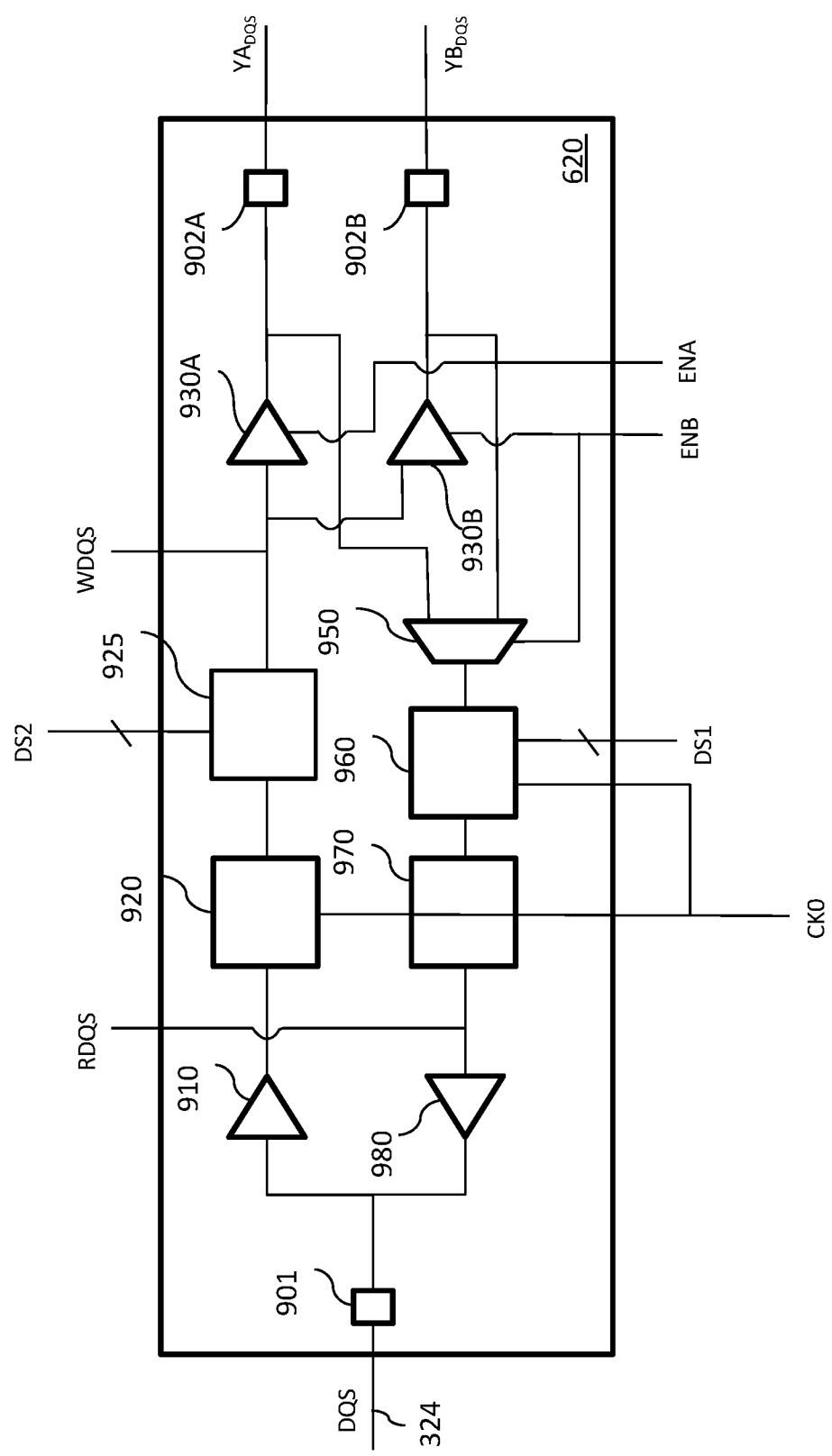


FIG. 9

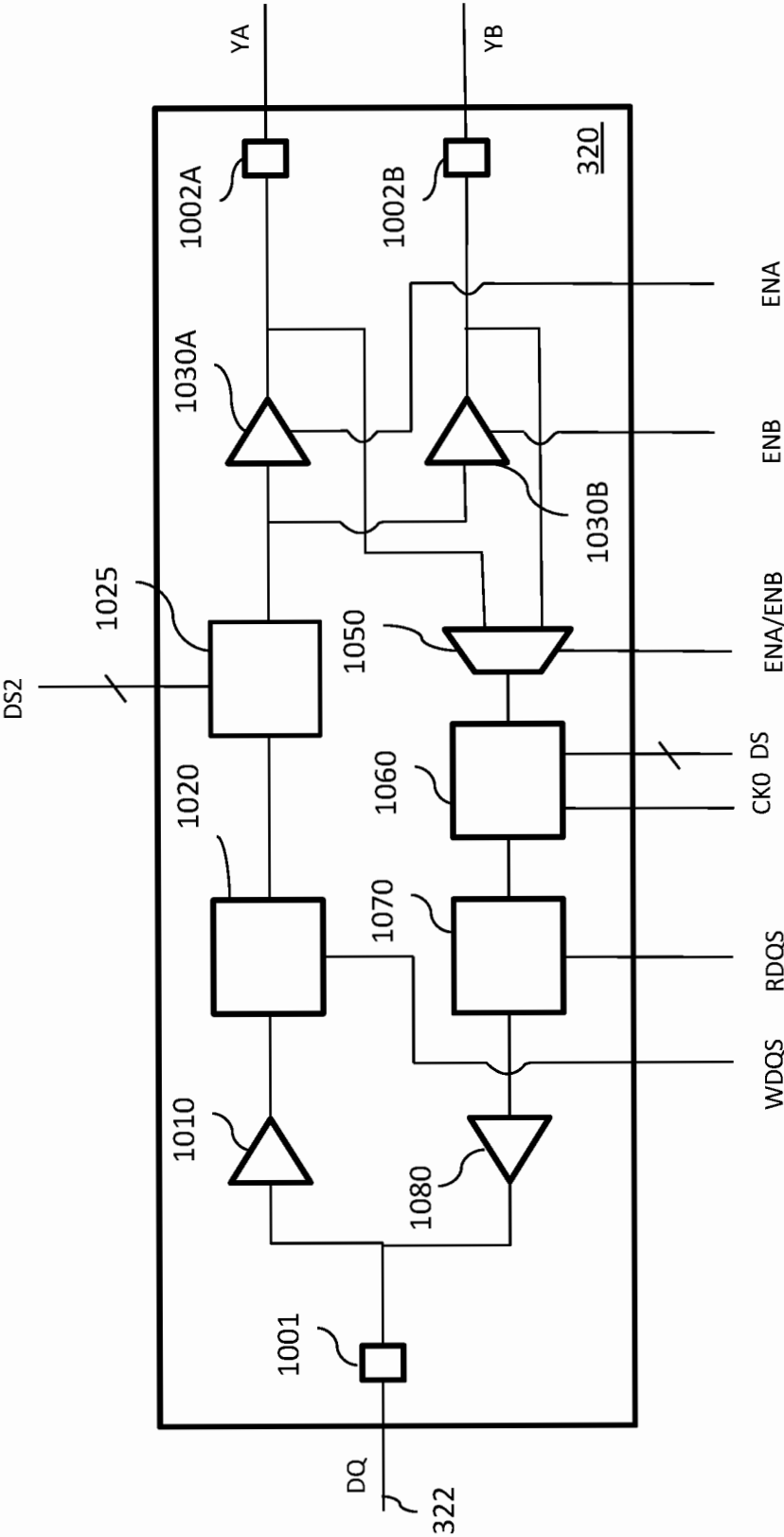


FIG. 10

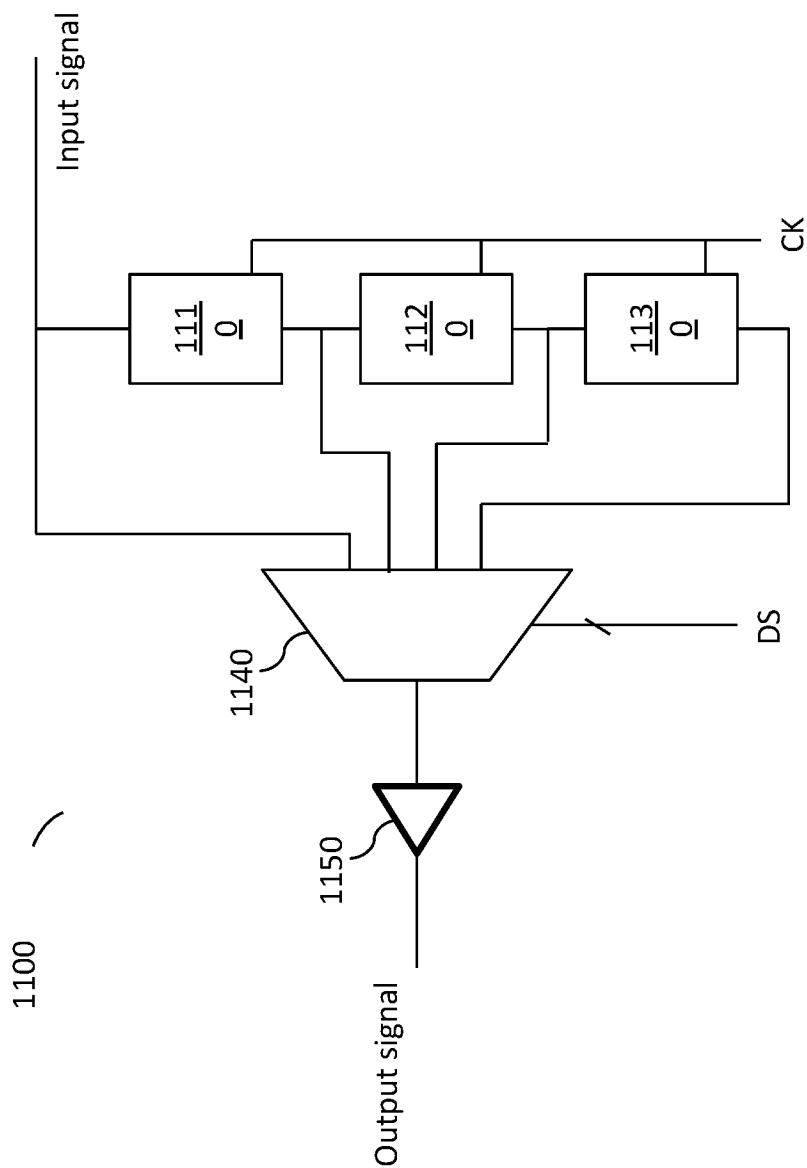


FIG. 11

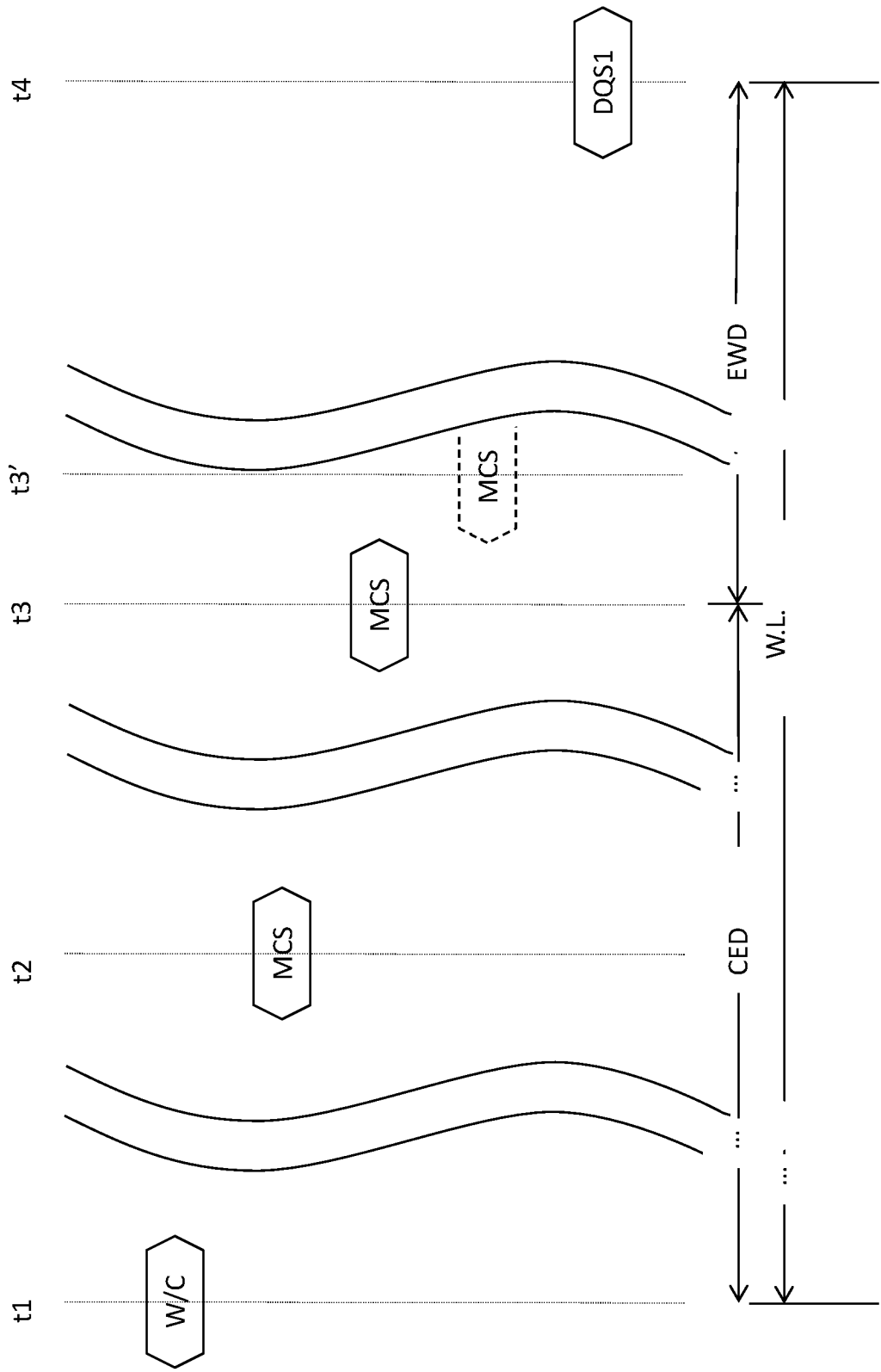


FIG. 12A

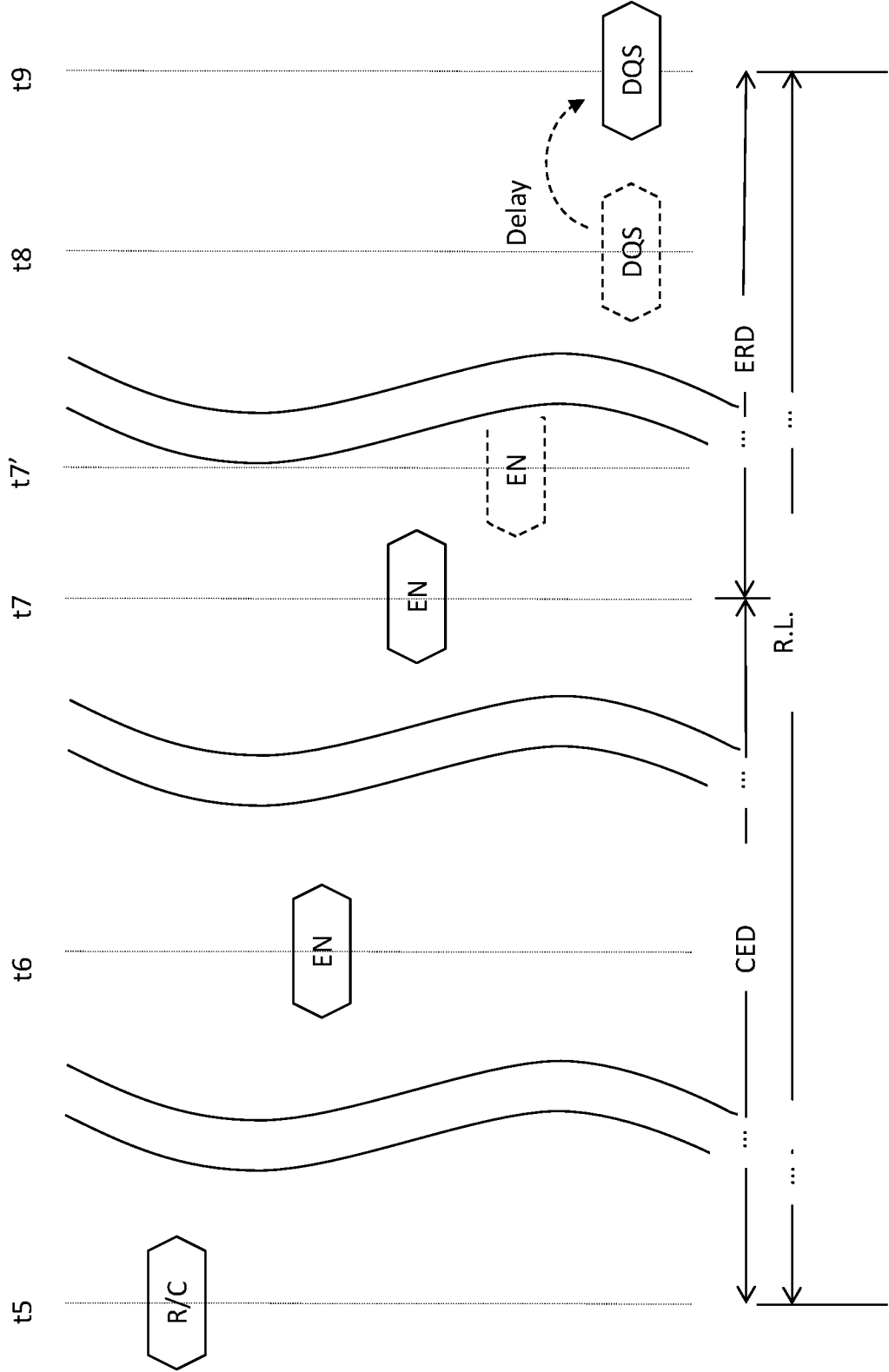


FIG. 12B

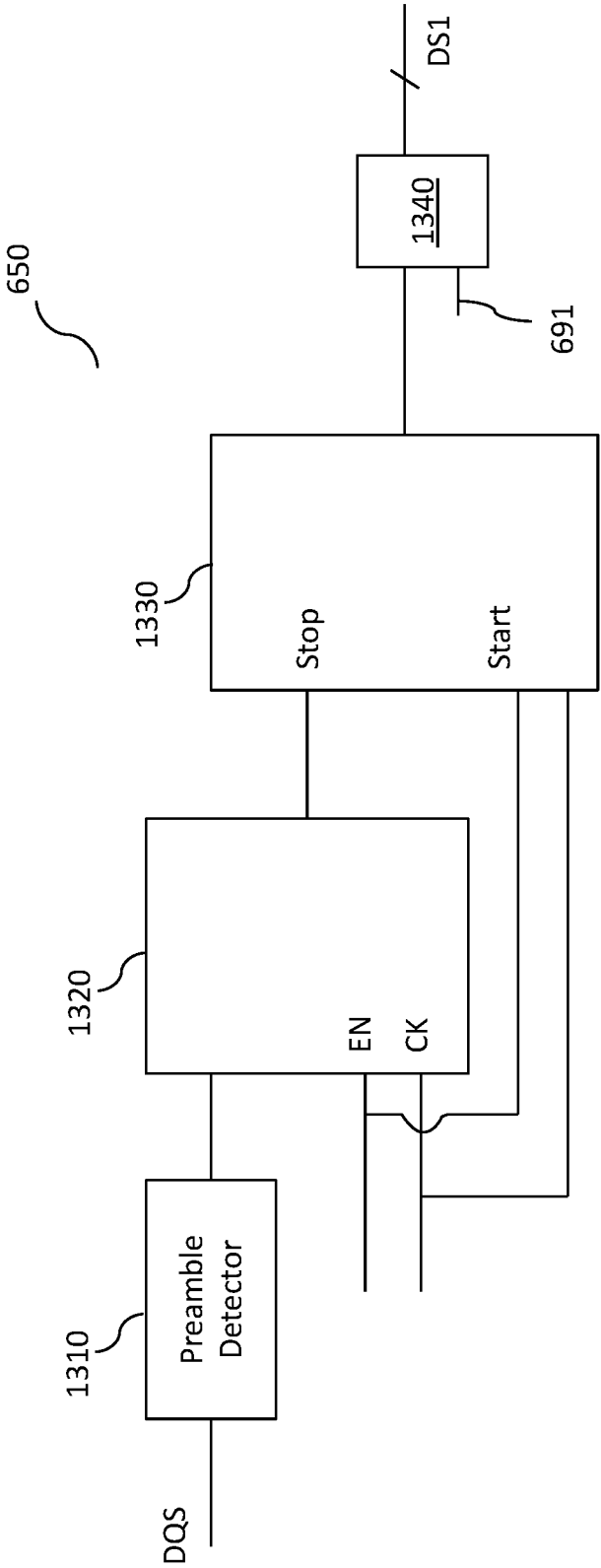


FIG. 13

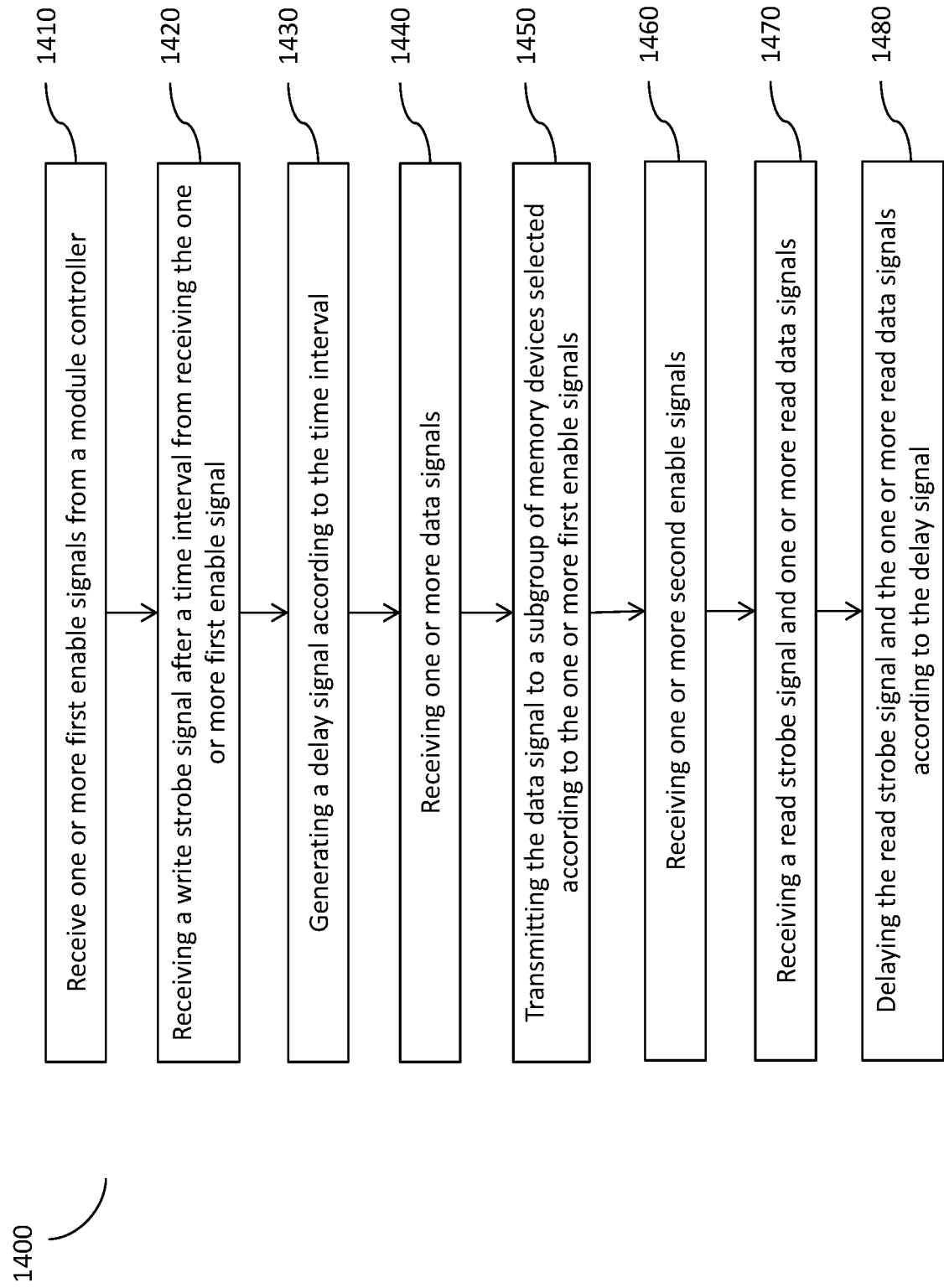


FIG. 14

**COMBINED DECLARATION AND ASSIGNMENT
FOR UTILITY OR DESIGN APPLICATION**

**Title of
Invention**

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

DECLARATION

As a below named inventor, I hereby declare that:

This declaration
is directed to:

☐ The attached application, or

☒ United States application or PCT international application
number 14/445,035 filed on July 28, 2014.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."

ASSIGNMENT

In consideration of the sum of One Dollar (\$1.00) and other good and valuable consideration paid, I hereby sell and assign to

Netlist, Inc.

(Name of Assignee)

of 175 Technology Dr., Suite #150, Irvine, CA 92618

(Address of Assignee)

(hereinafter designated as the Assignee) my entire right, title and interest for the United States as defined in 35 USC 100, for any invention set forth in the above-identified application.

I agree to execute all papers necessary in connection with this application for patent in the USPTO for the invention, and any continuation, divisional or reissue applications thereof and also to execute separate assignments in connection with such applications as the Assignee may deem necessary or expedient.

I agree to execute all papers necessary in connection with any post-grant proceeding which may occur in connection with this application or continuation, divisional or reissue thereof and to cooperate with the Assignee in every way possible in obtaining evidence and going forward with such post-grant proceeding.

I agree to execute all papers and documents and perform any act which may be necessary in connection with claims or provisions of the International Convention for Protection of Industrial Property or similar agreements.

I agree to perform all affirmative acts which may be necessary to obtain a grant of a valid United States patent to the

Assignee.

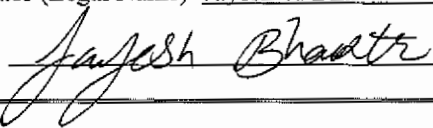
I hereby authorize and request the USPTO to issue any and all Letters Patents of the United States resulting from the application or any continuation, divisional or reissue applications thereof to the Assignee, as Assignee of my entire interest, and covenant that I have not executed, and will not execute, any agreement in conflict herewith.

First Inventor (Legal Name): Hyun Lee

Signature: 

Date: 3/31/2016

First Inventor (Legal Name): Jayesh R. Bhakta

Signature: 

Date: 3/31/2016



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/059,958

RECEIPT DATE / TIME
11/29/2022 09:41:51 PM ET

ATTORNEY DOCKET #
NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY Jamie Zheng

PATENT CENTER # 61233353

FILING DATE -

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun Lee

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 9

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NL073-20US03_ADS.pdf	9	Application Data Sheet	1227 KB
NL073-20US03_Appl_Transmittal.pdf	1	Transmittal of New Application	99 KB
NL073-20US03_Drawings.pdf	22	Drawings-only black and white line drawings	269 KB
NL073-20US03_Oath_Assign.pdf	2	Oath or Declaration filed	73 KB
NL073-20US03_373-Stmnt.pdf	1	Assignee showing of ownership per 37 CFR 3.73	58 KB

NL073- 20US03__Specification.pdf		31	-	300 KB
NL073- 20US03__Specification- SPEC.pdf	(1-28)	28	Specification	287 KB
NL073- 20US03__Specification- CLM.pdf	(29-30)	2	Claims	82 KB
NL073- 20US03__Specification- ABST.pdf	(31-31)	1	Abstract	89 KB
POA__Netlist__General.pdf		1	Power of Attorney	148 KB

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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	HORWICH, Jordan	Confirmation No.:	TBD
Serial No.:	Filed herewith	Art Unit	TBD
Filed:	November 29, 2022	Examiner:	TBD
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION</i>	Attorney Docket No.:	NL073-20US03

STATEMENT UNDER 37 C.F.R. § 3.73(c)

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Netlist, Inc.

states (state) that it is (they are) the assignee(s) of the entire right, title and interest in the patent application/patent identified above by virtue of an assignment from the inventor(s) of the patent application/patent identified above.

The assignment was recorded in the United States Patent and Trademark Office on _____ at Reel _____, Frame _____, or for which a copy thereof is attached.

The undersigned is authorized to act on behalf of the assignee(s).

Date: November 29, 2022

<u>/Jamie J. Zheng/</u>	51,167
Jamie J. Zheng	(Reg. No.)
Customer No. 79141	
(650) 521-4828	

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application a continuation of U.S. Patent Application Ser. No. 17/141,978, filed on January 5, 2021, entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 16/432,700, filed on Jun. 5, 2019 (now U.S. Pat. No. 10,884,923), entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Ser. No. 14/445,035, filed on Jul. 28, 2014 (now U.S. Pat. No. 10,324,841), entitled “Memory Module with Local Synchronization,” which claims priority to U.S. Provisional Patent Application No. 61/859,215, filed July 27, 2013, each of which is incorporated herein by reference in its entirety.

FIELD

[0002] The disclosure herein is related generally to memory modules, and more particularly to memory modules having distributed data buffers.

BACKGROUND

[0003] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a

memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

[0004] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0005] The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0006] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0007] For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules. The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0008] In some conventional memory systems, the memory controllers include leveling

mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0010] FIGS. 2A-2D are each a diagrams illustrating interactions among components in a a memory module according to certain embodiments.

[0011] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0012] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0013] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0014] FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0016] FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

[0017] FIG. 7A is a flowchart illustrating a method of operation for a memory module according

to certain embodiments.

[0018] FIG. 7B is a flowchart illustrating a method of training locally synchronized clock signals according to certain embodiments.

[0019] FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0020] FIG. 9 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0021] FIG. 10 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0022] FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0023] FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0024] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0025] FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

[0026] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g.,

read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

[0027] In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

[0028] FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and

one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

[0029] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0030] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

[0031] Furthermore, the memory devices 112 may include any type of memory devices. For

example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory (SDRAM) devices in compliance with, for example, the Double Data Rate Type 3 (DDR3) or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0032] As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0033] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0034] Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Patent No. 7,532,537, entitled “Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation,” issued on May 12, 2009, the entirety of which is incorporated herein by reference.

[0035] Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers 118, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer control signals are used to convey configuration or module self-test control information from the module control device 116 to the data buffers 118. In one embodiment, the data buffer control signals are transmitted to the data buffers 118 via respective data buffer control signal lines 230. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers 118 via the data buffer control signal lines and decoded/processed at the data buffers.

[0036] Module control device 116 transmits the module C/A signals to the memory devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers 118 via data buffer control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the data buffers 118 share a same set of data buffer control signal lines 230.

[0037] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices $M_{11}, \dots, M_{i1}, M_{i+1,1}, \dots, M_{N1}$, a second rank includes memory devices $M_{12}, \dots, M_{i2}, M_{i+1,2}, \dots, M_{N2}$, and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11}, M_{12}, M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0038] In certain embodiments, the data buffers 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers 118 is associated with the first group of memory devices M_{11}, M_{12}, M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory devices, data buffers ID- i among the data buffers 118 is associated with the i^{th} group of memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be associated with a same group of memory devices, or vice versa.

[0039] In one embodiment, each group or sets of memory devices are coupled to the associated data buffer 118 via a set of module data/strobe lines 210 and one or more clock signal lines 211. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0040] In one embodiment, the data buffers 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each data buffer 118 is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the data buffer control signals. Further, each data buffers 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0041] In one embodiment, the data buffers 118 are distributed across the memory module 110 or the module board 119 in positions corresponding to edge connectors 121 coupled to respective groups of system data/strobe signal lines 130. For example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

[0042] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0043] In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0044] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system 100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the memory devices 112.

[0045] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the

control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

[0046] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

[0047] For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of interface timing training may be required because the system memory controller 101 has very limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0048] Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very significant portion

of a data period. As a result, the clock skew issues cannot simply be addressed by pipelining the data paths.

[0049] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer 118 in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit 116 and the group of locally synchronized components.

[0050] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices 112, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller 101, and/or programmable phase adjustment for one or more control signal from the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data buffers,

as disclosed in commonly-owned U.S. Patent 8,001,434, issued August 16, 2011, entitled “Memory Board with Self-Testing Capability,” which is incorporated herein by reference in its entirety.

[0051] In certain embodiments, operations of the data buffers 118 are controlled by the data buffer control signals from the module control circuit 116, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers 118 to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers 118 are distributed across the memory module, the data buffer control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. Patent Application No. 13/952,599, filed July 27, 2013, entitled “Memory Module with Distributed Data Buffers and Method Of Operation,” the entirety of which is incorporated herein by reference.

[0052] Because the data buffers 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signals lines 230 from the module control device 116 to the farthest positioned data buffers 118, such as data buffer ID-1 and data buffer ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of data

buffer control signals may reach different data buffers 118 at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

[0053] With the data buffers 118 receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the data buffers 118. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used to adjust the timing of the transmission of read data to the MCH 101 during a subsequent read operation, as explained in more detail below.

[0054] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0055] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n

data (DQ) signal lines 322 each for transmitting one of a set of bits $DQ_0, DQ_1, \dots, DQ_{n-1}$, in a data signal and at least one strobe (DQS) signal line 324 for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line Y_{DQS} . When the subsets of memory devices are coupled to the associated data buffer 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

[0056] Each data buffer 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, YA_1, \dots, YA_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each data buffer 118 further includes an DB Control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS} .

[0057] The DB Control circuit 310 also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines 230, and outputs datapath control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit 310 also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective

DQ signal line 322 with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

[0058] In certain embodiments, the DB Control circuit 310 also provides one or more delay signals DS, which is used by the DQ routing circuits 320 to control the timing of read and/or write data output by the data buffer 118, as explained in further detail below. In certain embodiments, the DB Control circuit 310 regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer 118 and the respective group of memory devices form a group of locally synchronized devices on the memory module 110.

[0059] The memory devices 112 in each group of locally synchronized devices are coupled to the data buffer 118 in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices 112 in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer 118 via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M₁₁, M₁₂, M₁₃, and M₁₄ in the first group of memory devices can be coupled to the data buffer ID-1 via a same set of module data lines Y-1₀, Y-1₁, ..., Y-1_{n-1} and module strobe line Y-1_{DQS}. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0060] Alternatively, as shown in FIG. 4B, memory devices M₁₁ and M₁₃, which form a subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YA-1₀, YA-1₁, ..., YA-1_n and module strobe line YA-1_{DQS} and memory devices M₁₂ and M₁₄, which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YB-1₀, YB-1₁, ..., YB-1_n and module strobe line YB-1_{DQS}. Memory

devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board 119. Memory devices coupled to the same data buffers may be placed side-by-side, on opposite sides of the module boards 119, or stacked over each other, and/or over the associated data buffer.

[0061] Multiple memory devices having a data width that is less than a data width of the data buffers 118 may be used in place of one of the memory devices 112, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M_{11-1} and M_{11-2} may be used in place of the memory device M_{11} . Each of the two memory devices M_{11-1} and M_{11-2} has a data width of 4, and together they act like a memory device M_{11} of a data width of 8. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory circuit M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}.

[0062] In another embodiment, as shown in FIG. 5B, four memory devices M_{11-1} to M_{11-4} may be used as the memory device M_{11} . Each of the four memory devices M_{11-1} to M_{11-4} has a data width of 4, and together they act like a memory device M_{11} of a data width of 16. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory device M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}, and so on.

[0063] FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits 630 on each data buffer. In certain embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit 116 uses the DBCS to configure the programmable features in the data buffers 118

based on configuration information received from the MCH 101; in the test mode, the DBCS is used to control the self-test processes run by the data buffers; in the normal operation mode, the DBCS includes the one or more enable signals and one or more ODT signals.

[0064] FIG. 6B illustrates the DB Control circuit 310 in an data buffer 118. As shown, the DB Control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, a phase-locked loop circuit (PLL) 611 that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit 615 for adjusting the phase of the locally synchronized clock CK0, and a clock driver 618 that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to a corresponding system DQS signal line 324 and on another side to corresponding module DQS signal lines YADQS and YBDQS. The DB Control circuit 310 further includes receiver circuits 630 to receive respective ones of the DBCS.

[0065] The DB Control circuit 310 further includes a control processing circuit 640 that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more data path enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the control processing circuit 640, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

[0066] The strobe routing circuit 620 also buffers strobe signals received from either the MCH

101 or the memory devices 112, and outputs either a write strobe WDQS or read strobe RDQS, respectively, to the DQ routing circuits 320. In one embodiment, the DB Control circuit 310 further includes a dynamic delay control circuit 650 that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit 320 and the strobe routing circuit 620.

[0067] In certain embodiments, as shown in FIG. 6C, the control processing circuit 640 includes control logic 660 that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals 330. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals 681 to programmable registers 671 that provides static control of the timing of read data/strobe signals, configuration signals 682 to programmable registers 671 that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals 683 to programmable registers 673 that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

[0068] The programmable registers 671 is coupled via signal lines 691 to the dynamic delay control circuit 650, which also takes as input a DQS signal, clock signal CK, and one of the datapath enable signals EN, and output the delay signal DS1. The dynamic delay control circuit 650 is described in further detail below.

[0069] FIG. 7A is a flowchart illustrating a method 700 performed by the memory module 110 according to certain embodiments. As shown in FIG. 7A, the method includes receiving 710 clock signal together with system C/A signals from the MCH 101, and generating 715 module clock signal CK. The module clock signal CK is transmitted 720 to a plurality of distributed data buffers 118, which generates 725 locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits 730 its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform 735 memory operations in accordance with the respective locally synchronized clock

signals.

[0070] Since the module control circuit 116 generates the DBCS based on signals from the MCH 101, the MCH 101 can train the data buffers 118 to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit 116 generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH 101. As shown in FIG. 7B, a method 750 for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing 751 data to a set of memory locations in the memory devices 112, which is followed by performing a read operation by reading 752 data from the set of memory locations. The read data is compared 753 with the write data to determine 754 if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit 116 agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit 116 do not agree with the corresponding write data bits. The disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers 118 further away from the module control circuit 116 being excessively delayed during transmission, and proper phase adjustment can be programmed 755 into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

[0071] FIG. 8 illustrates a DQ or DQS routing circuit 320 or 620 according to an embodiment. As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 801 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 802 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS}, or YA/YA_{DQS} and YB/YB_{DQS}. The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 810 that buffers write data/strobe, a write data/strobe receiver 820 that samples the write data/strobe, and a write delay circuit 825

controlled by the delay signal DS2. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 830 that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

[0072] The DQS routing circuit further includes a plurality of read paths 850 that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit 860 by an amount controlled by the delay signal DS1, and sampled by a sampler circuit 870. The sampled read data/strobe is transmitted by transmitter 880 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 801.

[0073] FIG. 9 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 901 that is coupled to a corresponding DQS signal line 324, a second DQS pin 902A that is coupled to a corresponding module DQS line Y_{ADQS} , a third DQS pin 902B that is coupled to a corresponding module DQS line Y_{BDQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first DQS pin 901 and the second DQS pin 902A and a second write strobe path coupled between the first DQS pin 901 and the third DQS pin 902B. The first write strobe path includes a write strobe buffer 910 that buffers a write strobe, a write strobe receiver 920 that samples the write strobe according to the buffered module signal CK0 and a write delay circuit 925 controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 930A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{ADQS} . The second write strobe path includes the write strobe buffer 910, the write strobe receiver 920, the write delay circuit 825 controlled by the delay signal DS2, and a second write strobe transmitter 930B that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{BDQS} . The first and second write strobe transmitters, 930A and 930B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0074] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 901 and a selected one of the second and third DQS pins 902A and 902B. In the read strobe path, a select circuit 950 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 902A or a read strobe signal received via DQS pin 902B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 960 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 970 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 980 onto the corresponding strobe signal line 324 via the first DQS pin 901.

[0075] FIG. 10 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1001 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1002A that is coupled to a corresponding module DQ line Y_{ADQ} , a third DQ pin 1002B that is coupled to a corresponding module DQ line Y_{BDQ} . The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1001 and the second DQ pin 1002A and a second write data path coupled between the first DQ pin 1001 and the third DQ pin 1002B. The first write data path includes a write data buffer 1010, a write data receiver 1020 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, a write delay circuit 1025 controlled by the delay signal DS2, and a first write data transmitter 1030A that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{ADQ} . The second write data path includes the write data buffer 1010, the write data receiver 1020, the write delay circuit 825 controlled by the delay signal DS2, and a second write data transmitter 1030B that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{BDQ} . The first and second write data transmitters, 1130A and 1130B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0076] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1001 and a selected one of the second and third DQ pins 1002A and 1002B. In the read data path, a select circuit 1050 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1002A or a read data signal received via DQ pin 1002B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1060 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1070 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1080 onto the corresponding data signal line 130 via the first DQ pin 1001.

[0077] FIG. 11 illustrate a delay circuit 1100 that can be used as the delay circuit 825/860/925/960/1025/1060 according to an embodiment. As shown, the delay circuit 1100 includes a plurality of delay stages, such as delay stages 1110, 1120, and 1130, each delaying an input signal by a predetermined amount. The delay circuit 1100 further includes a select circuit 1140 (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

[0078] As stated above, the MCH 101 can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the interface between the data buffers and the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers 118 having programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

[0079] FIG. 12A is a timing diagram for a write operation according to one embodiment. As

shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by a data buffer 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle in which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the data buffer 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer 118 since both these signals are received by the data buffer. Based on such determination, the data buffer 118 can have knowledge of the time interval between t3 and t1, referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer 118 to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

[0080] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t5, the module control circuit 116 outputs one or more enable signals EN at time t6 in response to the read commands. The one or more enable signals are received by an data buffer 118 at time t7, which outputs at time t8 read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle which t3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers 118.

[0081] With knowledge of the time interval between t7 and t5, which should be about the same as the time interval between t3 and t1, i.e., the command-to-enable delay CED, in certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the

one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t9 by the data buffer to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t9 and t5 being consistent with a read latency R.L. associated with the system 100.

[0082] The time interval between t4 and t3, i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the DB Control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the dynamic delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers 671 in circuit 1340, which output the delay signal DS1.

[0083] Thus, as shown in FIG. 14, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1400. In the method, during a write operation, one or more data buffer control signals are received by an data buffer 118 from a module control circuit or module controller 116 (1410). The module controller 116 generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH 101. The one or more data buffer control signals are used to control the data buffer 118. For example, the one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer 118 from the MCH 101 (1420). In one embodiment, upon receiving the one or more first

enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0084] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the data buffer 118. The time interval CED can be used by the data buffer 118 to properly time the transmission of read data to the MCH 101.

[0085] As shown in FIG. 14, a delay signal DS is generated according to the time interval EWD (1430). Concurrent to receiving the write strobe signal DQS, the data buffer 118 also receives a set of write data signals DQ (1440). The received write data signals are transmitted to the subgroup of memory devices (1450), which are selected from the group of memory devices coupled to the data buffer 118 by the one or more first enable signals.

[0086] During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer 114 from the module controller 116 (1460). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the data buffer 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of memory devices (1470). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.

WE CLAIM:

1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:

a printed circuit board (PCB) having connectors formed along an edge thereof for connecting to respective ones of the C/A signal lines and the plurality of sets of data/strobe signal lines;

memory devices mounted on the PCB and organized in one or more ranks and in a plurality of groups, a respective group of the memory devices including at least one respective memory device in each of the one or more ranks and configured to communicate data/strobe signals with the memory controller via a respective set of the plurality of sets of data/strobe signal lines;

circuits mounted on the PCB and configurable to:

receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines;

generate a module clock signal and module C/A signals in response to the system clock and input C/A signals;

generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals having respective phase relationships with the module clock signal; and

output the module C/A signals to the memory devices, wherein the module C/A signals including one or more chip select signals corresponding, respectively, to the one or more ranks; and

output the plurality of local clock signals to respective groups of the memory devices, wherein a respective local clock signal having a respective programmable phase relationship with the module clock signal is output to a corresponding group of the

memory devices, the corresponding group of the memory devices including at least one corresponding memory device that performs a memory read or write operation by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines in accordance with the respective local clock signal and in response to module C/A signals received by the at least one corresponding memory device.

2. The memory module of claim 1, wherein the circuits include logic and configuration registers programmable by the logic to control the respective programmable phase relationships.

3. The memory module of claim 2, wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and

the memory module in the configuration mode is configurable to perform a set of operations and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

ABSTRACT OF THE DISCLOSURE

A memory module is operable in a computer system having a memory controller and a system bus and comprises memory devices organized in one or more ranks and in a plurality of groups, and circuits configurable to receive from the memory controller a system clock and input control and address (C/A) signals, generate a module clock signal and module C/A signals in response to the system clock and input C/A signals, generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, and output the plurality of local clock signals to respective groups of the memory devices. A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices that includes at least one corresponding memory device in each of the one or more ranks.

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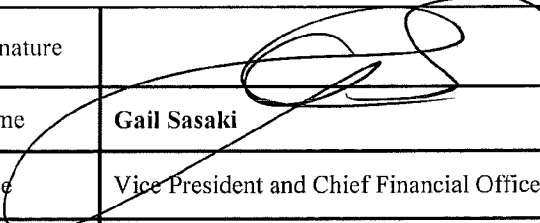
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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
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Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

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Application Information:

Title of the Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		
Attorney Docket Number	NL073-20US03	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

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Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

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Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

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This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

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	Continuation of	17/141978	2021-01-05		
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17/141978	Continuation of	16/432700	2019-06-05	10884923	2021-01-05
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16/432700	Continuation of	14/445035	2014-07-28	10324841	2019-06-18
Prior Application Status	Expired	Remove			
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
14/445035	Claims benefit of provisional	61/859215	2013-07-27		
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.					Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Remove			
Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button.			
Add			

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. ☐ NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.	
--	--

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.		
Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	NETLIST, INC.	
Mailing Address Information For Applicant:		
Address 1	111 Academy Way, Suite 100	
Address 2		
City	Irvine	State/Province
Country	US	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Signature:					Remove
NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).					
This Application Data Sheet must be signed by a patent practitioner if one or more of the applicants is a juristic entity (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, all joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of all joint inventor-applicants.					
See 37 CFR 1.4(d) for the manner of making signatures and certifications.					
Signature	/ Jamie J. Zheng /			Date (YYYY-MM-DD)	2022-11-29
First Name	Jamie J.	Last Name	Zheng	Registration Number	51167
Additional Signature may be generated within this form by selecting the Add button. Add					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	NL073-20US03
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/059,958	11/29/2022		0	NL073-20US03	3	1

CONFIRMATION NO. 5230

79141
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

FILING RECEIPT



0000000051727867

Date Mailed: 12/15/2022

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 17/141,978 01/05/2021 PAT 11,513,955
which is a CON of 16/432,700 06/05/2019 PAT 10,884,923
which is a CON of 14/445,035 07/28/2014 PAT 10,324,841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/13/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/059,958**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
18/059,958

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(i))	3 minus 20 =	0			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =					x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

		(Column 1)	(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY		
		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
AMENDMENT A	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
					TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE		
AMENDMENT B	Total (37 CFR 1.16(i))	*	Minus	**	=	x	=	OR	x	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x	=	OR	x	=
	Application Size Fee (37 CFR 1.16(s))							OR		
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							OR		
					TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE		

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/059,958	11/29/2022	Hyun Lee	NL073-20US03

CONFIRMATION NO. 5230

FORMALITIES LETTER



OC000000051727872

79141
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

Date Mailed: 12/15/2022

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

An application number and filing date have been accorded to the above-identified nonprovisional application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee has not been received.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) has not been received.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- **\$320** basic filing fee.
- **\$160** surcharge.
- **\$700** search fee.
- **\$800** examination fee.
- **\$(0)** previous unapplied payment amount.
- **\$1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be addressed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

REPLY VIA ELECTRONIC FILING

A reply may be submitted via electronic filing only by registered users of the appropriate USPTO patent electronic filing system and must include:

- A separate document having the document description, "**Applicant response to Pre-Exam Formalities Notice**"; and
- the item(s) for which a reply period is set forth above.

For more information about electronic filing, contact the Patent Electronic Business Center (EBC) 1-866-217-9197 or visit <https://www.uspto.gov/ebc>

NOTE: If the reply is not being submitted electronically, a copy of this notice must be included.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/rmohamed/

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Certificate of Mailing or Transmission under 37 CFR 1.8

I hereby certify that this correspondence is being:

1. ☐ Deposited with the United States Postal Service with sufficient postage as first class mail in an envelope addressed to:

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

on _____ ,
Date

2. ☐ Facsimile transmitted to the United States Patent and Trademark Office, or

OR

3. ☒ Transmitted via the USPTO's patent electronic filing system (Patent Center or EFS-Web).

/Jamie J. Zheng/

Signature

April 17, 2023

Date

Jamie J. Zheng

Typed or printed name

51167

Telephone number

Note: Each paper must have its own certificate of mailing or transmission, or this certificate must identify each submitted paper:

US Pat. Appl. No.18/059,958 - Response to Notice to File Missing Parts

A Federal agency may not conduct or sponsor, and a person is not required to respond to, nor shall a person be subject to a penalty for failure to comply with an information collection subject to the requirements of the Paperwork Reduction Act of 1995, unless the information collection has a currently valid OMB Control Number. The OMB Control Number for this information collection is 0651-0031. Public burden for this form is estimated to average 1.8 minutes per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the information collection. Send comments regarding this burden estimate or any other aspect of this information collection, including suggestions for reducing this burden to the Chief Administrative Officer, United States Patent and Trademark Office, P.O. Box 1450, Alexandria, VA 22313-1450 or email InformationCollection@uspto.gov. **DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronically filed April 17, 2023

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	TBA
Filed:	November 29, 2022	Examiner:	TBA
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		
		Attorney Docket No.:	NL073-20US03

**RESPONSE TO THE NOTICE TO FILE MISSING PARTS
OF NONPROVISIONAL APPLICATION**

Mail Stop: MISSING PARTS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Notice to File Missing Parts of Nonprovisional Application mailed December 15, 2022 (hereinafter, the "Notice"), a copy of which is enclosed, submitted herewith are the statutory basic filing, search and examination fees. The surcharge for the late payment of the statutory basic filing fee is also submitted herewith.

Applicant petitions for a two-month extension of time, which extends the due date for responding to the Notice from February 15, 2023, to April 17, 2023 (the first business day after Saturday, April 15, 2023).

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

Respectfully submitted,

Date: April 17, 2023

/ Jamie J. Zheng /
Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51,167
(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/059,958	11/29/2022	Hyun Lee	NL073-20US03

CONFIRMATION NO. 5230

FORMALITIES LETTER



OC000000051727872

79141
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

Date Mailed: 12/15/2022

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

An application number and filing date have been accorded to the above-identified nonprovisional application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee has not been received.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) has not been received.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required (including the fees noted above) are as follows. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- **\$320** basic filing fee.
- **\$160** surcharge.
- **\$700** search fee.
- **\$800** examination fee.
- **\$(0)** previous unapplied payment amount.
- **\$1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be addressed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

REPLY VIA ELECTRONIC FILING

A reply may be submitted via electronic filing only by registered users of the appropriate USPTO patent electronic filing system and must include:

- A separate document having the document description, "**Applicant response to Pre-Exam Formalities Notice**"; and
- the item(s) for which a reply period is set forth above.

For more information about electronic filing, contact the Patent Electronic Business Center (EBC) 1-866-217-9197 or visit <https://www.uspto.gov/ebc>

NOTE: If the reply is not being submitted electronically, a copy of this notice must be included.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/rmohamed/

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION #
18/059,958RECEIPT DATE / TIME
04/18/2023 01:47:10 AM ETATTORNEY DOCKET #
NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY Jamie Zheng

PATENT CENTER # 61936271

AUTHORIZED BY -

CUSTOMER # 79141

FILING DATE 11/29/2022

CORRESPONDENCE
ADDRESS -FIRST NAMED
INVENTOR Hyun Lee

Payment Information

PAYMENT METHOD
CARD / 2819PAYMENT TRANSACTION ID
E20234H248168301PAYMENT AUTHORIZED BY
Jamie Zheng

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1051	SURCHARGE- LATE FILING FEE, SEARCH FEE, EXAMINATION FEE, INVENTOR'S OATH OR DECLARATION, OR APPLICATION FILED WITHOUT AT LEAST ONE CLAIM OR BY REFERENCE	160.00	1	160.00
1111	UTILITY PATENT APPL. SEARCH FEE	700.00	1	700.00
1011	BASIC FILING FEE - UTILITY (PAPER FILING ALSO REQUIRES NON- ELECTRONIC FILING FEE UNDER 1.16(T))	320.00	1	320.00

1311	PATENT APPL. EXAMINATION FEE	800.00	1	800.00
1252	EXTENSION FOR RESPONSE WITHIN SECOND MONTH	638.00	1	638.00
TOTAL AMOUNT:				\$2,618.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #
18/059,958

RECEIPT DATE / TIME
04/18/2023 01:47:10 AM ET

ATTORNEY DOCKET #
NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY Jamie Zheng

PATENT CENTER # 61936271

FILING DATE 11/29/2022

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun Lee

CORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NL073- 20US03__MP__Response.pdf	4	Transmittal Letter	482 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
NL073- 20US03__MP__Response.pdf	114CCB70BC0E7027CF6B63E7308B6B87FB72BF1685C42AE9A ED4A2381EBC985F99E9C2325E0B69F4228522F7024551C4BE3 936BBA76C2D68B93E76E5564AA5F9

by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

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National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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UNITED STATES DEPARTMENT OF COMMERCE
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/059,958	11/29/2022		1980	NL073-20US03	3	1

79141
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

CONFIRMATION NO. 5230

UPDATED FILING RECEIPT



0000000056145853

Date Mailed: 04/20/2023

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 17/141,978 01/05/2021 PAT 11,513,955
which is a CON of 16/432,700 06/05/2019 PAT 10,884,923
which is a CON of 14/445,035 07/28/2014 PAT 10,324,841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/13/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/059,958**

Projected Publication Date: 07/27/2023

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
18/059,958

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(i))	3 minus 20 =	* 0			OR	x =	0
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x =		OR	x =	
Independent (37 CFR 1.16(h))	*	Minus ***	=	x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x =		OR	x =	
Independent (37 CFR 1.16(h))	*	Minus ***	=	x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230
79141	7590	05/31/2023	EXAMINER	
USCH Law, PC			AHMED, ZUBAIR	
3790 El Camino Real #1147				
Palo Alto, CA 94304				
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			05/31/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eofficeaction@appcoll.com
jzheng@uschlaw.com
patents@uschlaw.com

DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to original application filed on 11/29/2022. Claims 1-3 have been examined and are pending in this application.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See

MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

As enumerated in the table below, instant claims 1-3 are anticipated by claims 1, 4, and 17-19 of US Patent 10,324,841.

Status	Instant Application	US Patent 10,324,841
Anticipation	1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors formed along an edge thereof for connecting to respective ones of the C/A signal lines and the plurality of sets of data/strobe signal lines; memory devices mounted on the PCB and organized in one or more ranks and in a plurality of groups, a respective group of the memory devices including at least one respective memory device in each of the one or more ranks and configured to communicate data/strobe signals with the memory controller via a respective set of the plurality of sets of data/strobe signal lines; circuits mounted on the PCB and configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines; generate a module clock signal and module C/A signals in response to the system clock and input C/A signals; generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals	17. A memory module having a width of $N \times n$ bits and configured to communicate with a memory controller via one or more clock signal lines, a set of control signal lines, and N sets of data lines, where N and n are integers larger than one, comprising: a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals; N data buffers each n-bits wide and each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers configured to operatively couple to a respective set of the N sets of data lines, the respective data buffer generating a respective local clock signal in response to the module clock signal, the respective local clock signal being phase-locked with the module clock signal and having a respective programmable delay from the module clock signal; the respective data buffer including data paths and logic controlling the data paths in response to the data buffer control signals, at least one of the data paths including a programmable delay element controlled by the logic;

	having respective phase relationships with the module clock signal; and output the module C/A signals to the memory devices, wherein the module C/A signals including one or more chip select signals corresponding, respectively, to the one or more ranks; and output the plurality of local clock signals to respective groups of the memory devices, wherein a respective local clock signal having a respective programmable phase relationship with the module clock signal is output to a corresponding group of the memory devices, the corresponding group of the memory devices including at least one corresponding memory device that performs a memory read or write operation by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines in accordance with the respective local clock signal and in response to module C/A signals received by the at least one corresponding memory device.	and memory devices including N groups of memory devices corresponding to respective ones of the N data buffers, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data and data strobes with the memory controller via the data paths in the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal. 18. The memory module of claim 17, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between respective groups of the N groups of memory devices and the edge connectors.
Anticipation	2. The memory module of claim 1, wherein the circuits include logic and configuration registers programmable by the logic to control the respective programmable phase relationships.	19. The memory module of claim 17, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a programmable delay circuit to

		delay the local clock signal by the programmable delay, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.
Anticipation	3. The memory module of claim 2, wherein: the memory module is operable in at least a normal operation mode and a configuration mode; the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and the memory module in the configuration mode is configurable to perform a set of operations and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.	1. A memory module operable in a system with a memory controller, the system operating according to a system clock, the memory module having a data width of (N×n) bits, where N and n are integers larger than one, comprising: N data buffers each having a data width of n-bits, each of the N data buffers outputting a respective local clock signal; memory devices including a plurality of groups of memory devices, each respective group of memory devices configured to communicate data with the memory controller via a respective one of the N data buffers, the each respective group of memory devices receiving and transmitting data signals in accordance with the local clock signal output by the respective one of the data buffers; and a module control device configured to receive command/address (C/A) signals and a system clock from the memory controller and to output module C/A signals, data buffer control signals and a module clock signal, the module clock signal being derived from the system clock signal, the module control device transmitting the module C/A signals to at least some of the memory devices, the module control device transmitting the data buffer control signals and the module clock signal to the data buffers; wherein the module control device does not transmit

		any clock signal to the memory devices; and wherein the each of the N data buffers is configured to derive the respective local clock signal from the module clock signal received from the module control device. 4. The memory module of claim 3, wherein the respective one of the N data buffers includes one or more programmable registers controlling the programmable delay, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective one of the N data buffers is configured to program the one or more programmable registers during the configuration mode.
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As enumerated in the table below, instant claims 1-3 are anticipated by claims 1-3 and 6 of US Patent 10,884,923.

Status	Instant Application	US Patent 10,884,923
Anticipation	1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors formed along an edge thereof for connecting	1. A memory module configured to couple to a memory controller via one or more clock signal lines, a set of control signal lines and N sets of n-bit wide data/strobe signal lines, where N and n are integers larger than one, comprising: a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and

	to respective ones of the C/A signal lines and the plurality of sets of data/strobe signal lines; memory devices mounted on the PCB and organized in one or more ranks and in a plurality of groups, a respective group of the memory devices including at least one respective memory device in each of the one or more ranks and configured to communicate data/strobe signals with the memory controller via a respective set of the plurality of sets of data/strobe signal lines; circuits mounted on the PCB and configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines; generate a module clock signal and module C/A signals in response to the system clock and input C/A signals; generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals having respective phase relationships with the module clock signal; and output the module C/A signals to the memory devices, wherein the module C/A signals including one or more chip select signals corresponding, respectively, to the one or more ranks; and output the plurality of local clock signals to respective groups of the memory devices, wherein a	control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals; N data buffers configurable to receive the module clock signal and the data buffer control signals from the control circuit, wherein a respective data buffer of the N data buffers is configurable to operatively couple to a respective set of n-bit wide data/strobe signal lines, the respective data buffer including a n-bit wide data path and logic configured to control the data path in response to the data buffer control signals, the n-bit wide data path including at least one programmable delay element, wherein the respective data buffer is further configurable to generate a respective local clock signal in response to the module clock signal, and wherein the respective local clock signal is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal; and memory devices organized in one or more ranks and including N groups of memory devices corresponding to respective ones of the N data buffers, each group of memory devices including at least one memory device in each of the one or more ranks, wherein a respective group of the N groups of memory devices is configurable to receive the module C/A signals from the control circuit and a
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	respective local clock signal having a respective programmable phase relationship with the module clock signal is output to a corresponding group of the memory devices, the corresponding group of the memory devices including at least one corresponding memory device that performs a memory read or write operation by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines in accordance with the respective local clock signal and in response to module C/A signals received by the at least one corresponding memory device.	respective local clock signal from a corresponding data buffer of the N data buffers, and wherein the respective group of the N groups of memory devices is further configurable to communicate data and data strobe signals with the memory controller via the n-bit wide data path in the corresponding data buffer and in response to the module C/A signals from the control circuit and the respective local clock signal from the corresponding data buffer. 2. The memory module of claim 1, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between the N groups of memory devices and the edge connectors.
Anticipation	2. The memory module of claim 1, wherein the circuits include logic and configuration registers programmable by the logic to control the respective programmable phase relationships.	3. The memory module of claim 2, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a programmable delay circuit to delay the local clock signal by the programmable delay, and a clock driver to drive the local clock signal to a

		corresponding group of the memory devices.
Anticipation	3. The memory module of claim 2, wherein: the memory module is operable in at least a normal operation mode and a configuration mode; the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and the memory module in the configuration mode is configurable to perform a set of operations and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.	6. The memory module of claim 5, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer the data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective data buffer is configured to program the one or more programmable registers during the configuration mode.

As enumerated in the table below, instant claims 1-3 are anticipated by claims 1, 8 and 10 of US Patent 11,513,955.

Status	Instant Application	US Patent 11,513,955
Anticipation	1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors formed along an edge thereof for connecting to respective ones of the C/A signal lines and the plurality of sets of data/strobe signal lines;	1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising: a module control circuit configurable to receive from the memory controller a system clock via the one or more clock signal lines and input address and control (C/A) signals via the set of C/A signal lines, and to output a module

	memory devices mounted on the PCB and organized in one or more ranks and in a plurality of groups, a respective group of the memory devices including at least one respective memory device in each of the one or more ranks and configured to communicate data/strobe signals with the memory controller via a respective set of the plurality of sets of data/strobe signal lines; circuits mounted on the PCB and configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines; generate a module clock signal and module C/A signals in response to the system clock and input C/A signals; generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals having respective phase relationships with the module clock signal; and output the module C/A signals to the memory devices, wherein the module C/A signals including one or more chip select signals corresponding, respectively, to the one or more ranks; and output the plurality of local clock signals to respective groups of the memory devices, wherein a respective local clock signal having a respective programmable phase	clock signal and module C/A signals in response to the system clock and the input C/A signals; memory devices organized in one or more ranks; and buffer circuits configurable to receive the module clock signal from the module control circuit and to generate a plurality of local clock signals, wherein a respective local clock signal of the plurality of local clock signals is output by a respective buffer circuit to a respective group of the memory devices, the respective group of the memory devices including at least one respective memory device in each of the one or more ranks, the respective local clock signal having a respective programmable phase relationship with the module clock signal, the buffer circuits including logic and configuration registers, wherein respective configuration registers included in the respective buffer circuit are programmable by the logic to control the respective programmable phase relationship; and wherein: the memory module is operable in at least a normal operation mode and a configuration mode; the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to a memory read or write command received via the C/A signal lines; the memory module in the configuration mode is configurable to perform a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices
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	relationship with the module clock signal is output to a corresponding group of the memory devices, the corresponding group of the memory devices including at least one corresponding memory device that performs a memory read or write operation by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines in accordance with the respective local clock signal and in response to module C/A signals received by the at least one corresponding memory device.	and at least one read operation to read from the set of memory locations; and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations. 10. The memory module of claim 1, further comprising a printed circuit board, wherein the module control circuit, the buffer circuits and the memory devices are mounted on the printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and wherein the buffer circuits are distributed along the edge of the printed circuit board and between the memory devices and the edge connectors.
Anticipation	2. The memory module of claim 1, wherein the circuits include logic and configuration registers programmable by the logic to control the respective programmable phase relationships.	8. The memory module of claim 1, wherein the respective buffer circuit further includes a respective phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a respective programmable delay circuit to add a respective delay to the respective local clock signal, and a respective clock driver to drive the respective local clock signal to the respective group of the memory devices.
Anticipation	3. The memory module of claim 2, wherein: the memory module is operable in at least a normal operation mode and a	Claim 1.

	configuration mode; the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to memory read or write commands received via the C/A signal lines; and the memory module in the configuration mode is configurable to perform a set of operations and the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.	
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Allowable Subject Matter

Claims 1-3 would be allowed given that the outstanding double patenting rejections are overcome.

Reasons for Allowance

The following is an examiner's statement of reasons for allowance.

Any comments considered necessary by applicants must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Matsui et al. US 2004/0105292 ("Matsui") and Sutardja et al. US 2007/0168781 ("Sutardja") are the closest prior art. Matsui and Sutardja are also the closest prior art with respect to the parent applications. As with the parent applications, Matsui, alone or in combination with Sutardja, does not teach all of the claim limitations of independent claim 1. Accordingly, claim 1 is allowable.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

Application/Control Number: 18/059,958
Art Unit: 2132

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/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

<i>Notice of References Cited</i>	Application/Control No. 18/059,958		Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED		Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20040105292-A1	06-2004	Matsui, Yoshinori	G11C7/1093	365/189.05
*	B	US-20070168781-A1	07-2007	Sutardja; Sehat	G11C29/76	714/718
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/0802	05/24/2023	ZA
G11C 5/04, 7/1066, 7/1093, 29/023, 29/028	05/24/2023	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	05/24/2023	ZA
711	119	05/24/2023	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached PE2E search history	05/24/2023	ZA
Inventor search	05/24/2023	ZA
Assignee search	05/24/2023	ZA
NPL search	05/24/2023	ZA
The instant application is allowed in view of Board's decision rendered with respect to the parent application 14/445,035	09/27/2018	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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Bibliographic Data

Application No: 18/059,958

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /ZUBAIR AHMED/

Examiner's Signature

Initials

Title:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND
METHOD OF OPERATION

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
11/29/2022	711	2132	NL073-20US03
RULE			

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CONTINUING DATA

This application is a CON of 17141978 01/05/2021 PAT 11513955

17141978 is a CON of 16432700 06/05/2019 PAT 10884923

16432700 is a CON of 14445035 07/28/2014 PAT 10324841

14445035 has PRO of 61859215 07/27/2013

FOREIGN APPLICATIONS

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FILING FEE RECEIVED

\$2,618

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	("11513955").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:03 AM
L2	1	("10884923").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:03 AM
L3	3	("10324841").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:04 AM
L4	42437	(G06F12/0802 OR G11C5/04 OR G11C7/1066 OR G11C7/1093 OR G11C29/023 OR G11C29/028).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:51 PM
L5	197	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:56 PM
L6	120	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:57 PM
L7	3	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:57 PM
L8	0	L4 AND (memory near3 module) AND (clocks SAME	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU,	OR	ON	ON	2023/05/24 02:59 PM

		(command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH module near2 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	SG, TW, WO); EPO; JPO; IBM_TDB)				
L9	0	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:59 PM
L10	12	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:59 PM
L11	10	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:00 PM
L12	0	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:00 PM

L13	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L14	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L15	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L16	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L17	0	netlist AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:04 PM

L18	1674	AND dqs)) AND (generat\$3 WITH module near2 clock) (711/120 711/119).ccls.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:05 PM
L19	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:05 PM
L20	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM
L21	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM
L22	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



Search History

Report Run Date: 2023-05-24 UTC
Searches: 1
Markers: 0

Search #	Timestamp	Results	Page
1	2023-05-24 19:08 UTC	48,884,603	3



US Patents | US Designs | US Applications | EPO Patents | EPO Applications | China Patents | China Applications | Japan Patents | Japan Applications | Korea Patents | Korea Applications | WIPO Applications | Argentina Patents | Argentina Applications | Brazil Patents | Brazil Applications | Canada Patents | Canada Applications | Chile Patents | Chile Applications | Colombia Applications | Costa Rica Applications | Cuba Patents | Cuba Applications | Dominican Republic Applications | Ecuador Patents | Ecuador Applications | El Salvador Applications | Guatemala Applications | Honduras Applications | Mexico Patents | Mexico Applications | Nicaragua Patents | Panama Applications | Peru Applications | Trinidad & Tobago Patents | Uruguay Applications | Austria Patents | Austria Applications | Belarus Patents | Belgium Patents | Belgium Applications | Bosnia & Herzegovina Patents | Bosnia & Herzegovina Applications | Bulgaria Patents | Bulgaria Applications | Croatia Patents | Croatia Applications | Czech Republic Patents | Czech Republic Applications | Czechoslovakia Patents | Czechoslovakia Applications | Denmark Patents | Denmark Applications | Estonia Patents | Estonia Applications | EUIPO Patents | Finland Patents | Finland Applications | France Patents | France Applications | Germany Patents | Germany Applications | Great Britain Patents | Great Britain Applications | Greece Patents | Greece Applications | Hungary Patents | Hungary Applications | Iceland Patents | Iceland Applications | Ireland Patents | Ireland Applications | Italy Patents | Italy Applications | Latvia Patents | Latvia Applications | Lithuania Patents | Lithuania Applications | Luxembourg Patents | Luxembourg Applications | Malta Patents | Monaco Patents | Montenegro Patents | Montenegro Applications | Netherlands Patents | Netherlands Applications | Norway Patents | Norway Applications | Poland Patents | Poland Applications | Portugal Patents | Portugal Applications | Republic of Moldova Patents | Republic of Moldova Applications | Romania Patents | Romania Applications | San Marino Patents | San Marino Applications | Serbia Patents | Serbia Applications | Slovakia Patents | Slovakia Applications | Slovenia Patents | Spain Patents | Spain Applications | Sweden Patents | Sweden Applications | Switzerland Patents | Switzerland Applications | Ukraine Patents | Yugoslavia/Serbia and Montenegro Patents | Yugoslavia/Serbia and Montenegro Applications | Armenia Patents | Australia Patents | Australia Applications | Cyprus Patents | Gulf Cooperation Council Patents | Hong Kong Patents | India Patents | India Applications | Indonesia Patents | Indonesia Applications | Israel Patents | Israel Applications | Jordan Patents | Jordan Applications | Kyrgyzstan Patents | Macao Applications | Malaysia Patents | Mongolia Patents | New Zealand Patents | Philippines Patents | Philippines Applications | Saudi Arabia Patents | Saudi Arabia Applications | Singapore Patents | Singapore Applications | Taiwan Patents | Taiwan Applications | Tajikistan Patents | Tajikistan Applications | Thailand Patents | Thailand Applications | Uzbekistan Patents | Vietnam Patents | Algeria Patents | ARIPO Patents | ARIPO Applications | Egypt Patents | Kenya Patents | Malawi Patents | Morocco Patents | Morocco Applications | OAPI Patents | South Africa Patents | Tunisia Applications | Zambia Patents | Zimbabwe Patents | EAPO Patents | EAPO Applications | Georgia Patents | Georgia Applications | Kazakhstan Patents | Kazakhstan Applications | Russia Patents | Russia Applications | Turkey Patents | Turkey Applications | IEEE Xplore Publications: IEEE Periodicals | IEEE Xplore Publications: IEEE Conferences | IEEE Xplore Publications: IEEE Standards | IEEE Xplore Publications: IEEE Early Access | IEEE Xplore Publications: SMPTE Periodicals | IEEE Xplore Publications: SMPTE Conferences | IEEE Xplore Publications: SMPTE Standards | IEEE Xplore Publications: MIT Press eBooks | IEEE Xplore Publications: Wiley-IEEE eBooks | IEEE Xplore Publications: IBM Periodicals | IEEE Xplore Publications: URSI Periodicals | IEEE Xplore Publications: VDE Conferences | IEEE Xplore Publications: Periodicals from China | IP.com Prior Art Database: The IP.com Journal | IP.com Prior Art Database: Internet Society RFC | IP.com Prior Art Database: IBM TDB Archive | IP.com Prior Art Database: Legacy Journals | IP.com Prior Art Database: Software Patent Institute | OnePetro.org: Periodicals at OnePetro.org | OnePetro.org: Conferences at OnePetro.org | Other Literature: IBM Redbooks | Other Literature: PubMed Central | Other Literature: arXiv.org

1 2023-05-24 19:08 UTC | 48,884,603 results from

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Text

A memory module operable in a computer system having a memory controller and a system bus, the system bus

including one or more clock signal lines, a set of control/address (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) having connectors formed along an edge thereof for connecting to respective ones of the C/A signal lines and the plurality of sets of data/strobe signal lines; memory devices mounted on the PCB and organized in one or more ranks and in a plurality of groups, a respective group of the memory devices including at least one respective memory device in each of the one or more ranks and configured to communicate data/strobe signals with the memory controller via a respective set of the plurality of sets of data/strobe signal lines; circuits mounted on the PCB and configurable to: receive from the memory controller a system clock via the one or more clock signal lines and input control and address (C/A) signals via the set of C/A signal lines; generate a module clock signal and module C/A signals in response to the system clock and input C/A signals; generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, the plurality of local clock signals having respective phase relationships with the module clock signal; and output the module C/A signals to the memory devices, wherein the module C/A signals including one or more chip select signals corresponding, respectively, to the one or more ranks; and output the plurality of local clock signals to respective groups of the memory devices, wherein a respective local clock signal having a respective programmable phase relationship with the module clock signal is output to a corresponding group of the memory devices, the corresponding group of the memory devices including at least one corresponding memory device that performs a memory read or write operation by communicating data/strobe signals with the memory controller via a corresponding set of the plurality of sets of data/strobe signal lines in accordance with the respective local clock signal and in response to module C/A signals received by the at least one corresponding memory device.

Concept Modifiers: None

Filters: None



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
18/059,958	11/29/2022	Hyun Lee	NL073-20US03

CONFIRMATION NO. 5230

79141
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

PUBLICATION NOTICE



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Title:MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

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NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

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Electronically filed October 31, 2023

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>	Attorney Docket No.:	NL073-20US03

AMENDMENT AND RESPONSE TO NON-FINAL OFFICE ACTION

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

The enclosed Amendment is in response to the Non-Final Office Action dated May 31, 2023, for the above identified patent application.

Petition for Extension of Time under 37 CFR 1.136. It is respectfully requested that the time for response to the Office Action dated May 31, 2023, be extended for a period of two (2) month from August 31, 2023, to October 31, 2023.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Cancelled)
3. (Cancelled)
4. (New) A memory module operable in a computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines;
 - a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to adjust timing of one or more signals received or output by the each buffer circuit of the plurality of buffer circuits;
 - a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines;
 - memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks;wherein:

in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to communicate first data/strobe signals with the memory controller via a first buffer circuit of the plurality of buffer circuits;

in response to the buffer control signals, the first buffer circuit is configured to receive the first data/strobe signals and to communicate second data/strobe signals with the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines;

at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal of the first data/strobe signals received by the first buffer circuit based on a first delay and a second delay;

the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during a configuration mode, the memory module in the configuration mode being configured to program the one or more configuration registers in the each buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines; and

the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

5. (New) The memory module of claim 4, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

6. (New) The memory module of claim 4, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

during the configuration mode, the module control circuit output configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

7. (New) The memory module of claim 6, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.
8. (New) The memory module of claim 4, wherein the at least one signal includes one or more of a data signal and a strobe signal received from the at least one first memory device in the first group of memory devices.
9. (New) The memory module of claim 4, wherein the at least one delay control circuit in the first buffer circuit is configured to determine the second delay based on at least one signal received from the module control circuit and at least one signal received from the memory controller during a memory operation.
10. (New) the memory module of claim 4, wherein:
the system bus further includes one or more clock signal lines;
the module control circuit is further configurable to receive from the memory controller a system clock via the one or more clock signal lines and output a registered clock; and
the first buffer circuit is configured to receive the registered clock and to generate a first buffered clock, the first buffered clock having a programmable phase relationship with the registered clock determined by at least one of the one or more registers in the first buffer circuit.
11. (New) The memory module of claim 10, wherein the first buffer circuit is further configured to output the first buffered clock to the first group of memory devices.
12. (New) The memory module of claim 4, wherein the memory module is configurable to perform a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations; and
wherein at least one of the one or more configuration registers is programmed based on information derived from the set of training operations.
13. (New) The memory module of claim 4, wherein the at least one delay circuit in the first buffer circuit is configured to delay the at least one signal based on a combination of the first delay and the second delay.

14. (New) A method performed at a memory module in a computer system, the computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and memory devices mounted on the PCB and organized in ranks, the method comprising:

operating in a configuration mode, including:

receiving mode register command signals from the memory controller via the C/A signal lines; and

programming one or more configuration registers in each buffer circuit of a plurality of buffer circuits in response to the mode register command signals, wherein the plurality of buffer circuits are mounted on the PCB and each buffer circuit of the plurality of buffer circuits corresponds to a respective set of the plurality of sets of data/strobe signal lines, wherein the memory devices include a plurality of groups of memory devices, each group of the plurality of groups of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; and

performing a memory read or write operation, including:

receiving from the memory controller input address and control (C/A) signals via the set of C/A signal lines;

outputting registered C/A signals associated with the read or write operation to the memory devices, wherein in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to communicate first data/strobe signals with the memory controller via a first buffer circuit of the plurality of buffer circuits;

outputting buffer control signals associated with the memory read or write operation to the plurality of buffer circuits, wherein in response to the buffer control signals, the first buffer circuit is configured to receive the first data/strobe signals and to communicate second data/strobe signals with the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; and

delaying at least one signal of the first data/strobe signals received by the first buffer circuit based on a first delay and a second delay;

wherein:

the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during the configuration mode; and

the second delay is determined by at least one delay control circuit in the first buffer circuit while the memory module is not operating in the configuration mode.

15. (New) The method of claim 14, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

16. (New) The method of claim 14, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

the method further comprises, during the configuration mode, outputting configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

17. (New) The method of claim 16, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

18. (New) The method of claim 14, wherein the at least one signal includes one or more of a data signal and a strobe signal received from the at least one first memory device in the first group of memory devices.

19. (New) The method of claim 14, wherein the at least one delay control circuit is configured to determine the second delay information based on at least one signal received from the memory controller during a memory operation.
20. (New) the method of claim 14, wherein the system bus further includes one or more clock signal lines, the method further comprising:
- receiving from the memory controller a system clock via the one or more clock signal lines;
 - outputting a registered clock; and
 - generating a first buffered clock in the first buffer circuit, the first buffered clock having a programmable phase relationship with the registered clock based on a value stored in at least one of the one or more registers in the first buffer circuit.
21. (New) The method of claim 20, further comprising outputting by the first buffer circuit the respective buffered clock to the first group of memory devices.
22. (New) The method of claim 14, further comprising:
- performing a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations;
 - wherein at least one the one or more configuration registers in the first buffer circuit is programmed based on information derived from the set of training operations.
23. (New) The method of claim 14, wherein the at least one delay circuit is configured to delay the at least one signal based on a combination of the first delay information and the second delay information.

REMARKS

This amendment responds to the Non-Final Office Action dated May 31, 2023, in which claims 1-3 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 4, and 17-19 of US Patent 10,324,841.

REMARKS CONCERNING CLAIMS

Claims 1-3 are cancelled.

New claims 4-23 are added.

Support for the amendment can be found throughout the Specification and Drawings, for example, in paragraphs [0035], [0047], [0050], [0063], [0065], [0067]-[0068], and [0070]. Therefore, no new matter has been added.

REMARKS CONCERNING DOUBLE PATENTING

Claims 1-3 are rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1, 4, and 17-19 of US Patent 10,324,841.

Claims 1-3 are cancelled. Thus, the rejection thereto is moot. New claims 4-23 differ significantly in scope from original claims 1-3 and from any of the claims in US Patent 10,324,841. Reconsideration of the double patenting rejection is therefore respectfully requested.

New claims 4-23 are believed to be in condition for allowance.

CONCLUDING REMARKS

By responding in the foregoing remarks only to particular positions asserted by the examiner, the Applicant does not necessarily acquiesce in other positions that have not been explicitly addressed. In addition, the Applicant's arguments for the patentability of a claim should not be understood as implying that no other reasons for the patentability of that claim exist.

In light of the above amendments and remarks, the Applicant respectfully requests that the Examiner reconsider this application with a view towards allowance. The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: October 31, 2023

/Jamie J. Zheng/

51,167

Jamie J. Zheng

(Reg. No.)

Customer No. 79141

Electronically filed October 31, 2023

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>	Attorney Docket No.:	NL073-20US03

AMENDMENT AND RESPONSE TO NON-FINAL OFFICE ACTION

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

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The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Cancelled)
3. (Cancelled)
4. (New) A memory module operable in a computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines;
 - a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to adjust timing of one or more signals received or output by the each buffer circuit of the plurality of buffer circuits;
 - a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines;
 - memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks;wherein:

in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to communicate first data/strobe signals with the memory controller via a first buffer circuit of the plurality of buffer circuits;

in response to the buffer control signals, the first buffer circuit is configured to receive the first data/strobe signals and to communicate second data/strobe signals with the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines;

at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal of the first data/strobe signals received by the first buffer circuit based on a first delay and a second delay;

the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during a configuration mode, the memory module in the configuration mode being configured to program the one or more configuration registers in the each buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines; and

the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

5. (New) The memory module of claim 4, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

6. (New) The memory module of claim 4, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

during the configuration mode, the module control circuit output configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

7. (New) The memory module of claim 6, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.
8. (New) The memory module of claim 4, wherein the at least one signal includes one or more of a data signal and a strobe signal received from the at least one first memory device in the first group of memory devices.
9. (New) The memory module of claim 4, wherein the at least one delay control circuit in the first buffer circuit is configured to determine the second delay based on at least one signal received from the module control circuit and at least one signal received from the memory controller during a memory operation.
10. (New) the memory module of claim 4, wherein:
the system bus further includes one or more clock signal lines;
the module control circuit is further configurable to receive from the memory controller a system clock via the one or more clock signal lines and output a registered clock; and
the first buffer circuit is configured to receive the registered clock and to generate a first buffered clock, the first buffered clock having a programmable phase relationship with the registered clock determined by at least one of the one or more registers in the first buffer circuit.
11. (New) The memory module of claim 10, wherein the first buffer circuit is further configured to output the first buffered clock to the first group of memory devices.
12. (New) The memory module of claim 4, wherein the memory module is configurable to perform a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations; and
wherein at least one of the one or more configuration registers is programmed based on information derived from the set of training operations.
13. (New) The memory module of claim 4, wherein the at least one delay circuit in the first buffer circuit is configured to delay the at least one signal based on a combination of the first delay and the second delay.

14. (New) A method performed at a memory module in a computer system, the computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and memory devices mounted on the PCB and organized in ranks, the method comprising:

operating in a configuration mode, including:

receiving mode register command signals from the memory controller via the C/A signal lines; and

programming one or more configuration registers in each buffer circuit of a plurality of buffer circuits in response to the mode register command signals, wherein the plurality of buffer circuits are mounted on the PCB and each buffer circuit of the plurality of buffer circuits corresponds to a respective set of the plurality of sets of data/strobe signal lines, wherein the memory devices include a plurality of groups of memory devices, each group of the plurality of groups of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; and

performing a memory read or write operation, including:

receiving from the memory controller input address and control (C/A) signals via the set of C/A signal lines;

outputting registered C/A signals associated with the read or write operation to the memory devices, wherein in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to communicate first data/strobe signals with the memory controller via a first buffer circuit of the plurality of buffer circuits;

outputting buffer control signals associated with the memory read or write operation to the plurality of buffer circuits, wherein in response to the buffer control signals, the first buffer circuit is configured to receive the first data/strobe signals and to communicate second data/strobe signals with the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; and

delaying at least one signal of the first data/strobe signals received by the first buffer circuit based on a first delay and a second delay;

wherein:

the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during the configuration mode; and

the second delay is determined by at least one delay control circuit in the first buffer circuit while the memory module is not operating in the configuration mode.

15. (New) The method of claim 14, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

16. (New) The method of claim 14, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

the method further comprises, during the configuration mode, outputting configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

17. (New) The method of claim 16, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

18. (New) The method of claim 14, wherein the at least one signal includes one or more of a data signal and a strobe signal received from the at least one first memory device in the first group of memory devices.

19. (New) The method of claim 14, wherein the at least one delay control circuit is configured to determine the second delay information based on at least one signal received from the memory controller during a memory operation.
20. (New) the method of claim 14, wherein the system bus further includes one or more clock signal lines, the method further comprising:
- receiving from the memory controller a system clock via the one or more clock signal lines;
 - outputting a registered clock; and
 - generating a first buffered clock in the first buffer circuit, the first buffered clock having a programmable phase relationship with the registered clock based on a value stored in at least one of the one or more registers in the first buffer circuit.
21. (New) The method of claim 20, further comprising outputting by the first buffer circuit the respective buffered clock to the first group of memory devices.
22. (New) The method of claim 14, further comprising:
- performing a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations;
 - wherein at least one the one or more configuration registers in the first buffer circuit is programmed based on information derived from the set of training operations.
23. (New) The method of claim 14, wherein the at least one delay circuit is configured to delay the at least one signal based on a combination of the first delay information and the second delay information.

REMARKS

This amendment responds to the Non-Final Office Action dated May 31, 2023, in which claims 1-3 are rejected on the ground of nonstatutory double patenting as being unpatentable over claims 1, 4, and 17-19 of US Patent 10,324,841.

REMARKS CONCERNING CLAIMS

Claims 1-3 are cancelled.

New claims 4-23 are added.

Support for the amendment can be found throughout the Specification and Drawings, for example, in paragraphs [0035], [0047], [0050], [0063], [0065], [0067]-[0068], and [0070]. Therefore, no new matter has been added.

REMARKS CONCERNING DOUBLE PATENTING

Claims 1-3 are rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1, 4, and 17-19 of US Patent 10,324,841.

Claims 1-3 are cancelled. Thus, the rejection thereto is moot. New claims 4-23 differ significantly in scope from original claims 1-3 and from any of the claims in US Patent 10,324,841. Reconsideration of the double patenting rejection is therefore respectfully requested.

New claims 4-23 are believed to be in condition for allowance.

CONCLUDING REMARKS

By responding in the foregoing remarks only to particular positions asserted by the examiner, the Applicant does not necessarily acquiesce in other positions that have not been explicitly addressed. In addition, the Applicant's arguments for the patentability of a claim should not be understood as implying that no other reasons for the patentability of that claim exist.

In light of the above amendments and remarks, the Applicant respectfully requests that the Examiner reconsider this application with a view towards allowance. The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: October 31, 2023

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

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APPLICATION #
18/059,958

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NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY J amie Zheng

PATENT CENTER # 63129493

AUTHORIZED BY -

CUSTOMER # 79141

FILING DATE 11/29/2022

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Hyun Lee

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FILING DATE 11/29/2022

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun Lee

CORRESPONDENCE
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AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 4

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
Certificate_Transmission_sb0092.pdf	2	Transmittal Letter	529 KB
NL073-20US03_Amendment_NFOA.pdf	9	-	142 KB
NL073-20US03_Amendment_NFOA-A....pdf	(1-1) 1	Amendment/Request for Reconsideration-After Non-Final Rejection	105 KB
NL073-20US03_Amendment_NFOA-CLM.pdf	(2-7) 6	Claims	118 KB

NL073- 20US03_Amendment_NFO A-REM.pdf	(8-9)	2	Applicant Arguments/Remarks Made in an Amendment	102 KB
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Certificate_Transmission_sb009 2.pdf	24AC2A7F476A7C42CF4E6DE988933C77B9F3B9C4E2C030DD EEA7183A1EBC6BA92296D33F729B2CFAE58D0B0F0B00A9CA 2936DBC6B8E0E90F27BE235D853857E9
NL073- 20US03_Amendment_NFOA.pd f	403C33FA05E42ACA6C29C9C6A1A4F912575CA819B8CDE4330 5A728E092849681509323053AFBE802BB17F9A9B230C8095046 1C1E377DF432B14BEDEDBF6FA1EE
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$100 =	
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☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

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		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)	
AMENDMENT	11/01/2023	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA			
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$100 =	0	
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	x \$480 =	0	
	☐ Application Size Fee (37 CFR 1.16(s))							
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))							
						TOTAL ADD'L FEE	0	
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =		
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =		
	☐ Application Size Fee (37 CFR 1.16(s))							
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EXAMINER	
AHMED, ZUBAIR	
ART UNIT	PAPER NUMBER
2132	

DATE MAILED: 12/08/2023

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/08/2024

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230

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nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	03/08/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

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1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230
79141	7590	12/08/2023	EXAMINER	
USCH Law, PC 3790 El Camino Real #1147 Palo Alto, CA 94304			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
DATE MAILED: 12/08/2023				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 18/059,958	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed on 11/01/2023.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 4-23. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)	5. ☐ Examiner's Amendment/Comment
2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____.	6. ☒ Examiner's Statement of Reasons for Allowance
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____.	7. ☐ Other _____.
4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	

/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
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DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to amendment filed on 11/01/2023. In this amendment, Applicant canceled claims 1-3 and added new claims 4-23.

Response to Arguments

Applicant's arguments, filed 11/01/2023, have been fully considered and are persuasive. The double patenting rejections of claims 1-3 are moot. A NOTICE OF ALLOWANCE is being administered.

Allowable Subject Matter

Claims 4-23 are allowed.

Reasons for Allowance

The following is an examiner's statement of reasons for allowance.

Any comments considered necessary by applicants must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

After careful considerations, examination and search of the claimed invention, the closest prior art of record does not teach or anticipate the claimed feature as claimed in independent claim 4 "a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module

control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines” and “in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to communicate first data/strobe signals with the memory controller via a first buffer circuit of the plurality of buffer circuits” and “in response to the buffer control signals, the first buffer circuit is configured to receive the first data/strobe signals and to communicate second data/strobe signals with the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines” and “at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal of the first data/strobe signals received by the first buffer circuit based on a first delay and a second delay” and “the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during a configuration mode, the memory module in the configuration mode being configured to program the one or more configuration registers in the each buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines” and “the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode” in combination with the overall claimed limitations when interpreted in light of the specification.

Matsui et al. US 2004/0105292 (“Matsui”) and Sutardja et al. US 2007/0168781 (“Sutardja”) are the closest prior art. Matsui and Sutardja are also the closest prior art with respect to the parent applications. As with the parent applications, Matsui, alone or

in combination with Sutardja, does not teach all of the claim limitations of independent claim 4. Accordingly, claim 4 is allowable.

Independent claim 14 recites similar subject matter and this claim is also allowable for the aforementioned reason.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197

Application/Control Number: 18/059,958
Art Unit: 2132

Page 5

(toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

Issue Classification 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC						
Symbol					Type	Version
G06F	/	12	/	0802	F	2013-01-01
G11C	/	5	/	04	I	2013-01-01
G11C	/	29	/	023	I	2013-01-01
G11C	/	29	/	028	I	2013-01-01
G11C	/	7	/	1066	I	2013-01-01
G11C	/	7	/	1093	I	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	21 November 2023 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	22 November 2023 (Date)	O.G. Print Claim(s) 4	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	12	/ 0802
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	120

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
711	102	119				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	21 November 2023 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	22 November 2023 (Date)	O.G. Print Claim(s) 4	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1	7	10	16	19										
	2	8	11	17	20										
	3	9	12	18	21										
1	4	10	13	19	22										
2	5	11	14	20	23										
3	6	12	15												
4	7	13	16												
5	8	14	17												
6	9	15	18												

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	21 November 2023 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	22 November 2023 (Date)	O.G. Print Claim(s) 4	O.G. Print Figure 1

<i>Index of Claims</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS									
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47									
CLAIM		DATE							
Final	Original	05/24/2023	11/21/2023						
	1	✓	-						
	2	✓	-						
	3	✓	-						
1	4		=						
2	5		=						
3	6		=						
4	7		=						
5	8		=						
6	9		=						
7	10		=						
8	11		=						
9	12		=						
10	13		=						
11	14		=						
12	15		=						
13	16		=						
14	17		=						
15	18		=						
16	19		=						
17	20		=						
18	21		=						
19	22		=						
20	23		=						

<i>Search Notes</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/0802	11/21/2023	ZA
G11C 5/04, 7/1066, 7/1093, 29/023, 29/028	11/21/2023	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	11/21/2023	ZA
711	119	11/21/2023	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached PE2E search history	11/21/2023	ZA
Inventor search	11/21/2023	ZA
Assignee search	11/21/2023	ZA
NPL search	11/21/2023	ZA
The instant application is allowed in view of Board's decision rendered with respect to the parent application 14/445,035	09/27/2018	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G06F	120/12/0802	11/21/2023	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	("11513955").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:03 AM
L2	1	("10884923").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:03 AM
L3	3	("10324841").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:04 AM
L4	42437	(G06F12/0802 OR G11C5/04 OR G11C7/1066 OR G11C7/1093 OR G11C29/023 OR G11C29/028).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:51 PM
L5	197	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:56 PM
L6	120	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:57 PM
L7	3	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:57 PM
L8	0	L4 AND (memory near3 module) AND (clocks SAME	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU,	OR	ON	ON	2023/05/24 02:59 PM

		(command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH module near2 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	SG, TW, WO); EPO; JPO; IBM_TDB)				
L9	0	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:59 PM
L10	12	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:59 PM
L11	10	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:00 PM
L12	0	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:00 PM

L13	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L14	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L15	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L16	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L17	0	netlist AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:04 PM

L18	1674	AND dqs)) AND (generat\$3 WITH module near2 clock) (711/120 711/119).ccls.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:05 PM
L19	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:05 PM
L20	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM
L21	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM
L22	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.



Search History

Report Run Date: 2023-11-21 UTC

Searches: 1

Markers: 0

Search #	Timestamp	Results	Page
1	2023-11-21 12:24 UTC	58,302,995	3



US Patents | US Designs | US Applications | EPO Patents | EPO Applications | China Patents | China Applications | Japan Patents | Japan Applications | Korea Patents | Korea Applications | WIPO Applications | Argentina Patents | Argentina Applications | Brazil Patents | Brazil Applications | Canada Patents | Canada Applications | Chile Patents | Chile Applications | Colombia Applications | Costa Rica Applications | Cuba Patents | Cuba Applications | Dominican Republic Applications | Ecuador Patents | Ecuador Applications | El Salvador Applications | Guatemala Applications | Honduras Applications | Mexico Patents | Mexico Applications | Nicaragua Patents | Panama Applications | Peru Applications | Trinidad & Tobago Patents | Uruguay Applications | Austria Patents | Austria Applications | Belarus Patents | Belgium Patents | Belgium Applications | Bosnia & Herzegovina Patents | Bosnia & Herzegovina Applications | Bulgaria Patents | Bulgaria Applications | Croatia Patents | Croatia Applications | Czech Republic Patents | Czech Republic Applications | Czechoslovakia Patents | Czechoslovakia Applications | Denmark Patents | Denmark Applications | Estonia Patents | Estonia Applications | EUIPO Patents | Finland Patents | Finland Applications | France Patents | France Applications | Germany Patents | Germany Applications | Great Britain Patents | Great Britain Applications | Greece Patents | Greece Applications | Hungary Patents | Hungary Applications | Iceland Patents | Iceland Applications | Ireland Patents | Ireland Applications | Italy Patents | Italy Applications | Latvia Patents | Latvia Applications | Lithuania Patents | Lithuania Applications | Luxembourg Patents | Luxembourg Applications | Malta Patents | Monaco Patents | Montenegro Patents | Montenegro Applications | Netherlands Patents | Netherlands Applications | Norway Patents | Norway Applications | Poland Patents | Poland Applications | Portugal Patents | Portugal Applications | Republic of Moldova Patents | Republic of Moldova Applications | Romania Patents | Romania Applications | San Marino Patents | San Marino Applications | Serbia Patents | Serbia Applications | Slovakia Patents | Slovakia Applications | Slovenia Patents | Spain Patents | Spain Applications | Sweden Patents | Sweden Applications | Switzerland Patents | Switzerland Applications | Ukraine Patents | Yugoslavia/Serbia and Montenegro Patents | Yugoslavia/Serbia and Montenegro Applications | Armenia Patents | Australia Patents | Australia Applications | Cyprus Patents | Gulf Cooperation Council Patents | Hong Kong Patents | India Patents | India Applications | Indonesia Patents | Indonesia Applications | Israel Patents | Israel Applications | Jordan Patents | Jordan Applications | Kyrgyzstan Patents | Macao Applications | Malaysia Patents | Mongolia Patents | New Zealand Patents | Philippines Patents | Philippines Applications | Saudi Arabia Patents | Saudi Arabia Applications | Singapore Patents | Singapore Applications | Taiwan Patents | Taiwan Applications | Tajikistan Patents | Tajikistan Applications | Thailand Patents | Thailand Applications | Uzbekistan Patents | Vietnam Patents | Algeria Patents | ARIPO Patents | ARIPO Applications | Egypt Patents | Kenya Patents | Malawi Patents | Morocco Patents | Morocco Applications | OAPI Patents | South Africa Patents | Tunisia Applications | Zambia Patents | Zimbabwe Patents | EAPO Patents | EAPO Applications | Georgia Patents | Georgia Applications | Kazakhstan Patents | Kazakhstan Applications | Russia Patents | Russia Applications | Turkey Patents | Turkey Applications | IEEE Xplore Publications: IEEE Periodicals | IEEE Xplore Publications: IEEE Conferences | IEEE Xplore Publications: IEEE Standards | IEEE Xplore Publications: IEEE Early Access | IEEE Xplore Publications: SMPTE Periodicals | IEEE Xplore Publications: SMPTE Conferences | IEEE Xplore Publications: SMPTE Standards | IEEE Xplore Publications: MIT Press eBooks | IEEE Xplore Publications: Wiley-IEEE eBooks | IEEE Xplore Publications: IBM Periodicals | IEEE Xplore Publications: URSI Periodicals | IEEE Xplore Publications: VDE Conferences | IEEE Xplore Publications: Periodicals from China | IP.com Prior Art Database: The IP.com Journal | IP.com Prior Art Database: Internet Society RFC | IP.com Prior Art Database: IBM TDB Archive | IP.com Prior Art Database: Legacy Journals | IP.com Prior Art Database: Software Patent Institute | OnePetro.org: Periodicals at OnePetro.org | OnePetro.org: Conferences at OnePetro.org | Other Literature: IBM Redbooks | Other Literature: PubMed Central | Other Literature: arXiv.org

1 2023-11-21 12:24 UTC | 58,302,995 results from

De-Dup: None | Relevance Cut-off: None | Sort: Relevance

Main Concept: Text

A memory module operable in a computer system having a memory controller and a system bus, the system bus

including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising: a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines; a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to adjust timing of one or more signals received or output by the each buffer circuit of the plurality of buffer circuits; a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines; memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; wherein: in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to communicate first data/strobe signals with the memory controller via a first buffer circuit of the plurality of buffer circuits; in response to the buffer control signals, the first buffer circuit is configured to receive the first data/strobe signals and to communicate second data/strobe signals with the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal of the first data/strobe signals received by the first buffer circuit based on a first delay and a second delay; the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during a configuration mode, the memory module in the configuration mode being configured to program the one or more configuration registers in the each buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines; and the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

Concept Modifiers: None

Filters: None

Bibliographic Data

Application No: 18/059,958

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /ZUBAIR AHMED/

Examiner's Signature

Initials

Title:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND
METHOD OF OPERATION

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
11/29/2022	711	2132	NL073-20US03
RULE			

APPLICANTS

NETLIST, INC., Irvine, CA, UNITED STATES

INVENTORS

Hyun Lee, Ladera Ranch, CA, UNITED STATES

Jayesh R. Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 17141978 01/05/2021 PAT 11513955

17141978 is a CON of 16432700 06/05/2019 PAT 10884923

16432700 is a CON of 14445035 07/28/2014 PAT 10324841

14445035 has PRO of 61859215 07/27/2013

FOREIGN APPLICATIONS

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12/13/2022

STATE OR COUNTRY

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Palo Alto, CA 94304
UNITED STATES

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\$2,618

PE2E SEARCH - Search History (Prior Art)

There are no Prior Art searches to show.

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	23761	(G06F12/0802 OR G11C5/04 OR G11C7/1066 OR G11C7/1093 OR G11C29/023 OR G11C29/028).cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N2	138	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N3	84	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N4	2	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N5	0	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH module near2 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N6	4	N1 AND (memory near3 module) AND (clocks	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM

N7	4	SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N8	1689	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N9	0	N8 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N10	0	N8 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM
N11	0	N8 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM
N12	0	N8 AND (memory near3 module) AND (clocks SAME (command\$1address	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM

N13	0	OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) ((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM
N14	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM
N15	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM
N16	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM

N17	0	AND dqs)) AND (generat\$3 WITH module near2 clock) netlist AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM
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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230
79141	7590	12/29/2023		
USCH Law, PC 3790 El Camino Real #1147 Palo Alto, CA 94304			EXAMINER AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			12/29/2023	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eofficeaction@appcoll.com
jzheng@usclaw.com
patents@usclaw.com



UNITED STATES PATENT AND TRADEMARK OFFICE

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P.O. Box 1450
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www.uspto.gov

Application No. : 18059958
Applicant : Lee
Filing Date : 11/29/2022
Date Mailed : 12/29/2023

NOTICE TO FILE CORRECTED APPLICATION PAPERS

Notice of Allowance Mailed

This application has been accorded an Allowance Date and is being prepared for issuance. The application, however, is incomplete for the reasons below.

Applicant is given two (2) months from the mail date of this Notice within which to respond. This time period for reply is extendable under 37 CFR 1.136(a) for only TWO additional MONTHS.

The informalities requiring correction are indicated in the attachment(s). If the informality pertains to the abstract, specification (including claims) or drawings, the informality must be corrected with an amendment in compliance with 37 CFR 1.121 (or, if the application is a reissue application, 37 CFR 1.173). Such an amendment may be filed after payment of the issue fee if limited to correction of informalities noted herein. See Waiver of 37 CFR 1.312 for Documents Required by the Office of Patent Publication, 1280 Off. Gaz. Patent Office 918 (March 23, 2004). In addition, if the informality is not corrected until after payment of the issue fee, for purposes of 35 U.S.C. 154(b)(1)(iv), "all outstanding requirements" will be considered to have been satisfied when the informality has been corrected. A failure to respond within the above-identified time period will result in the application being ABANDONED.

See attachment(s).

*A copy of this notice **MUST** be returned with the reply. Please address response to
"Mail Stop Issue Fee, Commissioner for Patents,
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/Joanna Black/
Publication Branch
Office of Data Management
(571) 272-4200

IDENTIFICATION OF DRAWING DEFICIENCIES

- ☐ There is a hole or the image thereof within the illustration. FIG(s)
- ☐ The illustration is penetrated or traversed by a solid or broken line that is not intended to be part of the drawing, such as a dark line caused by a flaw in the copying process. FIG(s)
- ☐ An ink stamp or the image thereof obscures part of the illustration. FIG(s)
- ☐ The drawing is marred by black smudges, obliterations, or fax/copier marks (for example, speckles or dots in a substantial portion of the drawing). FIG(s)
- ☒ Figure numbers are duplicated or missing. FIG(s) 2D
- ☐ Drawing sheet or figure is missing. FIG(s)
- ☐ Numbers, letters, or reference characters in the drawing have been crossed out or are illegibly handwritten. FIG(s)
- ☐ The character of the lines, numbers, and letters is poor. FIG(s)
- ☐ The drawing's background shows that the original drawing was made on graph paper or other paper with a pattern or decoration. FIG(s)
- ☐ The FIG. number label is placed in a location that causes the drawing to be read upside down. FIG(s)
- ☐ Data, a reference number, or part of the drawing is truncated or missing, or a lead line has no reference number. FIG(s)
- ☐ The drawing and/or the FIG. label contain(s) foreign language. FIG(s)
- ☐ This utility application contains a photograph of a view that is capable of being illustrated as a line drawing. FIG(s)
- ☐ A petition under 37 CFR 1.84(a)(2) to accept color drawings has been granted, but the brief description of the drawings in the specification does not contain (or has not been amended to contain) the paragraph required by 37 CFR 1.84(a)(2)(iii).
- ☐ This reissue application contains added and/or amended drawings that are not labeled as "New" or "Amended" or "Canceled" as required by 37 CFR 1.173(b)(3). FIG(s)
- ☐ This Design reissue application contains a drawing that is labeled as "Canceled" but is not surrounded by brackets, or a drawing that is surrounded by brackets but is not labeled as "Canceled." See 37 CFR 1.173(b)(3). FIG(s)
- ☐ OTHER:
- ☐ COMMENTS:

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 18/059,958	Filing Date 11/29/2022	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	x \$100 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	x \$480 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	03/08/2024		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$100 = 0
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$480 = 0
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						0
AMENDMENT			CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/MYRTLE B LEIGH/
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".						
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		
		Attorney Docket No.:	NL073-20US03

CERTIFICATE OF ELECTRONIC TRANSMISSION

I hereby certify that this correspondence is being transmitted to the United States Patent and Trademark Office (USPTO) via the USPTO's Patent Center on:

MARCH 8, 2024 (DATE).

By: /Jamie J. Zheng/

Name: Jamie J. Zheng

Reg. No: 51167

AMENDMENT AND REQUEST FOR CONTINUED EXAMINATION

Mail Stop: Issue Fee

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

This Amendment is filed after the Notice of Allowance dated December 8, 2023, and is submitted with a Request for Continued Examination (RCE).

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

Amendments to the Specification

Please amend the Title as follows:

MEMORY MODULE WITH STATIC AND DYNAMIC TIMING ADJUSTMENT IN DATA
BUFFERS LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Please amend the Abstract as follows:

A memory module is operable in a computer system having a memory controller and a system bus and comprises memory devices organized in one or more ranks and in a plurality of groups, and circuits configurable to receive from the memory controller a system clock and input control and address (C/A) signals, and output registered C/A signals and buffer control signals. The memory module further comprises a plurality of buffer circuits. In response to the buffer control signals, each buffer circuit is configured to communicate first data/strobe signals with at least one memory device and to communicate second data/strobe signals with the memory controller. The buffer circuit includes at least one delay circuit configured to delay at least one signal of the first data/strobe signals based on a first delay and a second delay ~~generate a module clock signal and module C/A signals in response to the system clock and input C/A signals, generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, and output the plurality of local clock signals to respective groups of the memory devices.~~ A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices that includes at least one corresponding memory device in each of the one or more ranks

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Cancelled)
3. (Cancelled)
4. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines;
 - a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to adjust timing of one or more signals ~~received or output by~~ in the each buffer circuit of the plurality of buffer circuits;
 - a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines;
 - memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks;wherein:
 - in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to ~~communicate~~ receive or output first

data/strobe signals ~~with the memory controller via a first buffer circuit of the plurality of buffer circuits;~~

in response to the buffer control signals, ~~[[the]]~~ a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to ~~communicate~~ receive or output second data/strobe signals ~~with, the second data/strobe signals corresponding,~~ respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines;

at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal ~~of the first data/strobe signals received by in~~ the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

the memory module is operable in a configuration mode, and the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines during ~~[[a]]the configuration mode, the memory module in the configuration mode being configured to program the one or more configuration registers in the each buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines; and~~

the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

5. (Previously Presented) The memory module of claim 4, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

6. (Currently Amended) The memory module of claim 4, wherein

the buffer control signals include one or more mode signals indicating a read or write ~~operations~~ operation and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

~~during~~ in the configuration mode, the module control circuit is configured to output configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

7. (Previously Presented) The memory module of claim 6, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

8. (Currently Amended) The memory module of claim 4, wherein the at least one signal includes ~~one or more of a~~ at least one data signal ~~and a~~ and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

9. (Previously Presented) The memory module of claim 4, wherein the at least one delay control circuit in the first buffer circuit is configured to determine the second delay based on at least one signal received from the module control circuit and at least one signal received from the memory controller during a memory operation.

10. (Previously Presented) the memory module of claim 4, wherein:
the system bus further includes one or more clock signal lines;
the module control circuit is further configurable to receive from the memory controller a system clock via the one or more clock signal lines and output a registered clock; and
the first buffer circuit is configured to receive the registered clock and to generate a first buffered clock, the first buffered clock having a programmable phase relationship with the registered clock determined by at least one of the one or more registers in the first buffer circuit.

11. (Previously Presented) The memory module of claim 10, wherein the first buffer circuit is further configured to output the first buffered clock to the first group of memory devices.

12. (Previously Presented) The memory module of claim 4, wherein the memory module is configurable to perform a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations; and

wherein at least one of the one or more configuration registers is programmed based on information derived from the set of training operations.

13. (Previously Presented) The memory module of claim 4, wherein the at least one delay circuit in the first buffer circuit is configured to delay the at least one signal based on a combination of the first delay and the second delay.

14. (Currently Amended) A method performed at a memory module in a computer system, the computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and memory devices mounted on the PCB and organized in ranks, the method comprising:
operating in a configuration mode, including:
receiving mode register command signals from the memory controller via the C/A signal lines; and

programming one or more configuration registers in each buffer circuit of a plurality of buffer circuits in response to the mode register command signals, wherein the plurality of buffer circuits are mounted on the PCB and each buffer circuit of the plurality of buffer circuits corresponds to a respective set of the plurality of sets of data/strobe signal lines, wherein the memory devices include a plurality of groups of memory devices, each group of the plurality of groups of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; and

performing a memory read or write operation, including:
receiving from the memory controller input address and control (C/A) signals via the set of C/A signal lines;

outputting registered C/A signals associated with the read or write operation to the memory devices, wherein in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to ~~communicate~~ receive or output first data/strobe signals with the memory controller via a first ~~buffer circuit of the plurality of buffer circuits;~~

outputting buffer control signals associated with the memory read or write operation to the plurality of buffer circuits, wherein in response to the buffer control signals, ~~[[the]]~~ a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to ~~communicate~~ receive or output second data/strobe signals with, the second data/strobe signals corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; and

delaying at least one signal of the first data/strobe signals received by in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

wherein:

the first delay is programed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during the configuration mode; and

the second delay is determined by at least one delay control circuit in the first buffer circuit while the memory module is not operating in the configuration mode.

15. (Previously Presented) The method of claim 14, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

16. (Previously Presented) The method of claim 14, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

the method further comprises, during the configuration mode, outputting configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

17. (Previously Presented) The method of claim 16, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

18. (Currently Amended) The method of claim 14, wherein the at least one signal includes ~~one or more of a~~ at least one data signal ~~and a~~ and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

19. (Previously Presented) The method of claim 14, wherein the at least one delay control circuit is configured to determine the second delay information based on at least one signal received from the memory controller during a memory operation.

20. (Previously Presented) the method of claim 14, wherein the system bus further includes one or more clock signal lines, the method further comprising:

receiving from the memory controller a system clock via the one or more clock signal lines;

outputting a registered clock; and

generating a first buffered clock in the first buffer circuit, the first buffered clock having a programmable phase relationship with the registered clock based on a value stored in at least one of the one or more registers in the first buffer circuit.

21. (Previously Presented) The method of claim 20, further comprising outputting by the first buffer circuit the respective buffered clock to the first group of memory devices.

22. (Previously Presented) The method of claim 14, further comprising:

performing a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations;

wherein at least one the one or more configuration registers in the first buffer circuit is programmed based on information derived from the set of training operations.

23. (Previously Presented) The method of claim 14, wherein the at least one delay circuit is configured to delay the at least one signal based on a combination of the first delay information and the second delay information.

REMARKS

This amendment is submitted after the Notice of Allowance dated December 8, 2023, in which claims 4-23 are allowed, and is submitted with a Request for Continued Examination (RCE).

REMARKS CONCERNING THE TITLE AND ABSTRACT

The Title and Abstract have been amended to correspond with the claims. No new matter has been added.

REMARKS CONCERNING CLAIMS

Claims 4, 8, 14 and 18 are amended to address possible issues of clarity

Claim 6 is amended to address minor issues of formality.

No new matter has been added.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: March 8, 2024

/Jamie J. Zheng/

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APPLICATION #
18/059,958RECEIPT DATE / TIME
03/08/2024 10:39:13 PM Z ETATTORNEY DOCKET #
NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY Jamie Zheng

PATENT CENTER # 64618120

FILING DATE 11/29/2022

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun LeeCORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 22

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NL073- 20US03_RCE_SB30PC_202 40308.pdf	3	Request for Continued Examination (RCE)	1642 KB
NL073- 20US03_IDS_Transmittal_20 24-03-08.pdf	4	Information Disclosure Statement (IDS) Form (SB08)	136 KB
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IPR-2022- 00236_Decision_Rehearing. pdf	7	Non Patent Literature	110 KB
IPR-2022- 00236_Request_Rehearing.p df	12	Non Patent Literature	162 KB

IPR2022-00711_Request_Rehearing.pdf		14	Non Patent Literature	321 KB
IPR2022-00711_Oral_Hearing.pdf		75	Non Patent Literature	401 KB
IPR2023-00847_Prelim_SurReply.pdf		8	Non Patent Literature	123 KB
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JEDEC_DDR4_Proposal_DB_Ctrl_Word.pdf		27	Non Patent Literature	340 KB
IPR2023-00847POPR.pdf		33	Non Patent Literature	940 KB
IPR2022-00711_Decision_Rehearing.pdf		7	Non Patent Literature	109 KB
IPR2022-00711_FWD.pdf		34	Non Patent Literature	484 KB
IPR2022-00711_Notice_Appeal.pdf		48	Non Patent Literature	688 KB
JESD82-32A.pdf		200	Non Patent Literature	3770 KB
IPR2024-00370_Petition.pdf		135	Non Patent Literature	19597 KB
NL073-20US03_Amendment_RCE.pdf		9	-	187 KB
NL073-20US03_Amendment_RCE-AMSB.pdf	(1-1)	1	Amendment Submitted/Entered with Filing of Continued Prosecution Application (CPA)/Request for Continued Examination(RCE)	150 KB
NL073-20US03_Amendment_RCE-SPEC.pdf	(2-2)	1	Specification	95 KB
NL073-20US03_Amendment_RCE-CLM.pdf	(3-8)	6	Claims	105 KB
NL073-20US03_Amendment_RCE	(9-9)	1	Applicant Arguments/Remarks Made in an Amendment	101 KB

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NL073- 20US03_IDS2_List_2024-03- 08.pdf	7	Information Disclosure Statement (IDS) Form (SB08)	254 KB
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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		
		Attorney Docket No.:	NL073-20US03

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By: /Jamie J. Zheng/

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AMENDMENT AND REQUEST FOR CONTINUED EXAMINATION

Mail Stop: Issue Fee

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

This Amendment is filed after the Notice of Allowance dated December 8, 2023, and is submitted with a Request for Continued Examination (RCE).

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

Amendments to the Specification

Please amend the Title as follows:

MEMORY MODULE WITH STATIC AND DYNAMIC TIMING ADJUSTMENT IN DATA
BUFFERS LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Please amend the Abstract as follows:

A memory module is operable in a computer system having a memory controller and a system bus and comprises memory devices organized in one or more ranks and in a plurality of groups, and circuits configurable to receive from the memory controller a system clock and input control and address (C/A) signals, and output registered C/A signals and buffer control signals. The memory module further comprises a plurality of buffer circuits. In response to the buffer control signals, each buffer circuit is configured to communicate first data/strobe signals with at least one memory device and to communicate second data/strobe signals with the memory controller. The buffer circuit includes at least one delay circuit configured to delay at least one signal of the first data/strobe signals based on a first delay and a second delay ~~generate a module clock signal and module C/A signals in response to the system clock and input C/A signals, generate a plurality of local clock signals corresponding, respectively, to the plurality of groups of memory devices, and output the plurality of local clock signals to respective groups of the memory devices.~~ A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices that includes at least one corresponding memory device in each of the one or more ranks

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Cancelled)
3. (Cancelled)
4. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines;
 - a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to adjust timing of one or more signals ~~received or output by~~ in the each buffer circuit of the plurality of buffer circuits;
 - a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines;
 - memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks;wherein:
 - in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to ~~communicate~~ receive or output first

data/strobe signals with the memory controller via a first buffer circuit of the plurality of buffer circuits;

in response to the buffer control signals, ~~[[the]]~~ a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to ~~communicate~~ receive or output second data/strobe signals ~~with, the second data/strobe signals corresponding,~~ respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines;

at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal ~~of the first data/strobe signals received by in~~ the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

the memory module is operable in a configuration mode, and the first delay is programed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines during [[a]]the configuration mode, ~~the memory module in the configuration mode being configured to program the one or more configuration registers in the each buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines; and~~

the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

5. (Previously Presented) The memory module of claim 4, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

6. (Currently Amended) The memory module of claim 4, wherein

the buffer control signals include one or more mode signals indicating a read or write ~~operations~~ operation and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

~~during in~~ the configuration mode, the module control circuit is configured to output configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

7. (Previously Presented) The memory module of claim 6, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

8. (Currently Amended) The memory module of claim 4, wherein the at least one signal includes ~~one or more of a~~ at least one data signal ~~and a~~ and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

9. (Previously Presented) The memory module of claim 4, wherein the at least one delay control circuit in the first buffer circuit is configured to determine the second delay based on at least one signal received from the module control circuit and at least one signal received from the memory controller during a memory operation.

10. (Previously Presented) the memory module of claim 4, wherein:
the system bus further includes one or more clock signal lines;
the module control circuit is further configurable to receive from the memory controller a system clock via the one or more clock signal lines and output a registered clock; and
the first buffer circuit is configured to receive the registered clock and to generate a first buffered clock, the first buffered clock having a programmable phase relationship with the registered clock determined by at least one of the one or more registers in the first buffer circuit.

11. (Previously Presented) The memory module of claim 10, wherein the first buffer circuit is further configured to output the first buffered clock to the first group of memory devices.

12. (Previously Presented) The memory module of claim 4, wherein the memory module is configurable to perform a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations; and

wherein at least one of the one or more configuration registers is programmed based on information derived from the set of training operations.

13. (Previously Presented) The memory module of claim 4, wherein the at least one delay circuit in the first buffer circuit is configured to delay the at least one signal based on a combination of the first delay and the second delay.

14. (Currently Amended) A method performed at a memory module in a computer system, the computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and memory devices mounted on the PCB and organized in ranks, the method comprising:

operating in a configuration mode, including:

receiving mode register command signals from the memory controller via the C/A signal lines; and

programming one or more configuration registers in each buffer circuit of a plurality of buffer circuits in response to the mode register command signals, wherein the plurality of buffer circuits are mounted on the PCB and each buffer circuit of the plurality of buffer circuits corresponds to a respective set of the plurality of sets of data/strobe signal lines, wherein the memory devices include a plurality of groups of memory devices, each group of the plurality of groups of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; and

performing a memory read or write operation, including:

receiving from the memory controller input address and control (C/A) signals via the set of C/A signal lines;

outputting registered C/A signals associated with the read or write operation to the memory devices, wherein in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to ~~communicate~~ receive or output first data/strobe signals ~~with the memory controller via a first buffer circuit of the plurality of buffer circuits;~~

outputting buffer control signals associated with the memory read or write operation to the plurality of buffer circuits, wherein in response to the buffer control signals, ~~[[the]]~~ a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to ~~communicate~~ receive or output second data/strobe signals with, the second data/strobe signals corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; and

delaying at least one signal of the first data/strobe signals received by in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

wherein:

the first delay is programed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during the configuration mode; and

the second delay is determined by at least one delay control circuit in the first buffer circuit while the memory module is not operating in the configuration mode.

15. (Previously Presented) The method of claim 14, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

16. (Previously Presented) The method of claim 14, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

the method further comprises, during the configuration mode, outputting configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

17. (Previously Presented) The method of claim 16, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

18. (Currently Amended) The method of claim 14, wherein the at least one signal includes ~~one or more of a~~ at least one data signal and a ~~and/or at least one~~ strobe signal received from the at least one first memory device in the first group of memory devices.

19. (Previously Presented) The method of claim 14, wherein the at least one delay control circuit is configured to determine the second delay information based on at least one signal received from the memory controller during a memory operation.

20. (Previously Presented) the method of claim 14, wherein the system bus further includes one or more clock signal lines, the method further comprising:

receiving from the memory controller a system clock via the one or more clock signal lines;

outputting a registered clock; and

generating a first buffered clock in the first buffer circuit, the first buffered clock having a programmable phase relationship with the registered clock based on a value stored in at least one of the one or more registers in the first buffer circuit.

21. (Previously Presented) The method of claim 20, further comprising outputting by the first buffer circuit the respective buffered clock to the first group of memory devices.

22. (Previously Presented) The method of claim 14, further comprising:

performing a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations;

wherein at least one the one or more configuration registers in the first buffer circuit is programmed based on information derived from the set of training operations.

23. (Previously Presented) The method of claim 14, wherein the at least one delay circuit is configured to delay the at least one signal based on a combination of the first delay information and the second delay information.

REMARKS

This amendment is submitted after the Notice of Allowance dated December 8, 2023, in which claims 4-23 are allowed, and is submitted with a Request for Continued Examination (RCE).

REMARKS CONCERNING THE TITLE AND ABSTRACT

The Title and Abstract have been amended to correspond with the claims. No new matter has been added.

REMARKS CONCERNING CLAIMS

Claims 4, 8, 14 and 18 are amended to address possible issues of clarity

Claim 6 is amended to address minor issues of formality.

No new matter has been added.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date:	<u>March 8, 2024</u>	<u>/Jamie J. Zheng/</u>	<u>51,167</u>
		Jamie J. Zheng	(Reg. No.)
		Customer No. 79141	

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>	Attorney Docket No.:	NL073-20US03

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information which the Examiner may consider material to the examination of the subject U.S. patent application. It is requested that the Examiner make this information of record if it is deemed material to the examination of the application.

1. Enclosures accompanying this Information Disclosure Statement are:

- 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
- 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☒ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☒ all other information or portion which caused it to be listed on the PTO-1449.
- 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
- 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.

2. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
- ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);
 - ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☐ Before the mailing of the first Office action on the merits;
 - ☒ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.
3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b, 3c or 3d)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$260 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 3c. ☐ The \$130 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 3d. ☐ The \$120 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check Item 4a, and 4b, 4c or 4d)

- 4a. ☐ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$260 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 4c. ☐ The \$130 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 4d. ☐ The \$120 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.

5. ☐ Certification Statement (applicable if Item 3a or Item 4a is checked)

(Check either Item 5a, 5b, 5c or 5d)

- 5a. ☐ In accordance with 37 C.F.R. §1.97(e)(1), it is certified that each item of information contained in this information disclosure statement was first cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this Information Disclosure Statement.
- 5b. ☐ In accordance with 37 C.F.R. §1.97(e)(2), it is certified that no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 C.F.R. §1.56(c) more than three months prior to the filing of this information disclosure statement.
- 5c. ☐ Each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
- 5d. ☐ Pursuant to 37 C.F.R. §1.704(d), each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
6. ☒ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.
7. ☒ This application is a continuation application under 37 C.F.R. §1.53(b) or (d).

(Check appropriate Items 7a, 7b and/or 7c)

- 7a. ☒ Copies of publications listed on Forms PTO-1449 from prior application Serial Nos. 17/141,978 filed on January 5, 2021, 16/432,700, filed on Jun. 5, 2019, 14/445,035, filed on Jul. 28, 2014, of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7b. ☒ Copies of the publications listed on the attached Forms PTO-1449 that were not previously cited in prior application Serial Nos. 17/141,978 filed on January 5, 2021, 16/432,700, filed on Jun. 5, 2019, 14/445,035, filed on Jul. 28, 2014, are provided herewith.
8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on . A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be

considered as if properly filed on .

9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:

(Check Item 9a, 9b, or 9c)

- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Deposit Account No. 60-3916 (order no. NL073-20US03).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: March 8, 2024

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

51,167
(Reg. No.)

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>	
				Application Number	18/059,958
				Filing Date	November 29, 2022
				First Named Inventor	Hyun LEE
				Art Unit	2139
Examiner Name	Zubair AHMED				
Sheet	1	of	2	Attorney Docket Number	NL073-20US03

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date YYYY-MM-DD	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
	1.	9536579 B2	2017-01-03	Masaaki Iijima, et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
		Country Code ³ – Number ⁴ - Kind Code ⁵ (if known)				

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
	2.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2024-00370, IPR Petition re US Patent No. 10,268,608, filed January 10, 2024
	3.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847, Patent Owner's Preliminary Response, filed September 14, 2023
	4.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847, Petitioner's Authorized Reply to Patent Owner's Preliminary Response, filed September 26, 2023
	5.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847 Patent Owner Pre-Institution Sur-Reply, filed October 5, 2023
	6.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847, DECISION Granting Institution of Inter Partes Review, issued December 12, 2023
	7.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Oral Hearing Transcript, August 14, 2023
	8.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Final Written Decision Determining All Challenged Claims Unpatentable, issued October 17, 2023
	9.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Patent Owner's Request for Rehearing, filed October 16, 2023

Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
				Application Number		18/059,958
				Filing Date		November 29, 2022
				First Named Inventor		Hyun LEE
				Art Unit		2139
				Examiner Name		Zubair AHMED
Sheet	2	of	2	Attorney Docket Number		NL073-20US03

	10.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, DECISION Denying Patent Owner's Request on Rehearing of Final Written Decision, issued December 20, 2023
	11.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Patent Owner's Notice of Appeal, filed February 21, 2024
	12.	Inter Partes Review of US Patent No. 9824035, Case No. IPR2022-00236, Petitioners' Request for Rehearing of the Final Written Decision, filed July 20, 2023
	13.	Inter Partes Review of US Patent No. 9824035, Case No. IPR2022-00236, Granting Petitioner's Request on Rehearing of Final Written Decision, issued August 16, 2023
	14.	JEDEC Committee Ballot, Proposed DDR4 DB Buffer Control Words, JEDEC 2011
	15.	JEDEC Standard JESD82-32A, DDR4 Data Buffer Definition (DDR4DB02), November 2016

Examiner Signature		Date Considered	
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UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC PAYMENT RECEIPT

APPLICATION #
18/059,958

RECEIPT DATE / TIME
03/08/2024 10:39:13 PM Z ET

ATTORNEY DOCKET #
NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	5230	FILED BY	Jamie Zheng
PATENT CENTER #	64618120	AUTHORIZED BY	-
CUSTOMER #	79141	FILING DATE	11/29/2022
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Hyun Lee

Payment Information

PAYMENT METHOD CARD / 2172	PAYMENT TRANSACTION ID E202438M39469303	PAYMENT AUTHORIZED BY Jamie Zheng
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FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1801	REQUEST FOR CONTINUED EXAMINATION (RCE) - 1ST REQUEST (SEE 37 CFR 1.114)	1360.00	1	1360.00
TOTAL AMOUNT:				\$1,360.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C.

371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>	
				Application Number	18/059,958
				Filing Date	November 29, 2022
				First Named Inventor	Hyun LEE
				Art Unit	2139
				Examiner Name	Zubair AHMED
Sheet	1	of	7	Attorney Docket Number	NL073-20US03

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
	1.	5036221 A	1991-07-30	Brucculeri, et al.	
	2.	6072346 A	2000-06-06	Ghahremani	
	3.	6184701 B1	2001-02-06	Kim, et al.	
	4.	6530006 B1	2003-03-04	Dodd et al.	
	5.	6807077 B2	2004-10-19	Noda et al.	
	6.	6877079 B2	2005-04-05	Yoo et al.	
	7.	6889300 B2	2005-05-03	Davis et al.	
	8.	6906555 B2	2005-06-14	Ma	
	9.	7036053 B2	2006-04-25	Zumkehr	
	10.	7275173 B2	2007-09-25	Lindt	
	11.	7356639 B2	2008-04-08	Perego et al.	
	12.	7562271 B2	2009-07-14	Shaeffer et al.	
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	15.	8020022 B2	2011-09-13	Noriyuki Tokuhiko	
	16.	8407441 B2	2013-03-26	Giovannini et al.	
	17.	8552779 B2	2013-10-08	Jones, et al.	
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	20.	8599595 B2	2013-12-03	Stephens, Jr.	
	21.	8713379 B2	2014-04-29	Takefman, et al.	
	22.	9153298 B2	2015-10-06	Stephens, Jr.	
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	24.	9361955 B2	2016-06-07	Muralimanohar et al.	
	25.	20060277355 A1	12/7/06	Ellsberry, et al.	
	26.	20070008791 A1	1/11/07	Butt, et al.	
	27.	20080025122 A1	1/31/08	Schakel et al.	
	28.	20100309706 A1	12/9/10	Saito, et al.	
	29.	20100312925 A1	12/9/10	Osanai, et al.	
	30.	20100312956 A1	12/9/10	Hiraishi, et al.	
	31.	20110062999 A1	3/17/11	Nimalyar, et al.	
	32.	20120106228 A1	5/3/12	Lee	
	33.	20120296598 A1	11/22/12	Chafin et al.	

Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>	
				Application Number	18/059,958
				Filing Date	November 29, 2022
				First Named Inventor	Hyun LEE
				Art Unit	2139
Examiner Name	Zubair AHMED				
Sheet	2	of	7	Attorney Docket Number	NL073-20US03

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ – Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
	34.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, IPR Petition re US Patent No. 9,824,035, filed December 23, 2021
	35.	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, IPR Petition re US Patent No. 10,268,608, filed December 23, 2021
	36.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, 'Ex. 1003 - IPR2022-00236 Declaration of Donald Alpert,' filed December 23, 2021
	37.	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, 'Ex. 1003 - IPR2022-00237 Declaration of Donald Alpert,' filed December 23, 2021
	38.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Preliminary Response, filed April 25, 2022
	39.	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, Patent Owner's Preliminary Response, filed April 25, 2022
	40.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Exhibit List, filed April 25, 2022
	41.	Complaint for Patent Infringement, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Apr. 28, 2021), 74 pages.
	42.	Netlist, Inc.'s Disclosure of Asserted Claims and Preliminary Infringement Contentions to Defendants, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Sept. 10, 2021), 9 pages.
	43.	Micron's Preliminary Invalidity Contentions in Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Nov. 5, 2021), 38 pages.

Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>	
				Application Number	18/059,958
				Filing Date	November 29, 2022
				First Named Inventor	Hyun LEE
				Art Unit	2139
Examiner Name	Zubair AHMED				
Sheet	3	of	7	Attorney Docket Number	NL073-20US03

	44.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent Application Publication No. 2010/0312925 A1 to Osanai et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 57 pages.
	45.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 8,713,379 to Takefman et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	46.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 8,020,022 to Tokuhiko, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 31 pages.
	47.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 6,877,079 to Yoo et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	48.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 7,562,271 to Shaeffer et al. Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 50 pages.
	49.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 6,530,006 to Dodd et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 51 pages.
	50.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 9,361,955 to Muralimanohar et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 63 pages.
	51.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 2012/0106228 to Lee, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 39 pages.
	52.	Invalidity Chart for U.S. Patent No. 9,824,035 based on JEDEC Submission entitled "Proposed DDR4 DB BCOM Protocol," Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 22 pages.

Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
				Application Number	18/059,958	
				Filing Date	November 29, 2022	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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	53.	Invalidity Chart for U.S. Patent No.10,268,608 based on US Patent Application Publication No. 2010/0312925 A1 to Osanai et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 53 pages.
	54.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 8,713,379 to Takefman et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 35 pages.
	55.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 8,020,022 to Tokuhito, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 30 pages.
	56.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 6,877,079 to Yoo et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	57.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 7,562,271 to Shaeffer et al. Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 52 pages.
	58.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 6,530,006 to Dodd et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 51 pages.
	59.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 9,361,955 to Muralimanohar et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 49 pages.
	60.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 2012/0106228 to Lee, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 36 pages.
	61.	Invalidity Chart for U.S. Patent No. 10,268,608 based on JEDEC Submission entitled "Proposed DDR4 DB BCOM Protocol," Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 19 pages.
	62.	Obviousness Chart for U.S. Patent No. 9,824,035, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 14 pages.

Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>	
				Application Number	18/059,958
				Filing Date	November 29, 2022
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				Art Unit	2139
Sheet	5	of	7	Examiner Name	Zubair AHMED
				Attorney Docket Number	NL073-20US03

	63.	Obviousness Chart for U.S. Patent No.10,268,608, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 9 pages.
	64.	Micron's Preliminary Identification of Proposed Claim Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Dec. 23, 2021), 14 pages.
	65.	Netlist, Inc.'s Preliminary Proposed Claim Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Dec. 23, 2021), 7 pages.
	66.	Defendants' Identification of Extrinsic Evidence in Support of Their Preliminary Proposed Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Jan. 13, 2022), 21 pages.
	67.	Netlist, Inc.'s Preliminary Identification of Extrinsic Evidence, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Jan 13, 2022), 16 pages.
	68.	Declaration of Harold S. Stone, Ph.D., in Support of Defendants' Claim Construction Positions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 27 pages.
	69.	Declaration of Alan Jay Smith, Ph.D., in Support of Defendants' Claim Construction Positions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 26 pages.
	70.	Micron's Opening Claim Construction Brief, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 40 pages.
	71.	Joint Claim Construction Chart, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No 1:22-cv-00136-LY, Dkt. 66-1 (W.D. Tex. Apr. 14, 2022), 13 pages.
	72.	Excerpt – Modern Dictionary of Electronics 7E, 1999
	73.	JEDEC Solid State Technology Association, 1st Showing, Committee: JC-40.4, Committee Item Number: 0311.xx, 'Proposed DDR4 DB BCOM Protocol,' undated, 10 pages
	74.	JEDEC Standard Double Data Rate (DDR) SDRAM Specification, JESD79 (June 2000), 78 pages
Examiner Signature		Date Considered

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
				Application Number	18/059,958	
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				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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	75.	JEDEC Standard No. 21-C, PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification, Revision 1.3, January 2002, 83 pages.
	76.	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3, DDR3 SDRAM Standard, June 2007, 12 pages
	77.	JEDEC Standard JESD79-3F (Revision of JESD79-3E, July 2010), DDR3 SDRAM Standard, July 2012, 226 pages.
	78.	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3A, DDR3 SDRAM Specification, September 2007, 12 pages
	79.	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3C, DDR3 SDRAM, April 2008, 7 pages
	80.	Excerpt - JEDEC Standard No. 21C, DDR4 SDRAM Load Reduced DIMM Design Specification, August 2015, 2 pages
	81.	Inter Partes Review of US Patent No. 10860506, Case No. IPR2022-00711, IPR Petition re US Patent No. 10,860,506, filed March 22, 2022.
	82.	Inter Partes Review of US Patent No. 10860506, Case No. IPR2022-00711, 'Ex. 1003 - IPR2022-00711 Declaration of Robert Wedig,' filed March 22, 2022.
	83.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, IPR Petition re US Patent No. 9,128,632, filed January 20, 2017.
	84.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, 'Ex. 1003 - IPR2017-00730 Declaration of Trevor Mudge,' filed January 20, 2017.
	85.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Patent Owner's Preliminary Response, filed April 30, 2017.
	86.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Decision Denying Institution of Inter Partes Review, entered July 21, 2017.
	87.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, PETITIONERS' REQUEST FOR REHEARING, filed August 21, 2017.
	88.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Decision Denying Petitioner's Request for Rehearing, entered October 30, 2017.
	89.	Inter Partes Review of US Patent No. 9606907, Case No. IPR2018-00362, Final Written Decision, entered June 27, 2019.
	90.	Complaint, Netlist, Inc. v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., Samsung Semiconductor, Inc., Case No. 2:21-cv-463 (E.D. Tex. Dec. 20, 2021), 49 pages.
	91.	First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. v. Netlist, Inc., C.A. No. 21-1453 (RGA) (USDC, Delaware, Jan. 18, 2022), 122 pages.
	92.	Excerpt – Microsoft Computer Dictionary, Fifth Edition, 2002.

Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
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				Filing Date	November 29, 2022	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
Sheet	7	of	7	Attorney Docket Number		NL073-20US03

	93.	Altera, Quartus II Handbook Version 11.0 Volume 3: Verification, May 2011, 172 pages.
	94.	Bruce Jacob, et al., Memory Systems (Cache, DRAM, Disk), © 2008 by Elsevier, Inc., 296 pages.
	95.	Harold S. Stone, Microcomputer Interfacing, © 1982 by Addison-Wesley Publishing Company, Inc., 399 pages
	96.	JEDEC Standard JESD82-32A, DDR4 Data Buffer Definition (DDR4DB02), August 2019, 200 pages.
	97.	Declaration of Julie Carlson re “JEDEC Standard JESD79-3C (Revision of JESD79-3B, April 2008), DDR3 SDRAM, November 2008,” September 16, 2021, 335 pages
	98.	Declaration of Julie Carlson re “JEDEC Standard No. 21-C, PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification, Revision 1.3, January 2002,” September 16, 2021, 187 pages.
	99.	United States District Court for the District of Delaware, Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. v. Netlist Inc., C.A. No. 21-1453 (RGA), Exhibit 42 – JEDEC Presentation (March 2011) and Exhibit 43 – JEDEC Proposed DDR4 DB BCOM Protocol (undated), 44 pages

Examiner Signature		Date Considered	
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**REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL
(Submitted Only via EFS-Web)**

Application Number	18/059,958	Filing Date	2022-11-29	Docket Number (if applicable)	NL073-20US03	Art Unit	2132
First Named Inventor	Hyun Lee			Examiner Name	Zubair AHMED		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____☐ Other _____☒ Enclosed☒ Amendment/Reply☒ Information Disclosure Statement (IDS)☐ Affidavit(s)/ Declaration(s)☐ Other _____**MISCELLANEOUS**☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)☐ Other _____**FEES****The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.**☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
Deposit Account No 603916**SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED**☒ Patent Practitioner Signature☐ Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	/Jamie J. Zheng/	Date (YYYY-MM-DD)	2024-03-08
Name	Jamie J. Zheng	Registration Number	51167

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
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6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
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9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.



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USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

EXAMINER

AHMED, ZUBAIR

ART UNIT

PAPER NUMBER

2132

DATE MAILED: 04/09/2024

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	07/09/2024

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

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I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

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III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

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(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	07/09/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

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www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230
79141	7590	04/09/2024	EXAMINER	
USCH Law, PC 3524 Waverley St. Palo Alto, CA 94306			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
DATE MAILED: 04/09/2024				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Corrected Notice of Allowability	Application No. 18/059,958	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to RCE filed on 03/08/2024.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 4-23. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>03/08/2024, 03/08/2024, 03/08/2024</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
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DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to RCE filed on 03/08/2024. Claims 1-3 were canceled before.

Information Disclosure Statement

The information disclosure statements (IDSs) submitted on 03/08/2024, 03/08/2024, 03/08/2024 were filed after the mailing date of the Notice of Allowance on 12/08/2023. The submission is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statements are being considered by the examiner.

Allowable Subject Matter

Claims 4-23 are allowed.

Reasons for Allowance

This communication warrants no examiner's reason for allowance, as the record makes evident the reason for allowance, satisfying the record "as a whole" as required by rule 37 CFR 1.104 (e). Accordingly, the reason for allowance is in all probability evident from the record and no statement for examiner's reason for allowance is necessary (see MPEP 1302.14).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

Application/Control Number: 18/059,958
Art Unit: 2132

Page 4

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

<i>Search Notes</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/0802	11/21/2023	ZA
G11C 5/04, 7/1066, 7/1093, 29/023, 29/028	11/21/2023	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	11/21/2023	ZA
711	119	11/21/2023	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached PE2E search history	03/21/2024	ZA
Inventor search	11/21/2023	ZA
Assignee search	11/21/2023	ZA
NPL search	11/21/2023	ZA
The instant application is allowed in view of Board's decision rendered with respect to the parent application 14/445,035	09/27/2018	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G06F	120/12/0802	11/21/2023	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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Issue Classification 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	

CPC						
Symbol					Type	Version
G06F	/	12	/	0802	F	2013-01-01
G11C	/	5	/	04	I	2013-01-01
G11C	/	29	/	023	I	2013-01-01
G11C	/	29	/	028	I	2013-01-01
G11C	/	7	/	1066	I	2013-01-01
G11C	/	7	/	1093	I	2013-01-01

CPC Combination Sets					
Symbol				Type	Version
	/		/		

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	21 March 2024 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	25 March 2024 (Date)	O.G. Print Claim(s) 4	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	12	/ 0802
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	120

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
711	120	119				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	21 March 2024 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	25 March 2024 (Date)	O.G. Print Claim(s) 4	O.G. Print Figure 1

Issue Classification 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1	7	10	16	19										
	2	8	11	17	20										
	3	9	12	18	21										
1	4	10	13	19	22										
2	5	11	14	20	23										
3	6	12	15												
4	7	13	16												
5	8	14	17												
6	9	15	18												

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	21 March 2024 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	25 March 2024 (Date)	O.G. Print Claim(s) 4	O.G. Print Figure 1

<i>Index of Claims</i> 	Application/Control No. 18/059,958	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	05/24/2023	11/21/2023	03/21/2024						
	1	✓	-	-						
	2	✓	-	-						
	3	✓	-	-						
1	4		=	=						
2	5		=	=						
3	6		=	=						
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16	19		=	=						
17	20		=	=						
18	21		=	=						
19	22		=	=						
20	23		=	=						

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>	
				Application Number	18/059,958
				Filing Date	November 29, 2022
				First Named Inventor	Hyun LEE
				Art Unit	2139
Examiner Name	Zubair AHMED				
Sheet	1	of	2	Attorney Docket Number	NL073-20US03

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date YYYY-MM-DD	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
	1.	9536579 B2	2017-01-03	Masaaki Iijima, et al.	

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreing Patent Document	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶
		Country Code ³ – Number ⁴ - Kind Code ⁵ (if known)				

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
	2.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2024-00370, IPR Petition re US Patent No. 10,268,608, filed January 10, 2024
	3.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847, Patent Owner's Preliminary Response, filed September 14, 2023
	4.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847, Petitioner's Authorized Reply to Patent Owner's Preliminary Response, filed September 26, 2023
	5.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847 Patent Owner Pre-Institution Sur-Reply, filed October 5, 2023
	6.	Inter Partes Review of US Patent No. 10268608, Case No. IPR2023-00847, DECISION Granting Institution of Inter Partes Review, issued December 12, 2023
	7.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Oral Hearing Transcript, August 14, 2023
	8.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Final Written Decision Determining All Challenged Claims Unpatentable, issued October 17, 2023
	9.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Patent Owner's Request for Rehearing, filed October 16, 2023

Examiner Signature	/ZUBAIR AHMED/	Date Considered	03/21/2024
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
				Application Number	18/059,958	
				Filing Date	November 29, 2022	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
Sheet	2	of	2	Attorney Docket Number		NL073-20US03

	10.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, DECISION Denying Patent Owner's Request on Rehearing of Final Written Decision, issued December 20, 2023
	11.	Inter Partes Review of US Patent No.10860506, Case No. IPR2022-00711, Patent Owner's Notice of Appeal, filed February 21, 2024
	12.	Inter Partes Review of US Patent No. 9824035, Case No. IPR2022-00236, Petitioners' Request for Rehearing of the Final Written Decision, filed July 20, 2023
	13.	Inter Partes Review of US Patent No. 9824035, Case No. IPR2022-00236, Granting Petitioner's Request on Rehearing of Final Written Decision, issued August 16, 2023
	14.	JEDEC Committee Ballot, Proposed DDR4 DB Buffer Control Words, JEDEC 2011
	15.	JEDEC Standard JESD82-32A, DDR4 Data Buffer Definition (DDR4DB02), November 2016

Examiner Signature	/ZUBAIR AHMED/	Date Considered	03/21/2024
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Bibliographic Data

Application No: 18/059,958

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /ZUBAIR AHMED/

Examiner's Signature

Initials

Title:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND
METHOD OF OPERATION

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
11/29/2022	711	2132	NL073-20US03
RULE			

APPLICANTS

NETLIST, INC., Irvine, CA, UNITED STATES

INVENTORS

Hyun Lee, Ladera Ranch, CA, UNITED STATES

Jayesh R. Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 17141978 01/05/2021 PAT 11513955

17141978 is a CON of 16432700 06/05/2019 PAT 10884923

16432700 is a CON of 14445035 07/28/2014 PAT 10324841

14445035 has PRO of 61859215 07/27/2013

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

12/13/2022

STATE OR COUNTRY

UNITED STATES

ADDRESS

USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306
UNITED STATES

FILING FEE RECEIVED

\$2,618

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	("11513955").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:03 AM
L2	1	("10884923").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:03 AM
L3	3	("10324841").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 09:04 AM
L4	42437	(G06F12/0802 OR G11C5/04 OR G11C7/1066 OR G11C7/1093 OR G11C29/023 OR G11C29/028).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:51 PM
L5	197	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:56 PM
L6	120	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:57 PM
L7	3	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:57 PM
L8	0	L4 AND (memory near3 module) AND (clocks	(US-PGPUB; USPAT; USOCR; FIT (AU, CN,	OR	ON	ON	2023/05/24 02:59 PM

L9	0	SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH module near2 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:59 PM
L10	12	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 02:59 PM
L11	10	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:00 PM
L12	0	L4 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:00 PM

L13	0	AND pcb ((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L14	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L15	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L16	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:02 PM
L17	0	netlist AND (memory near3 module) AND (clocks SAME	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU,	OR	ON	ON	2023/05/24 03:04 PM

L18	1674	(command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near2 clock) (711/120 711/119).ccls.	SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:05 PM
L19	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:05 PM
L20	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM
L21	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM
L22	0	L18 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2023/05/24 03:06 PM
L23	36	("5036221" "6072346" "6184701" "6530006" "6807077" "6877079" "6889300" "6906555"	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO;	OR	ON	ON	2024/03/21 06:48 AM

L24	13	"7036053" "7275173" "7356639" "7562271" "7808849" "7890732" "8020022" "8407441" "8552779" "8565033" "8566516" "8599595" "8713379" "9153298" "9263103" "9361955" "20060277355").pn. ("20070008791" "20080025122" "20100309706" "20100312925" "20100312956" "20110062999" "20120106228" "20120296598" "9536579").pn.	JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2024/03/21 06:51 AM
L25	0	L23 AND (((memory OR storage) near3 module) OR dimm) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2024/03/21 06:52 AM
L26	1	L24 AND (((memory OR storage) near3 module) OR dimm) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2024/03/21 06:53 AM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Electronically filed March 8, 2024

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>	Attorney Docket No.:	NL073-20US03

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information which the Examiner may consider material to the examination of the subject U.S. patent application. It is requested that the Examiner make this information of record if it is deemed material to the examination of the application.

1. Enclosures accompanying this Information Disclosure Statement are:

- 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
- 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☒ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☒ all other information or portion which caused it to be listed on the PTO-1449.
- 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
- 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.

2. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
- ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);
 - ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☐ Before the mailing of the first Office action on the merits;
 - ☒ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.
3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b, 3c or 3d)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$260 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 3c. ☐ The \$130 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 3d. ☐ The \$120 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check Item 4a, and 4b, 4c or 4d)

- 4a. ☐ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$260 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 4c. ☐ The \$130 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.
- 4d. ☐ The \$120 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916.

5. ☐ Certification Statement (applicable if Item 3a or Item 4a is checked)

(Check either Item 5a, 5b, 5c or 5d)

- 5a. ☐ In accordance with 37 C.F.R. §1.97(e)(1), it is certified that each item of information contained in this information disclosure statement was first cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this Information Disclosure Statement.
- 5b. ☐ In accordance with 37 C.F.R. §1.97(e)(2), it is certified that no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 C.F.R. §1.56(c) more than three months prior to the filing of this information disclosure statement.
- 5c. ☐ Each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
- 5d. ☐ Pursuant to 37 C.F.R. §1.704(d), each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
6. ☒ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.
7. ☒ This application is a continuation application under 37 C.F.R. §1.53(b) or (d).

(Check appropriate Items 7a, 7b and/or 7c)

- 7a. ☒ Copies of publications listed on Forms PTO-1449 from prior application Serial Nos. 17/141,978 filed on January 5, 2021, 16/432,700, filed on Jun. 5, 2019, 14/445,035, filed on Jul. 28, 2014, of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7b. ☒ Copies of the publications listed on the attached Forms PTO-1449 that were not previously cited in prior application Serial Nos. 17/141,978 filed on January 5, 2021, 16/432,700, filed on Jun. 5, 2019, 14/445,035, filed on Jul. 28, 2014, are provided herewith.
8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on . A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be

considered as if properly filed on .

9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:

(Check Item 9a, 9b, or 9c)

- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Deposit Account No. 60-3916 (order no. NL073-20US03).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: March 8, 2024

/Jamie J. Zheng/

Jamie J. Zheng
Customer No. 79141

51,167
(Reg. No.)

/ZUBAIR AHMED/

03/21/2024

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>	
				Application Number	18/059,958
				Filing Date	November 29, 2022
				First Named Inventor	Hyun LEE
				Art Unit	2139
				Examiner Name	Zubair AHMED
Sheet	1	of	7	Attorney Docket Number	NL073-20US03

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
	1.	5036221 A	1991-07-30	Bruculeri, et al.	
	2.	6072346 A	2000-06-06	Ghahremani	
	3.	6184701 B1	2001-02-06	Kim, et al.	
	4.	6530006 B1	2003-03-04	Dodd et al.	
	5.	6807077 B2	2004-10-19	Noda et al.	
	6.	6877079 B2	2005-04-05	Yoo et al.	
	7.	6889300 B2	2005-05-03	Davis et al.	
	8.	6906555 B2	2005-06-14	Ma	
	9.	7036053 B2	2006-04-25	Zumkehr	
	10.	7275173 B2	2007-09-25	Lindt	
	11.	7356639 B2	2008-04-08	Perego et al.	
	12.	7562271 B2	2009-07-14	Shaeffer et al.	
	13.	7808849 B2	2010-10-05	Swain, et al.	
	14.	7890732 B2	2011-02-15	Sukegawa	
	15.	8020022 B2	2011-09-13	Noriyuki Tokuhito	
	16.	8407441 B2	2013-03-26	Giovannini et al.	
	17.	8552779 B2	2013-10-08	Jones, et al.	
	18.	8565033 B1	2013-10-22	Manohararajah	
	19.	8566516 B2	2013-10-22	Schakel, et al.	
	20.	8599595 B2	2013-12-03	Stephens, Jr.	
	21.	8713379 B2	2014-04-29	Takefman, et al.	
	22.	9153298 B2	2015-10-06	Stephens, Jr.	
	23.	9263103 B2	2016-02-16	Giovannini, et al.	
	24.	9361955 B2	2016-06-07	Muralimanohar et al.	
	25.	20060277355 A1	12/7/06	Ellsberry, et al.	
	26.	20070008791 A1	1/11/07	Butt, et al.	
	27.	20080025122 A1	1/31/08	Schakel et al.	
	28.	20100309706 A1	12/9/10	Saito, et al.	
	29.	20100312925 A1	12/9/10	Osanai, et al.	
	30.	20100312956 A1	12/9/10	Hiraishi, et al.	
	31.	20110062999 A1	3/17/11	Nimalyar, et al.	
	32.	20120106228 A1	5/3/12	Lee	
	33.	20120296598 A1	11/22/12	Chafin et al.	

Examiner Signature	/ZUBAIR AHMED/	Date Considered	
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				Electronically filed March 8, 2024	
				Application Number	18/059,958
				Filing Date	November 29, 2022
				First Named Inventor	Hyun LEE
				Art Unit	2139
Examiner Name	Zubair AHMED				
Sheet	2	of	7	Attorney Docket Number	NL073-20US03

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ – Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
	34.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, IPR Petition re US Patent No. 9,824,035, filed December 23, 2021
	35.	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, IPR Petition re US Patent No. 10,268,608, filed December 23, 2021
	36.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, 'Ex. 1003 - IPR2022-00236 Declaration of Donald Alpert,' filed December 23, 2021
	37.	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, 'Ex. 1003 - IPR2022-00237 Declaration of Donald Alpert,' filed December 23, 2021
	38.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Preliminary Response, filed April 25, 2022
	39.	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, Patent Owner's Preliminary Response, filed April 25, 2022
	40.	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Exhibit List, filed April 25, 2022
	41.	Complaint for Patent Infringement, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Apr. 28, 2021), 74 pages.
	42.	Netlist, Inc.'s Disclosure of Asserted Claims and Preliminary Infringement Contentions to Defendants, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Sept. 10, 2021), 9 pages.
	43.	Micron's Preliminary Invalidity Contentions in Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Nov. 5, 2021), 38 pages.

Examiner Signature	/ZUBAIR AHMED/	Date Considered	03/21/2024
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
				Application Number	18/059,958	
				Filing Date	November 29, 2022	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
Sheet	3	of	7	Attorney Docket Number		NL073-20US03

	44.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent Application Publication No. 2010/0312925 A1 to Osanai et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 57 pages.
	45.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 8,713,379 to Takefman et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	46.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 8,020,022 to Tokuhito, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 31 pages.
	47.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 6,877,079 to Yoo et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	48.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 7,562,271 to Shaeffer et al. Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 50 pages.
	49.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 6,530,006 to Dodd et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 51 pages.
	50.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 9,361,955 to Muralimanohar et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 63 pages.
	51.	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 2012/0106228 to Lee, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 39 pages.
	52.	Invalidity Chart for U.S. Patent No. 9,824,035 based on JEDEC Submission entitled "Proposed DDR4 DB BCOM Protocol," Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 22 pages.

Examiner Signature	/ZUBAIR AHMED/	Date Considered	03/21/2024
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
				Application Number	18/059,958	
				Filing Date	November 29, 2022	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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	53.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent Application Publication No. 2010/0312925 A1 to Osanai et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 53 pages.
	54.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 8,713,379 to Takefman et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 35 pages.
	55.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 8,020,022 to Tokuhito, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 30 pages.
	56.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 6,877,079 to Yoo et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	57.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 7,562,271 to Shaeffer et al. Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 52 pages.
	58.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 6,530,006 to Dodd et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 51 pages.
	59.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 9,361,955 to Muralimanohar et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 49 pages.
	60.	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 2012/0106228 to Lee, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 36 pages.
	61.	Invalidity Chart for U.S. Patent No. 10,268,608 based on JEDEC Submission entitled "Proposed DDR4 DB BCOM Protocol," Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 19 pages.
	62.	Obviousness Chart for U.S. Patent No. 9,824,035, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 14 pages.

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				Art Unit	2139
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	63.	Obviousness Chart for U.S. Patent No.10,268,608, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 9 pages.		
	64.	Micron’s Preliminary Identification of Proposed Claim Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Dec. 23, 2021), 14 pages.		
	65.	Netlist, Inc.’s Preliminary Proposed Claim Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Dec. 23, 2021), 7 pages.		
	66.	Defendants’ Identification of Extrinsic Evidence in Support of Their Preliminary Proposed Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Jan. 13, 2022), 21 pages.		
	67.	Netlist, Inc.’s Preliminary Identification of Extrinsic Evidence, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Jan 13, 2022), 16 pages.		
	68.	Declaration of Harold S. Stone, Ph.D., in Support of Defendants’ Claim Construction Positions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 27 pages.		
	69.	Declaration of Alan Jay Smith, Ph.D., in Support of Defendants’ Claim Construction Positions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 26 pages.		
	70.	Micron’s Opening Claim Construction Brief, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 40 pages.		
	71.	Joint Claim Construction Chart, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No 1:22-cv-00136-LY, Dkt. 66-1 (W.D. Tex. Apr. 14, 2022), 13 pages.		
	72.	Excerpt – Modern Dictionary of Electronics 7E, 1999		
	73.	JEDEC Solid State Technology Association, 1st Showing, Committee: JC-40.4, Committee Item Number: 0311.xx, ‘Proposed DDR4 DB BCOM Protocol,’ undated, 10 pages		
	74.	JEDEC Standard Double Data Rate (DDR) SDRAM Specification, JESD79 (June 2000), 78 pages		
Examiner Signature	/ZUBAIR AHMED/		Date Considered	03/21/2024

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
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				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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	75.	JEDEC Standard No. 21-C, PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification, Revision 1.3, January 2002, 83 pages.
	76.	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3, DDR3 SDRAM Standard, June 2007, 12 pages
	77.	JEDEC Standard JESD79-3F (Revision of JESD79-3E, July 2010), DDR3 SDRAM Standard, July 2012, 226 pages.
	78.	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3A, DDR3 SDRAM Specification, September 2007, 12 pages
	79.	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3C, DDR3 SDRAM, April 2008, 7 pages
	80.	Excerpt - JEDEC Standard No. 21C, DDR4 SDRAM Load Reduced DIMM Design Specification, August 2015, 2 pages
	81.	Inter Partes Review of US Patent No. 10860506, Case No. IPR2022-00711, IPR Petition re US Patent No. 10,860,506, filed March 22, 2022.
	82.	Inter Partes Review of US Patent No. 10860506, Case No. IPR2022-00711, 'Ex. 1003 - IPR2022-00711 Declaration of Robert Wedig,' filed March 22, 2022.
	83.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, IPR Petition re US Patent No. 9,128,632, filed January 20, 2017.
	84.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, 'Ex. 1003 - IPR2017-00730 Declaration of Trevor Mudge,' filed January 20, 2017.
	85.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Patent Owner's Preliminary Response, filed April 30, 2017.
	86.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Decision Denying Institution of Inter Partes Review, entered July 21, 2017.
	87.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, PETITIONERS' REQUEST FOR REHEARING, filed August 21, 2017.
	88.	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Decision Denying Petitioner's Request for Rehearing, entered October 30, 2017.
	89.	Inter Partes Review of US Patent No. 9606907, Case No. IPR2018-00362, Final Written Decision, entered June 27, 2019.
	90.	Complaint, Netlist, Inc. v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., Samsung Semiconductor, Inc., Case No. 2:21-cv-463 (E.D. Tex. Dec. 20, 2021), 49 pages.
	91.	First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. v. Netlist, Inc., C.A. No. 21-1453 (RGA) (USDC, Delaware, Jan. 18, 2022), 122 pages.
	92.	Excerpt - Microsoft Computer Dictionary, Fifth Edition, 2002.

Examiner Signature	/ZUBAIR AHMED/	Date Considered	03/21/2024
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed March 8, 2024</i>		
				Application Number		18/059,958
				Filing Date		November 29, 2022
				First Named Inventor		Hyun LEE
				Art Unit		2139
Examiner Name				Zubair AHMED		
Sheet	7	of	7	Attorney Docket Number		NL073-20US03

	93.	Altera, Quartus II Handbook Version 11.0 Volume 3: Verification, May 2011, 172 pages.
	94.	Bruce Jacob, et al., Memory Systems (Cache, DRAM, Disk), © 2008 by Elsevier, Inc., 296 pages.
	95.	Harold S. Stone, Microcomputer Interfacing, © 1982 by Addison-Wesley Publishing Company, Inc., 399 pages
	96.	JEDEC Standard JESD82-32A, DDR4 Data Buffer Definition (DDR4DB02), August 2019, 200 pages.
	97.	Declaration of Julie Carlson re “JEDEC Standard JESD79-3C (Revision of JESD79-3B, April 2008), DDR3 SDRAM, November 2008,” September 16, 2021, 335 pages
	98.	Declaration of Julie Carlson re “JEDEC Standard No. 21-C, PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification, Revision 1.3, January 2002,” September 16, 2021, 187 pages.
	99.	United States District Court for the District of Delaware, Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. v. Netlist Inc., C.A. No. 21-1453 (RGA), Exhibit 42 – JEDEC Presentation (March 2011) and Exhibit 43 – JEDEC Proposed DDR4 DB BCOM Protocol (undated), 44 pages

Examiner Signature	/ZUBAIR AHMED/	Date Considered	03/21/2024
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PE2E SEARCH - Search History (Prior Art)

There are no Prior Art searches to show.

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	23761	(G06F12/0802 OR G11C5/04 OR G11C7/1066 OR G11C7/1093 OR G11C29/023 OR G11C29/028).cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N2	138	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N3	84	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N4	2	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:19 AM
N5	0	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND (generat\$3 WITH module near2 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM

N6	4	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N7	4	N1 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N8	1689	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N9	0	N8 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:20 AM
N10	0	N8 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3 AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM
N11	0	N8 AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM
N12	0	N8 AND (memory near3 module) AND (clocks	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM

N13	0	SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) ((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock) AND pcb	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:21 AM
N14	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM
N15	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near4 (command\$1address OR c\$1a)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM
N16	0	((lee near2 hyun).in. OR (bhakta near3 jayesh).in) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM

N17	0	AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near2 clock) netlist AND (memory near3 module) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND (generat\$3 WITH module near2 clock)	(US-PGPUB; USPAT)	OR	ON	ON	2023/11/21 07:22 AM
N18	25	("5036221" "6072346" "6184701" "6530006" "6807077" "6877079" "6889300" "6906555" "7036053" "7275173" "7356639" "7562271" "7808849" "7890732" "8020022" "8407441" "8552779" "8565033" "8566516" "8599595" "8713379" "9153298" "9263103" "9361955" "20060277355").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/03/21 06:56 AM
N19	9	("20070008791" "20080025122" "20100309706" "20100312925" "20100312956" "20110062999" "20120106228" "20120296598" "9536579").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2024/03/21 06:57 AM
N20	0	N18 AND (((memory OR storage) near3 module) OR dimm) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT)	OR	ON	ON	2024/03/21 06:57 AM
N21	1	N19 AND (((memory OR storage) near3 module) OR dimm) AND (clocks SAME (command\$1address OR c\$1a) SAME (dq AND dqs)) AND pcb AND read\$3 AND writ\$3	(US-PGPUB; USPAT)	OR	ON	ON	2024/03/21 06:57 AM

Electronically filed July 8, 2024

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		
		Attorney Docket No.:	NL073-20US03

CERTIFICATE OF ELECTRONIC TRANSMISSION

I hereby certify that this correspondence is being transmitted to the United States Patent and Trademark Office (USPTO) via the USPTO's Patent Center on:

JULY 8, 2024 (DATE).

By: /Jamie J. Zheng/

Name: Jamie J. Zheng

Reg. No: 51167

AMENDMENT UNDER 37 C.F.R. § 1.312

Mail Stop: Issue Fee

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

This Amendment is filed under 37 C.F.R. § 1.312 and is filed after the Notice of Allowance dated April 9, 2024, in the above-referenced patent application..

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

Amendments to the Specification

Please amend the Abstract as follows:

A memory module is operable in a computer system ~~having a memory controller and a system bus and~~ comprises memory devices organized in one or more ranks ~~and in a plurality of groups,~~ and circuits configurable to receive from ~~the a~~ memory controller ~~a system clock and~~ input control and address (C/A) signals, and output registered C/A signals and buffer control signals. The memory module further comprises a plurality of buffer circuits. In response to the buffer control signals, each buffer circuit is configured to communicate first data/strobe signals with at least one memory device and to communicate second data/strobe signals with the memory controller. The buffer circuit includes at least one delay circuit configured to delay at least one signal of the first data/strobe signals based on a first delay and a second delay. The first delay is programmed into and stored in at least one configuration register during a configuration mode. The second delay is determined while the memory module is not operating in the configuration mode. ~~A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices that includes at least one corresponding memory device in each of the one or more ranks~~

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Cancelled)
3. (Cancelled)
4. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) including connectors ~~along an edge thereof for connecting~~ configured to connect to the set of C/A signal lines and the plurality of sets of data/strobe signal lines;
 - a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to ~~adjust timing of delay~~ one or more signals in the each buffer circuit of the plurality of buffer circuits;
 - a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines;
 - memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks;wherein:

in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to receive or output first data/strobe signals;

in response to the buffer control signals, a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to receive or output second data/strobe signals, the second data/strobe signals corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines;

at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

the memory module is operable in a configuration mode, and the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines during the configuration mode; and

the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

5. (Previously Presented) The memory module of claim 4, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

6. (Previously Presented) The memory module of claim 4, wherein

the buffer control signals include one or more mode signals indicating a read or write operation and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

in the configuration mode, the module control circuit is configured to output configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the

configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

7. (Previously Presented) The memory module of claim 6, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

8. (Previously Presented) The memory module of claim 4, wherein the at least one signal includes at least one data signal and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

9. (Previously Presented) The memory module of claim 4, wherein the at least one delay control circuit in the first buffer circuit is configured to determine the second delay based on at least one signal received from the module control circuit and at least one signal received from the memory controller during a memory operation.

10. (Previously Presented) the memory module of claim 4, wherein:
the system bus further includes one or more clock signal lines;
the module control circuit is further configurable to receive from the memory controller a system clock via the one or more clock signal lines and output a registered clock; and
the first buffer circuit is configured to receive the registered clock and to generate a first buffered clock, the first buffered clock having a programmable phase relationship with the registered clock determined by at least one of the one or more registers in the first buffer circuit.

11. (Previously Presented) The memory module of claim 10, wherein the first buffer circuit is further configured to output the first buffered clock to the first group of memory devices.

12. (Previously Presented) The memory module of claim 4, wherein the memory module is configurable to perform a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations; and
wherein at least one of the one or more configuration registers is programmed based on information derived from the set of training operations.

13. (Previously Presented) The memory module of claim 4, wherein the at least one delay circuit in the first buffer circuit is configured to delay the at least one signal based on a combination of the first delay and the second delay.

14. (Currently Amended) A method performed at a memory module in a computer system, the computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising a printed circuit board (PCB) including connectors ~~along an edge thereof for connecting~~ configured to connect to the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and memory devices mounted on the PCB and organized in ranks, the method comprising:

operating in a configuration mode, including:

receiving mode register command signals from the memory controller via the C/A signal lines; and

programming one or more configuration registers in each buffer circuit of a plurality of buffer circuits in response to the mode register command signals, wherein the plurality of buffer circuits are mounted on the PCB and each buffer circuit of the plurality of buffer circuits corresponds to a respective set of the plurality of sets of data/strobe signal lines, wherein the memory devices include a plurality of groups of memory devices, each group of the plurality of groups of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; and

performing a memory read or write operation, including:

receiving from the memory controller input address and control (C/A) signals via the set of C/A signal lines;

outputting registered C/A signals associated with the read or write operation to the memory devices, wherein in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to receive or output first data/strobe signals;

outputting buffer control signals associated with the memory read or write operation to the plurality of buffer circuits, wherein in response to the buffer control signals, a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to receive or output second data/strobe signals, the second data/strobe signals

corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; and

delaying at least one signal in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

wherein:

the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during the configuration mode; and

the second delay is determined by at least one delay control circuit in the first buffer circuit while the memory module is not operating in the configuration mode.

15. (Previously Presented) The method of claim 14, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

16. (Previously Presented) The method of claim 14, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

the method further comprises, during the configuration mode, outputting configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

17. (Previously Presented) The method of claim 16, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

18. (Previously Presented) The method of claim 14, wherein the at least one signal includes at least one data signal and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

19. (Currently Amended) The method of claim 14, wherein the at least one delay control circuit is configured to determine the second delay ~~information~~ based on at least one signal received from the memory controller during a memory operation.
20. (Previously Presented) the method of claim 14, wherein the system bus further includes one or more clock signal lines, the method further comprising:
- receiving from the memory controller a system clock via the one or more clock signal lines;
 - outputting a registered clock; and
 - generating a first buffered clock in the first buffer circuit, the first buffered clock having a programmable phase relationship with the registered clock based on a value stored in at least one of the one or more registers in the first buffer circuit.
21. (Previously Presented) The method of claim 20, further comprising outputting by the first buffer circuit the respective buffered clock to the first group of memory devices.
22. (Previously Presented) The method of claim 14, further comprising:
- performing a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations;
 - wherein at least one the one or more configuration registers in the first buffer circuit is programmed based on information derived from the set of training operations.
23. (Currently Amended) The method of claim 14, wherein ~~the at least one delay circuit is configured to delay~~ the at least one signal in the first buffer circuit is delayed by an amount based on a combination of the first delay ~~information~~ and the second delay ~~information~~.

REMARKS

This amendment is filed under 37 C.F.R. § 1.312, after the Notice of Allowance dated April 9, 2024, in which claims 4-23 are allowed.

REMARKS CONCERNING THE ABSTRACT

The Abstract was amended to correspond with the claims in the amendment submitted on March 8, 2024, with a request for continued examination (RCE). Due to clerical errors, some of the intended changes were left out. The errors are corrected in the amendment to the specification submitted herewith. Further, a few less important texts in the abstract are deleted to limit the word count to about 150. No new matter has been added.

REMARKS CONCERNING CLAIMS

Claims 4 and 14 are amended to address potential issues of clarity

Claims 19 and 23 are amended to address antecedent basis issues. For example, in claim 23, the recitation of “the at least one delay circuit is configured to delay” lacks antecedent basis and is thus replaced with similar recitation.

No new matter is added, and the amendments should not necessitate additional search and/or examination.

Entry of the amendments is therefore respectfully requested.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: July 8, 2024

/Jamie J. Zheng/

Jamie J. Zheng

Customer No. 79141

51,167

(Reg. No.)

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

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79141 7590 04/09/2024
USCH Law, PC
3524 Waverley St.
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I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

Jamie J. Zheng	(Typed or printed name)
/Jamie J. Zheng/	(Signature)
July 8, 2024	(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
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18/059,958 11/29/2022 Hyun Lee NL073-20US03 5230

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	07/09/2024

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 USCH Law, PC
2 _____
3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 60-3916

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Jamie J. Zheng/ Date July 8, 2024

Typed or printed name Jamie J. Zheng Registration No. 51167



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NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 04/09/2024
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

EXAMINER

AHMED, ZUBAIR

ART UNIT

PAPER NUMBER

2132

DATE MAILED: 04/09/2024

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	07/09/2024

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

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If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

Electronically filed July 8, 2024

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		
		Attorney Docket No.:	NL073-20US03

CERTIFICATE OF ELECTRONIC TRANSMISSION

I hereby certify that this correspondence is being transmitted to the United States Patent and Trademark Office (USPTO) via the USPTO's Patent Center on:

JULY 8, 2024 (DATE).

By: /Jamie J. Zheng/

Name: Jamie J. Zheng

Reg. No: 51167

AMENDMENT UNDER 37 C.F.R. § 1.312

Mail Stop: Issue Fee

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

This Amendment is filed under 37 C.F.R. § 1.312 and is filed after the Notice of Allowance dated April 9, 2024, in the above-referenced patent application..

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

Amendments to the Specification

Please amend the Abstract as follows:

A memory module is operable in a computer system ~~having a memory controller and a system bus and~~ comprises memory devices organized in one or more ranks ~~and in a plurality of groups,~~ and circuits configurable to receive from ~~the a~~ memory controller ~~a system clock and input control and address (C/A) signals, and output registered C/A signals and buffer control signals.~~ The memory module further comprises a plurality of buffer circuits. In response to the buffer control signals, each buffer circuit is configured to communicate first data/strobe signals with at least one memory device and to communicate second data/strobe signals with the memory controller. The buffer circuit includes at least one delay circuit configured to delay at least one signal of the first data/strobe signals based on a first delay and a second delay. The first delay is programmed into and stored in at least one configuration register during a configuration mode. The second delay is determined while the memory module is not operating in the configuration mode. ~~A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices that includes at least one corresponding memory device in each of the one or more ranks~~

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Cancelled)
3. (Cancelled)
4. (Currently Amended) A memory module operable in a computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:
 - a printed circuit board (PCB) including connectors ~~along an edge thereof for connecting~~ configured to connect to the set of C/A signal lines and the plurality of sets of data/strobe signal lines;
 - a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to ~~adjust timing of delay~~ adjust timing of delay one or more signals in the each buffer circuit of the plurality of buffer circuits;
 - a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines;
 - memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks;wherein:

in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to receive or output first data/strobe signals;

in response to the buffer control signals, a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to receive or output second data/strobe signals, the second data/strobe signals corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines;

at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

the memory module is operable in a configuration mode, and the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit in response to mode register command signals received from the memory controller via the set of C/A signal lines during the configuration mode; and

the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

5. (Previously Presented) The memory module of claim 4, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

6. (Previously Presented) The memory module of claim 4, wherein

the buffer control signals include one or more mode signals indicating a read or write operation and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

in the configuration mode, the module control circuit is configured to output configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the

configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

7. (Previously Presented) The memory module of claim 6, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

8. (Previously Presented) The memory module of claim 4, wherein the at least one signal includes at least one data signal and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

9. (Previously Presented) The memory module of claim 4, wherein the at least one delay control circuit in the first buffer circuit is configured to determine the second delay based on at least one signal received from the module control circuit and at least one signal received from the memory controller during a memory operation.

10. (Previously Presented) the memory module of claim 4, wherein:
the system bus further includes one or more clock signal lines;
the module control circuit is further configurable to receive from the memory controller a system clock via the one or more clock signal lines and output a registered clock; and
the first buffer circuit is configured to receive the registered clock and to generate a first buffered clock, the first buffered clock having a programmable phase relationship with the registered clock determined by at least one of the one or more registers in the first buffer circuit.

11. (Previously Presented) The memory module of claim 10, wherein the first buffer circuit is further configured to output the first buffered clock to the first group of memory devices.

12. (Previously Presented) The memory module of claim 4, wherein the memory module is configurable to perform a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations; and
wherein at least one of the one or more configuration registers is programmed based on information derived from the set of training operations.

13. (Previously Presented) The memory module of claim 4, wherein the at least one delay circuit in the first buffer circuit is configured to delay the at least one signal based on a combination of the first delay and the second delay.

14. (Currently Amended) A method performed at a memory module in a computer system, the computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising a printed circuit board (PCB) including connectors ~~along an edge thereof for connecting~~ configured to connect to the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and memory devices mounted on the PCB and organized in ranks, the method comprising:

operating in a configuration mode, including:

receiving mode register command signals from the memory controller via the C/A signal lines; and

programming one or more configuration registers in each buffer circuit of a plurality of buffer circuits in response to the mode register command signals, wherein the plurality of buffer circuits are mounted on the PCB and each buffer circuit of the plurality of buffer circuits corresponds to a respective set of the plurality of sets of data/strobe signal lines, wherein the memory devices include a plurality of groups of memory devices, each group of the plurality of groups of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; and

performing a memory read or write operation, including:

receiving from the memory controller input address and control (C/A) signals via the set of C/A signal lines;

outputting registered C/A signals associated with the read or write operation to the memory devices, wherein in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to receive or output first data/strobe signals;

outputting buffer control signals associated with the memory read or write operation to the plurality of buffer circuits, wherein in response to the buffer control signals, a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to receive or output second data/strobe signals, the second data/strobe signals

corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; and

delaying at least one signal in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

wherein:

the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during the configuration mode; and

the second delay is determined by at least one delay control circuit in the first buffer circuit while the memory module is not operating in the configuration mode.

15. (Previously Presented) The method of claim 14, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

16. (Previously Presented) The method of claim 14, wherein

the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and

the method further comprises, during the configuration mode, outputting configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

17. (Previously Presented) The method of claim 16, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

18. (Previously Presented) The method of claim 14, wherein the at least one signal includes at least one data signal and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

19. (Currently Amended) The method of claim 14, wherein the at least one delay control circuit is configured to determine the second delay ~~information~~ based on at least one signal received from the memory controller during a memory operation.
20. (Previously Presented) the method of claim 14, wherein the system bus further includes one or more clock signal lines, the method further comprising:
receiving from the memory controller a system clock via the one or more clock signal lines;
outputting a registered clock; and
generating a first buffered clock in the first buffer circuit, the first buffered clock having a programmable phase relationship with the registered clock based on a value stored in at least one of the one or more registers in the first buffer circuit.
21. (Previously Presented) The method of claim 20, further comprising outputting by the first buffer circuit the respective buffered clock to the first group of memory devices.
22. (Previously Presented) The method of claim 14, further comprising:
performing a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations;
wherein at least one the one or more configuration registers in the first buffer circuit is programmed based on information derived from the set of training operations.
23. (Currently Amended) The method of claim 14, wherein ~~the at least one delay circuit is configured to delay~~ the at least one signal in the first buffer circuit is delayed by an amount based on a combination of the first delay ~~information~~ and the second delay ~~information~~.

REMARKS

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Claims 4 and 14 are amended to address potential issues of clarity

Claims 19 and 23 are amended to address antecedent basis issues. For example, in claim 23, the recitation of “the at least one delay circuit is configured to delay” lacks antecedent basis and is thus replaced with similar recitation.

No new matter is added, and the amendments should not necessitate additional search and/or examination.

Entry of the amendments is therefore respectfully requested.

The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: July 8, 2024

/Jamie J. Zheng/

Jamie J. Zheng

Customer No. 79141

51,167

(Reg. No.)

UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov**ELECTRONIC ACKNOWLEDGEMENT RECEIPT**APPLICATION #
18/059,958RECEIPT DATE / TIME
07/08/2024 11:43:49 PM Z ETATTORNEY DOCKET #
NL073-20US03**Title of Invention**

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application InformationAPPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY Jamie Zheng

PATENT CENTER # 66279849

FILING DATE 11/29/2022

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun LeeCORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents**TOTAL DOCUMENTS: 5**

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NL073- 20US03_Issue_Fee.pdf	2	Issue Fee Payment (PTO-85B)	290 KB
NL073- 20US03_312_Amendment.pdf	9	-	182 KB
NL073- 20US03_312_Amendment- A.NA.pdf	(1-1) 1	Amendment after Notice of Allowance (Rule 312)	149 KB
NL073- 20US03_312_Amendment- ABST.pdf	(2-2) 1	Abstract	81 KB

NL073- 20US03__312__Amendment- CLM.pdf	(3-8)	6	Claims	115 KB
NL073- 20US03__312__Amendment- REM.pdf	(9-9)	1	Applicant Arguments/Remarks Made in an Amendment	102 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
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NL073- 20US03__312__Amendment- A.NA.pdf	A515F3517E378CFCAC3E5F4925EB4A0EA319DB832F710FB6E B3507DFF4F8F37C74ADC57C6CE1E8A731D06C83D17FF3CC4 BAEBE55C34B5DBD5AA00A412890CBB8
NL073- 20US03__312__Amendment- ABST.pdf	83864C2256B9F642E74F2573BADEC9850C918CFDF31D024E2 B5ED19320CF88CBDEEEAF2A9BBC2A136874EBFC0C58EB C261D1CE92E857D1DE795146A8806B81
NL073- 20US03__312__Amendment- CLM.pdf	A2C16F832DAF1AF872B789C096375E0CD4594DC2C8C0BC2D 120A08C8CC3F31F4354C73CFEF8973406D3F32CBCE5CB030 C02511992195347CF3BF441987882DD4
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APPLICATION #
18/059,958

RECEIPT DATE / TIME
07/08/2024 11:43:49 PM Z ET

ATTORNEY DOCKET #
NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY Jamie Zheng

PATENT CENTER # 66279849

AUTHORIZED BY -

CUSTOMER # 79141

FILING DATE 11/29/2022

CORRESPONDENCE ADDRESS -

FIRST NAMED INVENTOR Hyun Lee

Payment Information

PAYMENT METHOD
CARD / 2172

PAYMENT TRANSACTION ID
E202478N44315107

PAYMENT AUTHORIZED BY
Jamie Zheng

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1200.00	1	1200.00
TOTAL AMOUNT:				\$1,200.00

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230
79141	7590	07/19/2024	EXAMINER	
USCH Law, PC 3524 Waverley St. Palo Alto, CA 94306			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
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eofficeaction@appcoll.com
jzheng@uschlaw.com
patents@uschlaw.com

Response to Rule 312 Communication	Application No. 18/059,958	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

1. ☒ The amendment filed on 08 July 2024 under 37 CFR 1.312 has been considered, and has been:

a) ☐ entered.

b) ☐ entered as directed to matters of form not affecting the scope of the invention.

c) ☐ disapproved because the amendment was filed after the payment of the issue fee.
Any amendment filed after the date the issue fee is paid must be accompanied by a petition under 37 CFR 1.313(c)(1) and the required fee to withdraw the application from issue.

d) ☒ disapproved. See explanation below.

e) ☐ entered in part. See explanation below.

f) ☐ not entered because the supplemental or corrected Application Data sheet (ADS)

☐ was not accompanied by a petition to accept an unintentionally delayed claim under 37 CFR 1.55 or 27 CFR 1.78;
☐ did not identify the information being changed in accordance with 37 CFR 1.76(c)(2);
☐ was not properly signed in accordance with 37 CFR 1.76(e) (or 37 CFR 1.33(b) for applications filed prior to September 16, 2012).

The scope of independent claims 4 and 14 have been broadened after the amendment. The amendments to claims 19 and 23 change the scope of these claims.

/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
---	--

Electronically filed July 8, 2024

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		
		Attorney Docket No.:	NL073-20US03

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By: /Jamie J. Zheng/

Name: Jamie J. Zheng

Reg. No: 51167

AMENDMENT UNDER 37 C.F.R. § 1.312

Mail Stop: Issue Fee

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

This Amendment is filed under 37 C.F.R. § 1.312 and is filed after the Notice of Allowance dated April 9, 2024, in the above-referenced patent application..

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/29/2022	Hyun Lee	NL073-20US03	5230
79141	7590	08/05/2024	EXAMINER	
USCH Law, PC			AHMED, ZUBAIR	
3524 Waverley St.			ART UNIT	
Palo Alto, CA 94306			PAPER NUMBER	
			2132	
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Application No. : 18059958
Applicant : Lee
Filing Date : 11/29/2022
Date Mailed : 08/05/2024

NOTICE TO FILE CORRECTED APPLICATION PAPERS

Notice of Allowance Mailed

This application has been accorded an Allowance Date and is being prepared for issuance. The application, however, is incomplete for the reasons below.

Applicant is given two (2) months from the mail date of this Notice within which to respond. This time period for reply is extendable under 37 CFR 1.136(a) for only TWO additional MONTHS.

The informalities requiring correction are indicated in the attachment(s). If the informality pertains to the abstract, specification (including claims) or drawings, the informality must be corrected with an amendment in compliance with 37 CFR 1.121 (or, if the application is a reissue application, 37 CFR 1.173). Such an amendment may be filed after payment of the issue fee if limited to correction of informalities noted herein. See Waiver of 37 CFR 1.312 for Documents Required by the Office of Patent Publication, 1280 Off. Gaz. Patent Office 918 (March 23, 2004). In addition, if the informality is not corrected until after payment of the issue fee, for purposes of 35 U.S.C. 154(b)(1)(iv), "all outstanding requirements" will be considered to have been satisfied when the informality has been corrected. A failure to respond within the above-identified time period will result in the application being ABANDONED.

See attachment(s).

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- ☐ There is a hole or the image thereof within the illustration. FIG(s)
- ☐ The illustration is penetrated or traversed by a solid or broken line that is not intended to be part of the drawing, such as a dark line caused by a flaw in the copying process. FIG(s)
- ☐ An ink stamp or the image thereof obscures part of the illustration. FIG(s)
- ☐ The drawing is marred by black smudges, obliterations, or fax/copier marks (for example, speckles or dots in a substantial portion of the drawing). FIG(s)
- ☐ Figure numbers are duplicated or missing. FIG(s)
- ☐ Drawing sheet or figure is missing. FIG(s)
- ☐ Numbers, letters, or reference characters in the drawing have been crossed out or are illegibly handwritten. FIG(s)
- ☐ The character of the lines, numbers, and letters is poor. FIG(s)
- ☐ The drawing's background shows that the original drawing was made on graph paper or other paper with a pattern or decoration. FIG(s)
- ☐ The FIG. number label is placed in a location that causes the drawing to be read upside down. FIG(s)
- ☐ Data, a reference number, or part of the drawing is truncated or missing, or a lead line has no reference number. FIG(s)
- ☐ The drawing and/or the FIG. label contain(s) foreign language. FIG(s)
- ☐ This utility application contains a photograph of a view that is capable of being illustrated as a line drawing. FIG(s)
- ☐ A petition under 37 CFR 1.84(a)(2) to accept color drawings has been granted, but the brief description of the drawings in the specification does not contain (or has not been amended to contain) the paragraph required by 37 CFR 1.84(a)(2)(iii).
- ☐ This reissue application contains added and/or amended drawings that are not labeled as "New" or "Amended" or "Canceled" as required by 37 CFR 1.173(b)(3). FIG(s)
- ☐ This Design reissue application contains a drawing that is labeled as "Canceled" but is not surrounded by brackets, or a drawing that is surrounded by brackets but is not labeled as "Canceled." See 37 CFR 1.173(b)(3). FIG(s)
- ☒ OTHER: Figure label "FIG. 2D" appears on Sheet 4 of 22 and Sheet 5 of 22, improperly.
- ☐ COMMENTS:

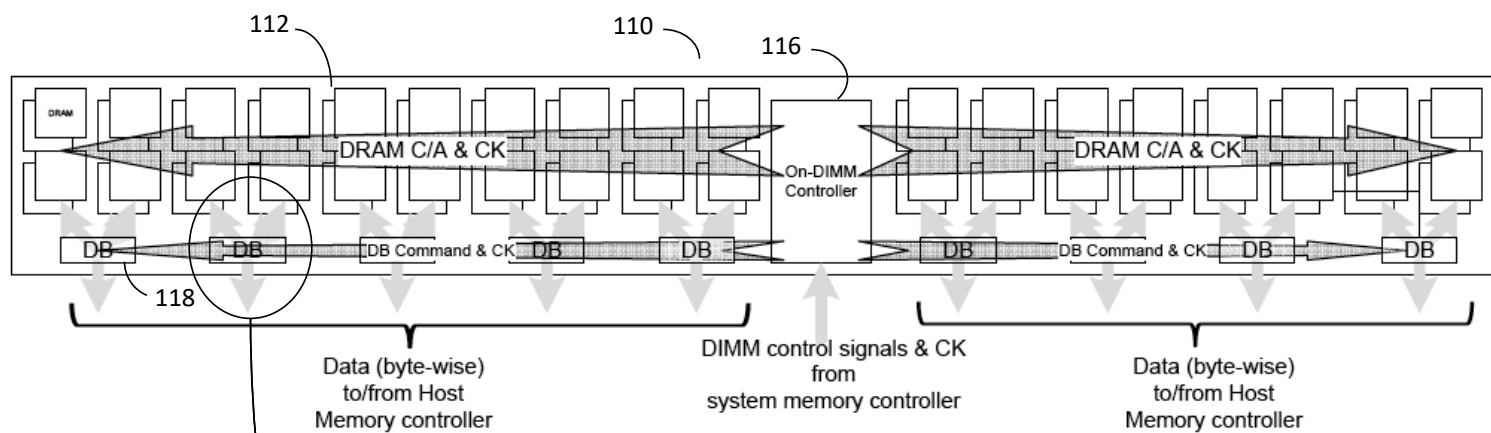


FIG. 2D

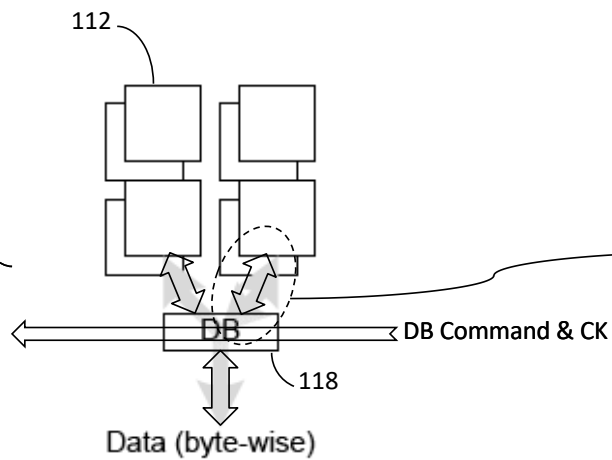


FIG. 2C

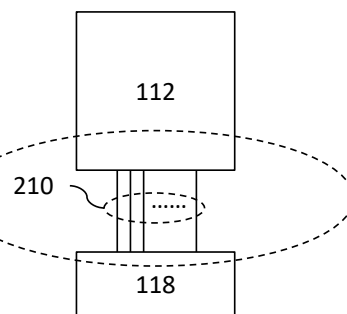


FIG. 2E

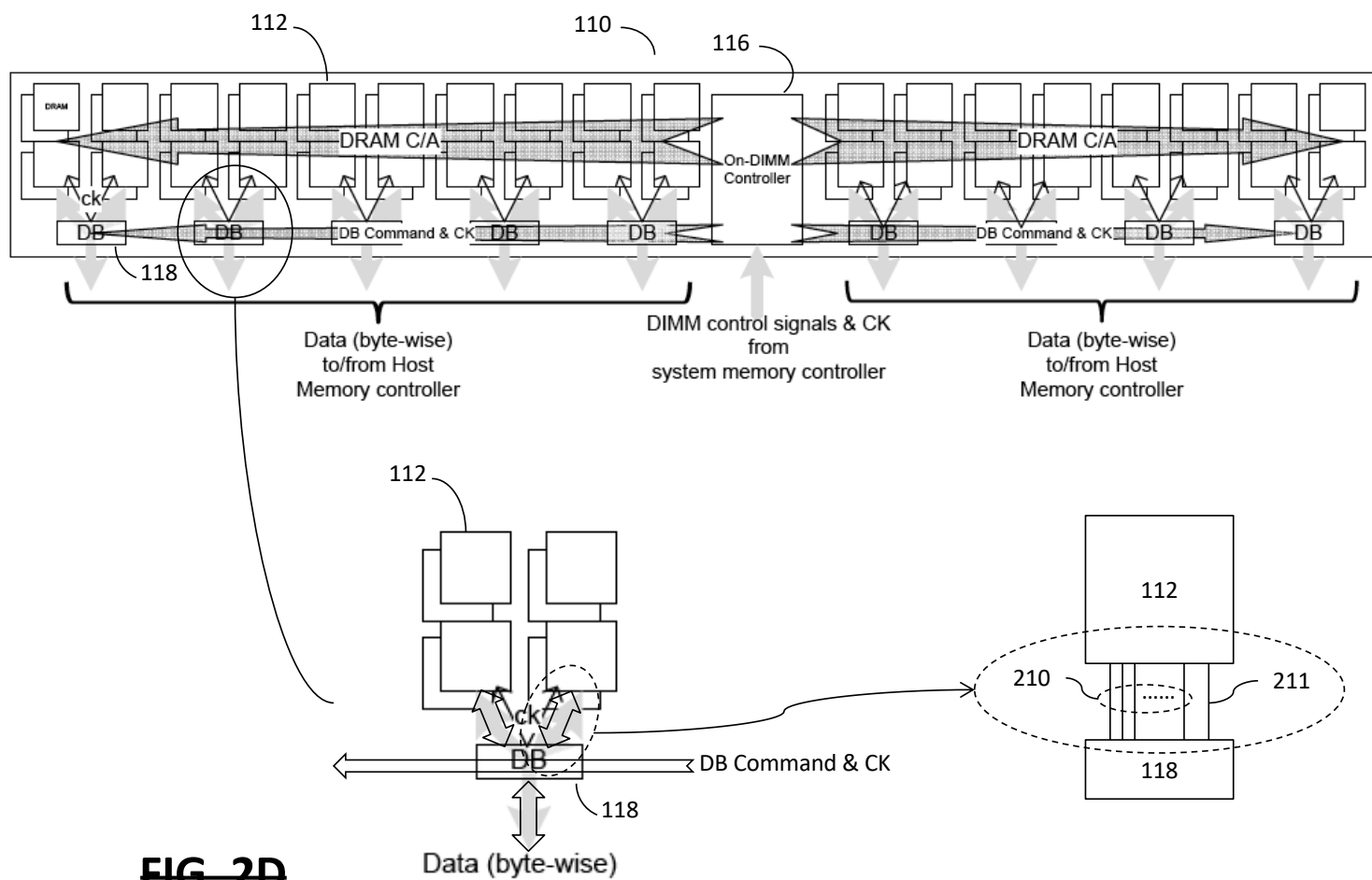


FIG. 2D

FIG. 2D

Placeholder Sheet for Supplemental Content

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Document Date: 09/30/2024

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Application Number: 18059958

Document Date: 09/30/2024

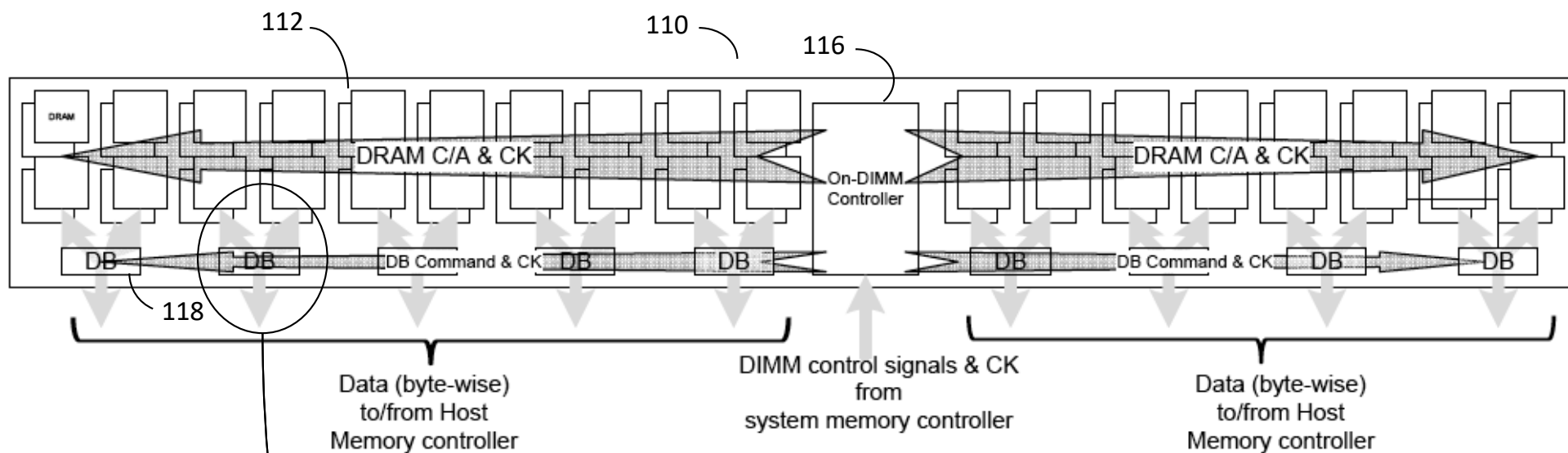
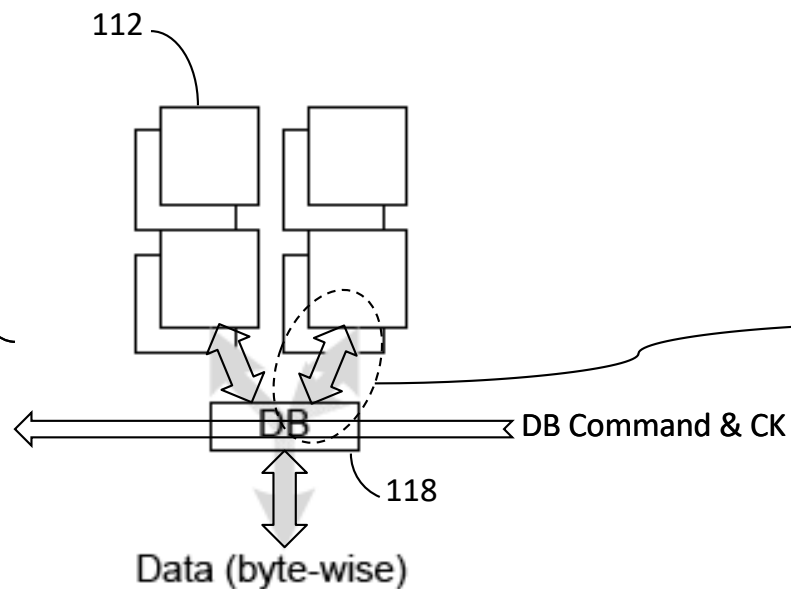
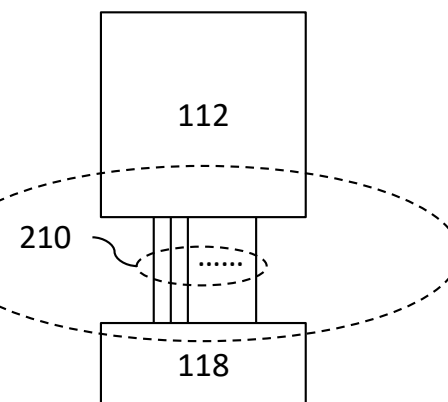
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**FIG. 2D****FIG. 2C****FIG. 2E**

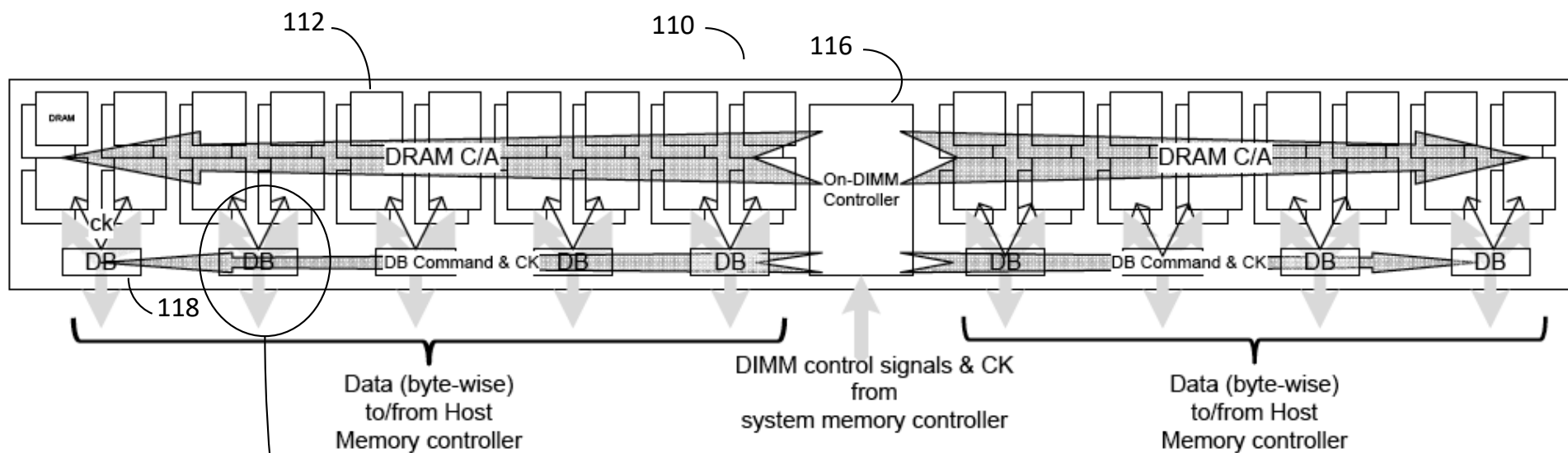


FIG. 2D

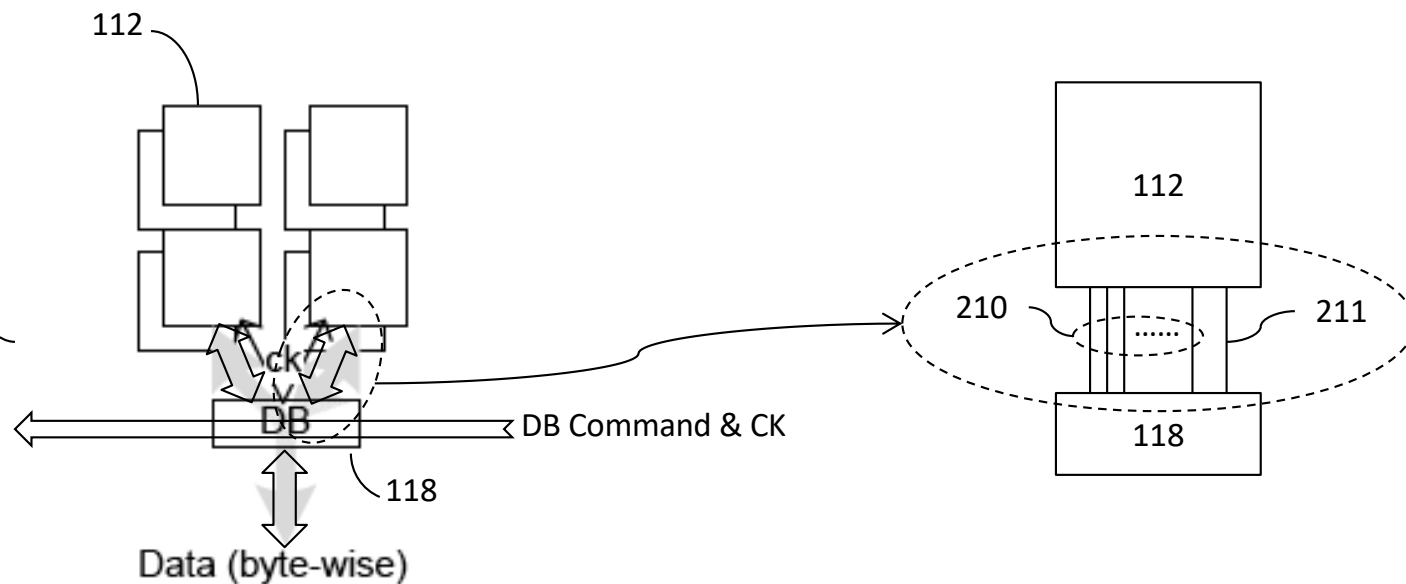


FIG. 2D

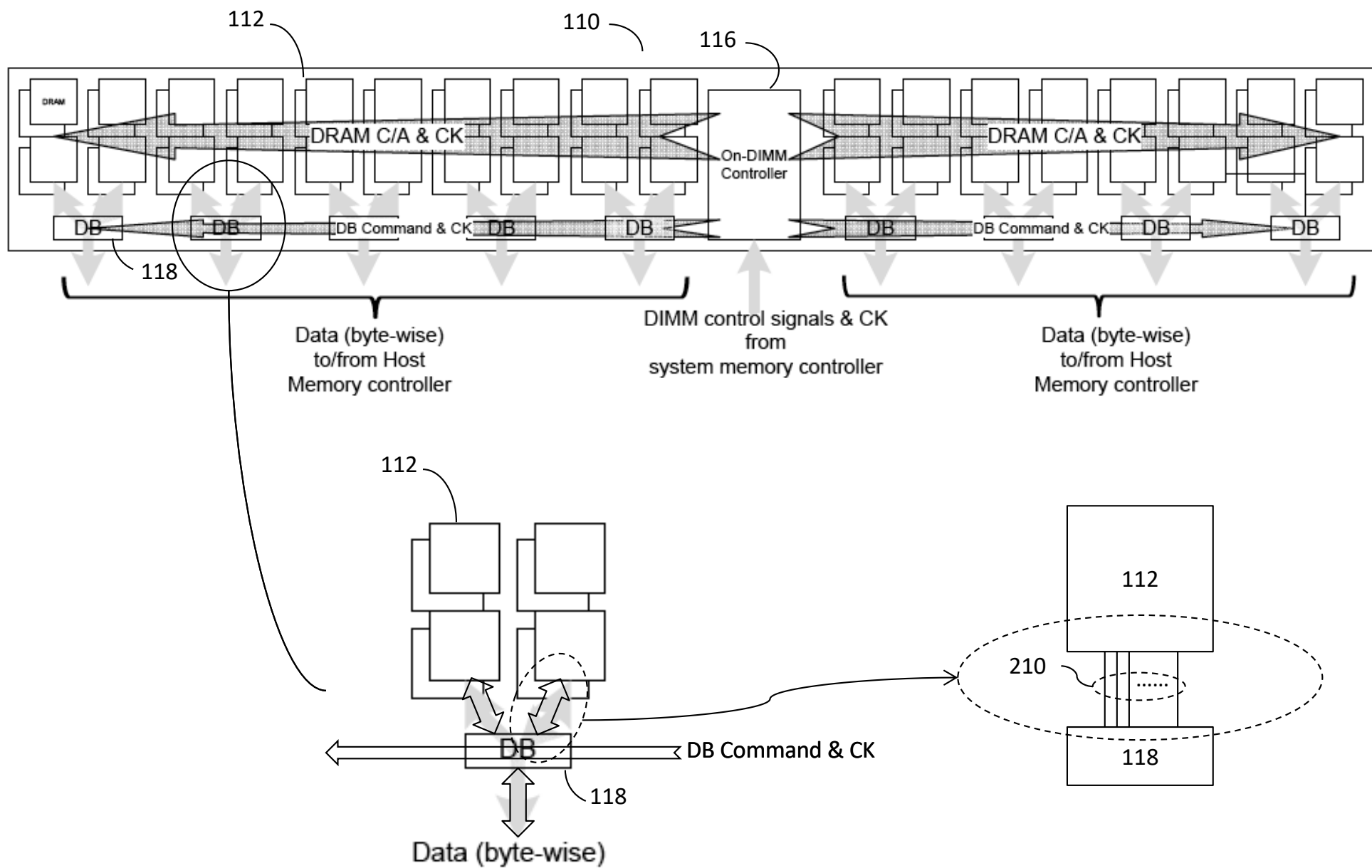


FIG. 2C

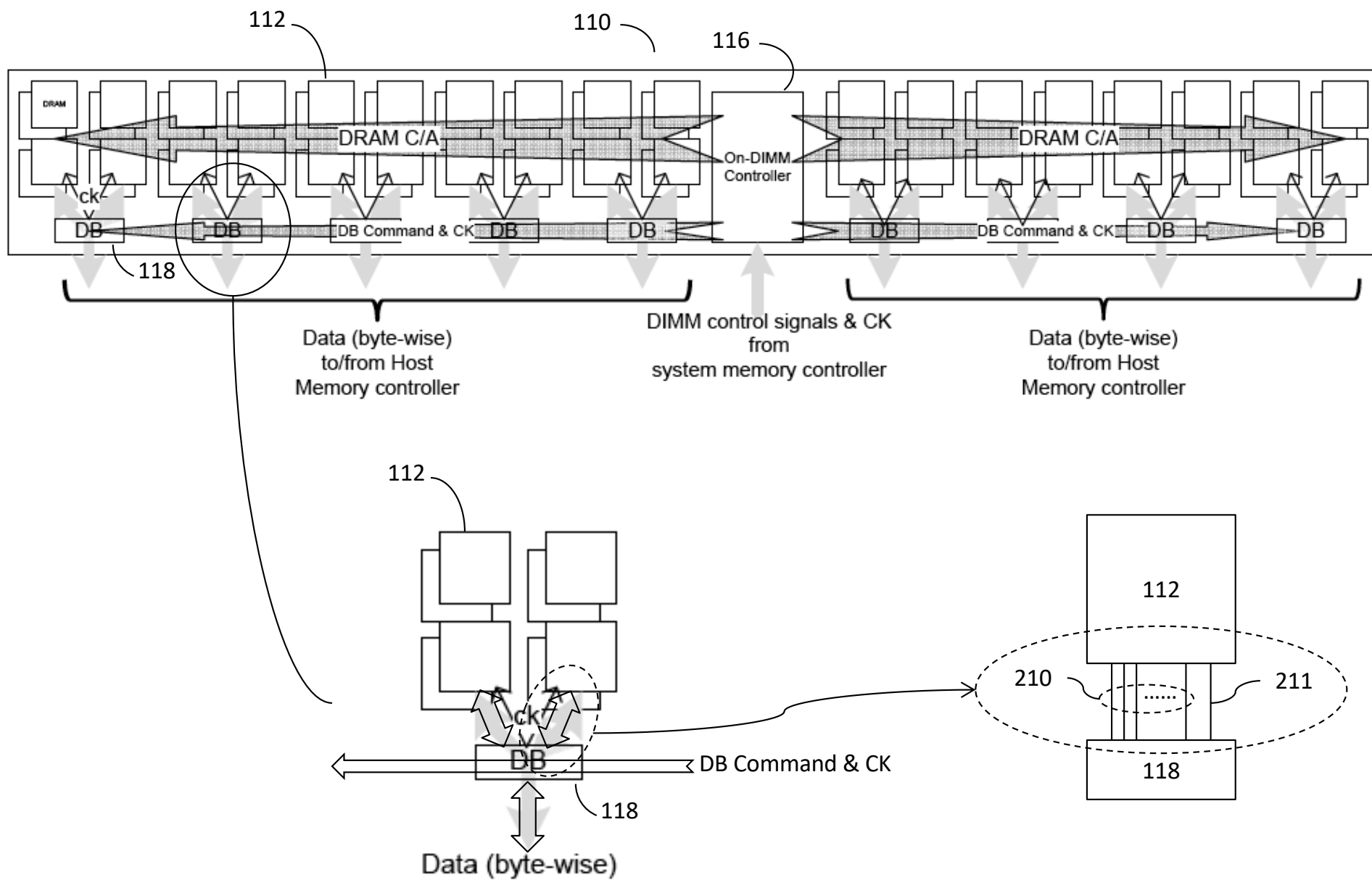


FIG. 2D

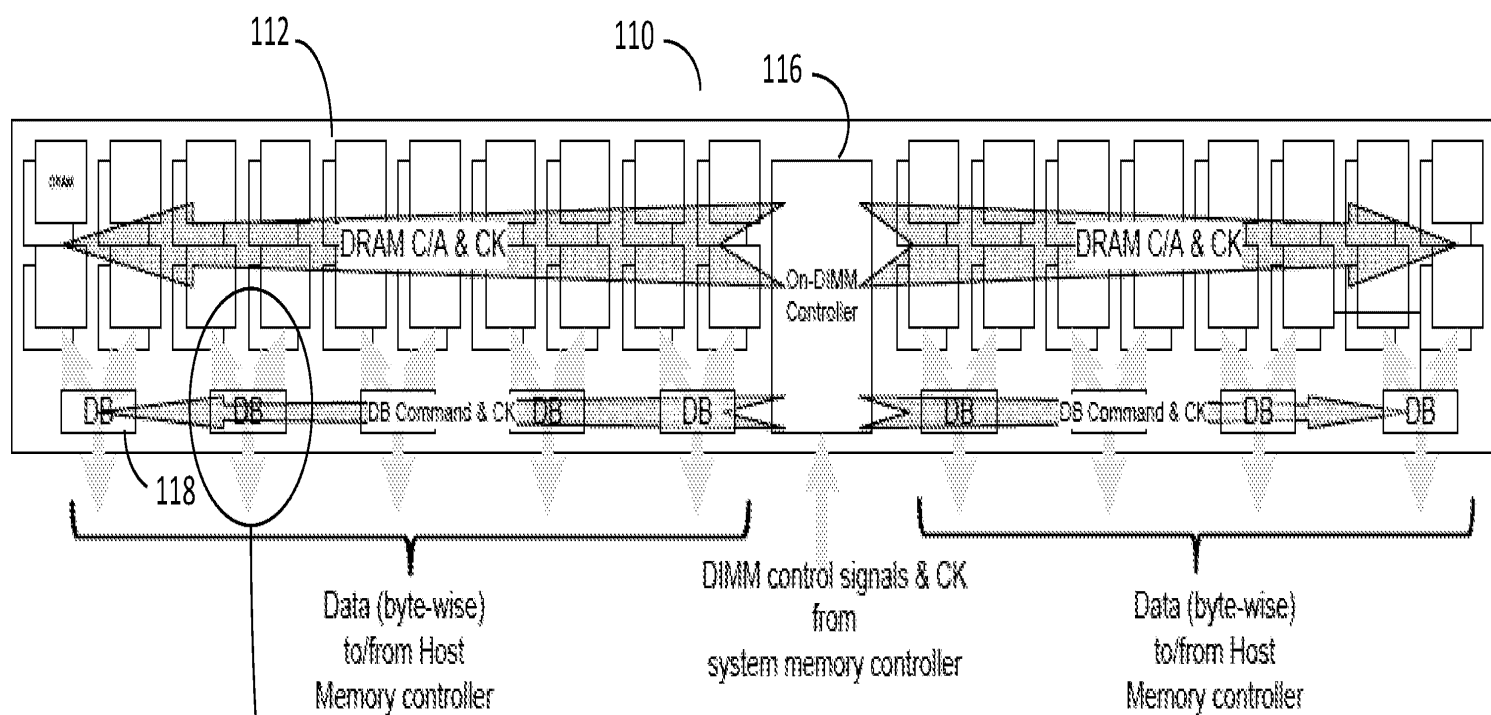


FIG. 2D

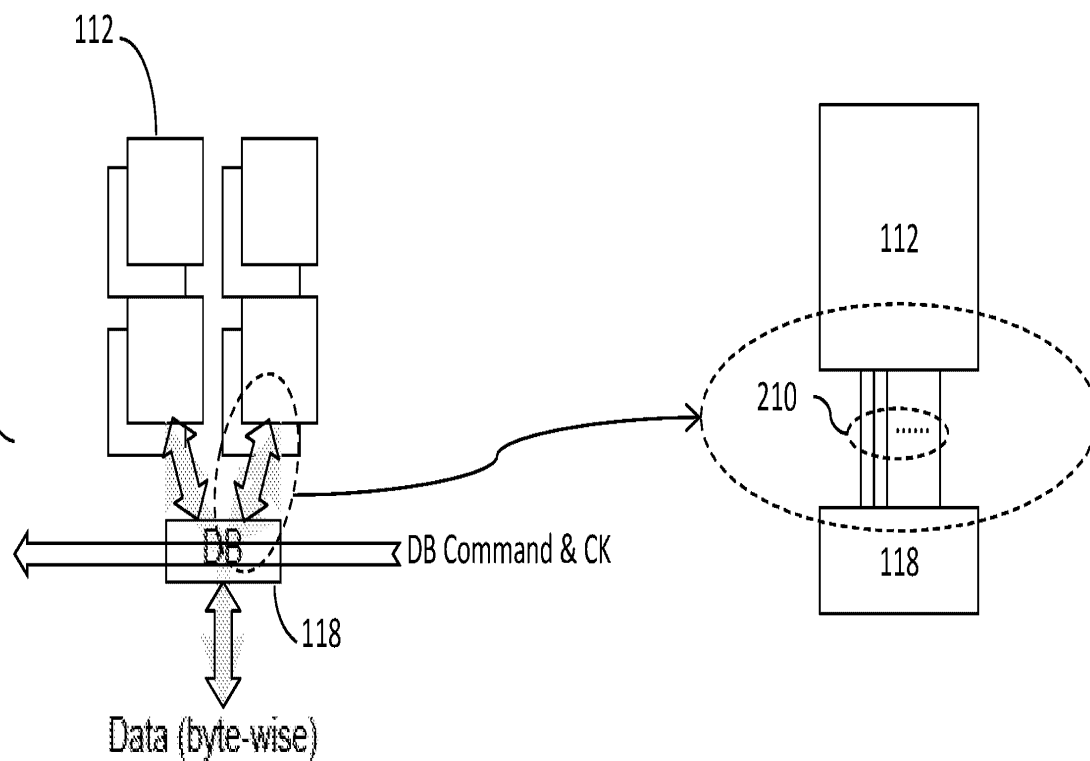


FIG. 2C

FIG. 2C

FIG. 2D



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79141	7590	08/05/2024		
USCH Law, PC 3524 Waverley St. Palo Alto, CA 94306			EXAMINER AHMED, ZUBAIR	
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Application No. : 18059958
Applicant : Lee
Filing Date : 11/29/2022
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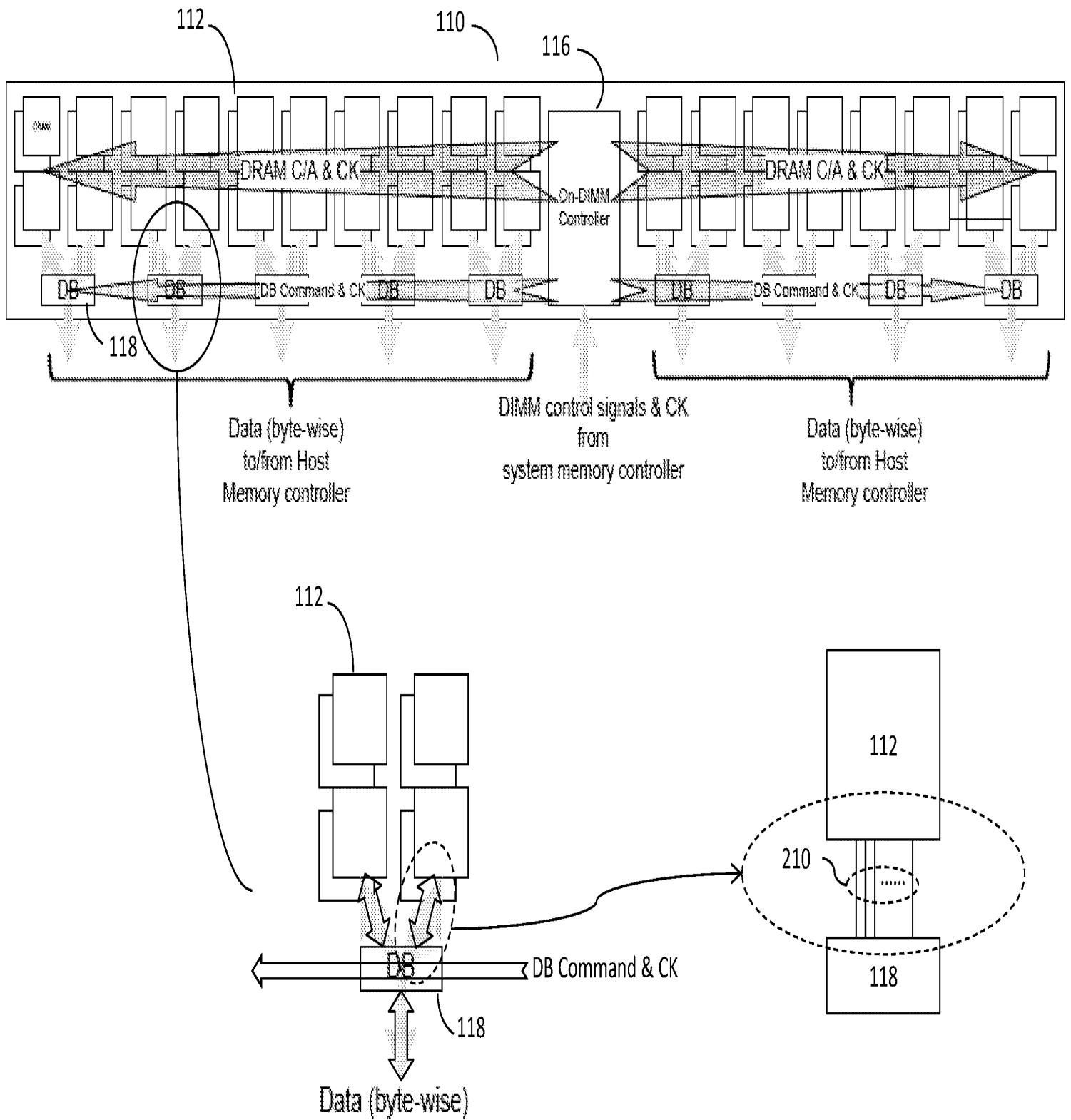


FIG. 2C

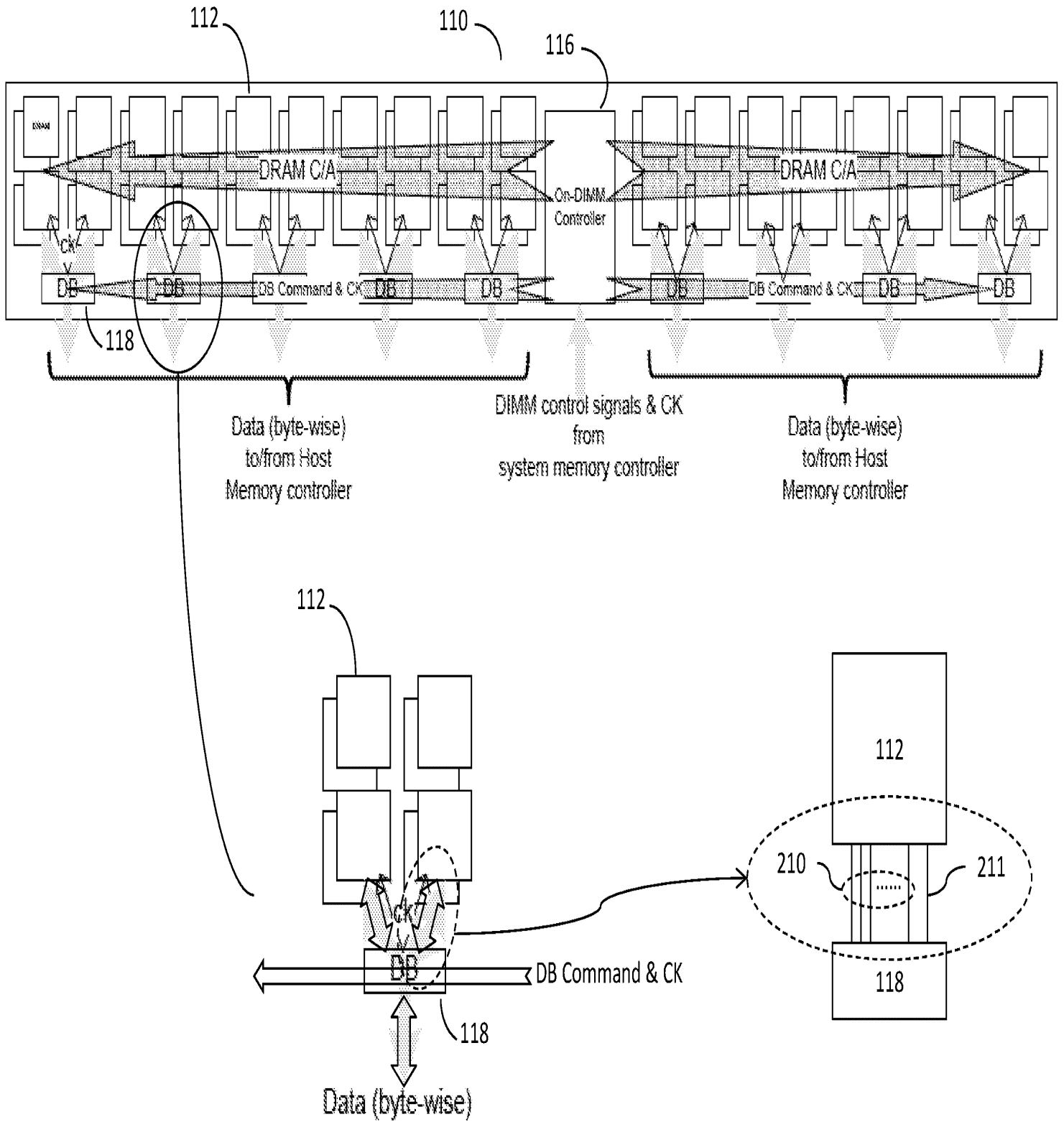


FIG. 2D

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APPLICATION #
18/059,958RECEIPT DATE / TIME
09/30/2024 10:59:05 PM Z ETATTORNEY DOCKET #
NL073-20US03

Title of Invention

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Application Information

APPLICATION TYPE Utility - Nonprovisional Application
under 35 USC 111(a)

PATENT # -

CONFIRMATION # 5230

FILED BY Jamie Zheng

PATENT CENTER # 67387880

FILING DATE 11/29/2022

CUSTOMER # 79141

FIRST NAMED
INVENTOR Hyun LeeCORRESPONDENCE
ADDRESS -

AUTHORIZED BY -

Documents

TOTAL DOCUMENTS: 4

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
NL073- 20US03_M327_20240805.pdf	3	Miscellaneous Incoming Letter	110 KB
NL073- 20US03_Annotated_Drawing _Sheets.pdf	2	Drawings-only black and white line drawings	136 KB
NL073- 20US03_Response_M327_N CAP.pdf	1	Miscellaneous Incoming Letter	104 KB
NL073- 20US03_Replacement_Draw ing_Sheets.pdf	2	Drawings-only black and white line drawings	136 KB

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NL073- 20US03_Annotated_Drawing_S heets.pdf	6A62F62726C990718945EFFDB803356B7A500BF831D51387D C8FBB58A3D43B99E4CACC543E45D6C17F46283BA7B2F8FBB F263B6711BA79A64A92F7A9B70739C6
NL073- 20US03_Response_M327_NC AP.pdf	77577CCA5848D9D2CF13D84F41FD3D83402A145C206EBA308 EEEE39EDE37F0B3C1A881884888C59C3149584BCB827B46 D6200DD093F65A98472EBB81E7EF2FA
NL073- 20US03_Replacement_Drawin g_Sheets.pdf	97BFDA2864813C8E9CF37F34657B74CCB3BC43A9475070D83 B5B466CACCD0EBB0AFC20966DA095215EFE21E92961DFFD3 A46B8080DD3CAA225F84FFEE17051BC

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Electronically filed September 30, 2024

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE	Confirmation No.:	5230
Serial No.:	18/059,958	Group Art Unit:	2132
Filed:	November 29, 2022	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		
		Attorney Docket No.:	NL073-20US03

RESPONSE TO NOTICE TO FILE CORRECTED APPLICATION PAPERS

Mail Stop: Issue Fee

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Notice to File Corrected Application Papers dated August 5, 2024 (the "Notice"), applicant submits herewith replacement drawing sheets 4-5 (of 22) containing FIGS. 2C-2D, respectively, to address the drawing deficiencies identified in the Notice. The corresponding annotated sheets showing changes made are also submitted herewith. A copy of the Notice is also submitted herewith for your easy reference.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US03).

The undersigned can be reached at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date:	<u>September 30, 2024</u>	<u>/Jamie J. Zheng/</u>	<u>51,167</u>
		Jamie J. Zheng	(Reg. No.)
		Customer No. 79141	



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
18/059,958	11/29/2022	2132	1980	NL073-20US03	3	1

79141
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

CONFIRMATION NO. 5230
REPLACEMENT FILING RECEIPT



0000000079073032

Date Mailed: 10/08/2024

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 17/141,978 01/05/2021 PAT 11,513,955
which is a CON of 16/432,700 06/05/2019 PAT 10,884,923
which is a CON of 14/445,035 07/28/2014 PAT 10,324,841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the Patent Prosecution Highway program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 12/13/2022

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 18/059,958**

Projected Publication Date: 07/27/2023

Non-Publication Request: No

Early Publication Request: No
Title

MEMORY MODULE WITH STATIC AND DYNAMIC TIMING ADJUSTMENT IN DATA BUFFERS

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/059,958	11/05/2024	12135644	NL073-20US03	5230

79141 7590 10/16/2024
USCH Law, PC
3524 Waverley St.
Palo Alto, CA 94306

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

NETLIST, INC., Irvine, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit [SelectUSA.gov](https://selectusa.gov).

United
States
of
America

To Promote the Progress

of Science and Useful Arts

The Director

*of the United States Patent and Trademark Office has received
an application for a patent for a new and useful invention. The title
and description of the invention are enclosed. The requirements
of law have been complied with, and it has been determined that
a patent on the invention shall be granted under the law.*

Therefore, this United States

Patent

grants to the person(s) having title to this patent the right to exclude others from making, using, offering for sale, or selling the invention throughout the United States of America or importing the invention into the United States of America, and if the invention is a process, of the right to exclude others from using, offering for sale or selling throughout the United States of America, products made by that process, for the term set forth in 35 U.S.C. 154(a)(2) or (c)(1), subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b). See the Maintenance Fee Notice on the inside of the cover.

Katherine Kelly Vidal

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE

Maintenance Fee Notice

If the application for this patent was filed on or after December 12, 1980, maintenance fees are due three years and six months, seven years and six months, and eleven years and six months after the date of this grant, or within a grace period of six months thereafter upon payment of a surcharge as provided by law. The amount, number and timing of the maintenance fees required may be changed by law or regulation. Unless payment of the applicable maintenance fee is received in the United States Patent and Trademark Office on or before the date the fee is due or within a grace period of six months thereafter, the patent will expire as of the end of such grace period.

Patent Term Notice

If the application for this patent was filed on or after June 8, 1995, the term of this patent begins on the date on which this patent issues and ends twenty years from the filing date of the application or, if the application contains a specific reference to an earlier filed application or applications under 35 U.S.C. 120, 121, 365(c), or 386(c), twenty years from the filing date of the earliest such application (“the twenty-year term”), subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b), and any extension as provided by 35 U.S.C. 154(b) or 156 or any disclaimer under 35 U.S.C. 253.

If this application was filed prior to June 8, 1995, the term of this patent begins on the date on which this patent issues and ends on the later of seventeen years from the date of the grant of this patent or the twenty-year term set forth above for patents resulting from applications filed on or after June 8, 1995, subject to the payment of maintenance fees as provided by 35 U.S.C. 41(b) and any extension as provided by 35 U.S.C. 156 or any disclaimer under 35 U.S.C. 253.

(12) **United States Patent**
Lee et al.

(10) **Patent No.:** **US 12,135,644 B2**
(45) **Date of Patent:** **Nov. 5, 2024**

(54) **MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION**

(71) Applicant: **NETLIST, INC.**, Irvine, CA (US)
(72) Inventors: **Hyun Lee**, Ladera Ranch, CA (US);
Jayesh R. Bhakta, Cerritos, CA (US)
(73) Assignee: **Netlist, Inc.**, Irvine, CA (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/059,958**
(22) Filed: **Nov. 29, 2022**

(65) **Prior Publication Data**
US 2023/0236970 A1 Jul. 27, 2023

Related U.S. Application Data
(63) Continuation of application No. 17/141,978, filed on Jan. 5, 2021, now Pat. No. 11,513,955, which is a (Continued)

(51) **Int. Cl.**
G06F 12/0802 (2016.01)
G11C 5/04 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **G06F 12/0802** (2013.01); **G11C 5/04** (2013.01); **G11C 7/1066** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC G06F 12/0802; G11C 5/04; G11C 7/1066;
G11C 7/1093; G11C 29/023; G11C 29/028
(Continued)

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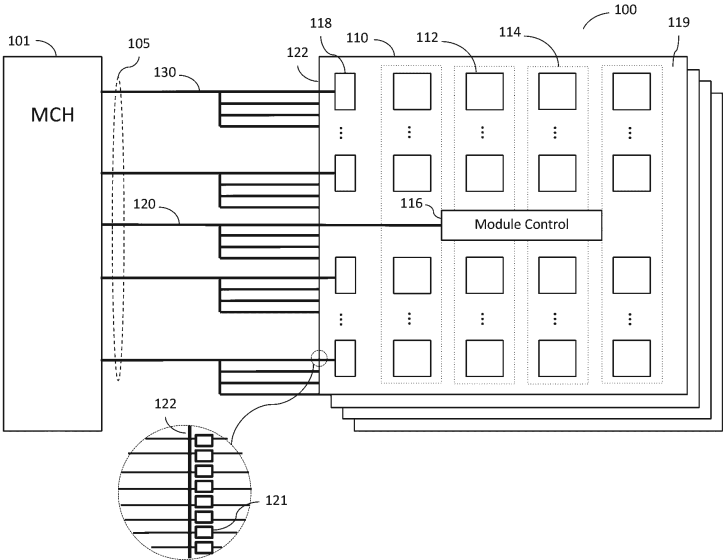
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(Continued)

Primary Examiner — David Yi
Assistant Examiner — Zubair Ahmed
(74) *Attorney, Agent, or Firm* — USCH Law, PC

(57) **ABSTRACT**
A memory module is operable in a computer system having a memory controller and a system bus and comprises memory devices organized in one or more ranks and in a plurality of groups, and circuits configurable to receive from the memory controller a system clock and input control and address (C/A) signals, and output registered C/A signals and buffer control signals. The memory module further comprises a plurality of buffer circuits. In response to the buffer control signals, each buffer circuit is configured to communicate first data/strobe signals with at least one memory device and to communicate second data/strobe signals with the memory controller.
The buffer circuit includes at least one delay circuit configured to delay at least one signal of the first data/strobe signals based on a first delay and a second delay. A respective local clock signal has a respective phase relationship with the module clock signal and is output to a corresponding group of the memory devices that includes at least one corresponding memory device in each of the one or more ranks.

20 Claims, 22 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/432,700, filed on Jun. 5, 2019, now Pat. No. 10,884,923, which is a continuation of application No. 14/445,035, filed on Jul. 28, 2014, now Pat. No. 10,324,841.

- (60) Provisional application No. 61/859,215, filed on Jul. 27, 2013.

- (51) **Int. Cl.**
GIIC 7/10 (2006.01)
GIIC 29/02 (2006.01)

- (52) **U.S. Cl.**
 CPC **GIIC 7/1093** (2013.01); **GIIC 29/023** (2013.01); **GIIC 29/028** (2013.01)

- (58) **Field of Classification Search**
 USPC 711/120, 119
 See application file for complete search history.

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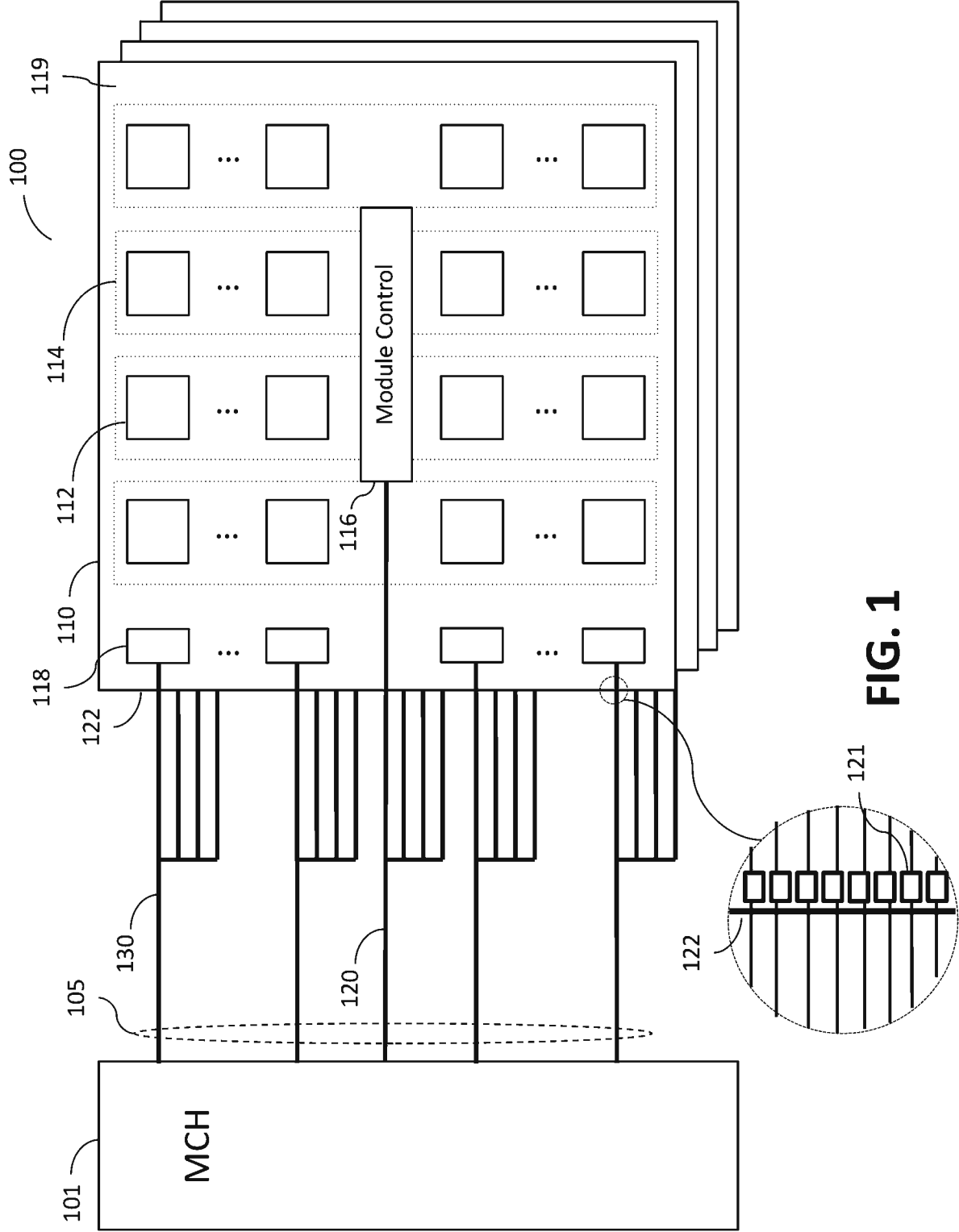
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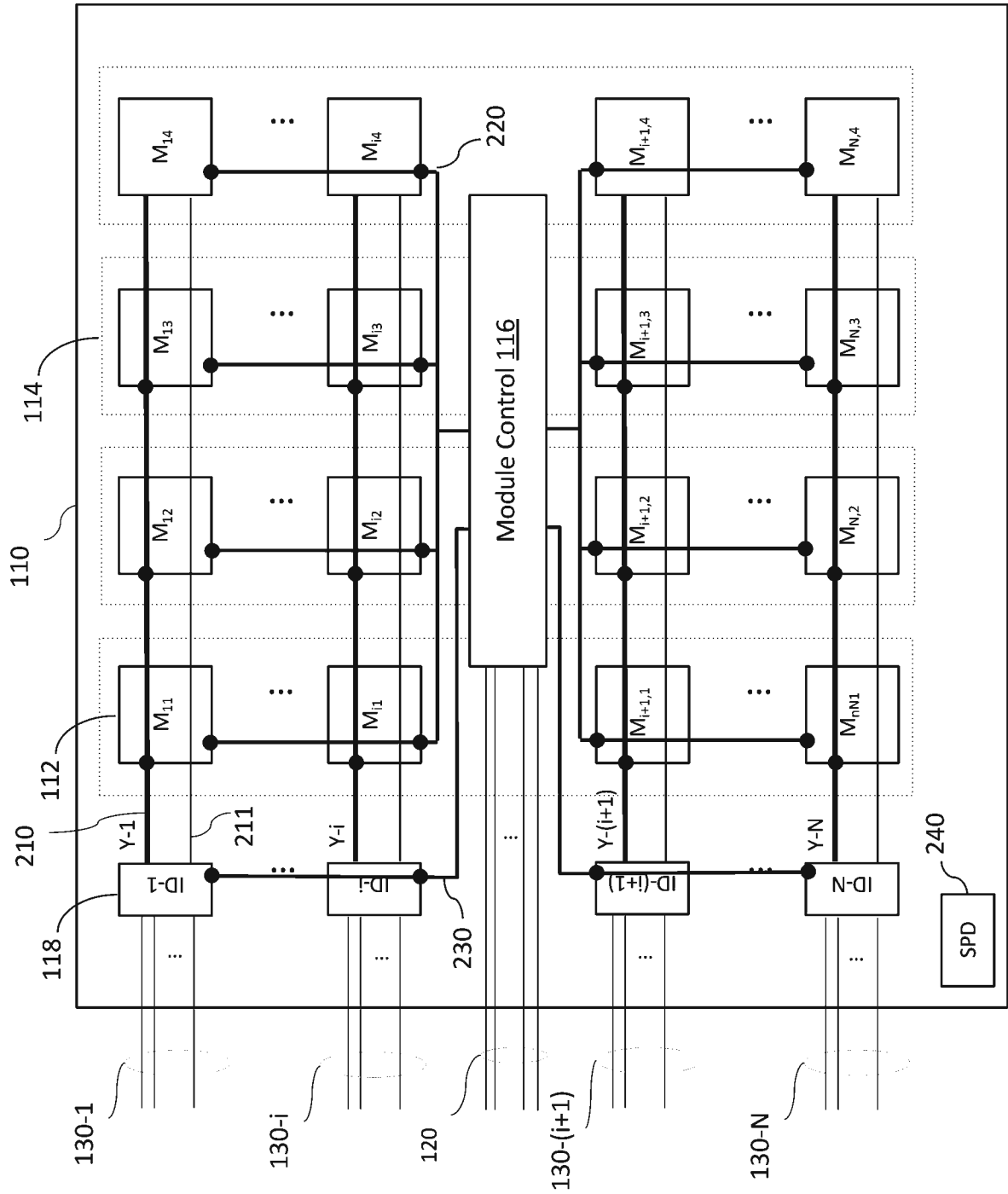


FIG. 2A

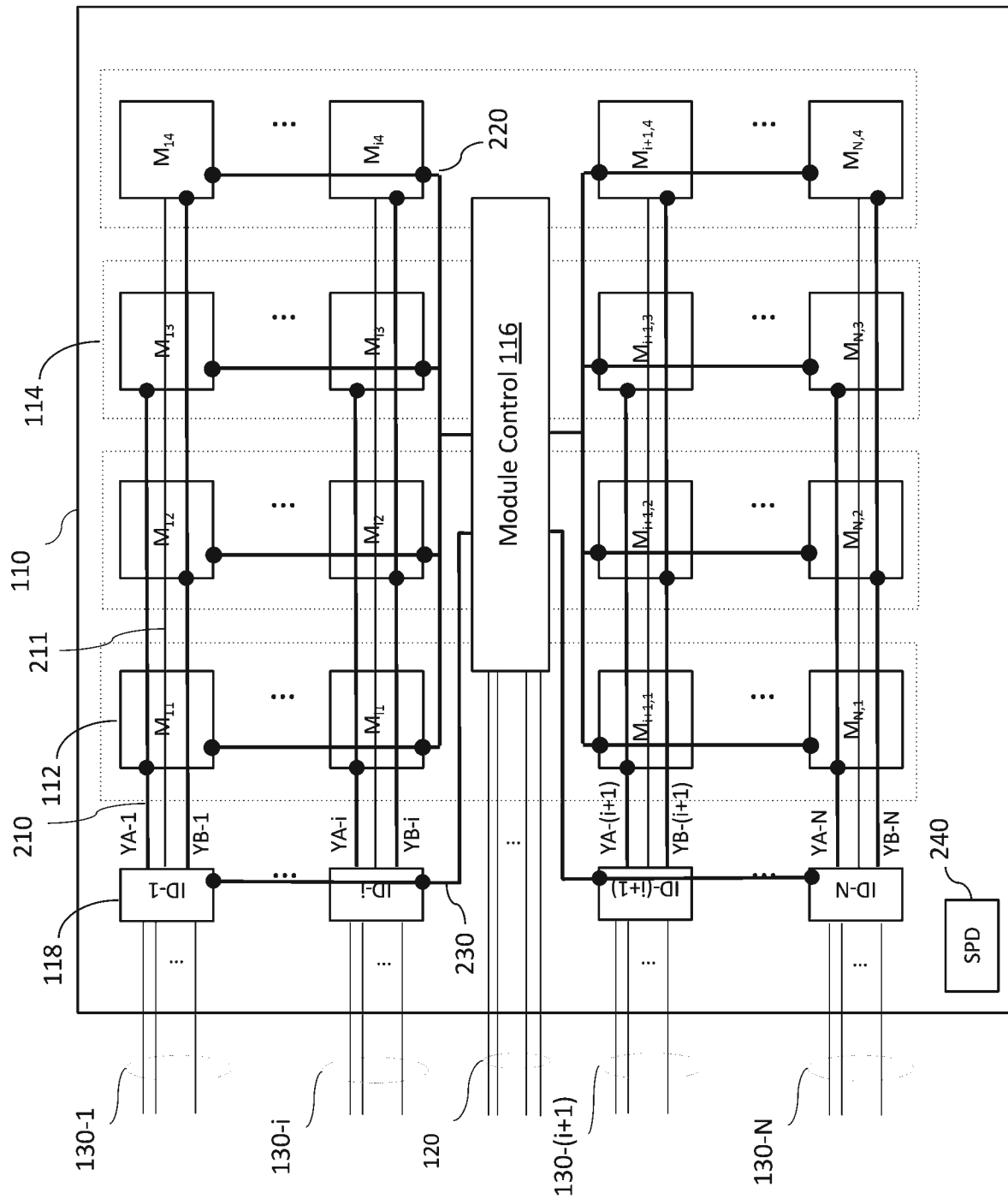


FIG. 2B

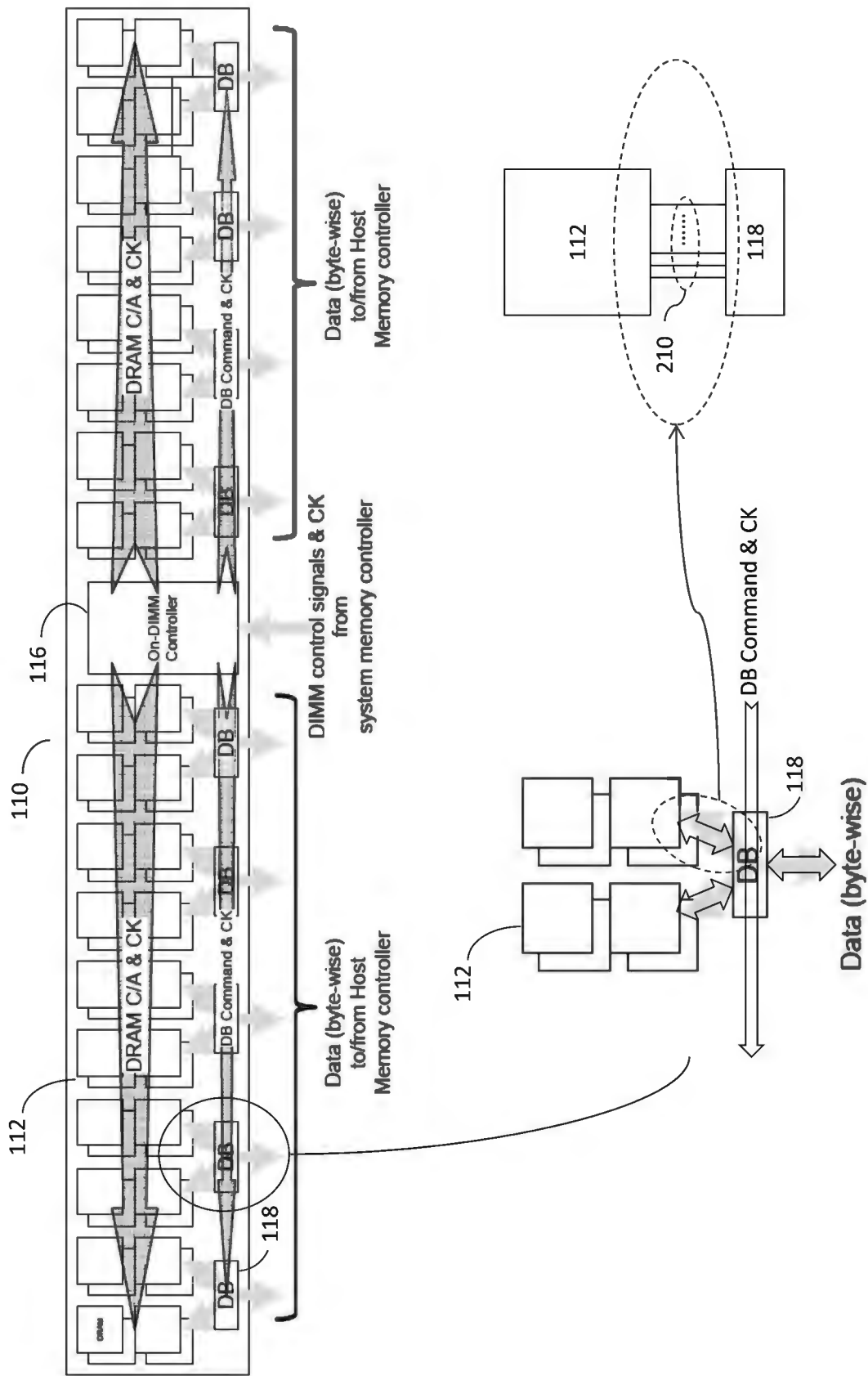


FIG. 2C

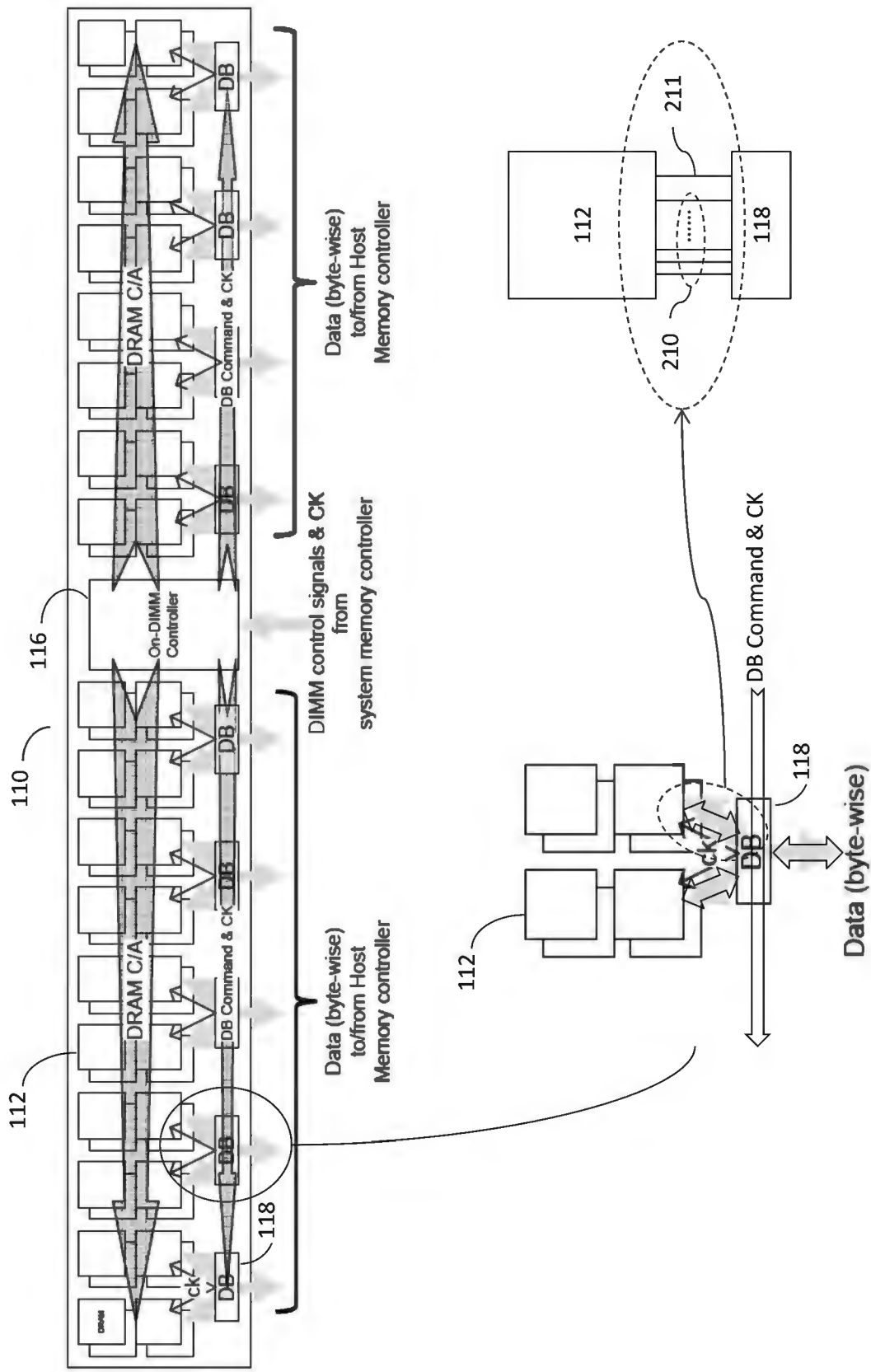


FIG. 2D

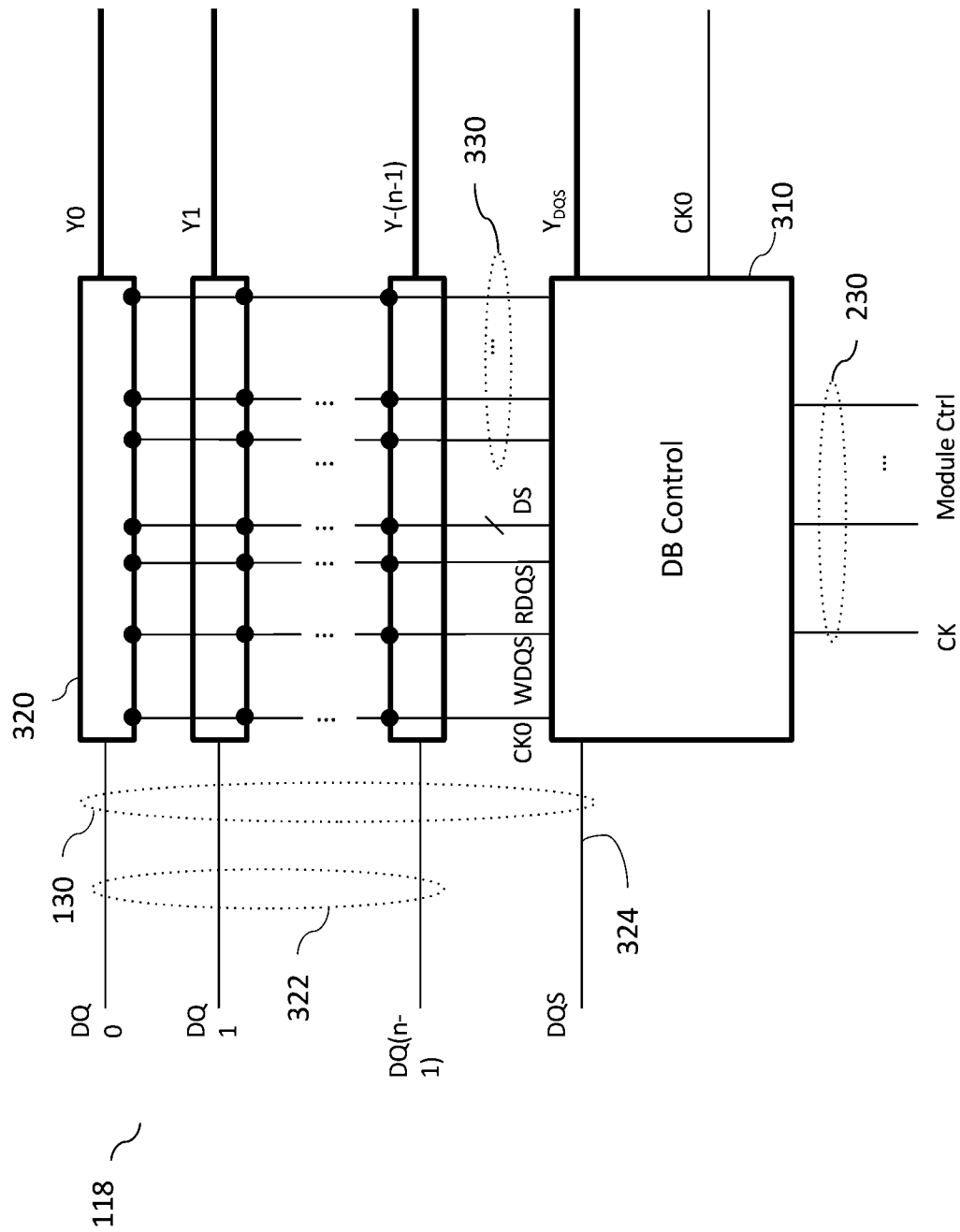


FIG. 3

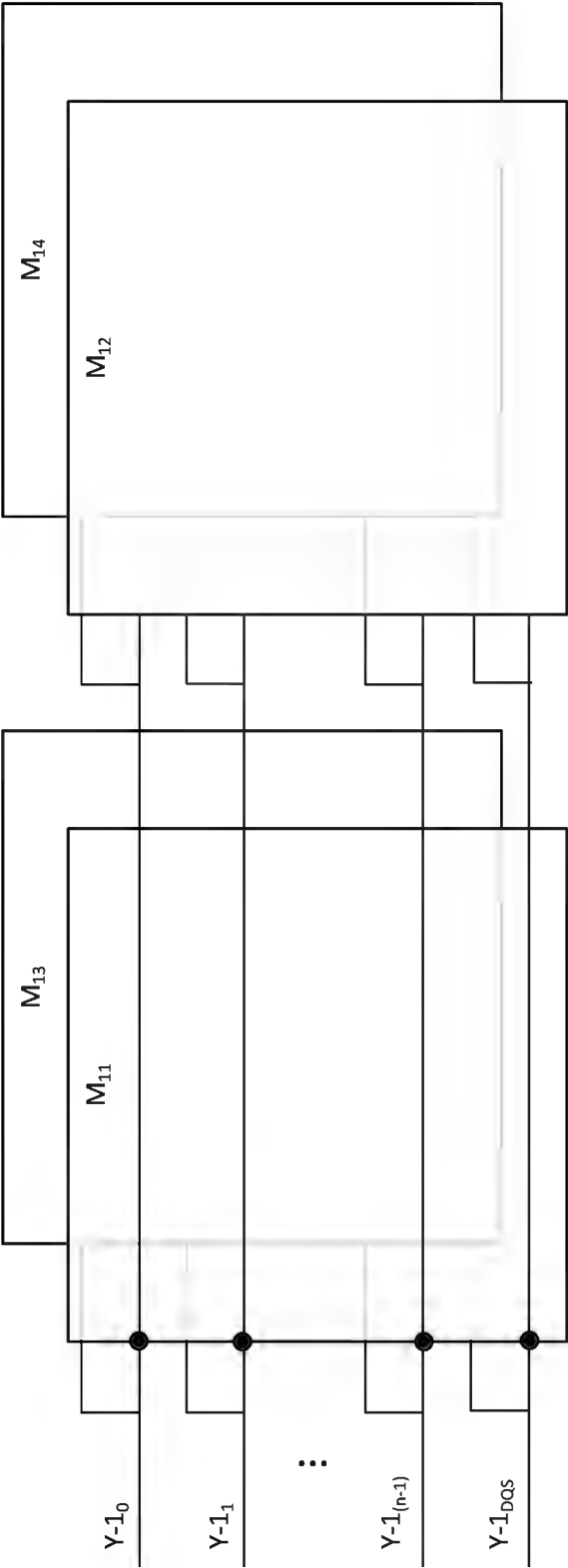


FIG. 4A

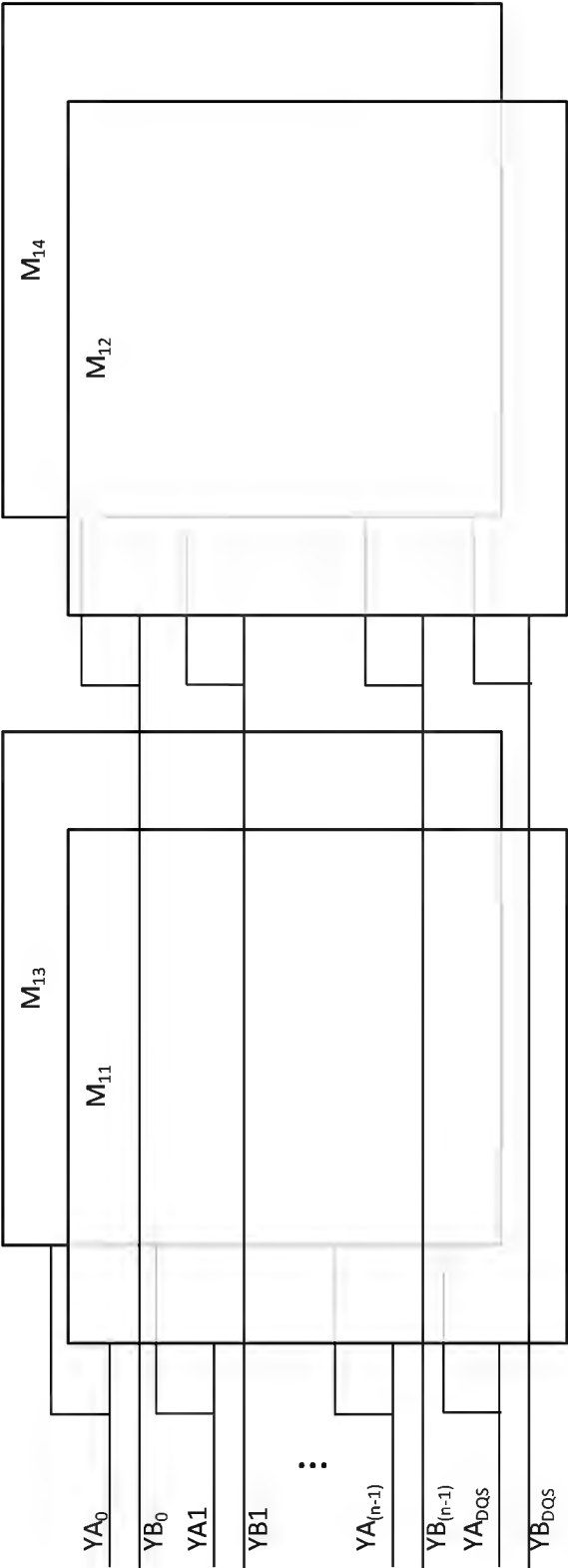


FIG. 4B

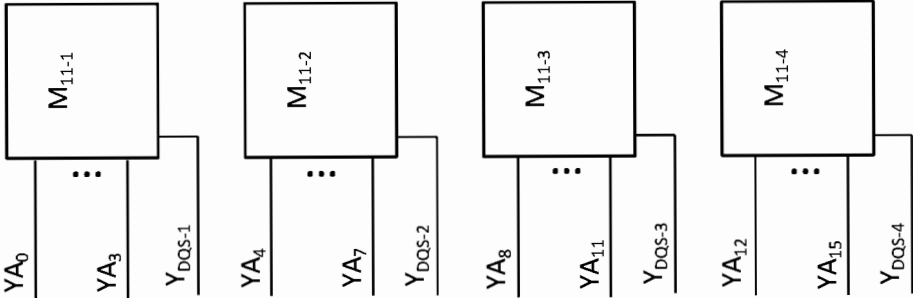


FIG. 5B

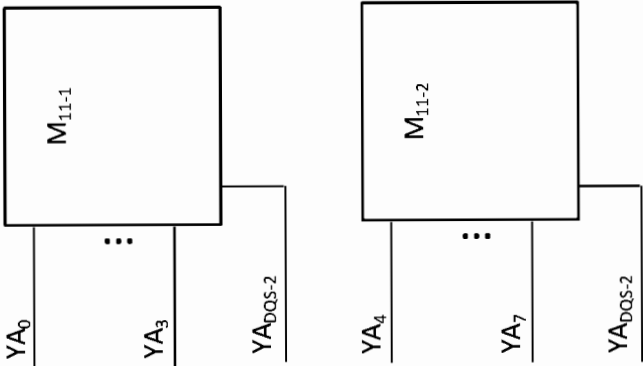


FIG. 5A

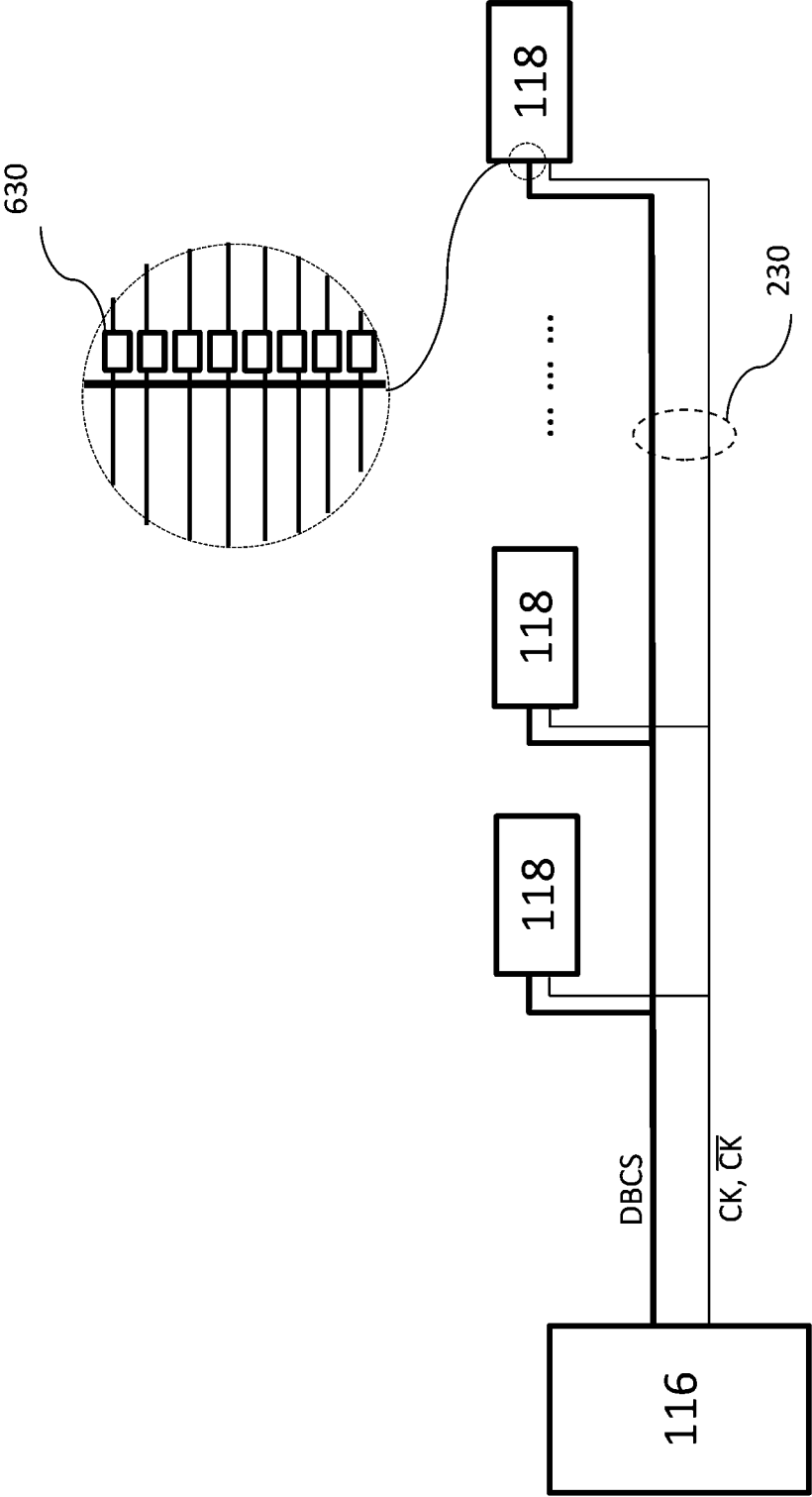


FIG. 6A

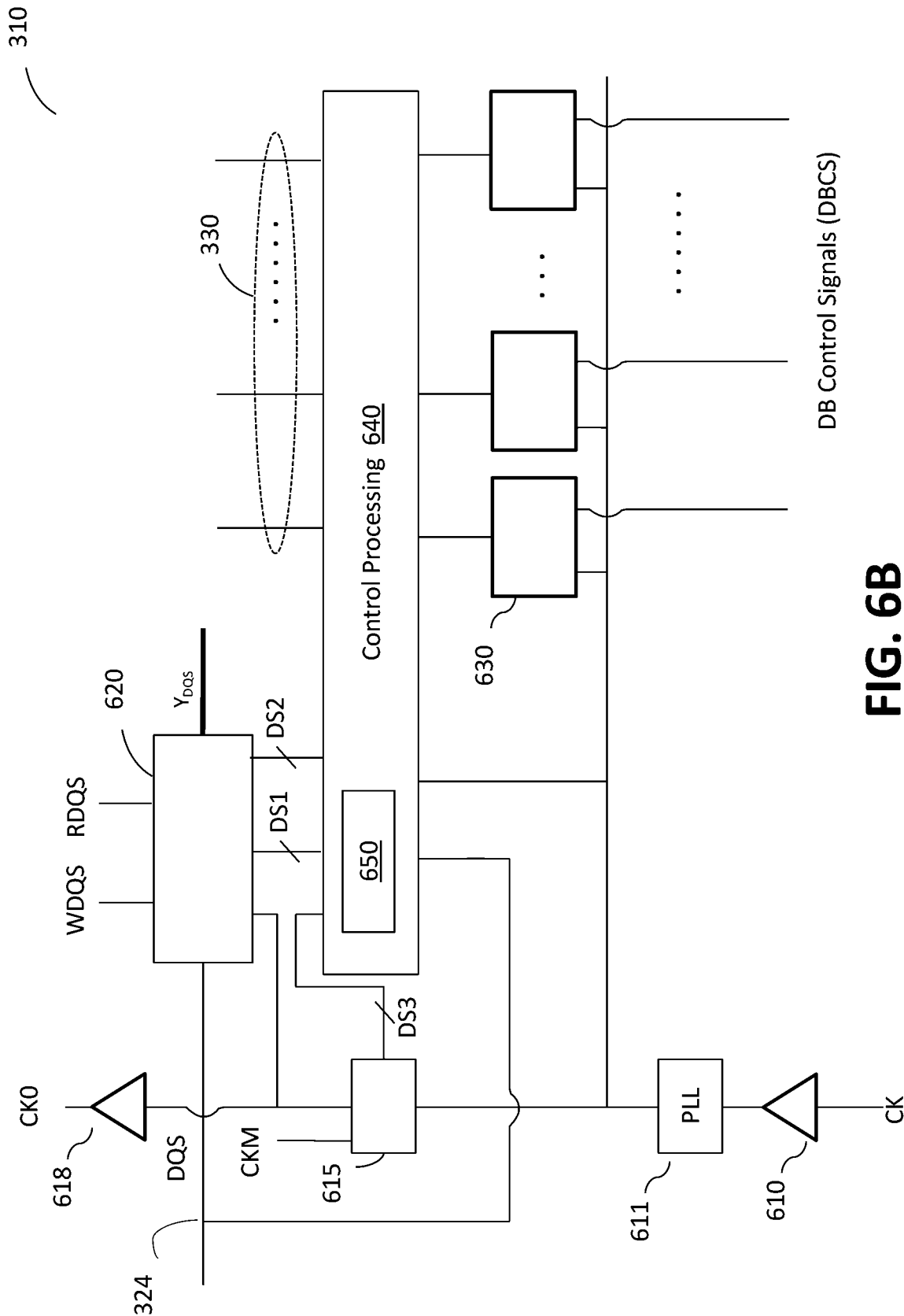
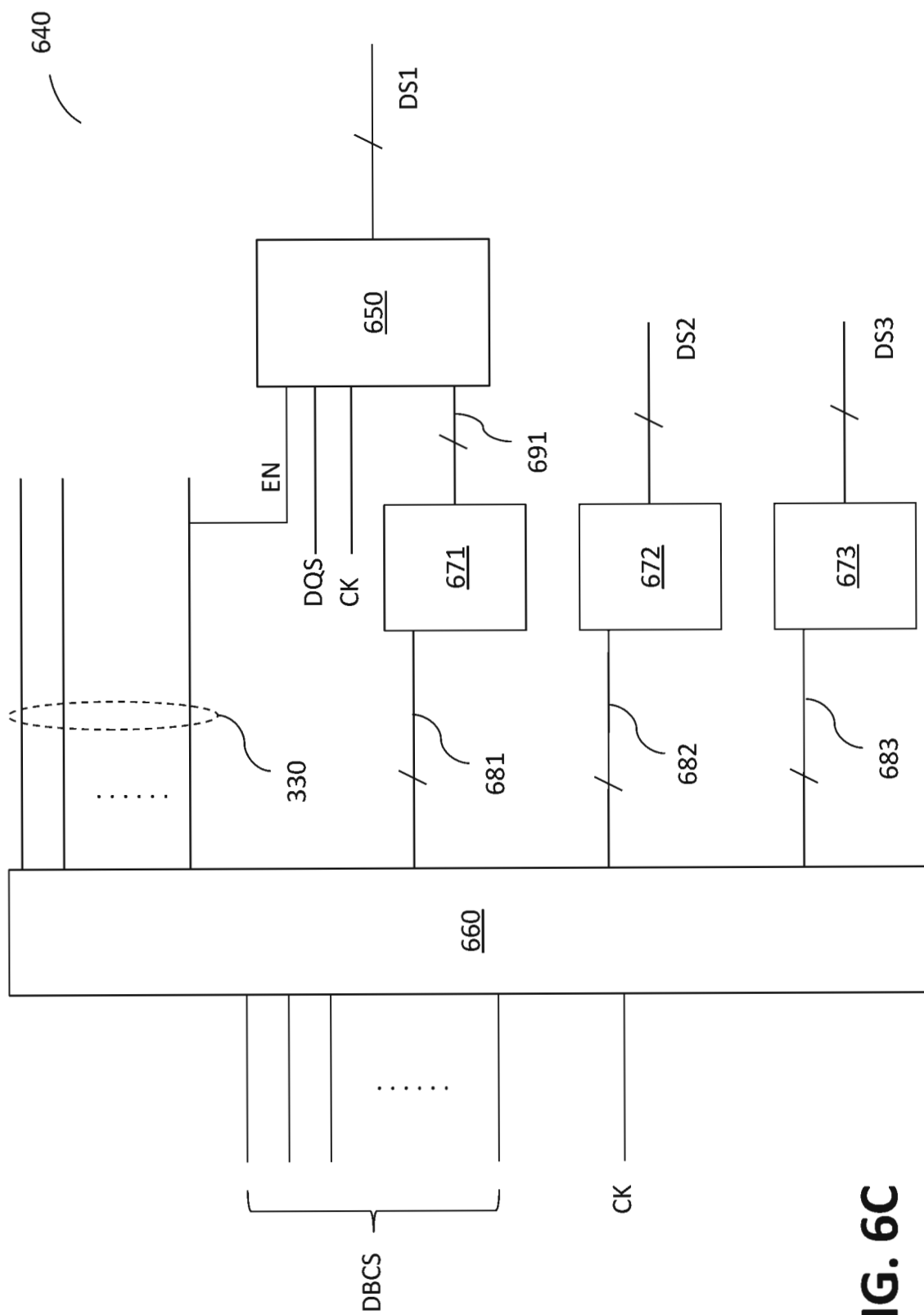
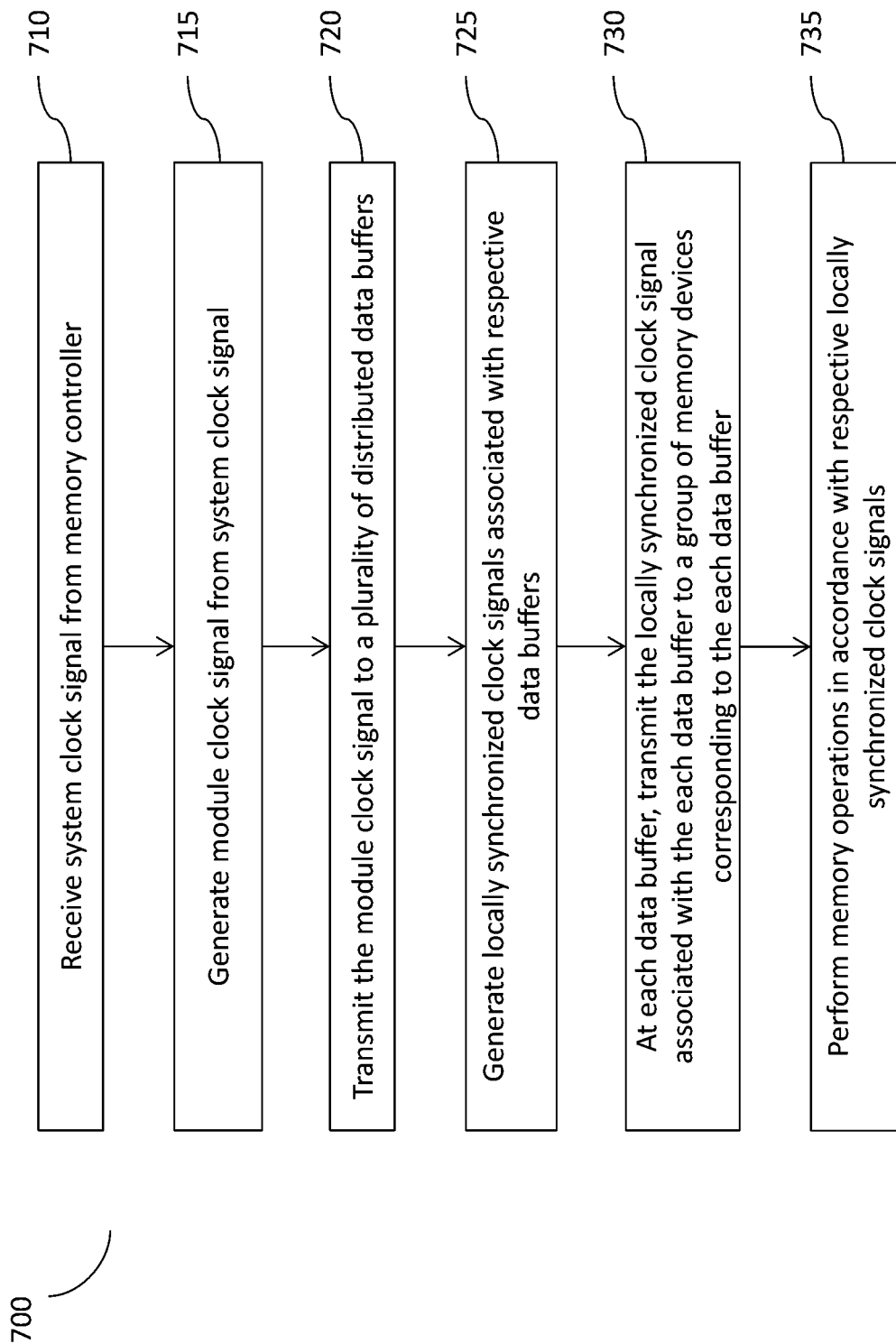
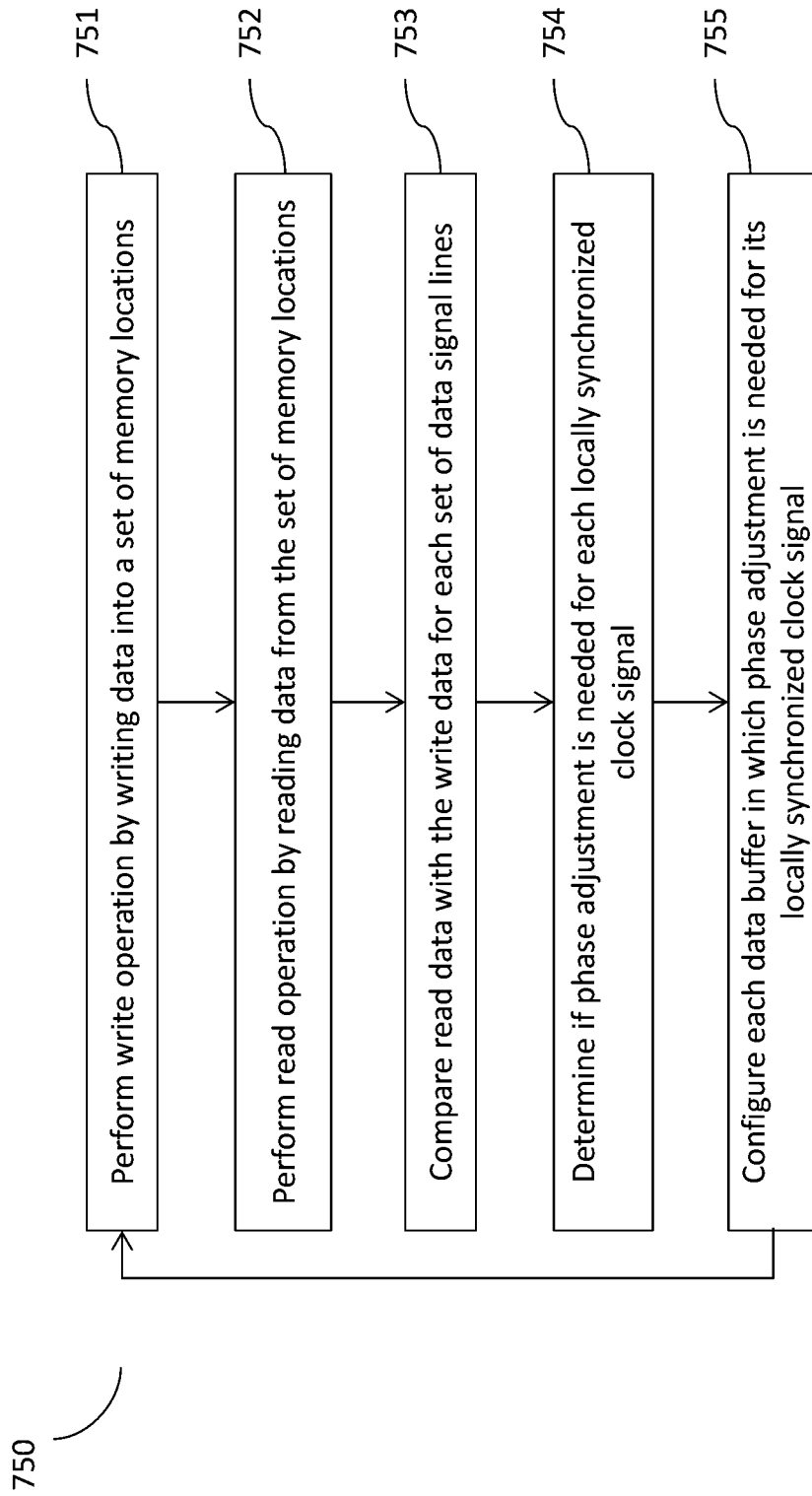


FIG. 6B



**FIG. 7A**

**FIG. 7B**

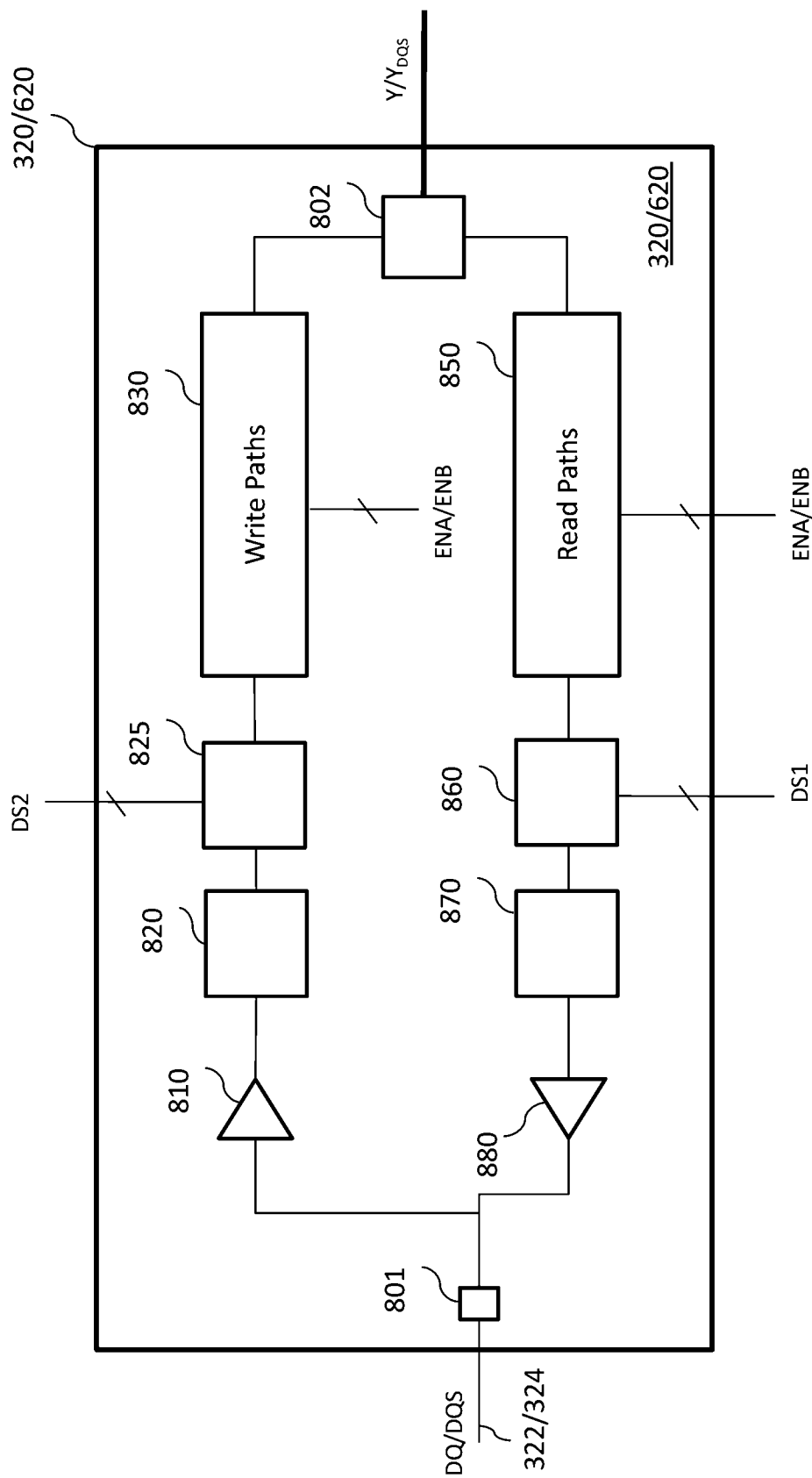


FIG. 8

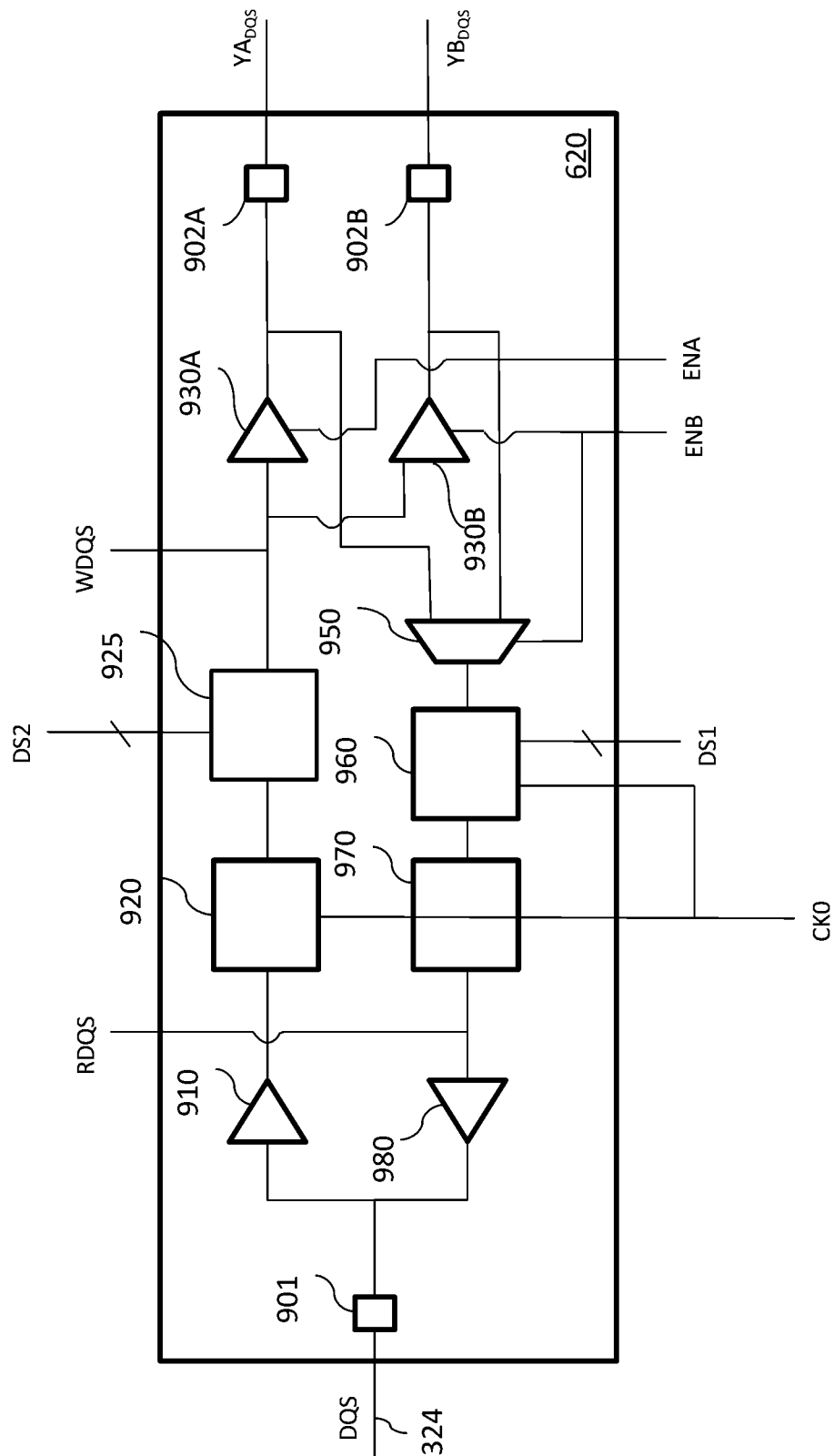


FIG. 9

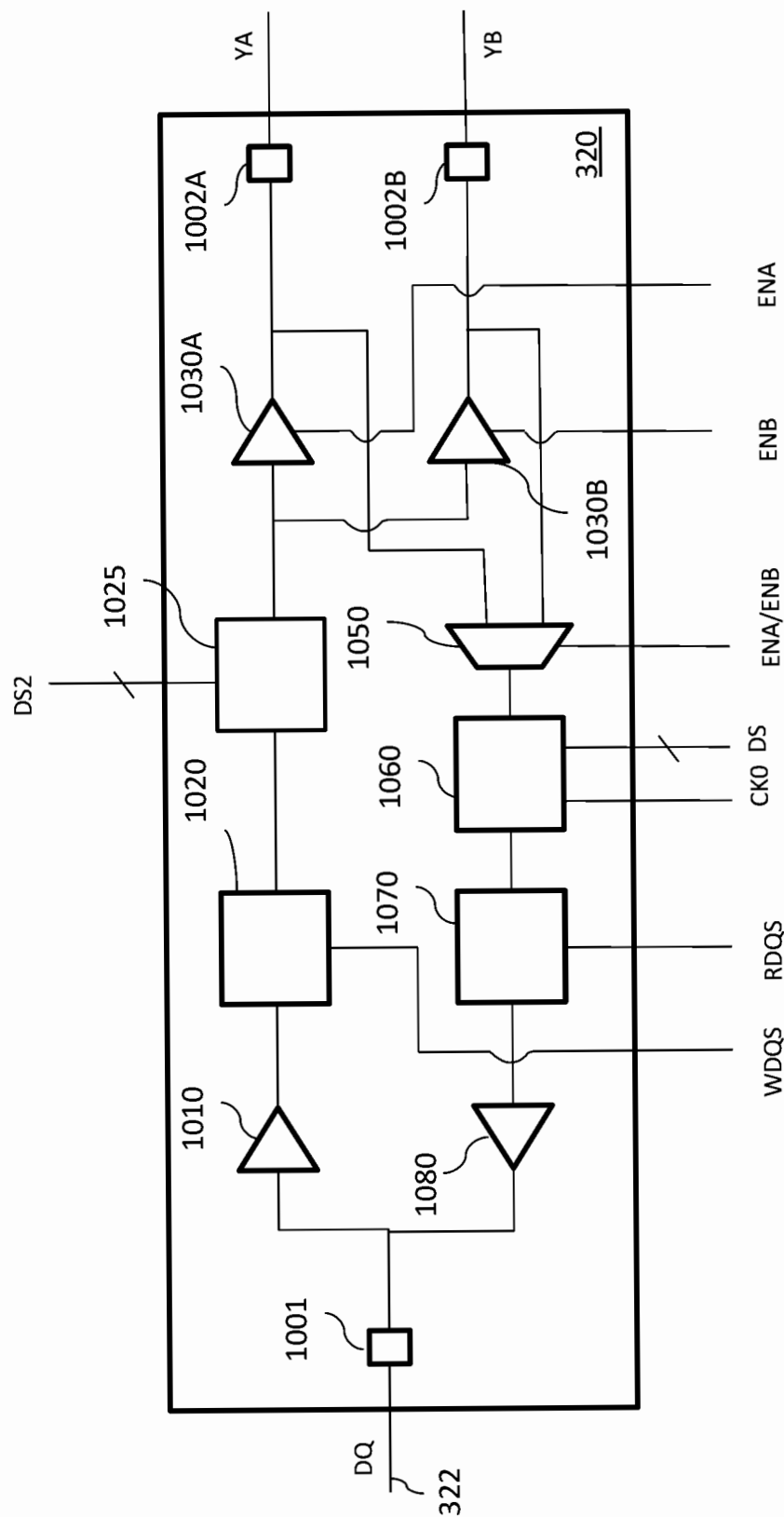


FIG. 10

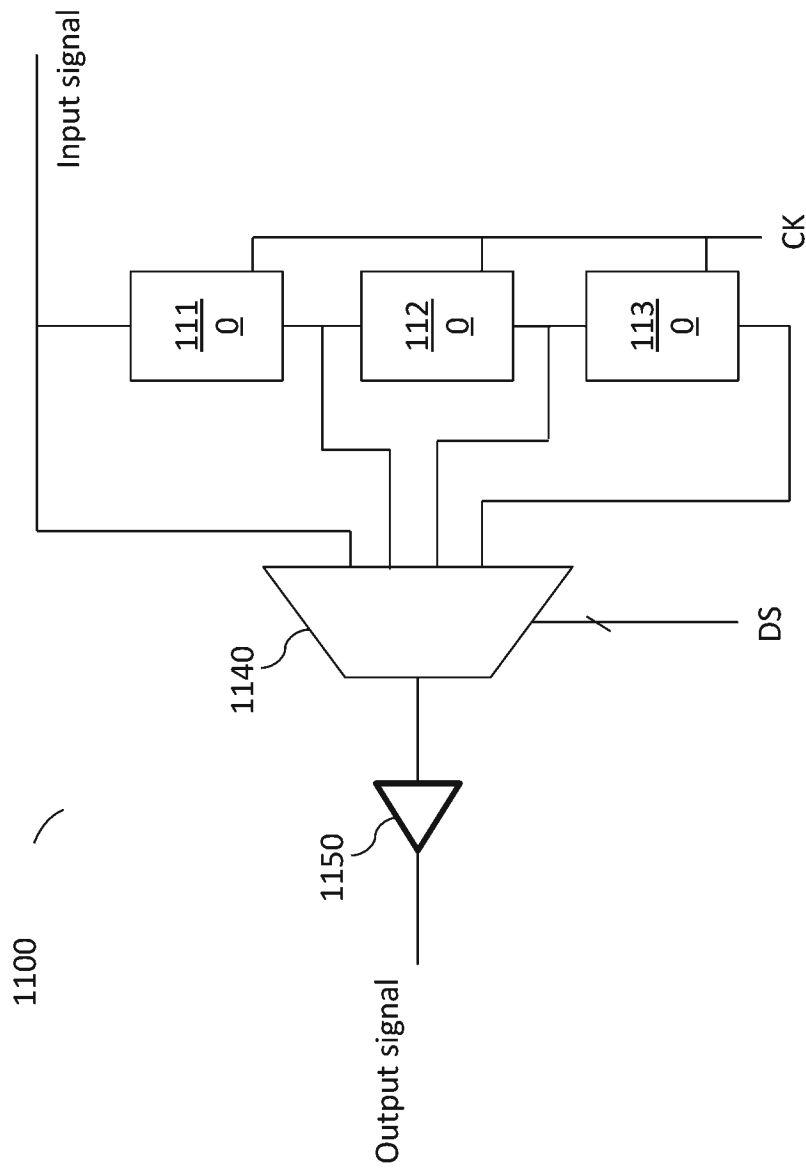


FIG. 11

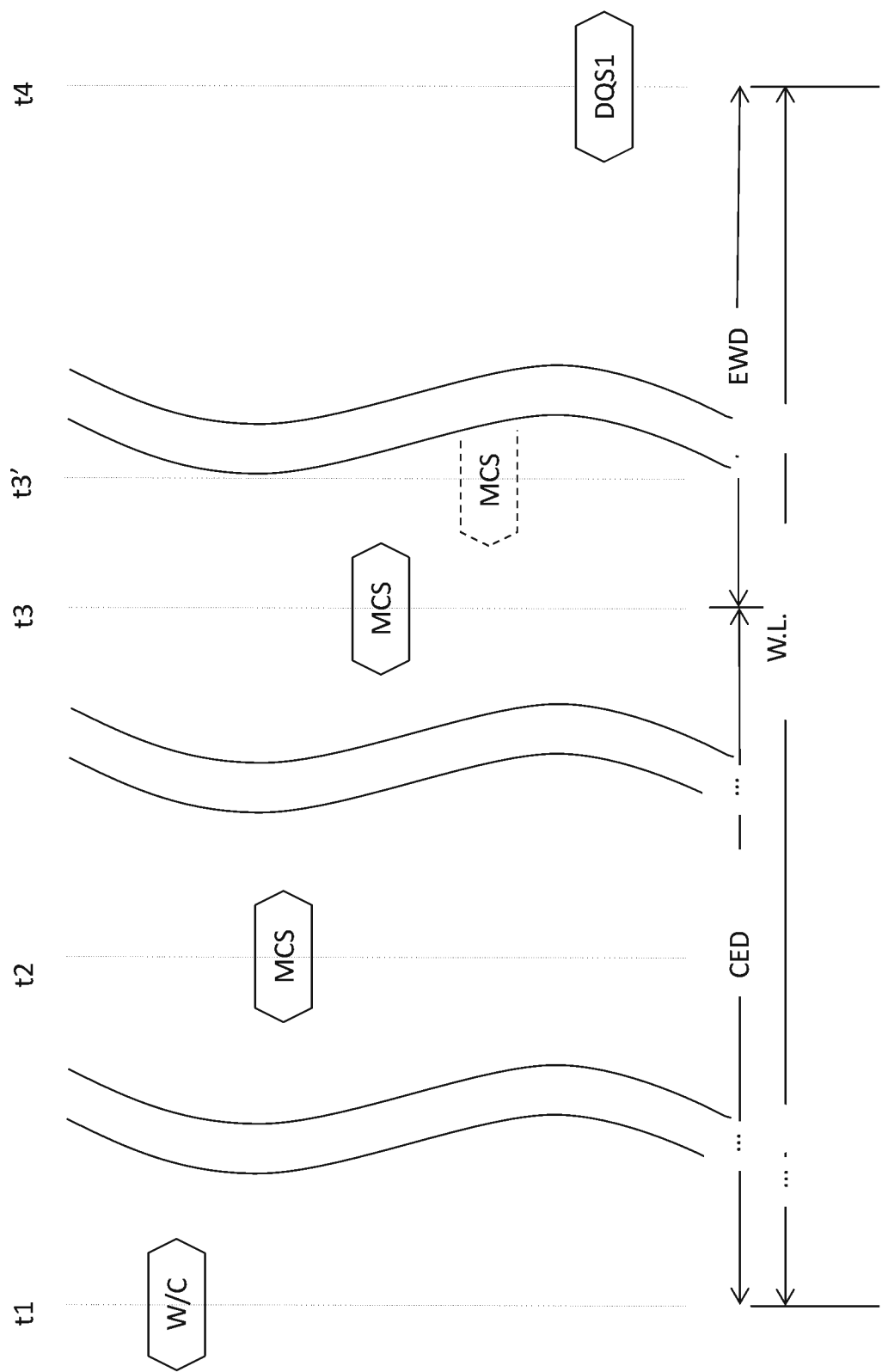


FIG. 12A

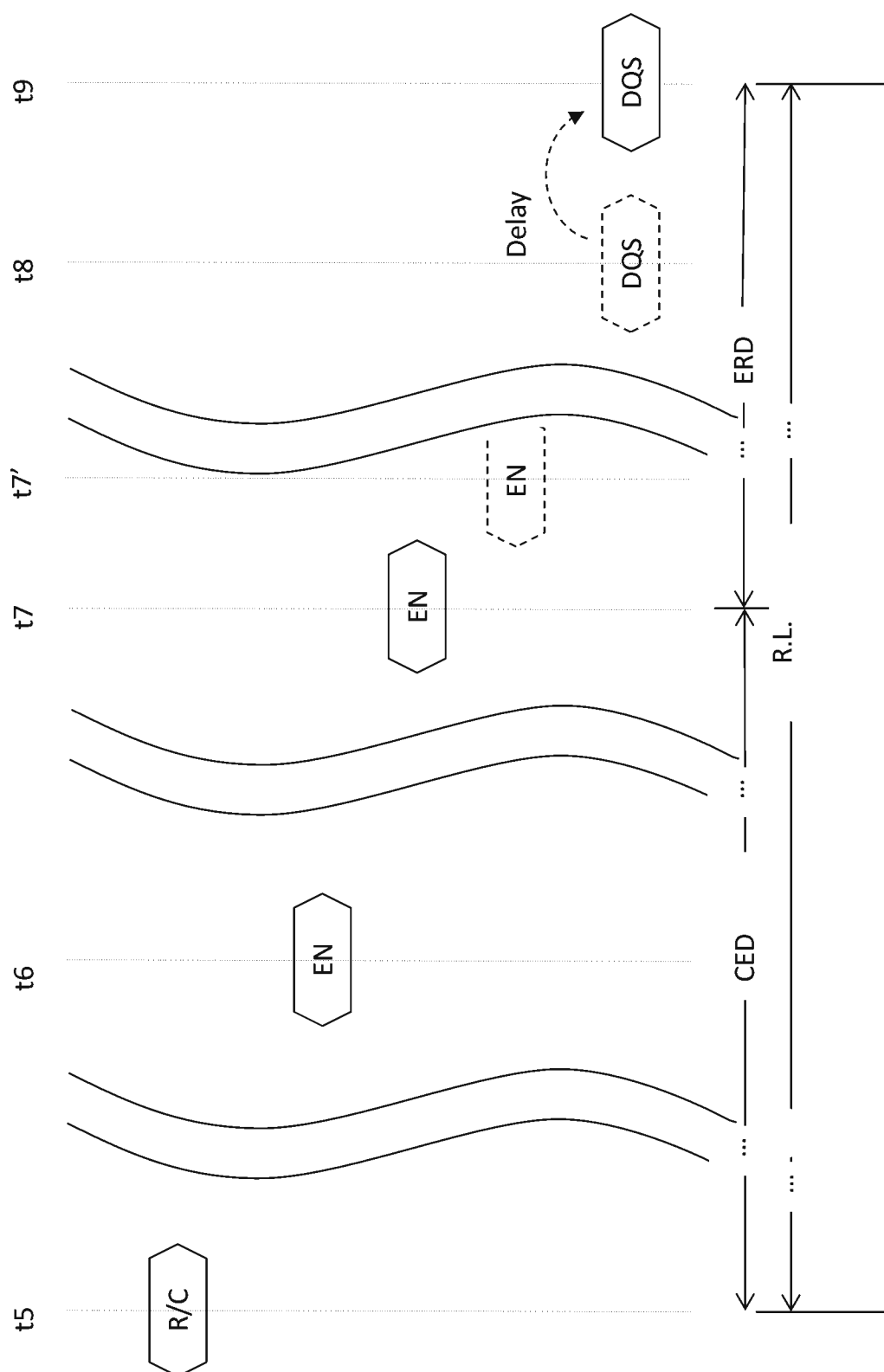


FIG. 12B

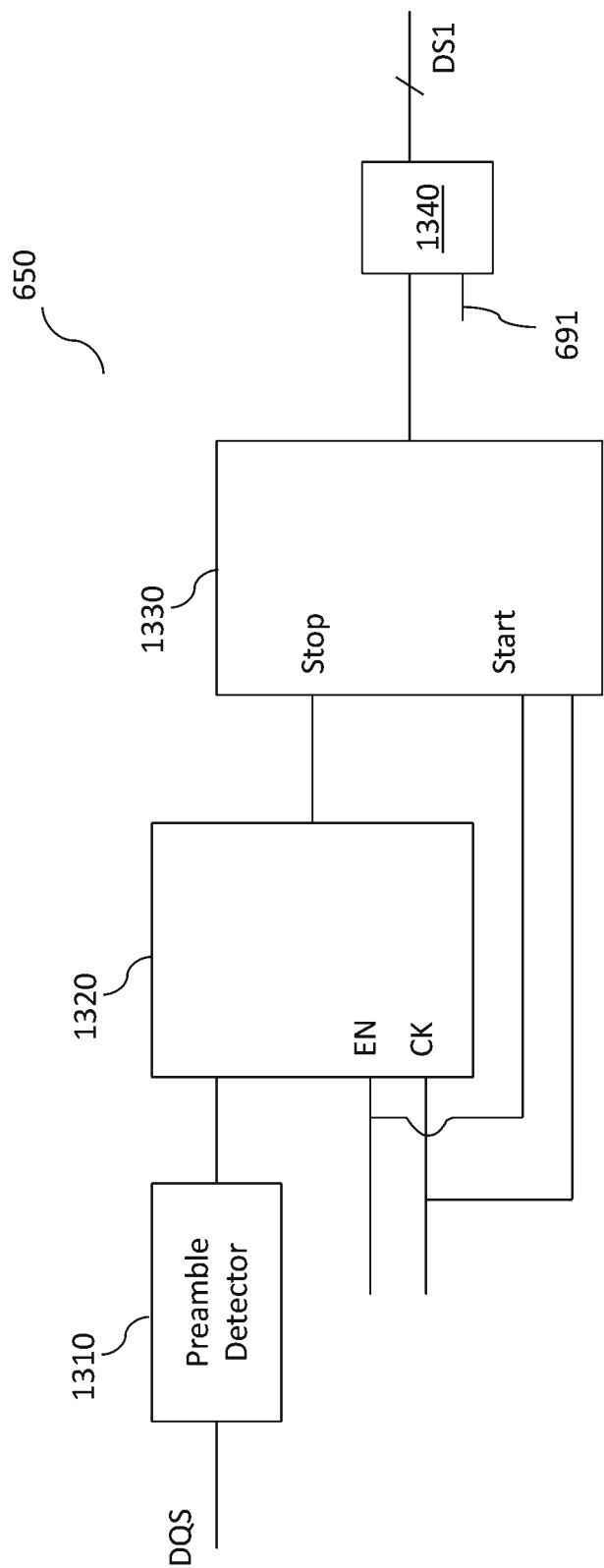
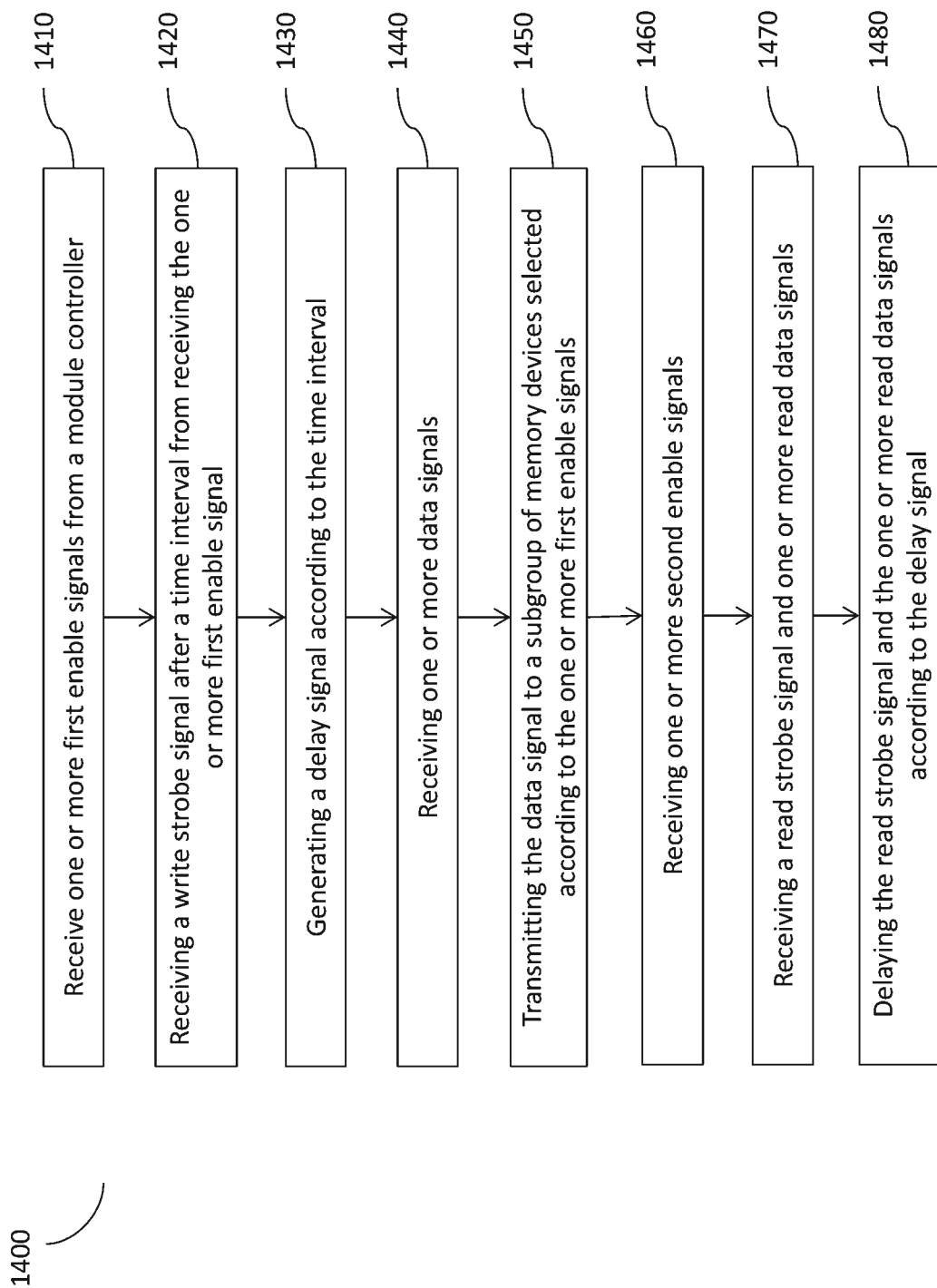


FIG. 13

**FIG. 14**

1

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

CROSS REFERENCE TO RELATED APPLICATIONS

The present application a continuation of U.S. patent application Ser. No. 17/141,978, filed on Jan. 5, 2021, entitled "Memory Module with Local Synchronization and Method of Operation," which is a continuation of U.S. patent application Ser. No. 16/432,700, filed on Jun. 5, 2019 (now U.S. Pat. No. 10,884,923), entitled "Memory Module with Local Synchronization and Method of Operation," which is a continuation of U.S. patent application Ser. No. 14/445,035, filed on Jul. 28, 2014 (now U.S. Pat. No. 10,324,841), entitled "Memory Module with Local Synchronization," which claims priority to U.S. Provisional Patent Application No. 61/859,215, filed Jul. 27, 2013, each of which is incorporated herein by reference in its entirety.

FIELD

The disclosure herein is related generally to memory modules, and more particularly to memory modules having distributed data buffers.

BACKGROUND

With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per

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memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules. The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

In some conventional memory systems, the memory controllers include leveling mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

FIGS. 2A-2D are each a diagrams illustrating interactions among components in a a memory module according to certain embodiments.

FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

FIG. 7A is a flowchart illustrating a method of operation for a memory module according to certain embodiments.

FIG. 7B is a flowchart illustrating a method of training locally synchronized clock signals according to certain embodiments.

FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

FIG. 9 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

FIG. 10 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g., read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or

different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

Furthermore, the memory devices 112 may include any type of memory devices. For example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory (SDRAM) devices in compliance with, for example, the Double Data Rate Type 3 (DDR3) or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and MCK, and the module clock signal may include a pair of complementary clock signals CK and CK.

Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent

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data from appearing at the output until needed during a read operation, and the system clock signal MCK.

Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Pat. No. 7,532,537, entitled "Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation," issued on May 12, 2009, the entirety of which is incorporated herein by reference.

Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers **118**, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer control signals are used to convey configuration or module self-test control information from the module control device **116** to the data buffers **118**. In one embodiment, the data buffer control signals are transmitted to the data buffers **118** via respective data buffer control signal lines **230**. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers **118** via the data buffer control signal lines and decoded/processed at the data buffers.

Module control device **116** transmits the module C/A signals to the memory devices **112** via module C/A signal lines **220**. The memory devices **112** operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers **118** via data buffer control signal lines **230**. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines **220**, and at least some of the data buffers **118** share a same set of data buffer control signal lines **230**.

As shown in FIGS. 2A and 2B, each rank **114** includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices $M_{11}, \dots, M_{i1}, M_{i+1,1}, \dots, M_{N1}$, a second rank includes memory devices $M_{12}, \dots, M_{i2}, M_{i+1,2}, \dots, M_{N2}$, and so on. In one embodiment, the memory devices **112** are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines **130** and including at least one memory device from each rank. For example, memory devices M_{11}, M_{12}, M_{13} , and M_{14} form an

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i^{th} group of a first group of memory devices, memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

In certain embodiments, the data buffers **118** are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines **130** and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers **118** is associated with the first group of memory devices M_{11}, M_{12}, M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines **130-1** and the first group of memory devices, data buffers ID-i among the data buffers **118** is associated with the i^{th} group of memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines **130-i** and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be associated with a same group of memory devices, or vice versa.

In one embodiment, each group or sets of memory devices are coupled to the associated data buffer **118** via a set of module data/strobe lines **210** and one or more clock signal lines **211**. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer **118** via a same set of module data/strobe lines **210** (as shown in FIG. 2A) or via respective subsets of module data/strobe lines **210** (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer **118** via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

In one embodiment, the data buffers **118** are in the data paths between the MCH **101** and the memory module **110** and include data buffers between the MCH **101** and the respective groups of memory devices. In one embodiment, each data buffer **118** is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the MCH **101** in response to the data buffer control signals. Further, each data buffers **118** is configured to isolate unselected subgroup(s) of memory devices from the MCH **101** during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

In one embodiment, the data buffers **118** are distributed across the memory module **110** or the module board **119** in positions corresponding to edge connectors **121** coupled to respective groups of system data/strobe signal lines **130**. For

example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least on data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system

100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the memory devices 112.

In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of interface timing training may be required because the system memory controller 101 has very limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very significant portion of a data period. As a result, the clock skew issues cannot simply be addressed by pipelining the data paths.

To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of

memory devices **112** and the system memory controller **101**. Each data buffer **118** and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer **118** in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit **116** and the group of locally synchronized components.

Thus, the memory module **110** in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module **110** and the system memory controller **101**. Also, signal interface between each data buffer **118** and the respective set of memory devices **112** can be synchronized. In one embodiment, each data buffer **118** has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices **112**, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller **101**, and/or programmable phase adjustment for one or more control signal from the control circuit **116**. The locally synchronized operation also makes it easier for each data buffer **118** to perform self-testing of the associated set of memory devices **112**, independent of the self-testing of other sets of memory devices performed by the other data buffers, as disclosed in commonly-owned U.S. Pat. No. 8,001,434, issued Aug. 16, 2011, entitled "Memory Board with Self-Testing Capability," which is incorporated herein by reference in its entirety.

In certain embodiments, operations of the data buffers **118** are controlled by the data buffer control signals from the module control circuit **116**, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers **118** to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit **116** based on the system clock signal MCK. The isolation circuits **118** buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers **118** are distributed across the memory module, the data buffer control signal lines **230** can stretch across the memory module **110**, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits **118** includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. patent application Ser. No. 13/952,599, filed Jul. 27, 2013, entitled "Memory Module with Distributed Data Buffers and Method Of Operation," the entirety of which is incorporated herein by reference.

Because the data buffers **118** are distributed across the memory module **110**, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signal lines **230** from the module control device **116** to the farthest positioned data buffers **118**, such as data buffer ID-1 and data buffer ID-(n-1) in the exam-

plary configuration shown in FIG. 2. In other words, a same set of data buffer control signals may reach different data buffers **118** at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70 ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

With the data buffers **118** receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH **101** from the data buffers **118**. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit **116** and a time when a write strobe or write data signal is received from the MCH **101**. This time interval is used to adjust the timing of the transmission of read data to the MCH **101** during a subsequent read operation, as explained in more detail below.

More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

In one embodiment, as shown in FIG. 3, each group of signal lines **130** include a set of n data (DQ) signal lines **322** each for transmitting one of a set of bits $DQ_0, DQ_1, \dots, DQ_{n-1}$, in a data signal and at least one strobe (DQS) signal line **324** for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line V_{DQS} . When the subsets of memory devices are coupled to the associated data buffer **118** via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

Each data buffer **118** includes a set of DQ routing circuits **320** coupled on one side to respective ones of the set of n DQ signal lines **322**, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, Y_1, \dots, Y_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each data buffer **118** further

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includes an DB Control circuit **310** coupled on one side to the at least one DQS signal line **324**, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS} .

The DB Control circuit **310** also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines **230**, and outputs datapath control signals **330** to the DQ routing circuits **320**, including, for example, one or more enable signals ENA and/or ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit **310** also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit **320** is configured to enable data communication between the respective DQ signal line **322** with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

In certain embodiments, the DB Control circuit **310** also provides one or more delay signals DS, which is used by the DQ routing circuits **320** to control the timing of read and/or write data output by the data buffer **118**, as explained in further detail below. In certain embodiments, the DB Control circuit **310** regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer **118** and the respective group of memory devices form a group of locally synchronized devices on the memory module **110**.

The memory devices **112** in each group of locally synchronized devices are coupled to the data buffer **118** in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices **112** in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer **118** via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M_{11} , M_{12} , M_{13} , and M_{14} in the first group of memory devices can be coupled to the data buffer ID-1 via a same set of module data lines $Y-1_0$, $Y-1_1$, . . . , $Y-1_{n-1}$ and module strobe line $Y-1_{DQS}$. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

Alternatively, as shown in FIG. 4B, memory devices M_{11} and M_{13} , which form a subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines $YA-1_0$, $YA-1_1$, . . . , $YA-1_n$ and module strobe line $YA-1_{DQS}$ and memory devices M_{12} and M_{14} , which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines $YB-1_0$, $YB-1_1$, . . . , $YB-1_n$ and module strobe line $YB-1_{DQS}$. Memory devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board **119**. Memory devices coupled to the same data buffers may be placed side-by-side, on opposite sides of the module boards **119**, or stacked over each other, and/or over the associated data buffer.

Multiple memory devices having a data width that is less than a data width of the data buffers **118** may be used in

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place of one of the memory devices **112**, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M_{11-1} and M_{11-2} may be used in place of the memory device M_{11} . Each of the two memory devices M_{11-1} and M_{11-2} has a data width of 4, and together they act like a memory device M_{11} of a data width of 8. Thus, memory device M_{11} is coupled to the data buffer ID-1 via module data lines $YA-1_0$, . . . , $YA-1_3$ and module strobe line $YA-1_{DQS-1}$ while memory circuit M_{11-2} is coupled to the data buffer ID-1 via module data lines $YA-1_4$, . . . , $YA-1_7$ and module strobe line $YA-1_{DQS-2}$.

In another embodiment, as shown in FIG. 5B, four memory devices M_{11-1} to M_{11-4} may be used as the memory device M_{11} . Each of the four memory devices M_{11-1} to M_{11-4} has a data width of 4, and together they act like a memory device M_{11} of a data width of 16. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines $YA-1_0$, . . . , $YA-1_3$ and module strobe line $YA-1_{DQS-1}$ while memory device M_{11-2} is coupled to the data buffer ID-1 via module data lines $YA-1_4$, . . . , $YA-1_7$ and module strobe line $YA-1_{DQS-2}$, and so on.

FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits **630** on each data buffer. In certain embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit **116** uses the DBCS to configure the programmable features in the data buffers **118** based on configuration information received from the MCH **101**; in the test mode, the DBCS is used to control the self-test processes run by the data buffers; in the normal operation mode, the DBCS includes the one or more enable signals and one or more ODT signals.

FIG. 6B illustrates the DB Control circuit **310** in an data buffer **118**. As shown, the DB Control circuit **310** includes a clock buffer **610** to receive the module clock signal CK from the module control device **116**, a phase-locked loop circuit (PLL) **611** that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit **615** for adjusting the phase of the locally synchronized clock CK0, and a clock driver **618** that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit **310** further includes a strobe routing circuit **620** that are coupled on one side to a corresponding system DQS signal line **324** and on another side to corresponding module DQS signal lines YA_{DQS} and YB_{DQS} . The DB Control circuit **310** further includes receiver circuits **630** to receive respective ones of the DBCS.

The DB Control circuit **310** further includes a control processing circuit **640** that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more data path enable signals ENA and/or ENB that are used by the DQ routing circuits **320** and the strobe routing circuit **620** to selectively enabling data communication between the MCH **101** and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the

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control processing circuit **640**, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

The strobe routing circuit **620** also buffers strobe signals received from either the MCH **101** or the memory devices **112**, and outputs either a write strobe WDQS or read strobe RDQS, respectively, to the DQ routing circuits **320**. In one embodiment, the DB Control circuit **310** further includes a dynamic delay control circuit **650** that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit **320** and the strobe routing circuit **620**.

In certain embodiments, as shown in FIG. 6C, the control processing circuit **640** includes control logic **660** that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals **330**. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals **681** to programmable registers **671** that provides static control of the timing of read data/strobe signals, configuration signals **682** to programmable registers **671** that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals **683** to programmable registers **673** that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

The programmable registers **671** is coupled via signal lines **691** to the dynamic delay control circuit **650**, which also takes as input a DQS signal, clock signal CK, and one of the datapath enable signals EN, and output the delay signal DS1. The dynamic delay control circuit **650** is described in further detail below.

FIG. 7A is a flowchart illustrating a method **700** performed by the memory module **110** according to certain embodiments. As shown in FIG. 7A, the method includes receiving **710** clock signal together with system C/A signals from the MCH **101**, and generating **715** module clock signal CK. The module clock signal CK is transmitted **720** to a plurality of distributed data buffers **118**, which generates **725** locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits **730** its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform **735** memory operations in accordance with the respective locally synchronized clock signals.

Since the module control circuit **116** generates the DBCS based on signals from the MCH **101**, the MCH **101** can train the data buffers **118** to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit **116** generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH **101**. As shown in FIG. 7B, a method **750** for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing **751** data to a set of memory locations in the memory devices **112**, which is followed by performing a read operation by reading **752** data from the set of memory locations. The read data is compared **753** with the write data

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to determine **754** if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit **116** agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit **116** do not agree with the corresponding write data bits. The disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers **118** further away from the module control circuit **116** being excessively delayed during transmission, and proper phase adjustment can be programmed **755** into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

FIG. 8 illustrates a DQ or DQS routing circuit **320** or **620** according to an embodiment. As shown, the DQ/DQS routing circuit **320/620** includes a DQ/DQS pin **801** that is coupled to the corresponding DQ/DQS signal line **322/324**, a set of one or more DQS pins **802** that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS}, or YA/YA_{DQS} and YB/YB_{DQS}. The DQ/DQS routing circuit **320/620** further includes a write strobe buffer **810** that buffers write data/strobe, a write data/strobe receiver **820** that samples the write data/strobe, and a write delay circuit **825** controlled by the delay signal DS2. The DQ/DQS routing circuit **320/620** further includes a plurality of write paths **830** that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

The DQS routing circuit further includes a plurality of read paths **850** that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit **860** by an amount controlled by the delay signal DS1, and sampled by a sampler circuit **870**. The sampled read data/strobe is transmitted by transmitter **880** onto the corresponding data/strobe signal line **322/324** via the DQ/DQS pin **801**.

FIG. 9 illustrates a DQS routing circuit **620** according to an embodiment. As shown, the DQS routing circuit **620** includes a first DQS pin **901** that is coupled to a corresponding DQS signal line **324**, a second DQS pin **902A** that is coupled to a corresponding module DQS line YA_{DQS}, a third DQS pin **902B** that is coupled to a corresponding module DQS line YB_{DQS}. The DQS routing circuit **620** further includes a first write strobe path coupled between the first DQS pin **901** and the second DQS pin **902A** and a second write strobe path coupled between the first DQS pin **901** and the third DQS pin **902B**. The first write strobe path includes a write strobe buffer **910** that buffers a write strobe, a write strobe receiver **920** that samples the write strobe according to the buffered module signal CK0 and a write delay circuit **925** controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits **320** as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter **930A** that transmits the write strobe to one or more memory devices **112** coupled to the module strobe line YA_{DQS}. The second write strobe path includes the write strobe buffer **910**, the write strobe receiver **920**, the write delay circuit **925** controlled by the delay signal DS2, and a second write strobe transmitter **930B** that transmits the write strobe to one or more memory devices **112** coupled to the module strobe line YB_{DQS}. The first and second write strobe transmitters, **930A** and **930B**, are controlled by two enable signals, ENA and ENB, respectively,

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such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

The DQS routing circuit further includes a read strobe path coupled between the first DQS pin **901** and a selected one of the second and third DQS pins **902A** and **902B**. In the read strobe path, a select circuit **950** (e.g., a multiplexor) selects either a read strobe signal received via DQS pin **902A** or a read strobe signal received via DQS pin **902B** based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit **960** by an amount controlled by the delay signal DS, and sampled by a sampler circuit **970** according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits **320** as the read strobe RDQS and is transmitted by transmitter **980** onto the corresponding strobe signal line **324** via the first DQS pin **901**.

FIG. **10** illustrates a DQ routing circuit **320** according to an embodiment. As shown, the DQ routing circuit **320** includes a first DQ pin **1001** that is coupled to a corresponding DQ signal line **130**, a second DQ pin **1002A** that is coupled to a corresponding module DQ line YA_{DQ} , a third DQ pin **1002B** that is coupled to a corresponding module DQ line YB_{DQ} . The DQ routing circuit **320** further includes a first write data path coupled between the first DQ pin **1001** and the second DQ pin **1002A** and a second write data path coupled between the first DQ pin **1001** and the third DQ pin **1002B**. The first write data path includes a write data buffer **1010**, a write data receiver **1020** that samples write data according to the write strobe WDQS from the DQS routing circuit **620**, a write delay circuit **1025** controlled by the delay signal DS2, and a first write data transmitter **1030A** that transmits the write data to one or more memory devices **112** coupled to the module data line YA_{DQ} . The second write data path includes the write data buffer **1010**, the write data receiver **1020**, the write delay circuit **825** controlled by the delay signal DS2, and a second write data transmitter **1030B** that transmits the write data to one or more memory devices **112** coupled to the module data line YB_{DQ} . The first and second write data transmitters, **1130A** and **1130B**, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

The DQ routing circuit further includes a read data path coupled between the first DQ pin **1001** and a selected one of the second and third DQ pins **1002A** and **1002B**. In the read data path, a select circuit **1050** (e.g., a multiplexor) selects either a read data signal received via DQ pin **1002A** or a read data signal received via DQ pin **1002B** based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit **1060** by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit **1070** according to the read strobe RDQS from the DQS routing circuit **620**, and transmitted by transmitter **1080** onto the corresponding data signal line **130** via the first DQ pin **1001**.

FIG. **11** illustrate a delay circuit **1100** that can be used as the delay circuit **825/860/925/960/1025/1060** according to an embodiment. As shown, the delay circuit **1100** includes a plurality of delay stages, such as delay stages **1110**, **1120**, and **1130**, each delaying an input signal by a predetermined amount. The delay circuit **1100** further includes a select circuit **1140** (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select

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circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

As stated above, the MCH **101** can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers **118** are in the data paths between the MCH **101** and the respective groups of memory devices **112**, the MCH **101** does not have direct control of the interface between the data buffers and the memory devices **112**. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers **118** having programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

FIG. **12A** is a timing diagram for a write operation according to one embodiment. As shown, after a write command W/C associated with the write operation is received by the module control circuit **116** at time **t1**, the module control circuit **116** outputs one or more enable signals EN at time **t2** in response to the write commands. The one or more enable signals are received by a data buffer **118** at time **t3**, which afterwards receives one or more strobe signal DQS from the MCH **101** at time **t4**. Note that the same enable signal may be received by another data buffer **118** at time **t3'**, which can be in a different cycle of the system clock MCK from the cycle in which **t3** is in. The time interval between **t4** and **t1** is consistent with a write latency W.L. associated with the system **100**, and is controllable by the MCH **101** and knowable to the data buffer **118**. The time interval between **t4** and **t3**, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer **118** since both these signals are received by the data buffer. Based on such determination, the data buffer **118** can have knowledge of the time interval between **t3** and **t1**, referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer **118** to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

FIG. **12B** is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit **116** at time **t5**, the module control circuit **116** outputs one or more enable signals EN at time **t6** in response to the read commands. The one or more enable signals are received by an data buffer **118** at time **t7**, which outputs at time **t8** read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer **118** at time **t3'**, which can be in a different cycle of the system clock MCK from the cycle which **t3** is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers **118**.

With knowledge of the time interval between **t7** and **t5**, which should be about the same as the time interval between **t3** and **t1**, i.e., the command-to-enable delay CED, in certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time **t9** by the data buffer to the MCH **101** via the respective group of data/strobe signal lines **130**, with the time interval between **t9** and **t5** being consistent with a read latency R.L. associated with the system **100**.

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The time interval between t_4 and t_3 , i.e., the enable to write data delay EWD, is determined by the delay control circuit **650** in the DB Control circuit **310**, as shown in FIG. **6**. According to one embodiment, as shown in FIG. **13**, the dynamic delay control circuit **650** includes a preamble detector **1310** to detect a write preamble in the DQS, a flip-flop circuit **1320** having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit **1330** having a Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit **1320**. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers **671** in circuit **1340**, which output the delay signal DS1.

Thus, as shown in FIG. **14**, in one embodiment, a memory module **110** operates in the memory system **100** according to a method **1400**. In the method, during a write operation, one or more data buffer control signals are received by an data buffer **118** from a module control circuit or module controller **116** (**1410**). The module controller **116** generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH **101**. The one or more data buffer control signals are used to control the data buffer **118**. For example, the one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer **118**. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer **118** from the MCH **101** (**1420**). In one embodiment, upon receiving the one or more first enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

Since the time interval between the arrival of the command signals from the MCH **101** and the arrival of the write data/strobe signal DQ/DQS from the MCH **101** is set according to a write latency parameter associated with the system **100**, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module **110** and the time when the one or more enable signals are received by the data buffer **118**. The time interval CED can be used by the data buffer **118** to properly time the transmission of read data to the MCH **101**.

As shown in FIG. **14**, a delay signal DS is generated according to the time interval EWD (**1430**). Concurrent to receiving the write strobe signal DQS, the data buffer **118** also receives a set of write data signals DQ (**1440**). The received write data signals are transmitted to the subgroup of memory devices (**1450**), which are selected from the group of memory devices coupled to the data buffer **118** by the one or more first enable signals.

During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer **114** from the module controller **116** (**1460**). The one or more second enable signals are generated by the module controller **116** in response to read command signals received from the MCH **101**, and are used by the data buffer **118** to select a subgroup of memory devices from which to receive read data. After-

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wards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of memory devices (**1470**). To properly time the transmission of the DQS and DQ signals to the MCH **101**, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system **100**.

We claim:

1. A memory module operable in a computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:

- a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines;
- a plurality of buffer circuits mounted on the PCB, each buffer circuit of the plurality of buffer circuits corresponding to a respective set of the plurality of sets of data/strobe signal lines and including one or more configuration registers, one or more delay control circuit, and one or more delay circuits, the one or more delay circuits to adjust timing of one or more signals in the each buffer circuit of the plurality of buffer circuits;
- a module control circuit mounted on the PCB and configurable to receive from the memory controller input address and control (C/A) signals associated with a read or write operation via the set of C/A signal lines, to output registered C/A signals associated with the read or write operation via first module control signal lines, and to output buffer control signals associated with the memory read or write operation via second module control signal lines;

- memory devices mounted on the PCB and organized in ranks, the memory devices including a plurality of groups of memory devices, a respective group of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks;

wherein:

- in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to receive or output first data/strobe signals;

- in response to the buffer control signals, a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to receive or output second data/strobe signals, the second data/strobe signals corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines;

- at least one delay circuit of the one or more delay circuits in the first buffer circuit is configured to delay at least one signal in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

- the memory module is operable in a configuration mode, and the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit in response to mode register command signals received

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from the memory controller via the set of C/A signal lines during the configuration mode; and
 the second delay is determined by at least one delay control circuit of the one or more delay control circuits in the first buffer circuit while the memory module is not operating in the configuration mode.

2. The memory module of claim 1, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

3. The memory module of claim 1, wherein the buffer control signals include one or more mode signals indicating a read or write operation and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and
 in the configuration mode, the module control circuit is configured to output configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

4. The memory module of claim 3, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

5. The memory module of claim 1, wherein the at least one signal includes at least one data signal and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

6. The memory module of claim 1, wherein the at least one delay control circuit in the first buffer circuit is configured to determine the second delay based on at least one signal received from the module control circuit and at least one signal received from the memory controller during a memory operation.

7. The memory module of claim 1, wherein:
 the system bus further includes one or more clock signal lines;
 the module control circuit is further configurable to receive from the memory controller a system clock via the one or more clock signal lines and output a registered clock; and
 the first buffer circuit is configured to receive the registered clock and to generate a first buffered clock, the first buffered clock having a programmable phase relationship with the registered clock determined by at least one of the one or more registers in the first buffer circuit.

8. The memory module of claim 7, wherein the first buffer circuit is further configured to output the first buffered clock to the first group of memory devices.

9. The memory module of claim 1, wherein the memory module is configurable to perform a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations; and
 wherein at least one of the one or more configuration registers is programmed based on information derived from the set of training operations.

10. The memory module of claim 1, wherein the at least one delay circuit in the first buffer circuit is configured to delay the at least one signal based on a combination of the first delay and the second delay.

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11. A method performed at a memory module in a computer system, the computer system having a memory controller and a system bus, the system bus including a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising a printed circuit board (PCB) including connectors along an edge thereof for connecting to the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and memory devices mounted on the PCB and organized in ranks, the method comprising:

operating in a configuration mode, including:

receiving mode register command signals from the memory controller via the C/A signal lines; and

programming one or more configuration registers in each buffer circuit of a plurality of buffer circuits in response to the mode register command signals, wherein the plurality of buffer circuits are mounted on the PCB and each buffer circuit of the plurality of buffer circuits corresponds to a respective set of the plurality of sets of data/strobe signal lines, wherein the memory devices include a plurality of groups of memory devices, each group of the plurality of groups of memory devices corresponding to a respective buffer circuit of the plurality of buffer circuits and including at least one respective memory device in each of the ranks; and

performing a memory read or write operation, including:

receiving from the memory controller input address and control (C/A) signals via the set of C/A signal lines;

outputting registered C/A signals associated with the read or write operation to the memory devices, wherein in response to the registered C/A signals, at least one first memory device in a first group of the plurality of groups of memory devices is configured to receive or output first data/strobe signals;

outputting buffer control signals associated with the memory read or write operation to the plurality of buffer circuits, wherein in response to the buffer control signals, a first buffer circuit of the plurality of buffer circuits is configured to output or receive the first data/strobe signals and to receive or output second data/strobe signals, the second data/strobe signals corresponding, respectively, to the first data/strobe signals and being communicated between the memory module and the memory controller via a first set of data/strobe signal lines of the plurality of sets of data/strobe signal lines; and

delaying at least one signal in the first buffer circuit based on a first delay and a second delay, each of the at least one signal being related to a corresponding signal in the first data/strobe signals or the second data/strobe signals;

wherein:

the first delay is programmed into and stored in at least one configuration register of the one or more configuration registers in the first buffer circuit during the configuration mode; and

the second delay is determined by at least one delay control circuit in the first buffer circuit while the memory module is not operating in the configuration mode.

12. The method of claim 11, wherein the at least one configuration register provides static control of the timing of the at least one signal and the at least one delay control circuit provides dynamic control of the timing of the at least one signal.

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13. The method of claim 11, wherein the buffer control signals include one or more mode signals indicating a read or write operations and one or more signals carrying information for controlling data paths in the plurality of buffer circuits; and the method further comprises, during the configuration mode, outputting configuration control signals to the buffer circuits in response to the mode register command signals, the configuration control signals including one or more mode signals indicating the configuration mode and one or more signals carrying configuration information for programming the one or more configuration registers.

14. The method of claim 13, wherein the each buffer circuit further includes logic configured to program the one or more configuration registers in response to the configuration information.

15. The method of claim 11, wherein the at least one signal includes at least one data signal and/or at least one strobe signal received from the at least one first memory device in the first group of memory devices.

16. The method of claim 11, wherein the at least one delay control circuit is configured to determine the second delay information based on at least one signal received from the memory controller during a memory operation.

17. The method of claim 11, wherein the system bus further includes one or more clock signal lines, the method further comprising:

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receiving from the memory controller a system clock via the one or more clock signal lines;

outputting a registered clock; and

generating a first buffered clock in the first buffer circuit, the first buffered clock having a programmable phase relationship with the registered clock based on a value stored in at least one of the one or more registers in the first buffer circuit.

18. The method of claim 17, further comprising outputting by the first buffer circuit the respective buffered clock to the first group of memory devices.

19. The method of claim 11, further comprising:

performing a set of training operations including at least one write operation to write a set of data into a set of memory locations on the memory module and at least one read operation to read from the set of memory locations;

wherein at least one the one or more configuration registers in the first buffer circuit is programmed based on information derived from the set of training operations.

20. The method of claim 11, wherein the at least one delay circuit is configured to delay the at least one signal based on a combination of the first delay information and the second delay information.

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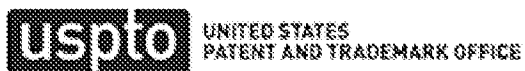
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		First Inventor	Hyun Lee
		Title	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
		Electronically filed	January 5, 2021
APPLICATION ELEMENTS <i>See MPEP Chapter 600 concerning utility patent application contents.</i>		Address to: Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450	
1. ☐ Fee Transmittal Form <i>(with duplicate for fee processing)</i> 2. ☐ Applicant claims Small Entity status, see 37 C.F.R. § 1.27 3. ☒ Specification [Total Pages 30] 4. ☒ Drawing(s) (35 USC § 113) [Total Sheets 22] 5. ☒ Oath or Declaration [Total Pages 2] a. ☐ Newly executed <i>(original or copy)</i> b. ☒ Copy from a prior application (37 CFR § 1.63(d)) 6. ☒ Application Data Sheet, see 37 C.F.R. § 1.76 7. ☐ CD-ROM or CD-R in duplicate, large table or Computer Program <i>(Appendix)</i> ☐ Landscape Table on CD 8. ☐ Nucleotide and/or Amino Acid Sequence Submission <i>(if applicable, all necessary)</i> a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statement verifying identity of above copies		ACCOMPANYING APPLICATION PARTS	
		9. ☐ Assignment Papers <i>(cover sheet & document(s))</i> Name of Assignee: 10. ☒ 37 CFR § 3.73(c) Statement a. ☒ Power of Attorney 11. ☐ English Translation Document <i>(if applicable)</i> 12. ☐ Information Disclosure Statement and PTO-1449 a. ☐ Copies of citations attached 13. ☐ Preliminary Amendment 14. ☐ Return Receipt Postcard <i>(MPEP 503)</i> 15. ☐ Certified Copy of Priority Document(s) <i>(if foreign priority is claimed)</i> 16. ☐ Non-Publication Request under 35 U.S.C. § 122 (b)(2)(B)(i) 17. ☐ Other:	
Note: (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 MUST be included in an Application Data Sheet (ADS). (2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).			
19. CORRESPONDENCE ADDRESS: Customer Number 79141			
Signature	/ Jamie J. Zheng /	Date	January 5, 2021
Name (Print/Type)	Jamie J. Zheng	Registration No. (Attorney/Agent)	51,167

Electronically filed January 5, 2021

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	To be assigned
Serial No.:	To be assigned	Art Unit	To be assigned
Filed:	Filed herewith	Examiner:	To be assigned
For:	Memory Module With Local Synchronization And Method Of Operation	Attorney Docket No.:	129980-5073-US02

STATEMENT UNDER 37 C.F.R. § 3.73(c)

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Netlist, Inc., a corporation, states that it is the assignee of the entire right, title and interest in the patent application/patent identified above by virtue of an assignment from the inventor(s) of the parent of the patent application/patent identified above.

The assignment was recorded in the United States Patent and Trademark Office on April 17, 2016 at Reel/Frame 038300/0101, or for which a copy thereof is attached.

The undersigned is authorized to act on behalf of the assignee.

Date: January 5, 2021

/ Jamie J. Zheng /

51,167

Jamie J. Zheng

(Reg. No.)

MORGAN, LEWIS & BOCKIUS LLP

1400 Page Mill Road

Palo Alto, CA 94304

(650) 843-4000

GENERAL POWER OF ATTORNEY BY APPLICANT FOR UNITED STATES PATENT

I hereby revoke all previous powers of attorney given in the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

I hereby appoint the practitioner associated with the following Customer Number

79141

as my/our attorney or agent, and to transact all business in the United States Patent and Trademark Office in connection with the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

Please recognize or change the **correspondence address** for the application referenced in the attached transmittal letter to the address associated with the above-mentioned **Customer Number**.

I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is currently being filed in this document)

SIGNATURE of Applicant for Patent

Signature

Date: January 9, 2013

Name

Gail Sasaki

Title

Vice President and Chief Financial Officer

Company

Netlist, Inc.

Note: Signature - this form must be signed by the Applicant or a person authorized to act on behalf of Applicant.

Submit multiple forms for more than one signature

Total of ___ forms are submitted.

**COMBINED DECLARATION AND ASSIGNMENT
FOR UTILITY OR DESIGN APPLICATION**

**Title of
Invention**

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

DECLARATION

As a below named inventor, I hereby declare that:

This declaration
is directed to:

☐ The attached application, or

☒ United States application or PCT international application
number 14/445,035 filed on July 28, 2014.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."

ASSIGNMENT

In consideration of the sum of One Dollar (\$1.00) and other good and valuable consideration paid, I hereby sell and assign to

Netlist, Inc.

(Name of Assignee)

of 175 Technology Dr., Suite #150, Irvine, CA 92618

(Address of Assignee)

(hereinafter designated as the Assignee) my entire right, title and interest for the United States as defined in 35 USC 100, for any invention set forth in the above-identified application.

I agree to execute all papers necessary in connection with this application for patent in the USPTO for the invention, and any continuation, divisional or reissue applications thereof and also to execute separate assignments in connection with such applications as the Assignee may deem necessary or expedient.

I agree to execute all papers necessary in connection with any post-grant proceeding which may occur in connection with this application or continuation, divisional or reissue thereof and to cooperate with the Assignee in every way possible in obtaining evidence and going forward with such post-grant proceeding.

I agree to execute all papers and documents and perform any act which may be necessary in connection with claims or provisions of the International Convention for Protection of Industrial Property or similar agreements.

I agree to perform all affirmative acts which may be necessary to obtain a grant of a valid United States patent to the

Assignee.

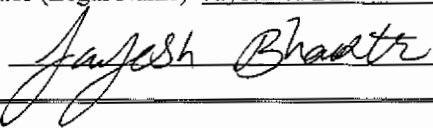
I hereby authorize and request the USPTO to issue any and all Letters Patents of the United States resulting from the application or any continuation, divisional or reissue applications thereof to the Assignee, as Assignee of my entire interest, and covenant that I have not executed, and will not execute, any agreement in conflict herewith.

First Inventor (Legal Name): Hyun Lee

Signature: 

Date: 3/31/2016

First Inventor (Legal Name): Jayesh R. Bhakta

Signature: 

Date: 3/31/2016

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyun		Lee		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Ladera Ranch	State/Province	CA	Country of Residence	US

Mailing Address of Inventor:

Address 1	175 Technology Dr.				
Address 2	Suite #150				
City	Irvine	State/Province	CA		
Postal Code	92618	Country	US		

Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jayesh	R.	Bhakta		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cerritos	State/Province	CA	Country of Residence	US

Mailing Address of Inventor:

Address 1	175 Technology Dr.				
Address 2	Suite #150				
City	Irvine	State/Province	CA		
Postal Code	92618	Country	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.

Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

☐ An Address is being provided for the correspondence information of this application.

Customer Number	79141		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		
Attorney Docket Number	129980-5073-US02	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	79141		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending		Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
	Continuation of	16/432700	2019-06-05		
Prior Application Status	Patented		Remove		
Application Number	Continuity Type	Prior Application Number	Filing Date (YYYY-MM-DD)	Patent Number	Issue Date (YYYY-MM-DD)
16/432700	Continuation of	14/445035	2014-07-28	10324841	2019-06-18
Prior Application Status	Expired		Remove		
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)		
14/445035	Claims benefit of provisional	61/859215	2013-07-27		
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.					Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)	Remove
Additional Foreign Priority Data may be generated within this form by selecting the Add button.				Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

☐	This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013. NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.
--------------------------	---

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	NETLIST, INC.	
Mailing Address Information For Applicant:		
Address 1	175 Technology Dr.	
Address 2	Suite #150	
City	Irvine	State/Province CA
Country	US	Postal Code 92618
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Signature:					Remove
NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).					
This Application Data Sheet must be signed by a patent practitioner if one or more of the applicants is a juristic entity (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, all joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of all joint inventor-applicants.					
See 37 CFR 1.4(d) for the manner of making signatures and certifications.					
Signature	/ Jamie J. Zheng /			Date (YYYY-MM-DD)	2021-01-05
First Name	Jamie J.	Last Name	Zheng	Registration Number	51167
Additional Signature may be generated within this form by selecting the Add button. Add					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US02
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application Serial No. 16/432,700, filed on June 5, 2019, entitled “Memory Module with Local Synchronization and Method of Operation,” which is a continuation of U.S. Patent Application Serial No. 14/445,035, filed on July 28, 2014 (now U.S. Patent 10,324,841), entitled “Memory Module with Local Synchronization,” which claims the benefit of U.S. Provisional Patent Application Serial No. 61/859,215, filed on July 27, 2013, entitled "Memory Module with Local Synchronization Using Distributed Data Buffers," each of which is herein fully incorporated by reference in its respective entirety.

FIELD

[0002] The disclosure herein is related generally to memory modules, and more particularly to a memory module with local synchronization and method of operation.

BACKGROUND

[0003] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a memory controller accesses the memory

module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

[0004] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0005] The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0006] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0007] For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules. The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0008] In some conventional memory systems, the memory controllers include leveling

mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0010] FIGS. 2A-2D are each a diagram illustrating interactions among components in a memory module according to certain embodiments.

[0011] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0012] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0013] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0014] FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0016] FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

[0017] FIG. 7A is a flowchart illustrating a method of operation for a memory module according to certain embodiments.

[0018] FIG. 7B is a flowchart illustrating a method of training locally synchronized clock signals according to certain embodiments.

[0019] FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0020] FIG. 9 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0021] FIG. 10 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0022] FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0023] FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0024] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0025] FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

[0026] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g., read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller

to the memory module. The C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

[0027] In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

[0028] FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120

and groups of system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

[0029] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0030] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

[0031] Furthermore, the memory devices 112 may include any type of memory devices. For example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory

(SDRAM) devices in compliance with, for example, the Double Data Rate Type 3 (DDR3) or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0032] As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0033] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0034] Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version

of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Patent No. 7,532,537, entitled “Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation,” issued on May 12, 2009, the entirety of which is incorporated herein by reference.

[0035] Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers 118, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer control signals are used to convey configuration or module self-test control information from the module control device 116 to the data buffers 118. In one embodiment, the data buffer control signals are transmitted to the data buffers 118 via respective data buffer control signal lines 230. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers 118 via the data buffer control signal lines and decoded/processed at the data buffers.

[0036] Module control device 116 transmits the module C/A signals to the memory devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers 118 via data buffer control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the data buffers 118 share a same set of data buffer control signal lines 230.

[0037] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices M_{11} , ..., M_{i1} , $M_{i+1,1}$, ..., M_{N1} , a second rank includes memory devices M_{12} , ..., M_{i2} , $M_{i+1,2}$, ..., M_{N2} , and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11} , M_{12} , M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0038] In certain embodiments, the data buffers 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers 118 is associated with the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory devices, data buffers ID- i among the data buffers 118 is associated with the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be associated with a same group of memory devices, or vice versa.

[0039] In one embodiment, each group or sets of memory devices are coupled to the

associated data buffer 118 via a set of module data/strobe lines 210 and one or more clock signal lines 211. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0040] In one embodiment, the data buffers 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each data buffer 118 is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the data buffer control signals. Further, each data buffers 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0041] In one embodiment, the data buffers 118 are distributed across the memory module 110 or the module board 119 in positions corresponding to edge connectors 121 coupled to respective groups of system data/strobe signal lines 130. For example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

[0042] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0043] In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0044] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system 100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the memory devices 112.

[0045] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the control signals to the module level physical memory domain,

and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

[0046] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

[0047] For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of interface timing training may be required because the system memory controller 101 has very limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0048] Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very

significant portion of a data period. As a result, the clock skew issues cannot simply be addressed by pipelining the data paths.

[0049] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer 118 in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit 116 and the group of locally synchronized components.

[0050] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices 112, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller 101, and/or programmable phase adjustment for one or more control signal from the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data buffers, as disclosed in commonly-owned U.S. Patent 8,001,434, issued August 16, 2011, entitled

“Memory Board with Self-Testing Capability,” which is incorporated herein by reference in its entirety.

[0051] In certain embodiments, operations of the data buffers 118 are controlled by the data buffer control signals from the module control circuit 116, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers 118 to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers 118 are distributed across the memory module, the data buffer control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. Patent Application No. 13/952,599, filed July 27, 2013, entitled “Memory Module with Distributed Data Buffers and Method Of Operation,” the entirety of which is incorporated herein by reference.

[0052] Because the data buffers 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signals lines 230 from the module control device 116 to the farthest positioned data buffers 118, such as data buffer ID-1 and data buffer ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of data buffer control signals may reach different data

buffers 118 at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

[0053] With the data buffers 118 receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the data buffers 118. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used to adjust the timing of the transmission of read data to the MCH 101 during a subsequent read operation, as explained in more detail below.

[0054] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0055] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n data (DQ) signal lines 322 each for transmitting one of a set of bits $DQ_0, DQ_1, \dots$,

DQ_{n-1}, in a data signal and at least one strobe (DQS) signal line 324 for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines Y₀, Y₁, ..., Y_{n-1} and at least one module strobe signal line Y_{DQS}. When the subsets of memory devices are coupled to the associated data buffer 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines YA₀, YA₁, ..., YA_n and at least one first module strobe signal line YA_{DQS}; and each subset of module data/strobe lines YB include a set of n second module data lines YB₀, YB₁, ..., YB_n and at least one second module strobe signal line YB_{DQS}.

[0056] Each data buffer 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines YA₀, YA₁, ..., YA_n and the second module data lines YB₀, YB₁, ..., YB_n. Each data buffer 118 further includes an DB Control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS}, or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS}.

[0057] The DB Control circuit 310 also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines 230, and outputs datapath control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit 310 also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective DQ signal line 322 with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

[0058] In certain embodiments, the DB Control circuit 310 also provides one or more delay signals DS, which is used by the DQ routing circuits 320 to control the timing of read and/or write data output by the data buffer 118, as explained in further detail below. In certain embodiments, the DB Control circuit 310 regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer 118 and the respective group of memory devices form a group of locally synchronized devices on the memory module 110.

[0059] The memory devices 112 in each group of locally synchronized devices are coupled to the data buffer 118 in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices 112 in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer 118 via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M₁₁, M₁₂, M₁₃, and M₁₄ in the first group of memory devices can be coupled to the data buffer ID-1 via a same set of module data lines Y-1₀, Y-1₁, ..., Y-1_{n-1} and module strobe line Y-1_{DQS}. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0060] Alternatively, as shown in FIG. 4B, memory devices M₁₁ and M₁₃, which form a subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YA-1₀, YA-1₁, ..., YA-1_n and module strobe line YA-1_{DQS} and memory devices M₁₂ and M₁₄, which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YB-1₀, YB-1₁, ..., YB-1_n and module strobe line YB-1_{DQS}. Memory devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board 119. Memory devices

coupled to the same data buffers may be placed side-by-side, on opposite sides of the module boards 119, or stacked over each other, and/or over the associated data buffer.

[0061] Multiple memory devices having a data width that is less than a data width of the data buffers 118 may be used in place of one of the memory devices 112, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M_{11-1} and M_{11-2} may be used in place of the memory device M_{11} . Each of the two memory devices M_{11-1} and M_{11-2} has a data width of 4, and together they act like a memory device M_{11} of a data width of 8. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1DQS-1 while memory circuit M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1DQS-2.

[0062] In another embodiment, as shown in FIG. 5B, four memory devices M_{11-1} to M_{11-4} may be used as the memory device M_{11} . Each of the four memory devices M_{11-1} to M_{11-4} has a data width of 4, and together they act like a memory device M_{11} of a data width of 16. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1DQS-1 while memory device M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1DQS-2, and so on.

[0063] FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits 630 on each data buffer. In certain embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit 116 uses the DBCS to configure the programmable features in the data buffers 118 based on configuration information received from the MCH 101; in the test mode, the DBCS is used to control the self-test processes

run by the data buffers; in the normal operation mode, the DBCS includes the one or more enable signals and one or more ODT signals.

[0064] FIG. 6B illustrates the DB Control circuit 310 in an data buffer 118. As shown, the DB Control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, a phase-locked loop circuit (PLL) 611 that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit 615 for adjusting the phase of the locally synchronized clock CK0, and a clock driver 618 that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to a corresponding system DQS signal line 324 and on another side to corresponding module DQS signal lines YADQS and YBDQS. The DB Control circuit 310 further includes receiver circuits 630 to receive respective ones of the DBCS.

[0065] The DB Control circuit 310 further includes a control processing circuit 640 that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more data path enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the control processing circuit 640, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

[0066] The strobe routing circuit 620 also buffers strobe signals received from either the MCH 101 or the memory devices 112, and outputs either a write strobe WDQS or read

strobe RDQS, respectively, to the DQ routing circuits 320. In one embodiment, the DB Control circuit 310 further includes a dynamic delay control circuit 650 that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit 320 and the strobe routing circuit 620.

[0067] In certain embodiments, as shown in FIG. 6C, the control processing circuit 640 includes control logic 660 that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals 330. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals 68 to programmable registers 671 that provides static control of the timing of read data/strobe signals, configuration signals 682 to programmable registers 671 that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals 683 to programmable registers 673 that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

[0068] The programmable registers 671 is coupled via signal lines 69 the dynamic delay control circuit 650, which also takes as input a DQS signal, clock signal CK, and one of the datapath enable signals EN, and output the delay signal DS1. The dynamic delay control circuit 650 is described in further detail below.

[0069] FIG. 7A is a flowchart illustrating a method 700 performed by the memory module 110 according to certain embodiments. As shown in FIG. 7A, the method includes receiving 710 clock signal together with system C/A signals from the MCH 101, and generating 715 module clock signal CK. The module clock signal CK is transmitted 720 to a plurality of distributed data buffers 118, which generates 725 locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits 730 its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform 735 memory operations in accordance with the respective locally synchronized clock signals.

[0070] Since the module control circuit 116 generates the DBCS based on signals from the MCH 101, the MCH 101 can train the data buffers 118 to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit 116 generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH 101. As shown in FIG. 7B, a method 750 for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing 751 data to a set of memory locations in the memory devices 112, which is followed by performing a read operation by reading 752 data from the set of memory locations. The read data is compared 753 with the write data to determine 754 if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit 116 agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit 116 do not agree with the corresponding write data bits. The disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers 118 further away from the module control circuit 116 being excessively delayed during transmission, and proper phase adjustment can be programmed 755 into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

[0071] FIG. 8 illustrates a DQ or DQS routing circuit 320 or 620 according to an embodiment. As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 801 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 802 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS}, or YA/YA_{DQS} and YB/YB_{DQS}. The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 810 that buffers write data/strobe, a write data/strobe receiver 820 that samples the write data/strobe, and a write delay circuit 825 controlled by the delay signal DS2. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 830

that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

[0072] The DQS routing circuit further includes a plurality of read paths 850 that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit 860 by an amount controlled by the delay signal DS1, and sampled by a sampler circuit 870. The sampled read data/strobe is transmitted by transmitter 880 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 801.

[0073] FIG. 9 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 901 that is coupled to a corresponding DQS signal line 324, a second DQS pin 902A that is coupled to a corresponding module DQS line Y_{ADQS} , a third DQS pin 902B that is coupled to a corresponding module DQS line Y_{BDQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first DQS pin 901 and the second DQS pin 902A and a second write strobe path coupled between the first DQS pin 901 and the third DQS pin 902B. The first write strobe path includes a write strobe buffer 910 that buffers a write strobe, a write strobe receiver 920 that samples the write strobe according to the buffered module signal CK0 and a write delay circuit 925 controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 930A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{ADQS} . The second write strobe path includes the write strobe buffer 910, the write strobe receiver 920, the write delay circuit 825 controlled by the delay signal DS2, and a second write strobe transmitter 930B that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{BDQS} . The first and second write strobe transmitters, 930A and 930B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0074] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 901 and a selected one of the second and third DQS pins 902A and 902B. In the read strobe path, a select circuit 950 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 902A or a read strobe signal received via DQS pin 902B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 960 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 970 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 980 onto the corresponding strobe signal line 324 via the first DQS pin 901.

[0075] FIG. 10 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1001 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1002A that is coupled to a corresponding module DQ line Y_{ADQ} , a third DQ pin 1002B that is coupled to a corresponding module DQ line Y_{BDQ} . The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1001 and the second DQ pin 1002A and a second write data path coupled between the first DQ pin 1001 and the third DQ pin 1002B. The first write data path includes a write data buffer 1010, a write data receiver 1020 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, a write delay circuit 1025 controlled by the delay signal DS2, and a first write data transmitter 1030A that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{ADQ} . The second write data path includes the write data buffer 1010, the write data receiver 1020, the write delay circuit 825 controlled by the delay signal DS2, and a second write data transmitter 1030B that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{BDQ} . The first and second write data transmitters, 1130A and 1130B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0076] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1001 and a selected one of the second and third DQ pins 1002A and 1002B. In the read data path, a select circuit 1050 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1002A or a read data signal received via DQ pin 1002B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1060 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1070 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1080 onto the corresponding data signal line 130 via the first DQ pin 1001.

[0077] FIG. 11 illustrate a delay circuit 1100 that can be used as the delay circuit 825/860/925/960/1025/1060 according to an embodiment. As shown, the delay circuit 1100 includes a plurality of delay stages, such as delay stages 1110, 1120, and 1130, each delaying an input signal by a predetermined amount. The delay circuit 1100 further includes a select circuit 1140 (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

[0078] As stated above, the MCH 101 can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the interface between the data buffers and the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers 118 having programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

[0079] FIG. 12A is a timing diagram for a write operation according to one embodiment. As shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by a data buffer 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle in which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the data buffer 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer 118 since both these signals are received by the data buffer. Based on such determination, the data buffer 118 can have knowledge of the time interval between t3 and t1, referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer 118 to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

[0080] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t5, the module control circuit 116 outputs one or more enable signals EN at time t6 in response to the read commands. The one or more enable signals are received by an data buffer 118 at time t7, which outputs at time t8 read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle which t3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers 118.

[0081] With knowledge of the time interval between t7 and t5, which should be about the same as the time interval between t3 and t1, i.e., the command-to-enable delay CED, in

certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t_9 by the data buffer to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t_9 and t_5 being consistent with a read latency R.L. associated with the system 100.

[0082] The time interval between t_4 and t_3 , i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the DB Control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the dynamic delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers 671 in circuit 1340, which output the delay signal DS1.

[0083] Thus, as shown in FIG. 14, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1400. In the method, during a write operation, one or more data buffer control signals are received by an data buffer 118 from a module control circuit or module controller 116 (1410). The module controller 116 generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH 101. The one or more data buffer control signals are used to control the data buffer 118. For example, the one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer 118 from the MCH 101 (1420). In one embodiment, upon receiving the one or more

first enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0084] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the data buffer 118. The time interval CED can be used by the data buffer 118 to properly time the transmission of read data to the MCH 101.

[0085] As shown in FIG. 14, a delay signal DS is generated according to the time interval EWD (1430). Concurrent to receiving the write strobe signal DQS, the data buffer 118 also receives a set of write data signals DQ (1440). The received write data signals are transmitted to the subgroup of memory devices (1450), which are selected from the group of memory devices coupled to the data buffer 118 by the one or more first enable signals.

[0086] During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer 114 from the module controller 116 (1460). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the data buffer 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of memory devices (1470). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.

WE CLAIM:

1. A memory module operable in a computer system having a memory controller and a system bus including one or more clock signal lines, a set of control signal lines and a plurality of sets of n -bit wide data/strobe signal lines, where n is an integer larger than one, comprising:

a control circuit configurable to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the set of control signal lines, and to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

memory devices organized in one or more ranks; and

a plurality of data buffers configurable to receive the module clock signal and the data buffer control signals from the control circuit; wherein:

a respective data buffer of the plurality of data buffers is coupled to a respective group of the memory devices and includes a respective n -bit-wide data path and logic configurable to control the respective n -bit-wide data path in response to the data buffer control signals, the respective group of the memory devices including at least one memory device in each of the one or more ranks, the respective n -bit-wide data path including at least one programmable delay element controlled by the logic;

the respective data buffer is configurable to generate, in response to receiving the module clock signal, a respective local clock signal that is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal; and

the respective group of the memory devices is configurable to receive the module C/A signals from the control circuit and the respective local clock signal, and to communicate respective data and data strobe signals with the memory controller in response to the module C/A signals and the respective local clock signal, the respective data and data strobe signals being communicated via the respective n -bit wide data path and a respective set of the plurality of sets of n -bit wide data/strobe signal lines.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

ABSTRACT OF THE DISCLOSURE

A memory module is operable in a memory system with a memory controller. The memory module comprises a module control device to receive command signals and a system clock from the memory controller and to output a module clock, module C/A signals and data buffer control signals. The module C/A signals are provided to memory devices organized in one or more ranks, while the data buffer control signals, together with the module clock, are provided to a plurality of buffer circuits corresponding to respective groups of memory devices and are used to control data paths in the buffer circuits. The plurality of buffer circuits include clock regeneration circuits to regenerate clock signals with programmable delays from the module clock. The regenerated clock signals are provided to respective groups of memory devices so as to locally sync the buffer circuits with respective groups of memory devices.

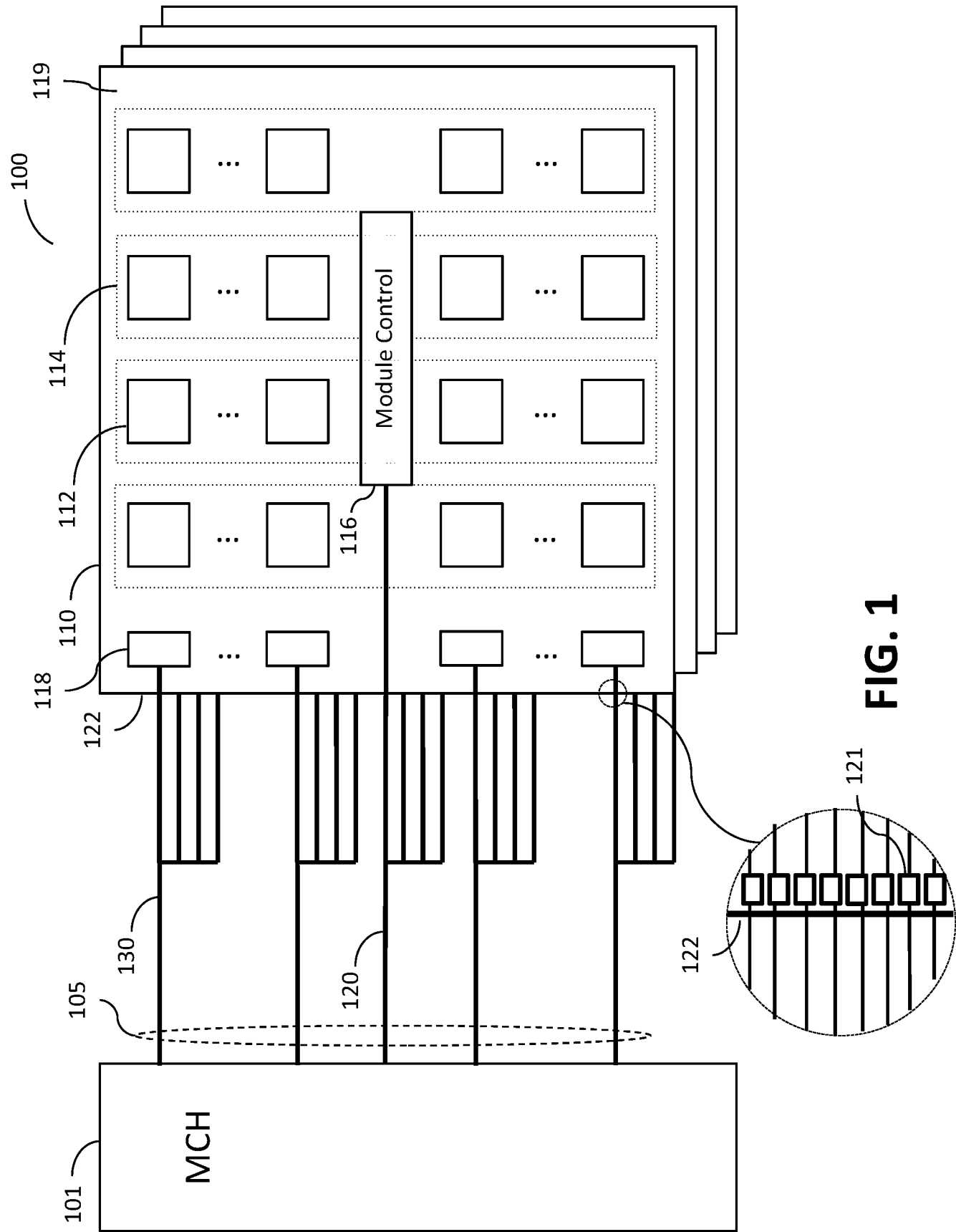


FIG. 1

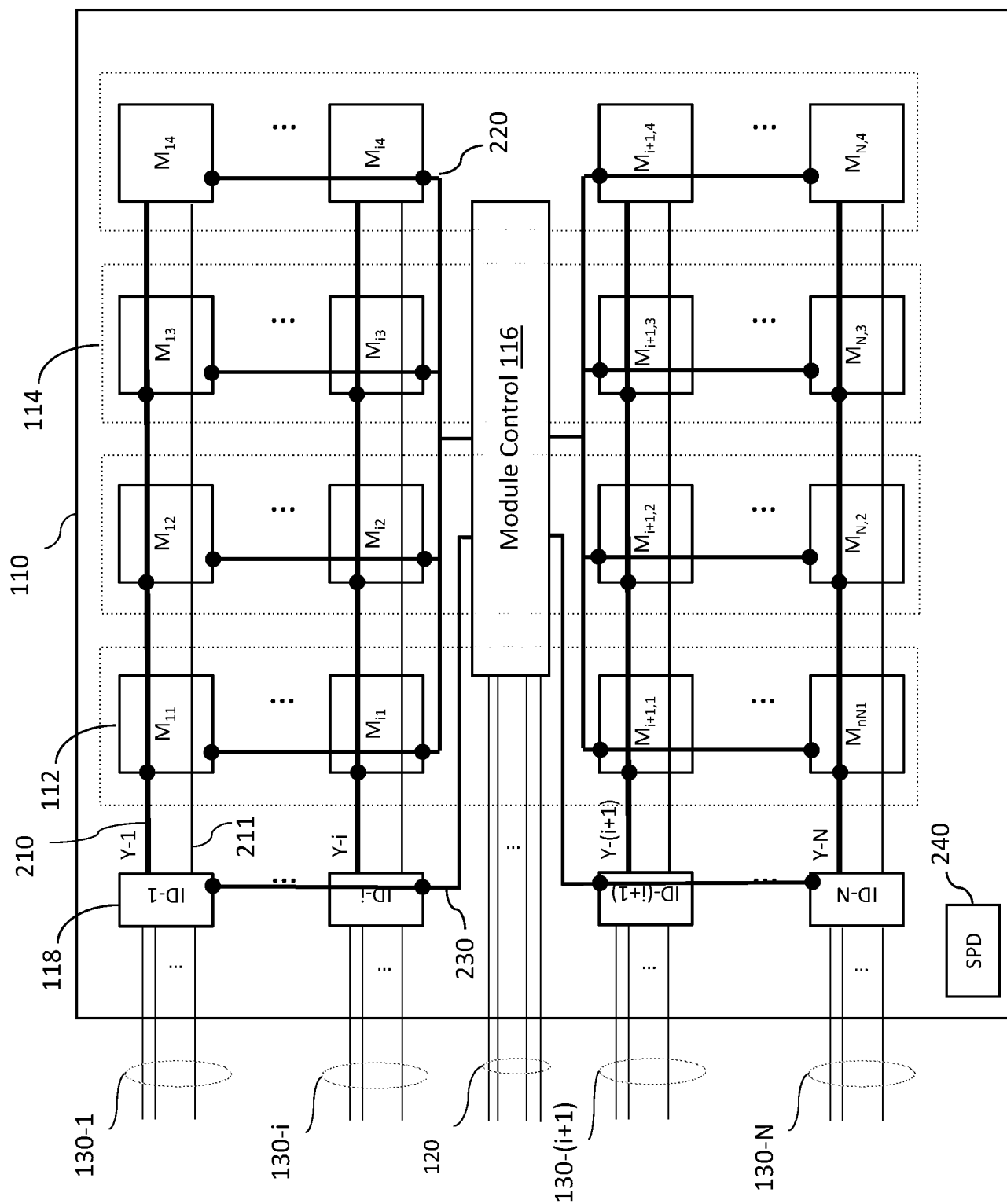


FIG. 2A

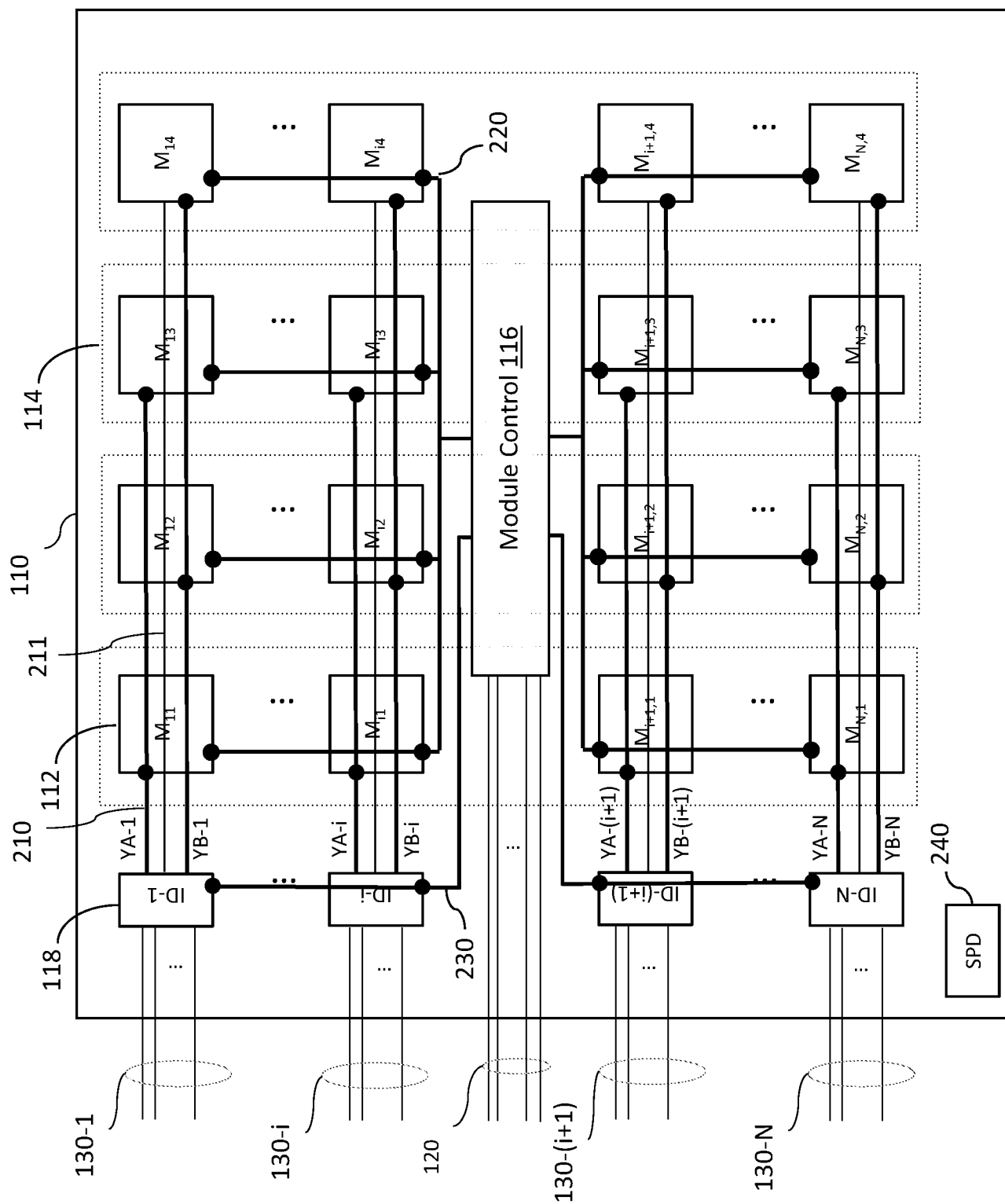


FIG. 2B

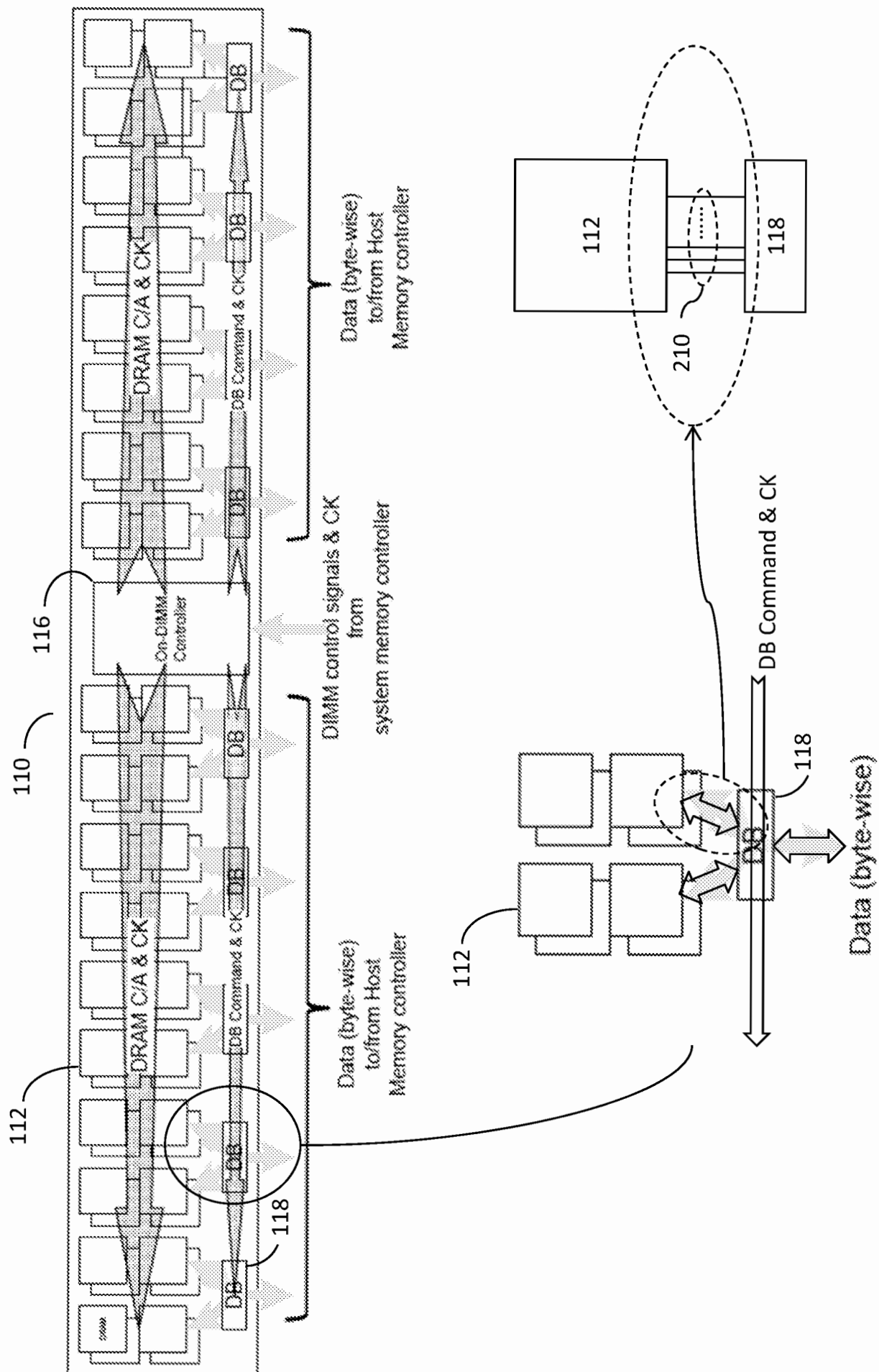


FIG. 2C

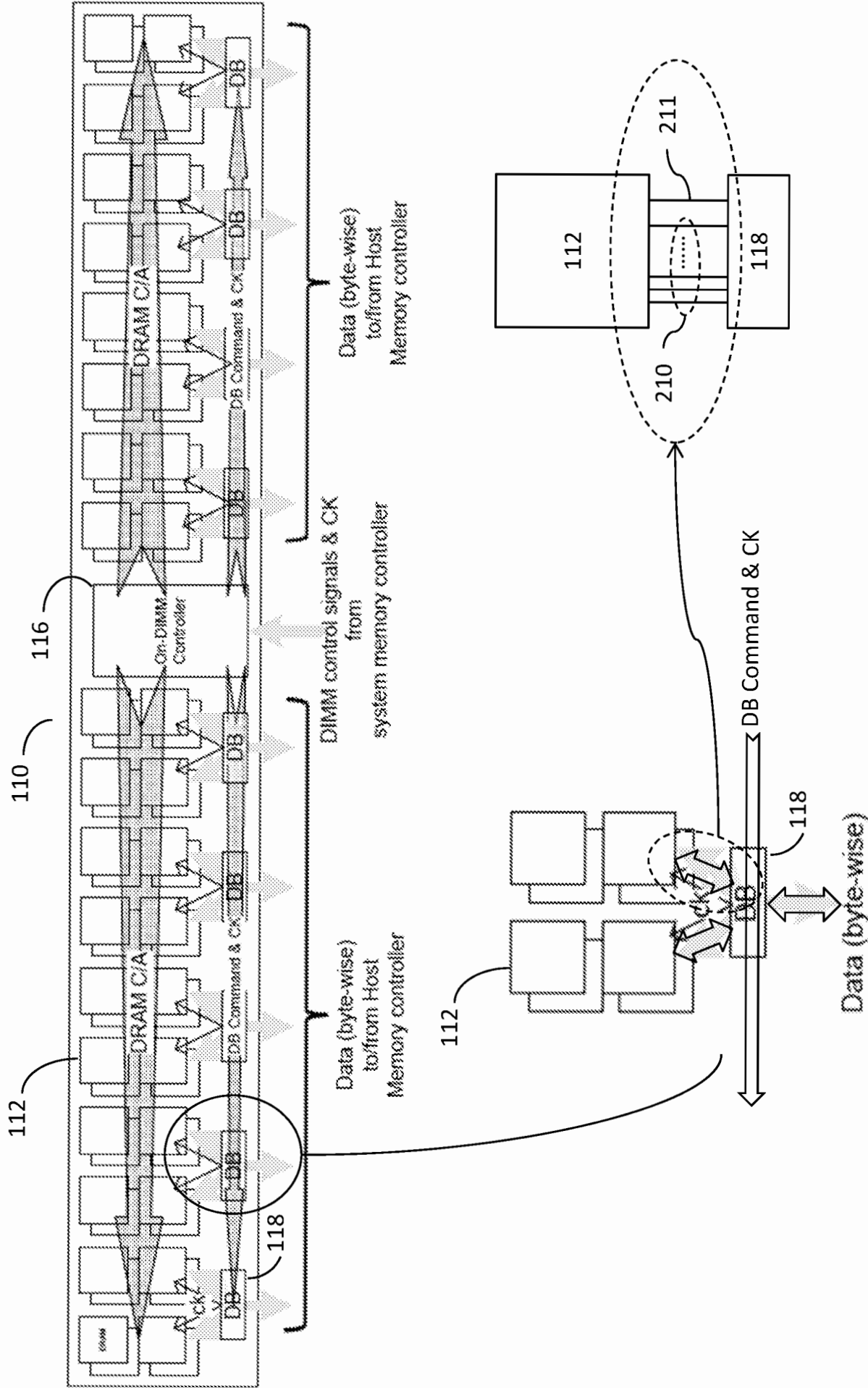


FIG. 2D

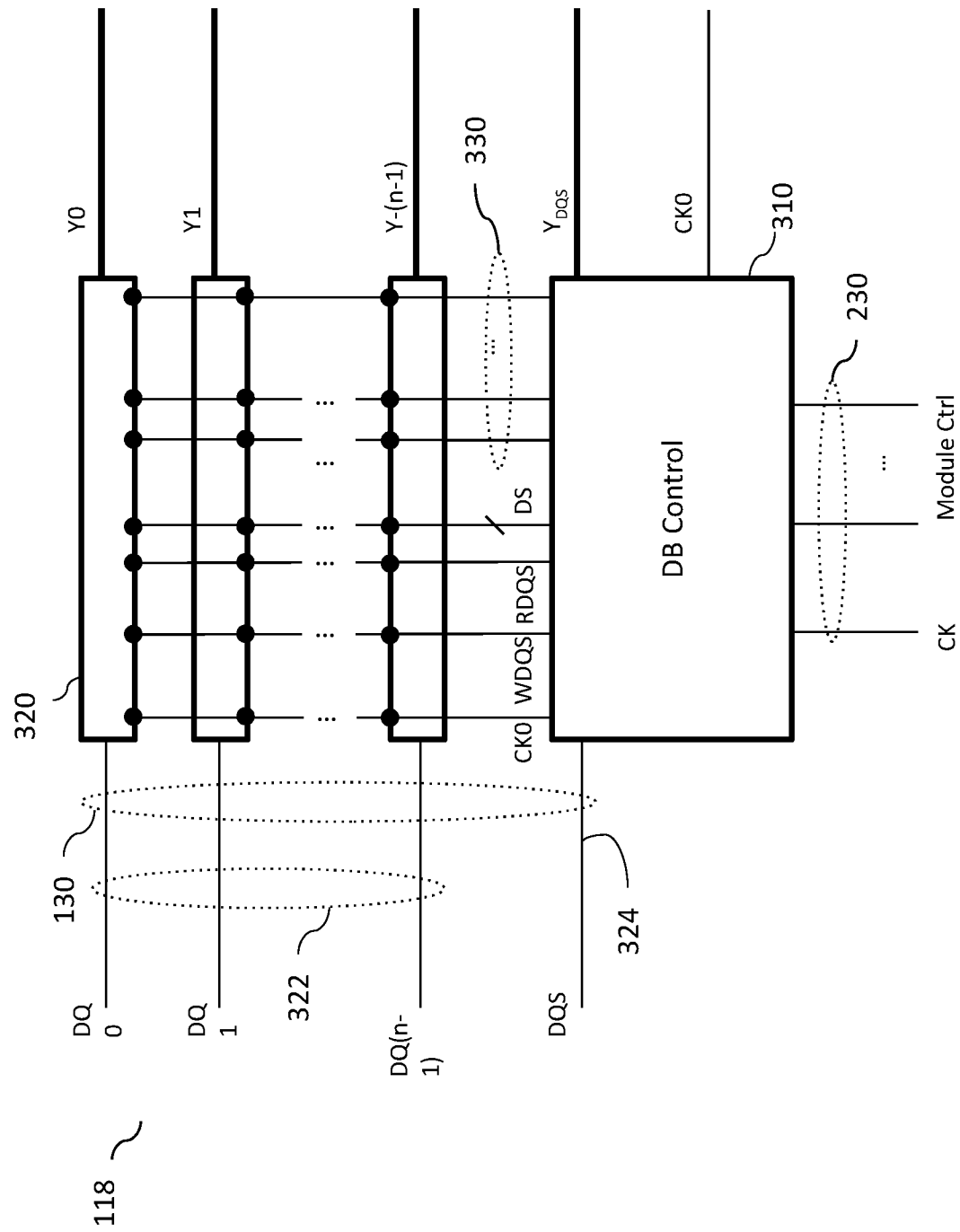


FIG. 3

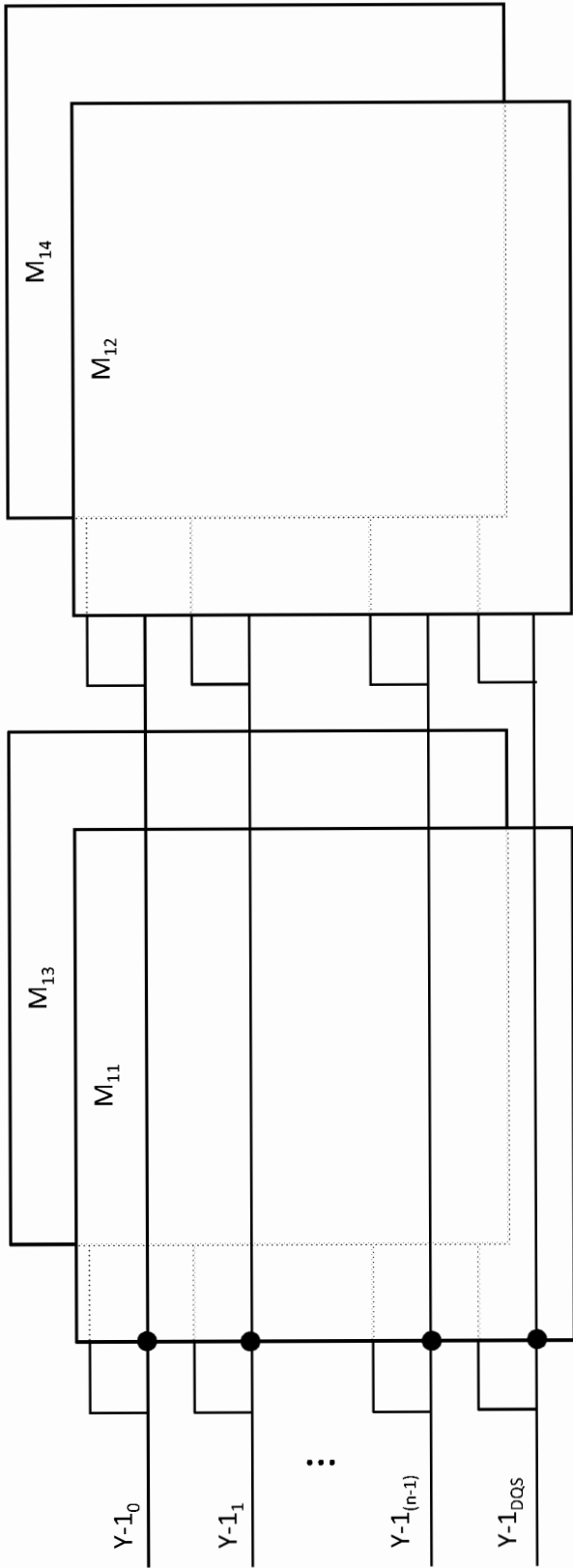


FIG. 4A

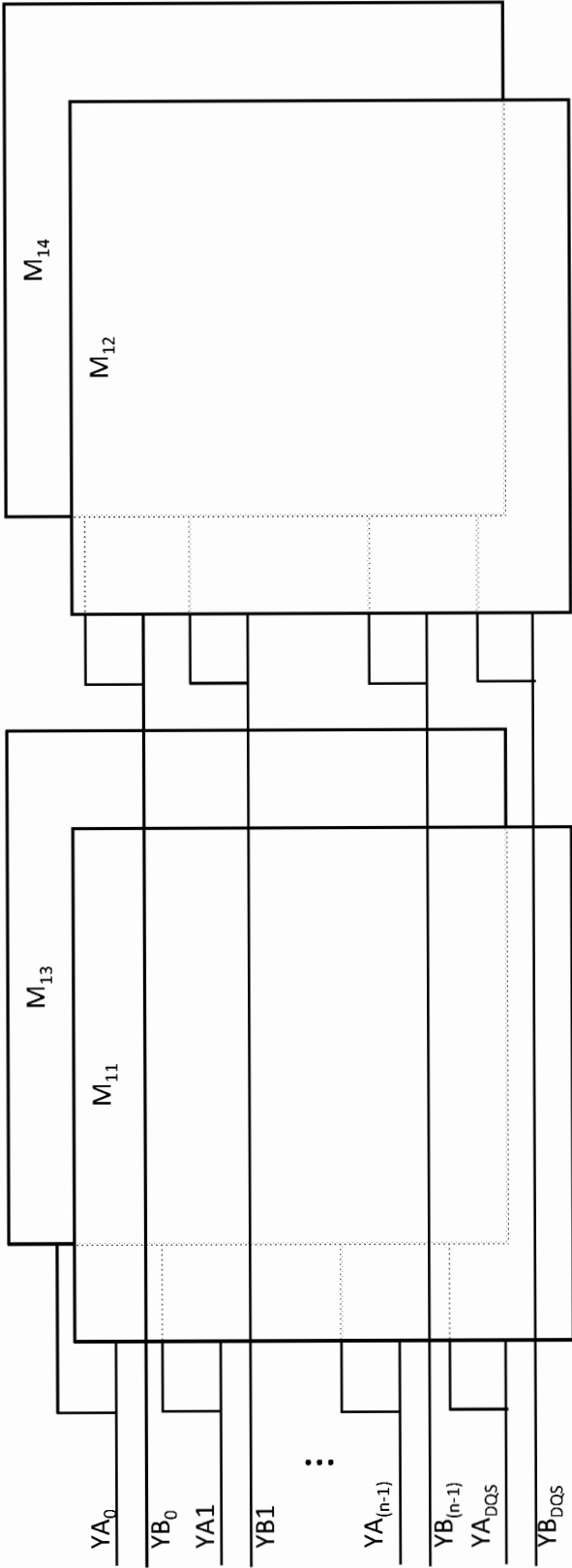


FIG. 4B

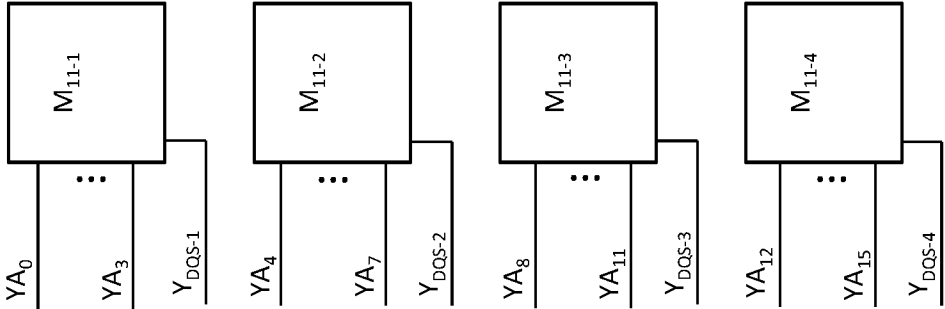


FIG. 5B

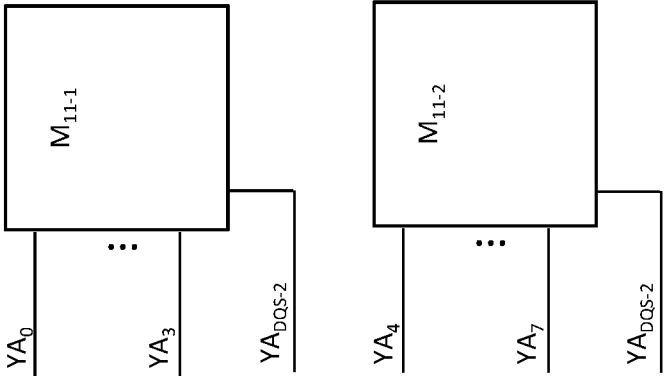


FIG. 5A

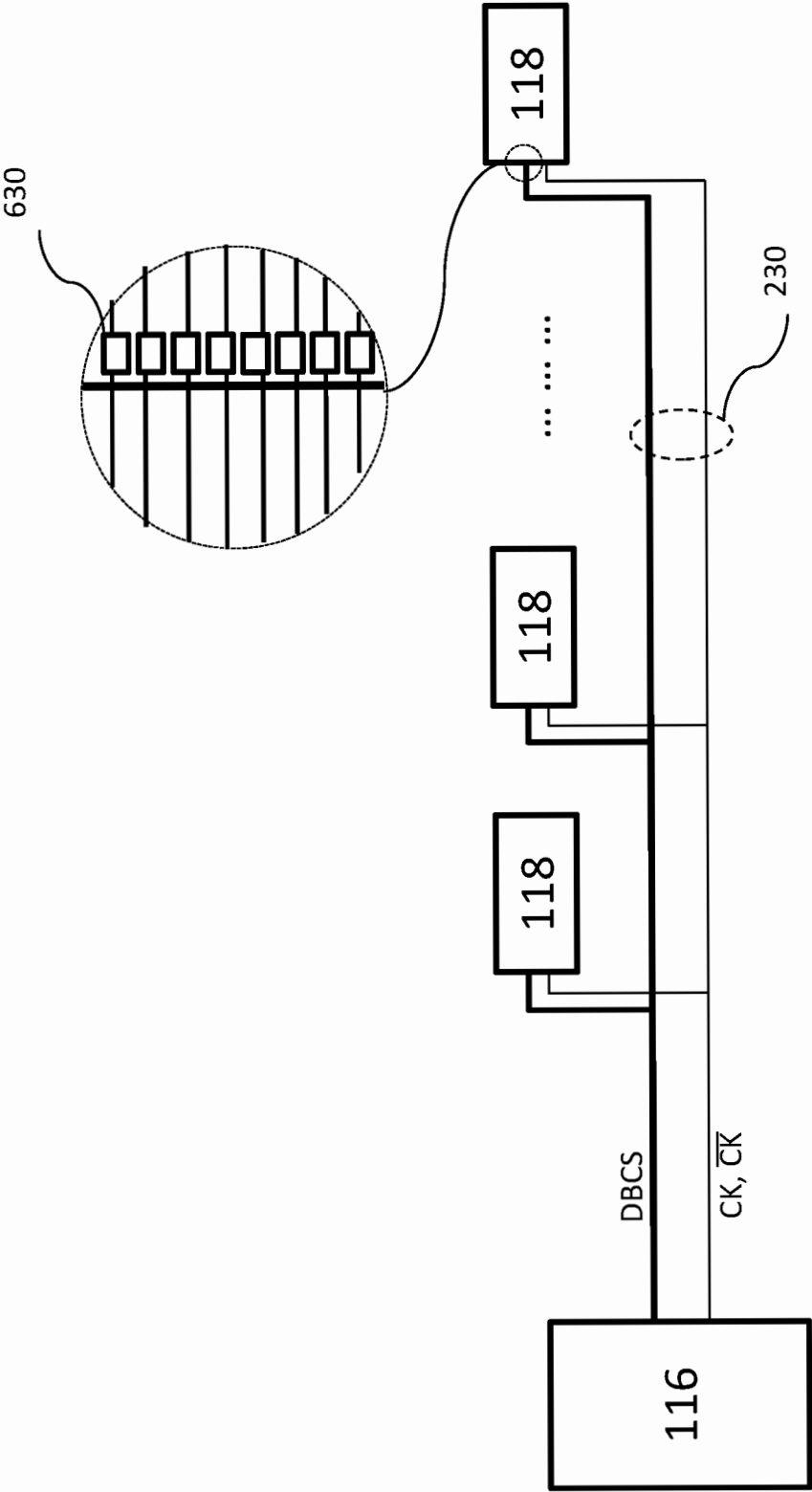


FIG. 6A

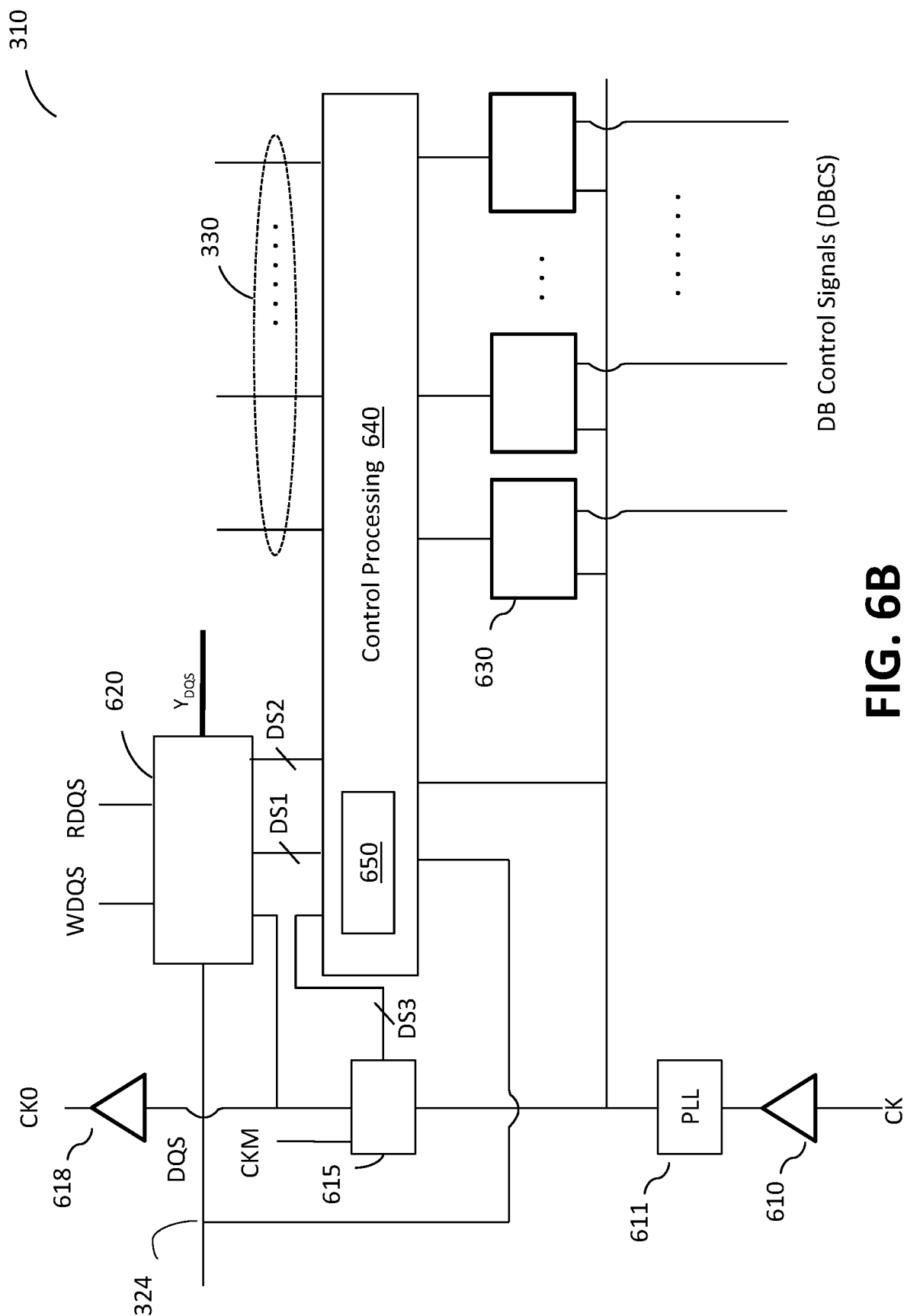


FIG. 6B

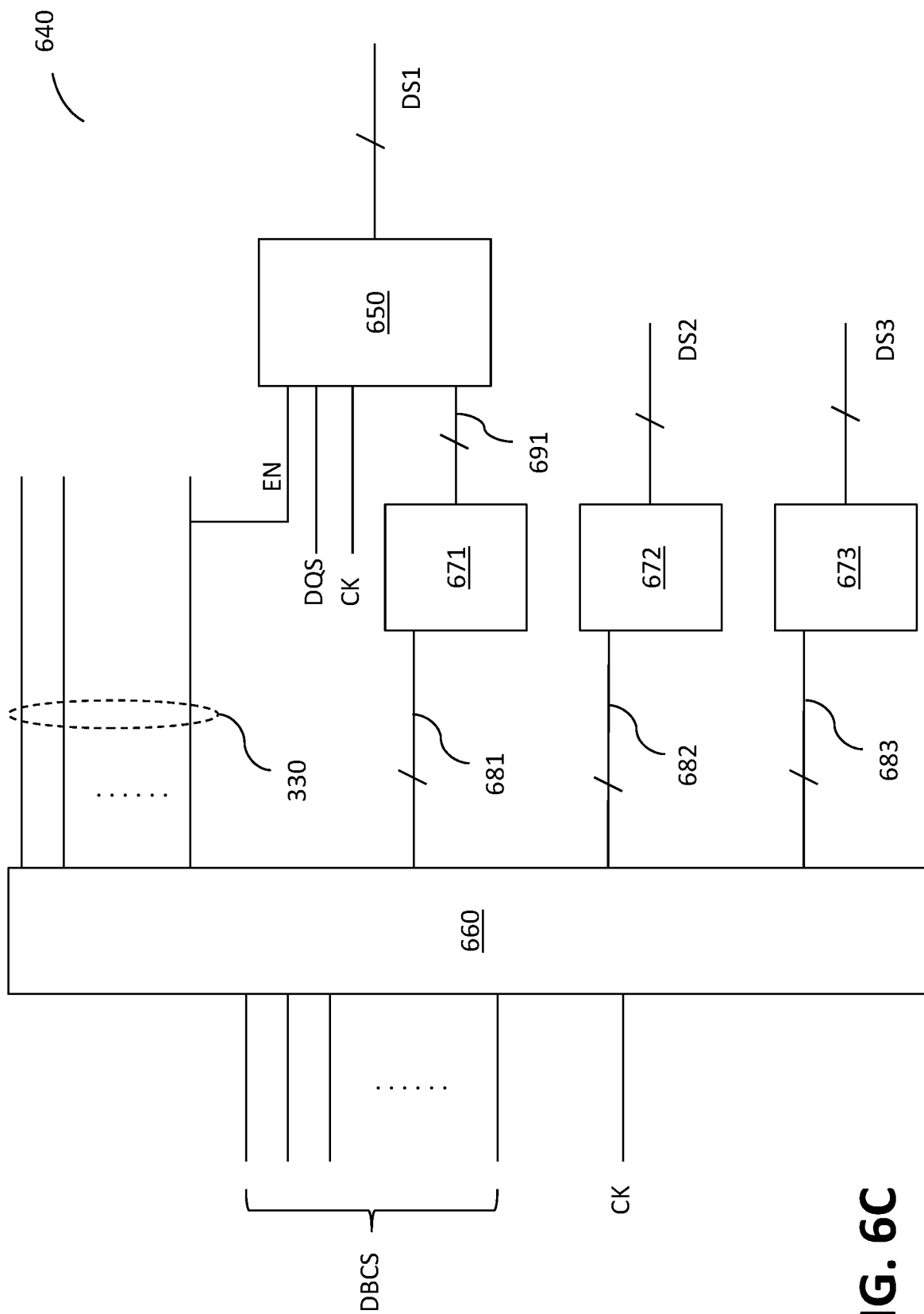
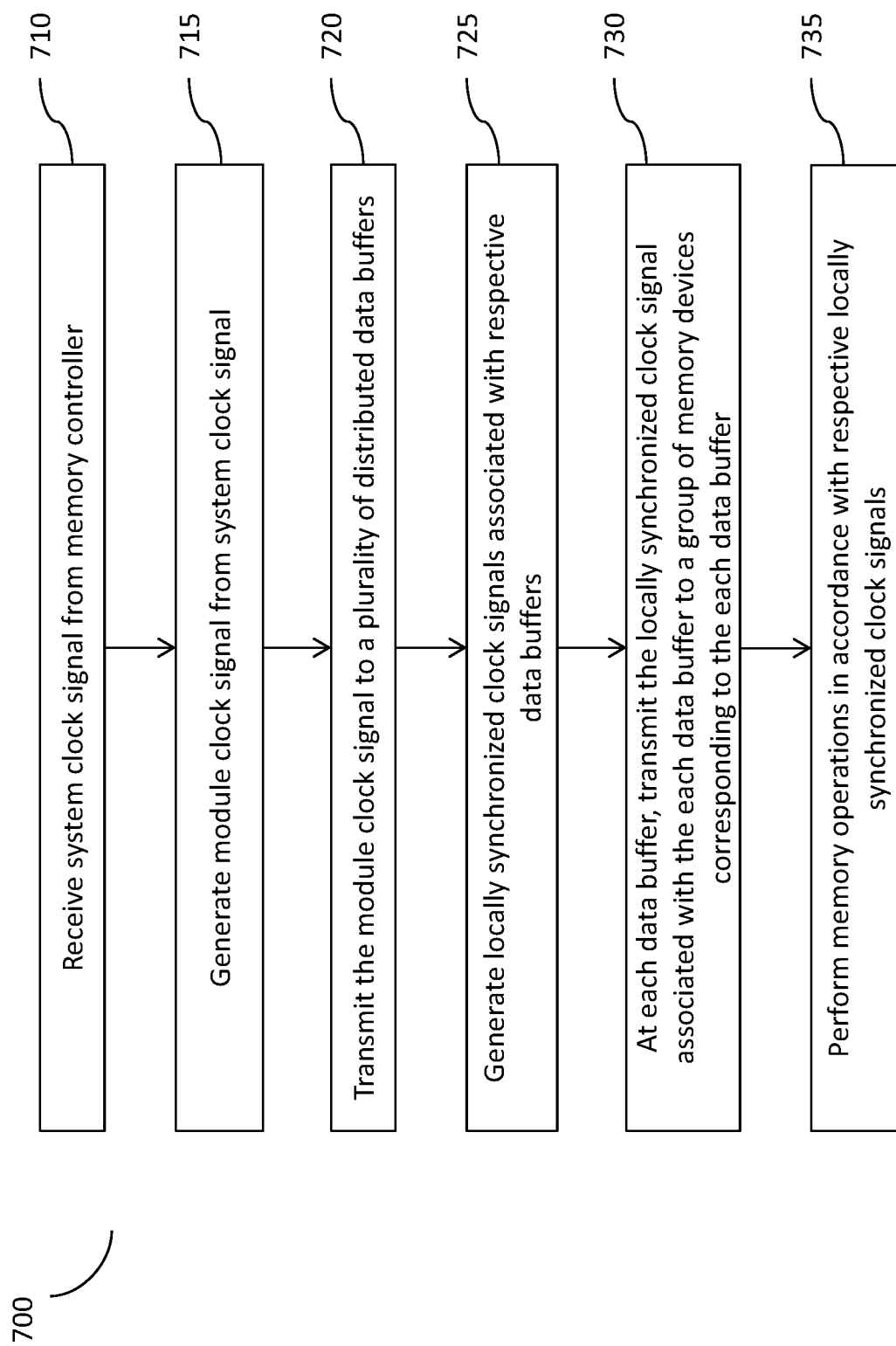
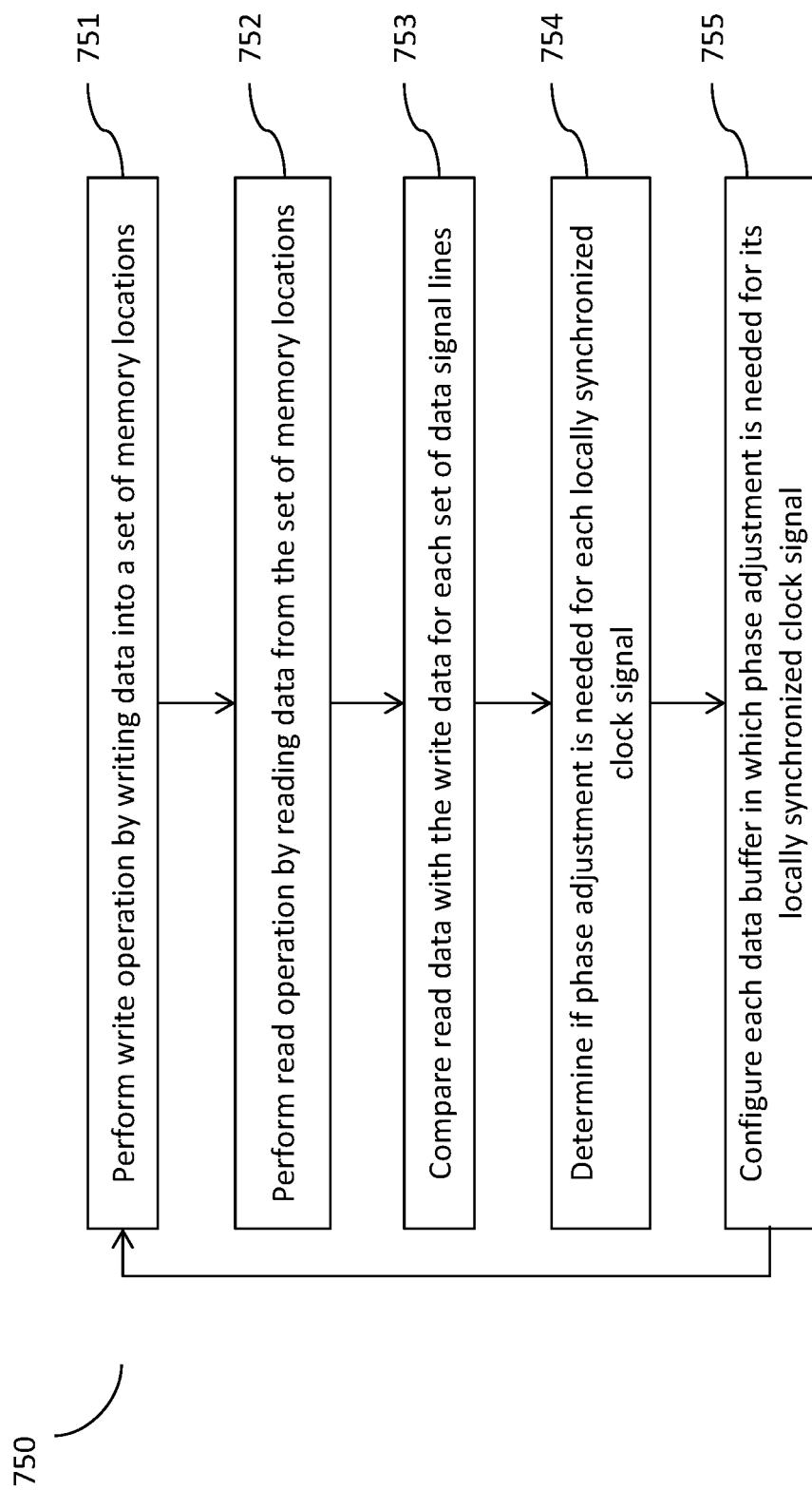


FIG. 6C

**FIG. 7A**

**FIG. 7B**

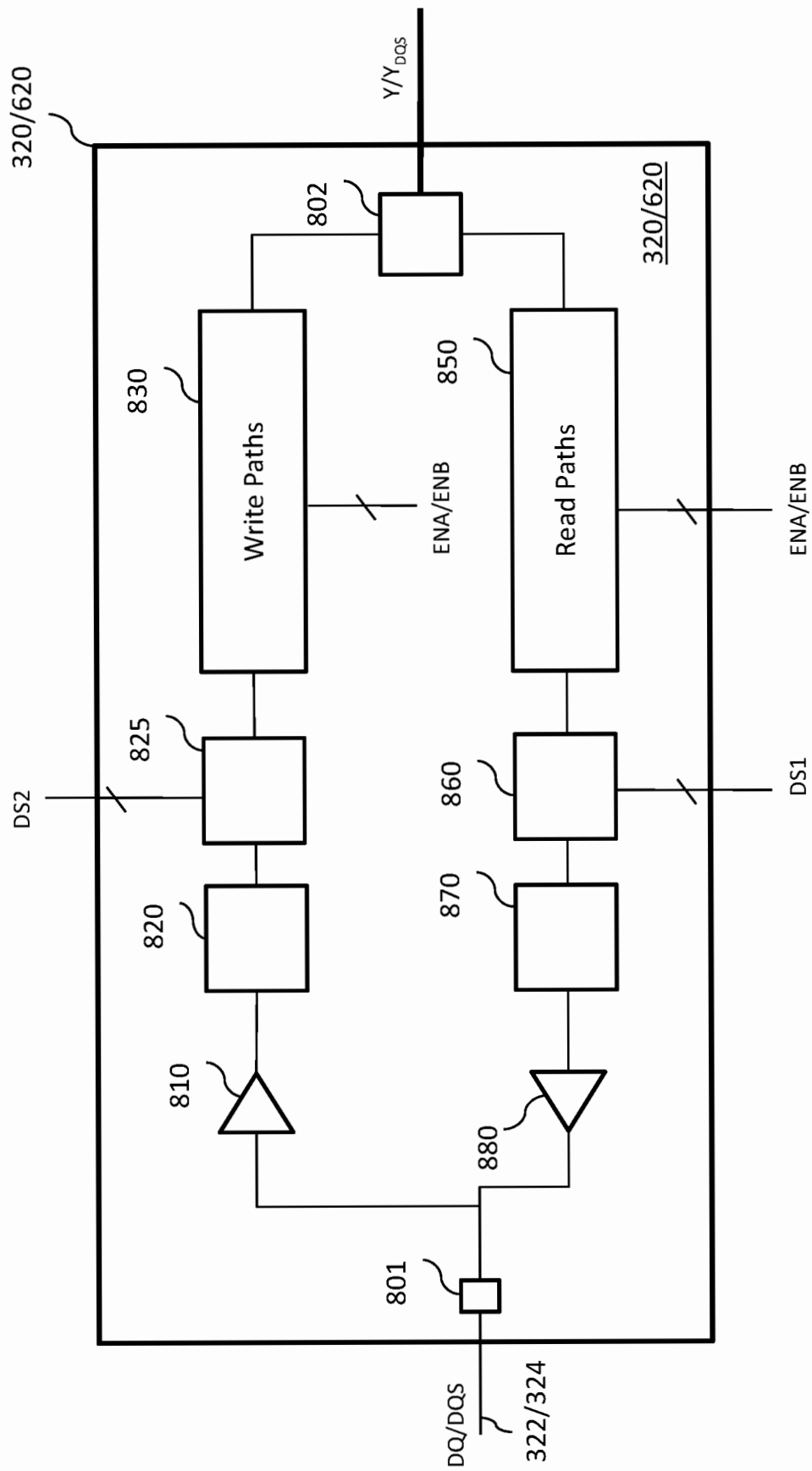


FIG. 8

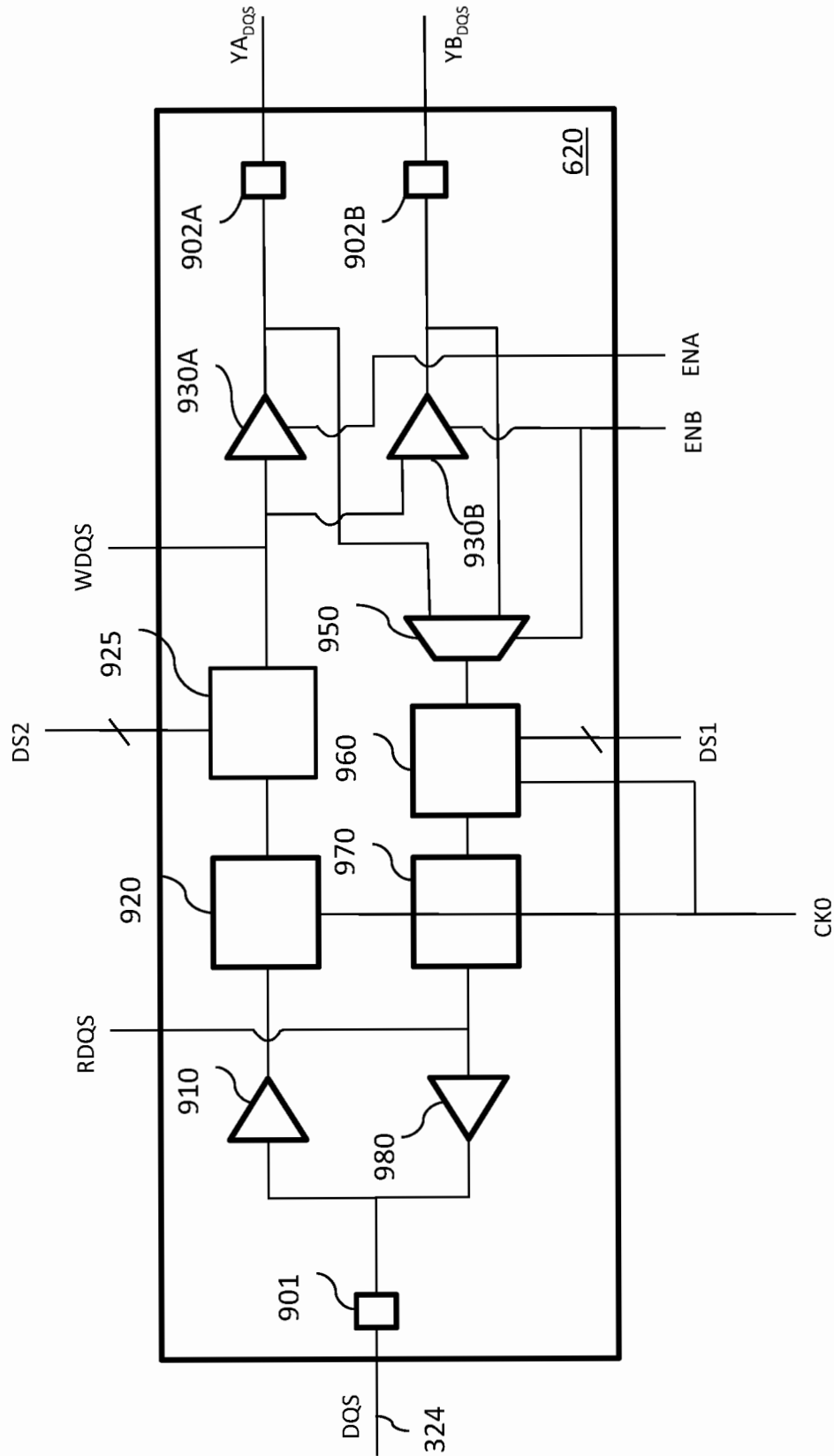


FIG. 9

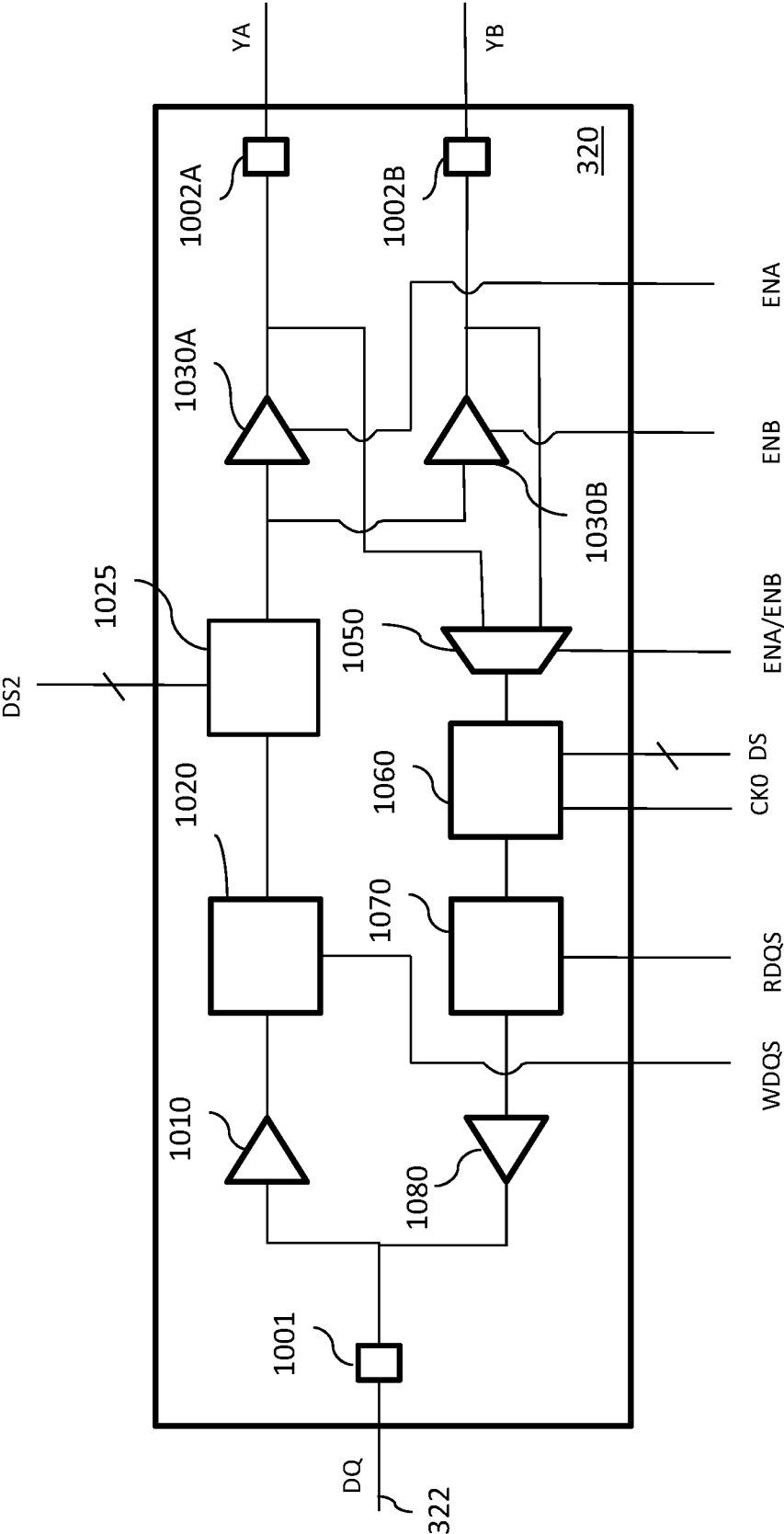


FIG. 10

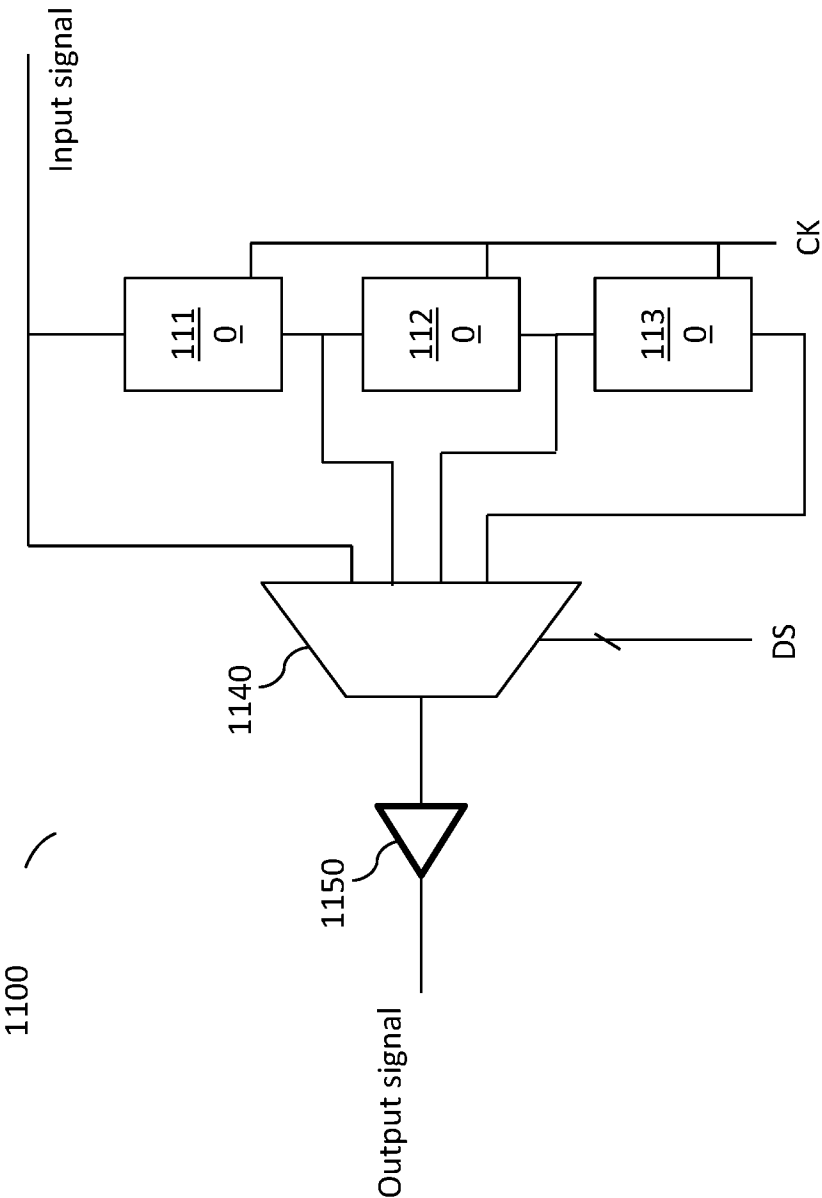


FIG. 11

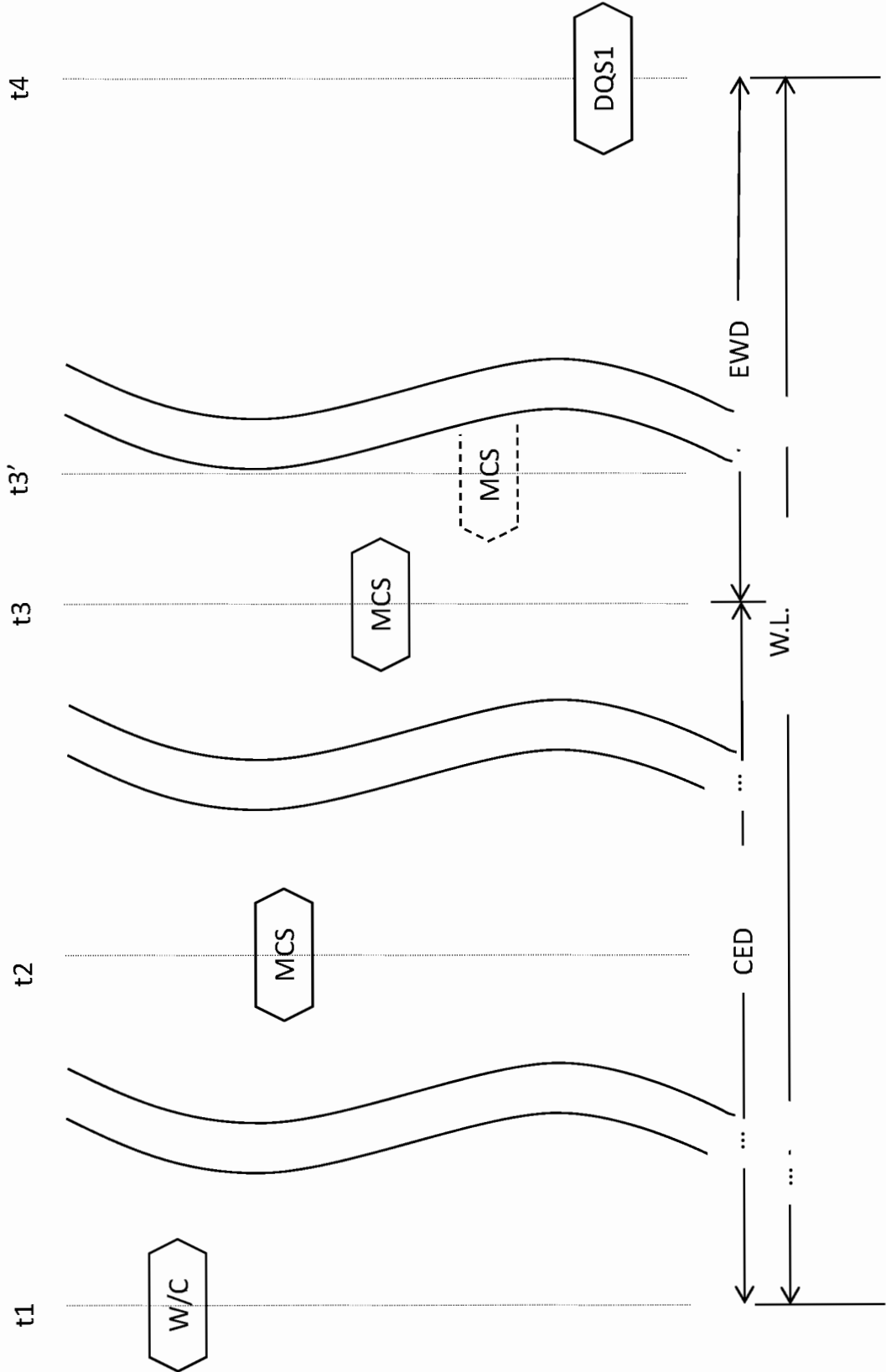


FIG. 12A

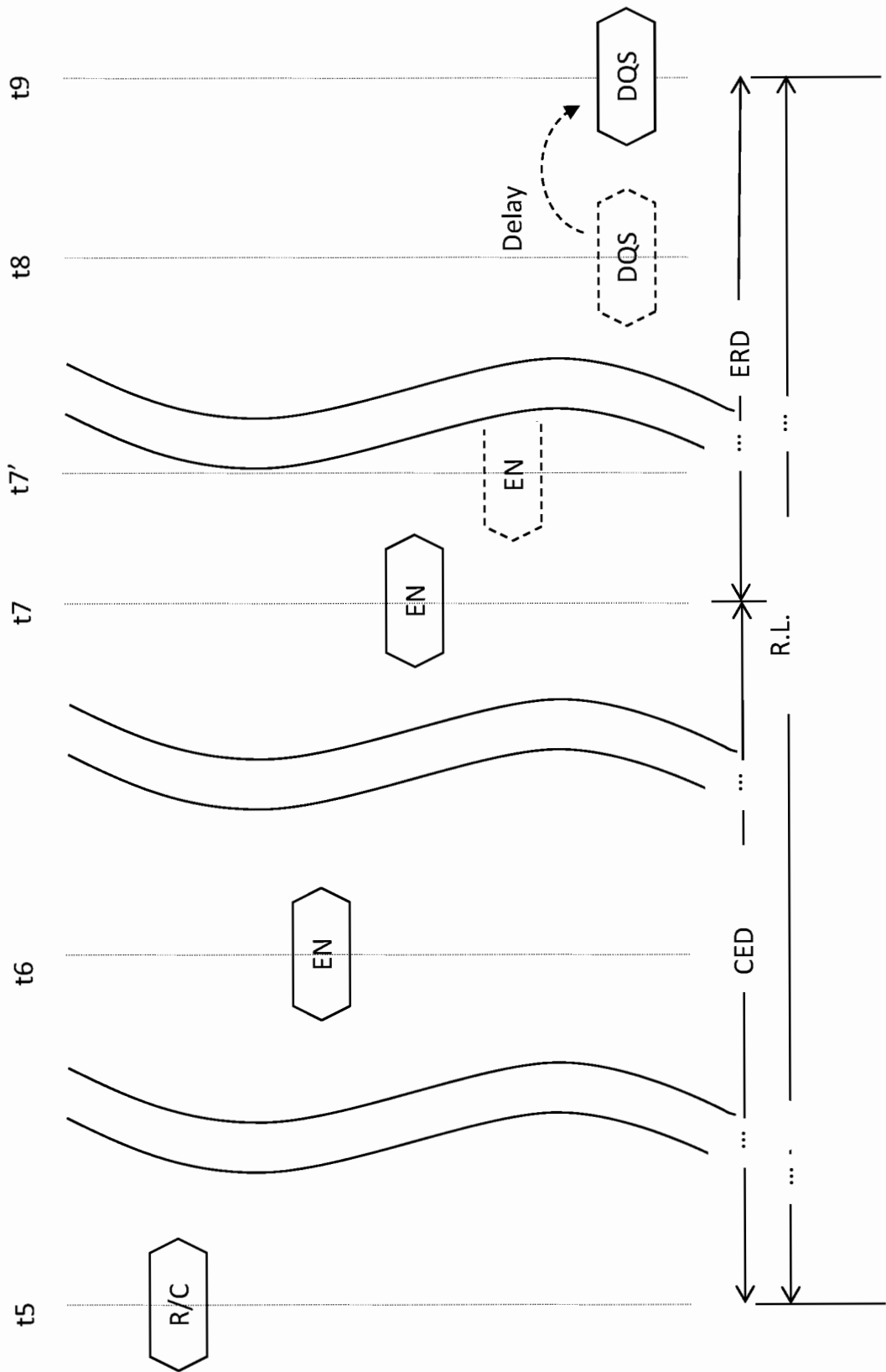


FIG. 12B

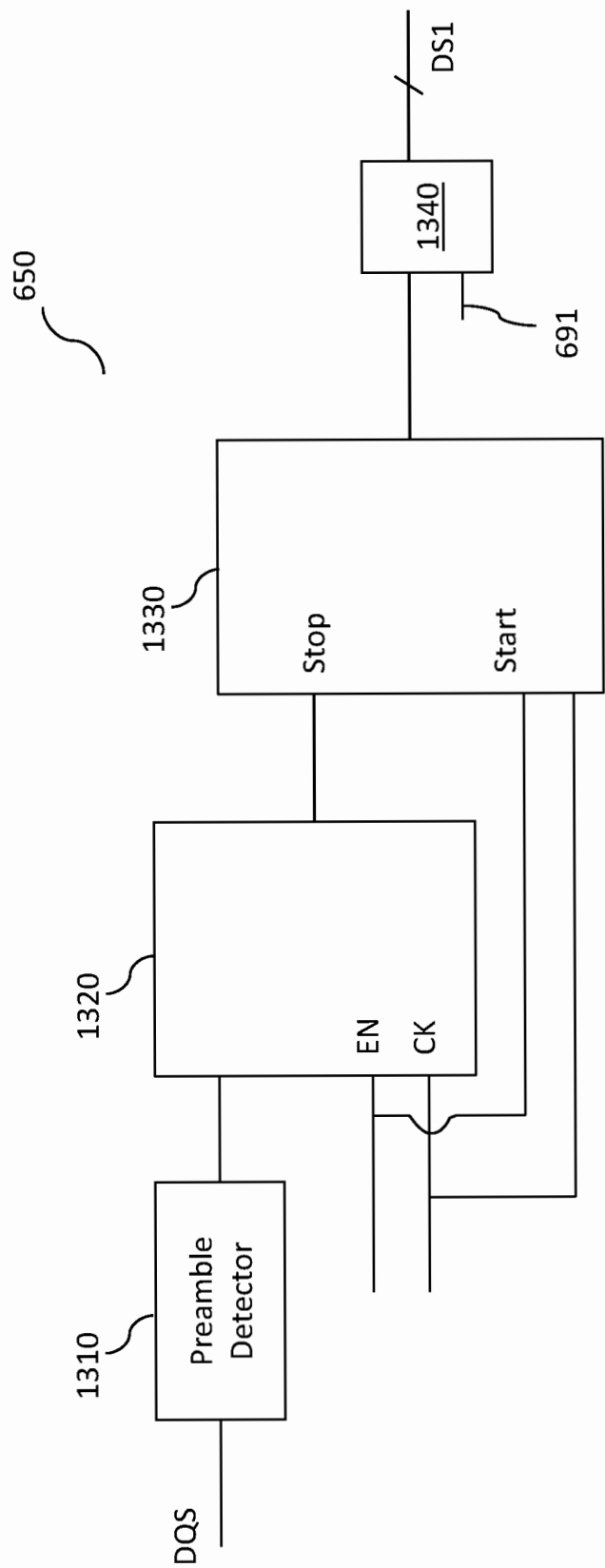
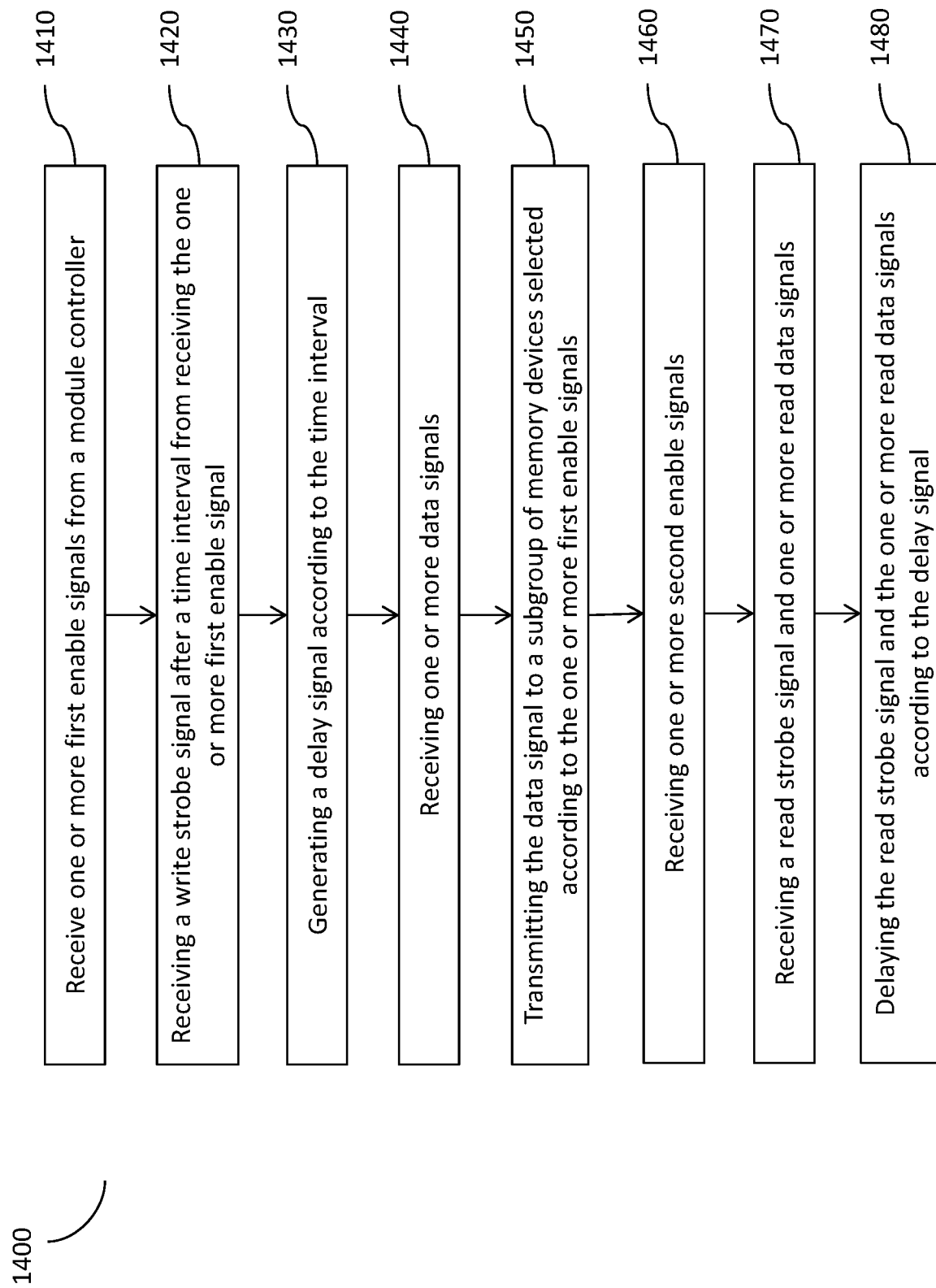


FIG. 13

**FIG. 14**

Electronic Acknowledgement Receipt

EFS ID:	41554741
Application Number:	17141978
International Application Number:	
Confirmation Number:	4088
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	129980-5073-US02
Receipt Date:	05-JAN-2021
Filing Date:	
Time Stamp:	17:50:25
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal of New Application	129980-5073-US02_TRANSMITTAL.pdf	147400	no	1
			8553687f4b07ed49f96a1b0f532238be5380e5d4		

Warnings:

Information:					
2		129980-5073-US02_STMT_POA.pdf	417954	yes	2
			ff6ada23ddcc52997e3c940d8af0b473c29f3ad3		
	Multipart Description/PDF files in .zip description				
	Document Description	Start	End		
	Assignee showing of ownership per 37 CFR 3.73	1	1		
	Power of Attorney	2	2		
Warnings:					
Information:					
3	Oath or Declaration filed	129980-5073-US02_DECL.pdf	74619	no	2
			f8cd2e30751d722412ad81f539bf1727130ca386		
Warnings:					
Information:					
4	Application Data Sheet	129980-5073-US02_ADS.pdf	1256390	no	9
			ae919278be61bae5b5a4ccd150c9245c6861a425		
Warnings:					
Information:					
5		129980-5073-US02_APPLN.pdf	263151	yes	30
			ff7ddcdffc66a50fa79e0b1bf3feef7263d6e644		
	Multipart Description/PDF files in .zip description				
	Document Description	Start	End		
	Specification	1	28		
	Claims	29	29		
	Abstract	30	30		
Warnings:					
Information:					
6	Drawings-only black and white line drawings	129980-5073-US02_DRAWINGS.pdf	1397603	no	22
			38beddb7368f01becd3d2e76d9701dae90042f3c		

Warnings:	
Information:	
Total Files Size (in bytes):	3557117
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.	

SCORE Placeholder Sheet for IFW Content

Application Number: 17141978

Document Date: 01/05/2021

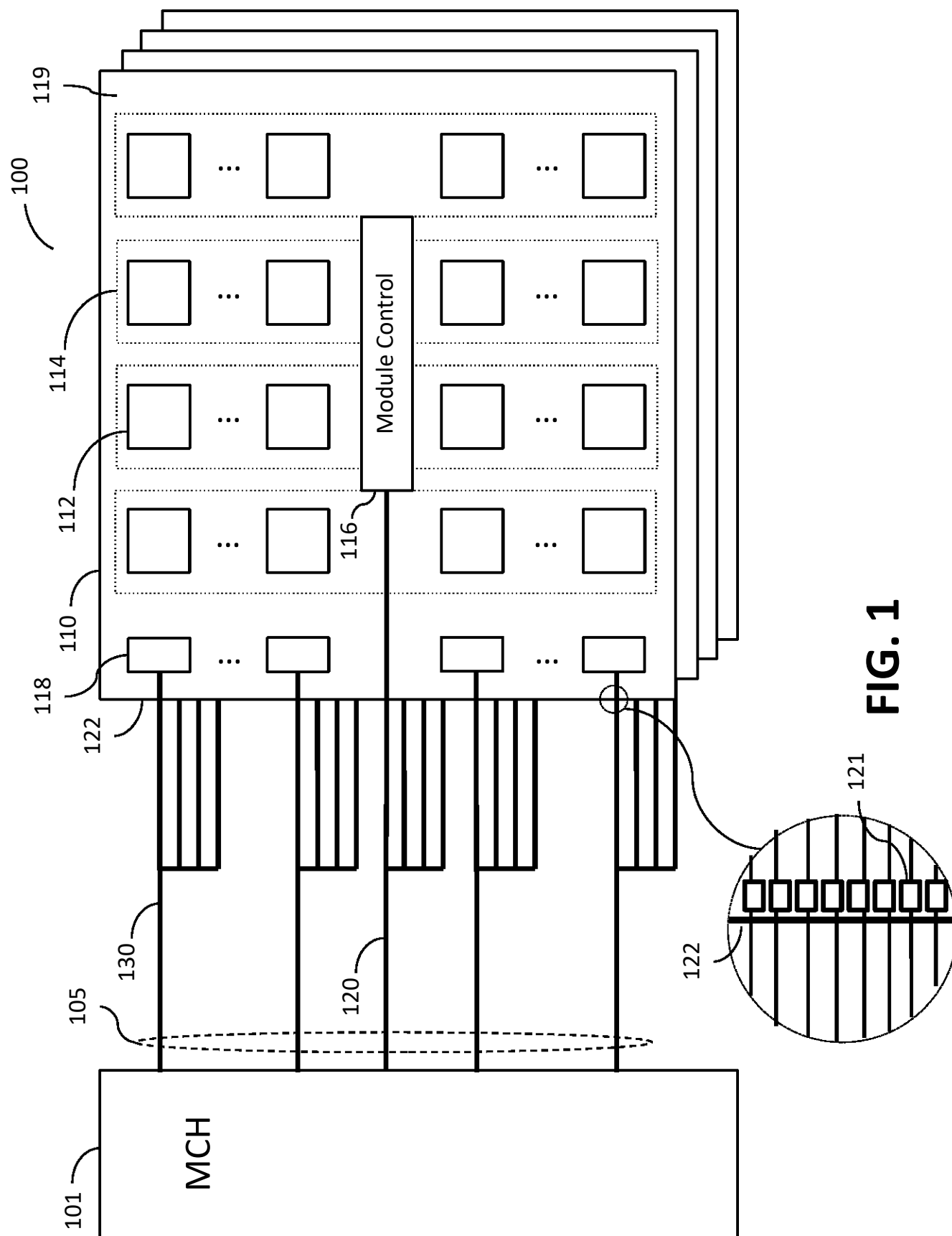
The presence of this form in the IFW record indicates that the following document type was received in electronic format on the date identified above. This content is stored in the SCORE database.

Since this was an electronic submission, there is no physical artifact folder, no artifact folder is recorded in PALM, and no paper documents or physical media exist. The TIFF images in the IFW record were created from the original documents that are stored in SCORE.

- Drawing

At the time of document entry (noted above):

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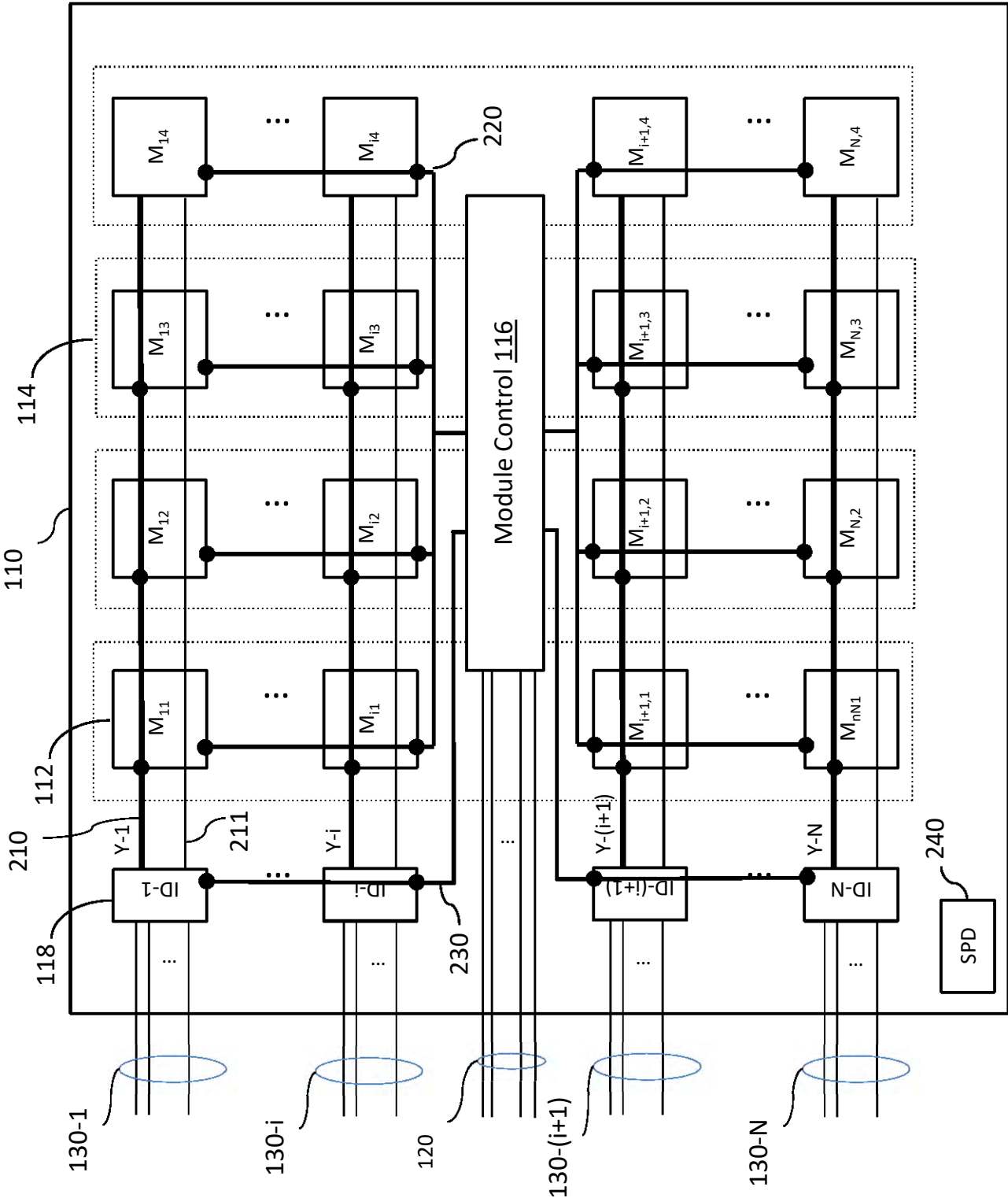


FIG. 2A

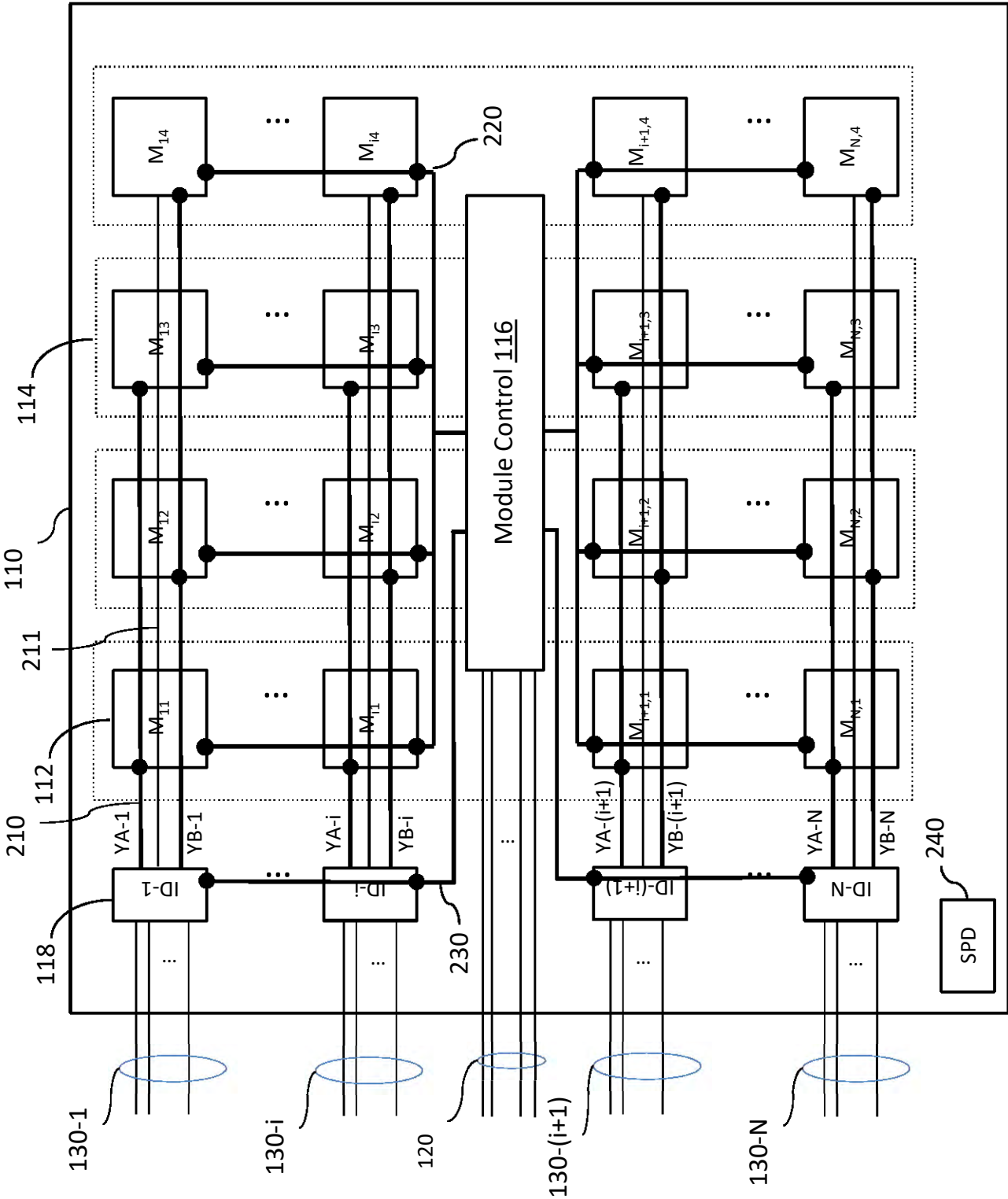
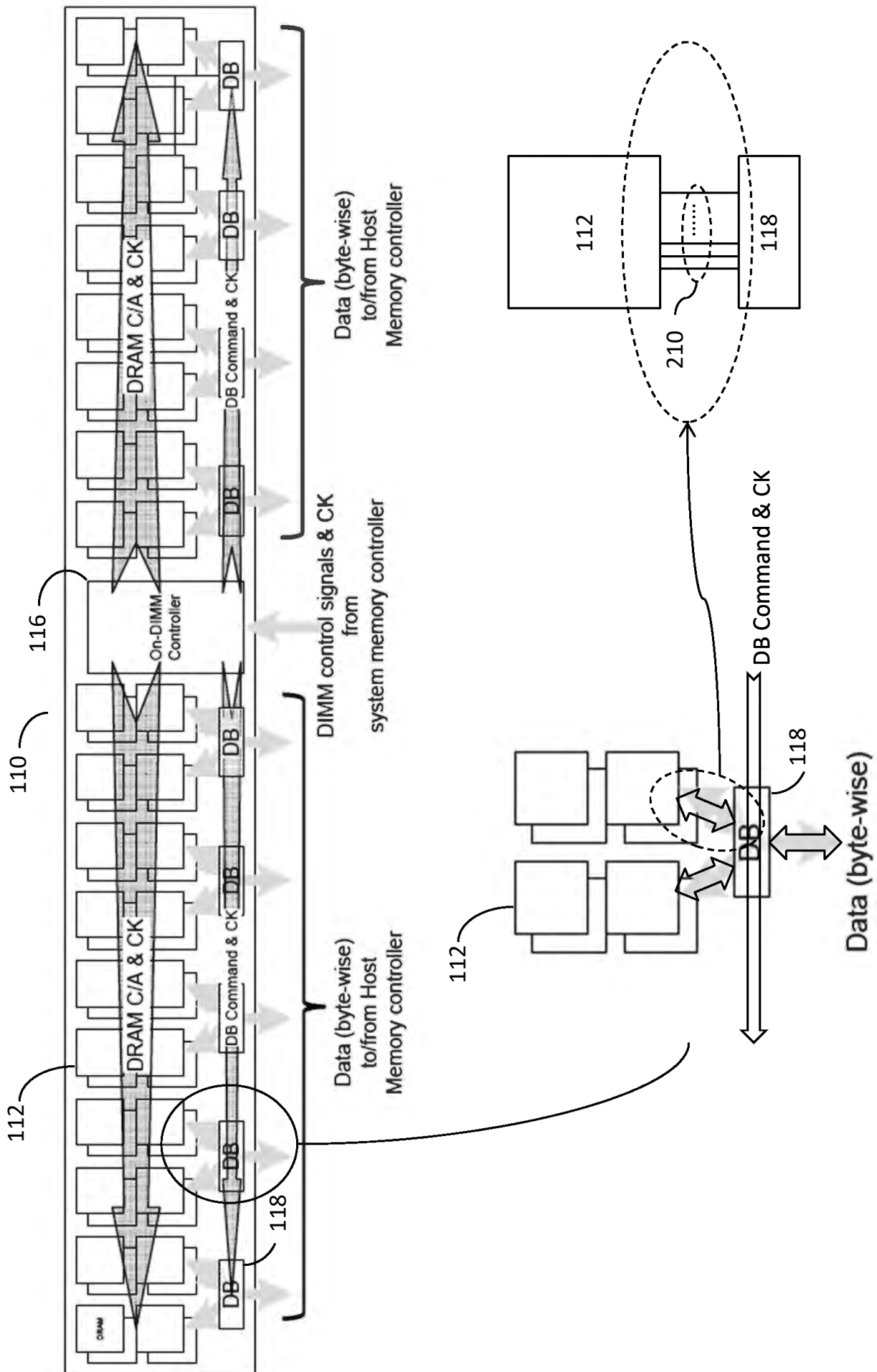


FIG. 2B

**FIG. 2C**

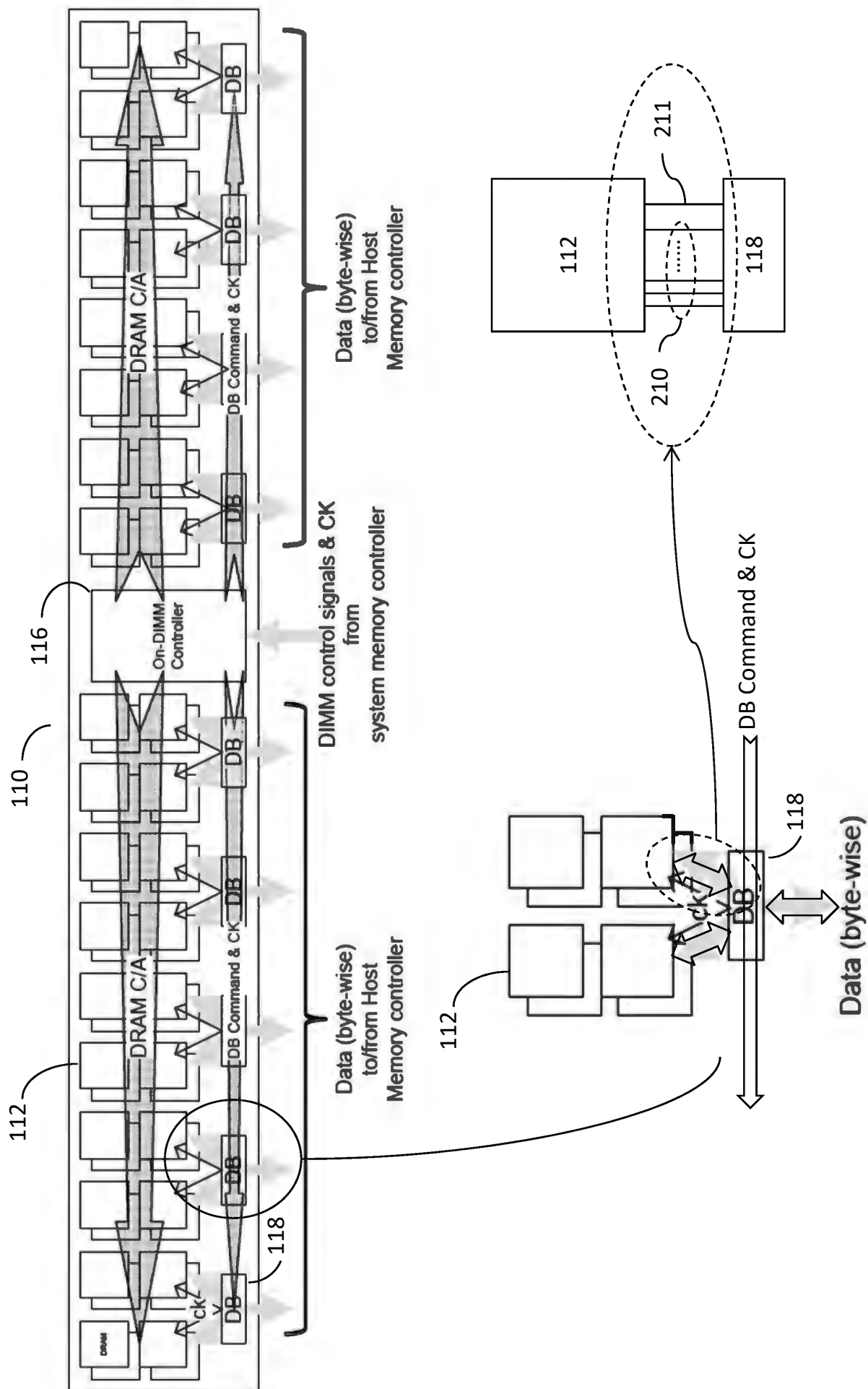


FIG. 2D

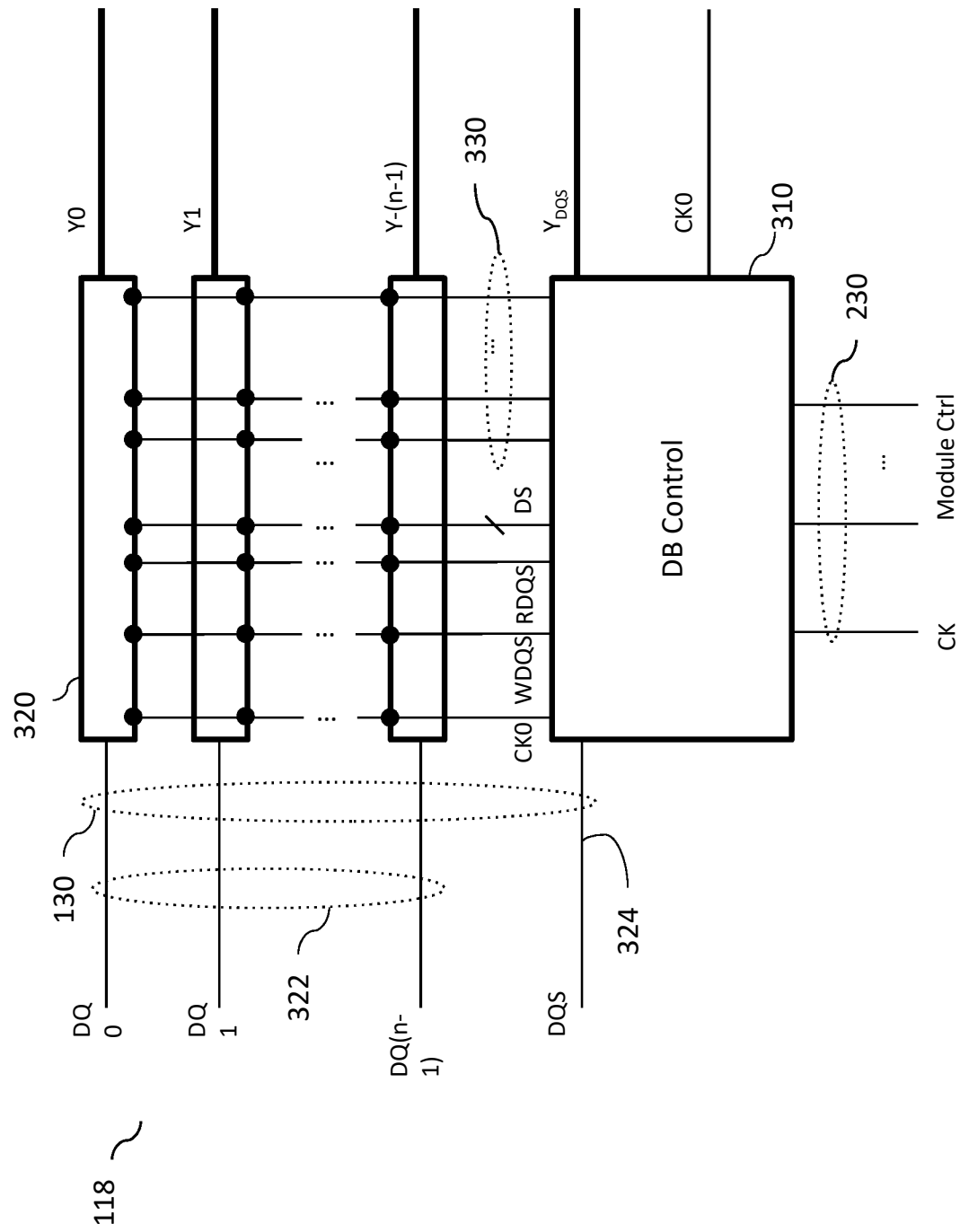
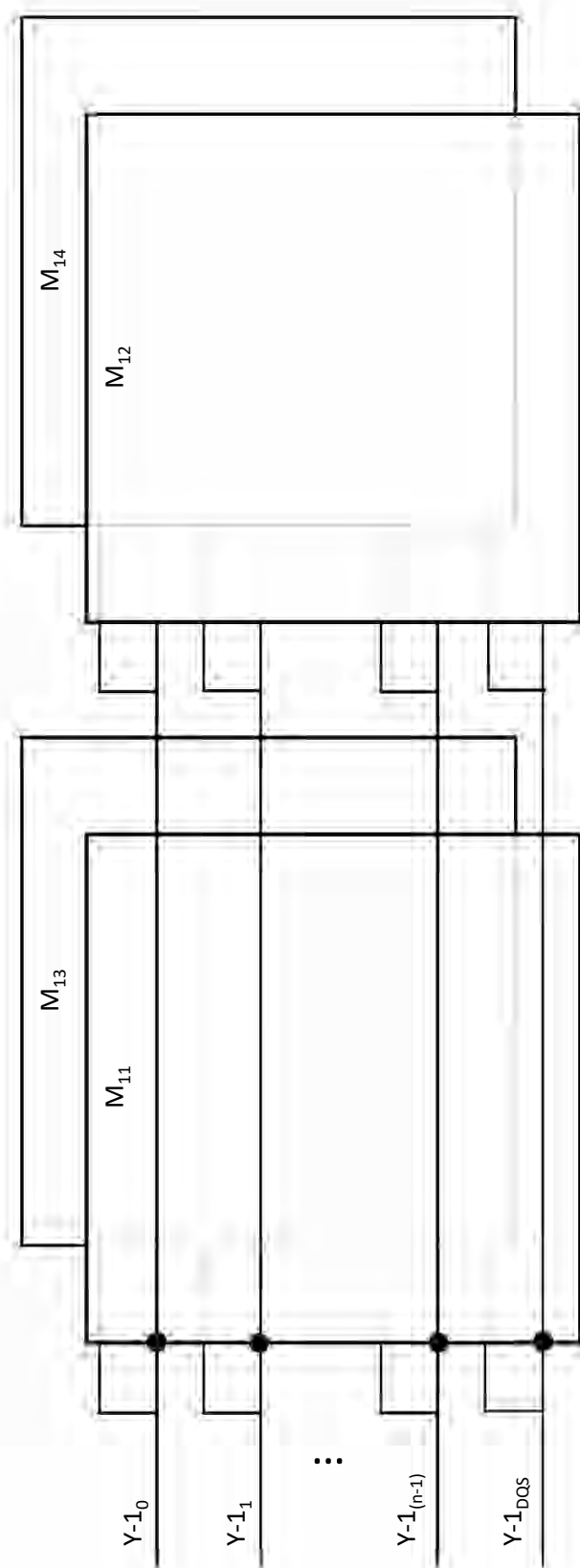


FIG. 3

**FIG. 4A**

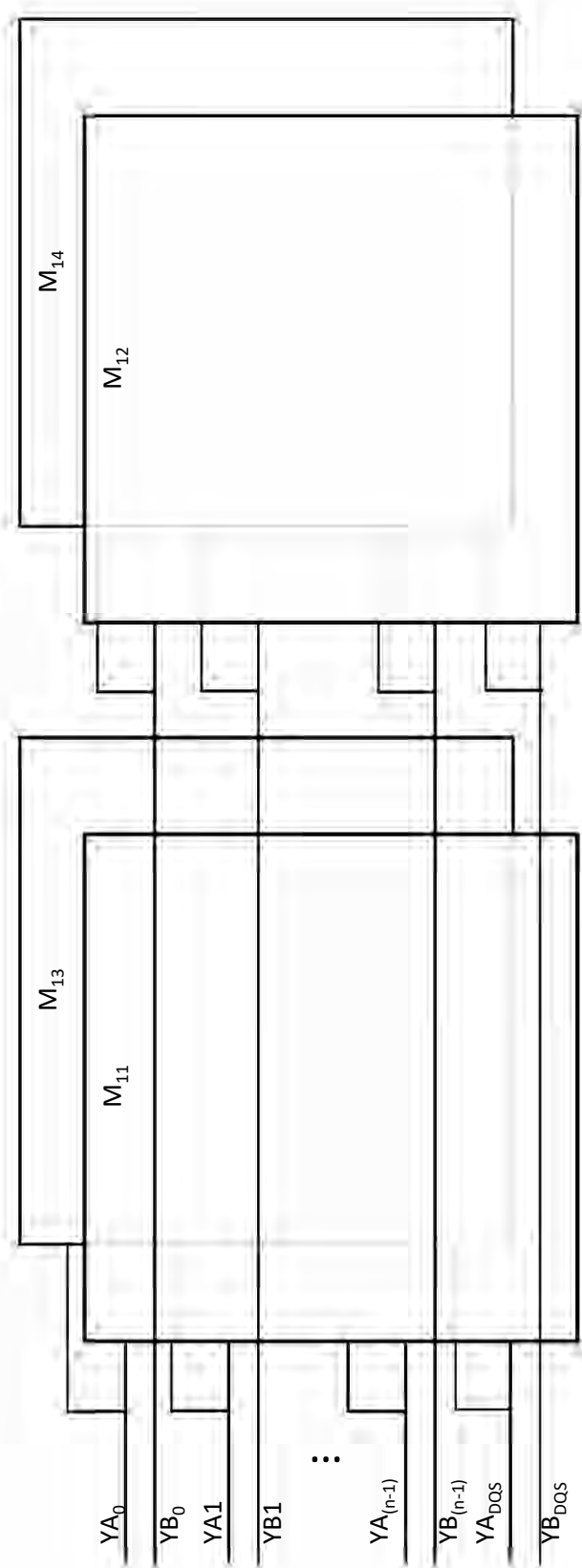


FIG. 4B

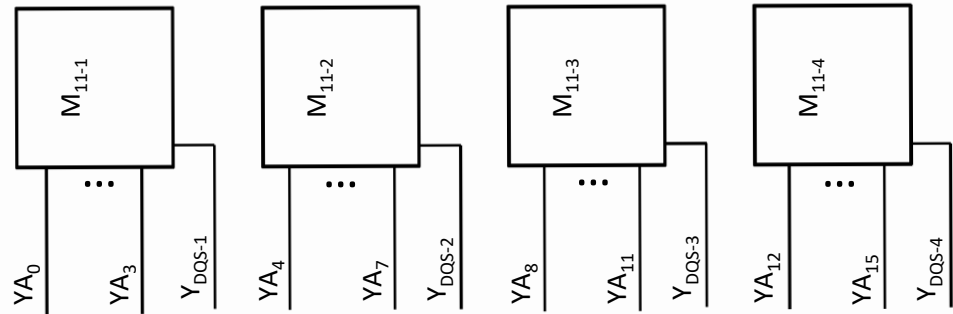


FIG. 5B

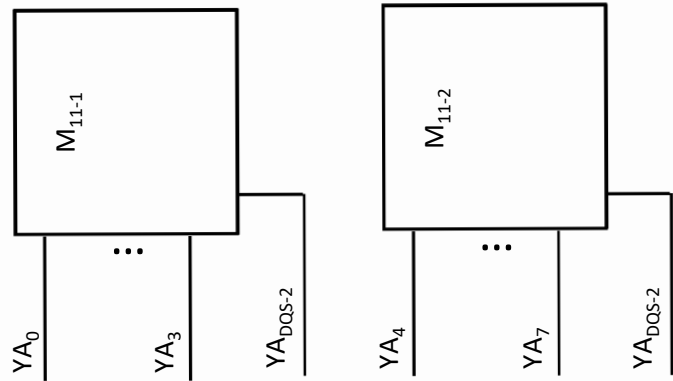
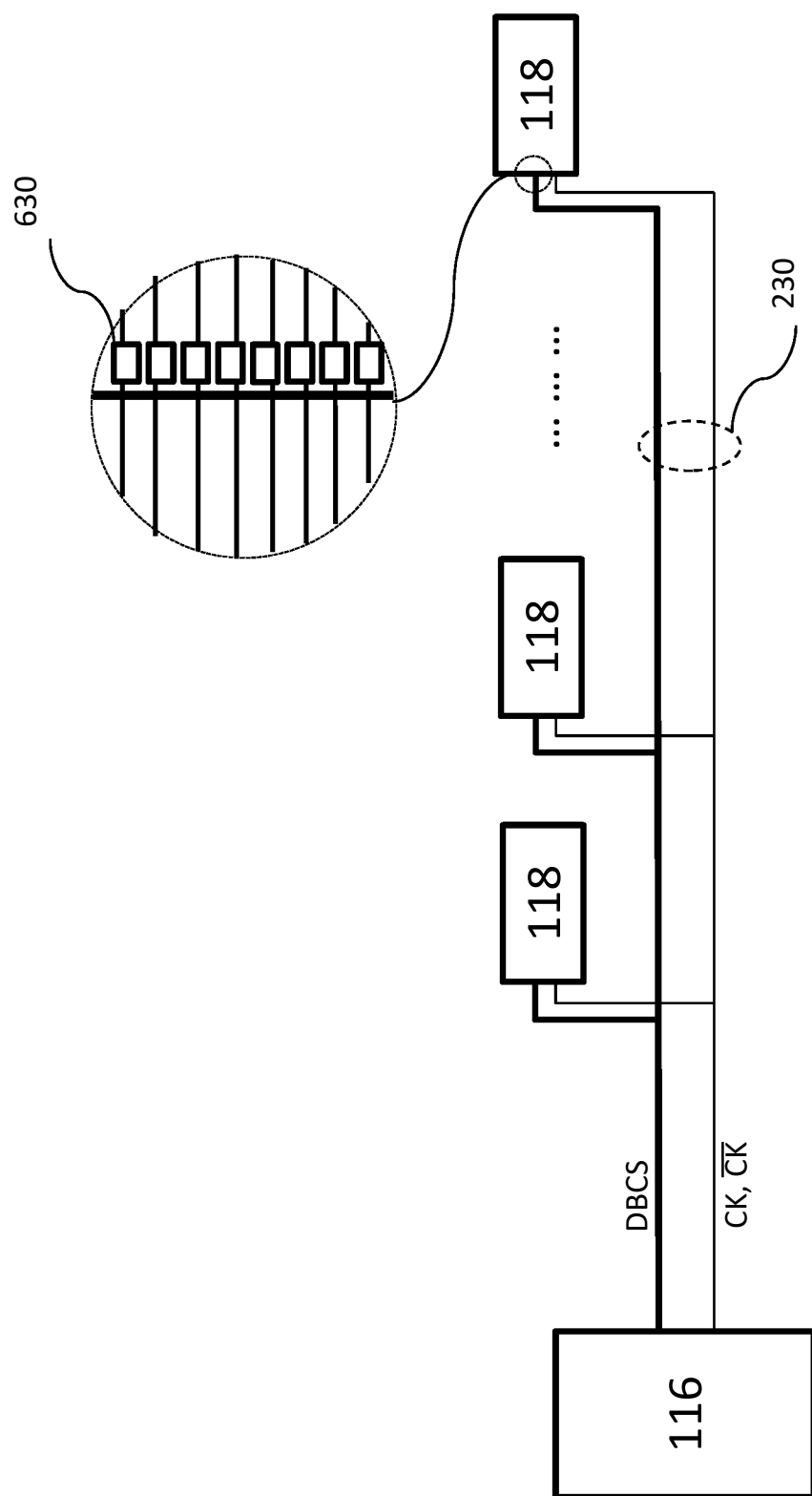


FIG. 5A

**FIG. 6A**

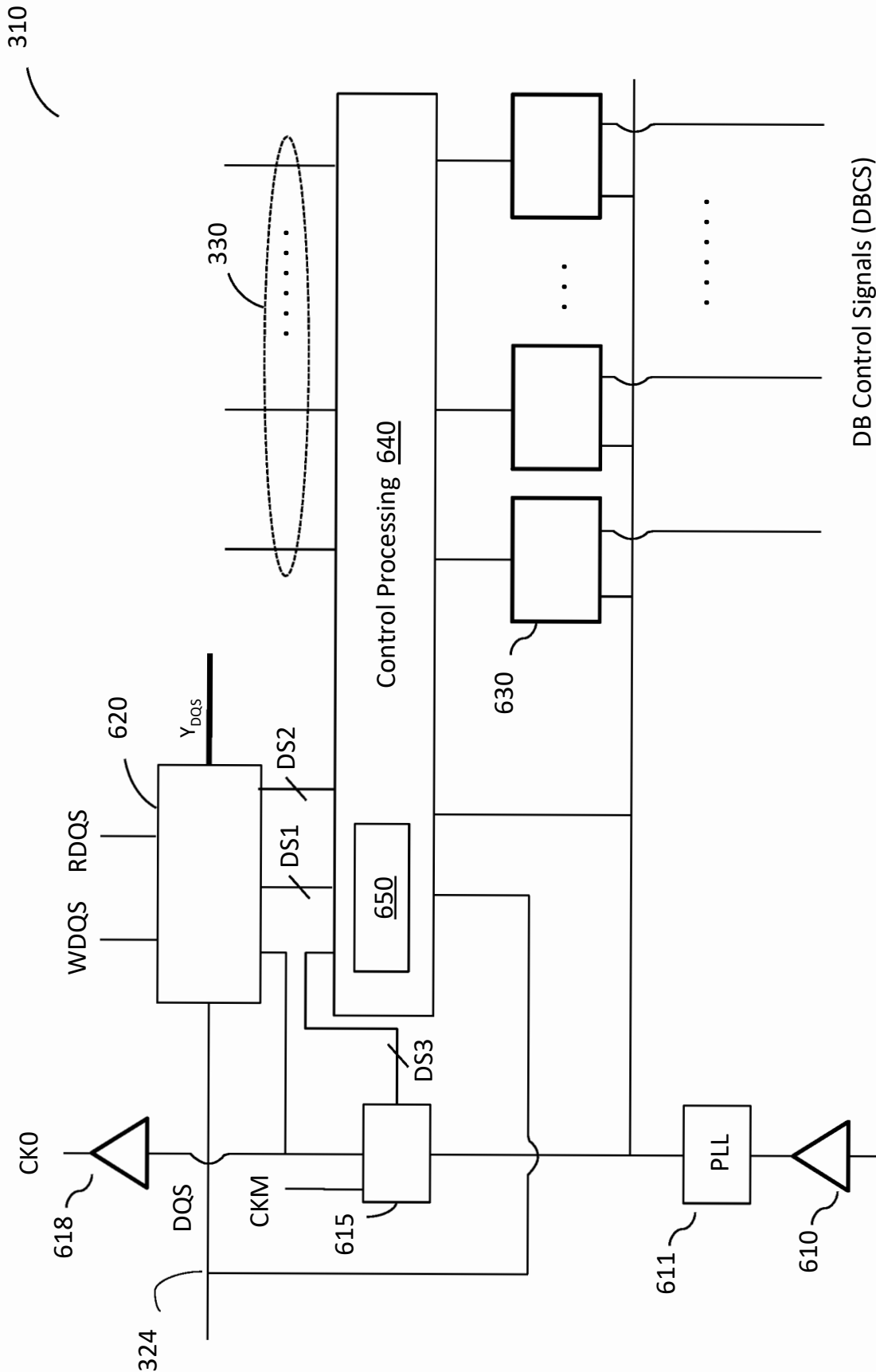


FIG. 6B

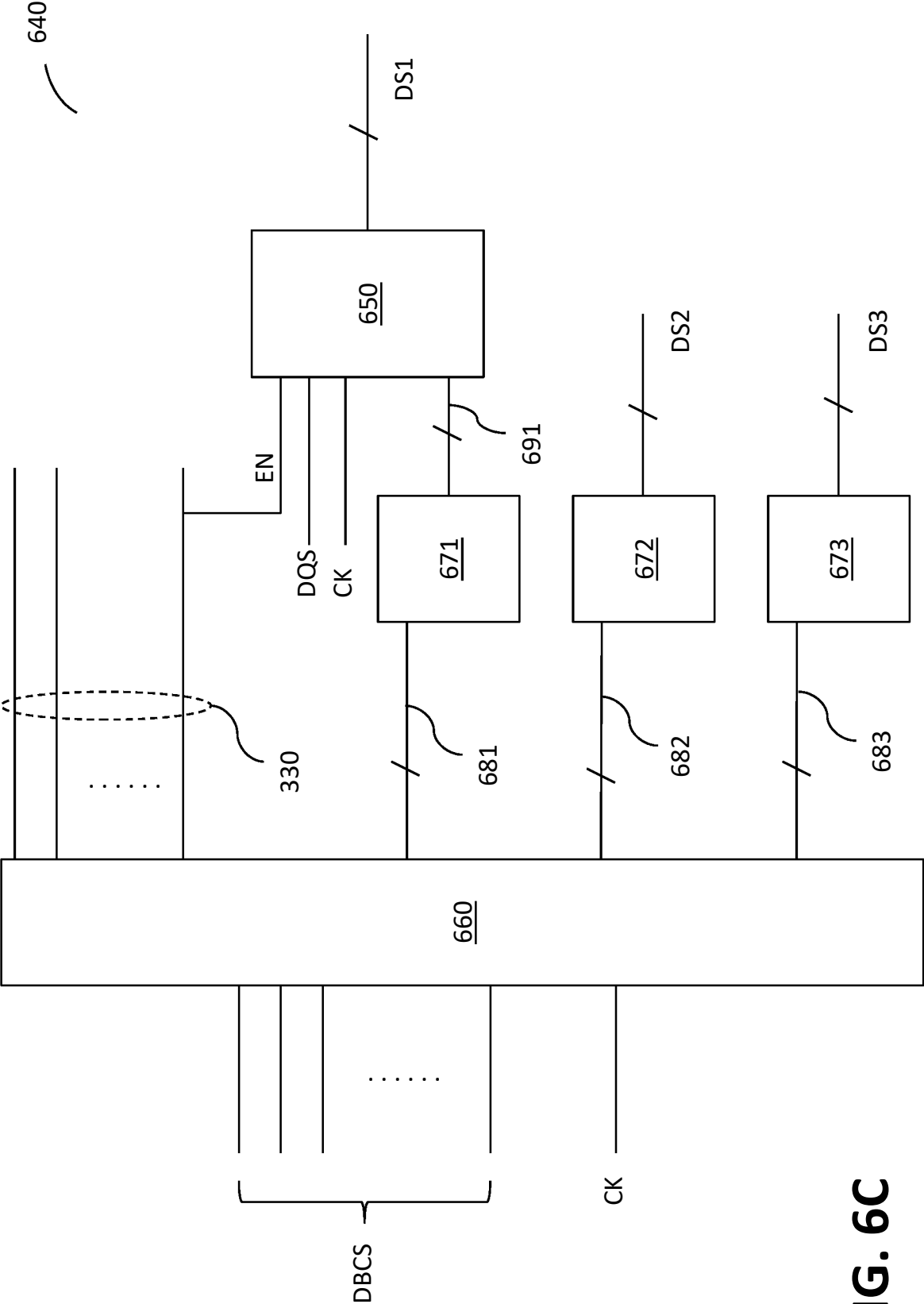
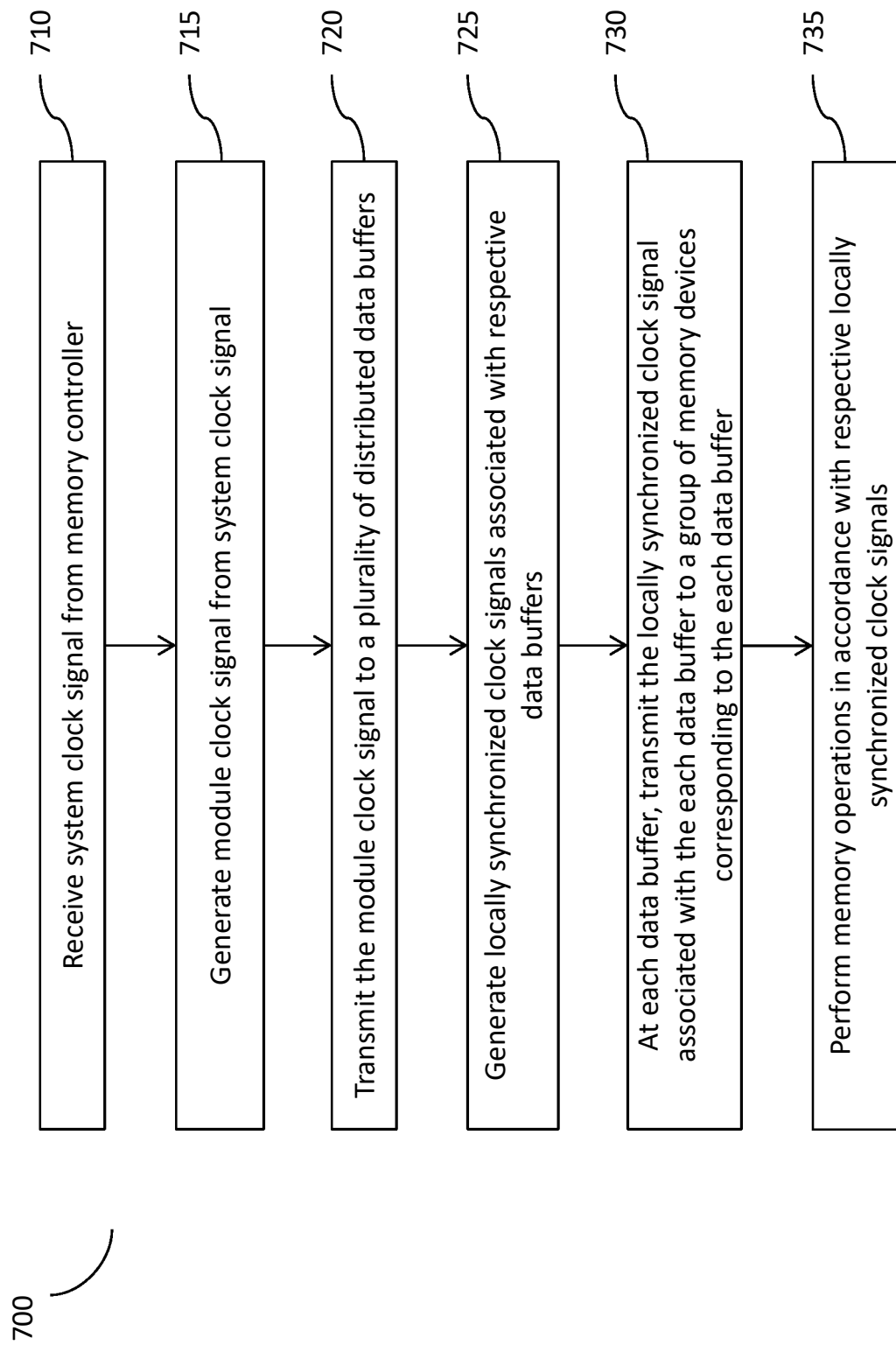
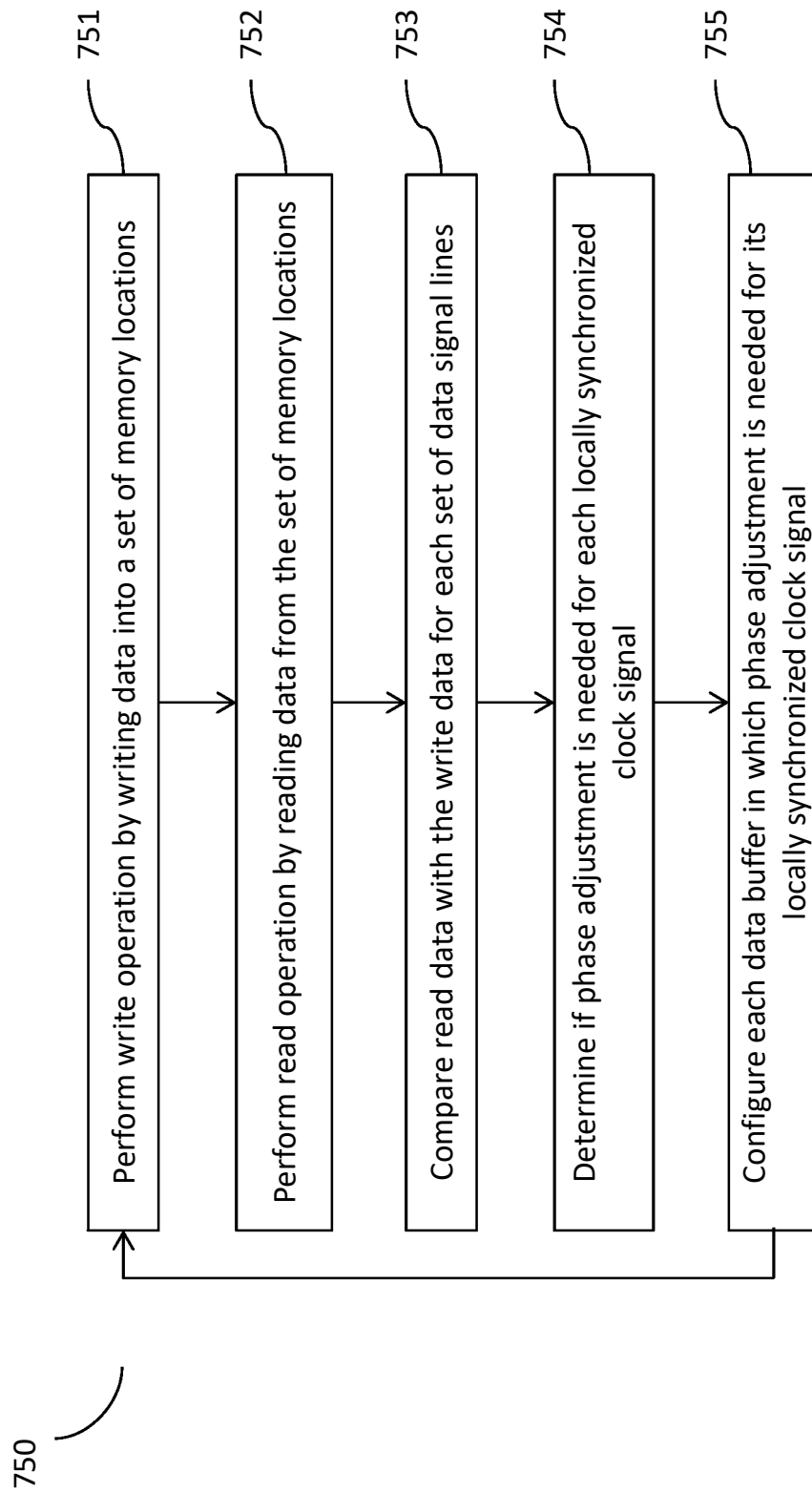


FIG. 6C

**FIG. 7A**

**FIG. 7B**

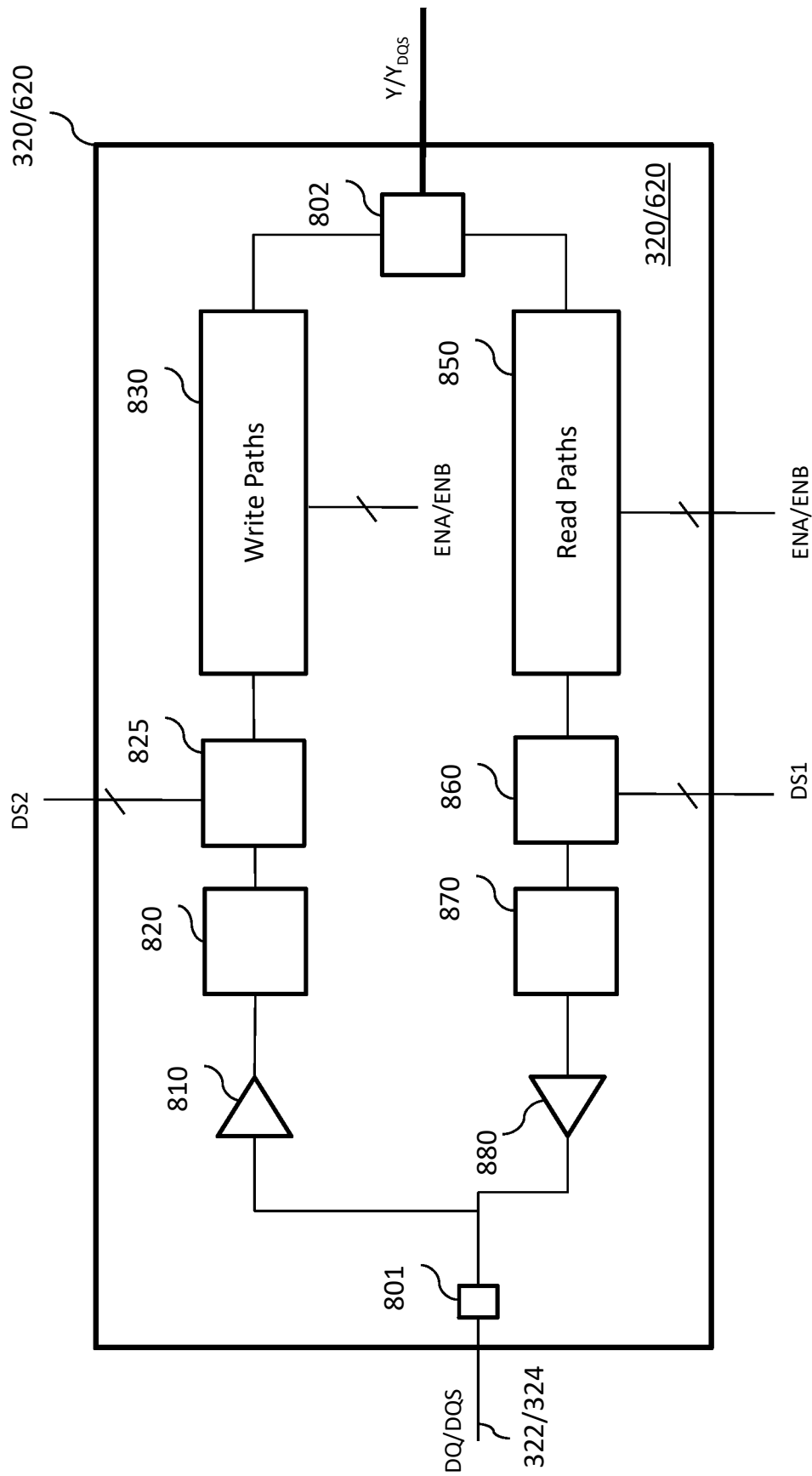
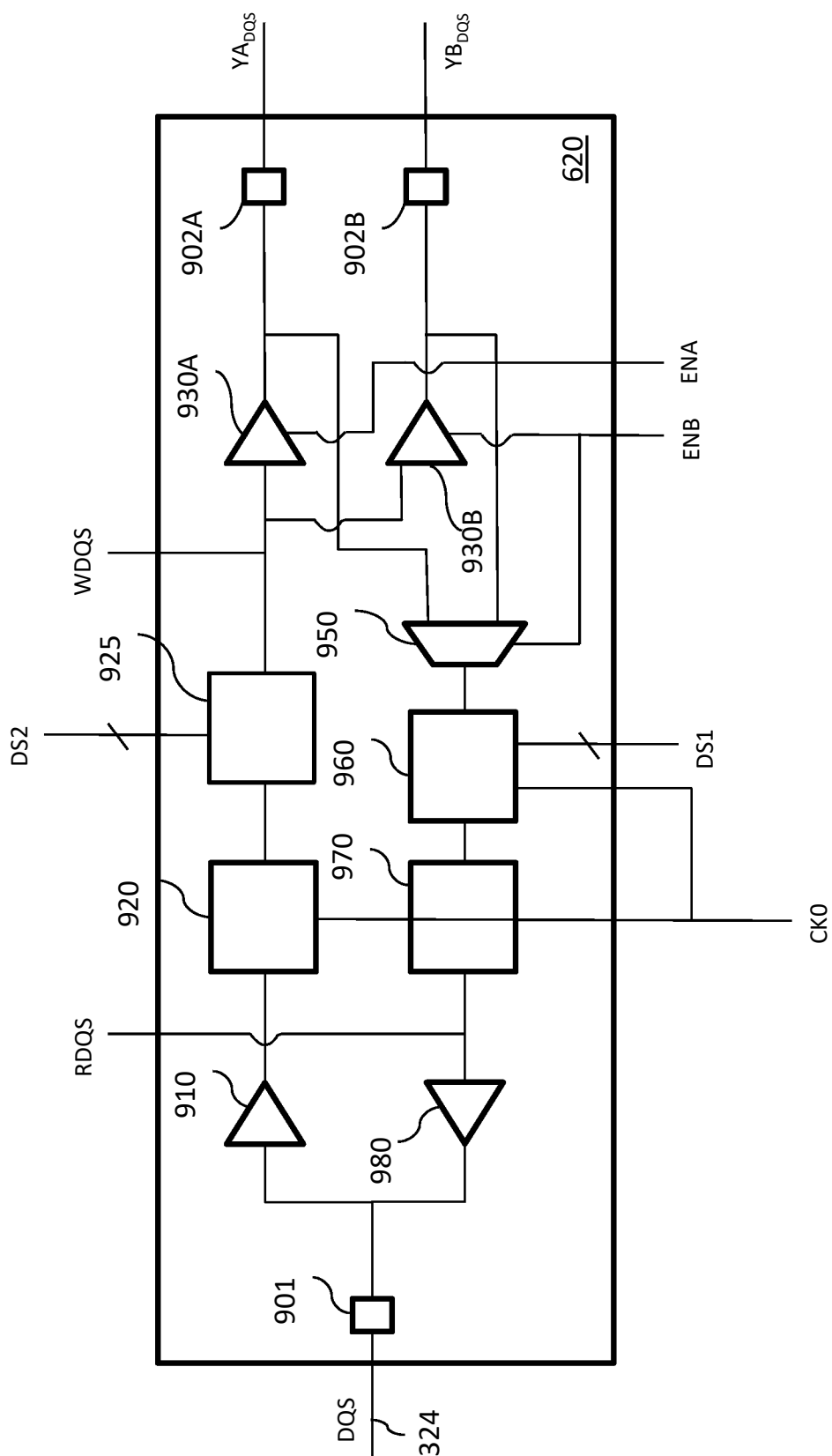
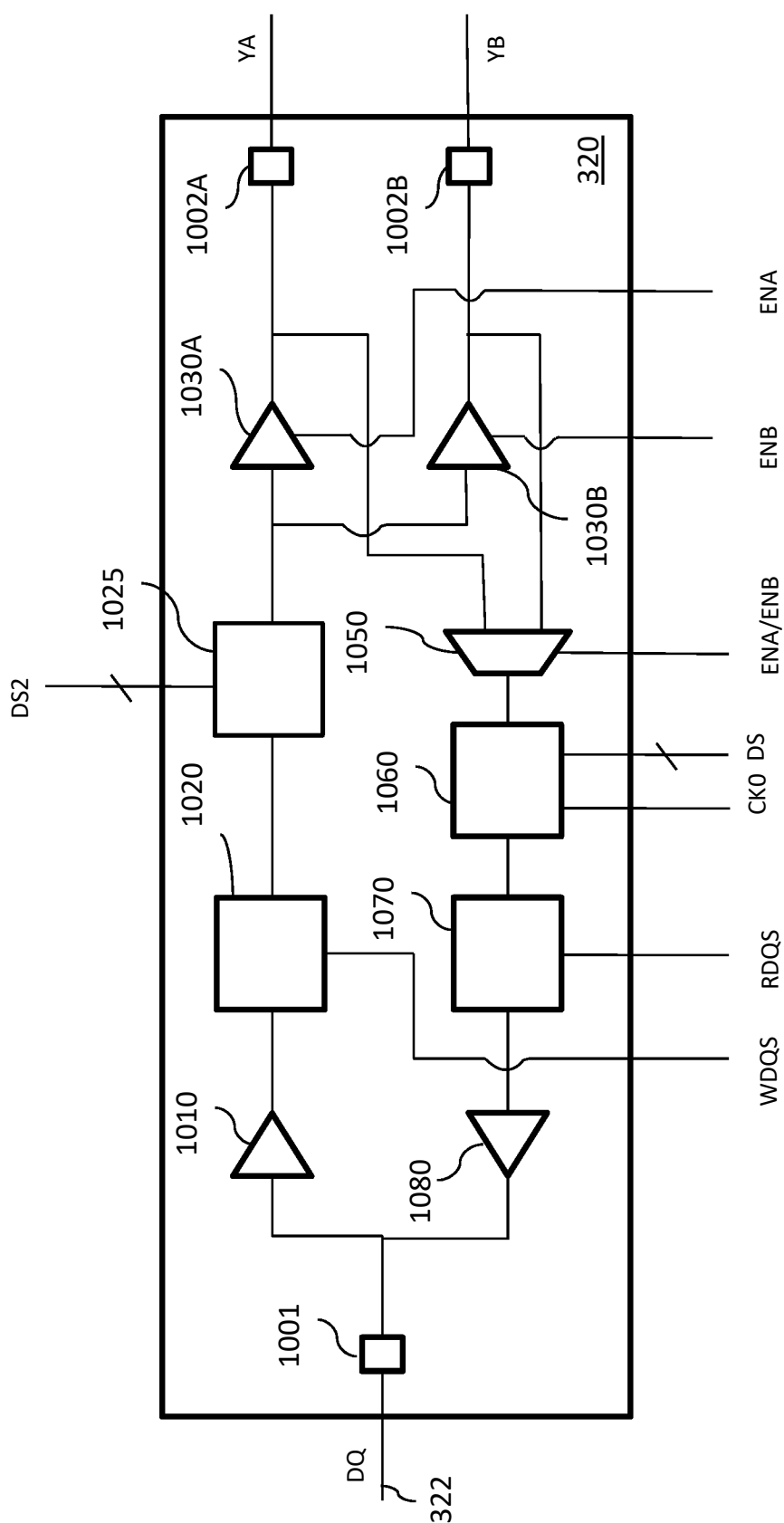
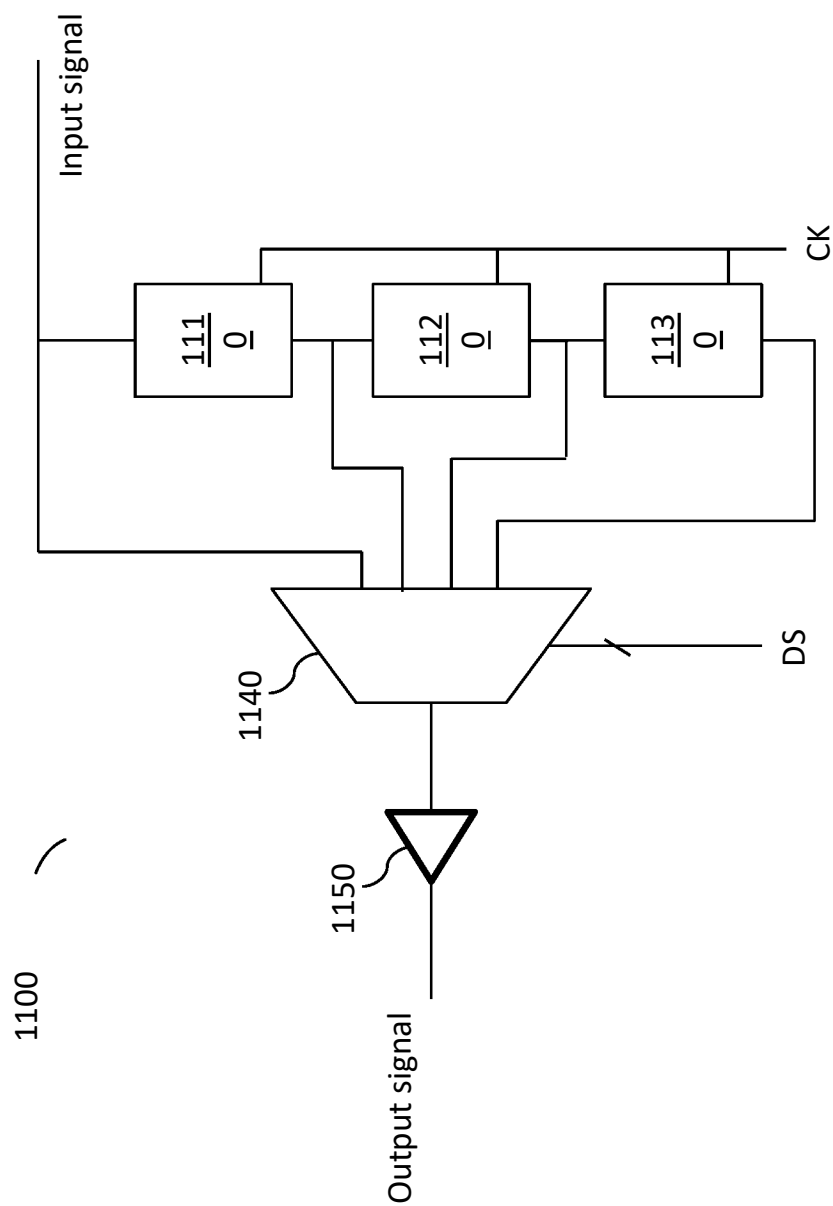


FIG. 8

**FIG. 9**

**FIG. 10**

**FIG. 11**

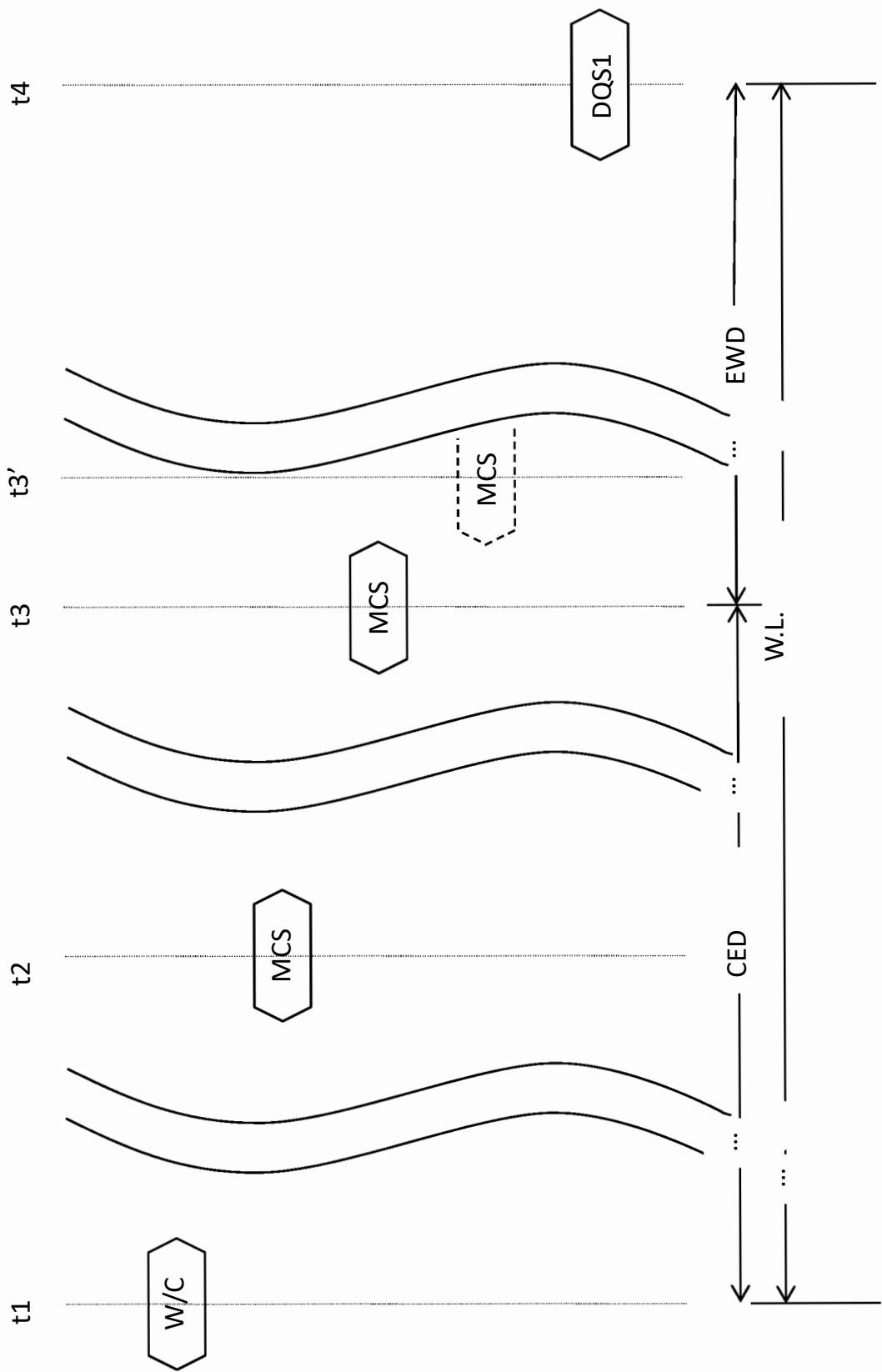


FIG. 12A

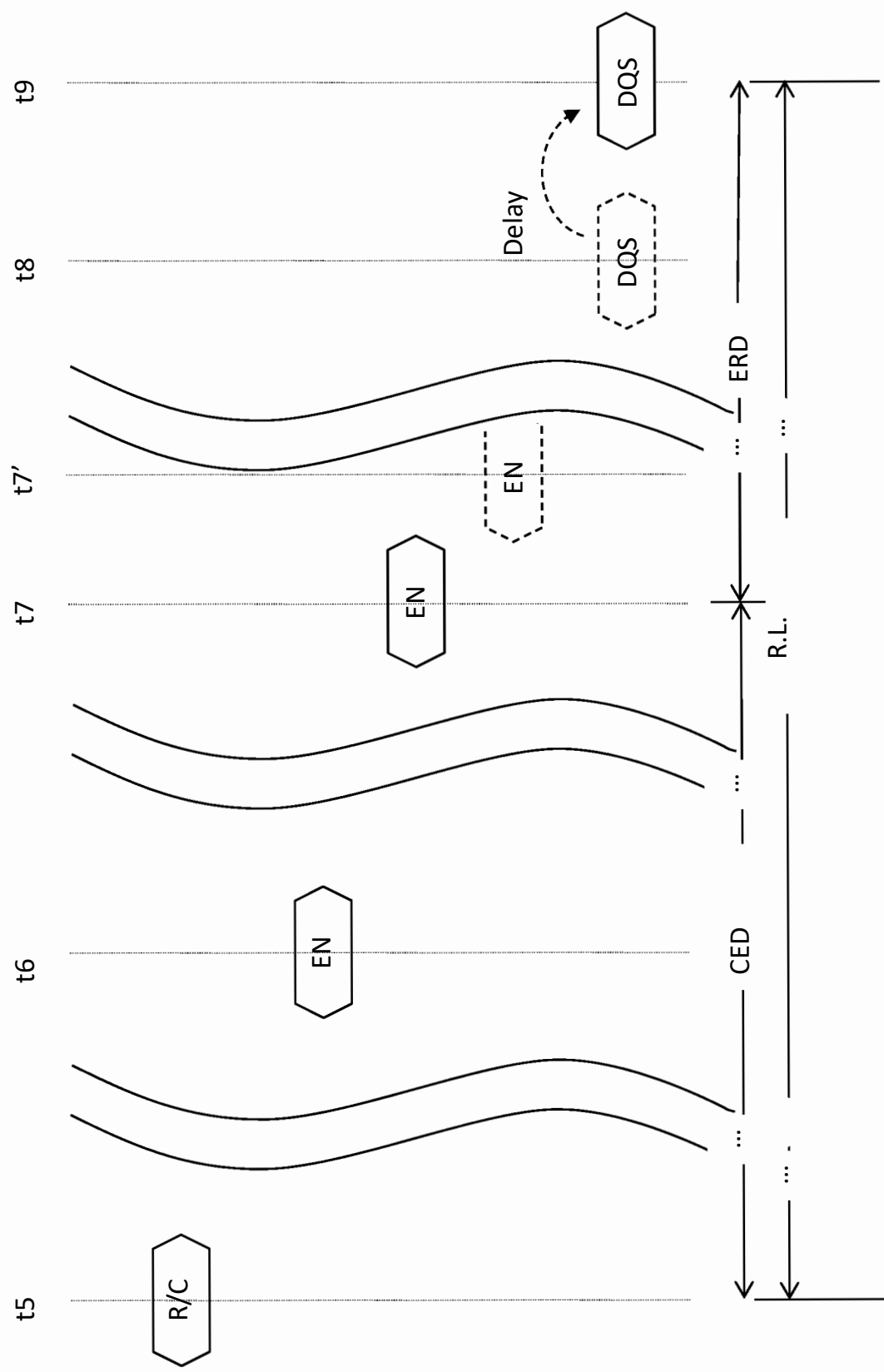


FIG. 12B

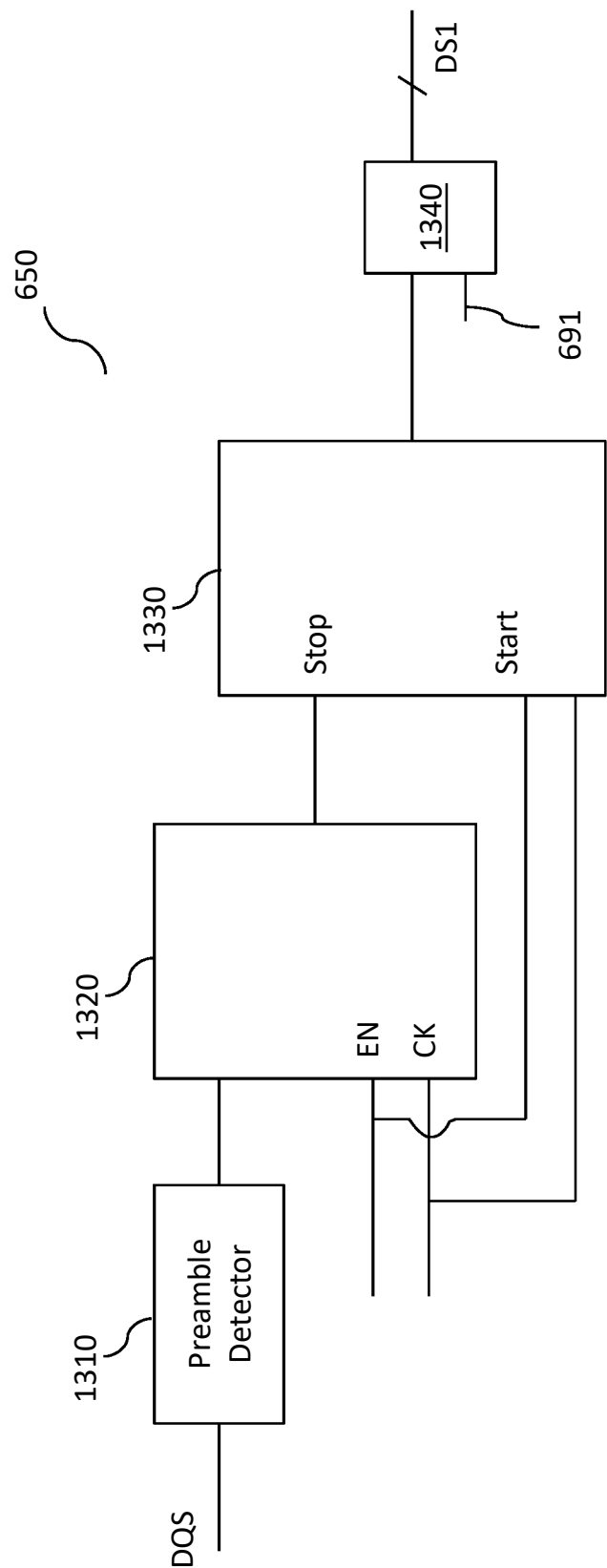
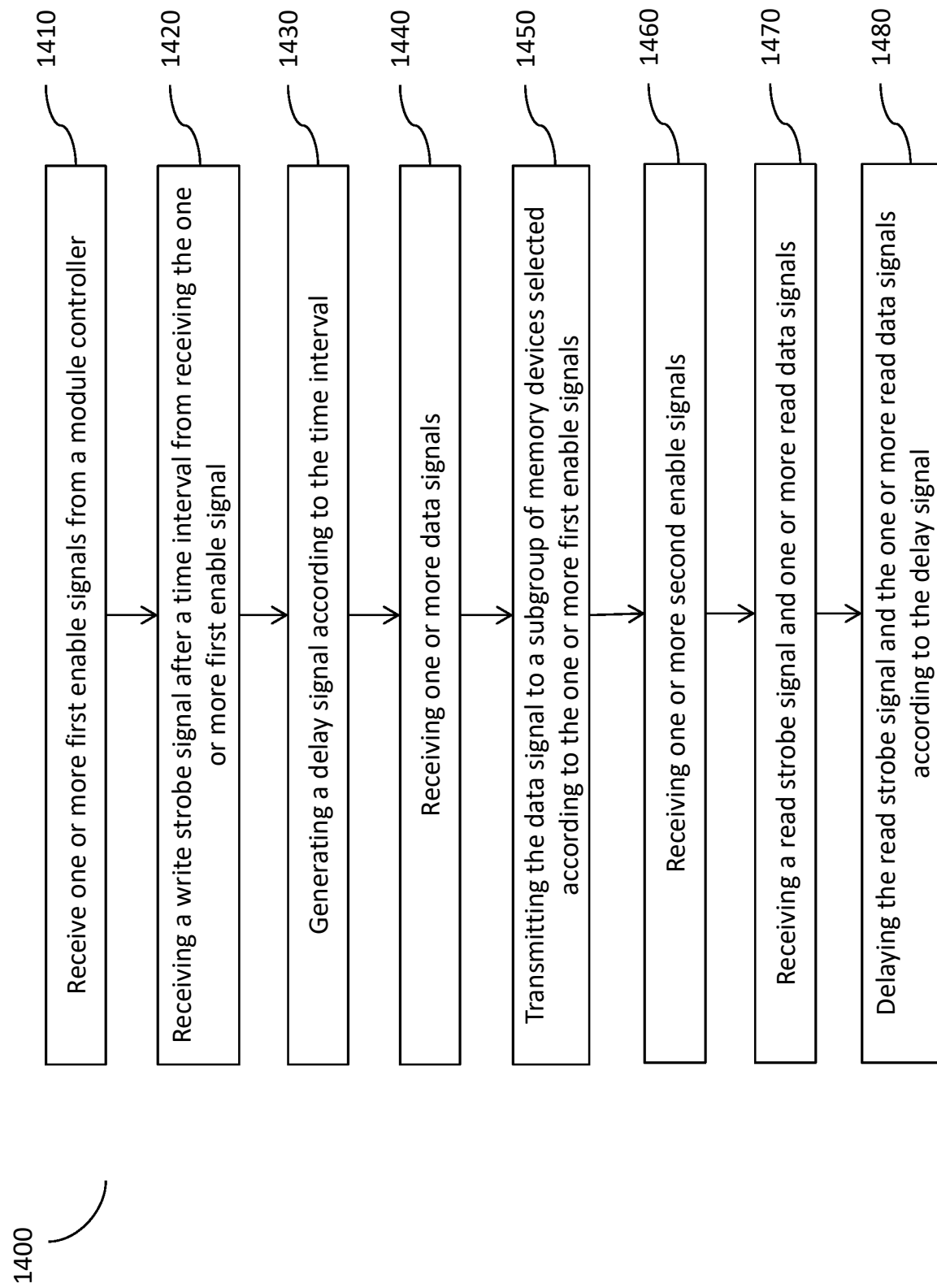


FIG. 13

**FIG. 14**



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/141,978	01/05/2021	Hyun Lee	129980-5073-US02

CONFIRMATION NO. 4088

FORMALITIES LETTER



OC000000122758477

79141
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

Date Mailed: 01/22/2021

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **320** basic filing fee.
- \$ **160** surcharge.
- \$ **700** search fee.
- \$ **800** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
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Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/Nguyen/



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/141,978	01/05/2021	2139	0.00	129980-5073-US02	1	1

CONFIRMATION NO. 4088

79141

Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

FILING RECEIPT



0000000122738478

Date Mailed: 01/22/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 16/432,700 06/05/2019 PAT 10884923
which is a CON of 14/445,035 07/28/2014 PAT 10324841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 01/19/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/141,978**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No
Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific

countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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NOT GRANTED

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/141,978

APPLICATION AS FILED - PART I

(Column 1)

(Column 2)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

FOR	NUMBER FILED	NUMBER EXTRA
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A
TOTAL CLAIMS (37 CFR 1.16(j))	1 minus 20 =	*
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))		

RATE(\$)	FEE(\$)
N/A	
N/A	
N/A	
TOTAL	

OR

RATE(\$)	FEE(\$)
N/A	320
N/A	700
N/A	800
x 100 =	0.00
x 480 =	0.00
	0.00
	0.00
TOTAL	1820

* If the difference in column 1 is less than zero, enter "0" in column 2.

APPLICATION AS AMENDED - PART II

(Column 1)

(Column 2)

(Column 3)

SMALL ENTITY

OR

OTHER THAN SMALL ENTITY

AMENDMENT A		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

(Column 1)

(Column 2)

(Column 3)

AMENDMENT B		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA
	Total (37 CFR 1.16(i))	*	Minus	**	=
	Independent (37 CFR 1.16(h))	*	Minus	***	=
	Application Size Fee (37 CFR 1.16(s))				
	FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))				

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

OR

RATE(\$)	ADDITIONAL FEE(\$)
x =	
x =	
TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	4088
Serial No.:	17/141,978	Art Unit:	2139
Filed:	01-05-2021	Examiner:	TBA
For:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION	Attorney Docket No:	129980-5073-US02

**RESPONSE TO THE NOTICE TO FILE MISSING PARTS
OF NONPROVISIONAL APPLICATION**

Mail Stop: MISSING PARTS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Notice to File Missing Parts of Nonprovisional Application mailed January 22, 2021, a copy of which is enclosed herein, submitted herewith are the statutory basic filing, search and examination fees. The surcharge for the late payment of the statutory basic filing fees is also submitted herewith.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. 129980-5073-US02).

Respectfully submitted,

Date: April 22, 2021

/ Jamie J. Zheng /
Jamie J. Zheng
Customer No. 79141
(650) 521-4828

51,167
(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

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United States Patent and Trademark Office
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/141,978	01/05/2021	Hyun Lee	129980-5073-US02

CONFIRMATION NO. 4088

FORMALITIES LETTER



OC000000122758477

79141
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

Date Mailed: 01/22/2021

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **320** basic filing fee.
- \$ **160** surcharge.
- \$ **700** search fee.
- \$ **800** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **1980** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/Nguyen/



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/141,978	01/05/2021	2139	0.00	129980-5073-US02	1	1

CONFIRMATION NO. 4088

79141

Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

FILING RECEIPT



0000000122738478

Date Mailed: 01/22/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 16/432,700 06/05/2019 PAT 10884923
which is a CON of 14/445,035 07/28/2014 PAT 10324841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 01/19/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/141,978**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No
Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific

countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop

technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/141,978

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(j))	1 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x 480 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x	=	OR	x	=
Independent (37 CFR 1.16(h))	*	Minus ***	=	x	=	OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x	=	OR	x	=
Independent (37 CFR 1.16(h))	*	Minus ***	=	x	=	OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

Electronic Patent Application Fee Transmittal

Application Number:	17141978			
Filing Date:	05-Jan-2021			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	129980-5073-US02			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	320	320
UTILITY SEARCH FEE	1111	1	700	700
UTILITY EXAMINATION FEE	1311	1	800	800
Pages:				
Claims:				
Miscellaneous-Filing:				
LATE FILING FEE FOR OATH OR DECLARATION	1051	1	160	160
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Extension - 1 month with \$0 paid	1251	1	220	220
Miscellaneous:				
Total in USD (\$)				2200

Electronic Acknowledgement Receipt

EFS ID:	42531422
Application Number:	17141978
International Application Number:	
Confirmation Number:	4088
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	129980-5073-US02
Receipt Date:	22-APR-2021
Filing Date:	05-JAN-2021
Time Stamp:	23:27:38
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 2200
RAM confirmation Number	E20214LN30041973
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	129980-5073US02_MP_Response.pdf	59099	no	1
			04abf0796001cd4ac9248960f0d6bdbaa4246af4		
Warnings:					
Information:					
2	Transmittal Letter	129980-5073-US02_NTFMP.pdf	283982	no	7
			6b830aca48fae38651c5113f197eb2622ce7070d		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	38930	no	2
			0f0db6951afc2cc60608915e0a33c4d2295f184		
Warnings:					
Information:					
Total Files Size (in bytes):			382011		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
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Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
17/141,978	01/05/2021	2139	1980	129980-5073-US02	1	1

CONFIRMATION NO. 4088

UPDATED FILING RECEIPT

79141
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304



Date Mailed: 04/27/2021

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

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Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

NETLIST, INC., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 16/432,700 06/05/2019 PAT 10884923
which is a CON of 14/445,035 07/28/2014 PAT 10324841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 01/19/2021

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 17/141,978**

Projected Publication Date: 08/05/2021

Non-Publication Request: No

Early Publication Request: No
Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

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countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

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This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

SelectUSA

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technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
17/141,978

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	320
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	700
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	800
TOTAL CLAIMS (37 CFR 1.16(j))	1 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	1 minus 3 =	*				x 480 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1820

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x	=	OR	x	=
Independent (37 CFR 1.16(h))	*	Minus ***	=	x	=	OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus **	=	x	=	OR	x	=
Independent (37 CFR 1.16(h))	*	Minus ***	=	x	=	OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
17/141,978	01/05/2021	Hyun Lee	NL073-20US02

CONFIRMATION NO. 4088

PUBLICATION NOTICE



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79141
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

Title:MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Publication No.US-2021-0240620-A1

Publication Date:08/05/2021

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088
79141	7590	02/10/2022	EXAMINER	
USCH Law, PC			AHMED, ZUBAIR	
3790 El Camino Real #1147				
Palo Alto, CA 94304				
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			02/10/2022	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eofficeaction@appcoll.com
jzheng@uschlaw.com
patents@uschlaw.com

DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to original application filed on 01/05/2021. Claim 1 is pending in this application.

Allowable Subject Matter

Claim 1 would be allowed provided that the outstanding double patenting rejection given in this Office Action is overcome.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory

double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Independent claim 1 of the instant application is anticipated by independent claim 1 of US Patent 10,884,923 as shown below in the table.

Status of Claim	Instant Application	US Patent 10,884,923
Anticipation	1. A memory module operable in a computer system having a memory controller and a system bus including one or more clock signal lines, a set of control signal lines and a plurality of sets of n-bit wide data/strobe signal lines, where n is an integer larger than one, comprising:	1. A memory module configured to couple to a memory controller via one or more clock signal lines, a set of control signal lines and N sets of n-bit wide data/strobe signal lines, where N and n are integers larger than one, comprising:

	a control circuit configurable to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the set of control signal lines, and to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals; memory devices organized in one or more ranks; and a plurality of data buffers configurable to receive the module clock signal and the data buffer control signals from the control circuit; wherein: a respective data buffer of the plurality of data buffers is coupled to a respective group of the memory devices and includes a respective n-bit-wide data path and logic configurable to control the respective n-bit-wide data path in response to the data buffer control signals, the respective group of the memory devices including at least one memory device in each of the one or more ranks, the respective n-bit-wide data path including at least one programmable delay element controlled by the logic;	a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals; N data buffers configurable to receive the module clock signal and the data buffer control signals from the control circuit, wherein a respective data buffer of the N data buffers is configurable to operatively couple to a respective set of n-bit wide data/strobe signal lines, the respective data buffer including a n-bit wide data path and logic configured to control the data path in response to the data buffer control signals, the n-bit wide data path including at least one programmable delay element, wherein the respective data buffer is further configurable to generate a respective local clock signal in response to the module clock signal, and wherein the respective local clock signal is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal; and
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	the respective data buffer is configurable to generate, in response to receiving the module clock signal, a respective local clock signal that is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal; and the respective group of the memory devices is configurable to receive the module C/A signals from the control circuit and the respective local clock signal, and to communicate respective data and data strobe signals with the memory controller in response to the module C/A signals and the respective local clock signal, the respective data and data strobe signals being communicated via the respective n-bit wide data path and a respective set of the plurality of sets of n-bit wide data/strobe signal lines.	memory devices organized in one or more ranks and including N groups of memory devices corresponding to respective ones of the N data buffers, each group of memory devices including at least one memory device in each of the one or more ranks, wherein a respective group of the N groups of memory devices is configurable to receive the module C/A signals from the control circuit and a respective local clock signal from a corresponding data buffer of the N data buffers, and wherein the respective group of the N groups of memory devices is further configurable to communicate data and data strobe signals with the memory controller via the n-bit wide data path in the corresponding data buffer and in response to the module C/A signals from the control circuit and the respective local clock signal from the corresponding data buffer.
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Reasons for Allowance

The instant application is a continuation of application 16/432,700, now US patent 10,884,923, which is a continuation of application 14/445,035, now US patent 10,324,841.

Matsui US 2004/0105292 ("Matsui") and Sutardja et al. US 2007/0168781 ("Sutardja") were the closest prior art with respect to the parent applications and continue to be the closest prior art for the instant application.

With respect to the parent application 14/445,035, PTAB (Patent Trial and Appeal Board) rendered a decision on 09/27/2018 indicating that the independent claims of the application were not obvious over Matsui in view of Sutardja. In view of the PTAB decision, application 14/445,035 was allowed. For the same reason, parent application 16/432,700 was also allowed.

Independent claim 1 of the instant application is anticipated by independent claim 1 of US Patent 10,884,923. Since Matsui and Sutardja continue to be the closest prior art, independent claim 1 of the instant application is conditionally indicated as allowable.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

<i>Notice of References Cited</i>	Application/Control No. 17/141,978		Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED		Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20040105292-A1	06-2004	Matsui, Yoshinori	G11C7/1048	365/189.05
*	B	US-20070168781-A1	07-2007	Sutardja; Sehat	G11C29/02	714/718
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

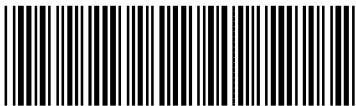
FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/0802, 13/4243, 13/4256, 1/10	02/03/2022	ZA
G11C 7/1066, 7/1093, 29/028, 5/04, 29/023	02/03/2022	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	02/03/2022	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached PE2E search history	02/03/2022	ZA
Inventor search	02/03/2022	ZA
Assignee search	02/03/2022	ZA
NPL search	02/03/2022	ZA
Reason for allowance - PTAB decision rendered on 07/27/2018 with respect to parent application 14/445,035	09/27/2018	ZA

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G06F	120/12/0802	02/03/2022	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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Search History

Report Run Date: 2022-02-03 UTC
Queries: 1
Markers: 0

Query #	Timestamp	Results	Page
1	2022-02-03 16:40 UTC	41,371,180	3



US Patents | US Designs | US Applications | EPO Patents | EPO Applications | China Patents | China Applications | Japan Patents | Japan Applications | Korea Patents | Korea Applications | WIPO Applications | Argentina Patents | Argentina Applications | Brazil Patents | Brazil Applications | Canada Patents | Canada Applications | Chile Patents | Chile Applications | Colombia Applications | Costa Rica Applications | Cuba Patents | Cuba Applications | Dominican Republic Applications | Ecuador Patents | Ecuador Applications | El Salvador Applications | Guatemala Applications | Honduras Applications | Mexico Patents | Mexico Applications | Nicaragua Patents | Panama Applications | Peru Applications | Trinidad & Tobago Patents | Uruguay Applications | Austria Patents | Austria Applications | Belarus Patents | Belgium Patents | Belgium Applications | Bosnia & Herzegovina Patents | Bosnia & Herzegovina Applications | Bulgaria Patents | Bulgaria Applications | Croatia Patents | Croatia Applications | Czech Republic Patents | Czech Republic Applications | Czechoslovakia Patents | Czechoslovakia Applications | Denmark Patents | Denmark Applications | Estonia Patents | Estonia Applications | EUIPO Patents | Finland Patents | Finland Applications | France Patents | France Applications | Germany Patents | Germany Applications | Great Britain Patents | Great Britain Applications | Greece Patents | Greece Applications | Hungary Patents | Hungary Applications | Iceland Patents | Iceland Applications | Ireland Patents | Ireland Applications | Italy Patents | Italy Applications | Latvia Patents | Latvia Applications | Lithuania Patents | Lithuania Applications | Luxembourg Patents | Malta Patents | Monaco Patents | Montenegro Patents | Montenegro Applications | Netherlands Patents | Netherlands Applications | Norway Patents | Norway Applications | Poland Patents | Poland Applications | Portugal Patents | Portugal Applications | Republic of Moldova Patents | Republic of Moldova Applications | Romania Patents | Romania Applications | San Marino Patents | San Marino Applications | Serbia Patents | Serbia Applications | Slovakia Patents | Slovakia Applications | Slovenia Patents | Spain Patents | Spain Applications | Sweden Patents | Sweden Applications | Switzerland Patents | Switzerland Applications | Ukraine Patents | Yugoslavia/Serbia and Montenegro Patents | Yugoslavia/Serbia and Montenegro Applications | Armenia Patents | Australia Patents | Australia Applications | Cyprus Patents | Gulf Cooperation Council Patents | Hong Kong Patents | India Patents | India Applications | Indonesia Patents | Indonesia Applications | Israel Applications | Jordan Patents | Kyrgyzstan Patents | Macao Applications | Malaysia Patents | Mongolia Patents | New Zealand Patents | Philippines Patents | Philippines Applications | Saudi Arabia Patents | Singapore Patents | Singapore Applications | Taiwan Patents | Taiwan Applications | Tajikistan Patents | Tajikistan Applications | Thailand Applications | Uzbekistan Patents | Vietnam Patents | Algeria Patents | ARIPO Patents | ARIPO Applications | Egypt Patents | Kenya Patents | Malawi Patents | Morocco Patents | Morocco Applications | OAPI Patents | South Africa Patents | Tunisia Applications | Zambia Patents | Zimbabwe Patents | EAPO Patents | EAPO Applications | Georgia Patents | Georgia Applications | Kazakhstan Patents | Kazakhstan Applications | Russia Patents | Russia Applications | Turkey Patents | Turkey Applications | IEEE Xplore Publications: IEEE Periodicals | IEEE Xplore Publications: IEEE Conferences | IEEE Xplore Publications: IEEE Standards | IEEE Xplore Publications: IEEE Early Access | IEEE Xplore Publications: SMPTE Periodicals | IEEE Xplore Publications: SMPTE Conferences | IEEE Xplore Publications: SMPTE Standards | IEEE Xplore Publications: MIT Press eBooks | IEEE Xplore Publications: Wiley-IEEE eBooks | IEEE Xplore Publications: IBM Periodicals | IEEE Xplore Publications: URSI Periodicals | IEEE Xplore Publications: VDE Conferences | IEEE Xplore Publications: Periodicals from China | IP.com Prior Art Database: The IP.com Journal | IP.com Prior Art Database: Internet Society RFC | IP.com Prior Art Database: IBM TDB Archive | IP.com Prior Art Database: Legacy Journals | IP.com Prior Art Database: Software Patent Institute | OnePetro.org: Periodicals at OnePetro.org | OnePetro.org: Conferences at OnePetro.org | Other Literature: IBM Redbooks | Other Literature: PubMed Central | Other Literature: arXiv.org

1 2022-02-03 16:40 UTC | 41,371,180 results from

De-Dup: None | Cut-off: None | Sort: Relevance

Main Concept: Text

A memory module operable in a computer system having a memory controller and a system bus including one or more clock signal lines, a set of control signal lines and a plurality of sets of n-bit wide data/strobe signal lines, where n is an

integer larger than one, comprising: a control circuit configurable to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the set of control signal lines, and to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals; memory devices organized in one or more ranks; and a plurality of data buffers configurable to receive the module clock signal and the data buffer control signals from the control circuit; wherein: a respective data buffer of the plurality of data buffers is coupled to a respective group of the memory devices and includes a respective n-bit-wide data path and logic configurable to control the respective n-bit-wide data path in response to the data buffer control signals, the respective group of the memory devices including at least one memory device in each of the one or more ranks, the respective n-bit-wide data path including at least one programmable delay element controlled by the logic; the respective data buffer is configurable to generate, in response to receiving the module clock signal, a respective local clock signal that is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal; and the respective group of the memory devices is configurable to receive the module C/A signals from the control circuit and the respective local clock signal, and to communicate respective data and data strobe signals with the memory controller in response to the module C/A signals and the respective local clock signal, the respective data and data strobe signals being communicated via the respective n-bit wide data path and a respective set of the plurality of sets of n-bit wide data/strobe signal lines.

Concept Modifiers: None

Filters: None

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	("10884923").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 09:26 AM
L2	3	("10324841").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 09:27 AM
L3	49265	(G06F12/0802 G06F13/4243 G11C7/1066 G11C7/1093 G11C29/028 G11C5/04 G06F13/4256 G11C29/023 G06F1/10).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:11 AM
L4	208	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:17 AM
L5	160	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:18 AM
L6	148	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:18 AM
L7	196	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO;	OR	ON	ON	2022/02/03 10:23 AM

L8	257	OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module) L3 AND (control\$4 SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:26 AM
L9	275	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:27 AM
L10	160	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:29 AM
L11	2	("20040105292").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:34 AM
L12	1	("20070168781").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:35 AM
L13	17	((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:37 AM
L14	14	((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO;	OR	ON	ON	2022/02/03 10:40 AM

L15	14	SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) ((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:40 AM
L16	12	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:42 AM
L17	12	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:43 AM
L18	20	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:46 AM
L19	19	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:47 AM

L20	19	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:47 AM
L21	1663	(711/120 711/119).ccls.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:50 AM
L22	1	L21 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:50 AM
L23	1	L21 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:51 AM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

Bibliographic Data

Application No: 17/141,978

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /ZUBAIR AHMED/

Examiner's Signature

Initials

Title:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND
METHOD OF OPERATION

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/05/2021	711	2132	NL073-20US02
RULE			

APPLICANTS

NETLIST, INC., Irvine, CA,

INVENTORS

Hyun Lee, Ladera Ranch, CA, UNITED STATES

Jayesh R. Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 16432700 06/05/2019 PAT 10884923

16432700 is a CON of 14445035 07/28/2014 PAT 10324841

14445035 has PRO of 61859215 07/27/2013

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

01/19/2021

STATE OR COUNTRY

UNITED STATES

ADDRESS

USCH Law, PC

3790 El Camino Real #1147

Palo Alto, CA 94304

UNITED STATES

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\$2,200

PE2E SEARCH - Search History (Prior Art)

There are no Prior Art searches to show.

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	26792	(G06F12/0802 G06F13/4243 G11C7/1066 G11C7/1093 G11C29/028 G11C5/04 G06F13/4256 G11C29/023 G06F1/10).cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:48 AM
N2	126	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:52 AM
N3	96	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:52 AM
N4	95	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM
N5	115	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM

N6	185	N1 AND (control\$4 SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM
N7	96	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:57 AM
N8	1652	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:57 AM
N9	1652	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:58 AM
N10	1	N9 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:59 AM
N11	1	N9 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:00 AM
N12	5	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:13 AM
N13	5	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:13 AM

N14	5	SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) ((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N15	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N16	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N17	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 17/141,978		Filing Date 01/05/2021		☐ To be Mailed			
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO											
APPLICATION AS FILED - PART I											
		(Column 1)		(Column 2)							
FOR		NUMBER FILED		NUMBER EXTRA		RATE (\$)		FEE (\$)			
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A		N/A		N/A					
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A		N/A		N/A					
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A		N/A		N/A					
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *				x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *				x \$480 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).									
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))											
* If the difference in column 1 is less than zero, enter "0" in column 2.						TOTAL					
APPLICATION AS AMENDED - PART II											
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT	06/10/2022	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0			x \$100 =		0	
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0			x \$480 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)		(Column 3)					
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR		PRESENT EXTRA		RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=			x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=			x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))										
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))										
								TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.								LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".								/ROSS W BROWN/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".											
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.											

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE, et al.	Confirmation No.:	4088
Serial No.:	17/141,978	Group Art Unit:	2139
Filed:	January 5, 2021	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>	Attorney Docket No.:	NL073-20US02

AMENDMENT AND RESPONSE TO NON-FINAL OFFICE ACTION

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Sir:

The enclosed Amendment is in response to the Non-Final Office Action dated February 10, 2022 for the above identified patent application.

Petition for Extension of Time under 37 CFR 1.136. It is respectfully requested that the time for response to the Office Action dated February 10, 2022, be extended for a period of one (1) month from May 10, 2022, to June 10, 2022.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US02).

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (New) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:
 - a module control circuit configurable to receive from the memory controller a system clock via the one or more clock signal lines and input address and control (C/A) signals via the set of C/A signal lines, and to output a module clock signal and module C/A signals in response to the system clock and the input C/A signals;
 - memory devices organized in one or more ranks; and
 - buffer circuits configurable to receive the module clock signal from the module control circuit and to generate a plurality of local clock signals, wherein a respective local clock signal of the plurality of local clock signals is output by a respective buffer circuit to a respective group of the memory devices, the respective group of the memory devices including at least one respective memory device in each of the one or more ranks, the respective local clock signal having a respective programmable phase relationship with the module clock signal, the buffer circuits including logic and configuration registers, wherein respective configuration registers included in the respective buffer circuit are programmable by the logic to control the respective programmable phase relationship; and
 - wherein:
 - the memory module is operable in at least a normal operation mode and a configuration mode;
 - the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to a memory read or write command received via the C/A signal lines;

the memory module in the configuration mode is configurable to perform a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices and at least one read operation to read from the set of memory locations; and

the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.

3. (New) The memory module of claim 2, wherein the respective group of the memory devices is configurable to receive the module C/A signals and the respective local clock signal, from the module control circuit, and to communicate respective data and data strobe signals with the memory controller in response to the module C/A signals and the respective local clock signal via a respective set of the plurality of sets of n-bit wide data/strobe signal lines.
4. (New) The memory module of claim 3, wherein the respective buffer circuit is configurable to generate the respective local clock signal based on values stored at the respective configuration registers.
5. (New) The memory module of claim 4, wherein the module clock signal received at a first buffer circuit is phase shifted from the module clock signal received at a second buffer circuit closer to the module control circuit than the first buffer circuit, and wherein values stored at first configuration registers in the first buffer circuit of the buffer circuits are different from values stored at second configuration registers at the second buffer circuit of the buffer circuits.
6. (New) The memory module of claim 2, wherein the respective configuration registers are programmed during a mode register write operation.
7. (New) The memory module of claim 6, wherein the logic is configured to program the respective configuration registers according to signals output from the module control circuit when the memory module is in the configuration mode.
8. (New) The memory module of claim 7, wherein the module control circuit is configurable to output the signals in response to mode register command signals received from the memory controller.

9. (New) The memory module of claim 2, wherein the respective buffer circuit further includes a respective phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a respective programmable delay circuit to add a respective delay to the respective local clock signal, and a respective clock driver to drive the respective local clock signal to the respective group of the memory devices.
10. (New) The memory module of claim 9, wherein the respective programmable delay circuit is controlled by the logic through the respective configuration registers.
11. (New) The memory module of claim 1, further comprising a printed circuit board, wherein the module control circuit, the plurality of buffer circuits and the memory devices are mounted to the printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and wherein the buffer circuits are distributed along the edge of the printed circuit board and between the memory devices and the edge connectors.
12. (New) A method of operating a memory module coupled to a memory controller via a system bus, the system bus including one or more clock signal lines, a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module including memory devices organized in one or more ranks, the method comprising:
- receiving a system clock from the memory controller via the one or more clock signal lines and input address and control (C/A) signals from the memory controller via the set of C/A signal lines;
 - outputting a module clock signal and module C/A signals in response to the system clock and the input C/A signals;
 - generating a plurality of local clock signals in response to the module clock signal; and
 - outputting a respective local clock signal of the plurality of local clock signals to a respective group of the memory devices, the respective group of the memory devices including at least one memory device in each of the one or more ranks, the respective local clock signal having a respective programmable phase relationship with the module clock signal;

wherein the memory module is operable in at least a normal operation mode and a configuration mode, the method further comprising

outputting or receiving data/strobe signals via the data/strobe signal lines in response to a memory read or write command received via the C/A signal lines when the memory module is in the normal operation mode;

performing a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices and at least one read operation to read from the set of memory locations when the memory module is in the configuration mode; and

programming the respective phase relationship based on information derived from the set of operations when the memory module is in the configuration mode;

wherein the memory module further comprises:

a module control circuit configurable to receiving the system clock from the memory controller via the one or more clock signal lines and the input address and control (C/A) signals from the memory controller via the set of C/A signal lines, and to output the module clock signal and the module C/A signals in response to the system clock and the input C/A signals;

a plurality of buffer circuits configurable to receive the module clock signal from the module control circuit and to generate the plurality of local clock signals, wherein each respective buffer circuit of the plurality of buffer circuits is coupled to a respective group of the memory devices and is configurable to output the respective local clock signal of the plurality of local clock signals to the respective group of the memory devices, wherein the plurality of buffer circuits including logic and configuration registers, and wherein respective configuration registers included in the respective buffer circuit are programmable by the logic to control the respective programmable phase relationship.

13. (New) The memory module of claim 12, further comprising, at the respective group of the memory devices:

receiving the module C/A signals and the respective local clock signal; and

communicating respective data and data strobe signals with the memory controller in response to the module C/A signals and the respective local clock signal via a respective set of the plurality of sets of n-bit wide data/strobe signal lines.

14. (New) The memory module of claim 13, further comprising generating the respective local clock signal using a respective delay circuit based on values stored at the respective configuration registers.
15. (New) The memory module of claim 14, wherein:
 - a first local clock signal is output to a first group of memory devices; and
 - a second local clock signal is output to a second group of memory devices further away from the module control circuit than the first group of memory device;
 - the first local clock signal is phase shifted by a first delay from the module clock signal;and
 - the second local clock signal is phase shifted by a second delay from the module clock signal, the second delay being different from the first delay.
16. (New) The memory module of claim 14, further comprising programming the respective configuration registers during a mode register write operation.
17. (New) The memory module of claim 16, wherein the respective configuration registers are programmed according to signals output from the module control circuit when the memory module is in the configuration mode.
18. (New) The memory module of claim 17, wherein the module control circuit is configurable to output the signals in response to mode register command signals received from the memory controller when the memory module is in the configuration mode.
19. (New) The memory module of claim 12, further comprising:
 - phase locking the respective local clock signal with the module clock signal;
 - adding a respective delay to the respective local clock signal; and
 - driving the respective local clock signal to the respective group of the memory devices.
20. (New) The memory module of claim 19, wherein the respective delay is controlled by logic through respective configuration registers.
21. (New) The method of 12, wherein the memory module further comprises:

a printed circuit board, wherein the module control circuit, the plurality of buffer circuits, and the memory devices are mounted to the printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and wherein the buffer circuits are distributed along the edge of the printed circuit board and between the memory devices and the edge connectors.

REMARKS

This amendment responds to the Non-Final Office Action dated February 10, 2022, in which claim 1 is rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1 of US Patent 10,884,923.

REMARKS CONCERNING CLAIMS

Claim 1 is cancelled.

New claims 2-21 are added.

Support for the amendment can be found throughout the Specification and Drawings, for example, in paragraphs [0057]-[0058], [0065]-[0067] and [0070]. Therefore, no new matter has been added.

REMARKS CONCERNING DOUBLE PATENTING

Claim 1 is rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1 of US Patent 10,884,923.

Claim 1 is cancelled. Thus, the rejection thereto is moot. New claims 2-21 differ significantly in scope from original claim 1 and should not be anticipated by any of the claims in US Patent 10,884,923. Reconsideration of the double patenting rejection is therefore respectfully requested.

New claims 2-21 are believed to be in condition for allowance.

CONCLUDING REMARKS

By responding in the foregoing remarks only to particular positions asserted by the examiner, the Applicant does not necessarily acquiesce in other positions that have not been explicitly addressed. In addition, the Applicant's arguments for the patentability of a claim should not be understood as implying that no other reasons for the patentability of that claim exist.

In light of the above amendments and remarks, the Applicant respectfully requests that the Examiner reconsider this application with a view towards allowance. The Examiner is invited to call the undersigned attorney at (650) 521-4828, if a telephone call could help resolve any remaining items.

Respectfully submitted,

Date: June 10, 2022

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

51,167
(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	17141978			
Filing Date:	05-Jan-2021			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NL073-20US02			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension - 1 month with \$0 paid	1251	1	220	220
Miscellaneous:				
Total in USD (\$)				220

Electronic Acknowledgement Receipt

EFS ID:	45932831
Application Number:	17141978
International Application Number:	
Confirmation Number:	4088
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NL073-20US02
Receipt Date:	10-JUN-2022
Filing Date:	05-JAN-2021
Time Stamp:	23:38:18
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 220
RAM confirmation Number	E202260N39003985
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NL073-20US02_Amendment_N FOA.pdf	163413	yes	9
			3c94a8fe137de94f1b3e273d0deda19ade8960df		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment/Request for Reconsideration-After Non-Final Rejection		1	1	
	Claims		2	7	
	Applicant Arguments/Remarks Made in an Amendment		8	9	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	38223	no	2
			9bda4e218872eb12b6159e9a823eb3d18e8706b2		
Warnings:					
Information:					
Total Files Size (in bytes):			201636		

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed July 10, 2022</i>		
				Application Number	17/141,978	
				Filing Date	January 5, 2021	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
Sheet	1	of	7	Attorney Docket Number		NL073-20US02

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
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				Art Unit	2139
				Examiner Name	Zubair AHMED
Sheet	2	of	7	Attorney Docket Number	NL073-20US02

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ – Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
	34	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, IPR Petition re US Patent No. 9,824,035, filed December 23, 2021
	35	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, IPR Petition re US Patent No. 10,268,608, filed December 23, 2021
	36	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, 'Ex. 1003 - IPR2022-00236 Declaration of Donald Alpert,' filed December 23, 2021
	37	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, 'Ex. 1003 - IPR2022-00237 Declaration of Donald Alpert,' filed December 23, 2021
	38	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Preliminary Response, filed April 25, 2022
	39	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, Patent Owner's Preliminary Response, filed April 25, 2022
	40	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Exhibit List, filed April 25, 2022
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				Examiner Name	Zubair AHMED	
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	44	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent Application Publication No. 2010/0312925 A1 to Osanai et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 57 pages.
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				Application Number	17/141,978	
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				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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				Application Number	17/141,978
				Filing Date	January 5, 2021
				First Named Inventor	Hyun LEE
				Art Unit	2139
				Examiner Name	Zubair AHMED
Sheet	5	of	7	Attorney Docket Number	NL073-20US02

	63	Obviousness Chart for U.S. Patent No.10,268,608, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 9 pages.
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	68	Declaration of Harold S. Stone, Ph.D., in Support of Defendants' Claim Construction Positions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 27 pages.
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				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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	76	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3, DDR3 SDRAM Standard, June 2007, 12 pages
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	91	First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. v. Netlist, Inc., C.A. No. 21-1453 (RGA) (USDC, Delaware, Jan. 8, 2022), 122 pages.

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				Art Unit	2139
				Examiner Name	Zubair AHMED
Sheet	7	of	7	Attorney Docket Number	NL073-20US02

Examiner Signature		Date Considered	
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Electronic Patent Application Fee Transmittal

Application Number:	17141978			
Filing Date:	05-Jan-2021			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NL073-20US02			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Miscellaneous:				
SUBMISSION- INFORMATION DISCLOSURE STMT	1806	1	260	260
Total in USD (\$)				260

Electronic Acknowledgement Receipt

EFS ID:	46152034
Application Number:	17141978
International Application Number:	
Confirmation Number:	4088
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NL073-20US02
Receipt Date:	10-JUL-2022
Filing Date:	05-JAN-2021
Time Stamp:	22:51:11
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$260
RAM confirmation Number	E202270M52571128
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Authorized User	

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File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	NL073-20US02_IDS_Transmittal_2022-07-10.pdf	122546	no	4
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18	Non Patent Literature	035_Invalidity_Chart_Tokuhito. pdf	375626	no	31
			c2f57f019c472d81fb5bdb8ee9b3da52c5dd08b		
Warnings:					
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19	Non Patent Literature	035_Invalidity_Chart_Yoo.pdf	7840344	no	48
			cf156d17ce2fa1a0db4cc1f89f577916dc90cab2		
Warnings:					
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20	Non Patent Literature	608_Invalidity_Chart_DDR4_P roposal.pdf	1014042	no	19
			9ac5c5f7c3913337c25d73662964e0759c0796b3		
Warnings:					
Information:					

21	Non Patent Literature	608_Invalidity_Chart_Dodd.pdf	441166	no	51
			97440776e33cb5a5d28cfded68567a16241baf17		
Warnings:					
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22	Non Patent Literature	608_Invalidity_Chart_Lee.pdf	304895	no	36
			4d672077d5c6a86f8ad68a90a31af64356ccb4b		
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23	Non Patent Literature	608_Invalidity_Chart_Muralimanohar.pdf	571150	no	49
			2100bd6c5191c9ac9fc8ed606aa6d1295ffaf558		
Warnings:					
Information:					
24	Non Patent Literature	608_Invalidity_Chart_Shaeffer.pdf	452017	no	52
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Warnings:					
Information:					
25	Non Patent Literature	608_Invalidity_Chart_Tokuhiro.pdf	831346	no	30
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Warnings:					
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26	Non Patent Literature	608_Invalidity_Chart_Takefman.pdf	345175	no	35
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27	Non Patent Literature	608_Invalidity_Chart_Yoo.pdf	6620949	no	48
			eeb749657ac2c4c670dc1e4845b3ecb09a86af39		
Warnings:					
Information:					

28	Non Patent Literature	035_Obviousness_Chart.pdf	137498	no	14
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Warnings:					
Information:					
29	Non Patent Literature	608_Obviousness_Chart.pdf	115958	no	9
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Information:					
30	Non Patent Literature	608_Invalidity_Chart_Osanai.pdf	577825	no	53
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31	Non Patent Literature	035_Invalidity_Chart_Osanai.pdf	603423	no	57
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32	Non Patent Literature	Claim_Construction_Micron.pdf	181786	no	14
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33	Non Patent Literature	Claim_Construction_Netlist.pdf	772289	no	7
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34	Non Patent Literature	Netlist_ID_Extrinsic_Evidence.pdf	268529	no	16
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35	Non Patent Literature	Micron_ID_Extrinsic_Evidence.pdf	192993	no	21
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36	Non Patent Literature	Stone_Declaration.pdf	300877	no	27
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37	Non Patent Literature	Smith_Declaration.pdf	294487	no	26
			ea8f77afe98c3b74c66181de94cdba505f3eff747		
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38	Non Patent Literature	Micron_Opening_CC_Brief.pdf	624888	no	40
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39	Non Patent Literature	Joint_CC.pdf	4410984	no	13
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40	Non Patent Literature	Ex-1015_Electronic_Dictionary.pdf	841001	no	6
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41	Non Patent Literature	DDR4_Proposal.pdf	772060	no	10
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42	Non Patent Literature	JEDEC_79.pdf	1790490	no	78
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43	Non Patent Literature	JEDEC_21C.pdf	1663358	no	83
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44	Non Patent Literature	JEDEC_79-3_Excerpt.pdf	2034001	no	12
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45	Non Patent Literature	JESD79-3F.pdf	10139699	no	226
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48	Non Patent Literature	JEDEC_21C_Excerpt.pdf	136349	no	2
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Warnings:					
Information:					

49	Non Patent Literature	IPR2022-00711_Petition.pdf	8248569	no	155
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50	Non Patent Literature	IPR2022-00711_Wedig_Declaration.pdf	9078629	no	273
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Information:					
51	Non Patent Literature	IPR2017-00730_Petition.pdf	1517599	no	62
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52	Non Patent Literature	IPR2017-00730_Mudge_Declaration.pdf	1544176	no	95
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Information:					
53	Non Patent Literature	IPR2017-00730_POPR.pdf	1035259	no	59
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54	Non Patent Literature	IPR2017-00730_Decision.pdf	438747	no	16
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55	Non Patent Literature	IPR2017-00730_Petitioner_Request.pdf	226847	no	16
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Information:					

56	Non Patent Literature	IPR2017-00730_Order_Denying_Rehearing.pdf	116362	no	7
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57	Non Patent Literature	IPR2018-00362_FWD.pdf	3402570	no	94
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58	Non Patent Literature	Netlist_Complaint_EDT_2021-12-20.pdf	4925913	no	49
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59	Non Patent Literature	Samsung_Amended_DJ_Complaint_2022-01-18.pdf	1220667	no	122
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Information:					
60	Information Disclosure Statement (IDS) Form (SB08)	NL073-20US02_IDS_List_2022-07-10.pdf	254495	no	7
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Warnings:					
Information:					
This is not an USPTO supplied IDS fillable form					
61	Fee Worksheet (SB06)	fee-info.pdf	38034	no	2
			74012e3d13d63755d7f77f3c8fb850804e17ac31		
Warnings:					
Information:					
Total Files Size (in bytes):			121831585		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE, et al.	Confirmation No.:	4088
Serial No.:	17/141,978	Group Art Unit:	2139
Filed:	January 5, 2021	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>	Attorney Docket No.:	NL073-20US02

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information which the Examiner may consider material to the examination of the subject U.S. patent application. It is requested that the Examiner make this information of record if it is deemed material to the examination of the application.

1. Enclosures accompanying this Information Disclosure Statement are:

- 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
- 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☒ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☒ all other information or portion which caused it to be listed on the PTO-1449.
- 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
- 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.

2. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
- ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);
 - ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☐ Before the mailing of the first Office action on the merits;
 - ☐ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.

3. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b, 3c or 3d)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$260 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916 (order no. NL049-19US02).
- 3c. ☐ The \$130 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916 (order no. NL049-19US02).
- 3d. ☐ The \$120 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916 (order no. NL049-19US02).

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check Item 4a, and 4b, 4c or 4d)

- 4a. ☐ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$260 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916 (order no. NL049-19US02).
- 4c. ☐ The \$130 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916 (order no. NL049-19US02).

- 4d. ☐ The \$120 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Deposit Account No. 60-3916 (order no. NL049-19US02).
5. ☐ Certification Statement (applicable if Item 3a or Item 4a is checked)
(Check either Item 5a, 5b, 5c or 5d)
- 5a. ☐ In accordance with 37 C.F.R. §1.97(e)(1), it is certified that each item of information contained in this information disclosure statement was first cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this Information Disclosure Statement.
- 5b. ☐ In accordance with 37 C.F.R. §1.97(e)(2), it is certified that no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 C.F.R. §1.56(c) more than three months prior to the filing of this information disclosure statement.
- 5c. ☐ Each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
- 5d. ☐ Pursuant to 37 C.F.R. §1.704(d), each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
6. ☒ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.
7. ☒ This application is a continuation application under 37 C.F.R. §1.53(b) or (d).
(Check appropriate Items 7a, 7b and/or 7c)
- 7a. ☐ A Petition to Withdraw from issue under 37 C.F.R. §1.313(b)(5) is concurrently filed herewith.
- 7b. ☐ Copies of publications listed on Form PTO-1449 from prior application Serial Nos. _____ filed on _____, of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7c. ☐ Copies of the publications listed on the attached Form PTO-1449 that were not previously cited in prior application Serial No. _____, filed on _____, are provided herewith.

8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on . A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be considered as if properly filed on .
9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:
- (Check Item 9a, 9b, or 9c)*
- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Deposit Account No. 60-3916 (order no. NL073-20US02).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: July 10, 2022

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

51,167
(Reg. No.)



UNITED STATES PATENT AND TRADEMARK OFFICE

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NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 07/13/2022
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

EXAMINER	
AHMED, ZUBAIR	
ART UNIT	PAPER NUMBER

2132

DATE MAILED: 07/13/2022

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	10/13/2022

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 07/13/2022
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	10/13/2022

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29☐ Applicant asserting small entity status. See 37 CFR 1.27☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088
79141	7590	07/13/2022	EXAMINER	
USCH Law, PC 3790 El Camino Real #1147 Palo Alto, CA 94304			AHMED, ZUBAIR	
			ART UNIT	PAPER NUMBER
			2132	
DATE MAILED: 07/13/2022				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 17/141,978	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to amendment filed on 06/10/2022.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2-21. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☒ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
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DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to amendment filed on 06/10/2022. Claim 1 was canceled and new claims 2-21 were added.

Response to Arguments

Applicant's arguments, filed 06/10/2022, with respect to claims 2-21 have been fully considered and are persuasive. The double patenting rejection of claim 1 is moot since the claim has been canceled in this amendment. A NOTICE OF ALLOWANCE is being administered.

Allowable Subject Matter

Claims 2-21 are allowed.

Examiner's Amendment

An Examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this Examiner's amendment was given in a telephone interview with Applicant's representative Jamie J. Zheng (Reg. No. 51,167 Tel. No. (650) 521-4828) on June 22, 2022.

Please cancel claim 1. Dependent claim 11 is amended herein.

11. (Currently Amended) The memory module of claim [[1]] 2, further comprising a printed circuit board, wherein the module control circuit, the plurality of buffer circuits

and the memory devices are mounted to the printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and wherein the buffer circuits are distributed along the edge of the printed circuit board and between the memory devices and the edge connectors.

Reasons for Allowance

The following is an examiner's statement of reasons for allowance.

Any comments considered necessary by applicants must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

After careful considerations, examination and search of the claimed invention, the closest prior art of record does not teach or anticipate the claimed feature as claimed in independent claim 2 "the buffer circuits including logic and configuration registers, wherein respective configuration registers included in the respective buffer circuit are programmable by the logic to control the respective programmable phase relationship" and "the memory module in the configuration mode is configurable to perform a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices and at least one read operation to read from the set of memory locations" and "the logic is further configurable to program the respective configuration registers based on information derived from the set of

operations” in combination with the overall claimed limitations when interpreted in light of the specification.

There are two prior art references with respect to the parent applications. These references are Matsui US 2004/0105292 (“Matsui”) and Sutardja et al. US 2007/0168781 (“Sutardja”). Before the Examiner discusses these two references, it is important to note that the claim limitation “the memory module in the configuration mode is configurable to perform a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices and at least one read operation to read from the set of memory locations” is being interpreted as the memory module performing write leveling and read leveling operations. Both Matsui and Sutardja are silent about write/read leveling operations.

In an updated search, a new reference Saito et al. US 2010/0309706 (“Saito”) was obtained. Saito teaches that a memory module 100 includes a plurality of data register buffers 300, para 0043 and FIG. 1. Each data register buffer 300 includes a write leveling circuit 322 and a read leveling circuit 323, para 0086 and FIG. 5. The write leveling circuit 322 is a circuit for performing a write leveling operation, and the read leveling circuit 323 is a circuit for performing a read leveling operation, para 0087. Therefore, Saito teaches the claim limitations “the memory module in the configuration mode is configurable to perform a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices and at least one read operation to read from the set of memory locations” and “the logic is further configurable to program the respective configuration registers based on information derived from the set of operations”. However, in Saito, the result of the write

leveling operation and the result of the read leveling operation are stored in a memory controller 12, paras 0154 and 0159. In contrast, in the claimed invention, “the buffer circuits including logic and configuration registers, wherein respective configuration registers included in the respective buffer circuit are programmable by the logic to control the respective programmable phase relationship” and “the logic is further configurable to program the respective configuration registers based on information derived from the set of operations”. In other words, the configuration registers in the buffer circuits store the results of the write/read leveling operations in the claimed invention (see paragraph [0065] of the instant filed specification).

Therefore, Saito alone or in combination with Matsui and Sutardja neither anticipates nor renders obvious independent claim 2. Therefore, claim 2 is allowable.

Independent claim 12 recites similar subject matter as claim 2 and is allowable for the aforementioned reason.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

<i>Notice of References Cited</i>	Application/Control No. 17/141,978		Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED		Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20100309706-A1	12-2010	Saito; Shunichi	G11C7/1078	365/191
*	B	US-20040105292-A1	06-2004	Matsui, Yoshinori	G11C7/10	365/189.05
*	C	US-20070168781-A1	07-2007	Sutardja; Sehat	G11C29/02	714/718
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	E					
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	G					
	H					
	I					
	J					
	K					
	L					
	M					

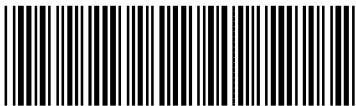
FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/0802, 13/4243, 13/4256, 1/10	06/22/2022	ZA
G11C 7/1066, 7/1093, 29/028, 5/04, 29/023	06/22/2022	ZA

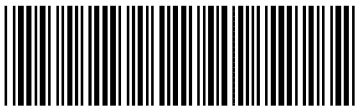
CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	06/22/2022	ZA
711	119	06/22/2022	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached PE2E search history	06/22/2022	ZA
Inventor search	06/22/2022	ZA
Assignee search	06/22/2022	ZA
NPL search	06/21/2022	ZA
Reason for allowance - PTAB decision rendered on 07/27/2018 with respect to parent application 14/445,035	09/27/2018	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

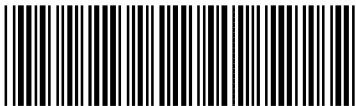
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711/G06F	120/12/0802	06/22/2022	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Index of Claims</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

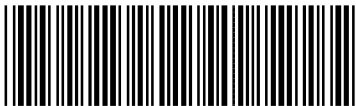
CLAIMS										
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	02/03/2022	06/22/2022							
	1	✓	-							
	2		=							
	3		=							
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Issue Classification 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	

CPC						
Symbol					Type	Version
G06F	/	12	/	0802	F	2013-01-01
G11C	/	5	/	04	I	2013-01-01
G11C	/	29	/	023	I	2013-01-01
G11C	/	29	/	028	I	2013-01-01
G11C	/	7	/	1066	I	2013-01-01
G11C	/	7	/	1093	I	2013-01-01

CPC Combination Sets					
Symbol				Type	Version
/	/	/	/		

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	22 June 2022 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	23 June 2022 (Date)	O.G. Print Claim(s) 2	O.G. Print Figure 1

<i>Issue Classification</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	12	/ 0802
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	120

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
711	120	119				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	22 June 2022 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	23 June 2022 (Date)	O.G. Print Claim(s) 2	O.G. Print Figure 1

Issue Classification 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
1	2	10	11	19	20										
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/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	22 June 2022 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	23 June 2022 (Date)	O.G. Print Claim(s) 2	O.G. Print Figure 1

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	("10884923").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 09:26 AM
L2	3	("10324841").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 09:27 AM
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L4	208	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:17 AM
L5	160	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:18 AM
L6	148	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:18 AM
L7	196	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO;	OR	ON	ON	2022/02/03 10:23 AM

L8	257	OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module) L3 AND (control\$4 SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:26 AM
L9	275	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:27 AM
L10	160	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:29 AM
L11	2	("20040105292").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:34 AM
L12	1	("20070168781").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:35 AM
L13	17	((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:37 AM
L14	14	((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO;	OR	ON	ON	2022/02/03 10:40 AM

L15	14	SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) ((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:40 AM
L16	12	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:42 AM
L17	12	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:43 AM
L18	20	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:46 AM
L19	19	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:47 AM

L20	19	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:47 AM
L21	1663	(711/120 711/119).ccls.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:50 AM
L22	1	L21 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:50 AM
L23	1	L21 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:51 AM
L24	75	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 11:05 AM
L25	41	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (mode WITH register)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 11:07 AM
L26	6	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (configur\$5 near4 mode WITH register)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 11:11 AM
L27	10	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU,	OR	ON	ON	2022/06/21 11:12 AM

L28	1	WITH normal near4 mode SAME data SAME strobe) AND (configur\$5 near4 register) ("20100309706").pn.	SG, TW, WO); EPO; JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 12:11 PM
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

PE2E SEARCH - Search History (Prior Art)

There are no Prior Art searches to show.

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	26792	(G06F12/0802 G06F13/4243 G11C7/1066 G11C7/1093 G11C29/028 G11C5/04 G06F13/4256 G11C29/023 G06F1/10).cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:48 AM
N2	126	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:52 AM
N3	96	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:52 AM
N4	95	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM
N5	115	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM

N6	185	N1 AND (control\$4 SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM
N7	96	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:57 AM
N8	1652	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:57 AM
N9	1652	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:58 AM
N10	1	N9 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:59 AM
N11	1	N9 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:00 AM
N12	5	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:13 AM
N13	5	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:13 AM

N14	5	SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) ((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N15	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N16	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N17	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM

Bibliographic Data

Application No: 17/141,978

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /ZUBAIR AHMED/

Examiner's Signature

Initials

Title:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND
METHOD OF OPERATION

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
01/05/2021	711	2132	NL073-20US02
RULE			

APPLICANTS

NETLIST, INC., Irvine, CA,

INVENTORS

Hyun Lee, Ladera Ranch, CA, UNITED STATES

Jayesh R. Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 16432700 06/05/2019 PAT 10884923

16432700 is a CON of 14445035 07/28/2014 PAT 10324841

14445035 has PRO of 61859215 07/27/2013

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

01/19/2021

STATE OR COUNTRY

UNITED STATES

ADDRESS

USCH Law, PC

3790 El Camino Real #1147

Palo Alto, CA 94304

UNITED STATES

FILING FEE RECEIVED

\$2,200

Integrated circuit **memory** devices that support selective mode **register** set commands and related **memory modules**, **memory** controllers, and methods

KH Lee, C Yoo, KH Kyung - US Patent 7,102,958, 2008 - Google Patents

... controlling a **memory module** including a plurality of **memory** devices coupled to a **memory** controller ... More particularly, a mode **register** set command may be provided from the **memory** ...

☆ Save  Cite Cited by 72 Related articles All 4 versions 

Memory module selection and reconfiguration apparatus in a data processing system

C Mantellina, D Zanzottara, M Gelmetti - US Patent 4,571,676, 1988 - Google Patents

... **memory registers** and defines for each **memory module** the ... several **memory modules** through multiplexer 21 and **register** ... for reading the working **memory** capacity, a suitable selection ...

☆ Save  Cite Cited by 69 Related articles All 2 versions 

Fully-buffered **memory-module** with error-correction code (ECC) controller in serializing advanced-**memory** buffer (AMB) that is transparent to motherboard **memory** ...

D Sun - US Patent 7,487,428, 2009 - Google Patents

... **Configuration** data for the ECC controller could be first programmed into a ... **memory** (SPD-EEPROM) on the **memory module**, and then copied to error-correction **configuration registers** ...

☆ Save  Cite Cited by 36 Related articles All 4 versions 

Load reduced **memory module** and **memory** system including the same

S Saito, T Sugano, A Hiraishi, F Osana... - US Patent ..., 2013 - Google Patents

... , a **memory module** called a Load Reduced **memory module** has been ... **memory module** is a **memory module** in which a **register** buffer is used instead of the AMB. Because the **register** ...

☆ Save  Cite Cited by 40 Related articles All 4 versions 

Memory module system having multiple **memory modules**

T Kubo, K Yasuda, H Iwamoto
2002
patents.google.com

[PDF] googleapis.com

Memory module, **memory** chip, and **memory** system

S Funaba, Y Nishio
2005
patents.google.com

[PDF] googleapis.com

Memory module including **module** data wirings available as a **memory** access data bus

S Funaba, Y Nakagome, M Horigu...
2003
patents.google.com

[PDF] googleapis.com

Apparatus for locating **memory modules** having different sizes within a **memory** space

JS Finnell, SC Steps - US Patent 4,654,787, 1987 - Google Patents

US4654787A - Apparatus for locating **memory modules** having different sizes within a **memory** space - Google Patents US4654787A - Apparatus for locating **memory modules** having ...

☆ Save  Cite Cited by 29 Related articles All 2 versions 

Memory systems and **memory** modules

MJS Smith, SN Rajan - US Patent 8,060,774, 2011 - Google Patents

... Further, the intelligent buffer chip and/or **register** chip on a **memory module** can self-test the **memory** and indicate correct operation via illumination of a second LED 109. Thus a failed ...

☆ Save  Cite Cited by 238 Related articles All 4 versions 

Memory module and **register** with minimized routing path

CN Yoon, Y Ahn, YJ Park, SJ Park - US Patent 7,652,949, 2010 - Google Patents

... **memory** devices or an odd number of pairs of **memory** devices. Generally, in the **memory module** having the **register**, the **register** is arranged at the central part of the **memory module** to ...

☆ Save  Cite Cited by 11 Related articles All 4 versions 

System and method for recognizing and **configuring** devices embedded on **memory modules**

IIMJ Alvarez, TA Dye, P Geiger - US Patent 7,032,158, 2006 - Google Patents

... as interleaving and striping on **memory modules**, detect error ... to **configure** the identified device-enhanced **memory modules** ... this **memory module** may be passed through (eg as **register** ...

☆ Save  Cite Cited by 92 Related articles All 4 versions 

Buffer circuit for a **memory module**

M Perner, N Hamzabegovic - US Patent 7,778,090, 2010 - Google Patents

... on an output request setting stored in a **configuration register** of the **configuration register** bank reads out information data the buffer circuit via the output pin of the **memory module**. ...

☆ Save  Cite Cited by 6 Related articles All 4 versions 

Memory module having **memory** chip and **register** buffer

F Osanai, T Sugano, M Nakamura - US Patent App. 13/431,498, 2012 - Google Patents

... The present invention relates to a **memory module**, and more particularly to a **memory module** that includes a **register** buffer for supplying command address signals to **memory** chips. ...

☆ Save  Cite Cited by 14 Related articles All 2 versions 



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088
79141	7590	07/20/2022	EXAMINER	
USCH Law, PC			AHMED, ZUBAIR	
3790 El Camino Real #1147				
Palo Alto, CA 94304				
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			07/20/2022	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eofficeaction@appcoll.com
jzheng@uschlaw.com
patents@uschlaw.com

Corrected Notice of Allowability	Application No. 17/141,978	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to IDS filed on 07/10/2022.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 2-21. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some* c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>07/10/2022</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
---	--

/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
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DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to IDS filed on 07/10/2022.

Information Disclosure Statement

The information disclosure statement (IDS) submitted on 07/10/2022 was filed before the mailing date of the Notice of Allowance on 07/13/2022. The submission is in compliance with the provisions of 37 CFR 1.97. Accordingly, the information disclosure statement is being considered by the examiner.

Allowable Subject Matter

Claims 2-21 are allowed.

Reasons for Allowance

This communication warrants no examiner's reason for allowance, as the record makes evident the reason for allowance, satisfying the record "as a whole" as required by rule 37 CFR 1.104 (e). Accordingly, the reason for allowance is in all probability evident from the record and no statement for examiner's reason for allowance is necessary (see MPEP 1302.14).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an

Art Unit: 2132

interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

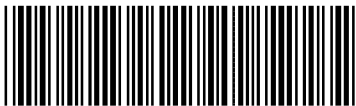
/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

<i>Index of Claims</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

CLAIMS										
☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47										
CLAIM		DATE								
Final	Original	02/03/2022	06/22/2022	07/17/2022						
	1	✓	-	-						
	2		=	=						
	3		=	=						
	4		=	=						
	5		=	=						
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	17		=	=						
	18		=	=						
	19		=	=						
	20		=	=						
	21		=	=						

<i>Search Notes</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F 12/0802, 13/4243, 13/4256, 1/10	06/22/2022	ZA
G11C 7/1066, 7/1093, 29/028, 5/04, 29/023	06/22/2022	ZA

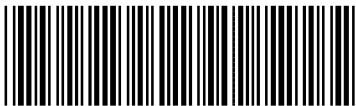
CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	06/22/2022	ZA
711	119	06/22/2022	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached PE2E search history	07/17/2022	ZA
Inventor search	06/22/2022	ZA
Assignee search	06/22/2022	ZA
NPL search	06/21/2022	ZA
Reason for allowance - PTAB decision rendered on 07/27/2018 with respect to parent application 14/445,035	09/27/2018	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 17/141,978	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G06F	120/12/0802	06/22/2022	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	1	("10884923").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 09:26 AM
L2	3	("10324841").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 09:27 AM
L3	49265	(G06F12/0802 G06F13/4243 G11C7/1066 G11C7/1093 G11C29/028 G11C5/04 G06F13/4256 G11C29/023 G06F1/10).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:11 AM
L4	208	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:17 AM
L5	160	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:18 AM
L6	148	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:18 AM
L7	196	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO;	OR	ON	ON	2022/02/03 10:23 AM

L8	257	OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	JPO; IBM_TDB)				
		L3 AND (control\$4 SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:26 AM
L9	275	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:27 AM
L10	160	L3 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:29 AM
L11	2	("20040105292").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:34 AM
L12	1	("20070168781").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:35 AM
L13	17	((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:37 AM
L14	14	((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO;	OR	ON	ON	2022/02/03 10:40 AM

L15	14	SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) ((Lee near2 Hyun) OR (Bhakta near2 Jayesh)) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	JPO; IBM_TDB) (US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:40 AM
L16	12	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:42 AM
L17	12	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:43 AM
L18	20	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:46 AM
L19	19	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:47 AM

L20	19	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:47 AM
L21	1663	(711/120 711/119).ccls.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:50 AM
L22	1	L21 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:50 AM
L23	1	L21 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/02/03 10:51 AM
L24	75	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 11:05 AM
L25	41	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (mode WITH register)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 11:07 AM
L26	6	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (configur\$5 near4 mode WITH register)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 11:11 AM
L27	10	L3 AND (mode WITH operation) AND ((read\$3 AND writ\$3)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU,	OR	ON	ON	2022/06/21 11:12 AM

L28	1	WITH normal near4 mode SAME data SAME strobe) AND (configur\$5 near4 register) ("20100309706").pn.	SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/06/21 12:11 PM
L29	32	("5036221" "6072346" "6184701" "6530006" "6807077" "6889300" "6906555" "7036053" "7275173" "7356639" "7562271" "7808849" "7890732" "8020022" "8407441" "8552779" "8565033" "8566516" "8599595" "8713379" "9153298" "9263103" "9361955" "20060277355").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:50 PM
L30	0	L29 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:51 PM
L31	0	L29 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (mode WITH register)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:51 PM
L32	0	L29 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:51 PM
L33	12	("20070008791" "20080025122" "20100309706" "20100312925" "20100312956" "20110062999" "20120106228" "20120296598").pn.	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:53 PM

L34	3	L33 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:53 PM
L35	0	L33 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (mode WITH register)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:55 PM
L36	3	L33 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:55 PM
L37	3	N22 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT; USOCR; FIT (AU, CN, EP, GB, JP, KR, RU, SG, TW, WO); EPO; JPO; IBM_TDB)	OR	ON	ON	2022/07/17 12:57 PM

PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

PE2E SEARCH - Search History (Prior Art)

There are no Prior Art searches to show.

PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	26792	(G06F12/0802 G06F13/4243 G11C7/1066 G11C7/1093 G11C29/028 G11C5/04 G06F13/4256 G11C29/023 G06F1/10).cpc.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:48 AM
N2	126	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:52 AM
N3	96	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:52 AM
N4	95	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM
N5	115	(control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM

N6	185	N1 AND (control\$4 SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) AND (memory near3 module)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:53 AM
N7	96	N1 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:57 AM
N8	1652	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:57 AM
N9	1652	(711/120 711/119).ccls.	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:58 AM
N10	1	N9 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 10:59 AM
N11	1	N9 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:00 AM
N12	5	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:13 AM
N13	5	((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:13 AM

N14	5	SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer) ((Lee near2 Hyun).in. OR (Bhakta near2 Jayesh).in.) AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N15	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N16	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N17	7	netlist AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/02/03 11:14 AM
N18	24	("5036221" "6072346" "6184701" "6530006" "6807077" "6889300" "6906555" "7036053" "7275173" "7356639" "7562271" "7808849" "7890732" "8020022" "8407441" "8552779" "8565033" "8566516" "8599595" "8713379" "9153298" "9263103" "9361955"	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:56 PM

N19	0	"20060277355").pn. N18 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:56 PM
N20	0	N18 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (mode WITH register)	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:57 PM
N21	0	N18 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:57 PM
N22	8	("20070008791" "20080025122" "20100309706" "20100312925" "20100312956" "20110062999" "20120106228" "20120296598").pn.	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:57 PM
N23	3	N22 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address OR c\$1A)) AND (data near2 buffer SAME memory) AND (PLL OR dll)	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:57 PM
N24	0	N22 AND (mode WITH operation) AND ((read\$3 AND writ\$3) WITH normal near4 mode SAME data SAME strobe) AND (mode WITH register)	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:58 PM
N25	3	N22 AND (control\$4 SAME (input\$4 AND output\$4) SAME clock SAME (command\$1address	(US-PGPUB; USPAT)	OR	ON	ON	2022/07/17 12:58 PM

		OR c\$1A)) AND (data near2 buffer SAME memory) AND ((PLL OR dll) SAME buffer)					
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed July 10, 2022</i>		
				Application Number	17/141,978	
				Filing Date	January 5, 2021	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
Sheet	1	of	7	Attorney Docket Number		NL073-20US02

PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code ²			
	1	5036221 A	1991-07-30	Bruculeri, et al.	
	2	6072346 A	2000-06-06	Ghahremani	
	3	6184701 B1	2001-02-06	Kim, et al.	
	4	6530006 B1	2003-03-04	Dodd et al.	
	5	6807077 B2	2004-10-19	Noda et al.	
	6	6877079 B2	2005-04-05	Yoo et al.	
	7	6889300 B2	2005-05-03	Davis et al.	
	8	6906555 B2	2005-06-14	Ma	
	9	7036053 B2	2006-04-25	Zumkehr	
	10	7275173 B2	2007-09-25	Lindt	
	11	7356639 B2	2008-04-08	Perego et al.	
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	13	7808849 B2	2010-10-05	Swain, et al.	
	14	7890732 B2	2011-02-15	Sukegawa	
	15	8020022 B2	2011-09-13	Noriyuki Tokuhito	
	16	8407441 B2	2013-03-26	Giovannini et al.	
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	19	8566516 B2	2013-10-22	Schakel, et al.	
	20	8599595 B2	2013-12-03	Stephens, Jr.	
	21	8713379 B2	2014-04-29	Takefman, et al.	
	22	9153298 B2	2015-10-06	Stephens, Jr.	
	23	9263103 B2	2016-02-16	Giovannini, et al.	
	24	9361955 B2	2016-06-07	Muralimanohar et al.	
	25	20060277355 A1	12/7/06	Ellsberry, et al.	
	26	20070008791 A1	1/11/07	Butt, et al.	
	27	20080025122 A1	1/31/08	Schakel et al.	
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	29	20100312925 A1	12/9/10	Osanai, et al.	
	30	20100312956 A1	12/9/10	Hiraishi, et al.	
	31	20110062999 A1	3/17/11	Nimalyar, et al.	
	32	20120106228 A1	5/3/12	Lee	
	33	20120296598 A1	11/22/12	Chafin et al.	

Examiner Signature	/ZUBAIR AHMED/	Date Considered	07/17/2022
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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed July 10, 2022</i>		
				Application Number	17/141,978	
				Filing Date	January 5, 2021	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
Sheet	2	of	7	Attorney Docket Number		NL073-20US02

FOREIGN PATENT DOCUMENTS						
Examiner Initials	Cite No.	Foreign Patent Document Country Code ³ – Number ⁴ - Kind Code ⁵ (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T ⁶

NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
	34	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, IPR Petition re US Patent No. 9,824,035, filed December 23, 2021
	35	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, IPR Petition re US Patent No. 10,268,608, filed December 23, 2021
	36	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, 'Ex. 1003 - IPR2022-00236 Declaration of Donald Alpert,' filed December 23, 2021
	37	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, 'Ex. 1003 - IPR2022-00237 Declaration of Donald Alpert,' filed December 23, 2021
	38	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Preliminary Response, filed April 25, 2022
	39	Inter Partes Review of US Patent No. 10268608, Case No. IPRIPR2022-00237, Patent Owner's Preliminary Response, filed April 25, 2022
	40	Inter Partes Review of US Patent No. 9824035, Case No. IPRIPR2022-00236, Patent Owner's Exhibit List, filed April 25, 2022
	41	Complaint for Patent Infringement, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Apr. 28, 2021), 74 pages.
	42	Netlist, Inc.'s Disclosure of Asserted Claims and Preliminary Infringement Contentions to Defendants, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Sept. 10, 2021), 9 pages.
	43	Micron's Preliminary Invalidity Contentions in Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Nov. 5, 2021), 38 pages.

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INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed July 10, 2022</i>		
				Application Number	17/141,978	
				Filing Date	January 5, 2021	
				First Named Inventor	Hyun LEE	
				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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	44	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent Application Publication No. 2010/0312925 A1 to Osanai et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 57 pages.
	45	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 8,713,379 to Takefman et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	46	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 8,020,022 to Tokuhito, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 31 pages.
	47	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 6,877,079 to Yoo et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	48	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 7,562,271 to Shaeffer et al. Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 50 pages.
	49	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 6,530,006 to Dodd et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 51 pages.
	50	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 9,361,955 to Muralimanohar et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 63 pages.
	51	Invalidity Chart for U.S. Patent No. 9,824,035 based on US Patent No. 2012/0106228 to Lee, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 39 pages.
	52	Invalidity Chart for U.S. Patent No. 9,824,035 based on JEDEC Submission entitled "Proposed DDR4 DB BCOM Protocol," Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 22 pages.

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				Art Unit	2139	
				Examiner Name	Zubair AHMED	
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	53	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent Application Publication No. 2010/0312925 A1 to Osanai et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 53 pages.
	54	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 8,713,379 to Takefman et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 35 pages.
	55	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 8,020,022 to Tokuhito, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 30 pages.
	56	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 6,877,079 to Yoo et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 48 pages.
	57	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 7,562,271 to Shaeffer et al. Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 52 pages.
	58	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 6,530,006 to Dodd et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 51 pages.
	59	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 9,361,955 to Muralimanohar et al., Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 49 pages.
	60	Invalidity Chart for U.S. Patent No. 10,268,608 based on US Patent No. 2012/0106228 to Lee, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 36 pages.
	61	Invalidity Chart for U.S. Patent No. 10,268,608 based on JEDEC Submission entitled "Proposed DDR4 DB BCOM Protocol," Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 19 pages.
	62	Obviousness Chart for U.S. Patent No. 9,824,035, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 14 pages.

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				Filing Date	January 5, 2021
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				Art Unit	2139
Examiner Name	Zubair AHMED				
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	63	Obviousness Chart for U.S. Patent No.10,268,608, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431 (W.D. Tex. Nov. 5, 2021), 9 pages.		
	64	Micron’s Preliminary Identification of Proposed Claim Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Dec. 23, 2021), 14 pages.		
	65	Netlist, Inc.’s Preliminary Proposed Claim Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Dec. 23, 2021), 7 pages.		
	66	Defendants’ Identification of Extrinsic Evidence in Support of Their Preliminary Proposed Constructions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Jan. 13, 2022), 21 pages.		
	67	Netlist, Inc.’s Preliminary Identification of Extrinsic Evidence, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Jan 13, 2022), 16 pages.		
	68	Declaration of Harold S. Stone, Ph.D., in Support of Defendants’ Claim Construction Positions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 27 pages.		
	69	Declaration of Alan Jay Smith, Ph.D., in Support of Defendants’ Claim Construction Positions, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 26 pages.		
	70	Micron’s Opening Claim Construction Brief, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No. 6:21-cv-00431-ADA (W.D. Tex. Feb. 3, 2022), 40 pages.		
	71	Joint Claim Construction Chart, Netlist, Inc. v. Micron Technology, Inc., Micron Semiconductor Products, Inc., and Micron Technology Texas LLC, Case No 1:22-cv-00136-LY, Dkt. 66-1 (W.D. Tex. Apr. 14, 2022), 13 pages.		
	72	Excerpt – Modern Dictionary of Electronics 7E, 1999		
	73	JEDEC Solid State Technology Association, 1st Showing, Committee: JC-40.4, Committee Item Number: 0311.xx, ‘Proposed DDR4 DB BCOM Protocol,’ undated, 10 pages		
	74	JEDEC Standard Double Data Rate (DDR) SDRAM Specification, JESD79 (June 2000), 78 pages		
Examiner Signature	/ZUBAIR AHMED/		Date Considered	07/17/2022

INFORMATION DISCLOSURE CITATION Substitute for Form 1449-PTO				<i>Electronically filed July 10, 2022</i>	
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				First Named Inventor	Hyun LEE
				Art Unit	2139
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	75	JEDEC Standard No. 21-C, PC2100 and PC1600 DDR SDRAM Registered DIMM Design Specification, Revision 1.3, January 2002, 83 pages.
	76	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3, DDR3 SDRAM Standard, June 2007, 12 pages
	77	JEDEC Standard JESD79-3F (Revision of JESD79-3E, July 2010), DDR3 SDRAM Standard, July 2012, 226 pages.
	78	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3A, DDR3 SDRAM Specification, September 2007, 12 pages
	79	Excerpt - JEDEC Standard DDR3 SDRAM JESD79-3C, DDR3 SDRAM, April 2008, 7 pages
	80	Excerpt - JEDEC Standard No. 21C, DDR4 SDRAM Load Reduced DIMM Design Specification, August 2015, 2 pages
	81	Inter Partes Review of US Patent No. 10860506, Case No. IPR2022-00711, IPR Petition re US Patent No. 10,860,506, filed March 22, 2022.
	82	Inter Partes Review of US Patent No. 10860506, Case No. IPR2022-00711, 'Ex. 1003 - IPR2022-00711 Declaration of Robert Wedig,' filed March 22, 2022.
	83	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, IPR Petition re US Patent No. 9,128,632, filed January 20, 2017.
	84	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, 'Ex. 1003 - IPR2017-00730 Declaration of Trevor Mudge,' filed January 20, 2017.
	85	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Patent Owner's Preliminary Response, filed April 30, 2017.
	86	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Decision Denying Institution of Inter Partes Review, entered July 21, 2017.
	87	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, PETITIONERS' REQUEST FOR REHEARING, filed August 21, 2017.
	88	Inter Partes Review of US Patent No. 9128632, Case No. IPR2017-00730, Decision Denying Petitioner's Request for Rehearing, entered October 30, 2017.
	89	Inter Partes Review of US Patent No. 9606907, Case No. IPR2018-00362, Final Written Decision, entered June 27, 2019.
	90	Complaint, Netlist, Inc. v. Samsung Electronics Co., Ltd., Samsung Electronics America, Inc., Samsung Semiconductor, Inc., Case No. 2:21-cv-463 (E.D. Tex. Dec. 20, 2021), 49 pages.
	91	First Amended Complaint for Declaratory Judgment of Non-Infringement and Unenforceability; Breach of Contract, Samsung Electronics Co., Ltd. and Samsung Semiconductor, Inc. v. Netlist, Inc., C.A. No. 21-1453 (RGA) (USDC, Delaware, Jan. 8, 2022), 122 pages.

Examiner Signature	/ZUBAIR AHMED/	Date Considered	07/17/2022
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				Art Unit	2139
				Examiner Name	Zubair AHMED
Sheet	7	of	7	Attorney Docket Number	NL073-20US02

Examiner Signature	/ZUBAIR AHMED/	Date Considered	07/17/2022
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Electronically filed October 12, 2022

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE, et al.	Confirmation No.:	4088
Serial No.:	17/141,978	Group Art Unit:	2139
Filed:	January 5, 2021	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		Attorney Docket No.: NL073-20US02

AMENDMENT UNDER 37 C.F.R. 1.312

Mail Stop: Amendment

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

The Amendment is filed after the Notice of Allowance dated July 13, 2022, and is submitted under 37 C.F.R. 1.312.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US02).

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Cancelled)
2. (Previously Presented) A memory module operable in a computer system having a memory controller and a system bus, the system bus including one or more clock signal lines, a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module comprising:

a module control circuit configurable to receive from the memory controller a system clock via the one or more clock signal lines and input address and control (C/A) signals via the set of C/A signal lines, and to output a module clock signal and module C/A signals in response to the system clock and the input C/A signals;

memory devices organized in one or more ranks; and

buffer circuits configurable to receive the module clock signal from the module control circuit and to generate a plurality of local clock signals, wherein a respective local clock signal of the plurality of local clock signals is output by a respective buffer circuit to a respective group of the memory devices, the respective group of the memory devices including at least one respective memory device in each of the one or more ranks, the respective local clock signal having a respective programmable phase relationship with the module clock signal, the buffer circuits including logic and configuration registers, wherein respective configuration registers included in the respective buffer circuit are programmable by the logic to control the respective programmable phase relationship; and

wherein:

the memory module is operable in at least a normal operation mode and a configuration mode;

the memory module in the normal operation mode is configurable to output or receive data/strobe signals via the data/strobe signal lines in response to a memory read or write command received via the C/A signal lines;

the memory module in the configuration mode is configurable to perform a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices and at least one read operation to read from the set of memory locations; and

the logic is further configurable to program the respective configuration registers based on information derived from the set of operations.

3. (Currently Amended) The memory module of claim 2, wherein the respective group of the memory devices is configurable to receive the module C/A signals and the respective local clock signal[[,]] from the module control circuit, and to communicate respective data and data strobe signals with the memory controller in response to the module C/A signals and the respective local clock signal via a respective set of the plurality of sets of ~~n-bit wide~~ data/strobe signal lines.

4. (Previously Presented) The memory module of claim 3, wherein the respective buffer circuit is configurable to generate the respective local clock signal based on values stored at the respective configuration registers.

5. (Previously Presented) The memory module of claim 4, wherein the module clock signal received at a first buffer circuit is phase shifted from the module clock signal received at a second buffer circuit closer to the module control circuit than the first buffer circuit, and wherein values stored at first configuration registers in the first buffer circuit of the buffer circuits are different from values stored at second configuration registers at the second buffer circuit of the buffer circuits.

6. (Currently Amended) The memory module of claim 2, wherein the respective configuration registers are programmed during [[a-]] one or more mode register write operations ~~operation~~.

7. (Previously Presented) The memory module of claim 6, wherein the logic is configured to program the respective configuration registers according to signals output from the module control circuit when the memory module is in the configuration mode.

8. (Currently Amended) The memory module of claim 7, wherein ~~the module control circuit is configurable to output the signals~~ output from the module control circuit are output in response to mode register command signals received from the memory controller.

9. (Previously Presented) The memory module of claim 2, wherein the respective buffer circuit further includes a respective phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a respective programmable delay circuit to add a respective delay to the respective local clock signal, and a respective clock driver to drive the respective local clock signal to the respective group of the memory devices.

10. (Previously Presented) The memory module of claim 9, wherein the respective programmable delay circuit is controlled by the logic through the respective configuration registers.

11. (Currently Amended) The memory module of claim 2, further comprising a printed circuit board, wherein the module control circuit, the ~~plurality of~~ buffer circuits and the memory devices are mounted ~~[[to]]~~ on the printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and wherein the buffer circuits are distributed along the edge of the printed circuit board and between the memory devices and the edge connectors.

12. (Currently Amended) A method of operating a memory module coupled to a memory controller via a system bus, the system bus including one or more clock signal lines, a set of address/control (C/A) signal lines and a plurality of sets of data/strobe signal lines, the memory module including memory devices organized in one or more ranks, the method comprising:

receiving a system clock from the memory controller via the one or more clock signal lines and input address and control (C/A) signals from the memory controller via the set of C/A signal lines;

outputting a module clock signal and module C/A signals in response to the system clock and the input C/A signals;

generating a plurality of local clock signals in response to the module clock signal; and

outputting a respective local clock signal of the plurality of local clock signals to a respective group of the memory devices, the respective group of the memory devices including at least one memory device in each of the one or more ranks, the respective local clock signal having a respective programmable phase relationship with the module clock signal;

wherein the memory module is operable in at least a normal operation mode and a configuration mode, the method further comprising

outputting or receiving data/strobe signals via the data/strobe signal lines in response to a memory read or write command received via the C/A signal lines when the memory module is in the normal operation mode;

performing a set of operations including at least one write operation to write a set of data into a set of memory locations in the memory devices and at least one read operation to read from the set of memory locations when the memory module is in the configuration mode; and

programming the respective programmable phase relationship based on information derived from the set of operations when the memory module is in the configuration mode;

wherein the memory module further comprises:

a module control circuit configurable to receiving the system clock from the memory controller via the one or more clock signal lines and the input address and control (C/A) signals from the memory controller via the set of C/A signal lines, and to output the module clock signal and the module C/A signals in response to the system clock and the input C/A signals;

a plurality of buffer circuits configurable to receive the module clock signal from the module control circuit and to generate the plurality of local clock signals, wherein each respective buffer circuit of the plurality of buffer circuits is coupled to a respective group of the memory devices and is configurable to output the respective local clock signal of the plurality of local clock signals to the respective group of the memory devices, wherein the plurality of buffer circuits including logic and configuration registers, and wherein respective configuration registers included in the respective buffer circuit are programmable by the logic to control the respective programmable phase relationship.

13. (Currently Amended) The memory module of claim 12, further comprising, at the respective group of the memory devices:

receiving the module C/A signals and the respective local clock signal; and

communicating respective data and data strobe signals with the memory controller in response to the module C/A signals and the respective local clock signal via a respective set of the plurality of sets of ~~n-bit wide~~ data/strobe signal lines.

14. (Previously Presented) The memory module of claim 13, further comprising generating the respective local clock signal using a respective delay circuit based on values stored at the respective configuration registers.

15. (Previously Presented) The memory module of claim 14, wherein:
a first local clock signal is output to a first group of memory devices; and
a second local clock signal is output to a second group of memory devices further away from the module control circuit than the first group of memory device;
the first local clock signal is phase shifted by a first delay from the module clock signal;
and
the second local clock signal is phase shifted by a second delay from the module clock signal, the second delay being different from the first delay.

16. (Currently Amended) The memory module of claim 14, further comprising programming the respective configuration registers during ~~[[a-]]~~ one or more mode register write operations ~~operation~~.

17. (Previously Presented) The memory module of claim 16, wherein the respective configuration registers are programmed according to signals output from the module control circuit when the memory module is in the configuration mode.

18. (Currently Amended) The memory module of claim 17, wherein ~~the module control circuit is configurable to output the signals~~ output from the module control circuit are output in response to mode register command signals received from the memory controller when the memory module is in the configuration mode.

19. (Previously Presented) The memory module of claim 12, further comprising:
phase locking the respective local clock signal with the module clock signal;
adding a respective delay to the respective local clock signal; and
driving the respective local clock signal to the respective group of the memory devices.

20. (Previously Presented) The memory module of claim 19, wherein the respective delay is controlled by logic through respective configuration registers.

21. (Currently Amended) The method of 12, wherein the memory module further comprises:
a printed circuit board, wherein the module control circuit, the plurality of buffer circuits, and the memory devices are mounted ~~[[τ]]~~on the printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of C/A signal lines and the plurality of sets of data/strobe signal lines, and wherein the buffer circuits are distributed along the edge of the printed circuit board and between the memory devices and the edge connectors.

REMARKS

This amendment is submitted under 37 C.F.R. 1.312 and responds to the Notice of Allowance dated July 13, 2022, in which claims 2-21 are allowed.

REMARKS CONCERNING CLAIMS

Claims 3, 6, 8, 11-13, 16, 18 and 21 are amended to address minor issues of consistency or clarity.

No new matter has been added and the amendments should not necessitate additional search or examination.

The Examiner is invited to call the undersigned attorney at (650) 521-4828 for any questions regarding this amendment and the application.

Respectfully submitted,

Date: October 12, 2022

/Jamie J. Zheng/
Jamie J. Zheng
Customer No. 79141

51,167
(Reg. No.)

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(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	10/13/2022

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 USCH Law, PC

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Netlist, Inc.

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 60-3916

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29☐ Applicant asserting small entity status. See 37 CFR 1.27☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Jamie J. Zheng/Date October 12, 2022Typed or printed name Jamie J. ZhengRegistration No. 51167



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 07/13/2022
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

EXAMINER	
AHMED, ZUBAIR	
ART UNIT	PAPER NUMBER

2132

DATE MAILED: 07/13/2022

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1200	\$0.00	\$0.00	\$1200	10/13/2022

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

Electronic Patent Application Fee Transmittal

Application Number:	17141978			
Filing Date:	05-Jan-2021			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng			
Attorney Docket Number:	NL073-20US02			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1200	1200

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1200

Electronic Acknowledgement Receipt

EFS ID:	46808345
Application Number:	17141978
International Application Number:	
Confirmation Number:	4088
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	NL073-20US02
Receipt Date:	13-OCT-2022
Filing Date:	05-JAN-2021
Time Stamp:	01:41:42
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$ 1200
RAM confirmation Number	E20220C242271324
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		NL073-20US02_312_Amendment.pdf	161862	yes	8
			8edf07bdab0525bd394596850ac617188fad87cf		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Amendment after Notice of Allowance (Rule 312)		1	1	
	Claims		2	7	
	Applicant Arguments/Remarks Made in an Amendment		8	8	
Warnings:					
Information:					
2	Issue Fee Payment (PTO-85B)	NL073-20US02_Issue_Fee.pdf	381002	no	2
			b3750234689004f4951162cd2800253671813ec4		
Warnings:					
Information:					
3	Fee Worksheet (SB06)	fee-info.pdf	37684	no	2
			adf8b505c253526d0f843546a3bc9d62ab55515e		
Warnings:					
Information:					
Total Files Size (in bytes):			580548		

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	01/05/2021	Hyun Lee	NL073-20US02	4088
79141	7590	10/21/2022	EXAMINER	
USCH Law, PC			AHMED, ZUBAIR	
3790 El Camino Real #1147				
Palo Alto, CA 94304				
			ART UNIT	PAPER NUMBER
			2132	
			NOTIFICATION DATE	DELIVERY MODE
			10/21/2022	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

eofficeaction@appcoll.com
jzheng@uschlaw.com
patents@uschlaw.com

Electronically filed October 12, 2022

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun LEE, et al.	Confirmation No.:	4088
Serial No.:	17/141,978	Group Art Unit:	2139
Filed:	January 5, 2021	Examiner:	Zubair AHMED
For:	<i>MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION</i>		Attorney Docket No.: NL073-20US02

AMENDMENT UNDER 37 C.F.R. 1.312

Mail Stop: Amendment

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

The Amendment is filed after the Notice of Allowance dated July 13, 2022, and is submitted under 37 C.F.R. 1.312.

The Commissioner is hereby authorized to charge any required fee(s) to Deposit Account No. 60-3916 (order no. NL073-20US02).



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/141,978	11/29/2022	11513955	NL073-20US02	4088

79141 7590 11/09/2022
USCH Law, PC
3790 El Camino Real #1147
Palo Alto, CA 94304

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

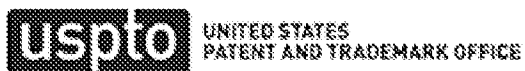
INVENTOR(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional inventors):

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

NETLIST, INC., Irvine, CA;

The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The USA offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to encourage and facilitate business investment. To learn more about why the USA is the best country in the world to develop technology, manufacture products, and grow your business, visit SelectUSA.gov.



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FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
17/141,978	01/05/2021	411112

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Maintenance Fee Address

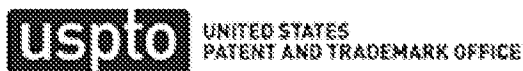
The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167



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Alexandria, Virginia 22313-1450
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CORRESPONDENCE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
17/141,978	01/05/2021	411112

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

- Correspondence Address

The address for Customer Number 207611 is:

207611 - Netlist, Inc.
111 Academy
Irvine , CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167

UTILITY PATENT APPLICATION TRANSMITTAL <i>(Only for new nonprovisional applications under 37 CFR § 1.53(b))</i>		Attorney Docket No.	129980-5073-US01		
		First Inventor	Hyun Lee		
		Title	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		
		Electronically filed	June 5, 2019		
APPLICATION ELEMENTS <i>See MPEP Chapter 600 concerning utility patent application contents.</i>		<i>Address to:</i> Commissioner for Patents P.O. Box 1450 Alexandria, VA 22313-1450			
1. ☐ Fee Transmittal Form <i>(with duplicate for fee processing)</i> 2. ☐ Applicant claims Small Entity status, see 37 C.F.R. § 1.27 3. ☒ Specification [Total Pages 34] 4. ☒ Drawing(s) <i>(35 USC § 113)</i> [Total Sheets 22] 5. ☒ Oath or Declaration [Total Pages 2] a. ☐ Newly executed <i>(original or copy)</i> b. ☒ Copy from a prior application <i>(37 CFR § 1.63(d))</i> 6. ☒ Application Data Sheet, see 37 C.F.R. § 1.76 7. ☐ CD-ROM or CD-R in duplicate, large table or Computer Program <i>(Appendix)</i> ☐ Landscape Table on CD 8. ☐ Nucleotide and/or Amino Acid Sequence Submission <i>(if applicable, all necessary)</i> a. ☐ Computer Readable Form (CRF) b. ☐ Specification Sequence Listing on i. ☐ CD-ROM or CD-R (2 copies); or ii. ☐ Paper c. ☐ Statement verifying identity of above copies		ACCOMPANYING APPLICATION PARTS 9. ☐ Assignment Papers <i>(cover sheet & document(s))</i> Name of Assignee: 10. ☒ 37 CFR § 3.73(c) Statement a. ☒ Power of Attorney 11. ☐ English Translation Document <i>(if applicable)</i> 12. ☐ Information Disclosure Statement and PTO-1449 a. ☐ Copies of citations attached 13. ☐ Preliminary Amendment 14. ☐ Return Receipt Postcard <i>(MPEP 503)</i> 15. ☐ Certified Copy of Priority Document(s) <i>(if foreign priority is claimed)</i> 16. ☐ Non-Publication Request under 35 U.S.C. § 122 (b)(2)(B)(i) 17. ☐ Other:			
Note: (1) Benefit claims under 37 CFR 1.78 and foreign priority claims under 1.55 MUST be included in an Application Data Sheet (ADS). (2) For applications filed under 35 U.S.C. 111, the application must contain an ADS specifying the applicant if the applicant is an assignee, person to whom the inventor is under an obligation to assign, or person who otherwise shows sufficient proprietary interest in the matter. See 37 CFR 1.46(b).					
19. CORRESPONDENCE ADDRESS: Customer Number 79141					
Signature	/Jamie J. Zheng/	Date	June 5, 2019		
Name (Print/Type)	Jamie J. Zheng	Registration No. (Attorney/Agent)	51,167		

GENERAL POWER OF ATTORNEY BY APPLICANT FOR UNITED STATES PATENT

I hereby revoke all previous powers of attorney given in the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

I hereby appoint the practitioner associated with the following Customer Number
:

79141

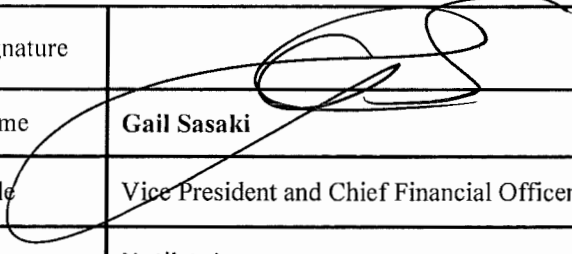
as my/our attorney or agent, and to transact all business in the United States Patent and Trademark Office in connection with the application referenced in the attached transmittal letter (form PTO/AIA/82A or equivalent).

Please recognize or change the **correspondence address** for the application referenced in the attached transmittal letter to the address associated with the above-mentioned **Customer Number**.

I am the Applicant:

- ☐ Inventor or Joint Inventor
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is currently being filed in this document)

SIGNATURE of Applicant for Patent

Signature		Date: January 9, 2013
Name	Gail Sasaki	
Title	Vice President and Chief Financial Officer	
Company	Netlist, Inc.	

Note: Signature - this form must be signed by the Applicant or a person authorized to act on behalf of Applicant.

Submit multiple forms for more than one signature
Total of ___ forms submitted.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	To be assigned
Serial No.:	To be assigned	Art Unit	To be assigned
Filed:	Filed herewith	Examiner:	To be assigned
For:	Memory Module With Local Synchronization And Method Of Operation	Attorney Docket No.:	129980-5073-US01

STATEMENT UNDER 37 C.F.R. § 3.73(c)

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

Netlist, Inc., a corporation, states that it is the assignee of the entire right, title and interest in the patent application/patent identified above by virtue of an assignment from the inventor(s) of the parent of the patent application/patent identified above.

The assignment was recorded in the United States Patent and Trademark Office on April 17, 2016 at Reel 038300, Frame 0101, or for which a copy thereof is attached.

The undersigned is authorized to act on behalf of the assignee.

Date: June 5, 2019

/Jamie J. Zheng/	51,167
Jamie J. Zheng	(Reg. No.)
MORGAN, LEWIS & BOCKIUS LLP	
1400 Page Mill Road	
Palo Alto, CA 94304	
(650) 843-4000	

**COMBINED DECLARATION AND ASSIGNMENT
FOR UTILITY OR DESIGN APPLICATION**

**Title of
Invention**

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

DECLARATION

As a below named inventor, I hereby declare that:

This declaration
is directed to:

☐ The attached application, or

☒ United States application or PCT international application
number 14/445,035 filed on July 28, 2014.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Note to Inventor: 37 C.F.R. § 1.63(c) states: "A person may not execute an oath or declaration for an application unless that person has reviewed and understands the contents of the application, including the claims, and is aware of the duty to disclose to the Office all information known to the person to be material to patentability as defined in § 1.56."

ASSIGNMENT

In consideration of the sum of One Dollar (\$1.00) and other good and valuable consideration paid, I hereby sell and assign to

Netlist, Inc.

(Name of Assignee)

of 175 Technology Dr., Suite #150, Irvine, CA 92618

(Address of Assignee)

(hereinafter designated as the Assignee) my entire right, title and interest for the United States as defined in 35 USC 100, for any invention set forth in the above-identified application.

I agree to execute all papers necessary in connection with this application for patent in the USPTO for the invention, and any continuation, divisional or reissue applications thereof and also to execute separate assignments in connection with such applications as the Assignee may deem necessary or expedient.

I agree to execute all papers necessary in connection with any post-grant proceeding which may occur in connection with this application or continuation, divisional or reissue thereof and to cooperate with the Assignee in every way possible in obtaining evidence and going forward with such post-grant proceeding.

I agree to execute all papers and documents and perform any act which may be necessary in connection with claims or provisions of the International Convention for Protection of Industrial Property or similar agreements.

I agree to perform all affirmative acts which may be necessary to obtain a grant of a valid United States patent to the

Assignee.

I hereby authorize and request the USPTO to issue any and all Letters Patents of the United States resulting from the application or any continuation, divisional or reissue applications thereof to the Assignee, as Assignee of my entire interest, and covenant that I have not executed, and will not execute, any agreement in conflict herewith.

First Inventor (Legal Name): Hyun Lee

Signature: 

Date: 3/31/2016

First Inventor (Legal Name): Jayesh R. Bhakta

Signature: 

Date: 3/31/2016

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

ABSTRACT OF THE DISCLOSURE

A memory module is operable in a memory system with a memory controller. The memory module comprises a module control device to receive command signals from the memory controller and to output module C/A signals and data buffer control signals. The module C/A signals are provided to memory devices organized in groups, each group including at least one memory device, while the data buffer control signals are provided to a plurality of buffer circuits to control data paths in the buffer circuits, a respective buffer circuit corresponding to a respective group of memory devices. The plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time. The plurality of buffer circuits include clock regeneration circuits to regenerate a clock signal received from the module control device and to provide regenerated clock signals to respective groups of memory devices.

WE CLAIM:

1. A memory module configured to couple to a memory controller via a $N \times n$ bit wide memory bus and one or more clock signal lines, the memory bus including a set of control signal lines and N sets of n -bit wide data/strobe signal lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers each configured to receive the module clock signal and the data buffer control signals from the module control circuit, a respective data buffer of the N data buffers being configured to operatively couple to a respective set of n -bit wide data/strobe signal lines, the respective data buffer including a n -bit wide data path and logic configured to control the data path in response to the data buffer control signals, the n -bit wide data path including at least one programmable delay element controlled by the logic, the respective data buffer being further configured to generate a respective local clock signal in response to the module clock signal, the respective local clock signal being phase-locked with the module clock signal and having a respective programmable delay from the module clock signal; and

memory devices organized in one or more ranks and including N groups of memory devices corresponding to respective ones of the N data buffers, each group of memory devices including at least one memory device in each of the one or more ranks, a respective group of the N groups of memory devices receiving the module C/A signals from the control circuit and a respective local clock signal from a corresponding one of the N data buffers, the respective group of the N groups of memory devices communicating data and data strobe signals with the memory controller via the n -bit wide data path in the corresponding one of the N data buffers and in response to the module C/A signals and the respective local clock signal.

2. The memory module of claim 1, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between the N groups of memory devices and the edge connectors.

3. The memory module of claim 2, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a programmable delay circuit to delay the local clock signal by the programmable delay, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.

4. The memory module of claim 3, wherein the programmable delay circuit is controlled by the logic.

5. The memory module of claim 4, wherein the respective data buffer includes one or more programmable registers programmable by the logic and configured to control the respective programmable delay.

6. The memory module of claim 5, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer the data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective data buffer is configured to program the one or more programmable registers during the configuration mode.

7. The memory module of claim 1, wherein the respective data buffer includes one or more programmable registers programmable by the logic and configured to control the respective programmable delay.

8. The memory module of claim 7, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer the data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective data buffer is configured to program the one or more programmable registers during the configuration mode.
9. The memory module of claim 7, wherein the respective data buffer includes one or more programmable registers configurable by the logic to time the data and data strobes being communicated via the respective data buffer.
10. The memory module of claim 8, wherein the respective data buffer is configured to program the one or more registers according to signals output from the control circuit during the configuration mode.
11. The memory module of claim 7, wherein the control circuit does not transmit any clock signals to the memory devices.
12. The memory module of claim 7, wherein the data buffer control signals are different from the module C/A signals.
13. The memory module of claim 7, wherein the memory devices are organized in one or more ranks and are each n -bit or $n/2$ -bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks.
14. The memory module of claim 13, wherein each of the memory devices is n -bits wide.
15. The memory module of claim 13, wherein each of the memory devices is $n/2$ bits wide.
16. A method of operating a memory module couple to a memory controller via a $N \times n$ bit wide memory bus and one or more clock signal lines, the memory bus including a set of control signal lines and N sets of n -bit wide data/strobe signal lines, where N and n are

integers larger than one, the memory module including memory devices organized in one or more ranks, the memory devices including N groups of memory devices corresponding to respective sets of the N sets of n -bit wide data/strobe signal lines, each group of memory devices including at least one memory device in each of the one or more ranks, the method comprising:

receiving a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines;

outputting a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

controlling N data paths in response to the data buffer control signals, each respective data path of the N data paths being n -bit wide and coupled between a respective set of n -bit wide data/strobe signal lines and a respective group of memory devices, wherein controlling the each respective data path include controlling a respective programmable delay element;

generating N local clock signals in response to the module clock signal, each respective local clock signal of the N local clock signals being phase-locked with the module clock signal and having a respective programmable delay from the module clock signal;

outputting the N local clock signals to respective groups of N groups of memory devices; and

performing memory read or write operations at a selected rank of the memory devices in response to the module C/A signals and the N local clock signals to communicate data and data strobe signals with the memory controller via the N data paths.

17. The memory module of claim 16, further comprising operating the memory module in a configuration mode, and programming one or more programmable registers in the configuration mode to control the respective programmable delay.

18. The memory module of claim 17, further comprising programming one or more additional programmable registers in the configuration mode to control timing of the data and data strobes.

19. The memory module of claim 16, wherein the module clock signal is not received by any of the memory devices.
20. The memory module of claim 16, wherein the data buffer control signals are different from the module C/A signals.

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. Patent Application Serial No. 14/445,035, filed on July 28, 2014, entitled “ Memory Module With Local Synchronization,” which claims the benefit of U.S. Provisional Patent Application Serial No. 61/859,215, filed on July 27, 2013, entitled "Memory Module With Local Synchronization Using Distributed Data Buffers," each of which is herein fully incorporated by reference in its respective entirety.

FIELD

[0002] The disclosure herein is related generally to memory modules, and more particularly to a memory module with local synchronization and method of operation.

BACKGROUND

[0003] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit

wide data path.

[0004] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0005] The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0006] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0007] For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules. The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0008] In some conventional memory systems, the memory controllers include leveling mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and

memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0010] FIGS. 2A-2D are each a diagrams illustrating interactions among components in a a memory module according to certain embodiments.

[0011] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0012] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0013] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0014] FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0016] FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

[0017] FIG. 7A is a flowchart illustrating a method of operation for a memory module according to certain embodiments.

[0018] FIG. 7B is a flowchart illustrating a method of training locally synchronized clock

signals according to certain embodiments.

[0019] FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0020] FIG. 9 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0021] FIG. 10 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0022] FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0023] FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0024] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0025] FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

[0026] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g., read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The C/A signals may include, for example, a row address strobe

signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

[0027] In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

[0028] FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of system data/strobe signal lines 130. Also as shown, each memory module

110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

[0029] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0030] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

[0031] Furthermore, the memory devices 112 may include any type of memory devices. For example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory (SDRAM) devices in compliance with, for example, the Double Data Rate Type 3 (DDR3)

or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0032] As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0033] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0034] Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered

version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Patent No. 7,532,537, entitled "Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation," issued on May 12, 2009, the entirety of which is incorporated herein by reference.

[0035] Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers 118, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer control signals are used to convey configuration or module self-test control information from the module control device 116 to the data buffers 118. In one embodiment, the data buffer control signals are transmitted to the data buffers 118 via respective data buffer control signal lines 230. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers 118 via the data buffer control signal lines and decoded/processed at the data buffers.

[0036] Module control device 116 transmits the module C/A signals to the memory

devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers 118 via data buffer control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the data buffers 118 share a same set of data buffer control signal lines 230.

[0037] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices $M_{11}, \dots, M_{i1}, M_{i+1,1}, \dots, M_{N,1}$, a second rank includes memory devices $M_{12}, \dots, M_{i2}, M_{i+1,2}, \dots, M_{N,2}$, and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11}, M_{12}, M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0038] In certain embodiments, the data buffers 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers 118 is associated with the first group of memory devices M_{11}, M_{12}, M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory devices, data buffer ID- i among the data buffers 118 is associated with the i^{th} group of memory devices M_{i1}, M_{i2}, M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be associated with a same group of memory devices, or vice versa.

[0039] In one embodiment, each group or sets of memory devices are coupled to the associated data buffer 118 via a set of module data/strobe lines 210 and one or more clock

signal lines 211. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0040] In one embodiment, the data buffers 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each data buffer 118 is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the data buffer control signals. Further, each data buffers 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0041] In one embodiment, the data buffers 118 are distributed across the memory module

110 or the module board 119 in positions corresponding to edge connectors 121 coupled to respective groups of system data/strobe signal lines 130. For example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

[0042] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such data include a number of row addresses, a number of column addresses, a data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0043] In certain embodiments, the SPD 240 and/or the control circuit 116 store module

configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0044] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system 100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the memory devices 112.

[0045] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level

physical memory domain.

[0046] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

[0047] For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of interface timing training may be required because the system memory controller 101 has very limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0048] Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very significant portion of a data period. As a result, the clock skew issues cannot simply be

addressed by pipelining the data paths.

[0049] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer 118 in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit 116 and the group of locally synchronized components.

[0050] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations, including, for example: programmable phase relationship between the clock it receives and the clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices 112, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller 101, and/or programmable phase adjustment for one or more control signal from the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data buffers, as disclosed in commonly-owned U.S. Patent 8,001,434, issued August 16, 2011, entitled "Memory Board with Self-Testing Capability," which is incorporated herein by reference in

its entirety.

[0051] In certain embodiments, operations of the data buffers 118 are controlled by the data buffer control signals from the module control circuit 116, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers 118 to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers 118 are distributed across the memory module, the data buffer control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. Patent Application No. 13/952,599, filed July 27, 2013, entitled "Memory Module with Distributed Data Buffers and Method Of Operation," the entirety of which is incorporated herein by reference.

[0052] Because the data buffers 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signals lines 230 from the module control device 116 to the farthest positioned data buffers 118, such as data buffer ID-1 and data buffer ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of data buffer control signals may reach different data buffers 118 at different times across more than one clock cycle of the system clock. For

example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

[0053] With the data buffers 118 receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the data buffers 118. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used to adjust the timing of the transmission of read data to the MCH 101 during a subsequent read operation, as explained in more detail below.

[0054] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0055] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n data (DQ) signal lines 322 each for transmitting one of a set of bits $DQ_0, DQ_1, \dots, DQ_{n-1}$, in a data signal and at least one strobe (DQS) signal line 324 for transmitting at least

one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line Y_{DQS} . When the subsets of memory devices are coupled to the associated data buffer 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

[0056] Each data buffer 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, YA_1, \dots, YA_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each data buffer 118 further includes an DB Control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second module strobe signal line YB_{DQS} .

[0057] The DB Control circuit 310 also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines 230, and outputs datapath control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit 310 also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective DQ signal line 322 with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

[0058] In certain embodiments, the DB Control circuit 310 also provides one or more

delay signals DS, which is used by the DQ routing circuits 320 to control the timing of read and/or write data output by the data buffer 118, as explained in further detail below. In certain embodiments, the DB Control circuit 310 regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer 118 and the respective group of memory devices form a group of locally synchronized devices on the memory module 110.

[0059] The memory devices 112 in each group of locally synchronized devices are coupled to the data buffer 118 in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices 112 in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer 118 via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M₁₁, M₁₂, M₁₃, and M₁₄ in the first group of memory devices can be coupled to the data buffer ID-1 via a same set of module data lines Y-1₀, Y-1₁, ..., Y-1_{n-1} and module strobe line Y-1_{DQS}. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0060] Alternatively, as shown in FIG. 4B, memory devices M₁₁ and M₁₃, which form a subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YA-1₀, YA-1₁, ..., YA-1_n and module strobe line YA-1_{DQS} and memory devices M₁₂ and M₁₄, which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines YB-1₀, YB-1₁, ..., YB-1_n and module strobe line YB-1_{DQS}. Memory devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board 119. Memory devices coupled to the same data buffers may be placed side-by-side, on opposite sides of the

module boards 119, or stacked over each other, and/or over the associated data buffer.

[0061] Multiple memory devices having a data width that is less than a data width of the data buffers 118 may be used in place of one of the memory devices 112, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M_{11-1} and M_{11-2} may be used in place of the memory device M_{11} . Each of the two memory devices M_{11-1} and M_{11-2} has a data width of 4, and together they act like a memory device M_{11} of a data width of 8. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory circuit M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}.

[0062] In another embodiment, as shown in FIG. 5B, four memory devices M_{11-1} to M_{11-4} may be used as the memory device M_{11} . Each of the four memory devices M_{11-1} to M_{11-4} has a data width of 4, and together they act like a memory device M_{11} of a data width of 16. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines YA-1₀, ..., YA-1₃ and module strobe line YA-1_{DQS-1} while memory device M_{11-2} is coupled to the data buffer ID-1 via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}, and so on.

[0063] FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits 630 on each data buffer. In certain embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit 116 uses the DBCS to configure the programmable features in the data buffers 118 based on configuration information received from the MCH 101; in the test mode, the DBCS is used to control the self-test processes run by the data buffers; in the normal operation mode, the DBCS includes the one or more

enable signals and one or more ODT signals.

[0064] FIG. 6B illustrates the DB Control circuit 310 in an data buffer 118. As shown, the DB Control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, a phase-locked loop circuit (PLL) 611 that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit 615 for adjusting the phase of the locally synchronized clock CK0, and a clock driver 618 that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to a corresponding system DQS signal line 324 and on another side to corresponding module DQS signal lines YADQS and YBDQS. The DB Control circuit 310 further includes receiver circuits 630 to receive respective ones of the DBCS.

[0065] The DB Control circuit 310 further includes a control processing circuit 640 that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more data path enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the control processing circuit 640, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

[0066] The strobe routing circuit 620 also buffers strobe signals received from either the MCH 101 or the memory devices 112, and outputs either a write strobe WDQS or read strobe RDQS, respectively, to the DQ routing circuits 320. In one embodiment, the DB

Control circuit 310 further includes a dynamic delay control circuit 650 that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit 320 and the strobe routing circuit 620.

[0067] In certain embodiments, as shown in FIG. 6C, the control processing circuit 640 includes control logic 660 that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals 330. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals 68 to programmable registers 671 that provides static control of the timing of read data/strobe signals, configuration signals 682 to programmable registers 671 that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals 683 to programmable registers 673 that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

[0068] The programmable registers 671 is coupled via signal lines 69 the dynamic delay control circuit 650, which also takes as input a DQS signal, clock signal CK, and one of the datapath enable signals EN, and output the delay signal DS1. The dynamic delay control circuit 650 is described in further detail below.

[0069] FIG. 7A is a flowchart illustrating a method 700 performed by the memory module 110 according to certain embodiments. As shown in FIG. 7A, the method includes receiving 710 clock signal together with system C/A signals from the MCH 101, and generating 715 module clock signal CK. The module clock signal CK is transmitted 720 to a plurality of distributed data buffers 118, which generates 725 locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits 730 its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform 735 memory operations in accordance with the respective locally synchronized clock signals.

[0070] Since the module control circuit 116 generates the DBCS based on signals from the MCH 101, the MCH 101 can train the data buffers 118 to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit 116 generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH 101. As shown in FIG. 7B, a method 750 for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing 751 data to a set of memory locations in the memory devices 112, which is followed by performing a read operation by reading 752 data from the set of memory locations. The read data is compared 753 with the write data to determine 754 if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit 116 agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit 116 do not agree with the corresponding write data bits. The disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers 118 further away from the module control circuit 116 being excessively delayed during transmission, and proper phase adjustment can be programmed 755 into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

[0071] FIG. 8 illustrates a DQ or DQS routing circuit 320 or 620 according to an embodiment. As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 801 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 802 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS}, or YA/YA_{DQS} and YB/YB_{DQS}. The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 810 that buffers write data/strobe, a write data/strobe receiver 820 that samples the write data/strobe, and a write delay circuit 825 controlled by the delay signal DS2. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 830

that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

[0072] The DQS routing circuit further includes a plurality of read paths 850 that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit 860 by an amount controlled by the delay signal DS1, and sampled by a sampler circuit 870. The sampled read data/strobe is transmitted by transmitter 880 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 801.

[0073] FIG. 9 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 901 that is coupled to a corresponding DQS signal line 324, a second DQS pin 902A that is coupled to a corresponding module DQS line Y_{ADQS} , a third DQS pin 902B that is coupled to a corresponding module DQS line Y_{BDQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first DQS pin 901 and the second DQS pin 902A and a second write strobe path coupled between the first DQS pin 901 and the third DQS pin 902B. The first write strobe path includes a write strobe buffer 910 that buffers a write strobe, a write strobe receiver 920 that samples the write strobe according to the buffered module signal CK0 and a write delay circuit 925 controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 930A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{ADQS} . The second write strobe path includes the write strobe buffer 910, the write strobe receiver 920, the write delay circuit 825 controlled by the delay signal DS2, and a second write strobe transmitter 930B that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{BDQS} . The first and second write strobe transmitters, 930A and 930B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0074] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 901 and a selected one of the second and third DQS pins 902A and 902B. In the read strobe path, a select circuit 950 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 902A or a read strobe signal received via DQS pin 902B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 960 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 970 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 980 onto the corresponding strobe signal line 324 via the first DQS pin 901.

[0075] FIG. 10 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1001 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1002A that is coupled to a corresponding module DQ line Y_{ADQ} , a third DQ pin 1002B that is coupled to a corresponding module DQ line Y_{BDQ} . The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1001 and the second DQ pin 1002A and a second write data path coupled between the first DQ pin 1001 and the third DQ pin 1002B. The first write data path includes a write data buffer 1010, a write data receiver 1020 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, a write delay circuit 1025 controlled by the delay signal DS2, and a first write data transmitter 1030A that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{ADQ} . The second write data path includes the write data buffer 1010, the write data receiver 1020, the write delay circuit 825 controlled by the delay signal DS2, and a second write data transmitter 1030B that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{BDQ} . The first and second write data transmitters, 1130A and 1130B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0076] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1001 and a selected one of the second and third DQ pins 1002A and 1002B. In the read data path, a select circuit 1050 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1002A or a read data signal received via DQ pin 1002B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1060 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1070 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1080 onto the corresponding data signal line 130 via the first DQ pin 1001.

[0077] FIG. 11 illustrate a delay circuit 1100 that can be used as the delay circuit 825/860/925/960/1025/1060 according to an embodiment. As shown, the delay circuit 1100 includes a plurality of delay stages, such as delay stages 1110, 1120, and 1130, each delaying an input signal by a predetermined amount. The delay circuit 1100 further includes a select circuit 1140 (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

[0078] As stated above, the MCH 101 can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the interface between the data buffers and the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers 118 having programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

[0079] FIG. 12A is a timing diagram for a write operation according to one embodiment. As shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by a data buffer 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle in which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the data buffer 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer 118 since both these signals are received by the data buffer. Based on such determination, the data buffer 118 can have knowledge of the time interval between t3 and t1, referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer 118 to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

[0080] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t5, the module control circuit 116 outputs one or more enable signals EN at time t6 in response to the read commands. The one or more enable signals are received by an data buffer 118 at time t7, which outputs at time t8 read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle which t3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers 118.

[0081] With knowledge of the time interval between t7 and t5, which should be about the same as the time interval between t3 and t1, i.e., the command-to-enable delay CED, in

certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t9 by the data buffer to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t9 and t5 being consistent with a read latency R.L. associated with the system 100.

[0082] The time interval between t4 and t3, i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the DB Control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the dynamic delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a Start input receiving the one of the data buffer control signals, a Stop input receiving an output of the flip-flop circuit 1320. Thus, the output of the counter circuit would indicate a time interval from when the write preamble is detected and when the one of the data buffer control signal is received. This output is combined with the output from the programmable registers 671 in circuit 1340, which output the delay signal DS1.

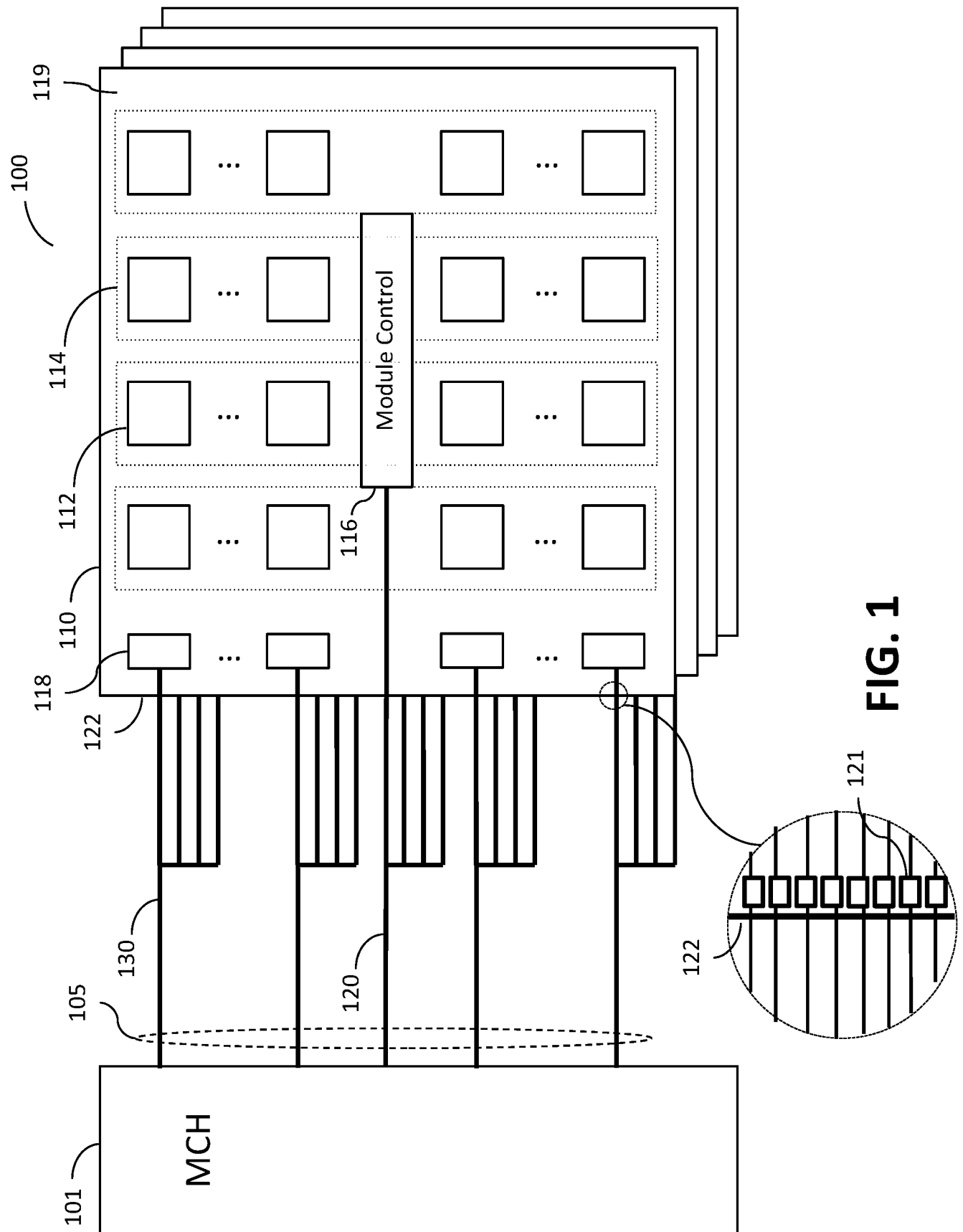
[0083] Thus, as shown in FIG. 14, in one embodiment, a memory module 110 operates in the memory system 100 according to a method 1400. In the method, during a write operation, one or more data buffer control signals are received by an data buffer 118 from a module control circuit or module controller 116 (1410). The module controller 116 generates the one or more data buffer control signals in response to C/A signals representing a write command from the MCH 101. The one or more data buffer control signals are used to control the data buffer 118. For example, the one or more data buffer control signals may include one or more first enable signals to enable a write path to allow write data be communicated to a selected subgroup of memory devices among the group of memory devices coupled to the data buffer 118. After a time interval from receiving the one or more first enable signals, write data DQ and write strobe DQS are received by the data buffer 118 from the MCH 101 (1420). In one embodiment, upon receiving the one or more

first enable signal, a counter is started, which is stopped when the write data DQ or write strobe DQS is received. Thus, a time interval EWD between receiving the one or more first enable signals and receiving the write strobe signal DQS is recorded.

[0084] Since the time interval between the arrival of the command signals from the MCH 101 and the arrival of the write data/strobe signal DQ/DQS from the MCH 101 is set according to a write latency parameter associated with the system 100, the time interval EWD can be used to ascertain a time interval CED between the time when a command signal is received by the memory module 110 and the time when the one or more enable signals are received by the data buffer 118. The time interval CED can be used by the data buffer 118 to properly time the transmission of read data to the MCH 101.

[0085] As shown in FIG. 14, a delay signal DS is generated according to the time interval EWD (1430). Concurrent to receiving the write strobe signal DQS, the data buffer 118 also receives a set of write data signals DQ (1440). The received write data signals are transmitted to the subgroup of memory devices (1450), which are selected from the group of memory devices coupled to the data buffer 118 by the one or more first enable signals.

[0086] During a read operation, another set of data buffer control signals including, for example, one or more second enable signals, are received by the data buffer 114 from the module controller 116 (1460). The one or more second enable signals are generated by the module controller 116 in response to read command signals received from the MCH 101, and are used by the data buffer 118 to select a subgroup of memory devices from which to receive read data. Afterwards, a read strobe signal DQS and a set of read data signal DQ are received from the selected subgroup of memory devices (1470). To properly time the transmission of the DQS and DQ signals to the MCH 101, the DQS and DQ signals are adjusted (e.g., delayed) according to the delay signal DS, such that the DQS and DQ signals follow a read command by a time interval consistent with a read latency parameter associated with the system 100.



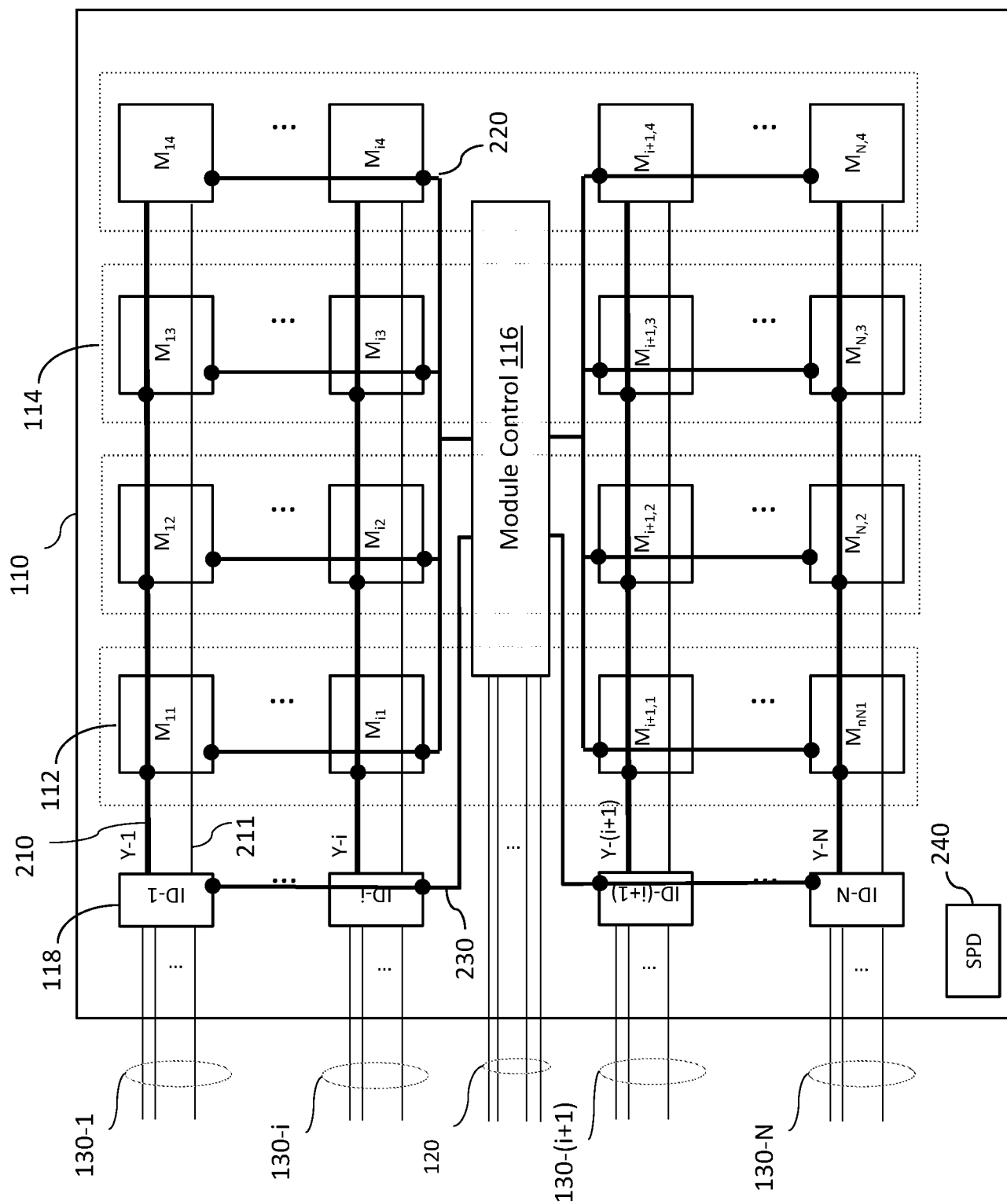


FIG. 2A

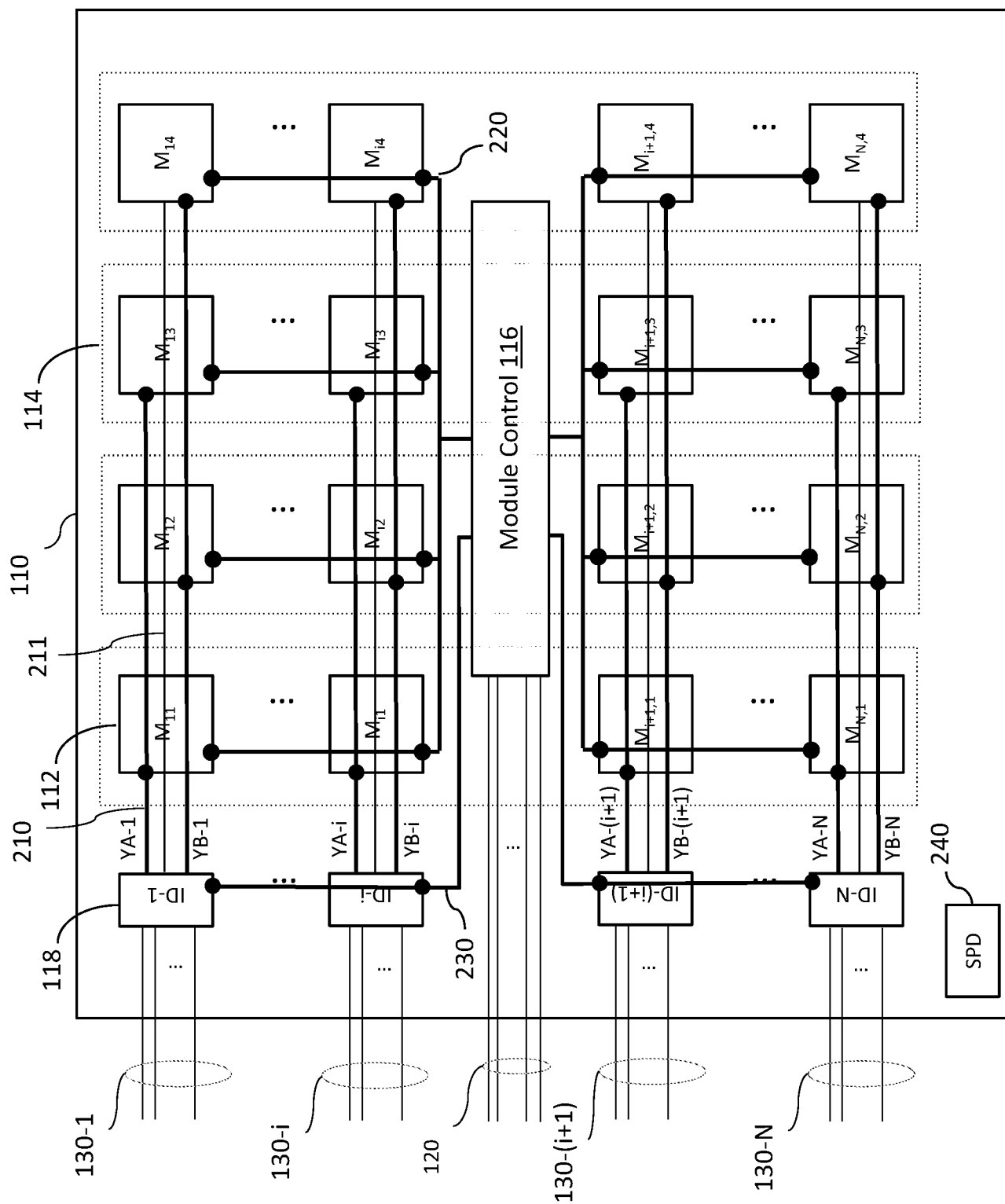


FIG. 2B

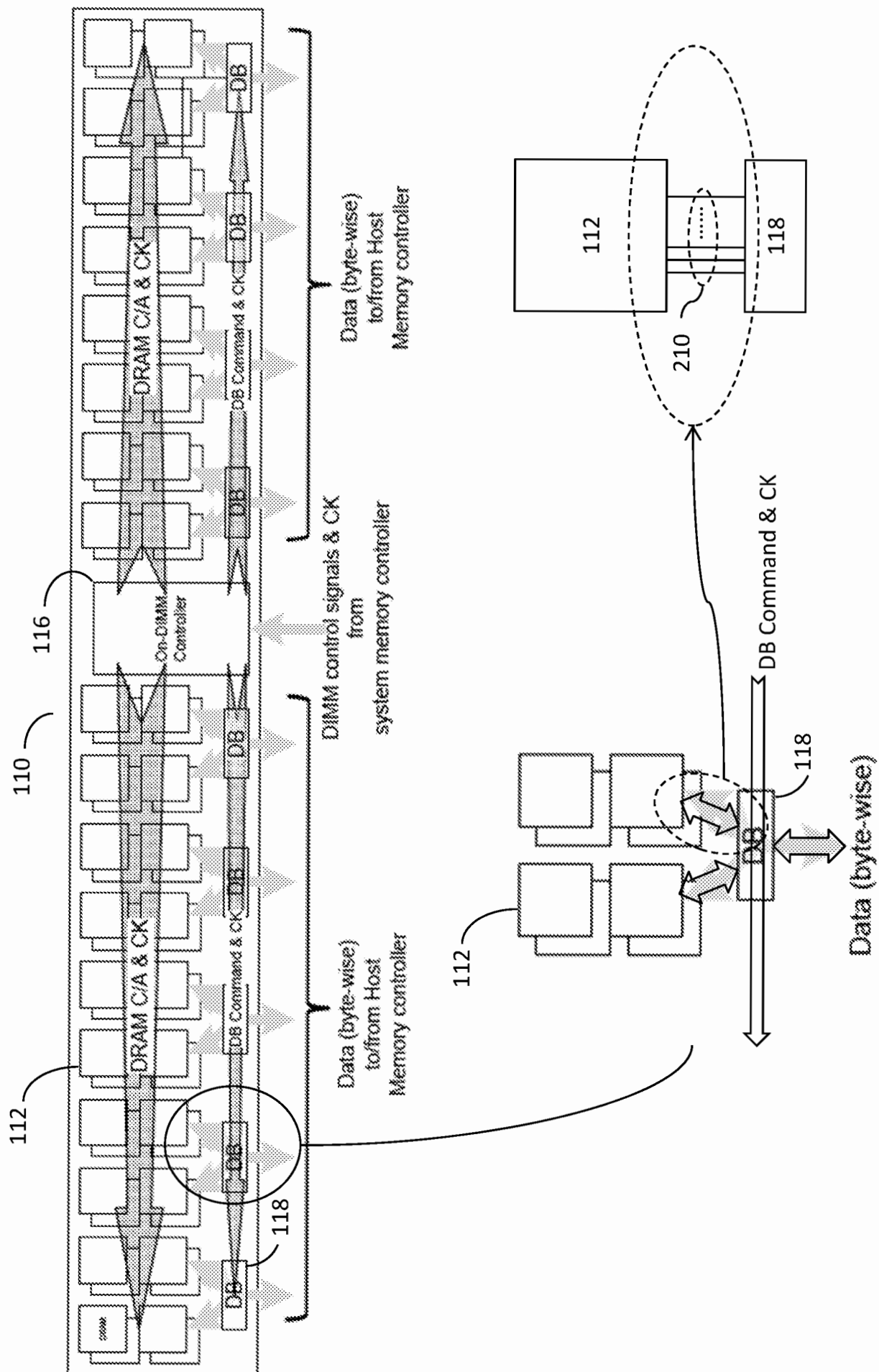


FIG. 2C

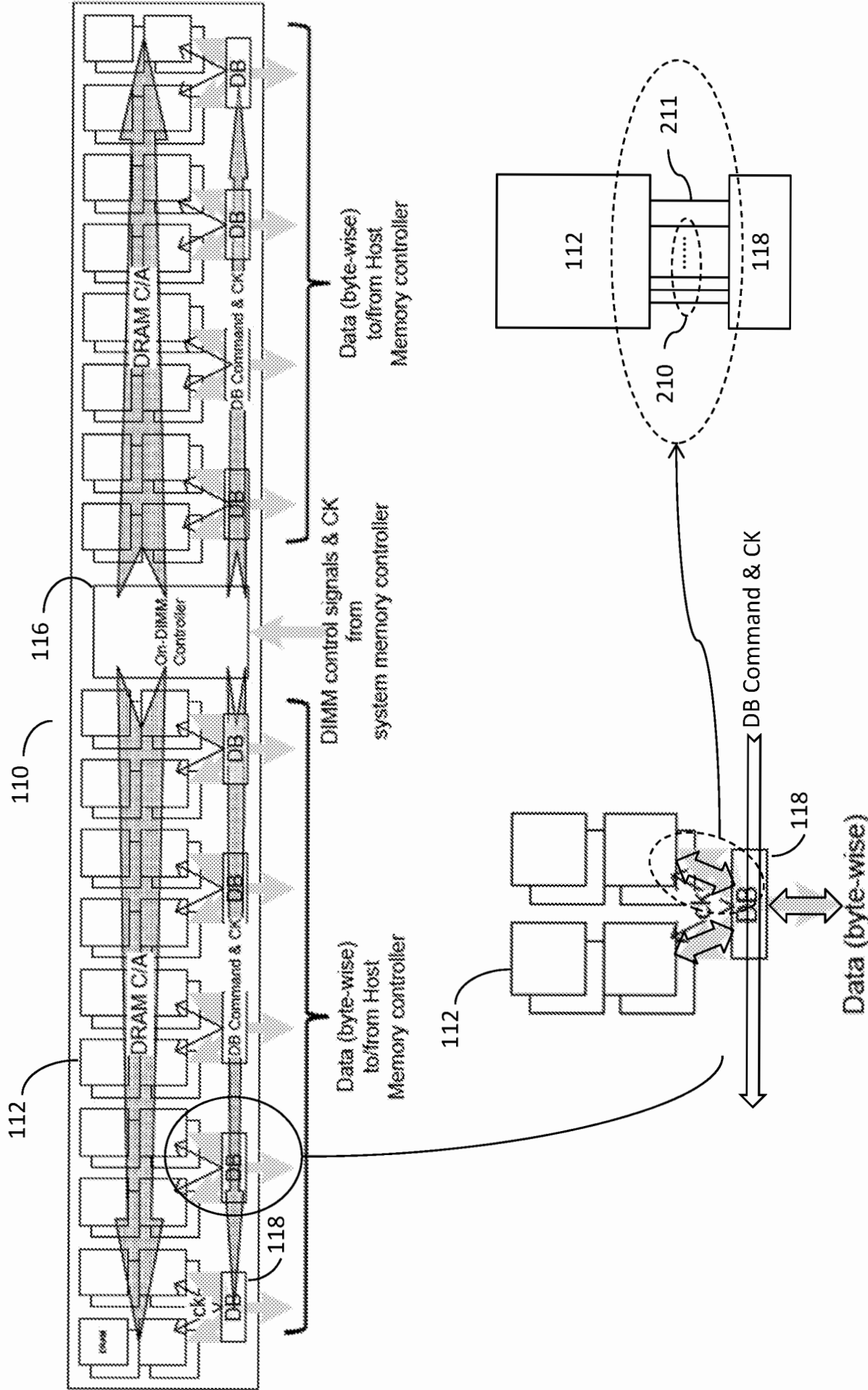
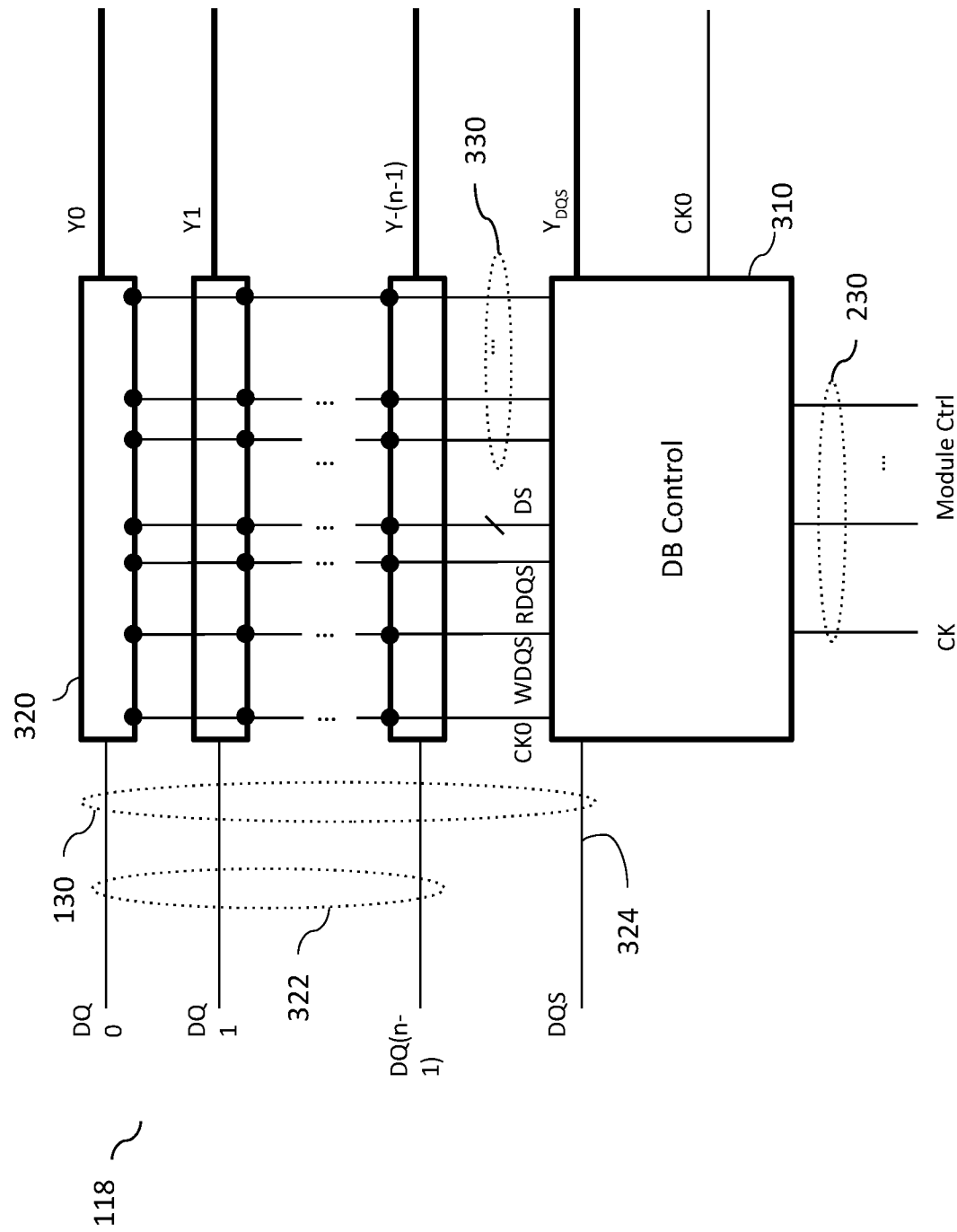


FIG. 2D

**FIG. 3**

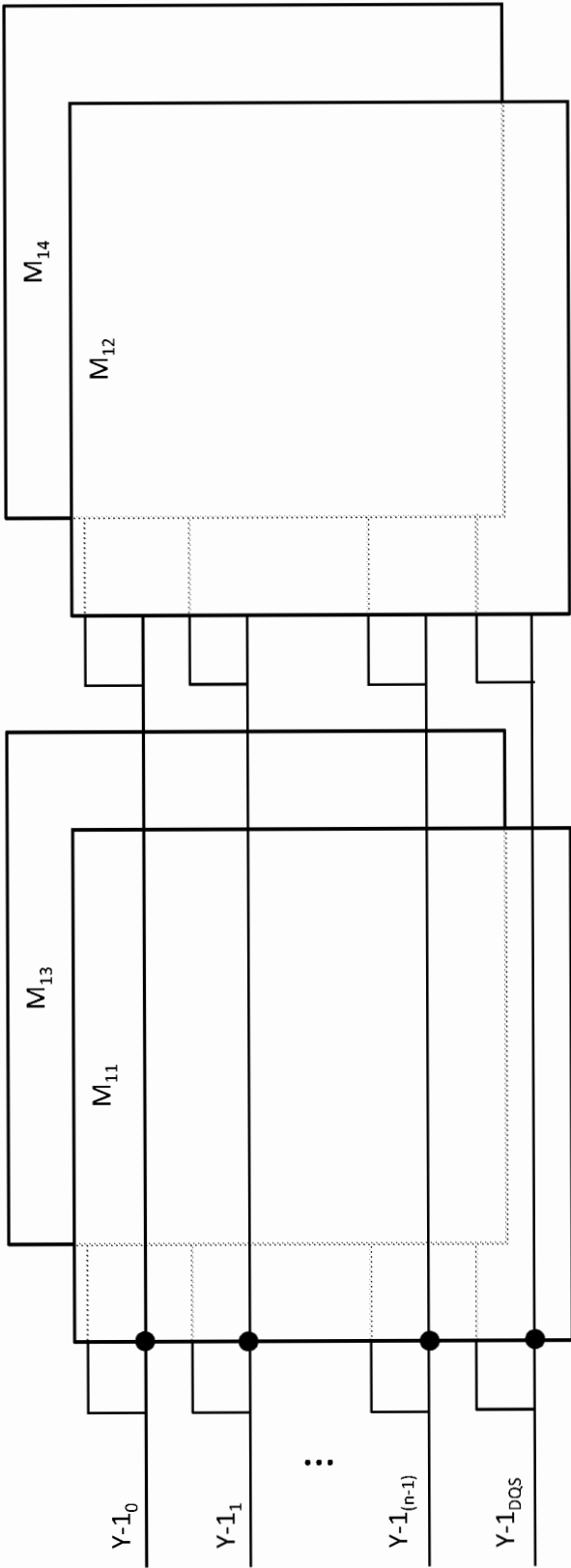


FIG. 4A

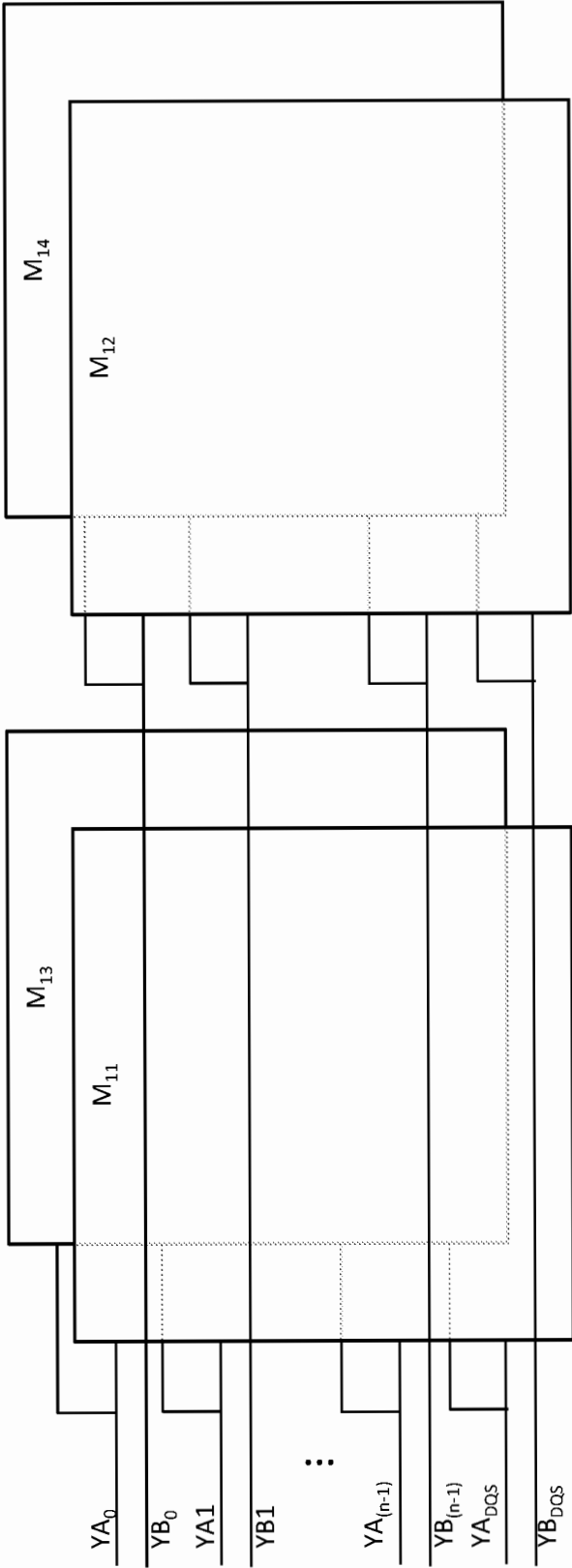


FIG. 4B

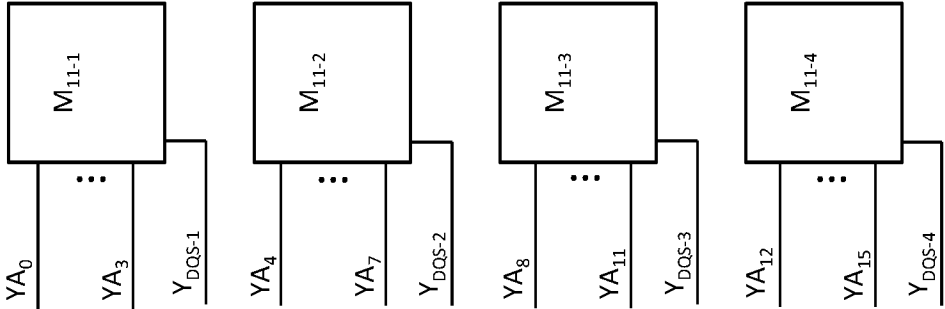


FIG. 5B

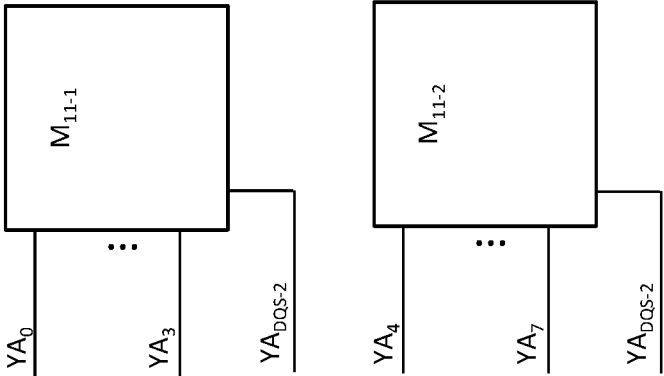


FIG. 5A

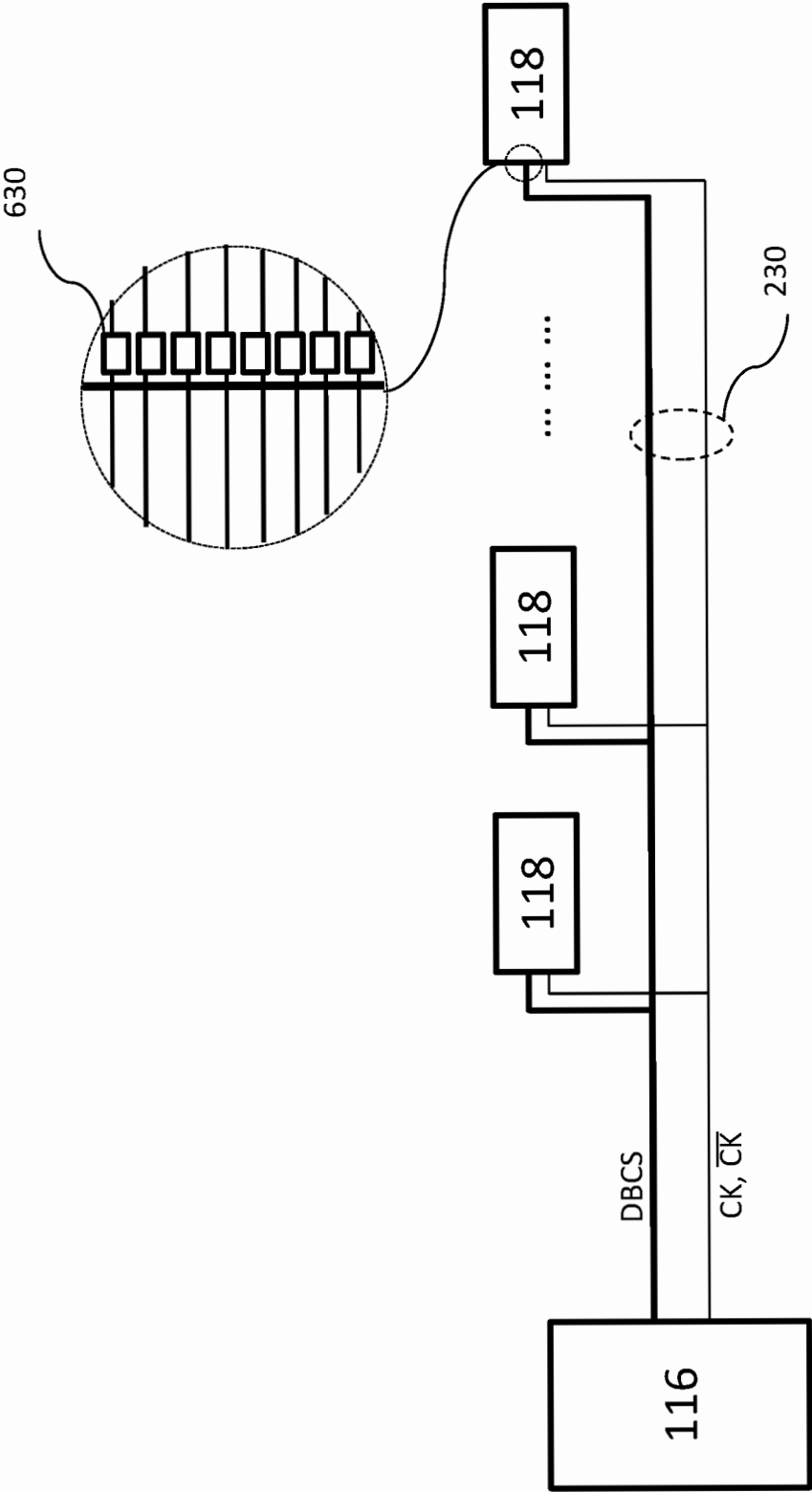
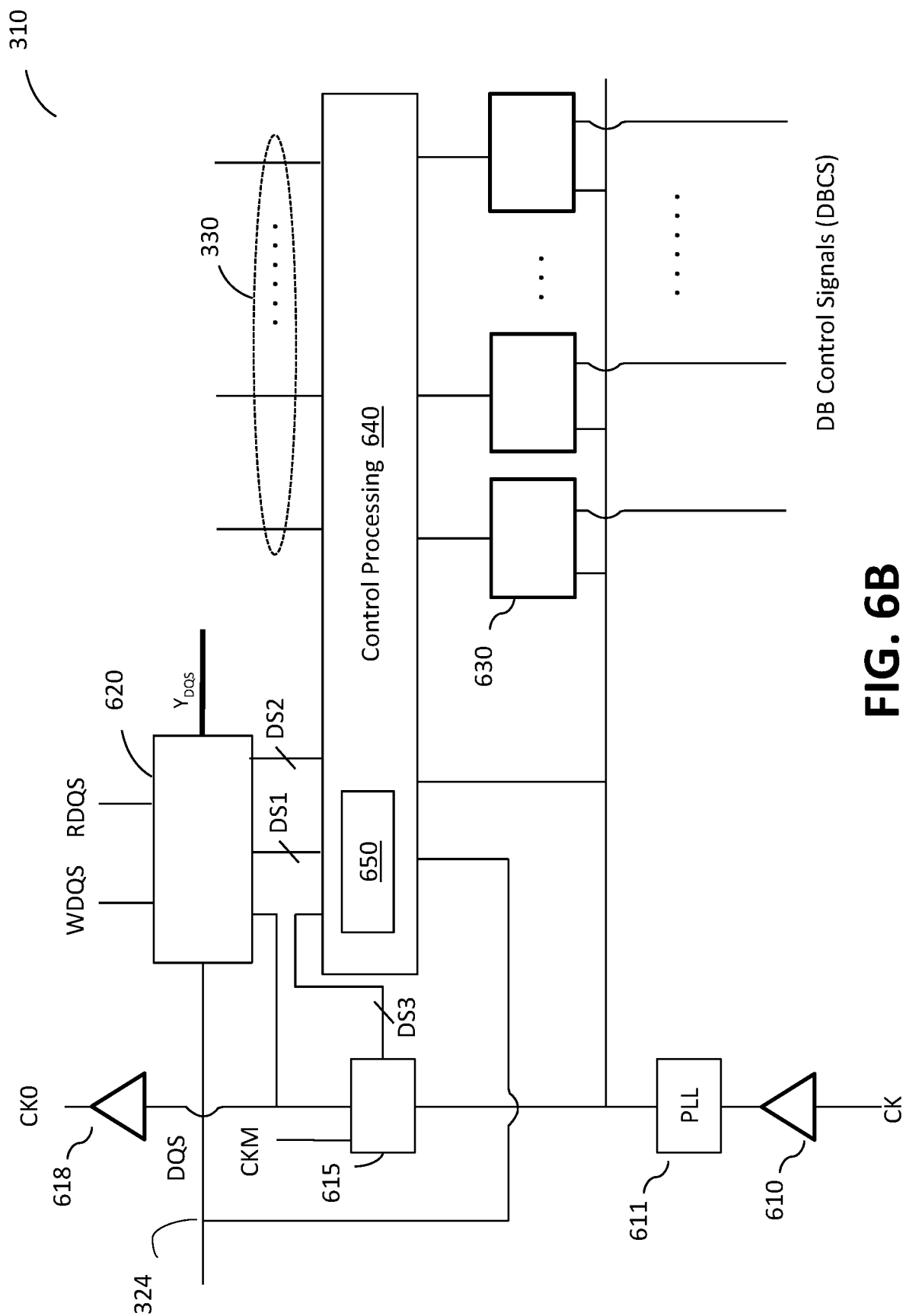


FIG. 6A

**FIG. 6B**

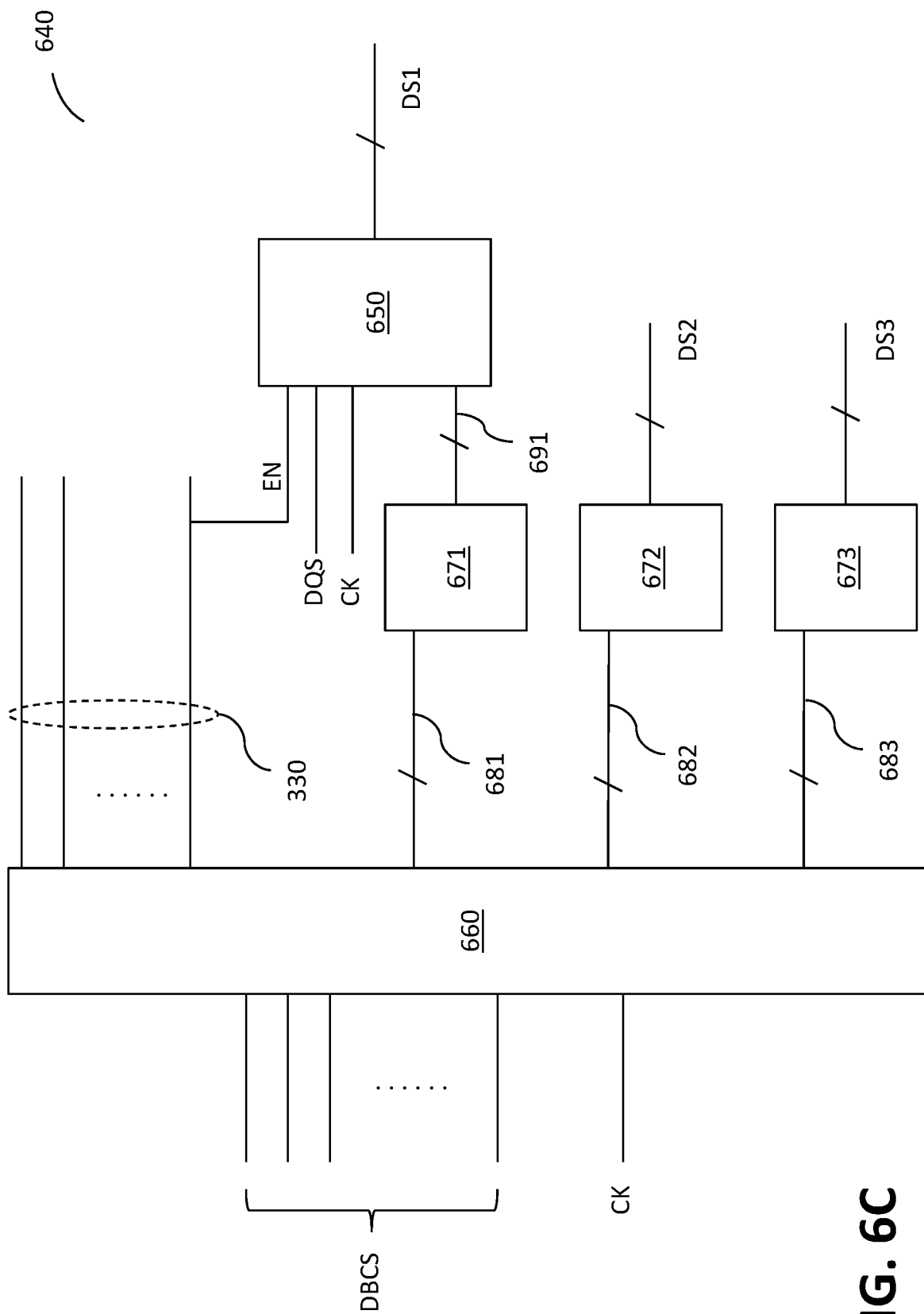
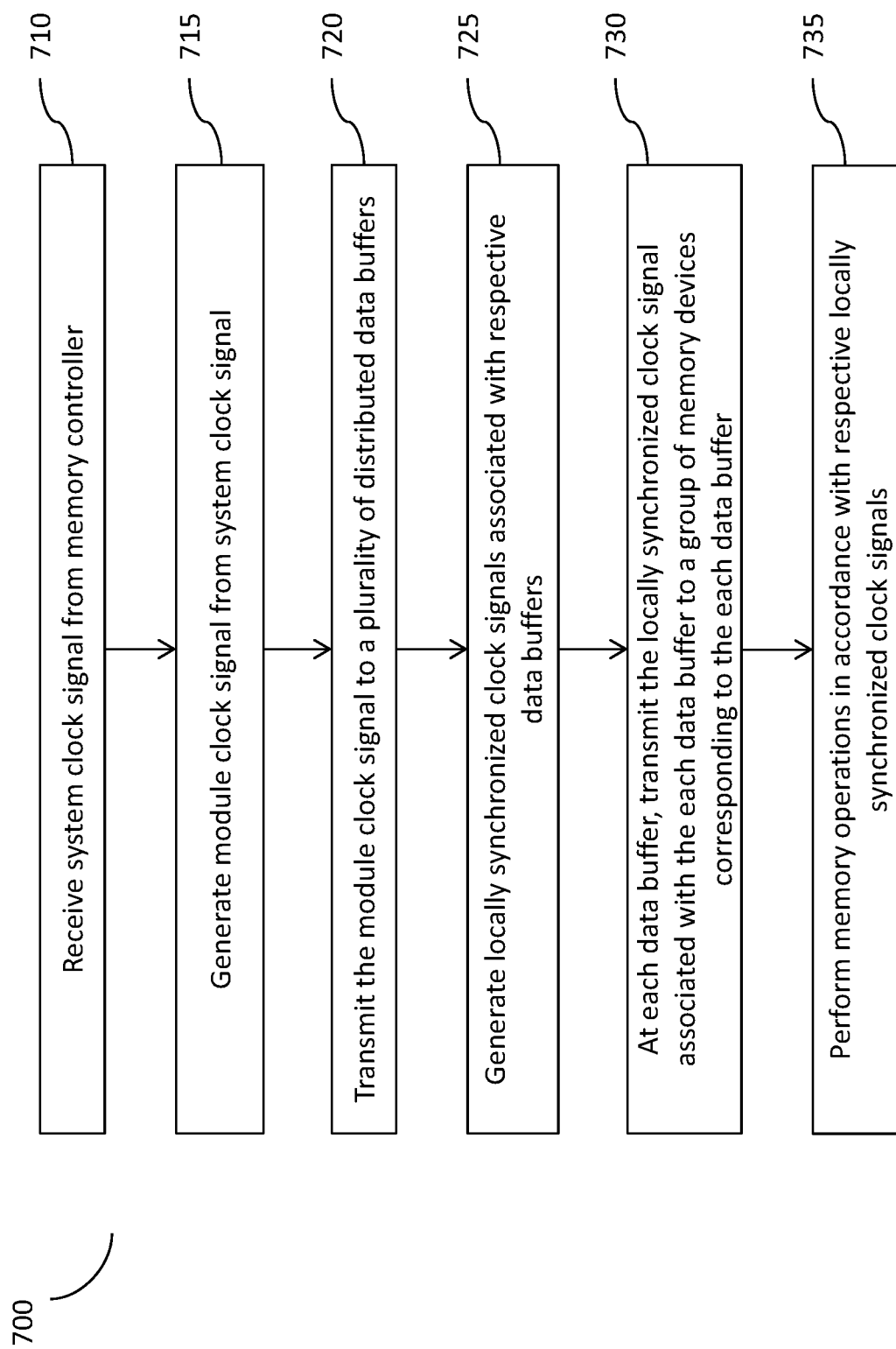
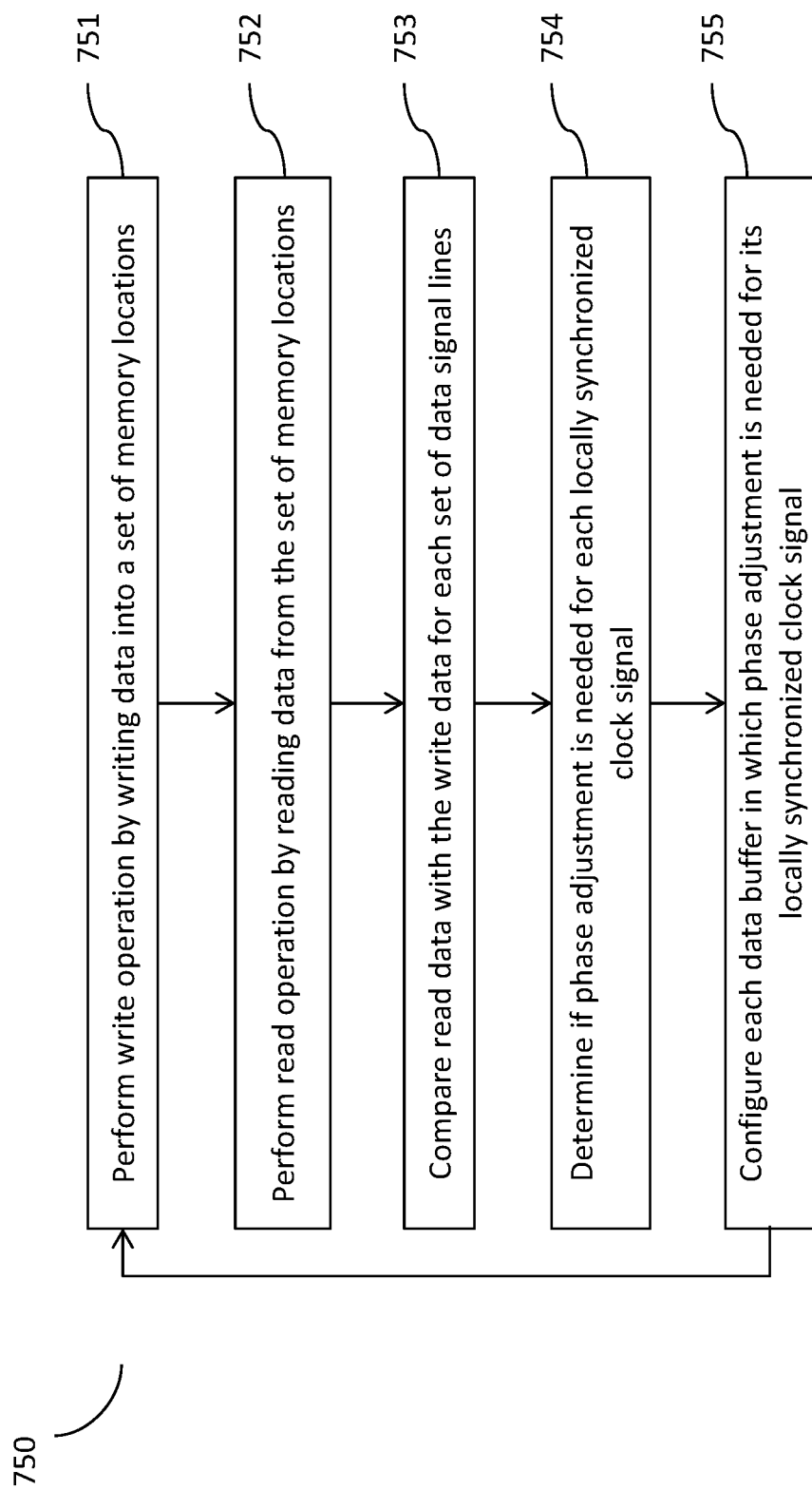


FIG. 6C

**FIG. 7A**

**FIG. 7B**

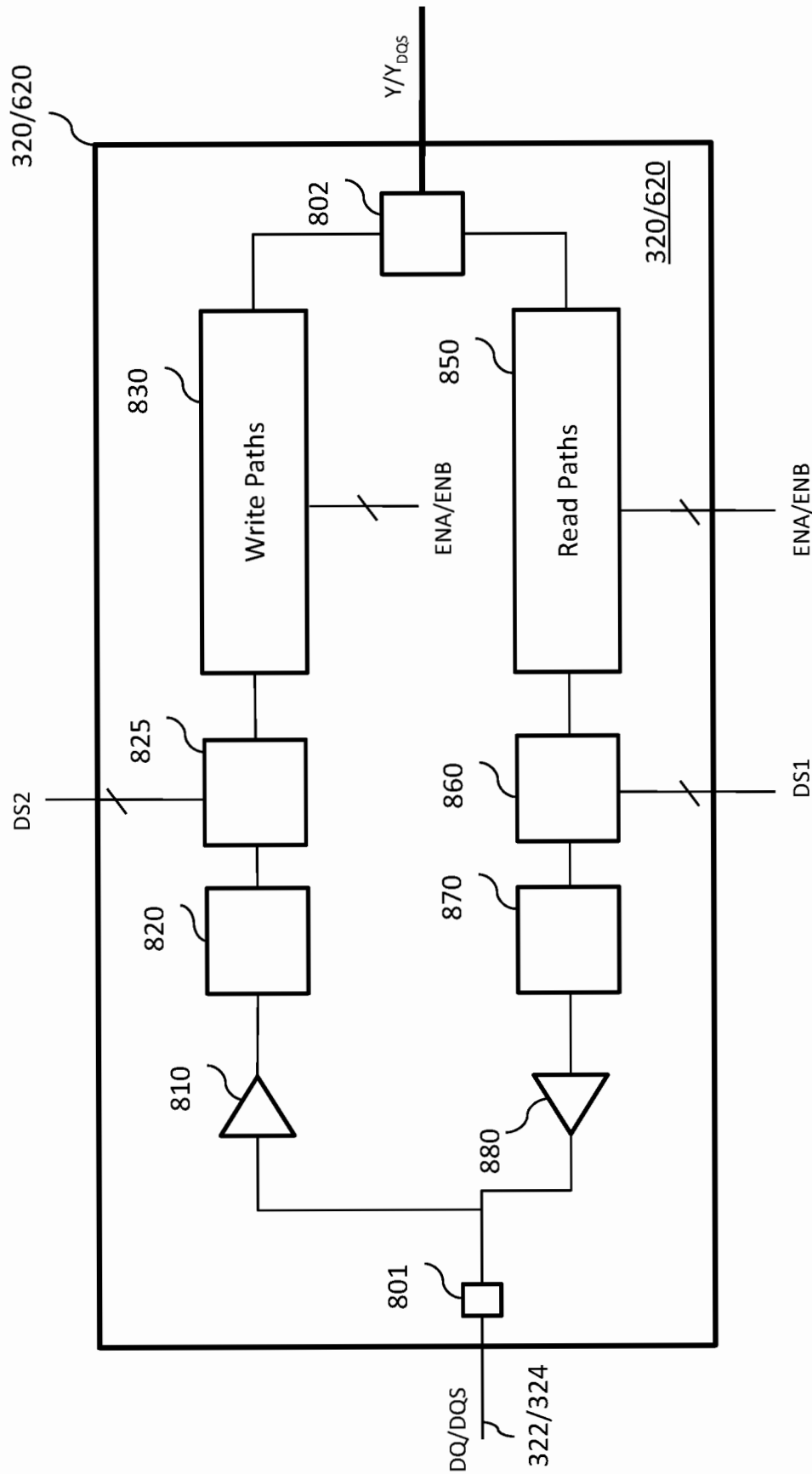


FIG. 8

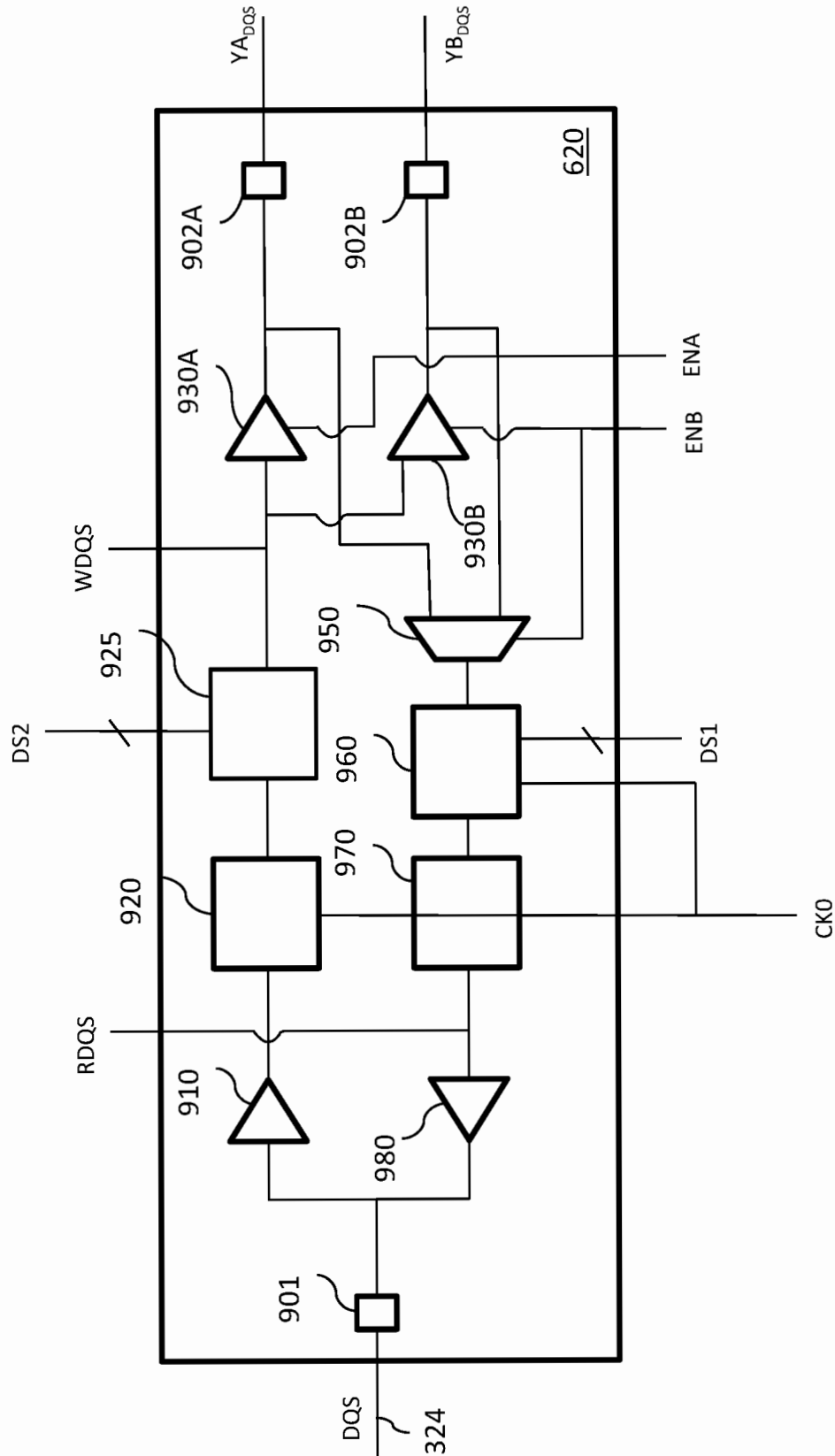


FIG. 9

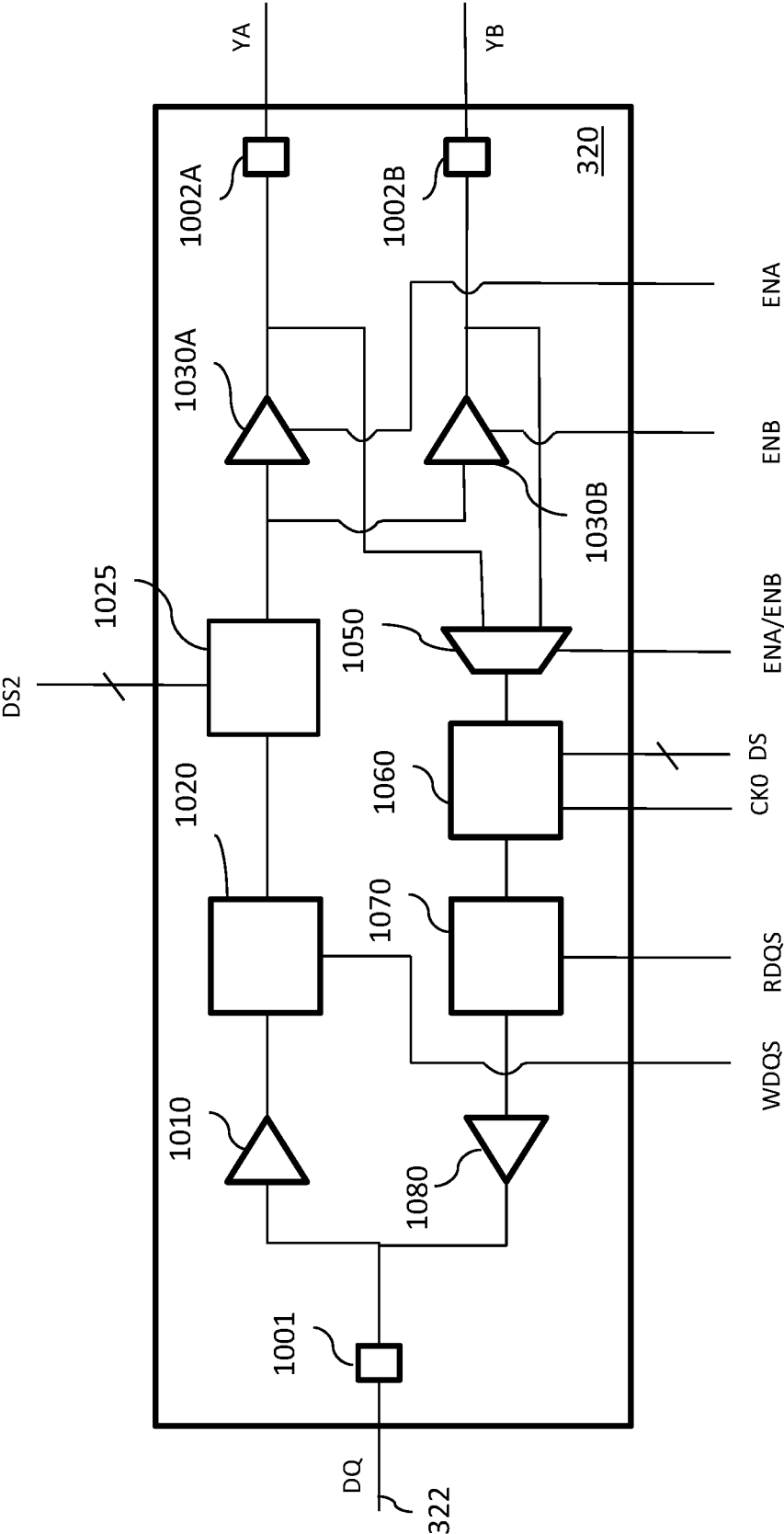
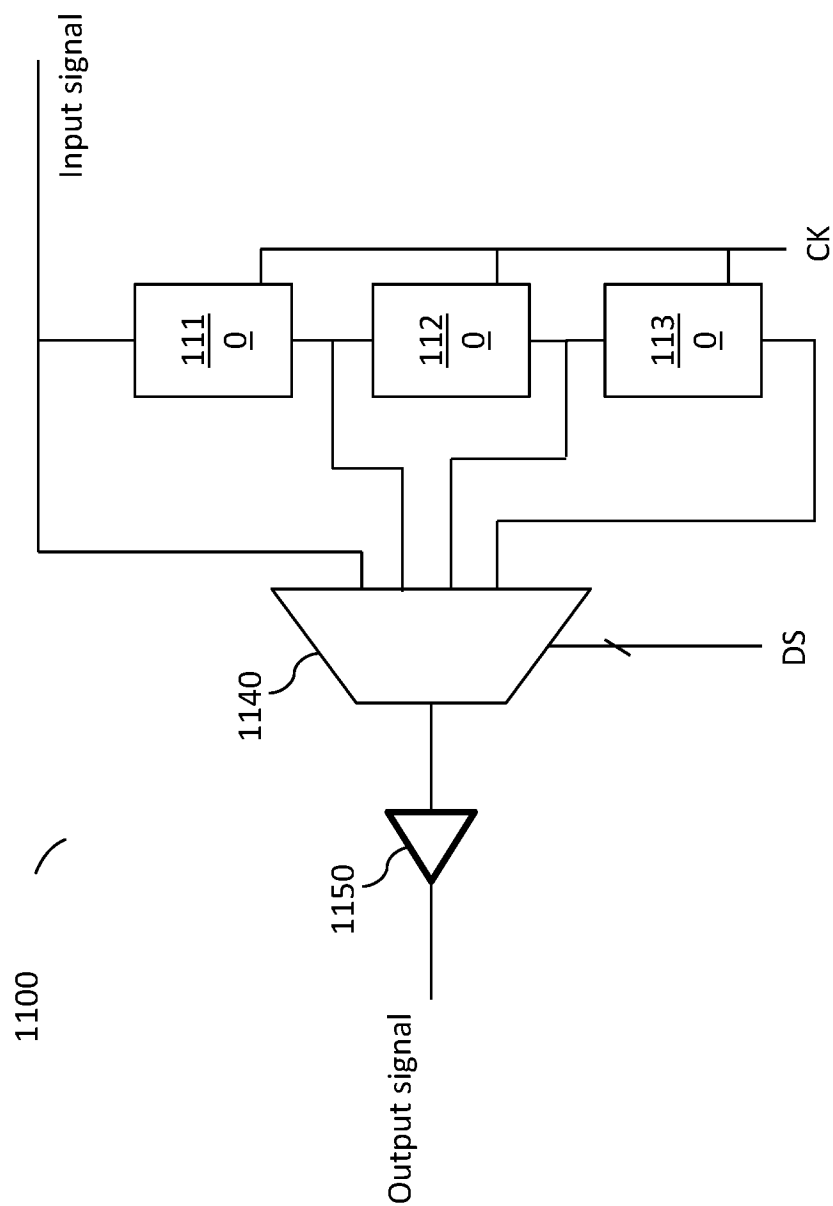


FIG. 10

**FIG. 11**

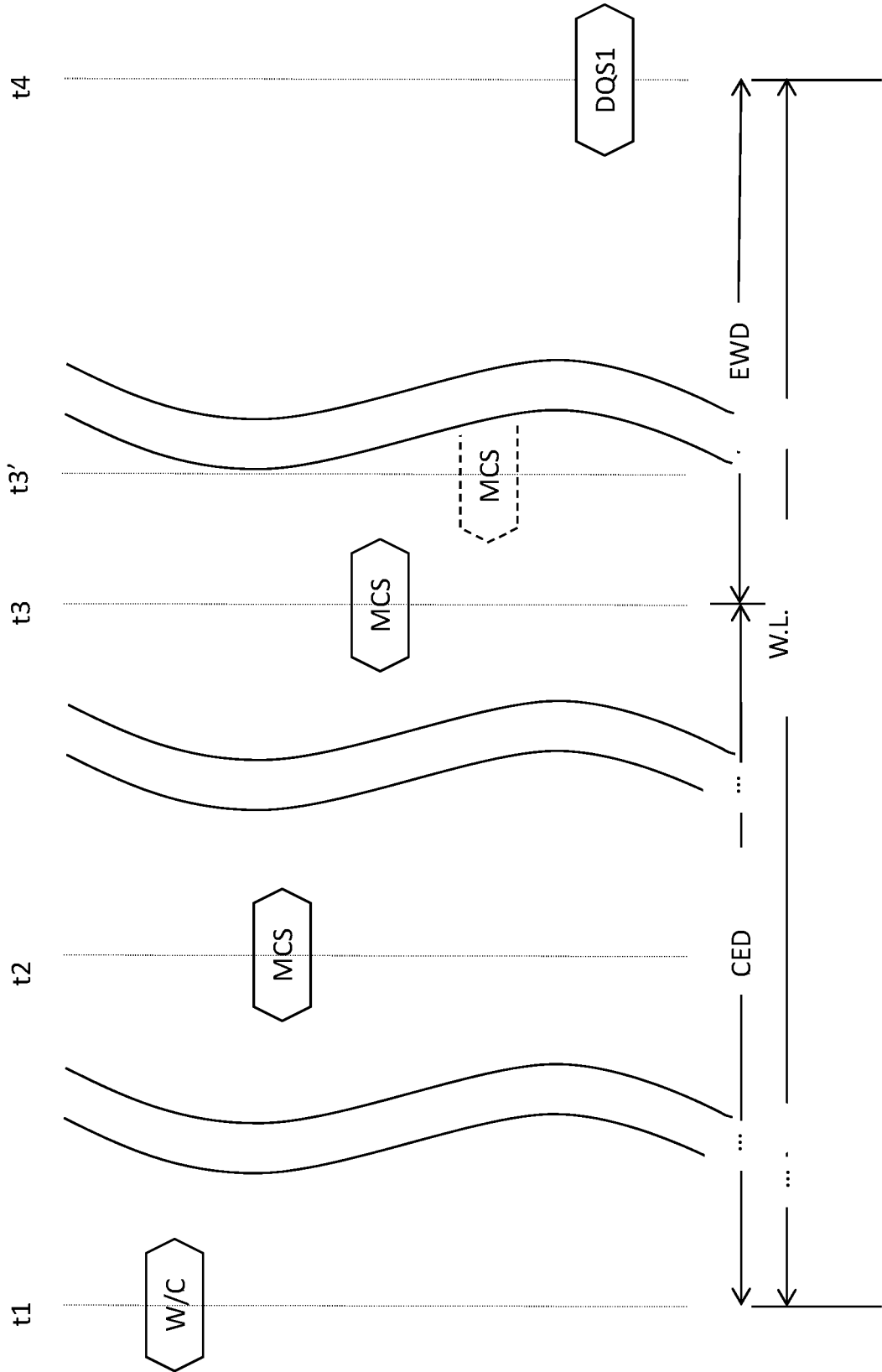


FIG. 12A

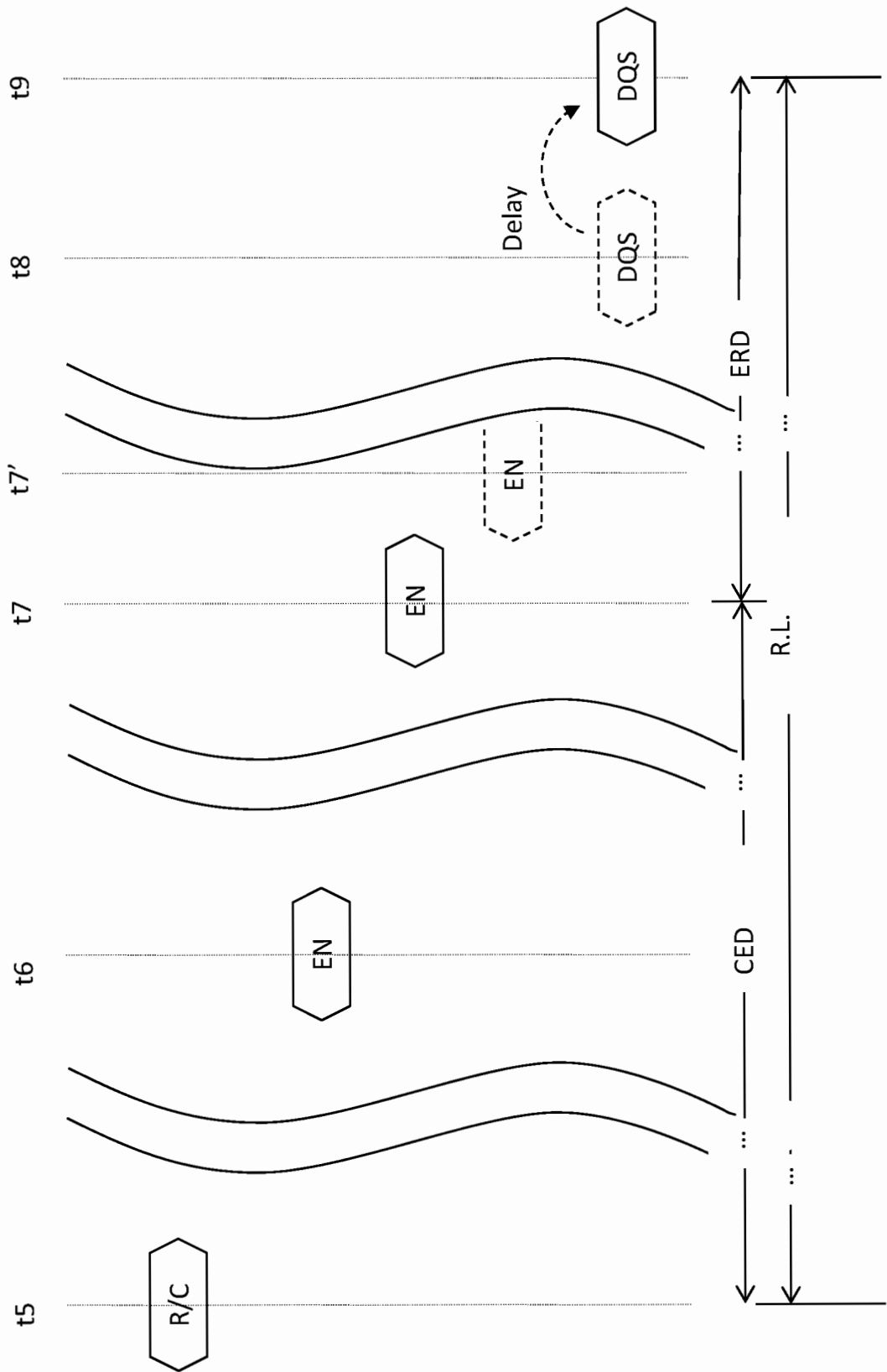


FIG. 12B

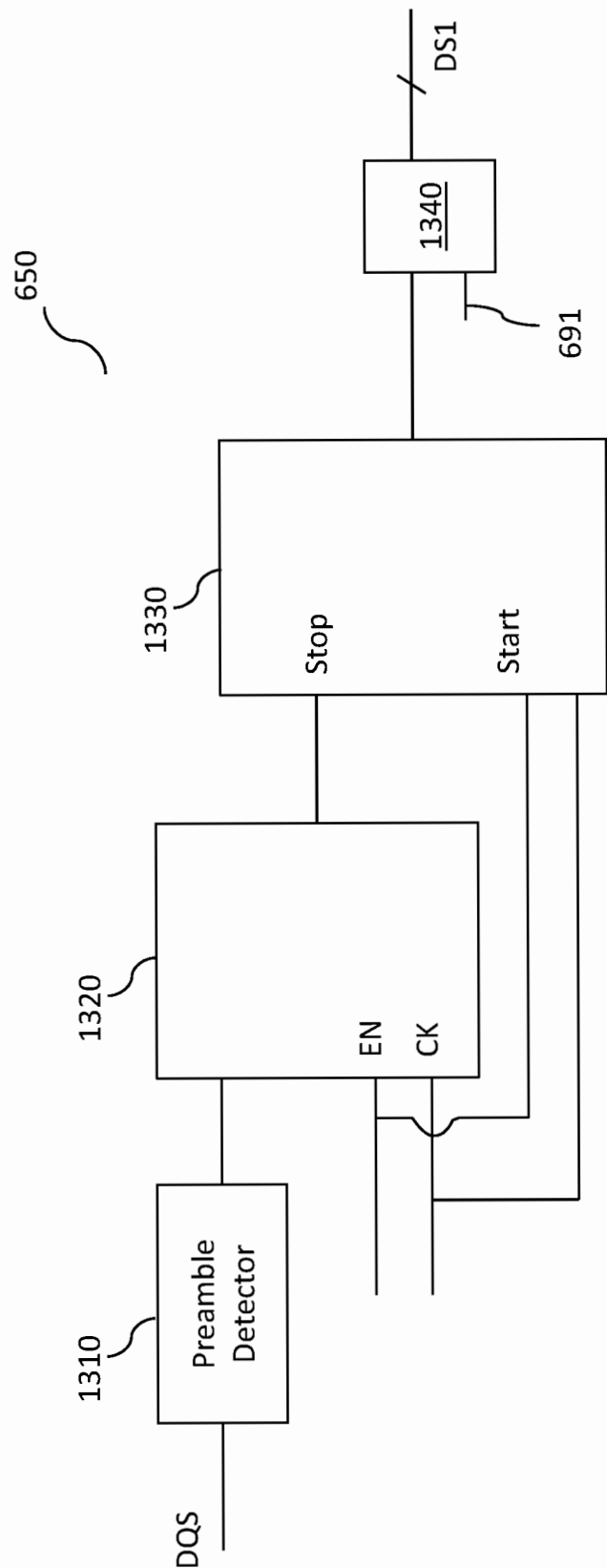
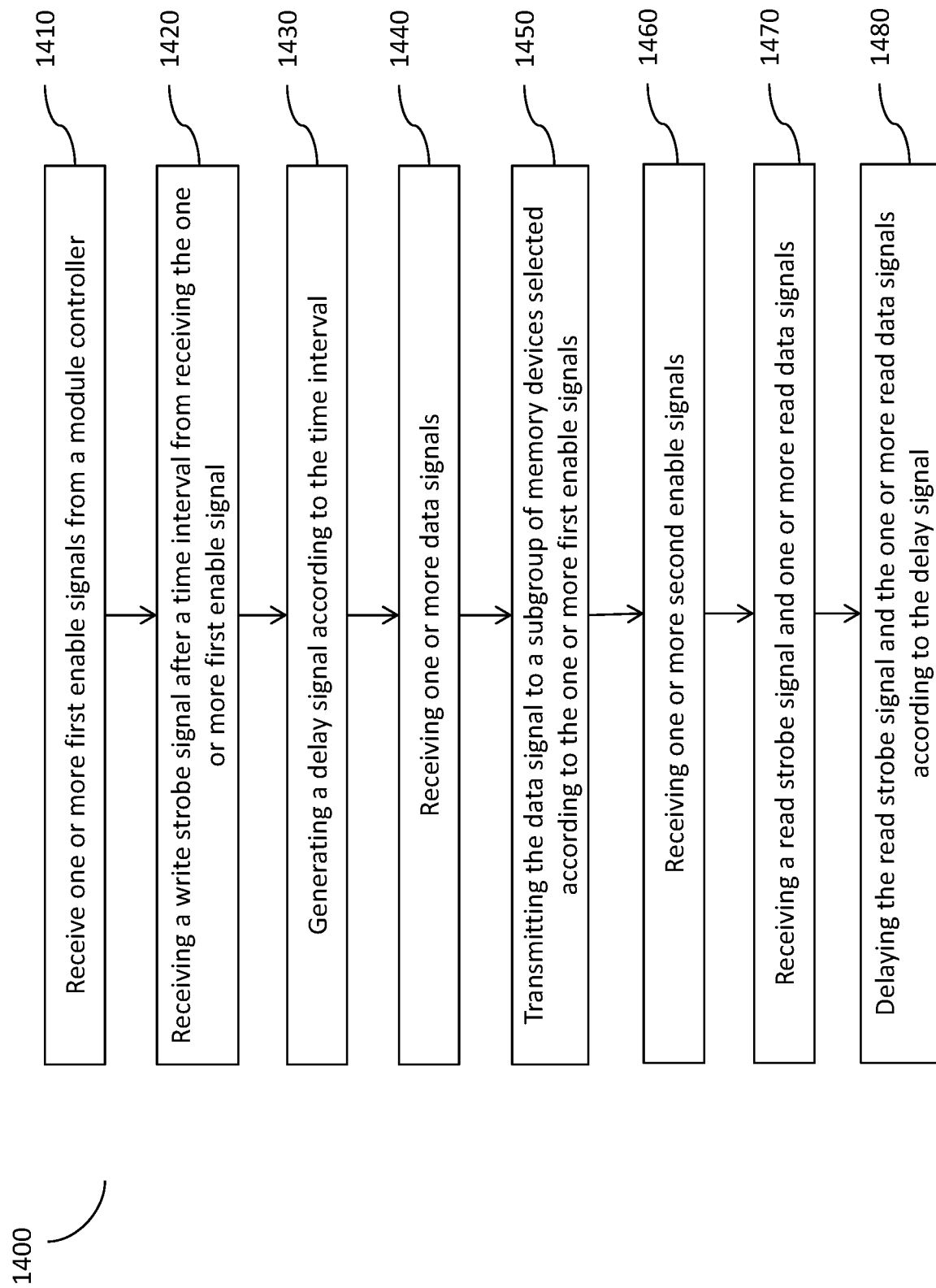


FIG. 13

**FIG. 14**

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US01
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐	Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)
--------------------------	---

Inventor Information:

Inventor	1				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Hyun		Lee		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Ladera Ranch	State/Province	CA	Country of Residence	US

Mailing Address of Inventor:

Address 1	21 Thalia				
Address 2					
City	Ladera Ranch	State/Province	CA		
Postal Code	92694	Country	US		

Inventor	2				Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Jayesh	R.	Bhakta		
Residence Information (Select One) ☒ US Residency ☐ Non US Residency ☐ Active US Military Service					
City	Cerritos	State/Province	CA	Country of Residence	US

Mailing Address of Inventor:

Address 1	12220 Rose Street				
Address 2					
City	Cerritos	State/Province	CA		
Postal Code	90703	Country	US		

All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button.

Add

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below.
For further information see 37 CFR 1.33(a).

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US01
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

☐ An Address is being provided for the correspondence information of this application.

Customer Number	79141		
Email Address		Add Email	Remove Email

Application Information:

Title of the Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		
Attorney Docket Number	129980-5073-US01	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	22	Suggested Figure for Publication (if any)	

Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	79141		

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US01
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Continuation of	14/445035	2014-07-28
Prior Application Status	Expired	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
14/445035	Claims benefit of provisional	61/859215	2013-07-27
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button.			Add

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Access Code ⁱ (if applicable)	Remove
Additional Foreign Priority Data may be generated within this form by selecting the Add button.				Add

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US01
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US01
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Applicant	1	Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section. Clear		
☒ Assignee	Legal Representative under 35 U.S.C. 117	Joint Inventor
Person to whom the inventor is obligated to assign.		Person who shows sufficient proprietary interest
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:		
Name of the Deceased or Legally Incapacitated Inventor:		
If the Applicant is an Organization check here. ☒		
Organization Name	Netlist, Inc.	
Mailing Address Information For Applicant:		
Address 1	175 Technology, Suite 150	
Address 2		
City	Irvine	State/Province
Country	US	Postal Code
Phone Number		Fax Number
Email Address		
Additional Applicant Data may be generated within this form by selecting the Add button. Add		

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US01
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
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Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Signature:

NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).

This Application Data Sheet **must** be signed by a patent practitioner if one or more of the applicants is a **juristic entity** (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, **all** joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of **all** joint inventor-applicants.

See 37 CFR 1.4(d) for the manner of making signatures and certifications.

Signature	/Jamie J. Zheng/		Date (YYYY-MM-DD)	2019-06-05
First Name	Jamie J.	Last Name	Zheng	Registration Number
				51167
Additional Signature may be generated within this form by selecting the Add button. Add				

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	129980-5073-US01
		Application Number	
Title of Invention	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Electronic Acknowledgement Receipt

EFS ID:	36209997
Application Number:	16432700
International Application Number:	
Confirmation Number:	7498
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073-US01
Receipt Date:	05-JUN-2019
Filing Date:	
Time Stamp:	18:08:24
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal of New Application	129980-5073-US01_TRANSMITTAL.pdf	145285	no	1
			e575ab27999c06e3760cbd5dc2a5a3e753abc6b5		

Warnings:

Information:					
2		129980-5073-US01_STMT_POA.pdf	122270	yes	2
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	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Power of Attorney		2	2	
	Assignee showing of ownership per 37 CFR 3.73		1	1	
Warnings:					
Information:					
3	Application Data Sheet	129980-5073-US01_Oath.pdf	75575	no	2
			e348a123a71193c78ea99ef5913a53f2792af8bc		
Warnings:					
Information:					
This is not an USPTO supplied ADS fillable form					
4		129980-5073-US01_APPLN.pdf	268316	yes	34
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	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Abstract		34	34	
	Claims		29	33	
	Specification		1	28	
Warnings:					
Information:					
5	Drawings-only black and white line drawings	129980-5073-US01_DRWGS.pdf	1682961	no	22
			3abfc9b19afecb7296ecac79172da6195fcaf5b8		
Warnings:					
Information:					

6	Application Data Sheet	129980-5073-US01_ADS.pdf	1823091	no	8
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

SCORE Placeholder Sheet for IFW Content

Application Number: 16432700

Document Date: 06/05/2019

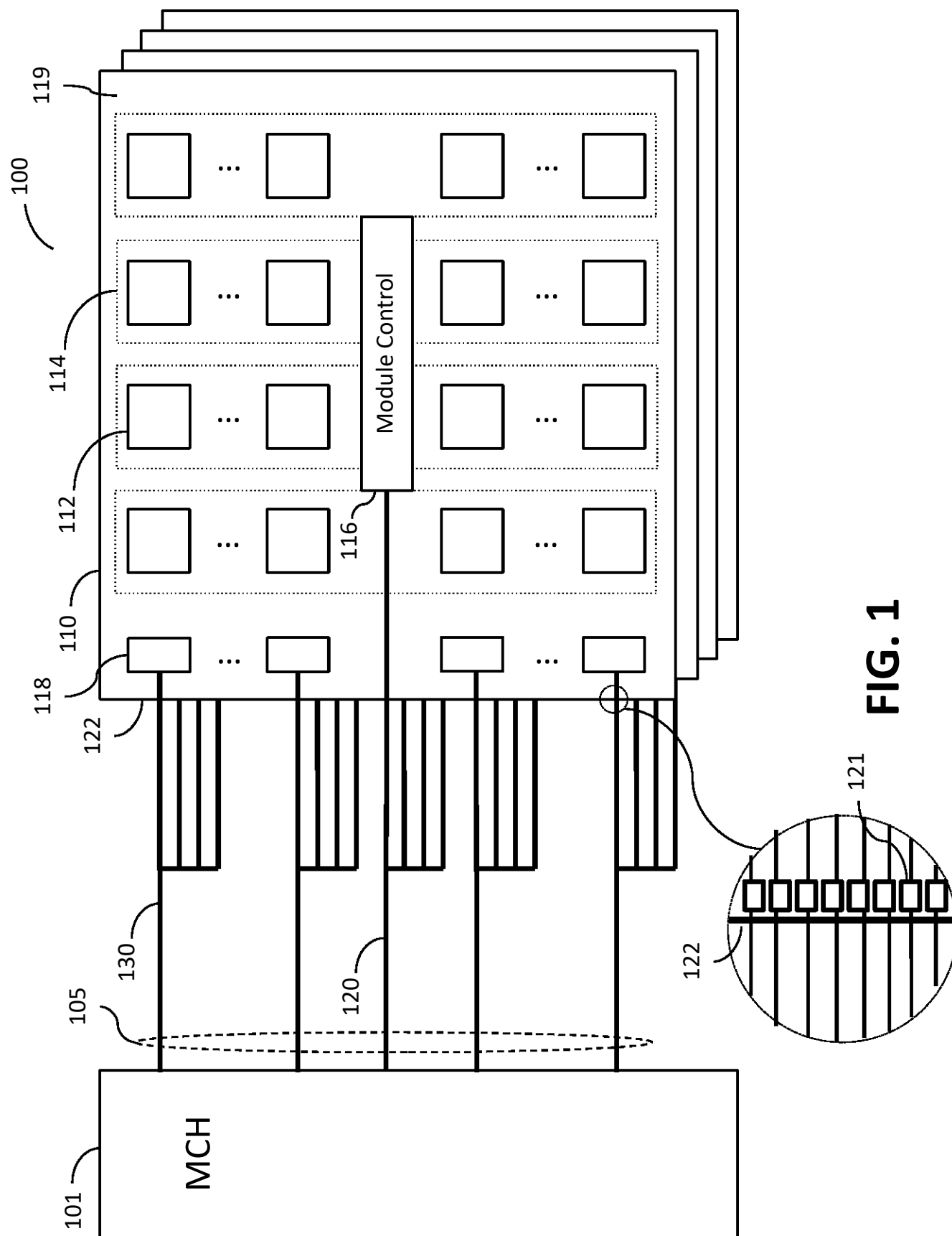
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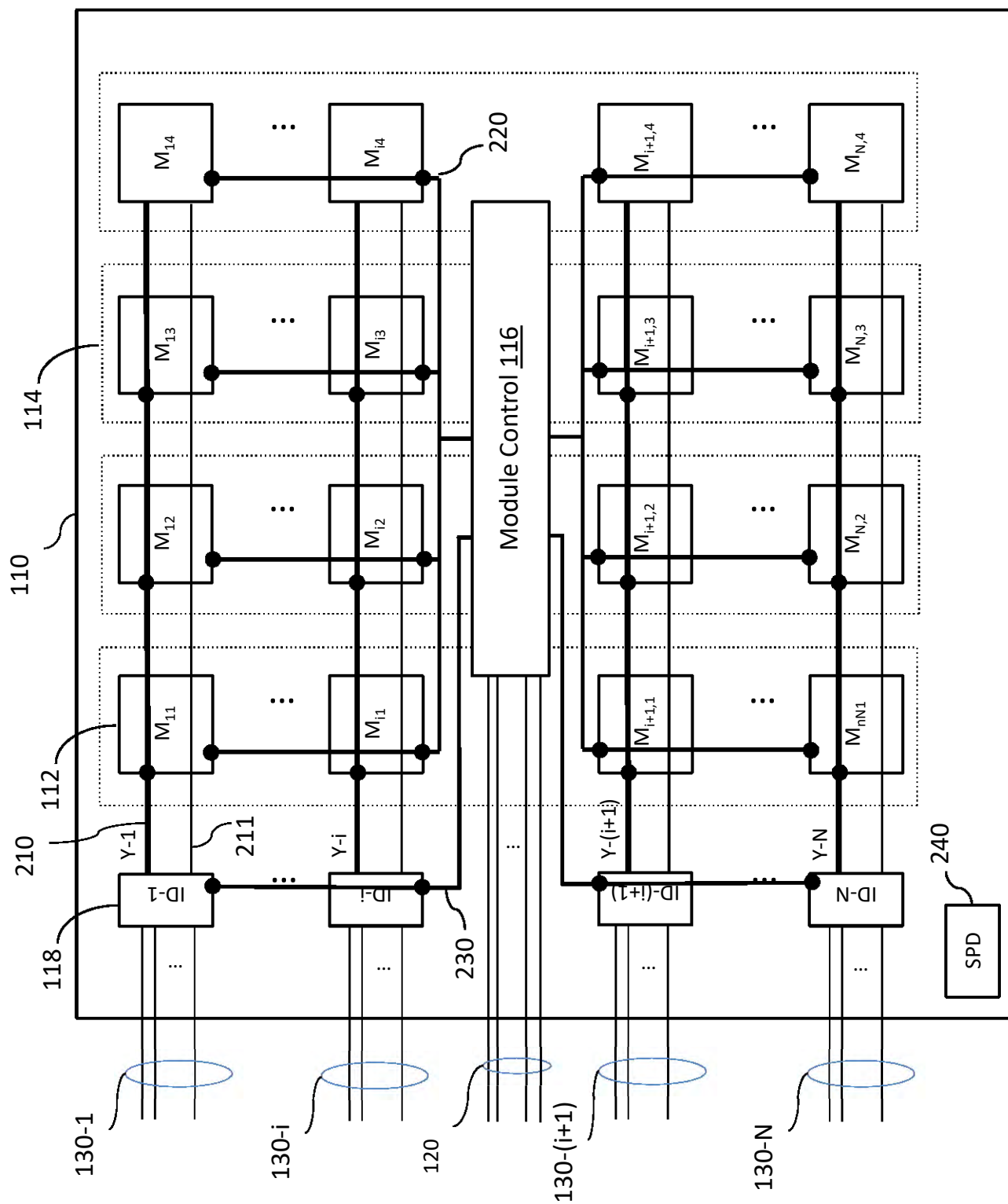


FIG. 2A

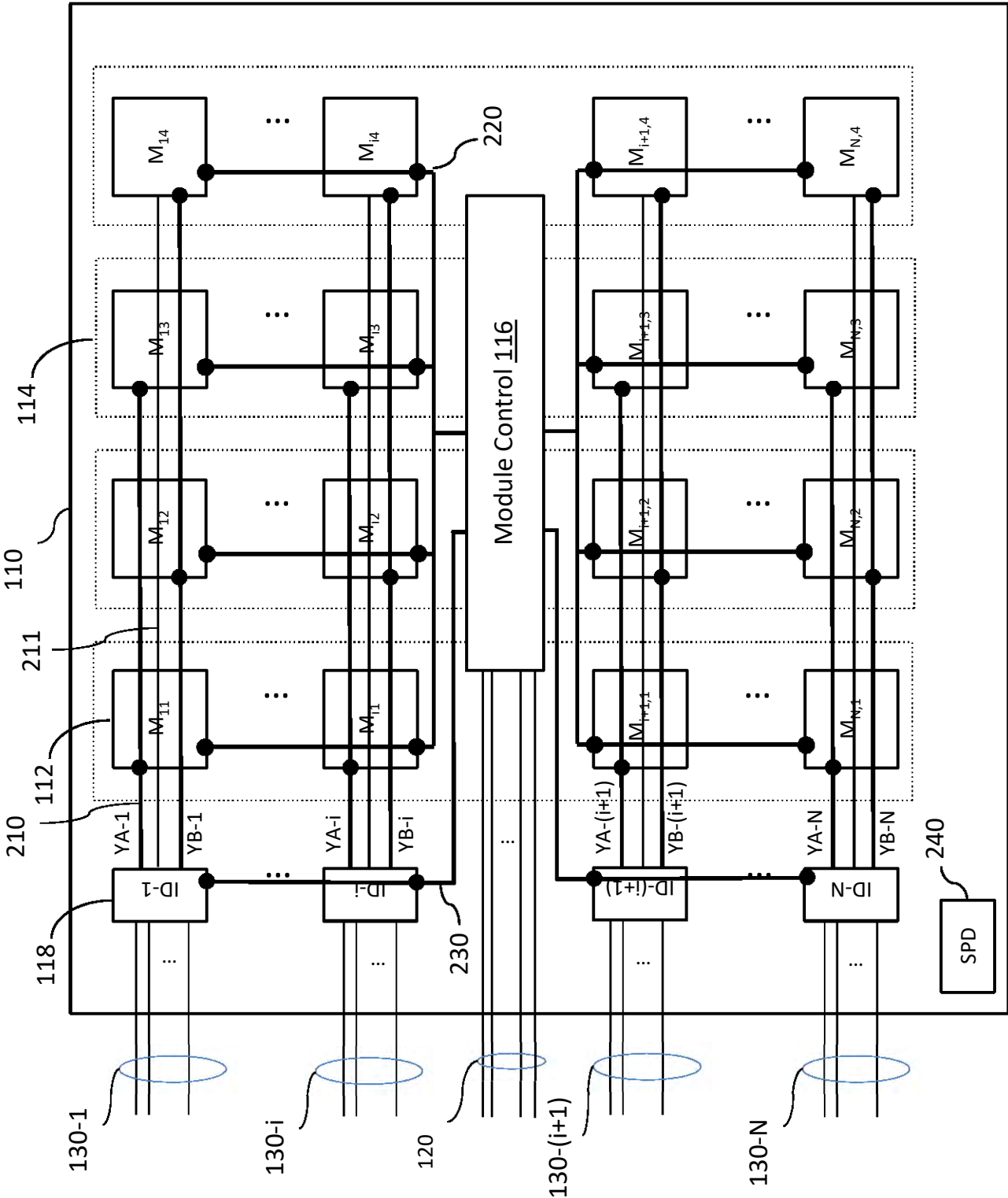
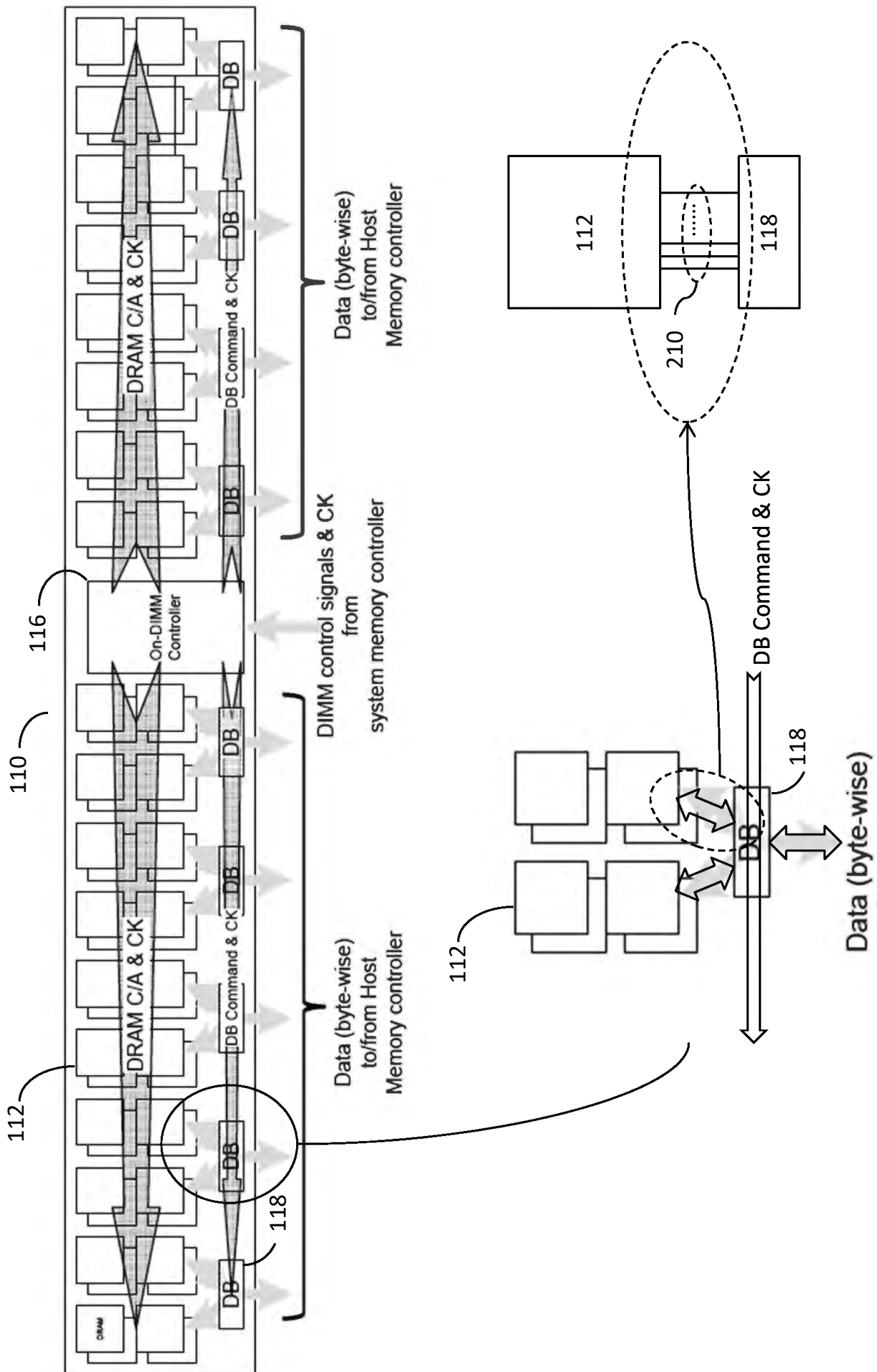


FIG. 2B

**FIG. 2C**

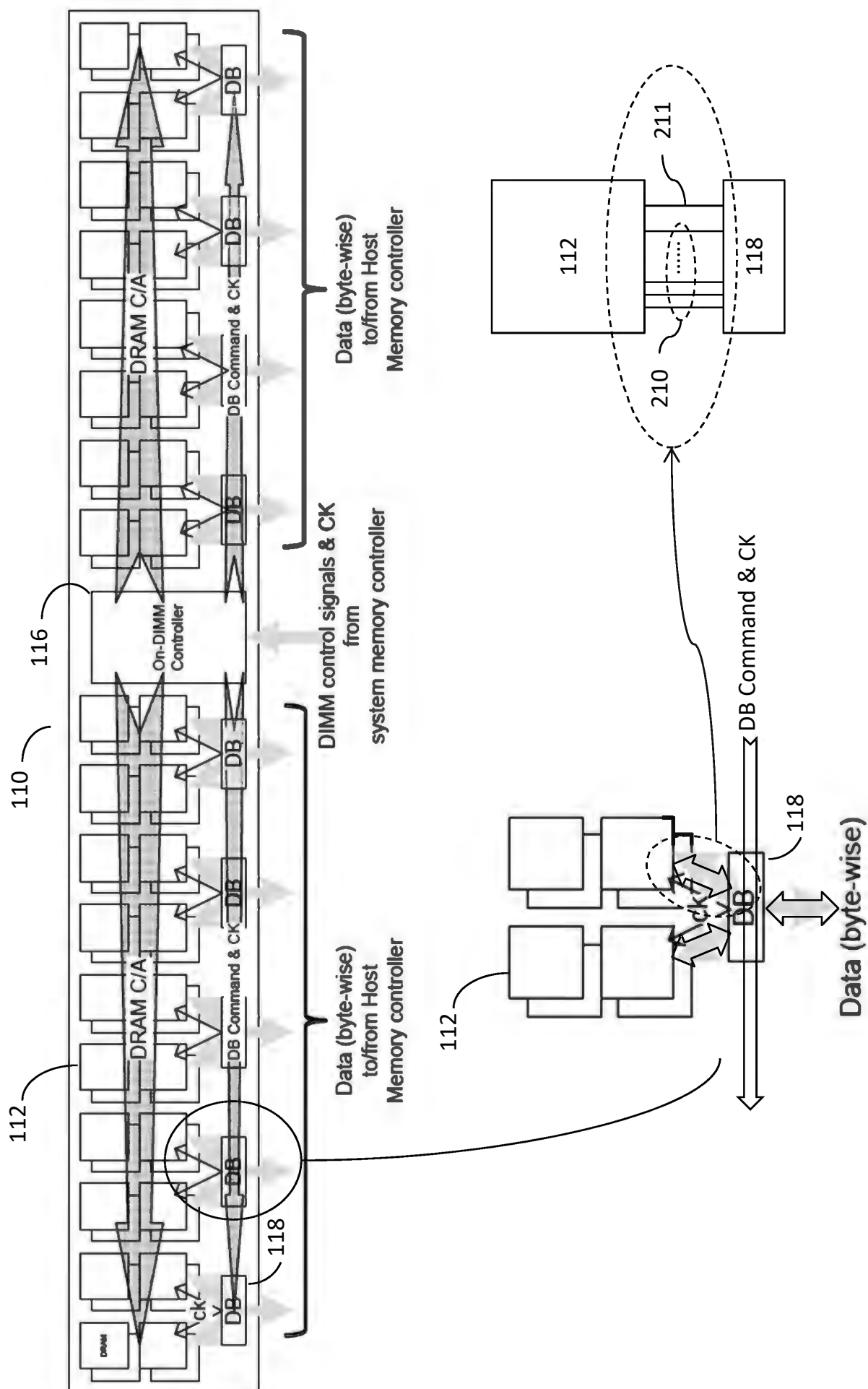


FIG. 2D

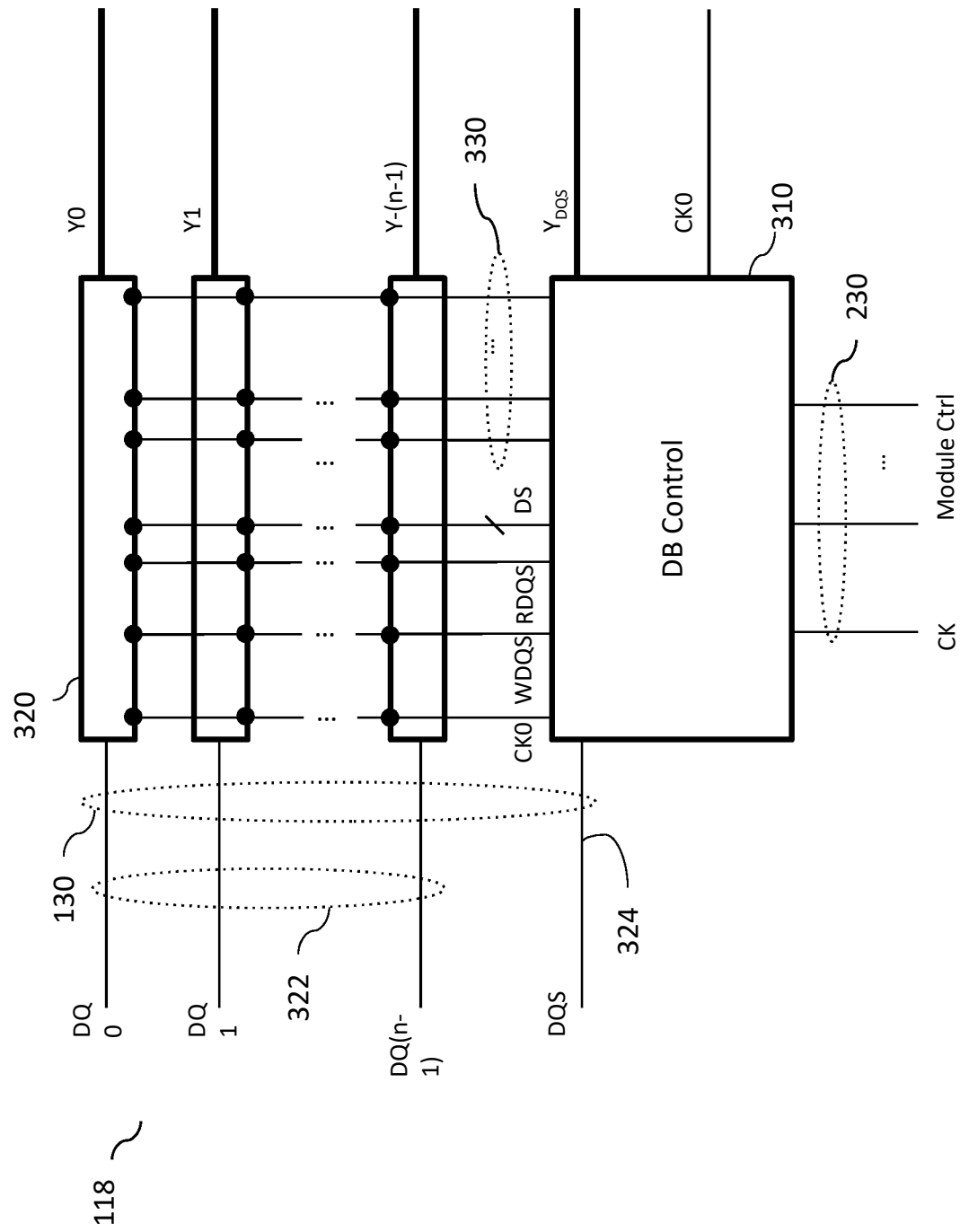
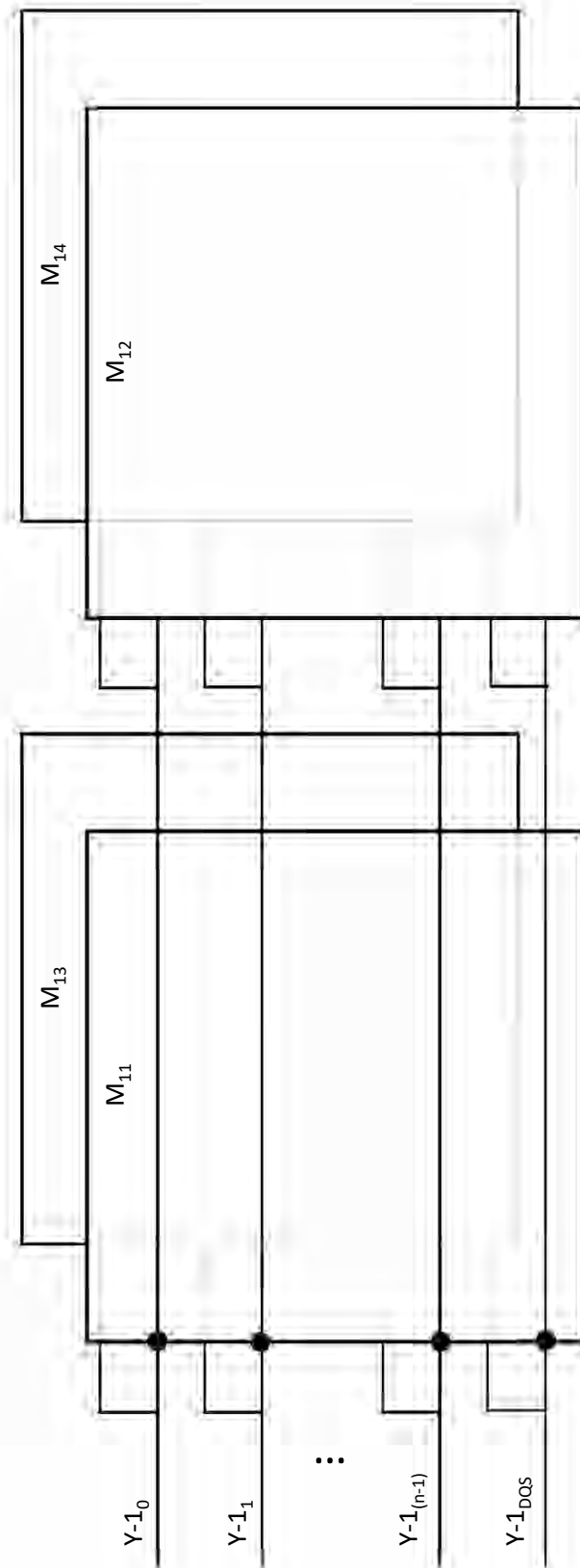


FIG. 3

**FIG. 4A**

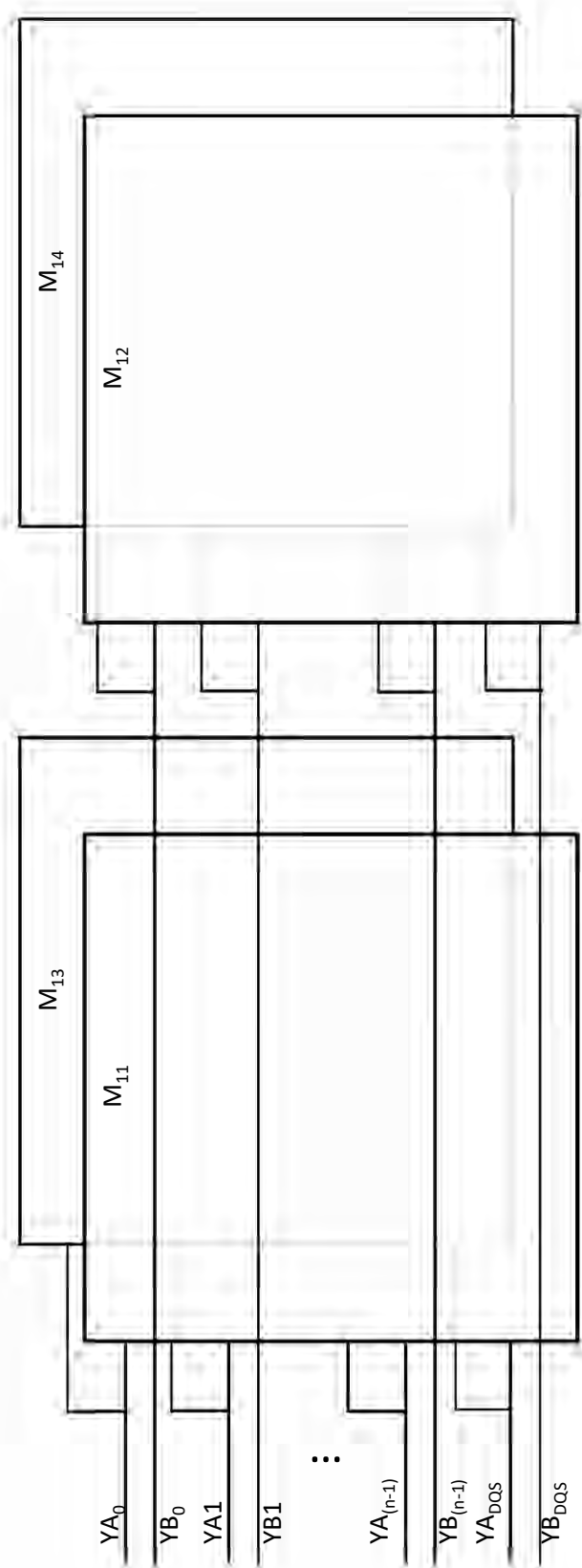


FIG. 4B

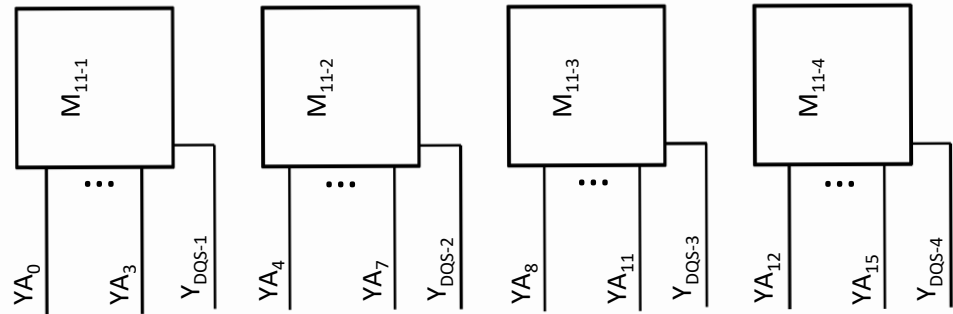


FIG. 5B

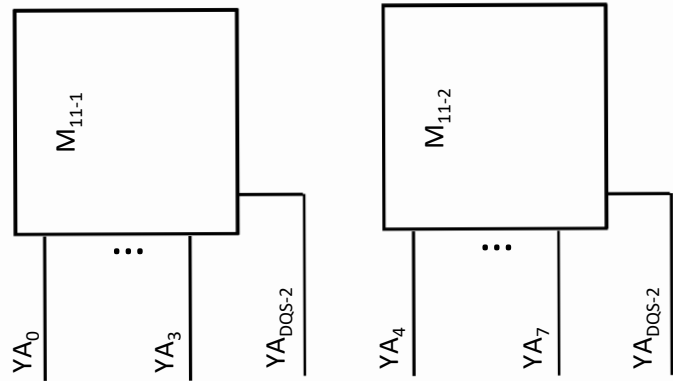
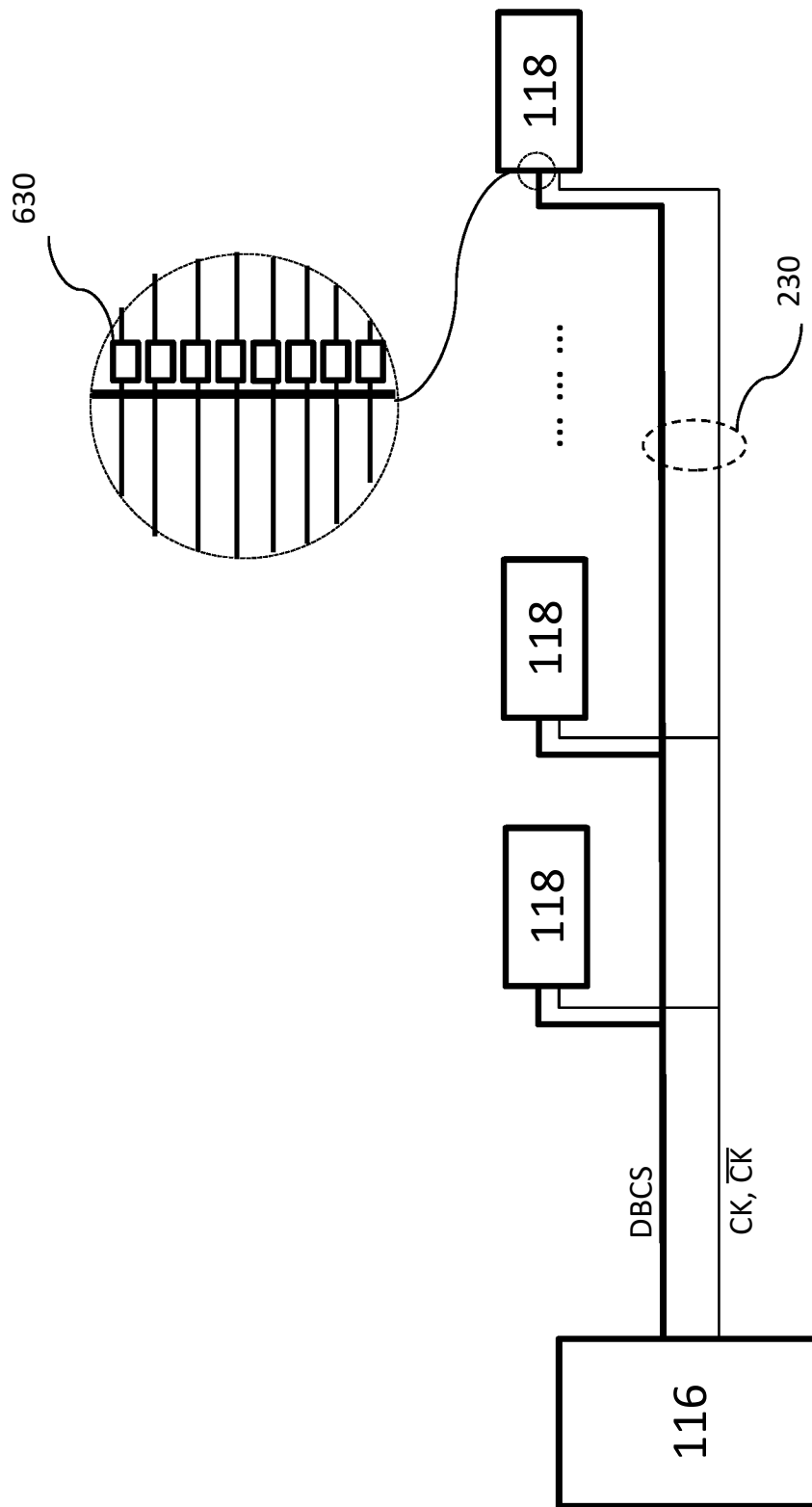


FIG. 5A

**FIG. 6A**

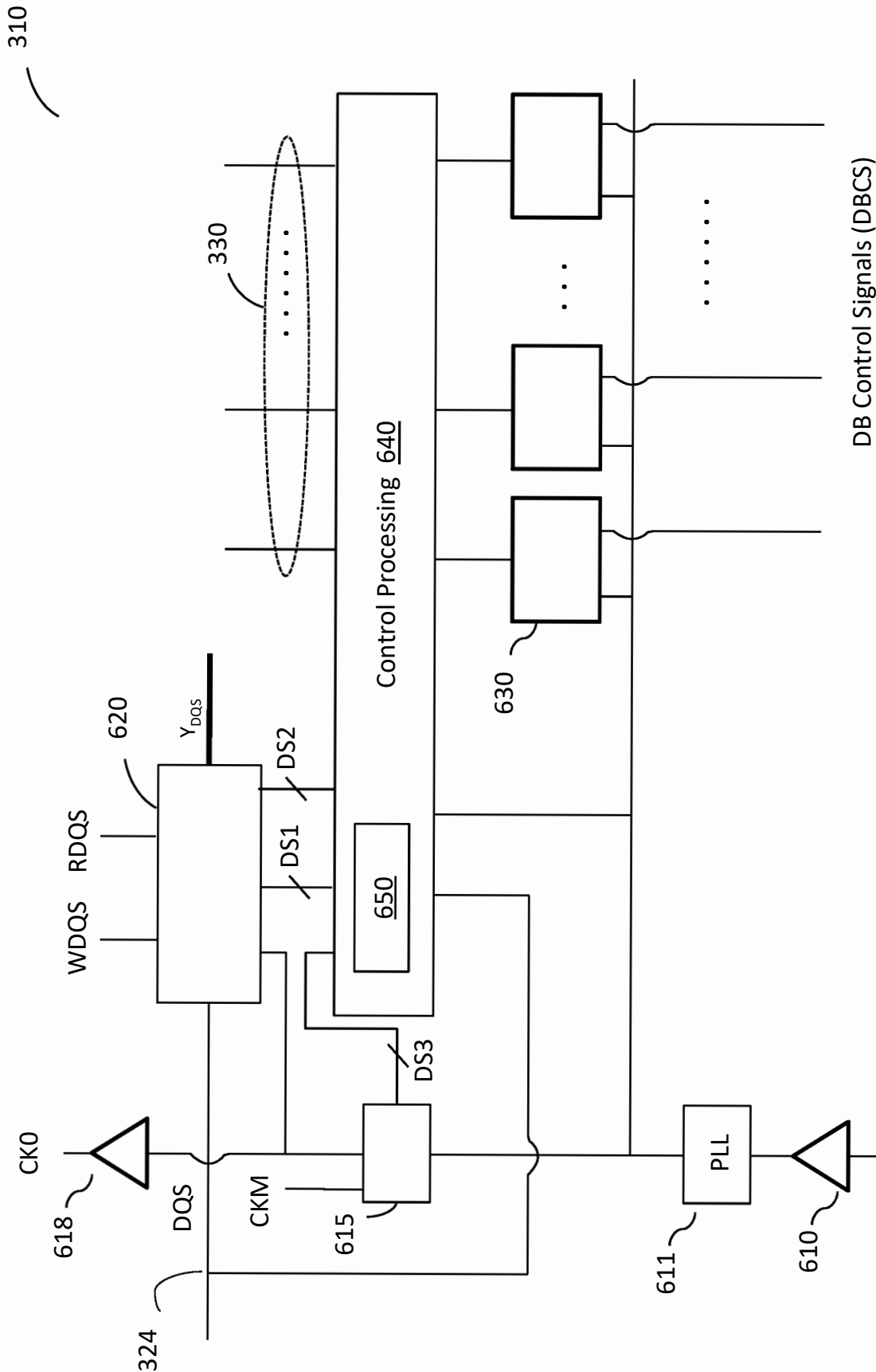


FIG. 6B

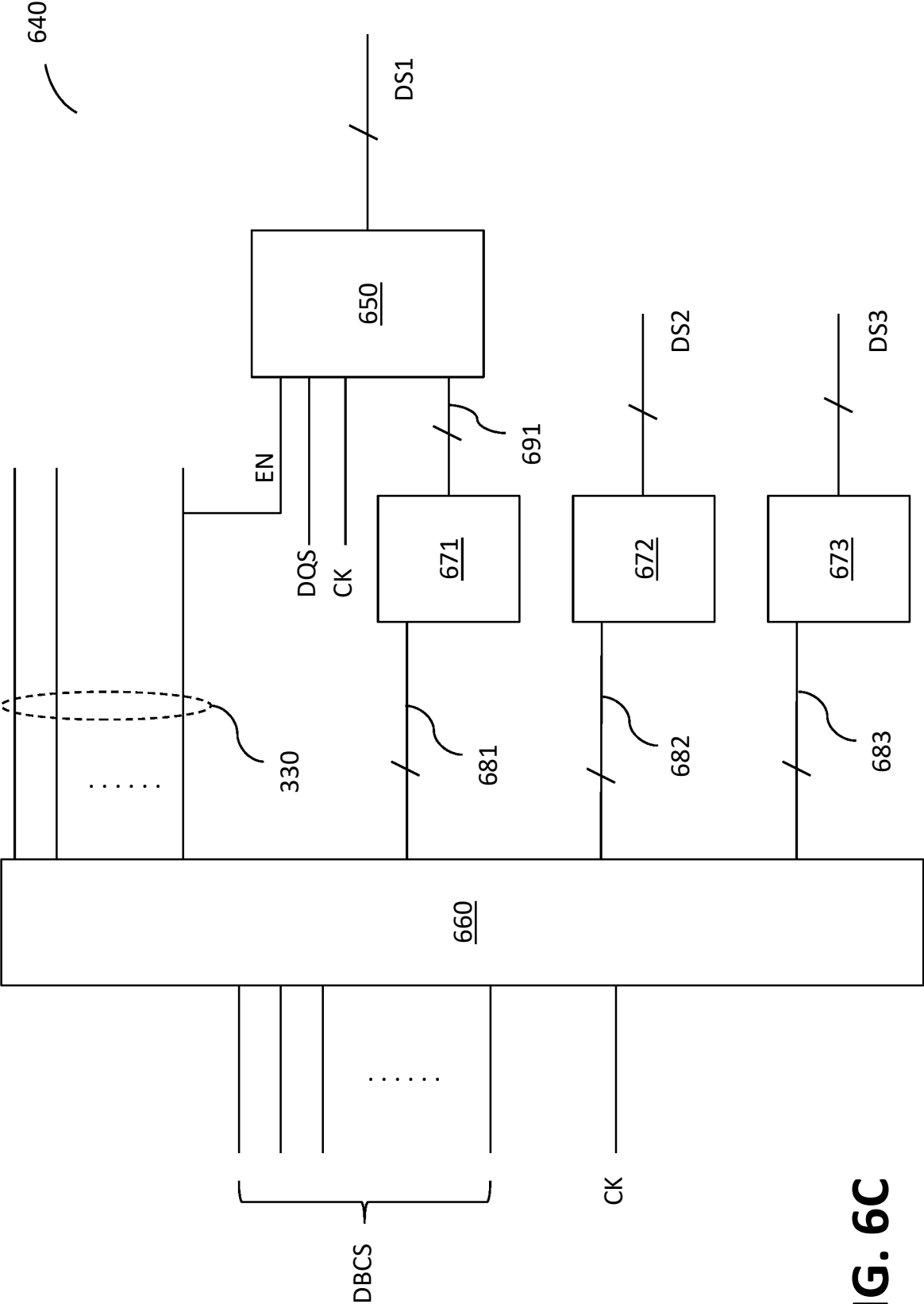
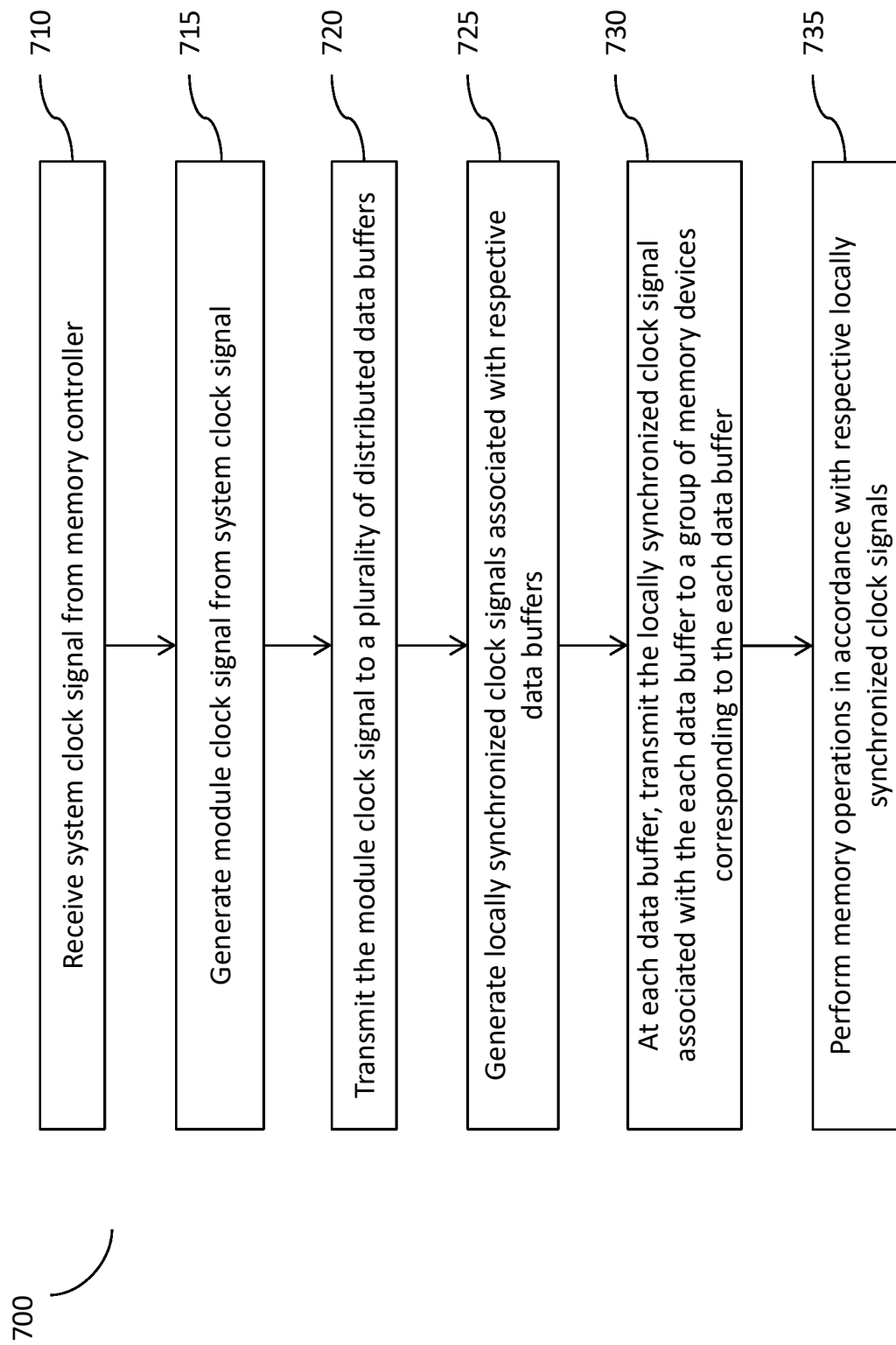
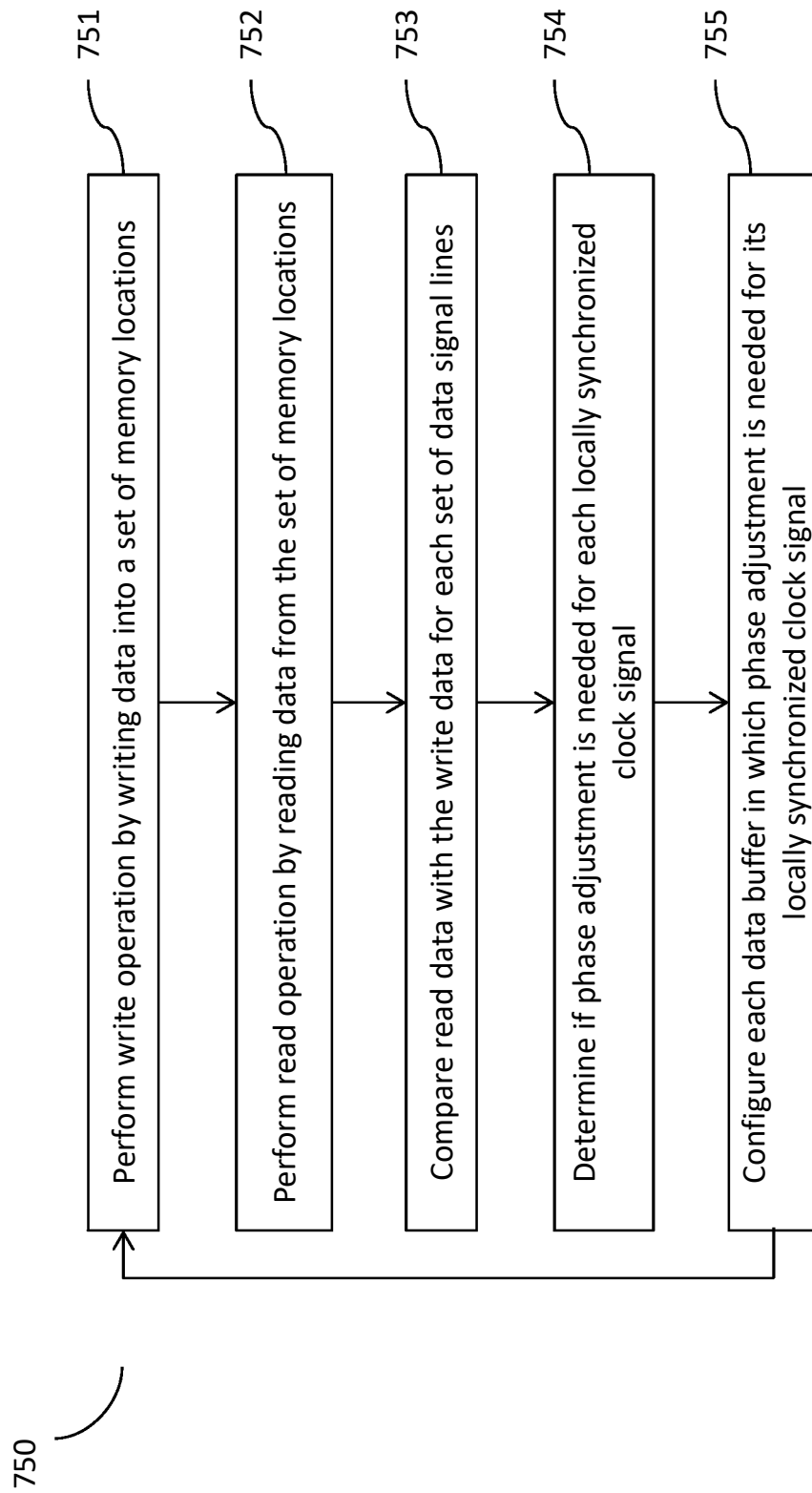


FIG. 6C

**FIG. 7A**

**FIG. 7B**

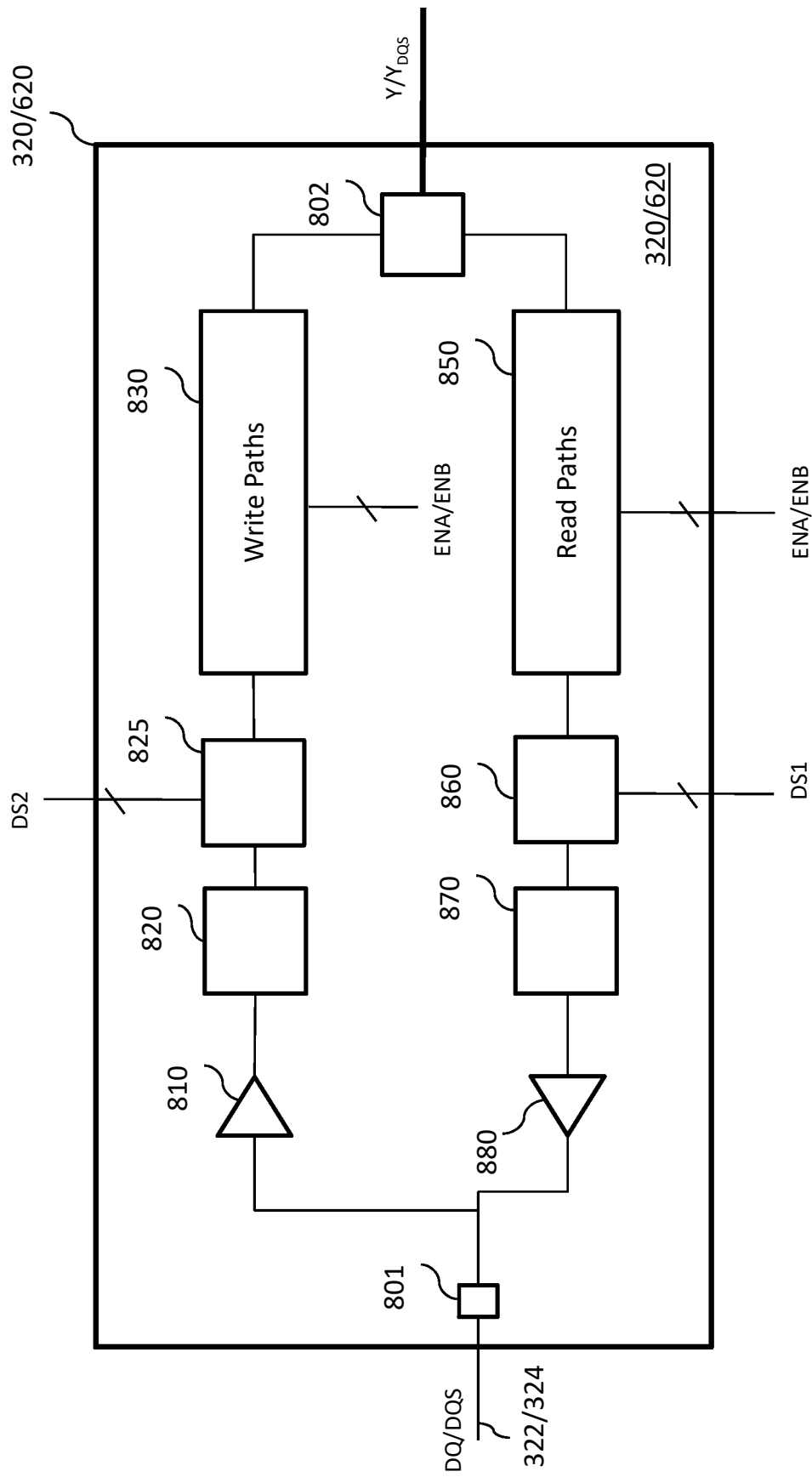
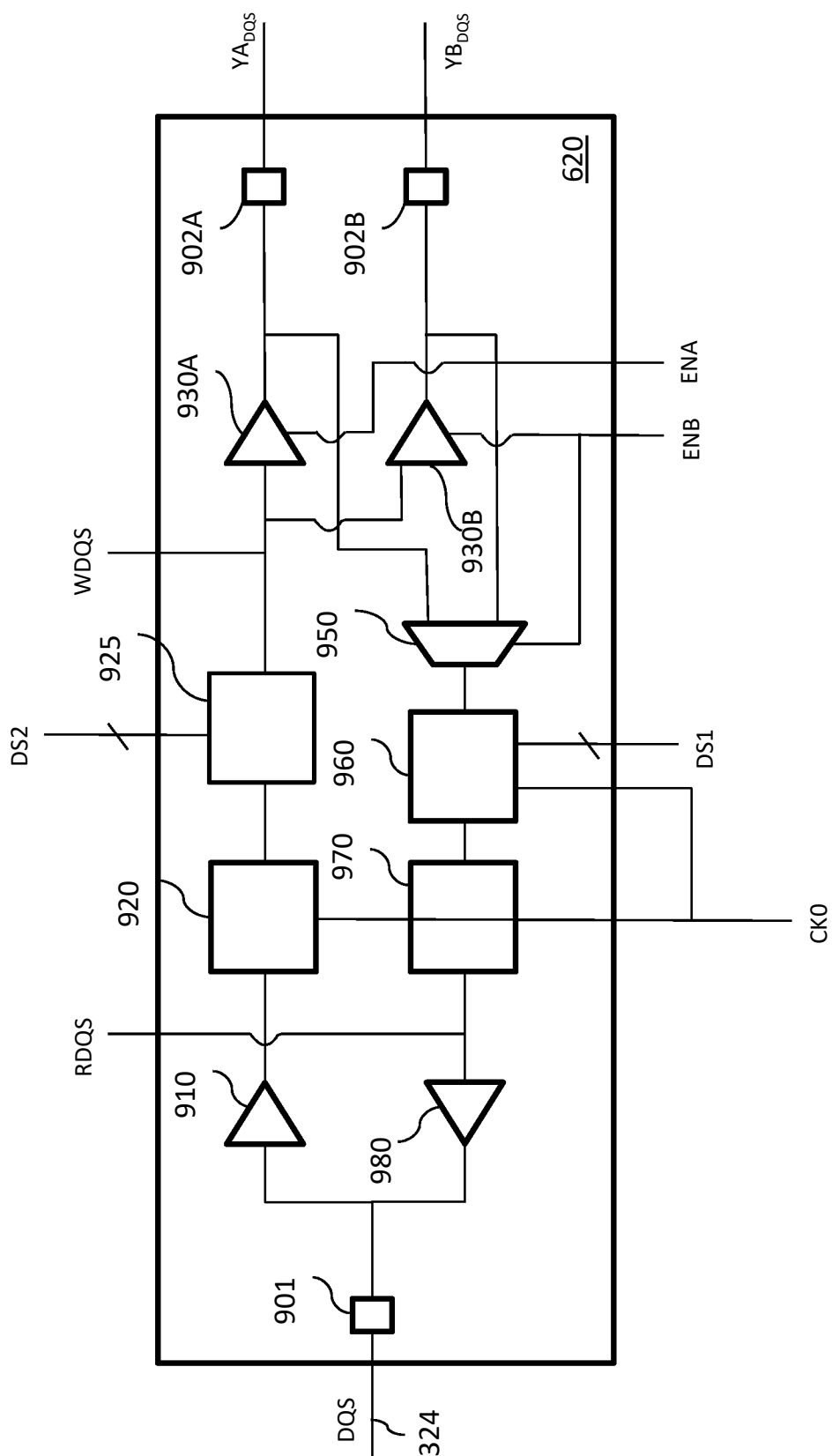


FIG. 8

**FIG. 9**

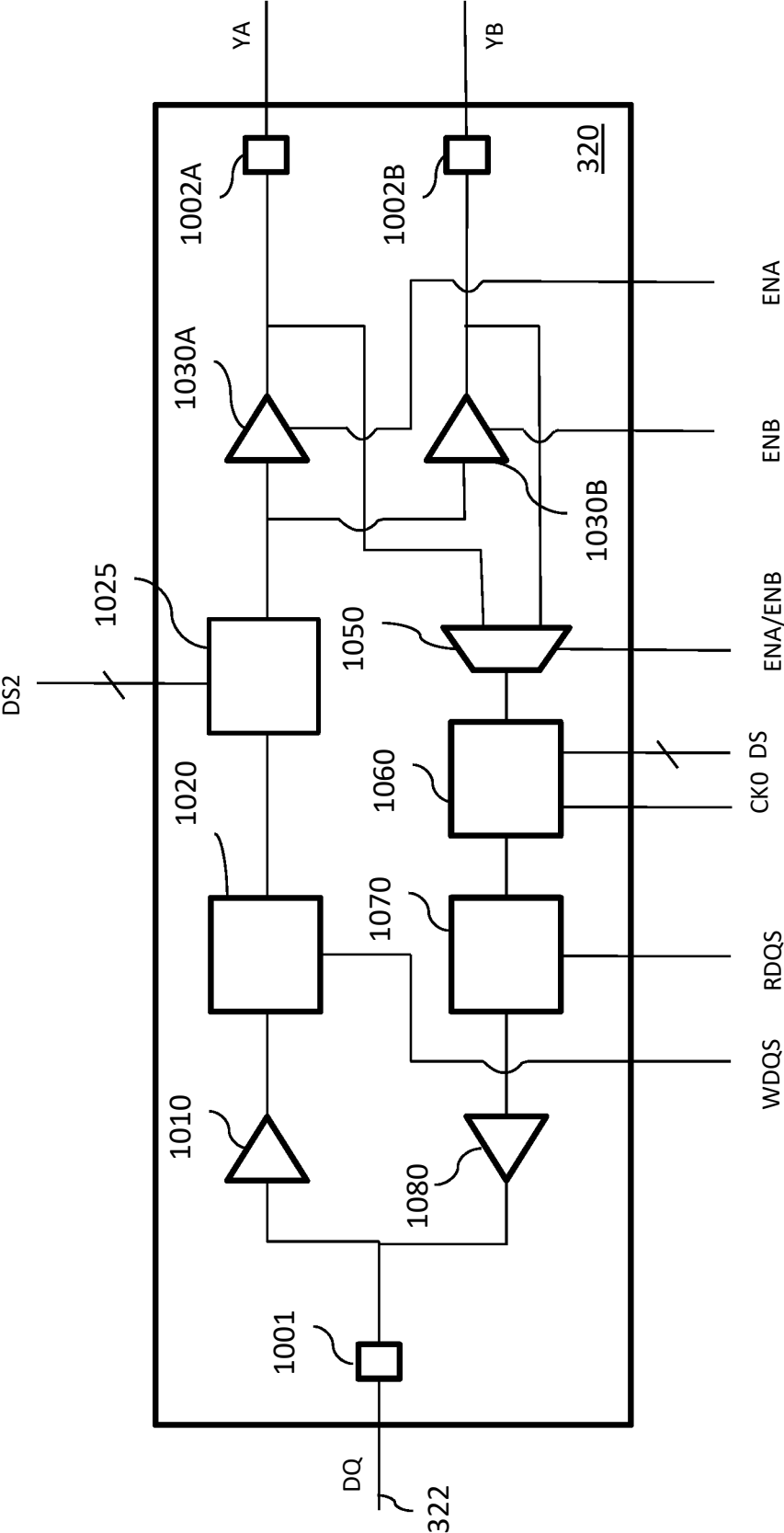


FIG. 10

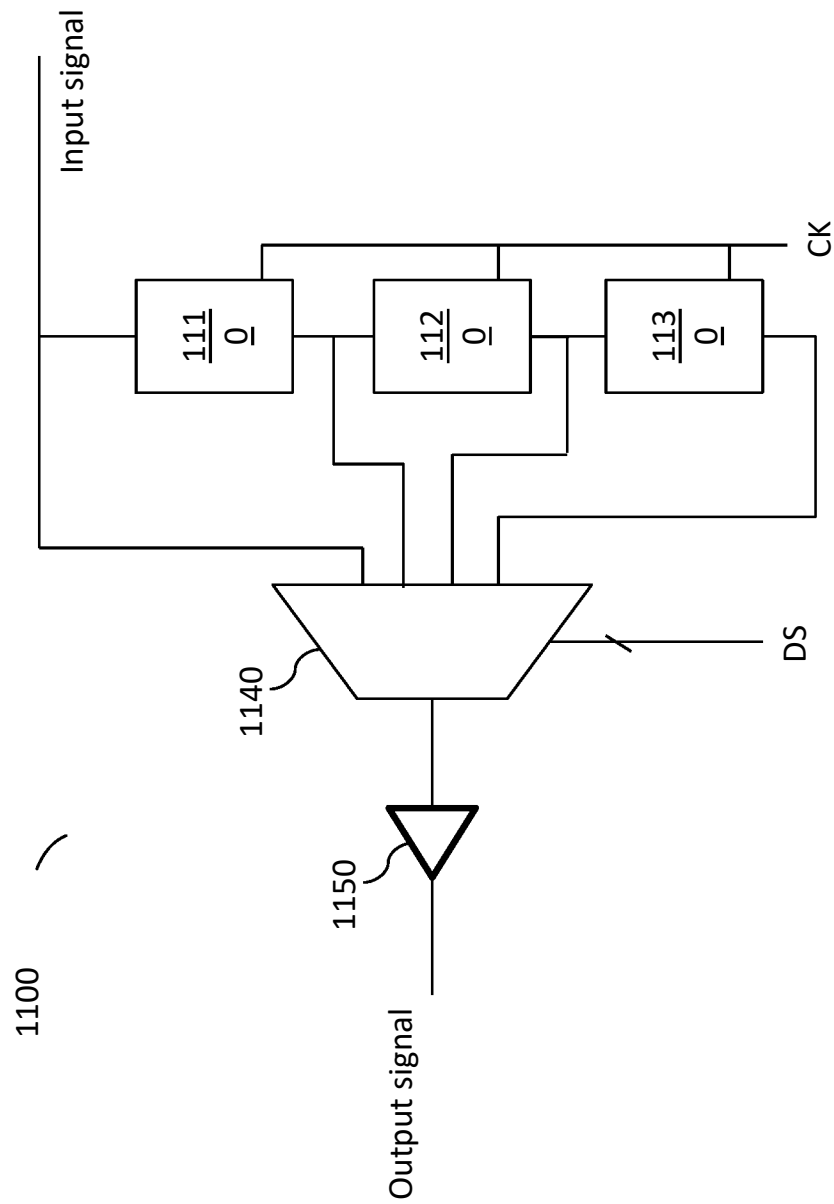


FIG. 11

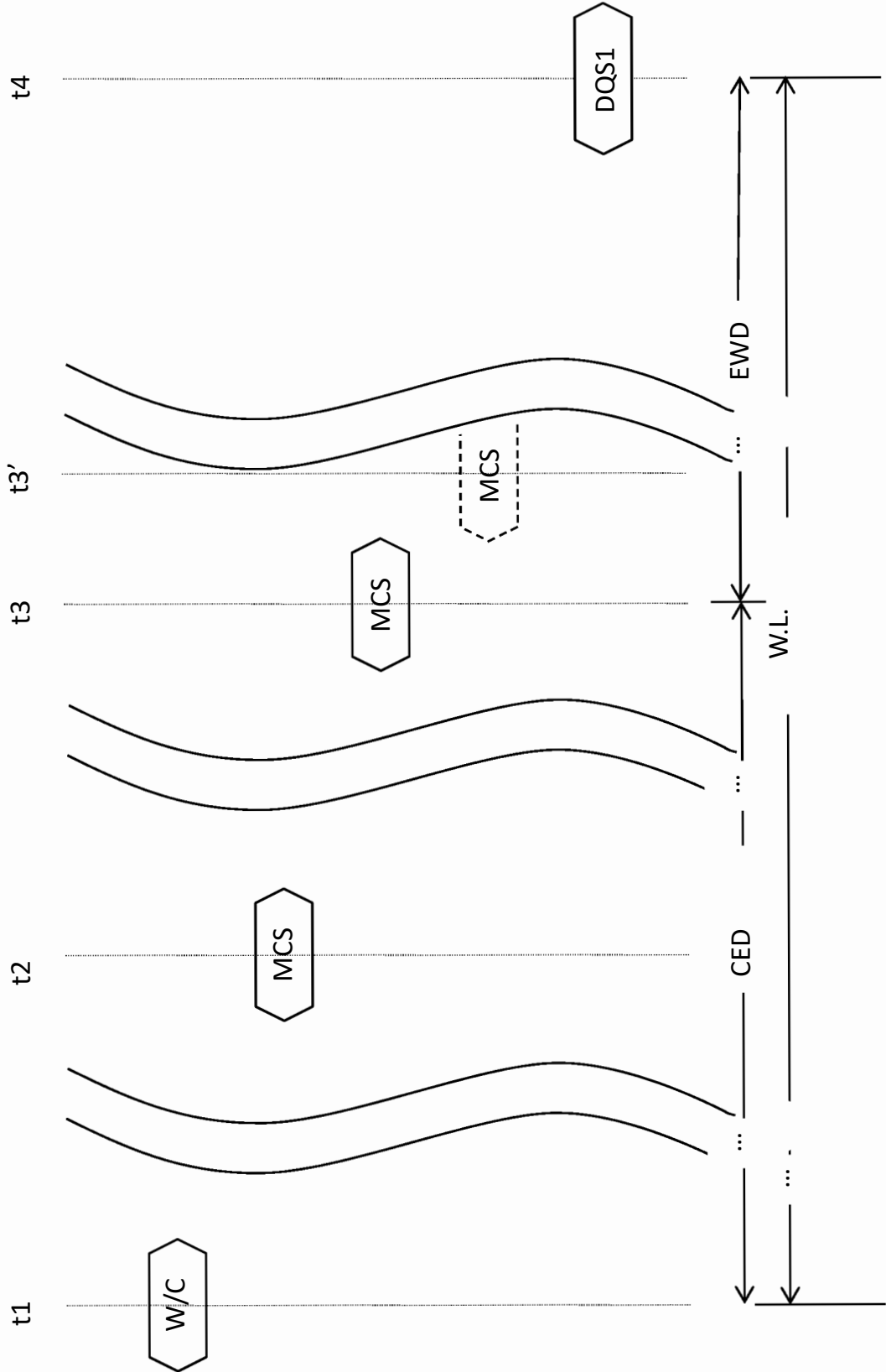


FIG. 12A

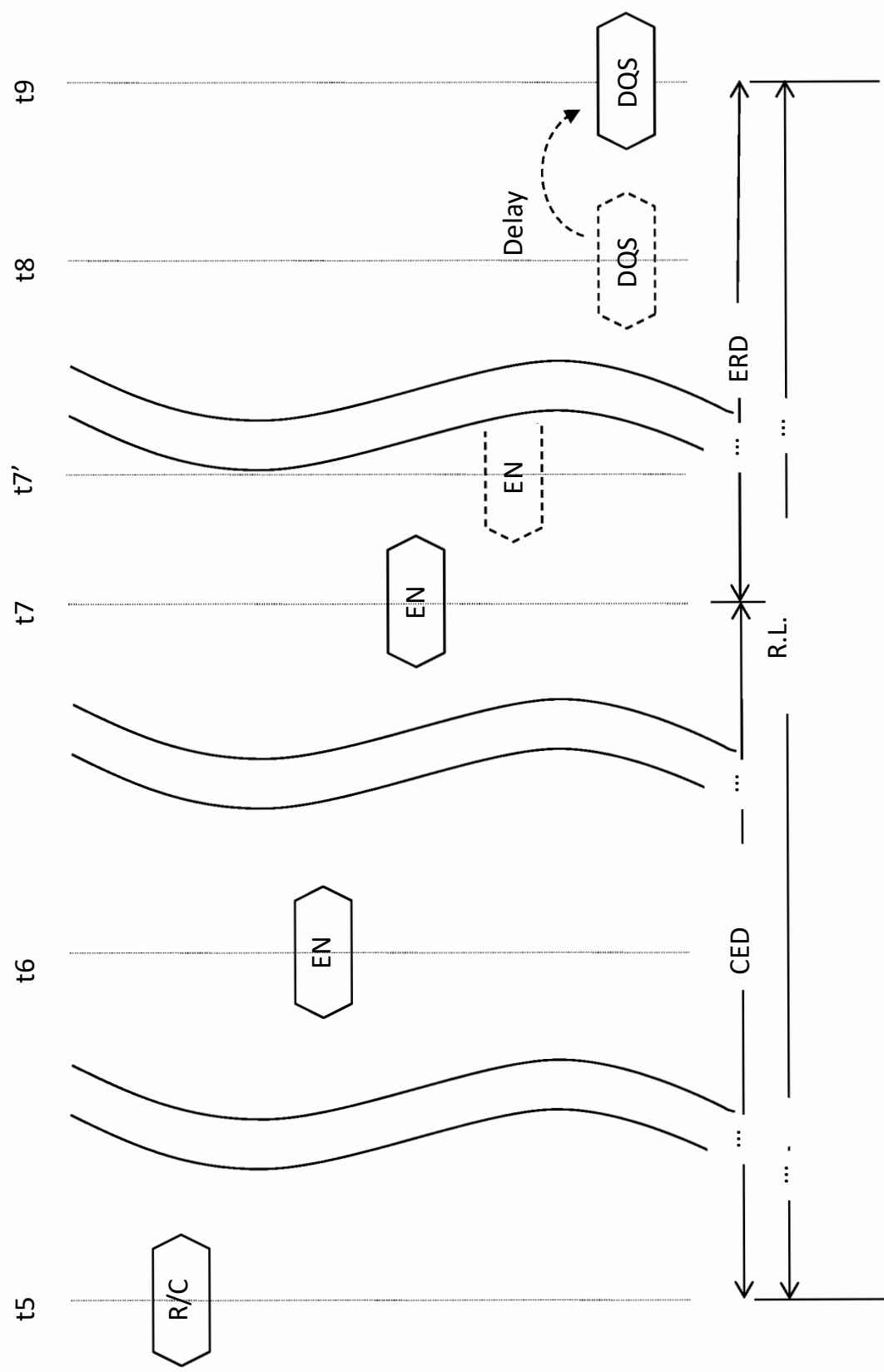


FIG. 12B

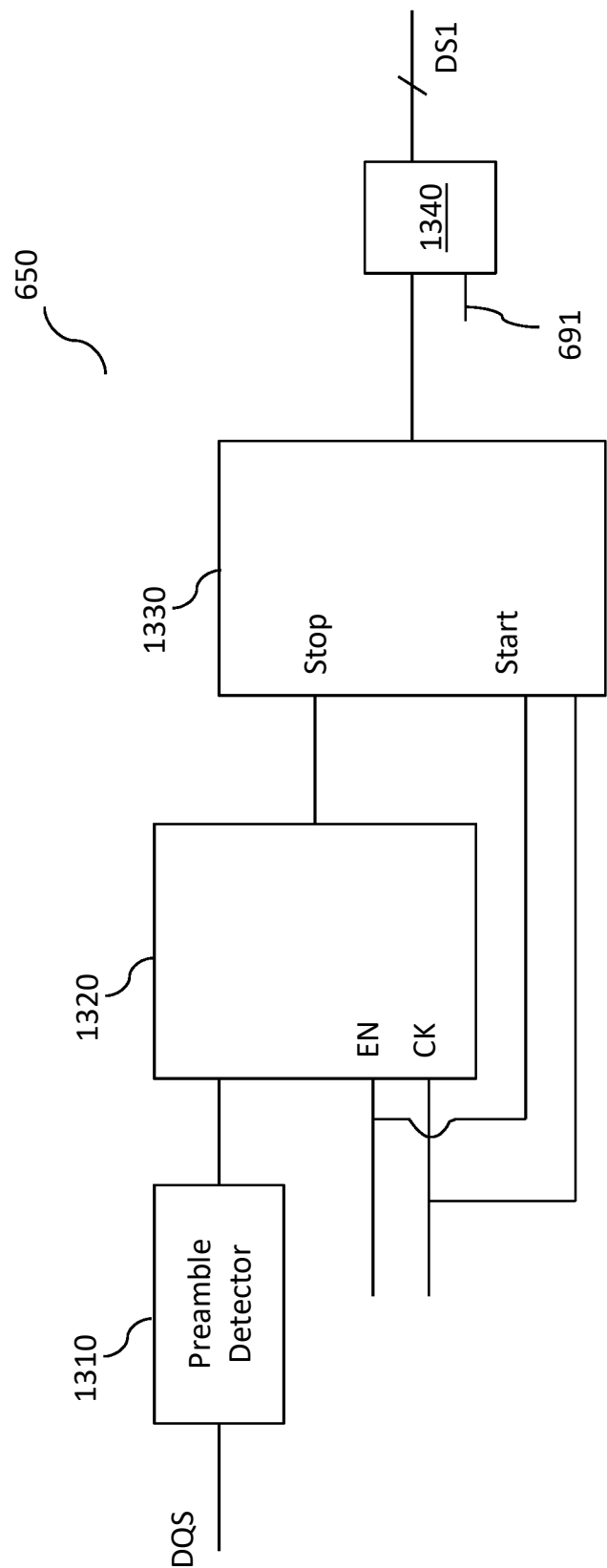
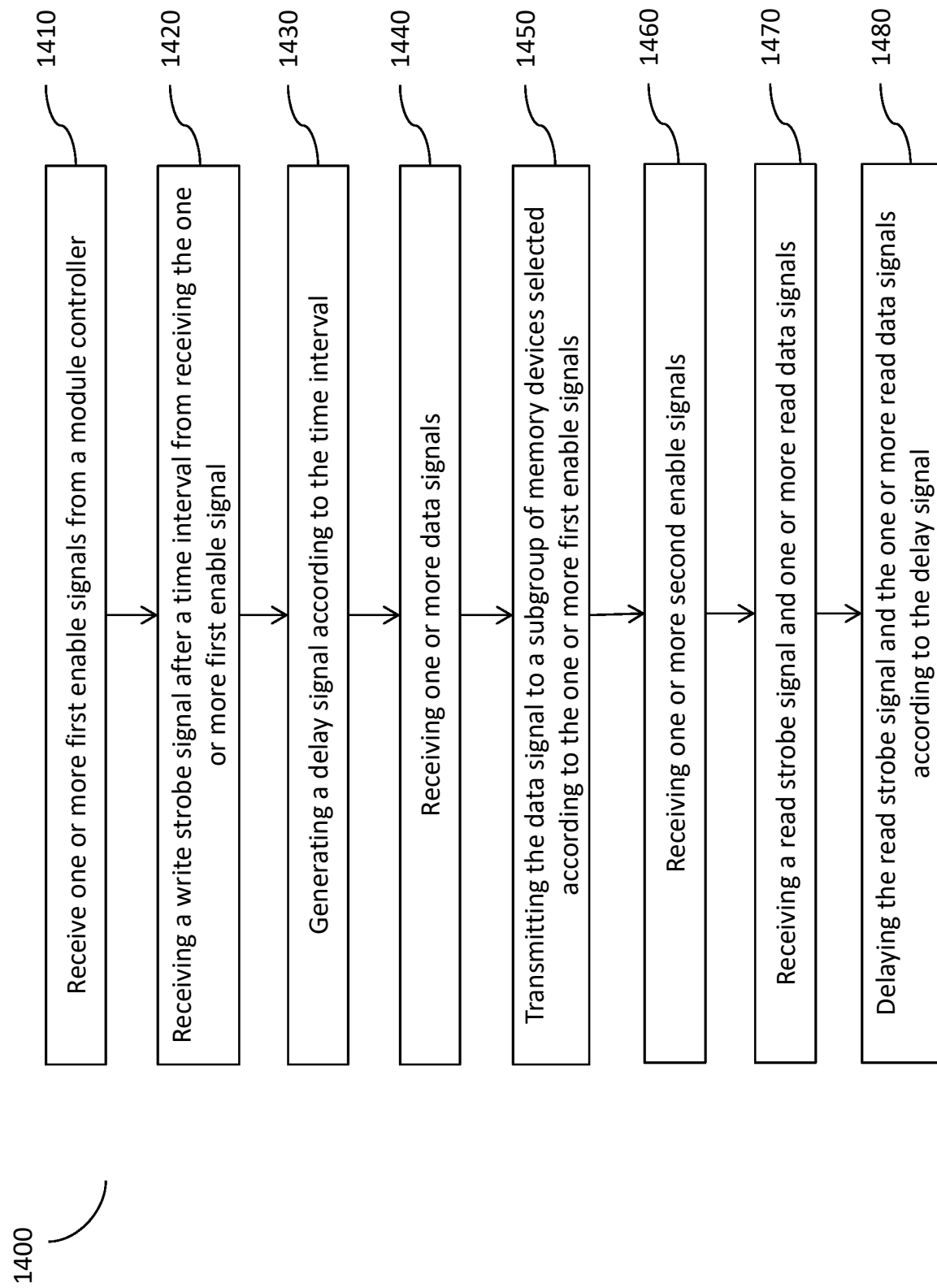


FIG. 13

**FIG. 14**



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/432,700	06/05/2019	2139	0.00	129980-5073-US01	20	2

CONFIRMATION NO. 7498

79141

Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

FILING RECEIPT



Date Mailed: 06/13/2019

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 14/445,035 07/28/2014 PAT 10324841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 06/12/2019

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/432,700**

Projected Publication Date: To Be Determined - pending completion of Missing Parts

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

PROTECTING YOUR INVENTION OUTSIDE THE UNITED STATES

Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

Almost every country has its own patent law, and a person desiring a patent in a particular country must make an application for patent in that country in accordance with its particular laws. Since the laws of many countries differ in various respects from the patent law of the United States, applicants are advised to seek guidance from specific foreign countries to ensure that patent rights are not lost prematurely.

Applicants also are advised that in the case of inventions made in the United States, the Director of the USPTO must issue a license before applicants can apply for a patent in a foreign country. The filing of a U.S. patent application serves as a request for a foreign filing license. The application's filing receipt contains further information and guidance as to the status of applicant's license for foreign filing.

Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

For information on preventing theft of your intellectual property (patents, trademarks and copyrights), you may wish to consult the U.S. Government website, <http://www.stopfakes.gov>. Part of a Department of Commerce initiative, this website includes self-help "toolkits" giving innovators guidance on how to protect intellectual property in specific countries such as China, Korea and Mexico. For questions regarding patent enforcement issues, applicants may call the U.S. Government hotline at 1-866-999-HALT (1-866-999-4258).

LICENSE FOR FOREIGN FILING UNDER
Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

GRANTED

The applicant has been granted a license under 35 U.S.C. 184, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" followed by a date appears on this form. Such licenses are issued in all applications where the conditions for issuance of a license have been met, regardless of whether or not a license may be required as set forth in 37 CFR 5.15. The scope and limitations of this license are set forth in 37 CFR 5.15(a) unless an earlier license has been issued under 37 CFR 5.15(b). The license is subject to revocation upon written notification. The date indicated is the effective date of the license, unless an earlier license of similar scope has been granted under 37 CFR 5.13 or 5.14.

This license is to be retained by the licensee and may be used at any time on or after the effective date thereof unless it is revoked. This license is automatically transferred to any related applications(s) filed under 37 CFR 1.53(d). This license is not retroactive.

The grant of a license does not in any way lessen the responsibility of a licensee for the security of the subject matter as imposed by any Government contract or the provisions of existing laws relating to espionage and the national security or the export of technical data. Licensees should apprise themselves of current regulations especially with respect to certain countries, of other agencies, particularly the Office of Defense Trade Controls, Department of State (with respect to Arms, Munitions and Implements of War (22 CFR 121-128)); the Bureau of Industry and Security, Department of Commerce (15 CFR parts 730-774); the Office of Foreign Assets Control, Department of Treasury (31 CFR Parts 500+) and the Department of Energy.

NOT GRANTED

No license under 35 U.S.C. 184 has been granted at this time, if the phrase "IF REQUIRED, FOREIGN FILING LICENSE GRANTED" DOES NOT appear on this form. Applicant may still petition for a license under 37 CFR 5.12, if a license is desired before the expiration of 6 months from the filing date of the application. If 6 months has lapsed from the filing date of this application and the licensee has not received any indication of a secrecy order under 35 U.S.C. 181, the licensee may foreign file the application pursuant to 37 CFR 5.15(b).

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The United States represents the largest, most dynamic marketplace in the world and is an unparalleled location for business investment, innovation, and commercialization of new technologies. The U.S. offers tremendous resources and advantages for those who invest and manufacture goods here. Through SelectUSA, our nation works to promote and facilitate business investment. SelectUSA provides information assistance to the international investor community; serves as an ombudsman for existing and potential investors; advocates on behalf of U.S. cities, states, and regions competing for global investment; and counsels U.S. economic development organizations on investment attraction best practices. To learn more about why the United States is the best country in the world to develop technology, manufacture products, deliver services, and grow your business, visit <http://www.SelectUSA.gov> or call +1-202-482-6800.



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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01

CONFIRMATION NO. 7498

FORMALITIES LETTER



79141
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

Date Mailed: 06/13/2019

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ **300** basic filing fee.
- \$ **160** surcharge.
- \$ **660** search fee.
- \$ **760** examination fee.
- \$(**0**) previous unapplied payment amount.
- \$ **1880** TOTAL FEE BALANCE DUE.

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

Mail Stop Missing Parts
Commissioner for Patents
P.O. Box 1450
Alexandria VA 22313-1450

Registered users of EFS-Web may alternatively submit their reply to this notice via EFS-Web, including a copy of this Notice and selecting the document description "Applicant response to Pre-Exam Formalities Notice".
<https://sportal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at **(571) 272-4000** or **(571) 272-4200** or **1-888-786-0101**.

/jmilani/

PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
16/432,700

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	300
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	660
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	760
TOTAL CLAIMS (37 CFR 1.16(j))	20 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*				x 460 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1720

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	7498
Serial No.:	16/432,700	Art Unit:	2139
Filed:	June 5, 2019	Examiner:	TBD
For:	Memory Module With Local Synchronization And Method Of Operation	Attorney Docket No:	129980-5073-US01

**RESPONSE TO THE NOTICE TO FILE MISSING PARTS
OF NONPROVISIONAL APPLICATION**

Mail Stop: MISSING PARTS
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In response to the Notice to File Missing Parts mailed June 13, 2019, enclosed is the Notice to File Missing Parts of Nonprovisional Application for the above-identified application. The fee for the late filing fee surcharge is estimated to be \$160.

Also enclosed please find a Fee Transmittal Form setting forth the calculation for the statutory basic filing, search and examination fees, together with the excess claim fees due in this matter. The fees calculated for the statutory basic filing, search and examination fees together with the excess claim fees and application size fee (if applicable) are estimated to be \$1720.

The Commissioner is hereby authorized to charge any required fee(s) to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073-US01).

Respectfully submitted,

Date: August 13, 2019

/ Jamie J. Zheng/ 51,167
Jamie J. Zheng (Reg. No.)
MORGAN, LEWIS & BOCKIUS LLP
1400 Page Mill Road
Palo Alto, CA 94304
(650) 843-4000



UNITED STATES PATENT AND TRADEMARK OFFICE

129980-5073-US01
650/12
UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22311-1450
www.uspto.gov

APPLICATION NUMBER	FILING (OR 371(c)) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01

CONFIRMATION NO. 7498

FORMALITIES LETTER



Date Mailed: 06/13/2019

79141

Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

NOTICE TO FILE MISSING PARTS OF NONPROVISIONAL APPLICATION

FILED UNDER 37 CFR 1.53(b)

Filing Date Granted

Items Required To Avoid Abandonment:

An application number and filing date have been accorded to this application. The item(s) indicated below, however, are missing. Applicant is given **TWO MONTHS** from the date of this Notice within which to file all required items below to avoid abandonment. Extensions of time may be obtained by filing a petition accompanied by the extension fee under the provisions of 37 CFR 1.136(a).

- The statutory basic filing fee is missing.
- The application search fee must be submitted.
- The application examination fee must be submitted.
- Surcharge as set forth in 37 CFR 1.16(f) must be submitted.

The surcharge is due for any one of:

- late submission of the basic filing fee, search fee, or examination fee,
- late submission of inventor's oath or declaration,
- filing an application that does not contain at least one claim on filing, or
- submission of an application filed by reference to a previously filed application.

SUMMARY OF FEES DUE:

The fee(s) required within **TWO MONTHS** from the date of this Notice to avoid abandonment is/are itemized below. No entity status discount is in effect. If applicant is qualified for small entity status, a written assertion of small entity status must be submitted to establish small entity status. (See 37 CFR 1.27). If applicant is qualified for micro entity status, an acceptable Certification of Micro Entity Status must be submitted to establish micro entity status. (See 37 CFR 1.29 and forms PTO/SB/15A and 15B.)

- \$ 300 basic filing fee.
- \$ 160 surcharge.
- \$ 660 search fee.
- \$ 760 examination fee.
- \$ (0) previous unapplied payment amount.
- \$ 1880 TOTAL FEE BALANCE DUE.

DOCKETED

FILE MISSING PARTS
DUE 13 JUN 2019
FRAI 133A07020
RECEIVED DOCKETING

JUN 13 2019

MORGAN LEWIS
PALO ALTO OFFICE

Replies must be received in the USPTO within the set time period or must include a proper Certificate of Mailing or Transmission under 37 CFR 1.8 with a mailing or transmission date within the set time period. For more information and a suggested format, see Form PTO/SB/92 and MPEP 512.

Replies should be mailed to:

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P.O. Box 1450
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<https://portal.uspto.gov/authenticate/AuthenticateUserLocalEPF.html>

For more information about EFS-Web please call the USPTO Electronic Business Center at 1-866-217-9197 or visit our website at <http://www.uspto.gov/ebc>.

If you are not using EFS-Web to submit your reply, you must include a copy of this notice.

Questions about the contents of this notice and the requirements it sets forth should be directed to the Office of Data Management, Application Assistance Unit, at (571) 272-4000 or (571) 272-4200 or 1-888-786-0101.

/jmilani/

UTILITY PATENT APPLICATION FEE TRANSMITTAL <i>(Only for new nonprovisional applications under 37 CFR § 1.53(b))</i>	Attorney Docket No.	129980-5073-US01
	First Inventor	Hyun Lee
	Title	Memory Module With Local Synchronization And Method Of Operation
	Electronically filed	August 13, 2019

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

BASIC FILING, SEARCH AND EXAMINATION FEES

Application Type	Filing Fees	Search Fees	Examination Fees	Fees Paid (\$)
Utility	\$ 300	\$ 660	\$ 760	\$ 1720

EXCESS CLAIM FEES

Type	No. Filed	Less	Extra	Extra Rate (\$)	Fee (\$)
Total Claims	20	- 20	0	\$ 100 each	0
Independent	2	- 3	0	\$ 460 each	0
Surcharge for late filing fee					160

APPLICATION SIZE FEE (Specification and Drawings)

Total Sheets	Extra Sheets	No. of each add'l 50 or fraction thereof	Fee (\$)
56 x .75- 100	0 / 50	0 (round up to whole no.) x \$400	0

Sub Total	1880
Applicant qualifies for the 50% Reduction for Independent Inventor, Nonprofit Organization or Small Business Concern minus electronic filing discount of \$75	
TOTAL FILING FEE	1880

Please charge the total filing fee and any other additional fees to Morgan, Lewis & Bockius LLP Deposit Account 50-0310 (order no. 129980-5073-US01).

Respectfully submitted,

Date: August 13, 2019

/ Jamie J. Zheng /

Jamie J. Zheng
MORGAN, LEWIS & BOCKIUS LLP
1400 Page Mill Road
Palo Alto, CA 94304
(650) 843-4000

51,167
(Reg. No.)

Electronic Patent Application Fee Transmittal

Application Number:	16432700			
Filing Date:	05-Jun-2019			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/S. Olivier			
Attorney Docket Number:	129980-5073-US01			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
UTILITY APPLICATION FILING	1011	1	300	300
UTILITY SEARCH FEE	1111	1	660	660
UTILITY EXAMINATION FEE	1311	1	760	760
Pages:				
Claims:				
Miscellaneous-Filing:				
LATE FILING FEE FOR OATH OR DECLARATION	1051	1	160	160
Petition:				

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1880

Electronic Acknowledgement Receipt

EFS ID:	36865376
Application Number:	16432700
International Application Number:	
Confirmation Number:	7498
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073-US01
Receipt Date:	13-AUG-2019
Filing Date:	05-JUN-2019
Time Stamp:	19:09:31
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	CARD
Payment was successfully received in RAM	\$1880
RAM confirmation Number	E20198CJ09574627
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

File Listing:					
Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980-5073US01_RESPONSE.pdf	790203	yes	5
			b48efcb88342bcf31ee963345f477f25c319d6cf		
	Multipart Description/PDF files in .zip description				
	Document Description		Start	End	
	Applicant Response to Pre-Exam Formalities Notice		1	4	
	Fee Worksheet (SB06)		5	5	
Warnings:					
Information:					
2	Fee Worksheet (SB06)	fee-info.pdf	37082	no	2
			9148046c3df057894402c091f710a1ad1ce51fab		
Warnings:					
Information:					
Total Files Size (in bytes):			827285		
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.					



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APPLICATION NUMBER	FILING or 371(c) DATE	GRP ART UNIT	FIL FEE REC'D	ATTY. DOCKET NO	TOT CLAIMS	IND CLAIMS
16/432,700	06/05/2019	2139	1880	129980-5073-US01	20	2

CONFIRMATION NO. 7498

UPDATED FILING RECEIPT

79141

Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304



Date Mailed: 08/16/2019

Receipt is acknowledged of this non-provisional utility patent application. The application will be taken up for examination in due course. Applicant will be notified as to the results of the examination. Any correspondence concerning the application must include the following identification information: the U.S. APPLICATION NUMBER, FILING DATE, NAME OF FIRST INVENTOR, and TITLE OF INVENTION. Fees transmitted by check or draft are subject to collection.

Please verify the accuracy of the data presented on this receipt. If an error is noted on this Filing Receipt, please submit a written request for a corrected Filing Receipt, including a properly marked-up ADS showing the changes with strike-through for deletions and underlining for additions. If you received a "Notice to File Missing Parts" or other Notice requiring a response for this application, please submit any request for correction to this Filing Receipt with your reply to the Notice. When the USPTO processes the reply to the Notice, the USPTO will generate another Filing Receipt incorporating the requested corrections provided that the request is grantable.

Inventor(s)

Hyun Lee, Ladera Ranch, CA;
Jayesh R. Bhakta, Cerritos, CA;

Applicant(s)

Netlist, Inc., Irvine, CA;

Power of Attorney: The patent practitioners associated with Customer Number 79141

Domestic Priority data as claimed by applicant

This application is a CON of 14/445,035 07/28/2014 PAT 10324841
which claims benefit of 61/859,215 07/27/2013

Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.) - None.

Foreign application information must be provided in an Application Data Sheet in order to constitute a claim to foreign priority. See 37 CFR 1.55 and 1.76.

Permission to Access Application via Priority Document Exchange: Yes

Permission to Access Search Results: Yes

Applicant may provide or rescind an authorization for access using Form PTO/SB/39 or Form PTO/SB/69 as appropriate.

If Required, Foreign Filing License Granted: 06/12/2019

The country code and number of your priority application, to be used for filing abroad under the Paris Convention, is **US 16/432,700**

Projected Publication Date: 11/21/2019

Non-Publication Request: No

Early Publication Request: No

Title

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Preliminary Class

711

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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Since the rights granted by a U.S. patent extend only throughout the territory of the United States and have no effect in a foreign country, an inventor who wishes patent protection in another country must apply for a patent in a specific country or in regional patent offices. Applicants may wish to consider the filing of an international application under the Patent Cooperation Treaty (PCT). An international (PCT) application generally has the same effect as a regular national patent application in each PCT-member country. The PCT process **simplifies** the filing of patent applications on the same invention in member countries, but **does not result** in a grant of "an international patent" and does not eliminate the need of applicants to file additional documents and fees in countries where patent protection is desired.

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Applicants may wish to consult the USPTO booklet, "General Information Concerning Patents" (specifically, the section entitled "Treaties and Foreign Patents") for more information on timeframes and deadlines for filing foreign patent applications. The guide is available either by contacting the USPTO Contact Center at 800-786-9199, or it can be viewed on the USPTO website at <http://www.uspto.gov/web/offices/pac/doc/general/index.html>.

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Title 35, United States Code, Section 184
Title 37, Code of Federal Regulations, 5.11 & 5.15

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PATENT APPLICATION FEE DETERMINATION RECORD

Substitute for Form PTO-875

Application or Docket Number
16/432,700

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	300
SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A			N/A	660
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	760
TOTAL CLAIMS (37 CFR 1.16(j))	20 minus 20 =	*			OR	x 100 =	0.00
INDEPENDENT CLAIMS (37 CFR 1.16(h))	2 minus 3 =	*				x 460 =	0.00
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0.00
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0.00
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	1720

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	*	Minus	**	=		OR	x	=
Independent (37 CFR 1.16(h))	*	Minus	***	=		OR	x	=
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".

*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".

The "Highest Number Previously Paid For" (Total or Independent) is the highest found in the appropriate box in column 1.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
Examiner Name	To be assigned				
Sheet	1	of	6	Attorney Docket Number	129980-5073US01

U.S. PATENT DOCUMENTS					
Examiner Initials	Cite No.	Document Number	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear
		Number - Kind Code			
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		5,655,153 A	08-05-1997	EMC Corp.	
		5,699,542 A	12-16-1997	Intel Corp.	

Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019		
				Application Number	16/432,700	
				Filing Date	June 5, 2019	
				First Named Inventor	Hyun Lee	
				Art Unit	2139	
				Examiner Name	To be assigned	
Sheet	2	of	6	Attorney Docket Number		129980-5073US01

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	2002/0038405 A1	03-28-2002	Leddige	
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Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
Examiner Name	To be assigned				
Sheet	3	of	6	Attorney Docket Number	129980-5073US01

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		6,400,637 B1	06-04-2002	Matsushita Electric Ind.	
		6,408,356 B1	06-18-2002	IBM	
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Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	4	of	6	Attorney Docket Number	129980-5073US01

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Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019		
				Application Number	16/432,700	
				Filing Date	June 5, 2019	
				First Named Inventor	Hyun Lee	
				Art Unit	2139	
				Examiner Name	To be assigned	
Sheet	5	of	6	Attorney Docket Number		129980-5073US01

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Examiner Signature		Date Considered	
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	6	of	6	Attorney Docket Number	129980-5073US01

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Examiner Initials	Cite No.	Foreign Patent Document Country Code – Number - Kind Code (if known)	Publication Date MM-DD-YYYY	Name of Patentee or Applicant of Cited Document	Pages, Columns, Lines, Where Relevant Passages or Relevant Figures Appear	T
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		EP 1816570 A2	08-08-2007	Rambus Inc.		

NON PATENT LITERATURE DOCUMENTS

Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published

Examiner Signature		Date Considered	
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Electronic Acknowledgement Receipt

EFS ID:	36916879
Application Number:	16432700
International Application Number:	
Confirmation Number:	7498
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/Lamesha Robertson
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073-US01
Receipt Date:	19-AUG-2019
Filing Date:	05-JUN-2019
Time Stamp:	18:04:27
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980_5073US01_IDS.pdf	822956	yes	10
			5d6a3164e9956751e333bd71c8db6e9fcdff31cc8		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Transmittal Letter	1	4
	Information Disclosure Statement (IDS) Form (SB08)	5	10
Warnings:			
Information:			
Total Files Size (in bytes):		822956	
This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503. <u>New Applications Under 35 U.S.C. 111</u> If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application. <u>National Stage of an International Application under 35 U.S.C. 371</u> If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course. <u>New International Application Filed with the USPTO as a Receiving Office</u> If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.			

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee et al	Confirmation No.:	7498
Serial No.:	16/432,700	Art Unit:	2133
Filed:	June 5, 2019	Examiner:	To be assigned
For:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION	Attorney Docket No.:	129980-5073US01

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information which the Examiner may consider material to the examination of the subject U.S. patent application. It is requested that the Examiner make this information of record if it is deemed material to the examination of the application.

1. Enclosures accompanying this Information Disclosure Statement are:
 - 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
 - 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☐ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☐ all other information or portion which caused it to be listed on the PTO-1449.
 - 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
 - 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.
2. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
 - ☒ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);

- ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
- ☐ Before the mailing of the first Office action on the merits;
- ☐ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.
3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b, 3c or 3d)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check Item 4a, and 4b, 4c or 4d)

- 4a. ☐ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 4c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 4d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

5. ☐ Certification Statement (applicable if Item 3a or Item 4a is checked)

(Check either Item 5a, 5b, 5c or 5d)

- 5a. ☐ In accordance with 37 C.F.R. §1.97(e)(1), it is certified that each item of information contained in this information disclosure statement was first cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this Information Disclosure Statement.
- 5b. ☐ In accordance with 37 C.F.R. §1.97(e)(2), it is certified that no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 C.F.R. §1.56(c) more than three months prior to the filing of this information disclosure statement.
- 5c. ☐ Each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
- 5d. ☐ Pursuant to 37 C.F.R. §1.704(d), each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
6. ☒ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.
7. ☒ This application is a continuation application under 37 C.F.R. §1.53(b) or (d).
(Check appropriate Items 7a, 7b and/or 7c)
- 7a. ☐ A Petition to Withdraw from issue under 37 C.F.R. §1.313(b)(5) is concurrently filed herewith.
- 7b. ☒ Copies of publications listed on Form PTO-1449 from prior application Serial No. 14/445,035, filed on July 28, 2014, of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7c. ☐ Copies of the publications listed on the attached Form PTO-1449 that were not previously cited in prior application Serial No. , filed on , are provided herewith.
8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f) supplements the Information Disclosure Statement filed on . A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be considered as if properly filed on .

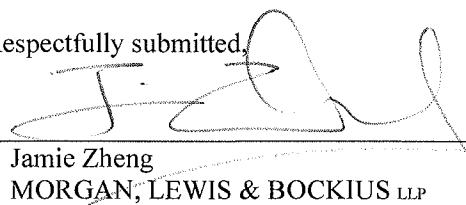
9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:

(Check Item 9a, 9b, or 9c)

- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073US01).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: August 19, 2019


51,167
(Reg. No.)
Jamie Zheng
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Palo Alto, CA 94304
(650) 843-4000



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01

CONFIRMATION NO. 7498

PUBLICATION NOTICE



OC000000112809522

79141
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

Title:MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

Publication No.US-2019-0354480-A1

Publication Date:11/21/2019

NOTICE OF PUBLICATION OF APPLICATION

The above-identified application will be electronically published as a patent application publication pursuant to 37 CFR 1.211, et seq. The patent application publication number and publication date are set forth above.

The publication may be accessed through the USPTO's publically available Searchable Databases via the Internet at www.uspto.gov. The direct link to access the publication is currently <http://www.uspto.gov/patft/>.

The publication process established by the Office does not provide for mailing a copy of the publication to applicant. A copy of the publication may be obtained from the Office upon payment of the appropriate fee set forth in 37 CFR 1.19(a)(1). Orders for copies of patent application publications are handled by the USPTO's Public Records Division. The Public Records Division can be reached by telephone at (571) 272-3150 or (800) 972-6382, by facsimile at (571) 273-3250, by mail addressed to the United States Patent and Trademark Office, Public Records Division, Alexandria, VA 22313-1450 or via the Internet.

In addition, information on the status of the application, including the mailing date of Office actions and the dates of receipt of correspondence filed in the Office, may also be accessed via the Internet through the Patent Electronic Business Center at www.uspto.gov using the public side of the Patent Application Information and Retrieval (PAIR) system. The direct link to access this status information is currently <https://portal.uspto.gov/pair/PublicPair>. Prior to publication, such status information is confidential and may only be obtained by applicant using the private side of PAIR.

Further assistance in electronically accessing the publication, or about PAIR, is available by calling the Patent Electronic Business Center at 1-866-217-9197.

Office of Data Management, Application Assistance Unit (571) 272-4000, or (571) 272-4200, or 1-888-786-0101

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/432,700		Filing Date 06/05/2019		☐ To be Mailed				
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO												
APPLICATION AS FILED - PART I												
		(Column 1)	(Column 2)									
FOR		NUMBER FILED	NUMBER EXTRA		RATE (\$)		FEE (\$)					
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A		N/A							
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))		N/A	N/A		N/A							
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A		N/A							
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *		x \$100 =								
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *		x \$460 =								
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).										
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))												
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL							
APPLICATION AS AMENDED - PART II												
		(Column 1)		(Column 2)	(Column 3)							
AMENDMENT	12/05/2019	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)				
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$100 =		0				
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$460 =		0				
	☐ Application Size Fee (37 CFR 1.16(s))											
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))											
						TOTAL ADD'L FEE		0				
		(Column 1)		(Column 2)	(Column 3)							
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)				
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =						
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =						
	☐ Application Size Fee (37 CFR 1.16(s))											
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))											
						TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE						
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/JESSICA T GAYNOR/						
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".												
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.												

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Electronically filed December 5, 2019

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	7498
Serial No.:	16/432,700	Art Unit:	2133
Filed:	June 5, 2019	Examiner:	Rutz, Jared Ian
For:	<i>Memory Module With Local Synchronization And Method Of Operation</i>	Attorney Docket No.:	129980-5073-US01

PRELIMINARY AMENDMENT

Mail Stop: Amendment

Commissioner for Patents

P.O. Box 1450

Alexandria, VA 22313-1450

Please enter the enclosed preliminary amendment.

The Commissioner is hereby authorized to charge any required fees or credit any overpayments to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073-US01).

Amendments to the Specification

Please revise the specification as shown below:

Please amend the abstract as follows:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

ABSTRACT OF THE DISCLOSURE

A memory module includes memory device groups, and a control circuit configurable to receive a system clock and input address and control (C/A) signals from a memory controller, and output a module clock, module C/A signals and data buffer control signals. The memory module further includes data buffers corresponding to respective memory device groups and configurable to receive the module clock and the data buffer control signals from the control circuit. A respective data buffer includes a n-bit wide data path and logic configured to control the data path in response to the data buffer control signals. The n-bit wide data path includes at least one programmable delay element controlled by the logic. The respective data buffer is further configurable to generate a respective local clock having a respective programmable delay from the module clock and to provide the respective local clock to a respective memory device group. ~~is operable in a memory system with a memory controller. The memory module comprises a module control device to receive command signals from the memory controller and to output module C/A signals and data buffer control signals. The module C/A signals are provided to memory devices organized in groups, each group including at least one memory device, while the data buffer control signals are provided to a plurality of buffer circuits to control data paths in the buffer circuits, a respective buffer circuit corresponding to a respective group of memory devices. The plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time. The plurality of buffer circuits include clock regeneration circuits to regenerate a clock signal received from the module control device and to provide regenerated clock signals to respective groups of memory devices.~~

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Currently Amended) A memory module configured to couple to a memory controller via one or more clock signal lines, a set of control signal lines and N sets of n -bit wide data/strobe signal lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers configurable to receive the module clock signal and the data buffer control signals from the control circuit, wherein a respective data buffer of the N data buffers is configurable to operatively couple to a respective set of n -bit wide data/strobe signal lines, the respective data buffer including a n -bit wide data path and logic configured to control the data path in response to the data buffer control signals, the n -bit wide data path including at least one programmable delay element, wherein the respective data buffer is further configurable to generate a respective local clock signal in response to the module clock signal, and wherein the respective local clock signal is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal; and

memory devices organized in one or more ranks and including N groups of memory devices corresponding to respective ones of the N data buffers, each group of memory devices including at least one memory device in each of the one or more ranks, wherein a respective group of the N groups of memory devices is configurable to receive the module C/A signals from the control circuit and a respective local clock signal from a corresponding data buffer of the N data buffers, and wherein the respective group of the N groups of memory devices is further configurable to communicate data and data strobe signals with the memory controller via the n -bit wide data path in the corresponding data buffer and in

response to the module C/A signals from the control circuit and the respective local clock signal from the corresponding data buffer.

2. (Original) The memory module of claim 1, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between the N groups of memory devices and the edge connectors.

3. (Original) The memory module of claim 2, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a programmable delay circuit to delay the local clock signal by the programmable delay, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.

4. (Original) The memory module of claim 3, wherein the programmable delay circuit is controlled by the logic.

5. (Original) The memory module of claim 4, wherein the respective data buffer includes one or more programmable registers programmable by the logic and configured to control the respective programmable delay.

6. (Original) The memory module of claim 5, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer the data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective data buffer is configured to program the one or more programmable registers during the configuration mode.

7. (Original) The memory module of claim 1, wherein the respective data buffer includes one or more programmable registers programmable by the logic and configured to control the respective programmable delay.

8. (Original) The memory module of claim 7, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer the data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective data buffer is configured to program the one or more programmable registers during the configuration mode.
9. (Currently Amended) The memory module of claim 1, wherein the respective data buffer includes one or more programmable registers configurable by the logic to time the data and data strobes being communicated via the respective data buffer.
10. (Currently Amended) The memory module of claim 9, wherein the respective data buffer is configured to program the one or more registers according to signals output from the control circuit during the configuration mode.
11. (Currently Amended) The memory module of claim 9, wherein the control circuit does not transmit any clock signals to the memory devices.
12. (Currently Amended) The memory module of claim 9, wherein the data buffer control signals are different from the module C/A signals.
13. (Currently Amended) The memory module of claim 9, wherein the memory devices are organized in one or more ranks and are each n -bit or $n/2$ -bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks.
14. (Original) The memory module of claim 13, wherein each of the memory devices is n -bits wide.
15. (Original) The memory module of claim 13, wherein each of the memory devices is $n/2$ bits wide.
16. (Currently Amended) A method of operating a memory module couple to a memory controller via one or more clock signal lines, a set of control signal lines and N sets of n -bit wide data/strobe signal lines, where N and n are integers larger than one, the memory module including memory devices organized in one or more ranks, the memory devices including N groups of memory devices corresponding to respective sets of the N sets of n -bit wide

data/strobe signal lines, each group of memory devices including at least one memory device in each of the one or more ranks, the method comprising:

receiving a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines;

outputting a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

controlling N data paths in response to the data buffer control signals, each respective data path of the N data paths being n-bit wide and coupled between a respective set of n-bit wide data/strobe signal lines and a respective group of memory devices, wherein controlling the each respective data path include controlling at least one respective programmable delay element;

generating N local clock signals in response to the module clock signal, each respective local clock signal of the N local clock signals being phase-locked with the module clock signal and having a respective programmable delay from the module clock signal;

outputting the N local clock signals to respective groups of N groups of memory devices; and

performing memory read or write operations at a selected rank of the memory devices in response to the module C/A signals and the N local clock signals to communicate data and data strobe signals with the memory controller via the N data paths.

17. (Currently Amended) The method of claim 16, further comprising operating the memory module in a configuration mode, and programming one or more programmable registers in the configuration mode to control the respective programmable delay.

18. (Currently Amended) The method of claim 17, further comprising programming one or more additional programmable registers in the configuration mode to control the at least one respective programmable delay element and timing of the data and data strobe signals.

19. (Currently Amended) The method of claim 16, wherein the module clock signal is not received by any of the memory devices.

20. (Currently Amended) The method of claim 16, wherein the data buffer control signals are different from the module C/A signals.

REMARKS CONCERNING CLAIMS

Claims 1, 9-13, and 16-20 are amended. No new matter is added.

CONCLUDING REMARKS

In light of the above amendments and remarks, the Applicant respectfully requests that the Examiner consider this application with a view towards allowance. **The Examiner is invited to call the undersigned attorney at (650) 843-7278, if a telephone call could help resolve any remaining items.**

Respectfully submitted,

Date: December 5, 2019

/Jamie J. Zheng/

51,167

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Electronic Acknowledgement Receipt

EFS ID:	37947187
Application Number:	16432700
International Application Number:	
Confirmation Number:	7498
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng/S. Olivier
Filer Authorized By:	Jamie Jie Zheng
Attorney Docket Number:	129980-5073-US01
Receipt Date:	05-DEC-2019
Filing Date:	05-JUN-2019
Time Stamp:	19:37:35
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980-5073US01_PRE_AMD. pdf	148275	yes	7
			91366b7e0cae3f71ef970ee0a79e770a384e77de		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Preliminary Amendment	1	1
	Abstract	2	2
	Claims	3	6
	Applicant Arguments/Remarks Made in an Amendment	7	7

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Total Files Size (in bytes):	148275
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee	Confirmation No.:	7498
Serial No.:	16/432,700	Art Unit:	2133
Filed:	June 5, 2019	Examiner:	Rutz, Jared Ian
For:	<i>Memory Module With Local Synchronization And Method Of Operation</i>	Attorney Docket No.:	129980-5073-US01

CORRECTED PRELIMINARY AMENDMENT

Mail Stop: Amendment

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

This Corrected Preliminary Amendment corrects and replaces the Preliminary Amendment filed on December 5, 2019, in which the amended claims were inadvertently presented without required markup showing amendments made.

The Commissioner is hereby authorized to charge any required fees or credit any overpayments to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073-US01).

Amendments to the Specification

Please revise the specification as shown below:

Please amend the abstract as follows:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

ABSTRACT OF THE DISCLOSURE

A memory module includes memory device groups, and a control circuit configurable to receive a system clock and input address and control (C/A) signals from a memory controller, and output a module clock, module C/A signals and data buffer control signals. The memory module further includes data buffers corresponding to respective memory device groups and configurable to receive the module clock and the data buffer control signals from the control circuit. A respective data buffer includes a n-bit wide data path and logic configured to control the data path in response to the data buffer control signals. The n-bit wide data path includes at least one programmable delay element controlled by the logic. The respective data buffer is further configurable to generate a respective local clock having a respective programmable delay from the module clock and to provide the respective local clock to a respective memory device group. ~~is operable in a memory system with a memory controller. The memory module comprises a module control device to receive command signals from the memory controller and to output module C/A signals and data buffer control signals. The module C/A signals are provided to memory devices organized in groups, each group including at least one memory device, while the data buffer control signals are provided to a plurality of buffer circuits to control data paths in the buffer circuits, a respective buffer circuit corresponding to a respective group of memory devices. The plurality of buffer circuits are distributed across a surface of the memory module such that each data buffer control signal arrives at the plurality of buffer circuits at different points in time. The plurality of buffer circuits include clock regeneration circuits to regenerate a clock signal received from the module control device and to provide regenerated clock signals to respective groups of memory devices.~~

Amendments to the Claims

The following listing of claims will replace all prior versions, and listings, of claims in the application:

Listing of Claims:

1. (Currently Amended) A memory module configured to couple to a memory controller via ~~a $N \times n$ -bit wide memory bus and~~ one or more clock signal lines, ~~the memory bus including~~ a set of control signal lines and N sets of n -bit wide data/strobe signal lines, where N and n are integers larger than one, comprising:

a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

N data buffers ~~each configured~~ configurable to receive the module clock signal and the data buffer control signals from the ~~module~~ control circuit, wherein a respective data buffer of the N data buffers ~~being configured~~ is configurable to operatively couple to a respective set of n -bit wide data/strobe signal lines, the respective data buffer including a n -bit wide data path and logic configured to control the data path in response to the data buffer control signals, the n -bit wide data path including at least one programmable delay element ~~controlled by the logic~~, wherein the respective data buffer ~~being~~ is further ~~configured~~ configurable to generate a respective local clock signal in response to the module clock signal, and wherein the respective local clock signal ~~being~~ is phase-locked with the module clock signal and ~~having~~ has a respective programmable delay from the module clock signal; and

memory devices organized in one or more ranks and including N groups of memory devices corresponding to respective ones of the N data buffers, each group of memory devices including at least one memory device in each of the one or more ranks, wherein a respective group of the N groups of memory devices ~~receiving~~ is configurable to receive the module C/A signals from the control circuit and a respective local clock signal from a corresponding ~~one~~ data buffer of the N data buffers, and wherein the respective group of the

~~N groups of memory devices [communicating]~~is further configurable to communicate data and data strobe signals with the memory controller via the n -bit wide data path in the corresponding ~~[one of the N]~~data ~~[buffers]~~buffer and in response to the module C/A signals from the control circuit and the respective local clock signal from the corresponding data buffer.

2. (Original) The memory module of claim 1, wherein the control circuit, the N data buffers and the memory devices are coupled to a same printed circuit board, the printed circuit board including connectors formed along an edge thereof for connecting to respective ones of the one or more clock signal lines, the set of control signal lines and the N sets of data lines, and wherein the N data buffers are distributed along the edge of the printed circuit board and disposed between the N groups of memory devices and the edge connectors.
3. (Original) The memory module of claim 2, wherein the respective data buffer includes a clock buffer to receive the module clock signal from the control circuit, a phase-locked loop circuit (PLL) to provide phase locking between the module clock signal and the respective local clock signal, a programmable delay circuit to delay the local clock signal by the programmable delay, and a clock driver to drive the local clock signal to a corresponding group of the memory devices.
4. (Original) The memory module of claim 3, wherein the programmable delay circuit is controlled by the logic.
5. (Original) The memory module of claim 4, wherein the respective data buffer includes one or more programmable registers programmable by the logic and configured to control the respective programmable delay.
6. (Original) The memory module of claim 5, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer the data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective data buffer is configured to program the one or more programmable registers during the configuration mode.

7. (Original) The memory module of claim 1, wherein the respective data buffer includes one or more programmable registers programmable by the logic and configured to control the respective programmable delay.
8. (Original) The memory module of claim 7, wherein the memory module is configurable to operate in at least a normal mode and a configuration mode, wherein the N data buffers are configured to buffer the data and data strobe signals between respective groups of memory devices and the memory controller during the normal mode, and wherein the respective data buffer is configured to program the one or more programmable registers during the configuration mode.
9. (Currently Amended) The memory module of claim ~~[7]~~1, wherein the respective data buffer includes one or more programmable registers configurable by the logic to time the data and data strobes being communicated via the respective data buffer.
10. (Currently Amended) The memory module of claim ~~[8]~~9, wherein the respective data buffer is configured to program the one or more registers according to signals output from the control circuit during the configuration mode.
11. (Currently Amended) The memory module of claim ~~[7]~~9, wherein the control circuit does not transmit any clock signals to the memory devices.
12. (Currently Amended) The memory module of claim ~~[7]~~9, wherein the data buffer control signals are different from the module C/A signals.
13. (Currently Amended) The memory module of claim ~~[7]~~9, wherein the memory devices are organized in one or more ranks and are each n -bit or $n/2$ -bit wide, and wherein each group of memory devices include at least one memory device from each of the one or more ranks.
14. (Original) The memory module of claim 13, wherein each of the memory devices is n -bits wide.
15. (Original) The memory module of claim 13, wherein each of the memory devices is $n/2$ bits wide.

16. (Currently Amended) A method of operating a memory module couple to a memory controller via ~~[a $N \times n$ -bit wide memory bus and]~~ one or more clock signal lines, ~~[the memory bus including]~~ a set of control signal lines and N sets of n -bit wide data/strobe signal lines, where N and n are integers larger than one, the memory module including memory devices organized in one or more ranks, the memory devices including N groups of memory devices corresponding to respective sets of the N sets of n -bit wide data/strobe signal lines, each group of memory devices including at least one memory device in each of the one or more ranks, the method comprising:

receiving a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines;

outputting a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals;

controlling N data paths in response to the data buffer control signals, each respective data path of the N data paths being n -bit wide and coupled between a respective set of n -bit wide data/strobe signal lines and a respective group of memory devices, wherein controlling the each respective data path include controlling ~~[a]~~at least one respective programmable delay element;

generating N local clock signals in response to the module clock signal, each respective local clock signal of the N local clock signals being phase-locked with the module clock signal and having a respective programmable delay from the module clock signal;

outputting the N local clock signals to respective groups of N groups of memory devices; and

performing memory read or write operations at a selected rank of the memory devices in response to the module C/A signals and the N local clock signals to communicate data and data strobe signals with the memory controller via the N data paths.

17. (Currently Amended) The ~~[memory module]~~method of claim 16, further comprising operating the memory module in a configuration mode, and programming one or more programmable registers in the configuration mode to control the respective programmable delay.

18. (Currently Amended) The ~~[memory module]~~method of claim 17, further comprising programming one or more additional programmable registers in the configuration mode to control the at least one respective programmable delay element and timing of the data and data ~~[strokes]~~strobe signals.

19. (Currently Amended) The ~~[memory module]~~method of claim 16, wherein the module clock signal is not received by any of the memory devices.

20. (Currently Amended) The ~~[memory module]~~method of claim 16, wherein the data buffer control signals are different from the module C/A signals.

REMARKS CONCERNING CLAIMS

This Corrected Preliminary Amendment corrects and replaces the Preliminary Amendment filed on December 5, 2019, in which the amended claims were inadvertently presented without required markup showing amendments made.

Claims 1, 9-13, and 16-20 are amended. No new matter is added.

CONCLUDING REMARKS

In light of the above amendments and remarks, the Applicant respectfully requests that the Examiner consider this application with a view towards allowance. **The Examiner is invited to call the undersigned attorney at (650) 843-7278, if a telephone call could help resolve any remaining items.**

Respectfully submitted,

Date: April 16, 2020

/Jamie J. Zheng/

51,167

Jamie J. Zheng

(Reg. No.)

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Electronic Acknowledgement Receipt

EFS ID:	39182544
Application Number:	16432700
International Application Number:	
Confirmation Number:	7498
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	129980-5073-US01
Receipt Date:	16-APR-2020
Filing Date:	05-JUN-2019
Time Stamp:	18:16:03
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		129980-5073US01_Corr_AMD.pdf	130801	yes	8
			3d22a772325a9468692c62204f684c821a8d9813		

	Multipart Description/PDF files in .zip description		
	Document Description	Start	End
	Preliminary Amendment	1	1
	Specification	2	2
	Claims	3	7
	Applicant Arguments/Remarks Made in an Amendment	8	8

Warnings:

Information:

Total Files Size (in bytes):	130801
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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 16/432,700		Filing Date 06/05/2019		☐ To be Mailed	
ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO									
APPLICATION AS FILED - PART I									
		(Column 1)	(Column 2)						
FOR		NUMBER FILED	NUMBER EXTRA		RATE (\$)		FEE (\$)		
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))		N/A	N/A		N/A				
☐ SEARCH FEE (37 CFR 1.16(k), (l), or (m))		N/A	N/A		N/A				
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))		N/A	N/A		N/A				
TOTAL CLAIMS (37 CFR 1.16(i))		minus 20 = *		x \$100 =					
INDEPENDENT CLAIMS (37 CFR 1.16(h))		minus 3 = *		x \$460 =					
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))		If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).							
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))									
* If the difference in column 1 is less than zero, enter "0" in column 2.					TOTAL				
APPLICATION AS AMENDED - PART II									
		(Column 1)		(Column 2)	(Column 3)				
AMENDMENT	04/16/2020	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	* 20	Minus	** 20	= 0	x \$100 =		0	
	Independent (37 CFR 1.16(h))	* 2	Minus	*** 3	= 0	x \$460 =		0	
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE		0	
		(Column 1)		(Column 2)	(Column 3)				
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)		ADDITIONAL FEE (\$)	
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$0 =			
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$0 =			
	☐ Application Size Fee (37 CFR 1.16(s))								
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))								
						TOTAL ADD'L FEE			
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE			
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/LAVINIA D JOHNSON/			
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".									
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.									

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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NOTICE OF ALLOWANCE AND FEE(S) DUE

79141 7590 09/04/2020
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

EXAMINER	
AHMED, ZUBAIR	
ART UNIT	PAPER NUMBER
2132	

DATE MAILED: 09/04/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01	7498

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	12/04/2020

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

79141 7590 09/04/2020
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01	7498

TITLE OF INVENTION: MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	12/04/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent) : ☐ Individual ☐ Corporation or other private group entity ☐ Government4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29☐ Applicant asserting small entity status. See 37 CFR 1.27☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01	7498
79141	7590	09/04/2020	EXAMINER	
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)			AHMED, ZUBAIR	
1400 Page Mill Road			ART UNIT	
Palo Alto, CA 94304			PAPER NUMBER	
			2132	
DATE MAILED: 09/04/2020				

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 16/432,700	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to preliminary amendment filed on 04/16/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>08/19/2019</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
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DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to preliminary amendment filed on 04/16/2020.

Allowable Subject Matter

Claims 1-20 are allowed.

Reasons for Allowance

The following is an examiner's statement of reasons for allowance.

Any comments considered necessary by applicants must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

After careful considerations, examination and search of the claimed invention, the closest prior art of record does not teach or anticipate the claimed feature as claimed in independent claim 1 "a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals" and "the respective data buffer including a n-bit wide data path and logic configured to control the data path in response to the data buffer control signals, the n-bit wide data path including at least one programmable delay element, wherein the respective data buffer is further configurable to generate a respective local clock signal

in response to the module clock signal, and wherein the respective local clock signal is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal” in combination with the overall claimed limitations when interpreted in light of the specification.

The closest prior arts are: Matsui US 2004/0105292 (hereinafter “Matsui”) and Sutardja et al. US 2007/0168781 (hereinafter “Sutardja”).

A decision on the parent application (Application No. 14/445,035 which is now US Patent 10,324,841) was rendered on 09/27/2018 by the Patent Trial and Appeal Board (PTAB). In their decision, the PTAB reversed the rejection of the independent claims of the parent application based on a combination of Matsui and Sutardja.

Independent claim 1 of the child application (the instant application) adds more specificity to the claim. In particular, it appears to the Examiner that Matsui and Sutardja, alone or in any combination, is silent about the above cited claim limitations.

Therefore, independent claim 1 is allowable.

Independent claim 16 has similar recitations as independent claim 1 and is allowable for the same reason.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an

interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

<i>Notice of References Cited</i>	Application/Control No. 16/432,700		Applicant(s)/Patent Under Reexamination Lee et al.	
	Examiner ZUBAIR AHMED		Art Unit 2132	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20040105292-A1	06-2004	Matsui, Yoshinori	G11C7/1048	365/63
*	B	US-20070168781-A1	07-2007	Sutardja; Sehat	G11C11/406	714/718
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Issue Classification 	Application/Control No. 16/432,700	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC						
Symbol					Type	Version
G06F	/	12	/	0802	F	2013-01-01
G11C	/	5	/	04	I	2013-01-01
G11C	/	29	/	023	I	2013-01-01
G11C	/	29	/	028	I	2013-01-01
G11C	/	7	/	1066	I	2013-01-01
G11C	/	7	/	1093	I	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	28 August 2020 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	01 September 2020 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 2C

Issue Classification 	Application/Control No. 16/432,700	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06F	/	12	/ 0802
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS
711	120

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					
711	120	119				

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	28 August 2020 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	01 September 2020 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 2C

Issue Classification 	Application/Control No. 16/432,700	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original

/ZUBAIR AHMED/ Examiner, Art Unit 2132 (Assistant Examiner)	28 August 2020 (Date)	Total Claims Allowed: 20	
/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132 (Primary Examiner)	01 September 2020 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 2C

Search Notes 	Application/Control No. 16/432,700	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F, 1/10, 13/4256, 13/4243, 12/0802	08/28/2020	ZA
G11C 5/04, 29/023, 29/028, 7/1093, 7/1066	08/28/2020	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	08/28/2020	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached EAST search history	08/28/2020	ZA
Inventor search	08/28/2020	ZA
Assignee search	08/28/2020	ZA
NPL search	08/28/2020	ZA
The instant application is allowed in view of the board's decision rendered with respect to the parent application 14/445,035	09/27/2018	ZA

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G06F	120/1/10	08/28/2020	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 16/432,700	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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Bibliographic Data

Application No: 16/432,700

Foreign Priority claimed: ☐ Yes ☒ No

35 USC 119 (a-d) conditions met: ☐ Yes ☒ No ☐ Met After Allowance

Verified and Acknowledged: /ZUBAIR AHMED/

Examiner's Signature

Initials

Title:

MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND
METHOD OF OPERATION

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
06/05/2019	711	2132	129980-5073-US01
RULE			

APPLICANTS

Netlist, Inc., Irvine, CA, UNITED STATES

INVENTORS

Hyun Lee, Ladera Ranch, CA, UNITED STATES

Jayesh R. Bhakta, Cerritos, CA, UNITED STATES

CONTINUING DATA

This application is a CON of 14445035 07/28/2014 PAT 10324841

14445035 has PRO of 61859215 07/27/2013

FOREIGN APPLICATIONS

IF REQUIRED, FOREIGN LICENSE GRANTED**

06/12/2019

STATE OR COUNTRY

UNITED STATES

ADDRESS

Morgan, Lewis & Bockius LLP (PA)(J. Zheng)

1400 Page Mill Road

Palo Alto, CA 94304

UNITED STATES

FILING FEE RECEIVED

\$1,880

EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S1	7	"10324841".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 10:12
S2	2	"20060224847".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:13
S3	3	"20070086269".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:18
S4	2	"6898683".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:34
S5	3	"20020078316".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:34
S6	53,804	(G11C5/04 G06F1/10 G06F13/4256 G11C29/023 G11C29/028 G06F13/4243 G06F12/0802 G11C7/1093 G11C7/1066).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:35

S7	2,686	S6 and (memory near2 module same buffer)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:36
S8	928	S6 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:36
S9	215	S6 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:39
S10	20	((Lee near2 Hyun) or (Bhakta near2 Jayesh)) and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:43
S11	30	netlist and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:45
S12	2,442	(711/120 711/119).ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:47
S13	2	S12 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:47

S14	9	S12 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:49
S15	37	S12 and (memory near2 module same buffer)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:50
S16	106	("4218740" "4249256" "4368515" "4392212" "4571676" "4592011" "4633429" "4670748" "4706214" "4866603" "4958322" "4961172" "4961204" "4980850" "5060188" "5247643" "5272664" "5345412" "5357478" "5388240" "5388072" "5392252" "5426753" "5463590" "5483497").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:55
S17	0	S16 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:56
S18	104	("5485589" "5495435" "5513135" "5532954" "5541448" "5572691" "5581498" "5590071" "5602999" "5638534" "5655153" "5699542" "5703826" "5702984" "5745914" "5764590" "5784705" "5805520" "5822251" "5926827" "5926839" "5953280" "5953215" "5959930" "5958025").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:03
S19	0	S18 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:03
S20	108	("5963464" "5966736" "5974473" "6011710" "6018787" "6044032" "6070227" "6097652" "6108745" "6115278" "6134638" "6151271"	US-PGPUB; USPAT; USOCR; FPRS; EPO;	OR	ON	2020/08/28 15:10

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S21	0	S20 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:10
S22	111	("20020039323" "6381140" "6400637" "6408356" "6414868" "6415374" "6438062" "6446158" "6446184" "6453381" "6470417" "6487102" "6502161" "6518794" "6530033" "6530007" "6553450" "6618320" "6621496" "6625687" "6636935" "6646949" "6658509" "6674684" "6681301").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:16
S23	0	S22 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:17
S24	77	("6683372" "6697888" "6705877" "6717885" "6721843" "6738880" "6742098" "20040105292" "6754797" "6754746" "6785189" "6788592" "6799252" "6807125" "6807650" "6813196" "6832303" "6832303" "6854042" "6880094" "6889304" "6912628" "6912615" "6925028" "6944694").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:29
S25	1	S24 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:29
S26	293,615	("6950366" "6954281" "6961281" "20050257109" "6981089" "6982893" "6982892" "6990043" "6996686" "7007130" "7007175" "7047361" "7046538" "7054179" "7065626" "7072231" "7073041" "7078793"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2020/08/28 15:36

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S27	573	("6950366" "6954281" "6961281" "20050257109" "6981089" "6982893" "6982892" "6990043" "6996686" "7007130" "7007175" "7047361" "7046538" "7054179" "7065626" "7072231" "7073041" "7078793" "20060179206" "7120727" "7124260" "20060233012" "7127584" "7130952" "7133960").pn."	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:36
S28	73	("6950366" "6954281" "6961281" "20050257109" "6981089" "6982893" "6982892" "6990043" "6996686" "7007130" "7007175" "7047361" "7046538" "7054179" "7065626" "7072231" "7073041" "7078793" "20060179206" "7120727" "7124260" "20060233012" "7127584" "7130952" "7133960").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:36
S29	0	S28 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:37
S30	62	("7133972" "20060259711" "7142461" "20060267172" "20060277355" "7149841" "7167967" "7181591" "20070058409" "20070070669" "7200021" "7227910" "20070168781" "7254036" "7266639" "7272709" "20070217559" "7281079" "7286436" "7289386" "20080025137" "20080037412" "7346750" "7356639" "20080046631").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:43
S31	2	S30 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:44
S32	56	("7346750" "7356639" "20080104352" "7437591" "20080291758" "7461182" "7471538" "20090103387" "20090116312" "7532537" "20090198924" "20090298503"	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO;	OR	ON	2020/08/28 15:52

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S33	0	S32 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:52
S34	43	("20110110168" "20110125966" "8001434" "8089795" "8130560" "8189328" "8214616" "8233303" "20120256639" "20120317352" "8565033" "20140029370" "9128632").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:56
S35	2	S34 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:57
S36	0	WO19922002879.pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:59
S37	0	EP1816570.pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 16:00

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				Application Number	16/432,700	
				Filing Date	June 5, 2019	
				First Named Inventor	Hyun Lee	
				Art Unit	2139	
				Examiner Name	To be assigned	
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				Art Unit	2139
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				Application Number		16/432,700
				Filing Date		June 5, 2019
				First Named Inventor		Hyun Lee
				Art Unit		2139
				Examiner Name		To be assigned
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				Art Unit	2139
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Examiner Signature	/ZUBAIR AHMED/	Date Considered	08/28/2020
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DB2/ 37107073.1

EAST Search History

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
S38	23,876	(G11C5/04 G06F1/10 G06F13/4256 G11C29/023 G11C29/028 G06F13/4243 G06F12/0802 G11C7/1093 G11C7/1066).cpc.	US-PGPUB; USPAT	OR	ON	2020/08/28 16:05
S39	2,296	S38 and (memory near2 module same buffer)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:05
S40	875	S38 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:06
S41	198	S38 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT	OR	ON	2020/08/28 16:06
S42	12	((Lee near2 Hyun) or (Bhakta near2 Jayesh)) and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT	OR	ON	2020/08/28 16:07
S43	21	netlist and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT	OR	ON	2020/08/28 16:08
S44	1,652	(711/120 711/119).ccls.	US-PGPUB; USPAT	OR	ON	2020/08/28 16:09
S45	1	S44 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT	OR	ON	2020/08/28 16:09
S46	8	S44 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:10
S47	33	S44 and (memory near2 module same buffer)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:10

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Main Concept Text

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a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals

(No Concept Modifiers)

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(No Filters)

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Options

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De-dup: None

Cut-off: None

Sort: Relevance

Content: Patent Publications (164)

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US Patents

US Designs

US Applications

EPO Patents

EPO Applications

China Patents

China Applications

Japan Patents

Japan Applications

Korea Patents

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Ecuador Patents
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El Salvador Applications
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Armenia Patents
Australia Patents
Australia Applications
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ARIPO Patents
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Egypt Patents
Kenya Patents
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Georgia Patents
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Russia Patents
Russia Applications
Turkey Patents
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				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	1	of	11	Attorney Docket Number	129980-5073US01

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Examiner Signature				Date Considered		

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				Examiner Name	To be assigned
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Examiner Signature		Date Considered	
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				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
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	Inter Partes Review of U.S. Pat. No. 7,881,150, Case No. IPR2015-01020, Exhibit 2002, "Videotaped Deposition of Srinivasan Jagannathan, Ph.D on Feb. 12, 2016," filed Feb. 22, 2016.
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	Inter Partes Review of U.S. Pat. No. 7,881,150, Case No. IPR2015-01020, Patent Owner's Objections to Petitioner's Demonstrative Exhibits, filed Jun. 23, 2016.
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		Inter Partes Review of U.S. Pat. No. 7,881,150, IPR Case No. IPR2014-00882, Decision—Institution of Inter Partes Review 37 C.F.R. § 42.108, issued Dec. 16, 2014.
		Inter Partes Review of U.S. Pat. No. 7,881,150, IPR Case No. IPR2014-00882, Decision—Institution of Inter Partes Review 37 C.F.R. § 42.108, issued Dec. 16, 2014.
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		Inter Partes Review of U.S. Pat. No. 7,881,150, IPR Case No. IPR2014-00882, Patent Owner's Preliminary Response Pursuant to 37 C.F.R. § 42.107, filed Oct. 7, 2014.
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		Inter Partes Review of U.S. Pat. No. 7,881,150, IPR Case No. IPR2014-01011, Exhibit 3002 to Decision—Institution of Inter Partes Review, Excerpts from IEEE Dictionary, issued Dec. 16, 2014.
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	Inter Partes Review of U.S. Pat. No. 8,081,536, Case No. IPR2015-01021, Petitioner's Response to Patent Owner's Submission, filed Jun. 13, 2016.
	Inter Partes Review of U.S. Pat. No. 8,081,536, Case No. IPR2015-01021, Petitioner's Updated Exhibit List, filed Jun. 24, 2016.
	Inter Partes Review of U.S. Pat. No. 8,081,536, Case No. IPR2015-01021, Supplemental Declaration of Dr. Jagannathan, filed May 19, 2016.
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		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F-1 to F-5 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G.1-G.6 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits G-1 to G-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Smart Storage Systems, Inc.'s Invalidity Contentions, dated Jun. 6, 2014.

Examiner Signature		Date Considered	
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Electronic Acknowledgement Receipt

EFS ID:	40471506
Application Number:	16432700
International Application Number:	
Confirmation Number:	7498
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	129980-5073-US01
Receipt Date:	03-SEP-2020
Filing Date:	05-JUN-2019
Time Stamp:	20:16:01
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Transmittal Letter	129980-5073-US01_IDS_2020-09-03.pdf	141004	no	4
			be4b32ea04dc47cad15dee0f60f9b1fd4ba3b3e		

Warnings:

Information:					
2	Non Patent Literature	2020-02-03_SK_hynix-Corrected_Opening_Brief.pdf	19012236	no	411
			68f46bdecf610ad3e10cc6fdea36477ad6d8d152		
Warnings:					
Information:					
3	Non Patent Literature	2020-03-23_Netlist_Opening_Response_Brief.pdf	3162767	no	80
			f121a25884336900414c4cd4af3a01ee66b7bbc0		
Warnings:					
Information:					
4	Non Patent Literature	ITC_Opinion.pdf	5969862	no	161
			ef735b67112ecafdf85bb8f9385eac3a5405d3009		
Warnings:					
Information:					
5	Non Patent Literature	IPR2018-00362-00365_Hearing_transcript.pdf	4664162	no	131
			dbd605c28d058261f21d33f241e0deef3143b45		
Warnings:					
Information:					
6	Non Patent Literature	IPR2018-00362-00363_FD.pdf	4903001	no	94
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Warnings:					
Information:					
7	Non Patent Literature	129980-5073-US01_160085CN-CHA-PI-CN_OA_Response.pdf	376206	no	12
			5525fbd831b75e8a4b91b97dbe4d9cb560921e23		
Warnings:					
Information:					
8	Non Patent Literature	129980-5073-US01_160085CN-CH-I_CN_OA.pdf	1605846	no	10
			011fab09819c415dd4b60d99acdb2815ab1603d9		
Warnings:					
Information:					

9	Non Patent Literature	129980-5073- US01_NL01P002WOEP.pdf	909944	no	18
			c691c4514f2b3023201bb2163f25e500b32c0c1f		

Warnings:

Information:

10	Information Disclosure Statement (IDS) Form (SB08)	129980-5073- US01_IDS_List_2020-09-03.pdf	249198	no	11
			d707179bb50d296addf734b4260480664da2dd39		

Warnings:

Information:

This is not an USPTO supplied IDS fillable form

Total Files Size (in bytes):	40994226
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

Application of:	Hyun Lee et al	Confirmation No.:	7498
Serial No.:	16/432,700	Art Unit:	2139
Filed:	June 5, 2019	Examiner:	To be assigned
For:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION	Attorney Docket No.:	129980-5073US01

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

Sir:

In accordance with the duty of disclosure provisions of 37 C.F.R. §1.56, there is hereby provided certain information which the Examiner may consider material to the examination of the subject U.S. patent application. It is requested that the Examiner make this information of record if it is deemed material to the examination of the application.

1. Enclosures accompanying this Information Disclosure Statement are:

- 1a. ☒ A list of all patents, publications, applications, or other information submitted for consideration by the office.
- 1b. A legible copy of:
 - ☐ Each foreign patent;
 - ☐ Each publication or that portion which caused it to be listed on the PTO-1449;
 - ☐ For each cited pending U.S. application, the application specification including the claims, and any drawing of the application, or portion of the application which caused it to be listed on the PTO-1449 including any claims directed to that portion;
 - ☐ all other information or portion which caused it to be listed on the PTO-1449.
- 1c. ☐ An English language copy of search report(s) from a counterpart foreign application or PCT International Search Report.
- 1d. ☐ Explanations of relevancy (ATTACHMENT 1(d), hereto) or English language abstracts of the non-English language publications.

2. ☒ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(b):
- ☐ Within three months of the filing date of a national application other than a continued prosecution application under §1.53(d);
 - ☐ Within three months of the date of entry of the national stage as set forth in §1.491 in an international application;
 - ☒ Before the mailing of the first Office action on the merits;
 - ☐ Before the mailing of a first Office action after the filing of a request for continued examination under §1.114.
3. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(c) after the period specified in 37 C.F.R. §1.97(b), but before the mailing date of any of a final action under 37 C.F.R. §1.113, a notice of allowance under 37 C.F.R. §1.311 or an action that otherwise closes prosecution in the application.

(Check either Item 3a, 3b, 3c or 3d)

- 3a. ☐ The Certification Statement in Item 5 below is applicable. Accordingly, no fee is required.
- 3b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 3d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

(Items 3b or 3c to be checked if any reference known for more than 3 months)

4. ☐ This Information Disclosure Statement is filed under 37 C.F.R. §1.97(d) after the period specified in 37 C.F.R. §1.97(c), but on or before the date of payment of the issue fee.

(Check Item 4a, and 4b, 4c or 4d)

- 4a. ☐ The Certification Statement in Item 5 below is applicable.
- 4b. ☐ The \$240 (large entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 4c. ☐ The \$120 (small entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).
- 4d. ☐ The \$60 (micro entity) fee set forth in 37 C.F.R. §1.17(p) in accordance with 37 C.F.R. §1.97(c) is to be charged to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no.).

5. ☐ Certification Statement (applicable if Item 3a or Item 4a is checked)
(Check either Item 5a, 5b, 5c or 5d)
- 5a. ☐ In accordance with 37 C.F.R. §1.97(e)(1), it is certified that each item of information contained in this information disclosure statement was first cited in a communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of this Information Disclosure Statement.
- 5b. ☐ In accordance with 37 C.F.R. §1.97(e)(2), it is certified that no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 C.F.R. §1.56(c) more than three months prior to the filing of this information disclosure statement.
- 5c. ☐ Each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
- 5d. ☐ Pursuant to 37 C.F.R. §1.704(d), each item of information contained in this information disclosure statement was cited in a communication from a foreign patent office in a counterpart application, and the communication was not **received** by any individual designated in 37 C.F.R. §1.56(c) more than thirty days prior to the filing of this information disclosure statement.
6. ☒ Copies of each cited U.S. patent and each U.S. patent application publication are not enclosed pursuant to the USPTO OG Notice dated 05 August 2003 waiving the requirement under 37 C.F.R. 1.98(a)(2)(i) for U.S. patent applications filed after June 30, 2003.
7. ☒ This application is a continuation application under 37 C.F.R. §1.53(b) or (d).
(Check appropriate Items 7a, 7b and/or 7c)
- 7a. ☐ A Petition to Withdraw from issue under 37 C.F.R. §1.313(b)(5) is concurrently filed herewith.
- 7b. ☒ Copies of publications listed on Form PTO-1449 from prior application Serial No. 14/445,035, filed on July 28, 2014, of which this application claims priority under 35 U.S.C. §120, are not being submitted pursuant to 37 C.F.R. §1.98(d).
- 7c. ☒ Copies of the publications listed on the attached Form PTO-1449 that were not previously cited in prior application Serial No. , filed on , are provided herewith.
8. ☐ This is a Supplemental Information Disclosure Statement. (Check Item 8a)
- 8a. ☐ This Supplemental Information Disclosure Statement under 37 C.F.R. §1.97(f)

supplements the Information Disclosure Statement filed on . A bona fide attempt was made to comply with 37 C.F.R. §1.98, but inadvertent omissions were made. These omissions have been corrected herein. Accordingly, additional time is requested so that this Supplemental Information Disclosure Statement can be considered as if properly filed on .

9. ☐ In accordance with 37 C.F.R. §1.98, a concise explanation of what is presently understood to be the relevance of each non-English language publication is:

(Check Item 9a, 9b, or 9c)

- 9a. ☐ satisfied because all non-English language publications were cited on the enclosed English language copy of the PCT International Search Report or the search report from a counterpart foreign application indicating the degree of relevance found by the foreign office.
- 9b. ☐ set forth in the application.
- 9c. ☐ enclosed as an attachment hereto.
10. ☒ The Commissioner is authorized to charge any additional fee required or credit any overpayment for this Information Disclosure Statement and/or Petition to Morgan, Lewis & Bockius LLP Deposit Account No. 50-0310 (order no. 129980-5073US01).
11. ☒ No admission is made that the information cited in this Statement is, or is considered to be, material to patentability nor a representation that a search has been made (other than a search report of a foreign counterpart application or PCT International Search Report if submitted herewith). 37 C.F.R. §§1.97(g) and (h).

Respectfully submitted,

Date: September 3, 2020

/Jamie J. Zheng/

51,167

Jamie Zheng

(Reg. No.)

MORGAN, LEWIS & BOCKIUS LLP

1400 Page Mill Road

Palo Alto, CA 94304

(650) 843-4000



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01	7498
79141	7590	10/21/2020	EXAMINER	
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)			AHMED, ZUBAIR	
1400 Page Mill Road			ART UNIT	PAPER NUMBER
Palo Alto, CA 94304			2132	
			NOTIFICATION DATE	DELIVERY MODE
			10/21/2020	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

padocketingdepartment@morganlewis.com
vskliba@morganlewis.com

Corrected Notice of Allowability	Application No. 16/432,700	Applicant(s) Lee et al.	
	Examiner ZUBAIR AHMED	Art Unit 2132	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to IDS filed on 09/03/2020.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-20. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892) 2. ☒ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date <u>09/03/2020</u> . 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date. _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
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/ZUBAIR AHMED/ Examiner, Art Unit 2132	/DAVID YI/ Supervisory Patent Examiner, Art Unit 2132
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DETAILED ACTION

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

This Office Action is responsive to IDS filed on 09/03/2020.

Information Disclosure Statement

The information disclosure statement filed 09/03/2020 fails to comply with 37 CFR 1.98(a)(2), which requires a legible copy of each cited foreign patent document; each non-patent literature publication or that portion which caused it to be listed; and all other information or that portion which caused it to be listed. It has been placed in the application file, but the information referred to therein has not been considered. Furthermore, the Chinese Patent Application No. 2014/800535993, filed July 28 2014, issued on March 12, 2018 and the Chinese Patent Application No. 2014/800535993, filed July 28, 2014, issued on March 12, 2018, filed February 22, 2019 were not considered by the Examiner because the content of the above two filings are in a foreign language and Applicant has not provided an English translation of the above two applications.

Allowable Subject Matter

Claims 1-20 are allowed.

Reasons for Allowance

This communication warrants no examiner's reason for allowance, as the record makes evident the reason for allowance, satisfying the record "as a whole" as required by rule 37 CFR 1.104 (e). Accordingly, the reason for allowance is in all probability

evident from the record and no statement for examiner's reason for allowance is necessary (see MPEP 1302.14).

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ZUBAIR AHMED whose telephone number is (571)272-1655. The examiner can normally be reached on 7:30AM - 5:00PM EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, DAVID X YI can be reached on (571) 270-7519. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <https://ppair-my.uspto.gov/pair/PrivatePair>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/ZUBAIR AHMED/
Examiner, Art Unit 2132

/DAVID YI/
Supervisory Patent Examiner, Art Unit 2132

<i>Search Notes</i> 	Application/Control No. 16/432,700	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

CPC - Searched*		
Symbol	Date	Examiner
G06F, 1/10, 13/4256, 13/4243, 12/0802	08/28/2020	ZA
G11C 5/04, 29/023, 29/028, 7/1093, 7/1066	08/28/2020	ZA

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner
711	120	08/28/2020	ZA

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
See attached EAST search history	10/15/2020	ZA
Inventor search	08/28/2020	ZA
Assignee search	08/28/2020	ZA
NPL search	10/11/2020	ZA
The instant application is allowed in view of the board's decision rendered with respect to the parent application 14/445,035	09/27/2018	ZA

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
711/G06F	120/1/10	08/28/2020	ZA

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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<i>Search Notes</i> 	Application/Control No. 16/432,700	Applicant(s)/Patent Under Reexamination Lee et al.
	Examiner ZUBAIR AHMED	Art Unit 2132

/ZUBAIR AHMED/ Examiner, Art Unit 2132	
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EAST Search History

EAST Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L1	8	"9606907".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/15 11:30
S1	7	"10324841".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 10:12
S2	2	"20060224847".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:13
S3	3	"20070086269".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:18
S4	2	"6898683".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:34
S5	3	"20020078316".pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 12:34

S6	53,804	(G11C5/04 G06F1/10 G06F13/4256 G11C29/023 G11C29/028 G06F13/4243 G06F12/0802 G11C7/1093 G11C7/1066).cpc.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:35
S7	2,686	S6 and (memory near2 module same buffer)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:36
S8	928	S6 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:36
S9	215	S6 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:39
S10	20	((Lee near2 Hyun) or (Bhakta near2 Jayesh)) and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:43
S11	30	netlist and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:45
S12	2,442	(711/120 711/119).ccls.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:47

S13	2	S12 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:47
S14	9	S12 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:49
S15	37	S12 and (memory near2 module same buffer)	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:50
S16	106	("4218740" "4249256" "4368515" "4392212" "4571676" "4592011" "4633429" "4670748" "4706214" "4866603" "4958322" "4961172" "4961204" "4980850" "5060188" "5247643" "5272664" "5345412" "5357478" "5388240" "5388072" "5392252" "5426753" "5463590" "5483497").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:55
S17	0	S16 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 14:56
S18	104	("5485589" "5495435" "5513135" "5532954" "5541448" "5572691" "5581498" "5590071" "5602999" "5638534" "5655153" "5699542" "5703826" "5702984" "5745914" "5764590" "5784705" "5805520" "5822251" "5926827" "5926839" "5953280" "5953215" "5959930" "5958025").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:03
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S36	0	WO19922002879.pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 15:59
S37	0	EP1816570.pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/08/28 16:00

S48	26	("6275900" "6832284" "6834012" "20050044301" "20060077731" "20110078406" "20040260905" "20120075319" "20140207871" "20140247832").pn.	US-PGPUB; USPAT; USOCR; FPRS; EPO; JPO; DERWENT; IBM_TDB	OR	ON	2020/10/11 11:25
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10/15/2020 11:45:55 AM

C:\Users\zahmed\Documents\EAST\Workspaces\16432700.wsp

EAST Search History

EAST Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	Time Stamp
L2	1	"9606907".pn.	US-PGPUB; USPAT	OR	ON	2020/10/15 11:45
S38	23,876	(G11C5/04 G06F1/10 G06F13/4256 G11C29/023 G11C29/028 G06F13/4243 G06F12/0802 G11C7/1093 G11C7/1066).cpc.	US-PGPUB; USPAT	OR	ON	2020/08/28 16:05
S39	2,296	S38 and (memory near2 module same buffer)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:05
S40	875	S38 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:06
S41	198	S38 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT	OR	ON	2020/08/28 16:06
S42	12	((Lee near2 Hyun) or (Bhakta near2 Jayesh)) and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT	OR	ON	2020/08/28 16:07
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S44	1,652	(711/120 711/119).ccls.	US-PGPUB; USPAT	OR	ON	2020/08/28 16:09
S45	1	S44 and (memory near2 module same buffer same clock same (command\$1address or c\$1a))	US-PGPUB; USPAT	OR	ON	2020/08/28 16:09
S46	8	S44 and (memory near2 module same buffer same clock)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:10
S47	33	S44 and (memory near2 module same buffer)	US-PGPUB; USPAT	OR	ON	2020/08/28 16:10
S49	10	("6275900" "6832284" "6834012" "20050044301" "20060077731" "20110078406" "20040260905" "20120075319" "20140207871" "20140247832").pn.	US-PGPUB; USPAT	OR	ON	2020/10/11 11:26

10/15/2020 11:46:05 AM

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Main Concept Text

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A memory module configured to couple to a memory controller via one or more clock signal lines, a set of control signal lines and N sets of n-bit wide data/strobe signal lines, where N and n are integers larger than one, comprising: a control circuit configured to receive a system clock from the memory controller via the one or more clock signal lines, and input address and control (C/A) signals from the memory controller via the control signal lines, the control circuit being further configured to output a module clock signal, module C/A signals and data buffer control signals in response to the system clock and the input C/A signals; N data buffers configurable to receive the module clock signal and the data buffer control signals from the control circuit, wherein a respective data buffer of the N data buffers is configurable to operatively couple to a respective set of n-bit wide data/strobe signal lines, the respective data buffer including a n-bit wide data path and logic configured to control the data path in response to the data buffer control signals, the n-bit wide data path including at least one programmable delay element, wherein the respective data buffer is further configurable to generate a respective local clock signal in response to the module clock signal, and wherein the respective local clock signal is phase-locked with the module clock signal and has a respective programmable delay from the module clock signal; and memory devices organized in one or more ranks and including N groups of memory devices corresponding to respective ones of the N data buffers, each group of memory devices including at least one memory device in each of the one or more ranks, wherein a respective group of the N groups of memory devices is configurable to receive the module C/A signals from the control circuit and a respective local clock signal from a corresponding data buffer of the N data buffers, and wherein the respective group of the N groups of memory devices is further configurable to communicate data and data strobe signals with the memory controller via the n-bit wide data path in the corresponding data buffer and in response to the module C/A signals from the control circuit and the respective local clock signal from the corresponding data buffer.

(No Concept Modifiers)

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(No Filters)

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Options

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De-dup: None

Cut-off: None

Sort: Relevance

Content: Patent Publications (164)

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US Patents
US Designs
US Applications
EPO Patents
EPO Applications
China Patents
China Applications
Japan Patents
Japan Applications
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Honduras Applications
Mexico Patents
Mexico Applications
Nicaragua Patents
Panama Applications
Peru Applications
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Norway Patents
Norway Applications
Poland Patents
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Republic of Moldova Patents
Republic of Moldova Applications

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San Marino Patents
San Marino Applications
Serbia Patents
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Sweden Applications
Switzerland Patents
Switzerland Applications
Ukraine Patents
Yugoslavia/Serbia and Montenegro Patents
Yugoslavia/Serbia and Montenegro Applications
Armenia Patents
Australia Patents
Australia Applications
Cyprus Patents
Gulf Cooperation Council Patents
Hong Kong Patents
India Patents
India Applications
Indonesia Patents
Indonesia Applications
Israel Applications
Jordan Patents
Kyrgyzstan Patents
Macao Applications
Malaysia Patents
Mongolia Patents
New Zealand Patents
Philippines Patents
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Saudi Arabia Patents
Singapore Patents
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Taiwan Applications
Tajikistan Patents
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Thailand Applications
Uzbekistan Patents
Vietnam Patents
Algeria Patents

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Kenya Patents
Malawi Patents
Morocco Patents
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OAPI Patents
South Africa Patents
Tunisia Applications
Zambia Patents
Zimbabwe Patents
EAPO Patents
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Georgia Patents
Georgia Applications
Kazakhstan Patents
Kazakhstan Applications
Russia Patents
Russia Applications
Turkey Patents
Turkey Applications

INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed September 3, 2020	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	1	of	11	Attorney Docket Number	129980-5073US01

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		EP 1988473 B1	11-05-2006	Hofmeister, Helge		
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Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published			
		NETLIST, Inc., United States Court of Appeals for the Federal Circuit, SK Hynix Inc., Corrected Principal Brief of Appeals, Document 26, Case: 19-2340, filed 03FEB2020, 284 pgs.			
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		Response to First Office Action in Chinese Patent Application No. 2014800535993, filed July 28, 2014, issued on March 12, 2018, filed February 22, 2019.			

Examiner Signature	/ZUBAIR AHMED/	Date Considered	10/15/2020
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed September 3, 2020	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	2	of	11	Attorney Docket Number	129980-5073US01

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Examiner Signature	/ZUBAIR AHMED/	Date Considered	10/15/2020
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed September 3, 2020	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	3	of	11	Attorney Docket Number	129980-5073US01

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	Inter Partes Review Case No. 2014-00883, Patent Owner's Request for Oral Hearing, filed Jul. 3, 2015.		
	Inter Partes Review Case No. 2014-00883, Petitioner Diablo Technologies, Inc.'s Reply to Patent Owner Netlist, Inc.'s Response, filed Jun. 19, 2015.		
	Inter Partes Review Case No. 2014-00883, Petitioner's Request for Oral Argument, filed Jul. 1, 2015.		
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Examiner Signature	/ZUBAIR AHMED/	Date Considered	10/15/2020

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	Inter Partes Review of U.S. Pat. No. 8,081,536, Case No. IPR2015-01021, Patent Owner Motion for Observations, filed Jun. 2, 2016.
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	U.S. District Court Northern District of California, Case No. 4:13-cv-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Diablo Technologies, Inc.'s Invalidity Contentions, dated Jun. 6, 2014.
	U.S. District Court Northern District of California, Case No. 4:13-cv-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Diablo Technologies, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
	U.S. District Court Northern District of California, Case No. 4:13-cv-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibit H to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.

Examiner Signature	/ZUBAIR AHMED/	Date Considered	10/15/2020
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed September 3, 2020			
				Application Number		16/432,700	
				Filing Date		June 5, 2019	
				First Named Inventor		Hyun Lee	
				Art Unit		2139	
				Examiner Name		To be assigned	
Sheet	11	of	11	Attorney Docket Number		129980-5073US01	

		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits D-1 to D-6 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits E.1-E.7 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F.1-F.5 to "Smart Storage Systems, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Exhibits F-1 to F-5 to "Diablo Technologies, Inc.'s Invalidity Contentions," dated Jun. 6, 2014.
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		U.S. District Court Northern District of California, Case No. 4:13-CV-05889-YGR, Netlist, Inc. v. Smart Storage Systems, Inc., and Diablo Technologies, Inc., Smart Storage Systems, Inc.'s Invalidity Contentions, dated Jun. 6, 2014.

Examiner Signature	/ZUBAIR AHMED/	Date Considered	10/15/2020
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PART B - FEE(S) TRANSMITTAL

Electronically filed December 2, 2020

Complete and send this form with applicable fee(s), to: MailMail Stop ISSUE FEE
Commissioner of Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450
(571) 273-2885or Fax

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advanced orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS**MORGAN, LEWIS & BOCKIUS LLP**
1400 Page Mill Road
Palo Alto, CA 94304

Note: A certificate of mailing below can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing.

Certificate of Mailing

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being facsimile transmitted to the USPTO (571) 273-2885, on the date indicated below.

(Depositor's name)

(Signature)

(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTY'S DOCKET NO.	CONFIRMATION NO.
16/432,700	06/05/2019	Hyun Lee	129980-5073-US01	7498

TITLE OF INVENTION MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION

APPLN. TYPE	ENTITY STATUS	ISSUE FEE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	1000	0	1000	12/04/2020

EXAMINER	ART UNIT	CLASS-SUB CLASS
AHMED, ZUBAIR	2132	711-120000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).
☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.
☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47) attached. Use of a Customer Number is required.

2. For printing on the patent front page, list
(1) the names of up to 3 registered patent attorneys or agents OR, alternatively,
(2) the name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed

1. **Morgan, Lewis & Bockius LLP**

2. _____

3. _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document has been filed for recordation as set forth in 37 CFR 3.11. Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE:

NETLIST, INC.

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Irvine, CA

Please check the appropriate assignee category or categories (will not be printed on the patent) ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. The following fee(s) are submitted:☒ Issue Fee☐ Advanced Order - # of Copies _____**4b. Payment of Fee(s): (Please first reapply any previously paid issue fee shown above.)**☒ Electronic Payment via Financial Manager☐ Non-electronic payment by credit card. Form PTO-2038 is attached.☒ The Director is hereby authorized to charge the required fee(s), or credit any overpayment, to Deposit Account Number 50-0310 (order no. 129980-5073-US01).**5. Change in Small Entity Status** (from status indicated above)☐ a. Applicant certifying micro entity status. See 37 CFR 1.29.☐ Applicant asserting small entity status. See 37 CFR 1.27.☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity (see form PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be sign in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature: / Jamie J. Zheng /

Date: December 2, 2020

Printed Name: Jamie J. Zheng

Registration No. 51,167

Electronic Patent Application Fee Transmittal

Application Number:	16432700			
Filing Date:	05-Jun-2019			
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION			
First Named Inventor/Applicant Name:	Hyun Lee			
Filer:	Jamie Jie Zheng/S. Olivier			
Attorney Docket Number:	129980-5073-US01			
Filed as Large Entity				
Filing Fees for Utility under 35 USC 111(a)				
Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Basic Filing:				
Pages:				
Claims:				
Miscellaneous-Filing:				
Petition:				
Patent-Appeals-and-Interference:				
Post-Allowance-and-Post-Issuance:				
UTILITY APPL ISSUE FEE	1501	1	1000	1000

Description	Fee Code	Quantity	Amount	Sub-Total in USD(\$)
Extension-of-Time:				
Miscellaneous:				
Total in USD (\$)				1000

Electronic Acknowledgement Receipt

EFS ID:	41277339
Application Number:	16432700
International Application Number:	
Confirmation Number:	7498
Title of Invention:	MEMORY MODULE WITH LOCAL SYNCHRONIZATION AND METHOD OF OPERATION
First Named Inventor/Applicant Name:	Hyun Lee
Customer Number:	79141
Filer:	Jamie Jie Zheng
Filer Authorized By:	
Attorney Docket Number:	129980-5073-US01
Receipt Date:	02-DEC-2020
Filing Date:	05-JUN-2019
Time Stamp:	23:51:26
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	yes
Payment Type	DA
Payment was successfully received in RAM	\$ 1000
RAM confirmation Number	E2020B2N51503073
Deposit Account	
Authorized User	

The Director of the USPTO is hereby authorized to charge indicated fees and credit any overpayment as follows:

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File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1	Issue Fee Payment (PTO-85B)	129980-5073US01_Issue_Fee.pdf	110815	no	1
			f995104ef7623aa1826854f7eb6171fa0c54c7d7		

Warnings:

Information:

2	Fee Worksheet (SB06)	fee-info.pdf	30322	no	2
			16815576d3c7379dc7e8acb35ff4fa7398ea84d7		

Warnings:

Information:

Total Files Size (in bytes):			141137
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This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
Examiner Name	To be assigned				
Sheet	4	of	6	Attorney Docket Number	129980-5073US01

Change(s) applied
to document,
/J.G./
10/22/2020

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	7,167,967 B2	01-23-2007	Buffalo Inc.	Bungo et al.
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Examiner Signature	/ZUBAIR AHMED/	Date Considered	08/28/2020
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DB2/ 37107073.1

INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
Examiner Name	To be assigned				
Sheet	3	of	6	Attorney Docket Number	129980-5073US01

Change(s) applied
to document,
/J.G./
10/22/2020

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Examiner Signature	/ZUBAIR AHMED/	Date Considered	08/28/2020
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INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019		
				Application Number		16/432,700
				Filing Date		June 5, 2019
				First Named Inventor		Hyun Lee
				Art Unit		2139
				Examiner Name		To be assigned
Sheet	5	of	6	Attorney Docket Number		129980-5073US01

Change(s) applied
to document,
/J.G./
10/22/2020

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		2011/0110168 A1	05-12-2011	Samsung Electronics	Sung et al.
		2011/0125966 A1	05-26-2011	Smart Modular Tech.	Amidi et al.

Examiner Signature	/ZUBAIR AHMED/	Date Considered	08/28/2020
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DB2/ 37107073.1

INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	6	of	6	Attorney Docket Number	129980-5073US01

Change(s) applied
to document,
/J.G./
10/22/2020

		8,001,434 B1	08-16-2011	Netlist, Inc.	Lee et al.
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Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published

Examiner Signature	/ZUBAIR AHMED/	Date Considered	08/28/2020
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DB2/ 37107073.1

INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed September 3, 2020	
				Application Number	16/432,700
				Filing Date	June 5, 2019
				First Named Inventor	Hyun Lee
				Art Unit	2139
				Examiner Name	To be assigned
Sheet	1	of	11	Attorney Docket Number	129980-5073US01

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NON PATENT LITERATURE DOCUMENTS		
Examiner Initials	Cite No.	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc.), date, page(s), volume-issue number(s), publisher, city and/or country where published
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Examiner Signature	/ZUBAIR AHMED/	Date Considered	10/15/2020
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Change(s) applied
to document,
/J.G./
10/22/2020

INFORMATION DISCLOSURE STATEMENT BY APPLICANT Substitute for Form 1449-PTO				Electronically filed August 19, 2019		
				Application Number		16/432,700
				Filing Date		June 5, 2019
				First Named Inventor		Hyun Lee
				Art Unit		2139
				Examiner Name		To be assigned
Sheet	2	of	6	Attorney Docket Number		129980-5073US01

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				Application Number		16/432,700
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				First Named Inventor		Hyun Lee
				Art Unit		2139
Examiner Name		To be assigned				
Sheet	1	of	6	Attorney Docket Number		129980-5073US01

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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
16/432,700	01/05/2021	10884923	129980-5073-US01	7498

79141 7590 12/16/2020
Morgan, Lewis & Bockius LLP (PA)(J. Zheng)
1400 Page Mill Road
Palo Alto, CA 94304

ISSUE NOTIFICATION

The projected patent number and issue date are specified above.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 30 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Application Information Retrieval (PAIR) WEB site (<http://pair.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Data Management (ODM) at (571)-272-4200.

APPLICANT(s) (Please see PAIR WEB site <http://pair.uspto.gov> for additional applicants):

Hyun Lee, Ladera Ranch, CA;
Netlist, Inc., Irvine, CA;
Jayesh R. Bhakta, Cerritos, CA;

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FEE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
16/432,700	06/05/2019	411120

The following fields have been changed to Customer Number 207611 on 05/27/2025 via Patent Center in view of the certification copied below that authorized the change.

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207611 - Netlist, Inc.
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Irvine, CA 92617

I certify, in accordance with 37 CFR 1.4(d)(4), that I am :

An attorney or Agent of Record registered to practice before the Patent and Trademark Office who has been given power of attorney in this application

Signature	Name	Registration #
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CORRESPONDENCE ADDRESS CHANGE

APPLICATION #	FILING OR 371(C) DATE	REQUEST ID
16/432,700	06/05/2019	411120

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Signature	Name	Registration #
/Jamie J. Zheng/	Jamie Zheng	51167

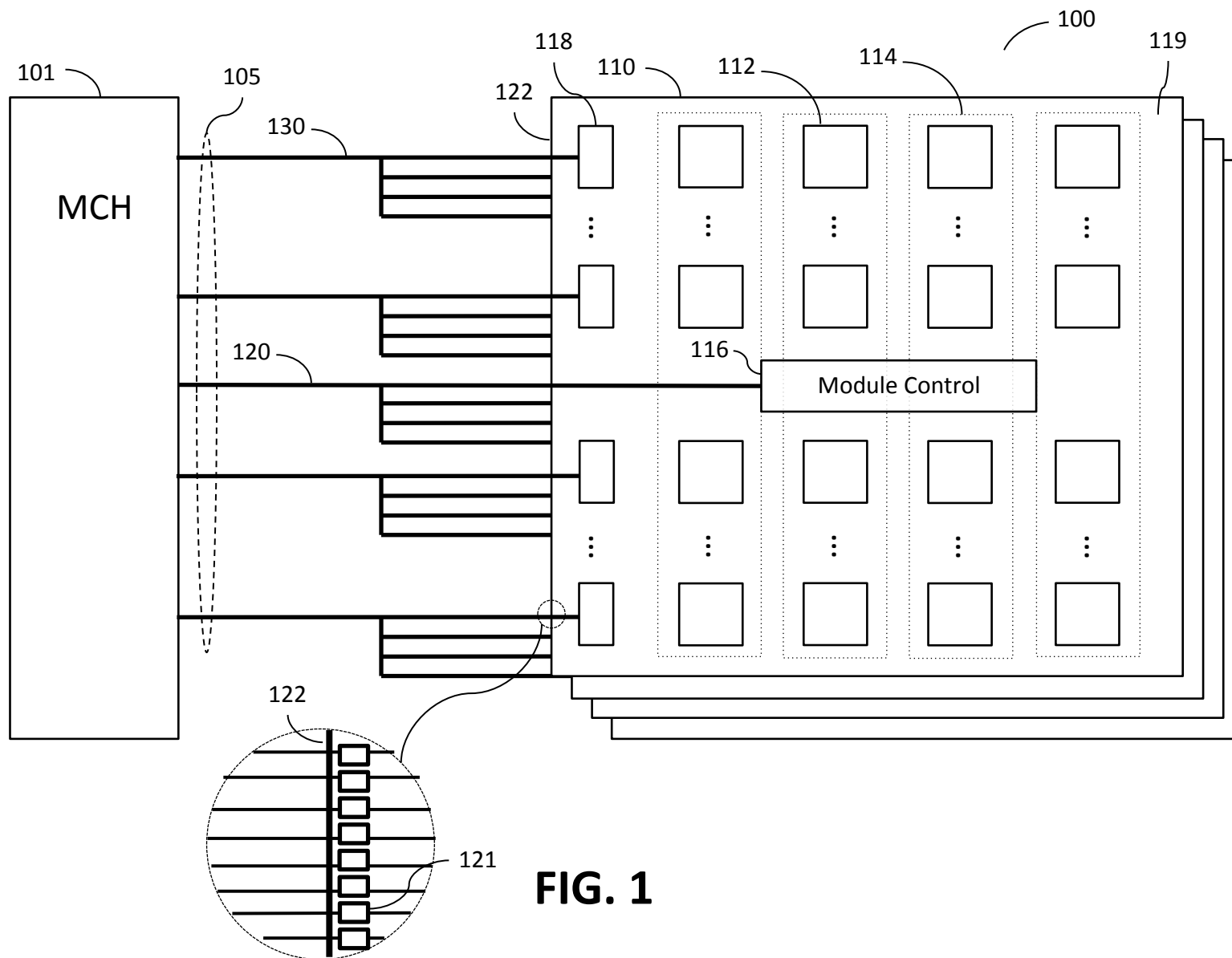


FIG. 1

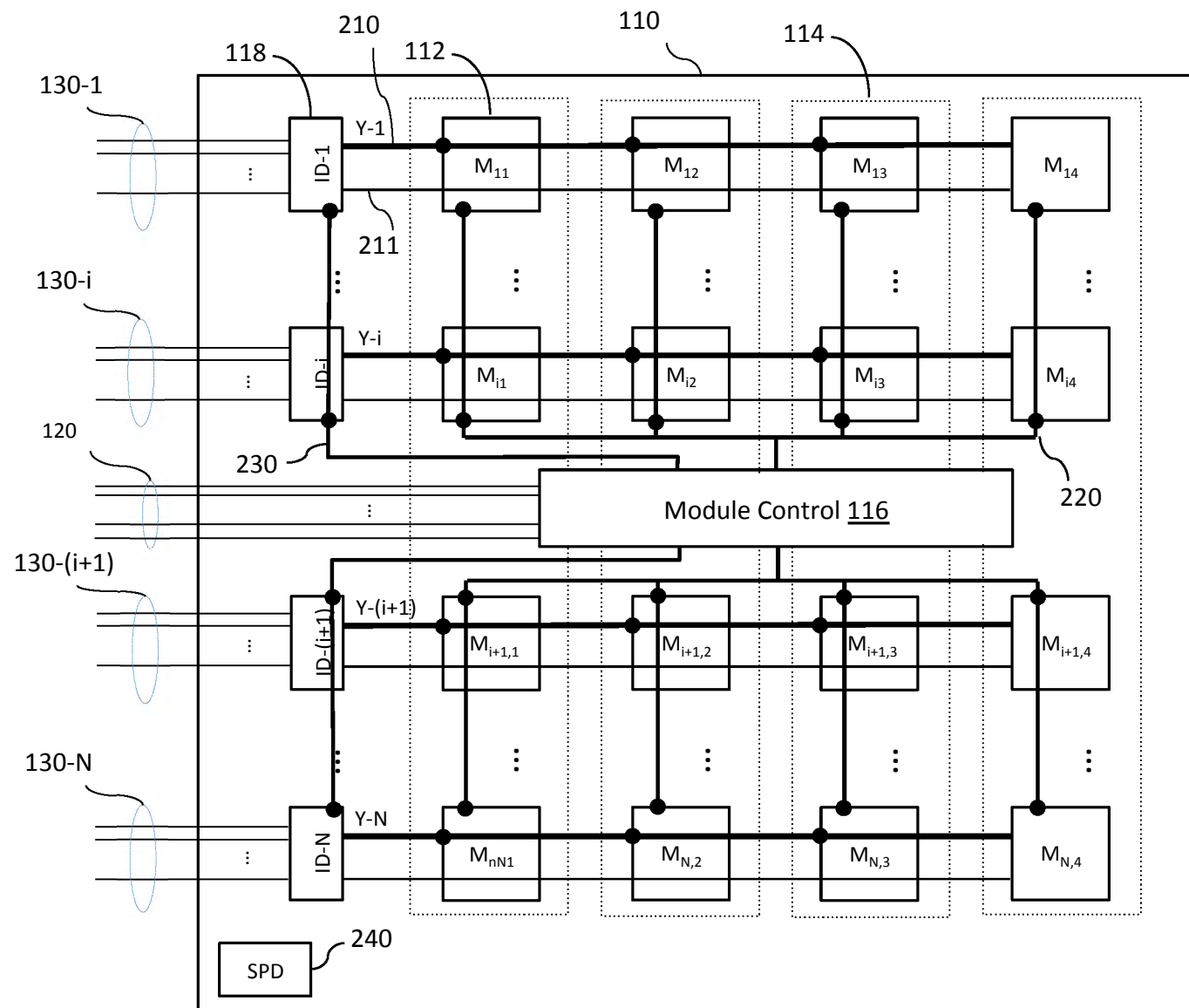


FIG. 2A

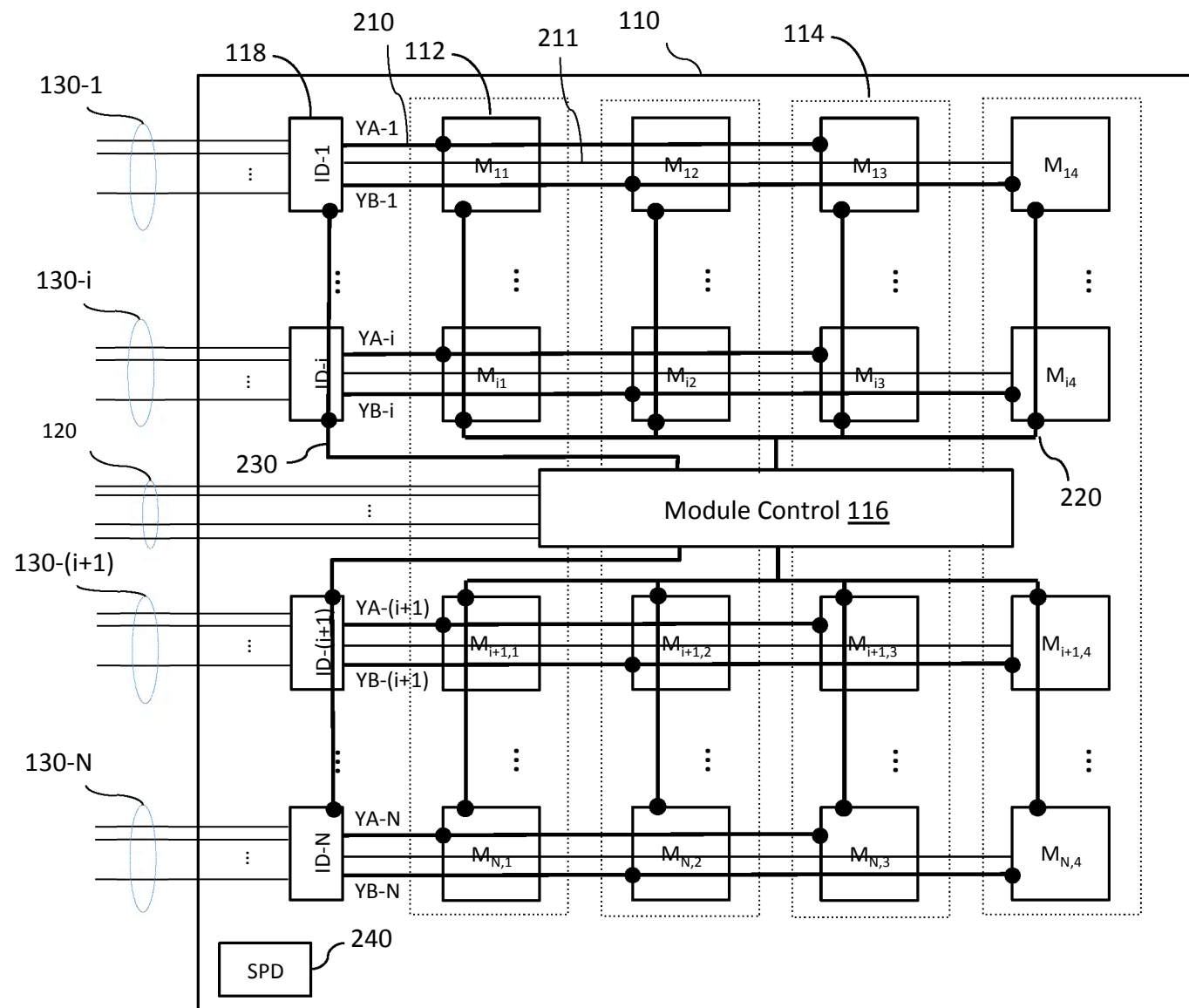


FIG. 2B

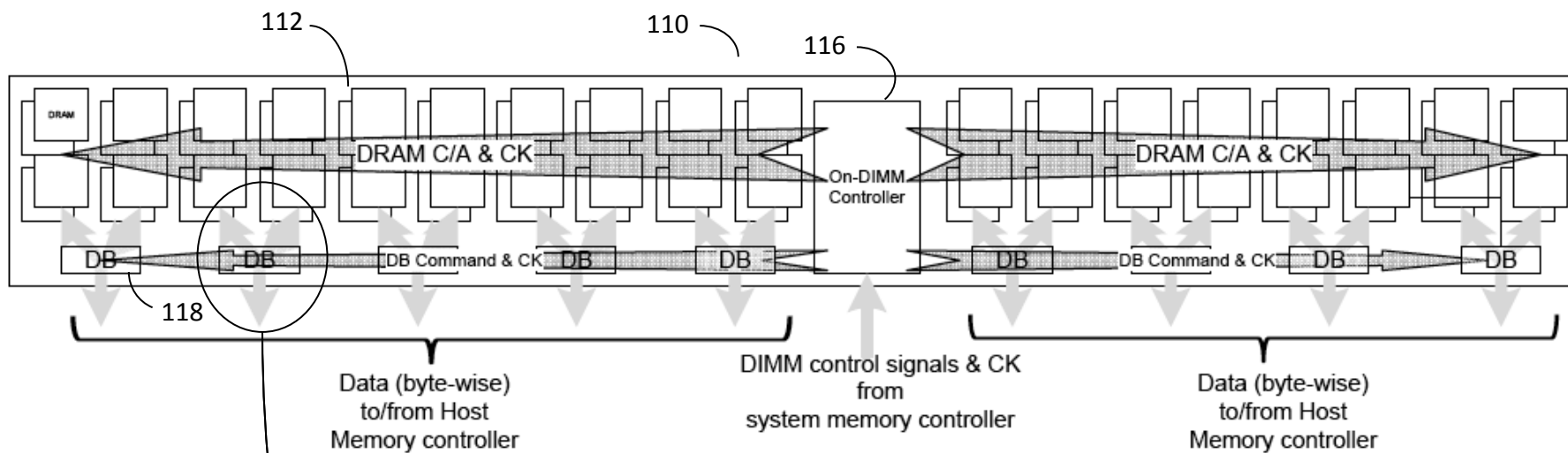


FIG. 2D

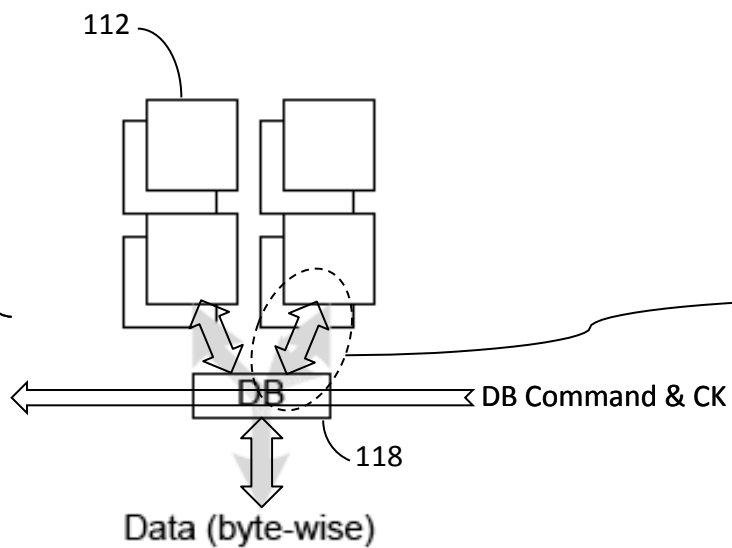
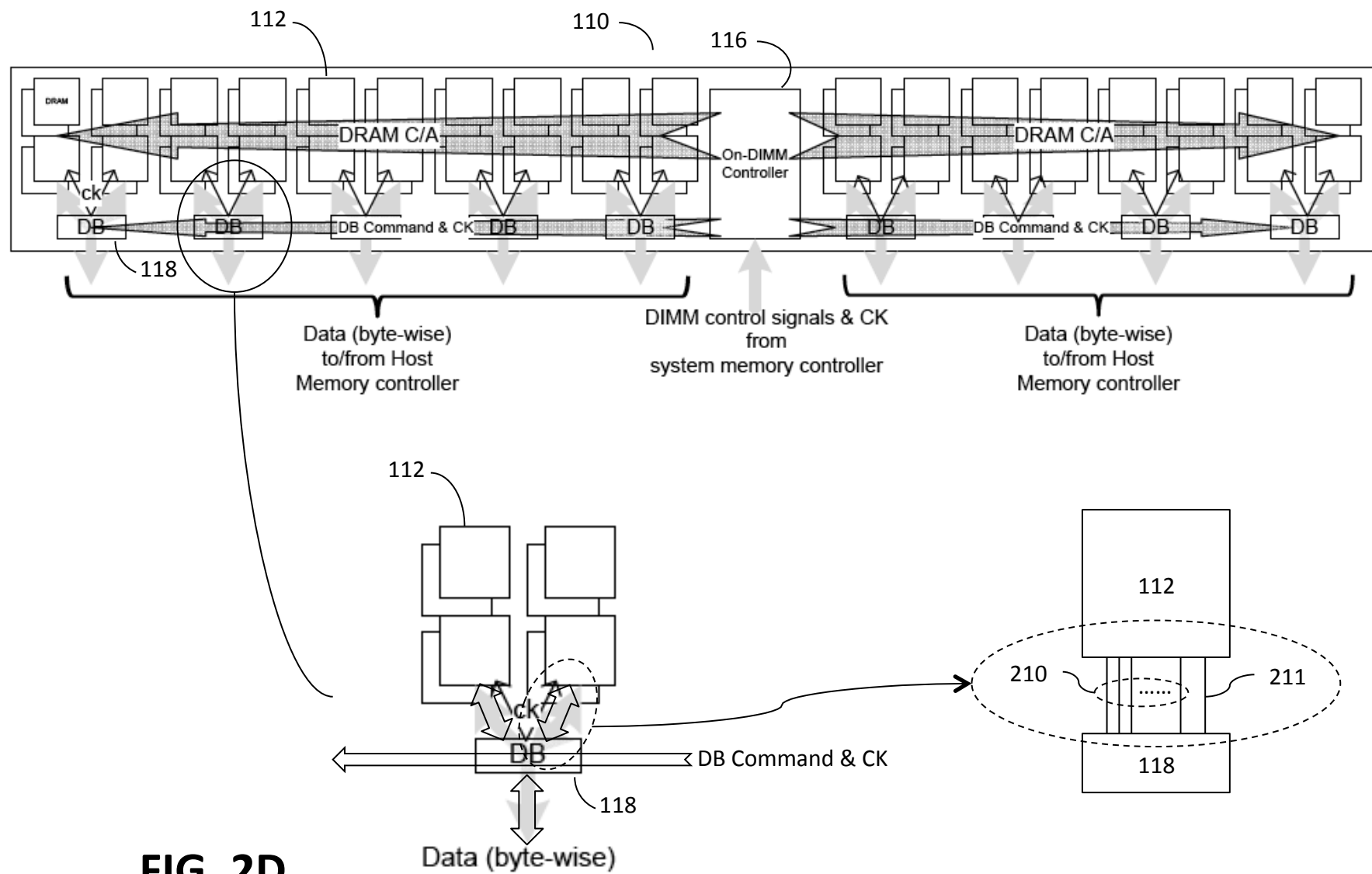


FIG. 2C



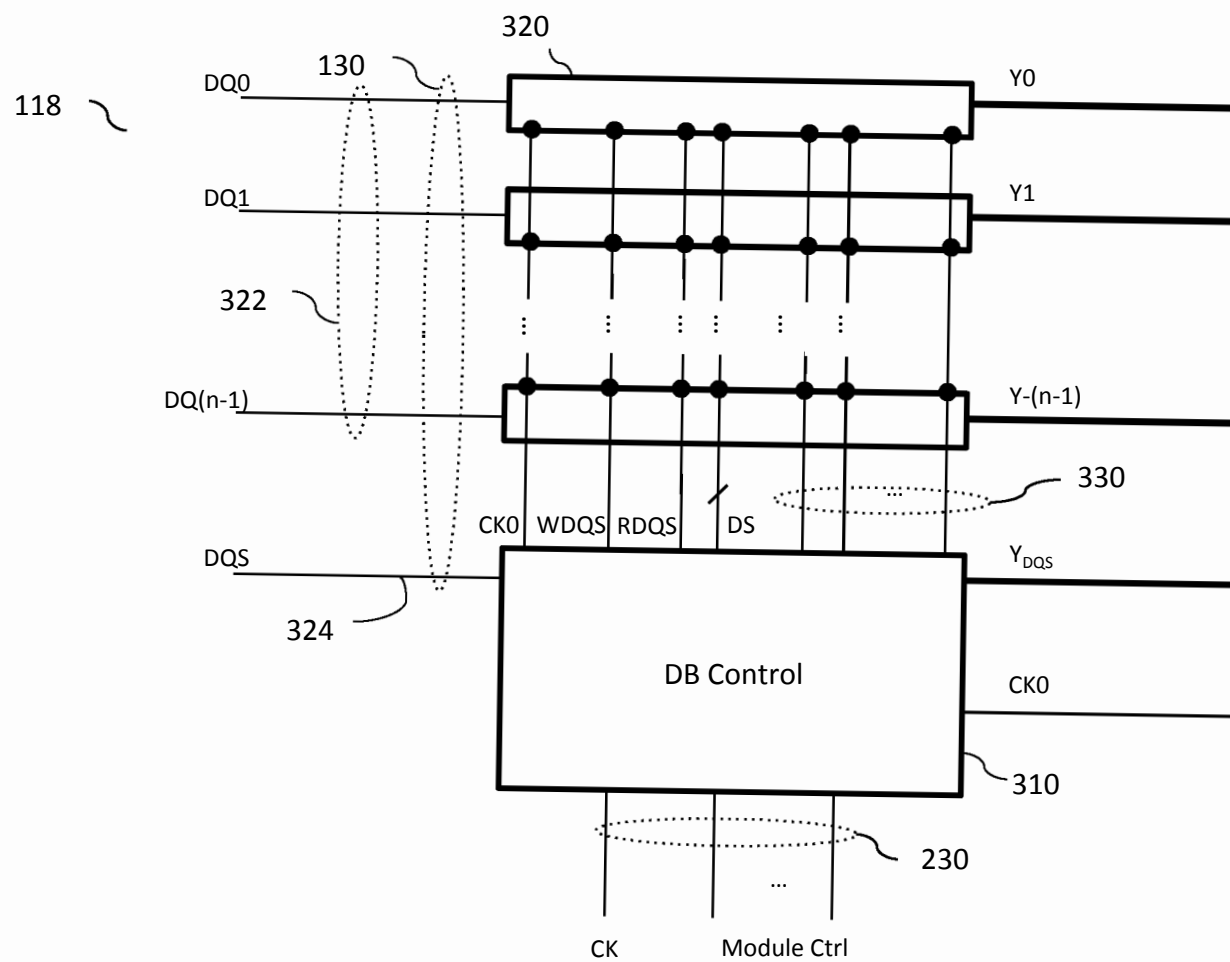


FIG. 3

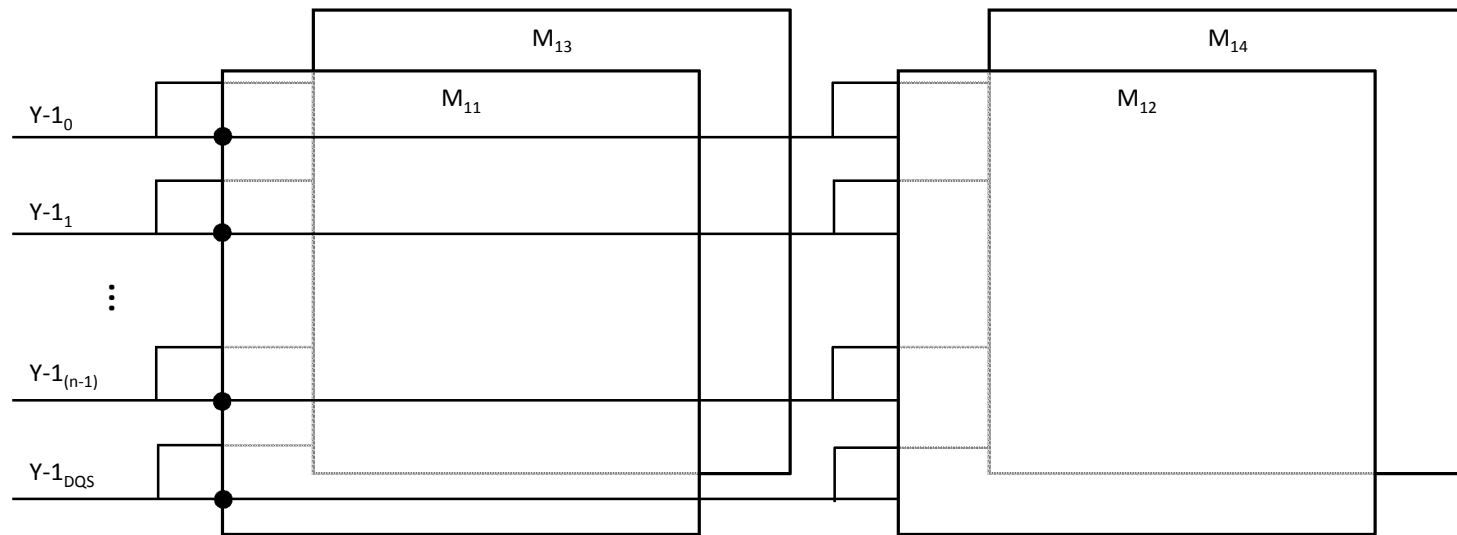


FIG. 4A

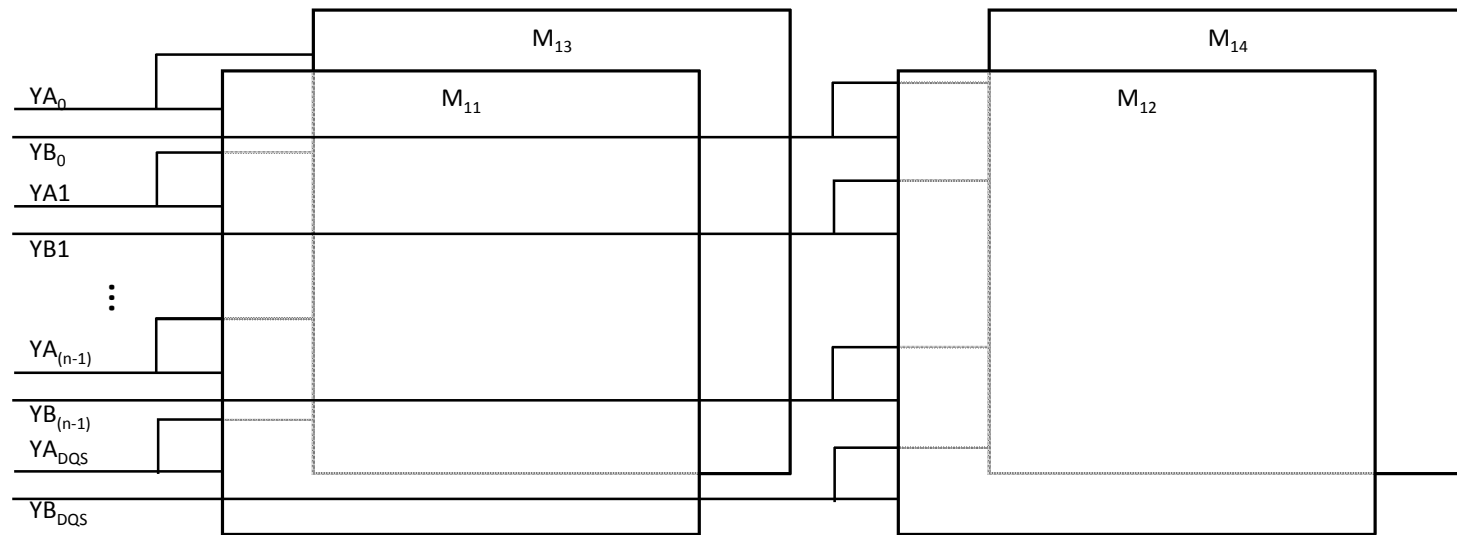


FIG. 4B

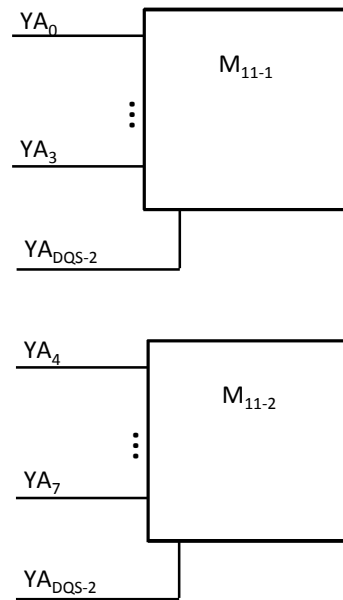


FIG. 5A

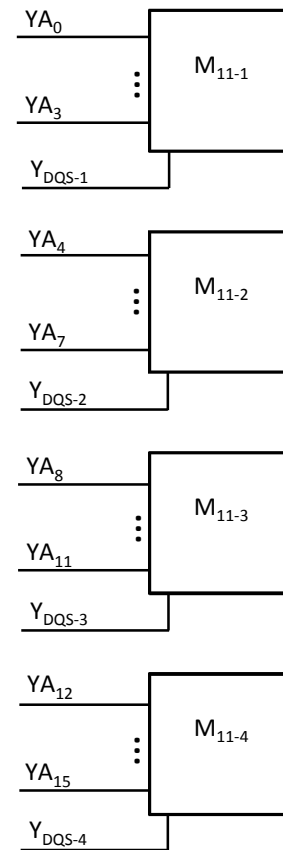


FIG. 5B

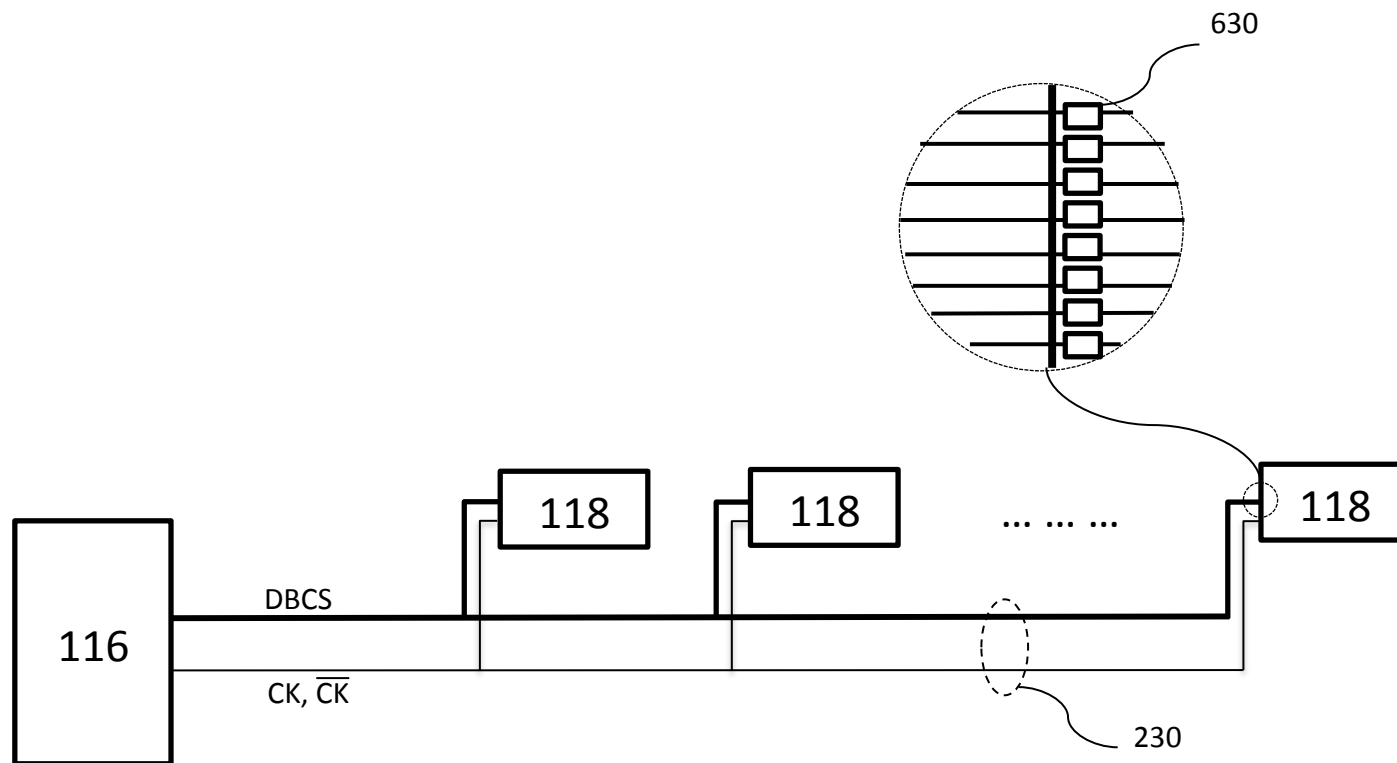
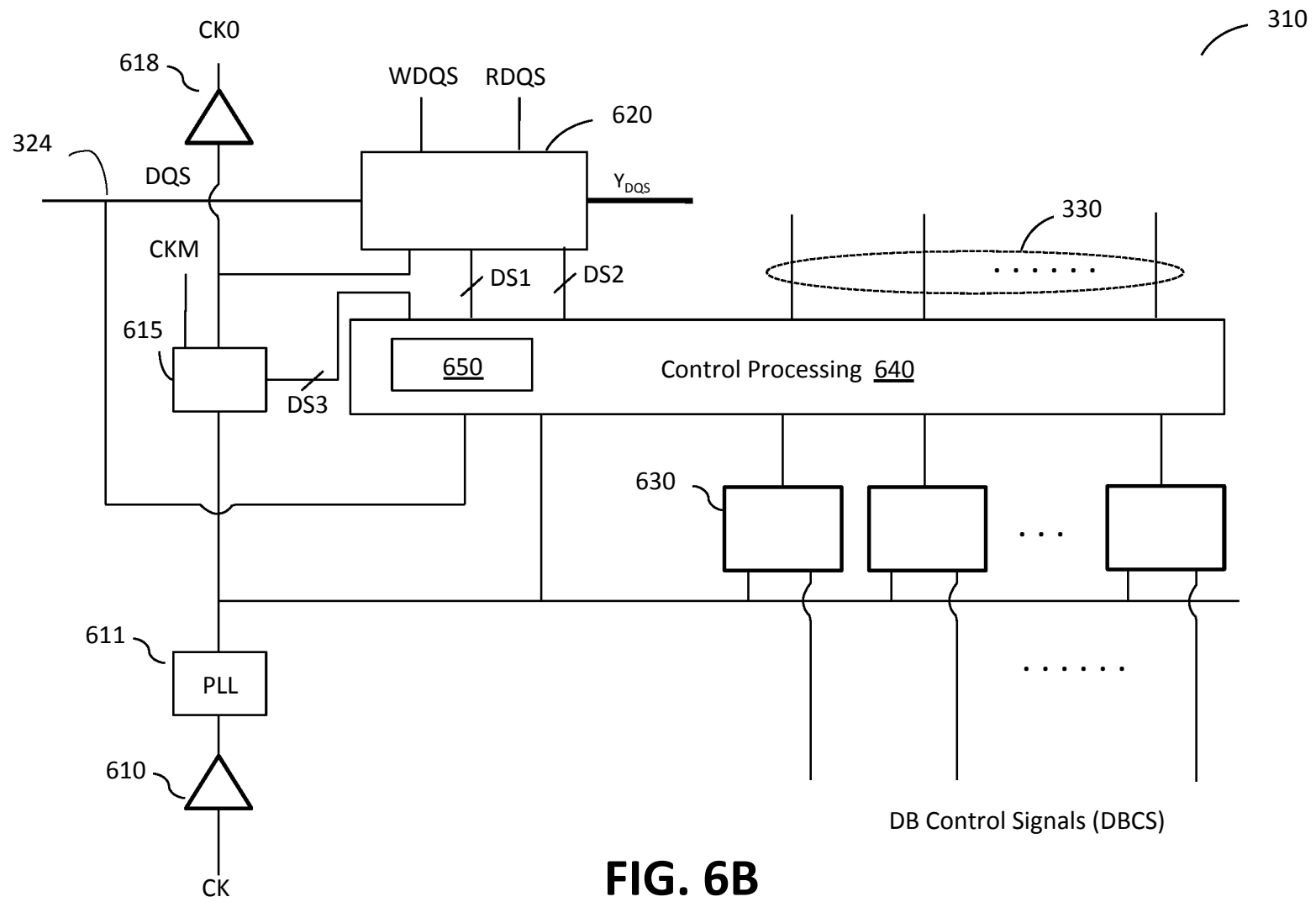
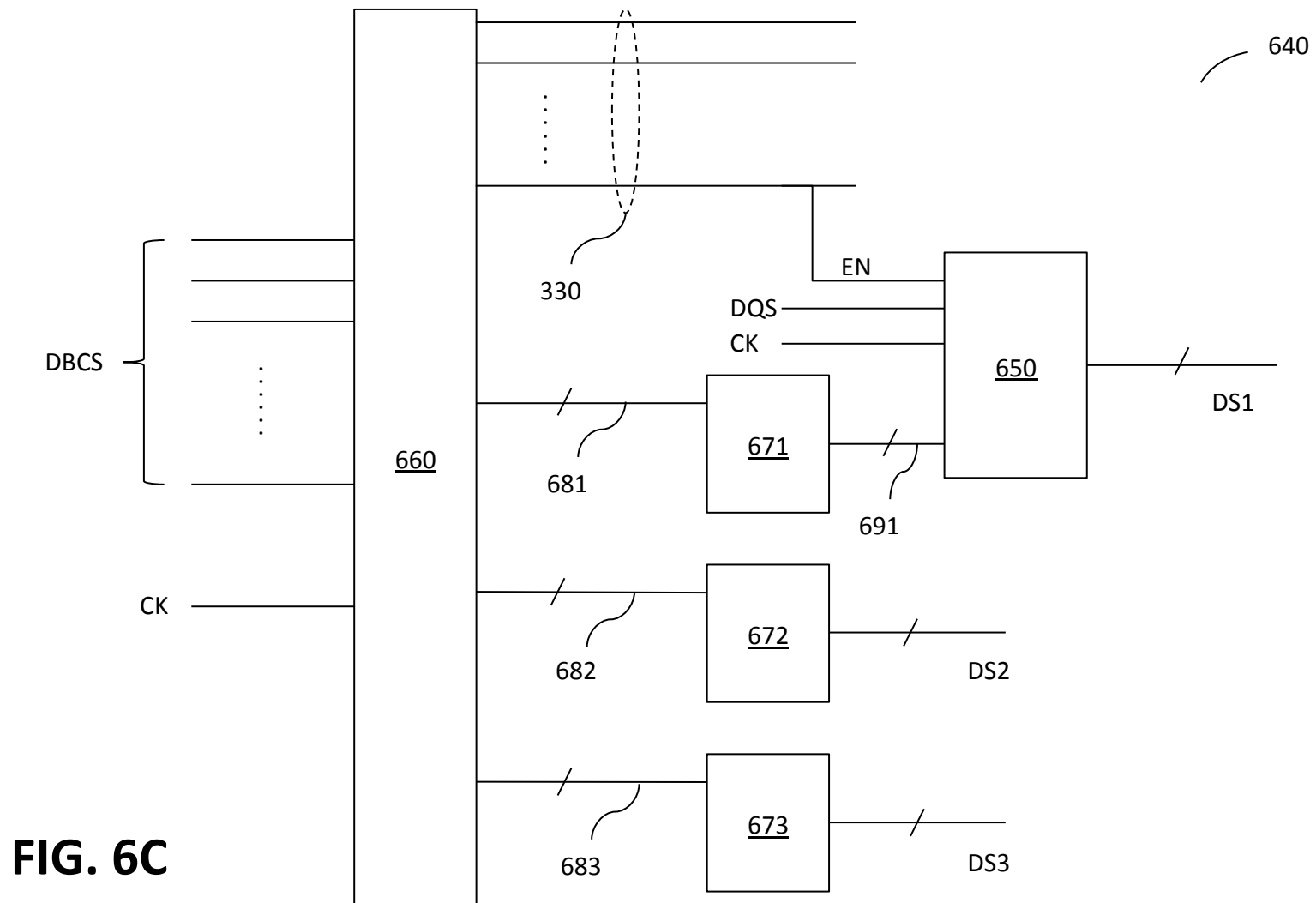


FIG. 6A





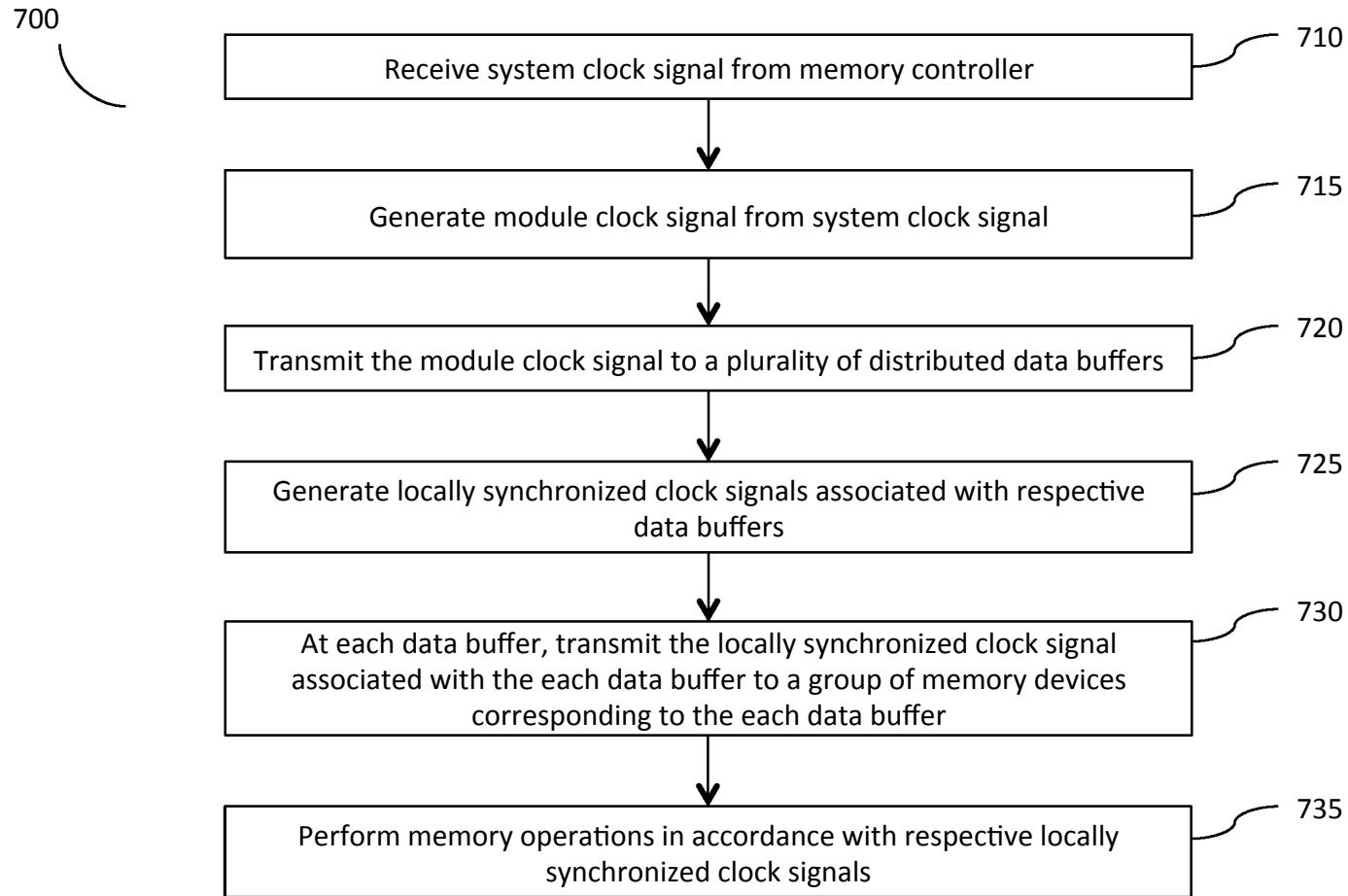


FIG. 7A

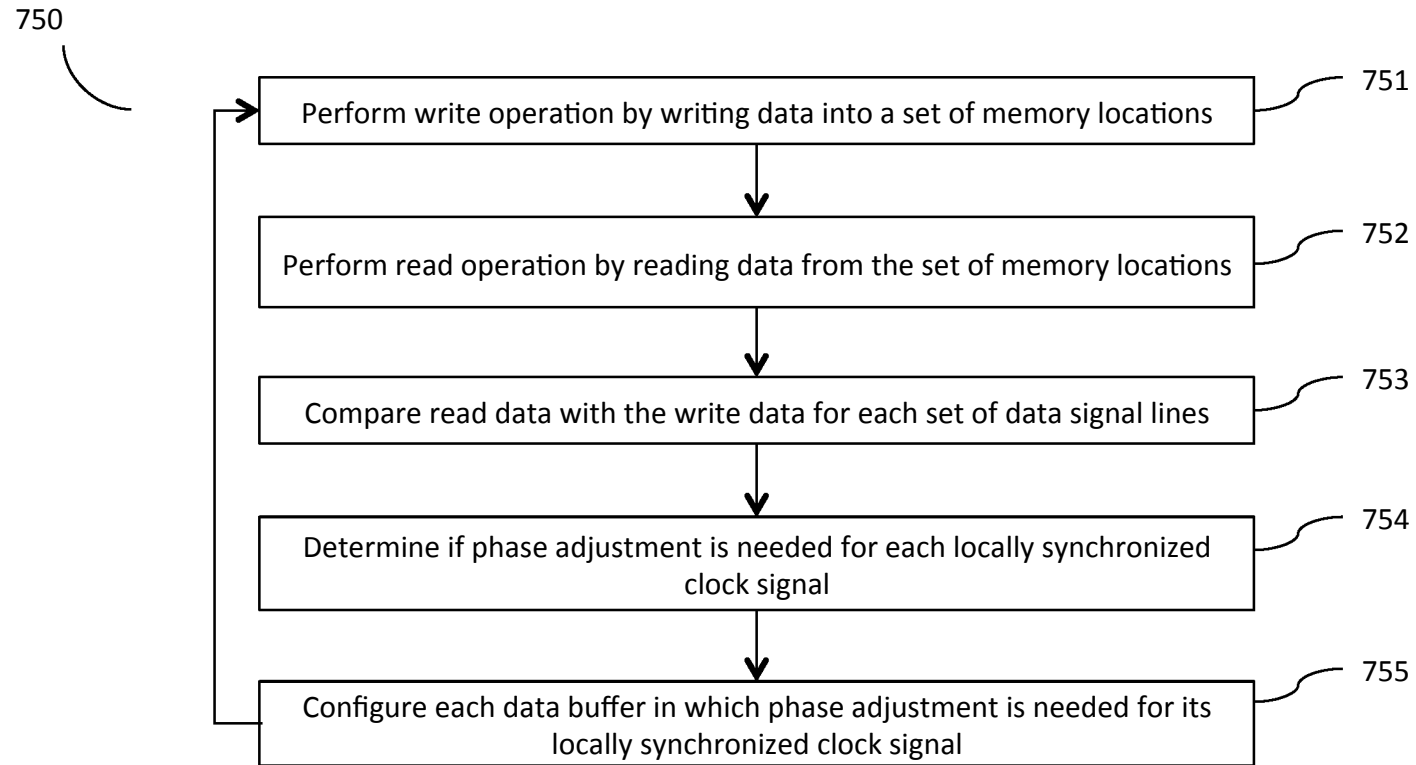


FIG. 7B

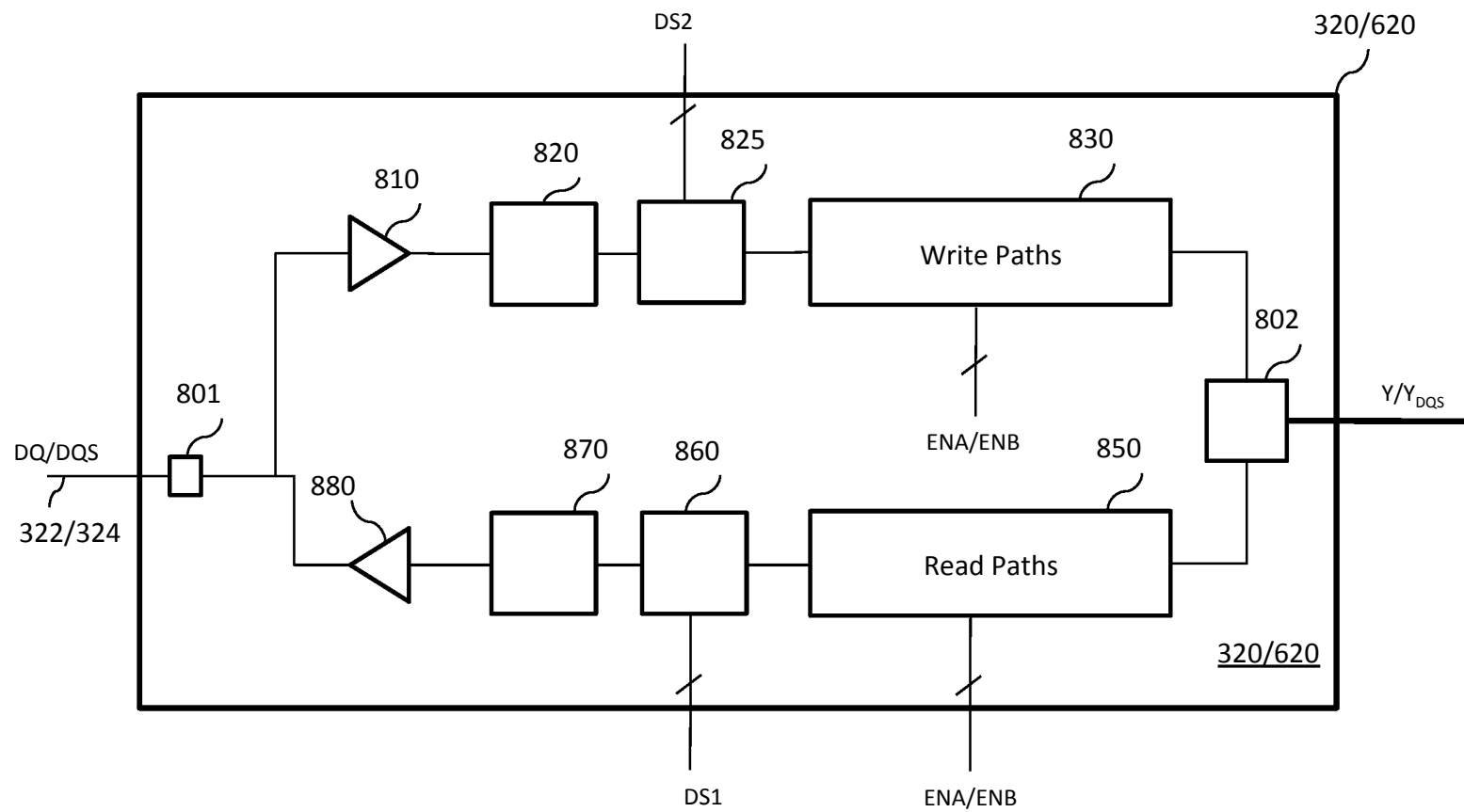


FIG. 8

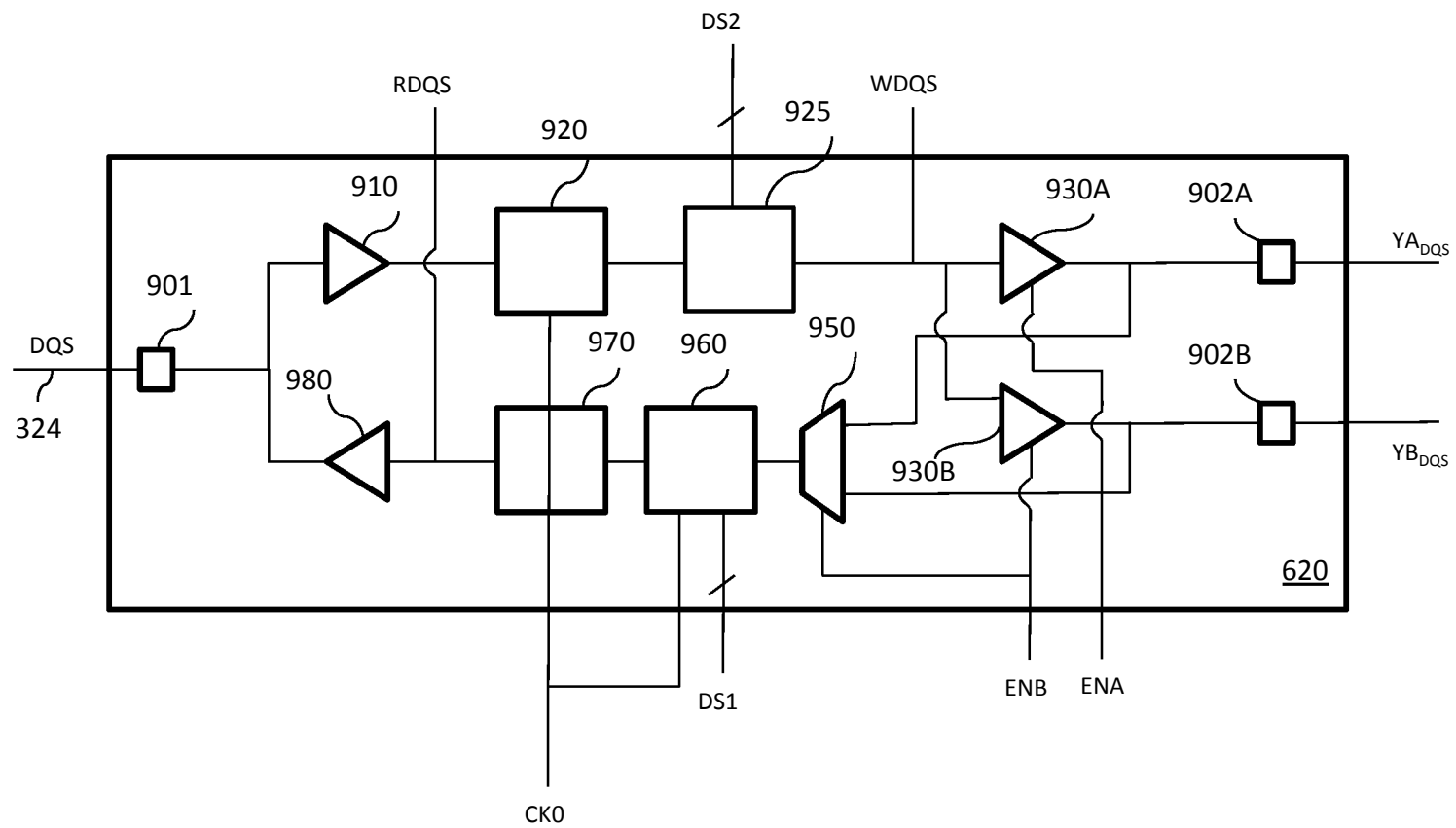


FIG. 9

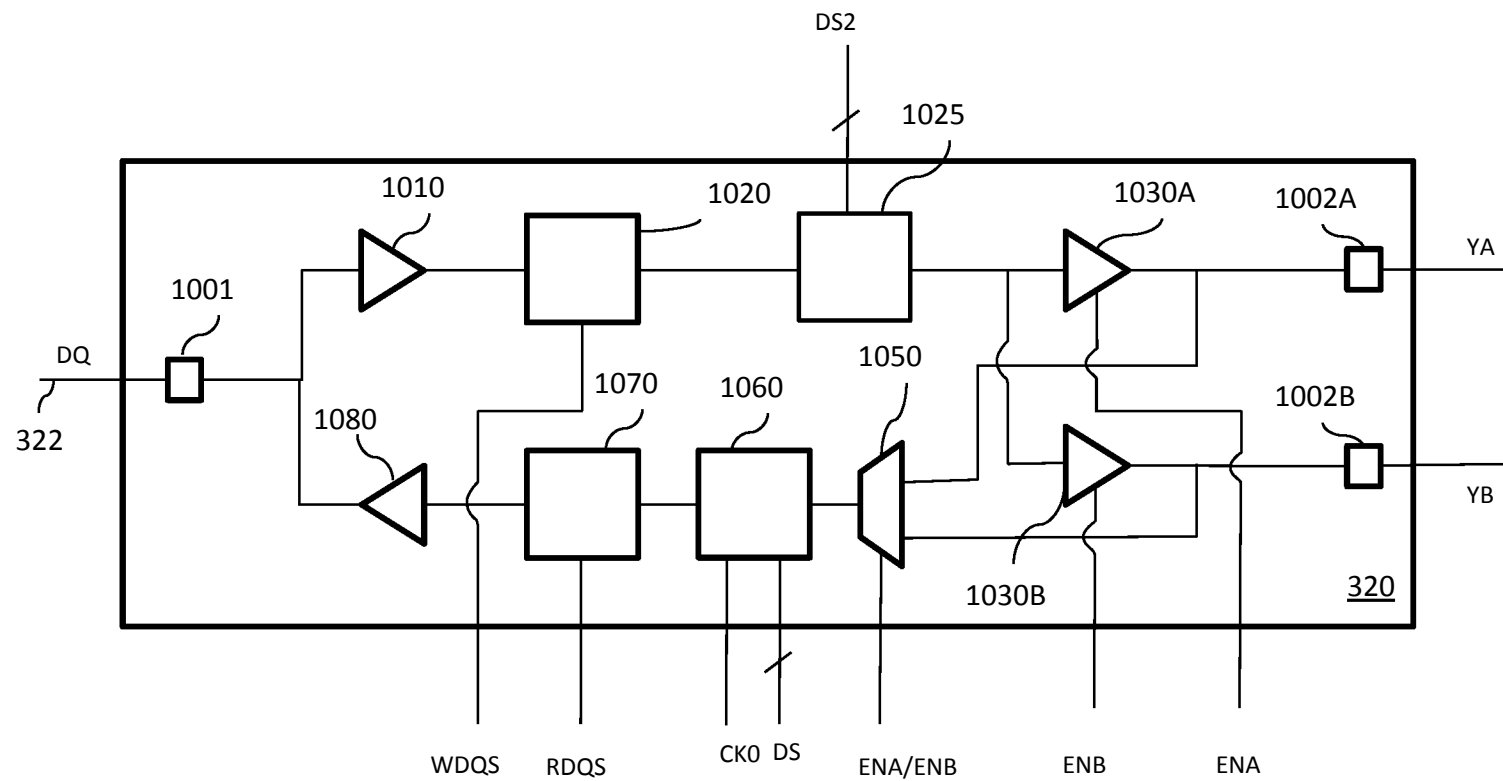


FIG. 10

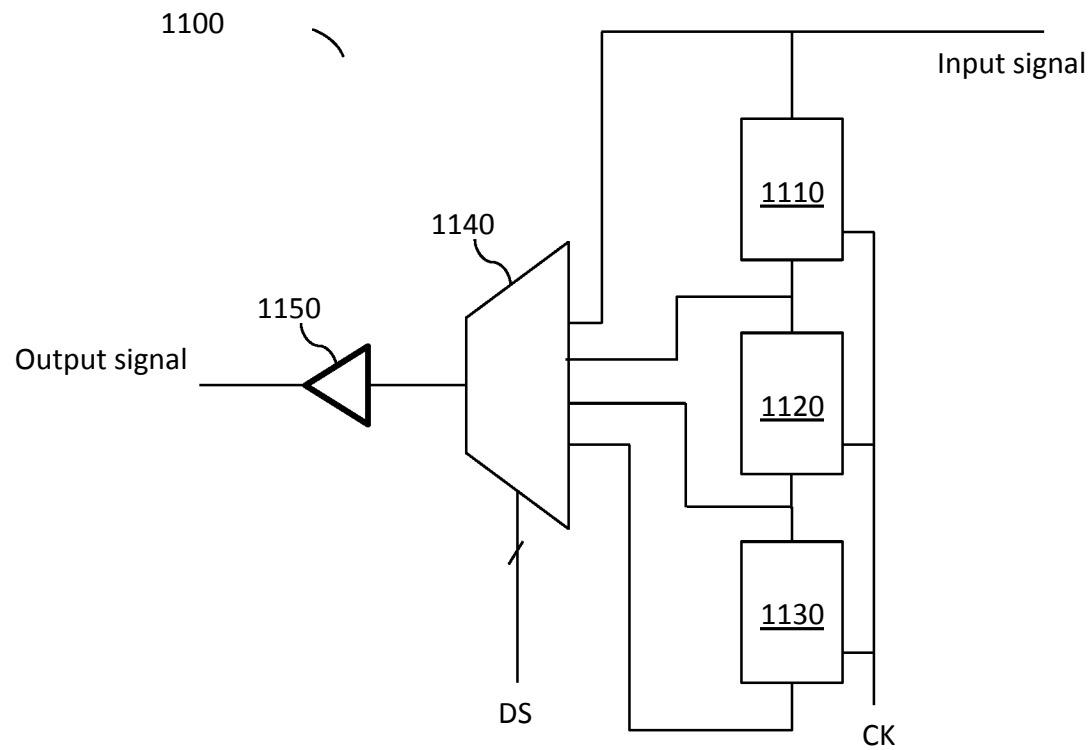


FIG. 11

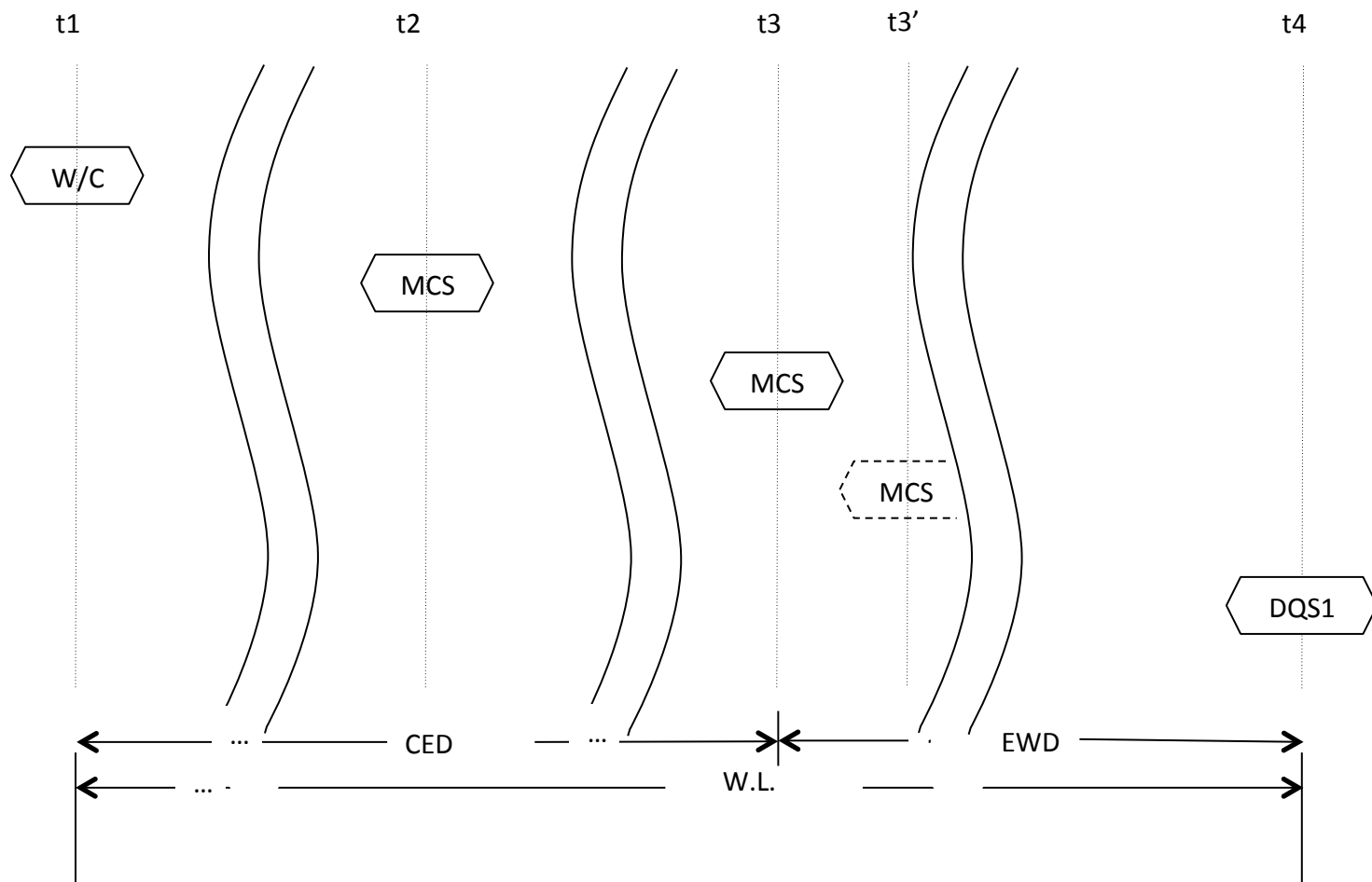


FIG. 12A

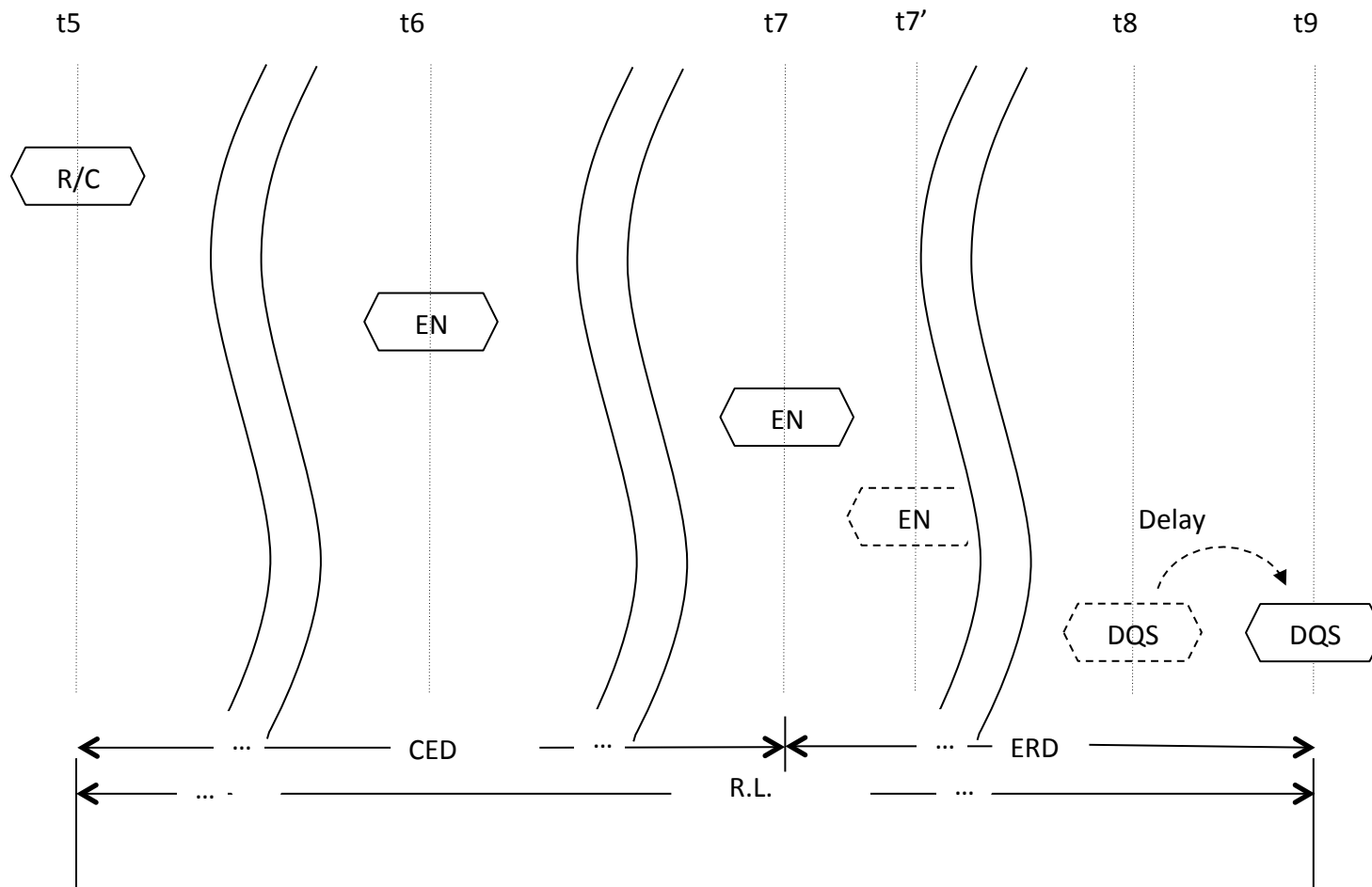


FIG. 12B

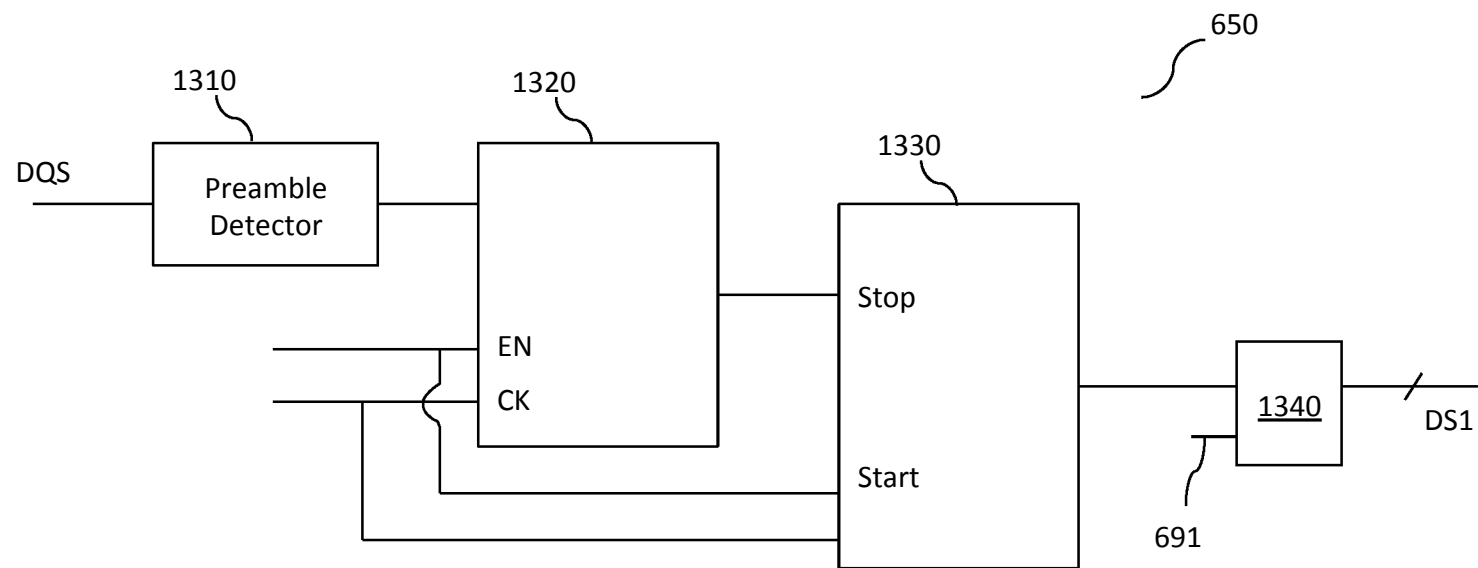


FIG. 13

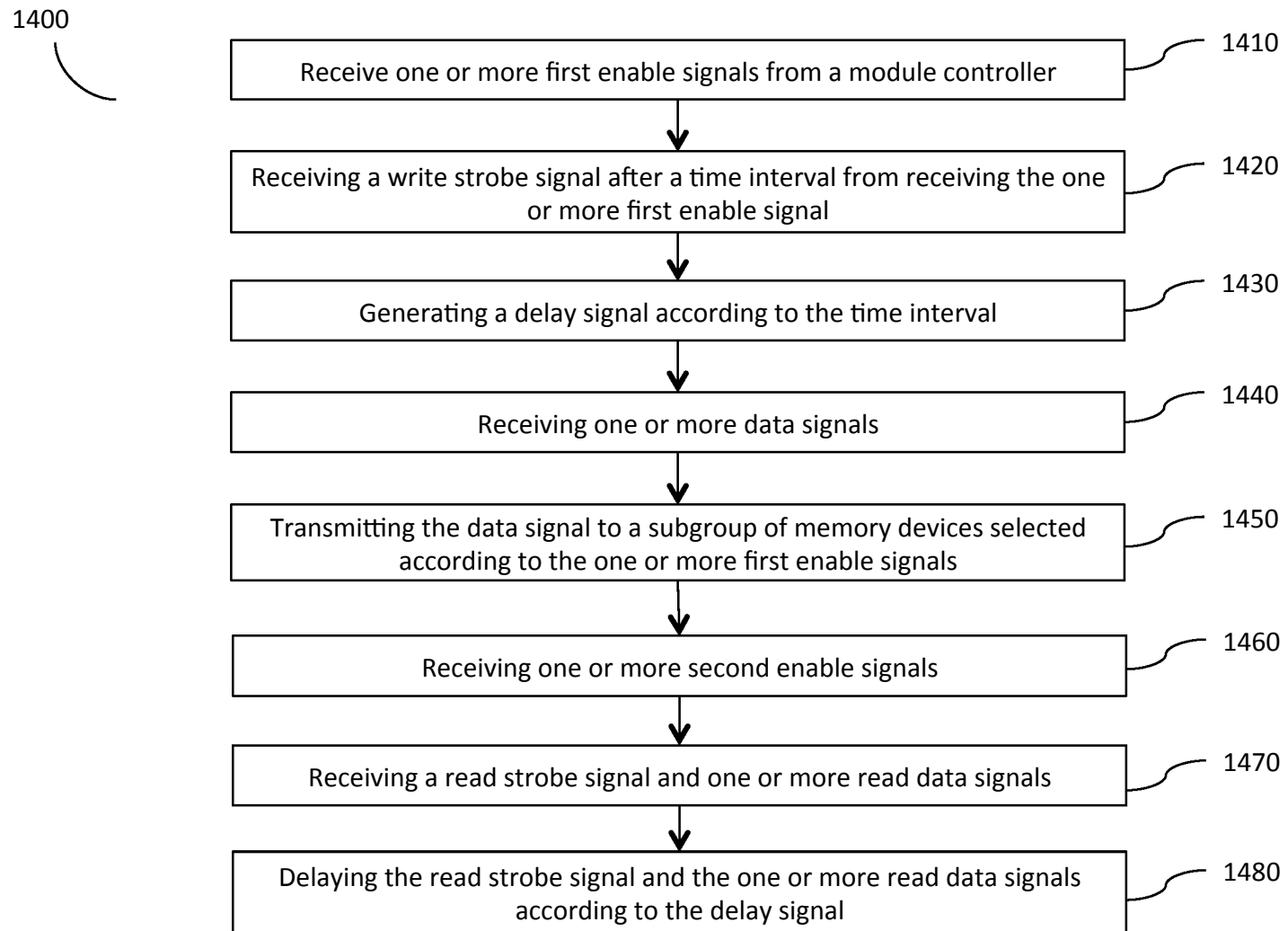


FIG. 14

MEMORY MODULE WITH LOCAL SYNCHRONIZATION

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority to U.S. Provisional Patent Application No. 61/859,215, filed July 27, 2013, which is incorporated herein by reference in its entirety.

FIELD

[0002] The disclosure herein is related generally to memory modules, and more particularly to memory modules having distributed data buffers.

BACKGROUND

[0003] With recent advancement of information technology and widespread use of the Internet to store and process information, more and more demands are placed on the acquisition, processing, storage and dissemination of vocal, pictorial, textual and numerical information by microelectronics-based combination of computing and communication means. In a typical computer or server system, memory modules are used to store data or information. A memory module usually includes multiple memory devices, such as dynamic random access memory devices (DRAM) or synchronous dynamic random access memory devices (SDRAM), packaged individually or in groups, and/or mounted on a printed circuit board (PCB). A processor or a memory controller accesses the memory module via a memory bus, which, for a single-in-line memory module (SIMM), can have a 32-bit wide data path, or for a dual-in-line memory module (DIMM), can have a 64-bit wide data path.

[0004] The memory devices of a memory module are generally organized in ranks, with each rank of memory devices generally having a bit width. For example, a memory module in which

each rank of the memory module is 64 bits wide is described as having an "x64" or "by 64" organization. Similarly, a memory module having 72-bit-wide ranks is described as having an "x72" or "by 72" organization.

[0005] The memory capacity or memory density of a memory module increases with the number of memory devices on the memory module. The number of memory devices of a memory module can be increased by increasing the number of memory devices per rank or by increasing the number of ranks.

[0006] In certain conventional memory modules, the ranks are selected or activated by control signals from a processor or memory controller during operation. Examples of such control signals include, but are not limited to, rank-select signals, also called chip-select signals. Most computer and server systems support a limited number of ranks per memory module, which limits the memory density of the memory modules that can be used in these computer and server systems.

[0007] For memory devices in such a memory module to be properly accessed, distribution of control signals and a control clock signal in the memory module is subject to strict constraints. In some conventional memory modules, control wires are routed so that there is an equal length of signal path to each memory component, in order to eliminate variation of the timing of the control signals and the control clock signal between different memory devices in the memory modules. The balancing of the length of the wires to each memory devices compromises system performance, limits the number of memory devices, and complicates their connections.

[0008] In some conventional memory systems, the memory controllers include leveling mechanisms for write and/or read operations to compensate for unbalanced wire lengths and memory device loading on the memory module. As memory operating speed and memory density continue to increase, however, such leveling mechanisms are also insufficient to insure proper timing of the control and/or data signals received and/or transmitted by the memory

modules.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a diagram illustrating a memory system including at least one memory module according to one embodiment.

[0010] FIGS. 2A-2D are each a diagrams illustrating interactions among components in a a memory module according to certain embodiments.

[0011] FIG. 3 is a diagram illustrating one of a plurality of data buffers in a memory module according to one embodiment.

[0012] FIGS. 4A-4B are each a diagram illustrating data and data strobe signal lines coupled to memory devices in a memory module according to certain embodiments.

[0013] FIGS. 5A-5B are diagrams illustrating different numbers of memory devices that can be coupled to each data buffer in a memory module according to certain embodiments.

[0014] FIG. 6A is a diagram illustrating data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments.

[0015] FIG. 6B is a diagram illustrating a control circuit in a data buffer according to certain embodiments.

[0016] FIG. 6C is a diagram illustrating a control processing circuit in a data buffer according to certain embodiments.

[0017] FIG. 7A is a flowchart illustrating a method of operation for a memory module according to certain embodiments.

[0018] FIG. 7B is a flowchart illustrating a method of training locally synchronized clock

signals according to certain embodiments.

[0019] FIG. 8 is a diagram illustrating a DQ or DQS routing circuit in a data buffer according to an embodiment.

[0020] FIG. 9 a diagram illustrating a DQS routing circuit having a delay circuit in a data buffer according to an embodiment.

[0021] FIG. 10 a diagram illustrating a DQ routing circuit having a delay circuit in a data buffer according to an embodiment.

[0022] FIG. 11 is a diagram illustrating a delay circuit in a DQ or DQS routing circuit according to an embodiment.

[0023] FIGS. 12A and 12B are timing diagrams associated with a write operation and a read operation, respectively, performed by a memory module according to one embodiment.

[0024] FIG. 13 is a diagram illustrating a delay control circuit in a data buffer according to certain embodiments.

[0025] FIG. 14 is a flowchart illustrating a method for data edge alignment according to embodiments.

DESCRIPTION OF EMBODIMENTS

[0026] A memory module according to one embodiment includes memory devices organized in groups, a module control device, and data buffers (DB). The data buffers are sometimes referred to herein as buffer circuits, isolation circuits, isolation devices or load reduction devices. The memory module is operable to perform memory operations in response to memory commands (e.g., read, write, refresh, precharge, etc.), each of which is represented by a set of control/address (C/A) signals transmitted by the memory controller to the memory module. The

C/A signals may include, for example, a row address strobe signal (/RAS), a column address strobe signal (/CAS), a write enable signal (/WE), an output enable signal (/OE), one or more chip select signals, row/column address signals, and bank address signals. The memory controller may also transmit a system clock signal to the memory module. In one embodiment, the C/A signals and the system clock signal are received by the module control device, which generates a set of module control/address (C/A) signals and a set of data buffer control signals in response to each memory command from the memory controller. The module C/A signals are transmitted by the module control device to the memory devices via module C/A signal lines, and the data buffer control signals (referred sometimes herein as DBCS) are transmitted by the module control device to the buffer circuits via data buffer control signal lines.

[0027] In certain embodiments, the buffer circuits are associated with respective groups of memory devices and are distributed across the memory module at positions corresponding to the data/strobe signal lines associated with the respective groups of memory devices. Thus, during certain high speed operations, each data buffer control signal may arrive at different buffer circuits at different points of time across more than one clock cycle of the system clock. Also, each buffer circuit associated with a respective group of memory devices is in the data paths between the respective group of memory devices and the memory controller. Thus, the memory controller does not have direct control of the memory devices. In one embodiment, each group of memory devices include at least two subgroups, each subgroup including at least one memory device. Each buffer circuit is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the memory controller in response to the data buffer control signals. The memory module can have more ranks of memory devices than what is supported by the memory controller.

[0028] FIG. 1 shows a system 100 including a system or host memory controller (MCH) 101 and one or more memory modules 110 coupled to the MCH by a memory bus 105, according to one embodiment. As shown, the memory bus includes C/A signal lines 120 and groups of

system data/strobe signal lines 130. Also as shown, each memory module 110 has a plurality of memory devices 112 organized in a plurality of ranks 114. Each memory module 110 further includes a module control circuit (module controller or module control device) 116 coupled to the MCH 101 via the C/A signal lines 120, and a plurality of buffer circuits or data buffers 118 coupled to the MCH 101 via respective groups of system data/strobe signal lines 130. In one embodiment, the memory devices 112, the module control circuit 116 and the data buffers 118 can be mounted on a same side or different sides of a printed circuit board (module board) 119, which has connectors 121 formed along an edge 122 thereof for connecting to respective ones of the C/A signal lines 120 and groups of system data/strobe signal lines 130.

[0029] In the context of the present description, a rank refers to a set of memory devices that are selectable by a same chip select signal from the memory controller. The number of ranks of memory devices in a memory module 110 may vary. For example, as shown, each memory module 110 may include four ranks of memory devices 112. In another embodiment, the memory module 110 may include 2 ranks of memory devices. In yet another embodiment, the memory module may include six or more ranks of memory devices 112.

[0030] In the context of the present description, a memory controller refers to any device capable of sending instructions or commands, or otherwise controlling the memory devices 112. Additionally, in the context of the present description, a memory bus refers to any component, connection, or groups of components and/or connections, used to provide electrical communication between a memory module and a memory controller. For example, in various embodiments, the memory bus 105 may include printed circuit board (PCB) transmission lines, module connectors, component packages, sockets, and/or any other components or connections that provide connections for signal transmission.

[0031] Furthermore, the memory devices 112 may include any type of memory devices. For example, in one embodiment, the memory devices 112 may include dynamic random access memory (DRAM) devices, such as Synchronous dynamic random access memory (SDRAM)

devices in compliance with, for example, the Double Data Rate Type 3 (DDR3) or Double Data Rate Type 4 (DDR4) standard. Additionally, in one embodiment, each memory module 110 may include a dual in-line memory module (DIMM).

[0032] As shown in FIG. 2A, which illustrates one memory module 110 according to an embodiment, the buffer circuits are coupled to respective groups of the system data/strobe signal lines 130 and are distributed across the memory module at positions corresponding to the respective groups of system data/strobe signal lines 130. The module control device 116 is coupled to the system C/A signal lines 120 and receives system memory commands represented by a set of system control/address (C/A) signals from the MCH 101 via signal lines 120. The module control device 116 generates module C/A signals and data buffer control signals based on memory commands from the system. The module control device 116 also received a system clock MCK and generates a module clock signal CK in response to the system clock signal MCK. The MCK signal may include a pair of complementary clock signals, MCK and $\overline{\text{MCK}}$, and the module clock signal may include a pair of complementary clock signals CK and $\overline{\text{CK}}$.

[0033] Examples of the system C/A signals include, but are not limited to, Chip Select (or /CS) signal, which is used to select a rank of memory devices to be accessed during a memory (read or write) operation; Row Address Strobe (or /RAS) signal, which is used mostly to latch a row address and to initiate a memory cycle; Column Address Strobe (or /CAS) signal, which is used mostly to latch a column address and to initiate a read or write operation; address signals, including bank address signals and row/column address signals, which are used to select a memory location on a memory device or chip; Write Enable (or /WE) signal, which is used to specify a read operation or a write operation, Output Enable (or /OE) signal, which is used to prevent data from appearing at the output until needed during a read operation, and the system clock signal MCK.

[0034] Examples of module C/A signals include, but are not limited to one or more module /CS signals; a module /RAS signal, which can be, for example, a registered version of the system /RAS signal; a module /CAS signal, which can be, for example, a registered version of the system /CAS signal; module address signals, which can be, for example, registered versions of some or all of the address signals; a module /WE signal, which can be, for example, a registered version of the system /WE signal; a module /OE signal, which can be, for example a registered version of the system /OE signal. In certain embodiments, the module C/A signals may also include the module clock signal CK. The one or more module C/S signals can be derived from the system /CS signals and one or more other system C/A signals, such as one or more bank address signals and/or one or more row/column address signals, as exemplified in commonly owned U.S. Patent No. 7,532,537, entitled “Memory Module with a Circuit Providing Load Isolation and Memory Domain Translation,” issued on May 12, 2009, the entirety of which is incorporated herein by reference.

[0035] Examples of data buffer control signals include, but are not limited to, one or more mode signals, which specify a mode of operation (e.g., configuration mode, test mode or normal operation mode) for the data buffers 118, and other data buffer control signals whose definitions depend on the mode of operation. For example, during the normal operation mode the other data buffer control signals may include one or more enable signals, one or more ODT signals, and the module clock signal. In the normal operation mode, the one or more enable signals are used by each data buffer to select one or more memory devices to communicate data with the memory controller; and the one or more ODT signals are used by the data buffers to set up on-die termination for the data/strobe signals. In the configuration mode or test mode, the other data buffer control signals are used to convey configuration or module self-test control information from the module control device 116 to the data buffers 118. In one embodiment, the data buffer control signals are transmitted to the data buffers 118 via respective data buffer control signal lines 230. Alternatively, the data buffer control signals can be packetized before being transmitted to the data buffers 118 via the data buffer control signal lines and

decoded/processed at the data buffers.

[0036] Module control device 116 transmits the module C/A signals to the memory devices 112 via module C/A signal lines 220. The memory devices 112 operate in response to the module C/A signals to receive write data or output read data as if the module C/A signals were from a memory controller. The module control device transmits the data buffer control signals together with the module clock signal CK to the data buffers 118 via data buffer control signal lines 230. As shown in FIG. 2, at least some of the memory devices in a same rank share a same set of module C/A signal lines 220, and at least some of the data buffers 118 share a same set of data buffer control signal lines 230.

[0037] As shown in FIGS. 2A and 2B, each rank 114 includes N memory devices, where N is an integer larger than one. For example, a first rank includes memory devices M_{11} , ..., M_{i1} , $M_{i+1,1}$, ..., M_N , a second rank includes memory devices M_{12} , ..., M_{i2} , $M_{i+1,2}$, ..., $M_{N,2}$, and so on. In one embodiment, the memory devices 112 are also organized in groups or sets, with each group corresponding to a respective group of system data/strobe signal lines 130 and including at least one memory device from each rank. For example, memory devices M_{11} , M_{12} , M_{13} , and M_{14} form a first group of memory devices, memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} form an i^{th} group of memory devices, and so on.

[0038] In certain embodiments, the data buffers 118 are associated with respective groups of memory devices and are coupled between respective groups of system data/strobe signal lines 130 and the respective groups of memory devices. For example, data buffer ID-1 among the data buffers 118 is associated with the first group of memory devices M_{11} , M_{12} , M_{13} , and M_{14} and is coupled between the group of system data/strobe signal lines 130-1 and the first group of memory devices, data buffers ID- i among the data buffers 118 is associated with the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} and is coupled between the group of system data/strobe signal lines 130- i and the i^{th} group of memory devices, and so on. The association of data buffers with groups of memory devices may not be exclusive, i.e., multiple data buffers may be

associated with a same group of memory devices, or vice versa.

[0039] In one embodiment, each group or sets of memory devices are coupled to the associated data buffer 118 via a set of module data/strobe lines 210 and one or more clock signal lines 211. Each group or set of memory devices is organized in subgroups or subsets, with each subgroup or subset including at least one memory device. The subgroups in a group of memory devices may be coupled to the associated data buffer 118 via a same set of module data/strobe lines 210 (as shown in FIG. 2A) or via respective subsets of module data/strobe lines 210 (as shown in FIG. 2B). For example, as shown in FIG. 2B, in the first group of memory devices, memory devices M_{11} and/or M_{13} form a first subgroup, and memory devices M_{12} and/or M_{14} form a second subgroup; in the i^{th} group of memory devices, memory devices M_{i1} and/or M_{i3} form a first subgroup, and memory devices M_{i2} and/or M_{i4} form a second subgroup; and so on. The first subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer 118 via an associated first subset of module data/strobe lines YA, and the second subgroup of at least one memory device in each group of memory devices is coupled to the associated data buffer via an associated second subset of module data/strobe lines YB, as shown. For example, memory devices M_{11} and/or M_{13} form the first subgroup are/is coupled to the data buffer ID-1 via the corresponding first subset of module data/strobe lines YA-1, and memory devices M_{12} and/or M_{14} form the second subgroup are/is coupled to the data buffer ID-1 via the corresponding second subset of module data/strobe lines YA-2.

[0040] In one embodiment, the data buffers 118 are in the data paths between the MCH 101 and the memory module 110 and include data buffers between the MCH 101 and the respective groups of memory devices. In one embodiment, each data buffer 118 is configured to enable a selected subgroup in the respective group of memory devices to communicate data with the MCH 101 in response to the data buffer control signals. Further, each data buffers 118 is configured to isolate unselected subgroup(s) of memory devices from the MCH 101 during

write operations, so that the MCH sees a load on each data line that is less than a load associated with the respective group of memory devices. In one embodiment, the MCH sees only a load associated with one memory device on each data/strobe signal line during write operations.

[0041] In one embodiment, the data buffers 118 are distributed across the memory module 110 or the module board 119 in positions corresponding to edge connectors 121 coupled to respective groups of system data/strobe signal lines 130. For example, as shown in FIGS. 2A and 2B, data buffer ID-1 is disposed in a first position corresponding to the first group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the first group of system data/strobe signal lines 130-1. Likewise, data buffer ID- i is disposed in an i^{th} position separate from the first position and corresponding to the i^{th} group of memory devices M_{i1} , M_{i2} , M_{i3} , and M_{i4} , or to the i^{th} group of system data/strobe signal lines 130- i . In one embodiment, the first position is between the first group of memory devices and an edge portion of the module board 119 where connections to the first group of data/strobe signal lines 130-1 are disposed, and the i^{th} position is between the i^{th} group of memory devices and an edge portion of the module board 119 where connections (not shown) to the i^{th} group of data/strobe signal lines 130- i are disposed. In one embodiment, the data buffers 118 are distributed along the edge 122 of the memory module 110. In one embodiment, each data buffer 118 is in a separate integrated circuit device package either by itself or together with at least some of the respective group of memory devices. In one embodiment, the module data/strobe signal lines 210, the module C/A signal lines 220, and the data buffer control signal lines 230 include signal traces formed on and/or in the module board 119.

[0042] As an option, memory module 110 may further include a serial-presence detect (SPD) device 240, which may include electrically erasable programmable read-only memory (EEPROM) for storing data that characterize various attributes of the memory module 110. Examples of such data include a number of row addresses, a number of column addresses, a

data width of the memory devices, a number of ranks on the memory module 110, a memory density per rank, a number of memory device on the memory module 110, and a memory density per memory device, etc. A basic input/output system (BIOS) of system 100 can be informed of these attributes of the memory module 110 by reading from the SPD 240 and can use such data to configure the MCH 101 properly for maximum reliability and performance.

[0043] In certain embodiments, the SPD 240 and/or the control circuit 116 store module configuration information, such as: memory space translation code, memory address mapping function code, input and output signals timing control information for the control circuit 116, electrical and logical level control information for input and output signals for the control circuit 116, the data buffers 118, etc. In certain embodiments, the SPD 240 contains a system view of the module 110 which can be different from an actual physical construction of the module 110. For example, the SPD 240 stores at least one memory operation parameter that is different from a corresponding memory operation parameter in a system memory controller setting. The SPD 240 may also store at least one data buffer operation parameter that is different from a corresponding parameter in the system memory controller setting.

[0044] Thus, in certain embodiment, in the memory module 110, C/A signals representing a memory command are received and buffered by the module control circuit 116, so that the MCH sees only the module control circuit 116 as far as the C/A signals are concerned. Write data and strobe signals from the controller are received and buffered by the data buffers 118 before being transmitted to the memory devices 112 by the data buffers 118. On the other hand, read data and strobe signals from the memory devices are received and buffered by the data buffers before being transmitted to the MCH via the system data/strobe signal lines 130. Thus, MCH 101 does not directly operate or control the memory devices 112. As far as data/strobe signals are concerned, the MCH 101 mainly sees the data buffers 118, and the system 100 depends on the data buffers 118 to properly time the transmission of the data and strobe signals between the MCH 101 and the memory devices 112.

[0045] In certain embodiments, the memory module 110 is a dual in-line memory module (DIMM) and the memory devices are double data rate (DDR) dynamic random access memory devices (DRAM). In certain embodiments, the control circuit 116 includes a DDR register, and logic for memory space translation between a system memory domain and a module level physical memory domain. Such translation may produce address mapping, proper interface timing for the control signals to the module level physical memory domain, and a proper interface electrical and logical level for the control signals to the module level physical memory domain.

[0046] As shown in FIG. 2C, in certain embodiments, the control circuit 116 transmits registered C/A and clock signals to the memory devices 112 in a fly-by configuration, and transmits data buffer control signals and registered clock signal to the data buffers 118 in a fly-by configuration. In a fly-by signal routing configuration, signals are routed such that they connect to each a row of devices in a daisy chain topology. Thus, signals routed in a fly-by configuration would arrive skewed in time at different devices. As the speed of memory operations increase, issues can also arise with respect to signal alignment for input, output delay variation due to process, voltage and temperature (PVT) variations, lack of synchronization with system memory controller interface, and phase drift accumulation during operation, etc. Electrical interface calibration drift during operation caused by charge build up and timing interface calibration drift during operation caused by environment change can also create issues.

[0047] For example, to offset the skew in the registered C/A signals, certain memory devices (e.g., the DDR3 SDRAM) have a training mode that permits the MCH to effectively zero-out the skew and re-align the data in time by adding delay to the data paths. In certain embodiments, however, load reduction mechanism in the data buffers 118 would provide a single data bus interface for the memory devices, which hides the data interfaces between the memory devices 112 and the data buffers 118 from the system memory controller 101. Thus, a long sequence of

interface timing training may be required because the system memory controller 101 has very limited control over the interface between the memory devices 112 and the data buffers 118. Furthermore, interface signal alignment-drift after the initial training would not be easily detected by the system memory controller 101, which may cause silent system failure.

[0048] Moreover, clock skew amongst the memory devices 112 and the data buffers 118 can cause synchronization issues. As the speed of memory operation increase, data period can become so short that variation of signal propagation time through I/Os becomes a very significant portion of a data period. As a result, the clock skew issues cannot simply be addressed by pipelining the data paths.

[0049] To address at least some of the above issues, in certain embodiments, as shown in FIG. 2D, the control circuit 116 transmits registered C/A signals to the memory devices 112 in a fly-by arrangement, and transmits the data buffer control signals and the registered clock signal to the data buffers 118, also in a fly-by arrangement. The memory devices 112 do not receive the registered clock signal from the control circuit 116. Instead, each data buffer 118 regenerates the clock that is used by a respective set of memory devices 112. Each Data buffer 118 is thus responsible for providing a correct data timing interface between the respective set of memory devices 112 and the system memory controller 101. Each data buffer 118 and its associated respective set of memory devices together form a group of locally synchronized components. The data buffer 118 in each group of locally synchronized components is also responsible for providing the correct control signal timing between the control circuit 116 and the group of locally synchronized components.

[0050] Thus, the memory module 110 in FIG. 2D allows a locally synchronized operation for each group of locally synchronized components, which can correspond to a nibble or a byte of a DDR data bus between the memory module 110 and the system memory controller 101. Also, signal interface between each data buffer 118 and the respective set of memory devices 112 can be synchronized. In one embodiment, each data buffer 118 has a set of configurable operations,

including, for example: programmable phase relationship between the clock it receives and the clock it regenerates, programmable phase adjustment for the data and data-strobe signals sent to the memory devices 112, programmable phase adjustment for the data and data-strobe signals sent to the system memory controller 101, and/or programmable phase adjustment for one or more control signal from the control circuit 116. The locally synchronized operation also makes it easier for each data buffer 118 to perform self-testing of the associated set of memory devices 112, independent of the self-testing of other sets of memory devices performed by the other data buffers, as disclosed in commonly-owned U.S. Patent 8,001,434, issued August 16, 2011, entitled “Memory Board with Self-Testing Capability,” which is incorporated herein by reference in its entirety.

[0051] In certain embodiments, operations of the data buffers 118 are controlled by the data buffer control signals from the module control circuit 116, which generates the data buffer control signals according to the C/A signals received from the MCH. Thus, the data buffer control signals need to be properly received by the data buffers 118 to insure their proper operation. In one embodiment, the data buffer control signals are transmitted together with the registered clock signal, or the module clock signal CK, which is generated by the module control circuit 116 based on the system clock signal MCK. The isolation circuits 118 buffers the module clock signal, which is used to time the sampling of the data buffer control signals. Since the data buffers 118 are distributed across the memory module, the data buffer control signal lines 230 can stretch across the memory module 110, over a distance of several centimeters. As the data buffer control signals travel over such a distance, they can become misaligned with the module clock signal, resulting in metastability in the received data buffer control signals. Therefore, in one embodiment, the isolation circuits 118 includes metastability detection circuits to detect metastability condition in the data buffer control signals and signal adjustment circuits to adjust the data buffer control signals and/or the module clock signal to mitigate any metastability condition in the data buffer control signals, as disclosed in commonly-owned U.S. Patent Application No. 13/952,599, filed July 27, 2013, entitled “Memory Module with

Distributed Data Buffers and Method Of Operation,” the entirety of which is incorporated herein by reference.

[0052] Because the data buffers 118 are distributed across the memory module 110, during high speed operations, it may take more than one clock cycle time of the system clock MCK for the data buffer control signals to travel along the data buffer control signals lines 230 from the module control device 116 to the farthest positioned data buffers 118, such as data buffer ID-1 and data buffer ID-($n-1$) in the exemplary configuration shown in FIG. 2. In other words, a same set of data buffer control signals may reach different data buffers 118 at different times across more than one clock cycle of the system clock. For example, when the clock frequency of the system clock is higher than 800 MHz, the clock cycle time is less than about 1.2 ns. With a signal travel speed of about 70ps per centimeter of signal line, a data buffer control signal would travel about 15 cm during one clock cycle. When the clock frequency increases to 1600 MHz, a data buffer control signal would travel less than 8 cm during one clock cycle. Further, a data buffer control signal line can have multiple data buffer control signals on the line at the same time. Thus, before one data buffer control signal reaches an end of the signal line, another data buffer control signal can appear on the same signal line.

[0053] With the data buffers 118 receiving data buffer control signals at different times across more than one clock cycle, the data buffer control signals alone are not sufficient to time the transmission of read data signals to the MCH 101 from the data buffers 118. In one embodiment, the data buffers have programmable control of the timing of read/write data signals and the phase of the locally synchronized clock signal for each group of locally synchronized devices. In certain embodiments, each data buffer can further determine individually, during a write operation, a time interval between a time when one or more data buffer control signals are received from the module control circuit 116 and a time when a write strobe or write data signal is received from the MCH 101. This time interval is used to adjust the timing of the transmission of read data to the MCH 101 during a subsequent read operation,

as explained in more detail below.

[0054] More illustrative information will now be set forth regarding various optional configurations, architectures, and features with which the foregoing framework may or may not be implemented, per the desires of the user. It should be strongly noted that the following information is set forth for illustrative purposes and should not be construed as limiting in any manner. Any of the following features may be optionally incorporated with or without the exclusion of other features described.

[0055] In one embodiment, as shown in FIG. 3, each group of signal lines 130 include a set of n data (DQ) signal lines 322 each for transmitting one of a set of bits $DQ_0, DQ_1, \dots, DQ_{n-1}$, in a data signal and at least one strobe (DQS) signal line 324 for transmitting at least one strobe signal DQS. Each set of module data/strobe lines Y include a set of n module data signal lines $Y_0, Y_1, \dots, Y_{n-1}$ and at least one module strobe signal line Y_{DQS} . When the subsets of memory devices are coupled to the associated data buffer 118 via respective subsets of memory devices, each set of module data/strobe lines Y may include multiple subsets of module data/strobe lines, such as the subsets of module data/strobe lines YA and YB shown in FIG. 2B. Each subset of module data/strobe lines YA include a set of n first module data lines $YA_0, YA_1, \dots, YA_n$ and at least one first module strobe signal line YA_{DQS} ; and each subset of module data/strobe lines YB include a set of n second module data lines $YB_0, YB_1, \dots, YB_n$ and at least one second module strobe signal line YB_{DQS} .

[0056] Each data buffer 118 includes a set of DQ routing circuits 320 coupled on one side to respective ones of the set of n DQ signal lines 322, and on another side to respective ones of the respective set of n module data lines, or respective ones of the respective subsets of module data lines, such as the first module data lines $YA_0, YA_1, \dots, YA_n$ and the second module data lines $YB_0, YB_1, \dots, YB_n$. Each data buffer 118 further includes an DB Control circuit 310 coupled on one side to the at least one DQS signal line 324, on another side to the one or more module strobe signal lines Y_{DQS} , or the first module strobe signal line YA_{DQS} and second

module strobe signal line YB_{DQS} .

[0057] The DB Control circuit 310 also receives the module clock signal CK and the data buffer control signals via the data buffer control signal lines 230, and outputs datapath control signals 330 to the DQ routing circuits 320, including, for example, one or more enable signals ENA and/or ENB, and some or all of the other received, decoded, and/or otherwise processed data buffer control signals. In certain embodiments, the DB Control circuit 310 also outputs one or more delay signals DS, a read DQS signal RDQS, a write DQS signal WDQS, and a buffer clock signal CK0. Each DQ routing circuit 320 is configured to enable data communication between the respective DQ signal line 322 with a selected subgroup of one or more memory devices in response to the data buffer control signals, as explained in more detail below.

[0058] In certain embodiments, the DB Control circuit 310 also provides one or more delay signals DS, which is used by the DQ routing circuits 320 to control the timing of read and/or write data output by the data buffer 118, as explained in further detail below. In certain embodiments, the DB Control circuit 310 regenerates clock signal CK0 from the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 is phase-locked to the module clock signal CK. In certain embodiments, the regenerated clock signal CK0 has a programmable delay from the module clock signal CK. In certain embodiments, the regenerated CK0 is provided to the respective group of memory devices so that the data buffer 118 and the respective group of memory devices form a group of locally synchronized devices on the memory module 110.

[0059] The memory devices 112 in each group of locally synchronized devices are coupled to the data buffer 118 in the each group of locally synchronized devices via a same set of module data/strobe signal lines. Or the memory devices 112 in each group of locally synchronized devices include subgroups of memory devices that are coupled to the data buffer 118 via different subsets of module data/strobe signal lines. For example, as shown in FIG. 4A, memory devices M_{11} , M_{12} , M_{13} , and M_{14} in the first group of memory devices can be coupled to the data

buffer ID-1 via a same set of module data lines $Y-1_0, Y-1_1, \dots, Y-1_{n-1}$ and module strobe line $Y-1_{DQS}$. In such embodiment, a subgroup in the group of memory devices can be selected by the data buffers to communicated data with the MCH based on the phases of the data/strobe signals, which can be different with respect to different subgroups of memory devices.

[0060] Alternatively, as shown in FIG. 4B, memory devices M_{11} and M_{13} , which form a subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines $YA-1_0, YA-1_1, \dots, YA-1_n$ and module strobe line $YA-1_{DQS}$ and memory devices M_{12} and M_{14} , which form another subgroup in the first group of memory devices, are coupled to the data buffer ID-1 via the module data lines $YB-1_0, YB-1_1, \dots, YB-1_n$ and module strobe line $YB-1_{DQS}$. Memory devices coupled to the same data buffers can be disposed on a same side or different sides of the memory board 119. Memory devices coupled to the same data buffers may be placed side-by-side, on opposite sides of the module boards 119, or stacked over each other, and/or over the associated data buffer.

[0061] Multiple memory devices having a data width that is less than a data width of the data buffers 118 may be used in place of one of the memory devices 112, which has the same data width as that of the data buffers. For example, as shown in FIG. 5A, two memory devices M_{11-1} and M_{11-2} may be used in place of the memory device M_{11} . Each of the two memory devices M_{11-1} and M_{11-2} has a data width of 4, and together they act like a memory device M_{11} of a data width of 8. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines $YA-1_0, \dots, YA-1_3$ and module strobe line $YA-1_{DQS-1}$ while memory circuit M_{11-2} is coupled to the data buffer ID-1 via module data lines $YA-1_4, \dots, YA-1_7$ and module strobe line $YA-1_{DQS-2}$.

[0062] In another embodiment, as shown in FIG. 5B, four memory devices M_{11-1} to M_{11-4} may be used as the memory device M_{11} . Each of the four memory devices M_{11-1} to M_{11-4} has a data width of 4, and together they act like a memory device M_{11} of a data width of 16. Thus, memory device M_{11-1} is coupled to the data buffer ID-1 via module data lines $YA-1_0, \dots, YA-1_3$ and module strobe line $YA-1_{DQS-1}$ while memory device M_{11-2} is coupled to the data buffer ID-1

via module data lines YA-1₄, ..., YA-1₇ and module strobe line YA-1_{DQS-2}, and so on.

[0063] FIG. 6A is a diagram illustrating clock signals and data buffer control signals from a module control device to a plurality of data buffers in a memory module according to certain embodiments. As shown, the data buffer control signals may include, for example, eight signals received respectively by eight signal receiving circuits 630 on each data buffer. In certain embodiments, one or more of the eight signals are to specify a mode of operation (e.g., normal operation, configuration or test mode). The definitions of the other data buffer control signals are different in different modes of operation. For example, in the configuration mode, the module control circuit 116 uses the DBCS to configure the programmable features in the data buffers 118 based on configuration information received from the MCH 101; in the test mode, the DBCS is used to control the self-test processes run by the data buffers; in the normal operation mode, the DBCS includes the one or more enable signals and one or more ODT signals.

[0064] FIG. 6B illustrates the DB Control circuit 310 in an data buffer 118. As shown, the DB Control circuit 310 includes a clock buffer 610 to receive the module clock signal CK from the module control device 116, a phase-locked loop circuit (PLL) 611 that provides phase locking between the input clock CK and the locally synchronized clock CK0, a clock phase adjustment circuit 615 for adjusting the phase of the locally synchronized clock CK0, and a clock driver 618 that drives the locally synchronized clock CK0 to the corresponding group of memory devices. The DB Control circuit 310 further includes a strobe routing circuit 620 that are coupled on one side to a corresponding system DQS signal line 324 and on another side to corresponding module DQS signal lines YA_{DQS} and YB_{DQS}. The DB Control circuit 310 further includes receiver circuits 630 to receive respective ones of the DBCS.

[0065] The DB Control circuit 310 further includes a control processing circuit 640 that decodes and/or otherwise processes the data buffer control signals. In the normal operation mode, the decoded/processed data buffer control signals may include, for example, one or more

data path enable signals ENA and/or ENB that are used by the DQ routing circuits 320 and the strobe routing circuit 620 to selectively enabling data communication between the MCH 101 and one of the subgroups in the respective group of memory devices, with which the data buffer is associated. In a configuration mode, the decoded/processed data buffer control signals are used to program configuration registers in the control processing circuit 640, which in turn control one or more delay signals DS1, DS2, DS3. The delay signal DS1 is used to control delay circuits in the read paths in the DQ/DQS routing circuits, the delay signal DS2 is used to control delay circuits in the write paths in the DQ/DQS routing circuits the data buffer, and the delay signal DS3 is used to control the phase of the locally synchronized clock CK0.

[0066] The strobe routing circuit 620 also buffers strobe signals received from either the MCH 101 or the memory devices 112, and outputs either a write strobe WDQS or read strobe RDQS, respectively, to the DQ routing circuits 320. In one embodiment, the DB Control circuit 310 further includes a dynamic delay control circuit 650 that provide dynamic adjustment of the timing of read data/strobe signals in the DQ routing circuit 320 and the strobe routing circuit 620.

[0067] In certain embodiments, as shown in FIG. 6C, the control processing circuit 640 includes control logic 660 that receives the data buffer control signals and the module clock signal CK and outputs in the normal operation mode the data path control signals 330. In a configuration mode, the control logic outputs configuration signals for programming configuration registers in the data buffer. For example, the control logic outputs configuration signals 681 to programmable registers 671 that provides static control of the timing of read data/strobe signals, configuration signals 682 to programmable registers 671 that controls the delay signals DS2, which used to control the timing of write data/strobe signals, and configuration signals 683 to programmable registers 673 that controls the delay signals DS3, which is used to control the phase of the locally synchronized clock signal CK0.

[0068] The programmable registers 671 is coupled via signal lines 691 to the dynamic delay

control circuit 650, which also takes as input a DQS signal, clock signal CK, and one of the datapath enable signals EN, and output the delay signal DS1. The dynamic delay control circuit 650 is described in further detail below.

[0069] FIG. 7A is a flowchart illustrating a method 700 performed by the memory module 110 according to certain embodiments. As shown in FIG. 7A, the method includes receiving 710 clock signal together with system C/A signals from the MCH 101, and generating 715 module clock signal CK. The module clock signal CK is transmitted 720 to a plurality of distributed data buffers 118, which generates 725 locally synchronized clock signals CK0 for each group of locally synchronized devices. Each data buffer then transmits 730 its locally synchronized clock signals CK0 to the memory devices corresponding to the each data buffer, and the memory devices perform 735 memory operations in accordance with the respective locally synchronized clock signals.

[0070] Since the module control circuit 116 generates the DBCS based on signals from the MCH 101, the MCH 101 can train the data buffers 118 to properly time the read/write data signals and to properly adjust the phase of the locally synchronized clock signals CK0. In certain embodiments, the module control circuit 116 generates the DBCS for setting the registers in the delay control circuit based on mode register command signals from the MCH 101. As shown in FIG. 7B, a method 750 for training the data buffers to provide a proper amount of phase shift to the locally synchronized clock signals CK0 includes performing a write operation by writing 751 data to a set of memory locations in the memory devices 112, which is followed by performing a read operation by reading 752 data from the set of memory locations. The read data is compared 753 with the write data to determine 754 if phase adjustment is needed for each locally synchronized clock signals CK0. For example, if the read data bits from data signal lines closer to the module control circuit 116 agree with the corresponding write data bits, while read data bits from the data signal lines further away from the module control circuit 116 do not agree with the corresponding write data bits. The

disagreement is likely caused by the locally synchronized clock signals CK0 in the data buffers 118 further away from the module control circuit 116 being excessively delayed during transmission, and proper phase adjustment can be programmed 755 into the data buffers to add a phase-shift to the clock signals CK0 to compensate the skew among the data buffers. The programming can be done using, for example, a mode register write operation. The above process can be repeated until proper clock signal phase adjustment is achieved.

[0071] FIG. 8 illustrates a DQ or DQS routing circuit 320 or 620 according to an embodiment. As shown, the DQ/DQS routing circuit 320/620 includes a DQ/DQS pin 801 that is coupled to the corresponding DQ/DQS signal line 322/324, a set of one or more DQS pins 802 that is coupled to a corresponding module DQ/DQS line(s) Y/Y_{DQS} , or Y_A/Y_{ADQS} and Y_B/Y_{BDQS} . The DQ/DQS routing circuit 320/620 further includes a write strobe buffer 810 that buffers write data/strobe, a write data/strobe receiver 820 that samples the write data/strobe, and a write delay circuit 825 controlled by the delay signal DS2. The DQ/DQS routing circuit 320/620 further includes a plurality of write paths 830 that are selectable or can be selectively enabled by one or more of the data buffer control signals, such as the enable signals ENA and ENB.

[0072] The DQS routing circuit further includes a plurality of read paths 850 that are selectable in accordance with the one or more of the data buffer control signals. Output from the selected read path is delayed in a delay circuit 860 by an amount controlled by the delay signal DS1, and sampled by a sampler circuit 870. The sampled read data/strobe is transmitted by transmitter 880 onto the corresponding data/strobe signal line 322/324 via the DQ/DQS pin 801.

[0073] FIG. 9 illustrates a DQS routing circuit 620 according to an embodiment. As shown, the DQS routing circuit 620 includes a first DQS pin 901 that is coupled to a corresponding DQS signal line 324, a second DQS pin 902A that is coupled to a corresponding module DQS line Y_{ADQS} , a third DQS pin 902B that is coupled to a corresponding module DQS line Y_{BDQS} . The DQS routing circuit 620 further includes a first write strobe path coupled between the first

DQS pin 901 and the second DQS pin 902A and a second write strobe path coupled between the first DQS pin 901 and the third DQS pin 902B. The first write strobe path includes a write strobe buffer 910 that buffers a write strobe, a write strobe receiver 920 that samples the write strobe according to the buffered module signal CK0 and a write delay circuit 925 controlled by the delay signal DS2. The sampled write strobe is provided to the DQ routing circuits 320 as the write strobe WDQS. The first write strobe path further includes a first write strobe transmitter 930A that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{ADQS} . The second write strobe path includes the write strobe buffer 910, the write strobe receiver 920, the write delay circuit 825 controlled by the delay signal DS2, and a second write strobe transmitter 930B that transmits the write strobe to one or more memory devices 112 coupled to the module strobe line Y_{BDQS} . The first and second write strobe transmitters, 930A and 930B, are controlled by two enable signals, ENA and ENB, respectively, such that the first write strobe path and the second write strobe path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0074] The DQS routing circuit further includes a read strobe path coupled between the first DQS pin 901 and a selected one of the second and third DQS pins 902A and 902B. In the read strobe path, a select circuit 950 (e.g., a multiplexor) selects either a read strobe signal received via DQS pin 902A or a read strobe signal received via DQS pin 902B based on one or both of the enable signals ENA or ENB. The selected read strobe signal is delayed in a delay circuit 960 by an amount controlled by the delay signal DS, and sampled by a sampler circuit 970 according to the buffered module clock signal CK0. The sampled read strobe is provided to the DQ routing circuits 320 as the read strobe RDQS and is transmitted by transmitter 980 onto the corresponding strobe signal line 324 via the first DQS pin 901.

[0075] FIG. 10 illustrates a DQ routing circuit 320 according to an embodiment. As shown, the DQ routing circuit 320 includes a first DQ pin 1001 that is coupled to a corresponding DQ signal line 130, a second DQ pin 1002A that is coupled to a corresponding module DQ line

Y_{ADQ}, a third DQ pin 1002B that is coupled to a corresponding module DQ line Y_{BDQ}. The DQ routing circuit 320 further includes a first write data path coupled between the first DQ pin 1001 and the second DQ pin 1002A and a second write data path coupled between the first DQ pin 1001 and the third DQ pin 1002B. The first write data path includes a write data buffer 1010, a write data receiver 1020 that samples write data according to the write strobe WDQS from the DQS routing circuit 620, a write delay circuit 1025 controlled by the delay signal DS2, and a first write data transmitter 1030A that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{ADQ}. The second write data path includes the write data buffer 1010, the write data receiver 1020, the write delay circuit 825 controlled by the delay signal DS2, and a second write data transmitter 1030B that transmits the write data to one or more memory devices 112 coupled to the module data line Y_{BDQ}. The first and second write data transmitters, 1030A and 1030B, are controlled by two enable signals, ENA and ENB, respectively. Thus, the first write data path and the second write data path can be selectively enabled/disabled by the enable signals, ENA and ENB.

[0076] The DQ routing circuit further includes a read data path coupled between the first DQ pin 1001 and a selected one of the second and third DQ pins 1002A and 1002B. In the read data path, a select circuit 1050 (e.g., a multiplexor) selects either a read data signal received via DQ pin 1002A or a read data signal received via DQ pin 1002B based on one or both of the enable signals ENA or ENB. The selected read data signal is delayed in a delay circuit 1060 by an amount controlled by the delay signal DS. The delayed read data signal is then sampled by a receiver circuit 1070 according to the read strobe RDQS from the DQS routing circuit 620, and transmitted by transmitter 1080 onto the corresponding data signal line 130 via the first DQ pin 1001.

[0077] FIG. 11 illustrate a delay circuit 1100 that can be used as the delay circuit 825/860/925/960/1025/1060 according to an embodiment. As shown, the delay circuit 1100 includes a plurality of delay stages, such as delay stages 1110, 1120, and 1130, each delaying

an input signal by a predetermined amount. The delay circuit 1100 further includes a select circuit 1140 (e.g., a multiplexor) that selects from among the outputs from the delay stages according to the delay signal DS (DS1, DS2, or DS3) it receives. The output of the select circuit is thus delayed from the input signal by an amount controlled by the delay signal DS.

[0078] As stated above, the MCH 101 can control the timing of read/write signals by programming the registers that control the DS1 and DS2 signals. However, since the data buffers 118 are in the data paths between the MCH 101 and the respective groups of memory devices 112, the MCH 101 does not have direct control of the interface between the data buffers and the memory devices 112. Thus, conventional read/write leveling techniques are not sufficient for managing read/write data timing. In one embodiment, in addition to the data buffers 118 having programmable control of timing of the read/write signals and the phase of the clock signal for each group of locally synchronized components, the data buffers also include signal alignment mechanism to dynamically control the timing of the transmission of read data/strobe signals, as discussed further below.

[0079] FIG. 12A is a timing diagram for a write operation according to one embodiment. As shown, after a write command W/C associated with the write operation is received by the module control circuit 116 at time t1, the module control circuit 116 outputs one or more enable signals EN at time t2 in response to the write commands. The one or more enable signals are received by a data buffer 118 at time t3, which afterwards receives one or more strobe signal DQS from the MCH 101 at time t4. Note that the same enable signal may be received by another data buffer 118 at time t3', which can be in a different cycle of the system clock MCK from the cycle in which t3 is in. The time interval between t4 and t1 is consistent with a write latency W.L. associated with the system 100, and is controllable by the MCH 101 and knowable to the data buffer 118. The time interval between t4 and t3, referred to hereafter as an enable-to-write data delay EWD, can be determined by the data buffer 118 since both these signals are received by the data buffer. Based on such determination, the data buffer 118 can

have knowledge of the time interval between t_3 and t_1 , referred to hereafter as a command-to-enable delay CED, which can be used by the data buffer 118 to generate an adjusted clock signal and to properly time transmission of read data to the MCH, as explained further below.

[0080] FIG. 12B is a timing diagram for a read operation according to one embodiment. As shown, after a read command R/C associated with the read operation is received by the module control circuit 116 at time t_5 , the module control circuit 116 outputs one or more enable signals EN at time t_6 in response to the read commands. The one or more enable signals are received by an data buffer 118 at time t_7 , which outputs at time t_8 read data signals (not shown) and one or more strobe signal DQS received from the respective group of memory devices. Note that the same enable signal may be received by another data buffer 118 at time t_3' , which can be in a different cycle of the system clock MCK from the cycle which t_3 is in. Thus, the enable signals alone cannot be used to time the transmission of the read signals by the data buffers 118.

[0081] With knowledge of the time interval between t_7 and t_5 , which should be about the same as the time interval between t_3 and t_1 , i.e., the command-to-enable delay CED, in certain embodiments, the data buffer can add a proper amount of delay to the read data signals and the one or more DQS signal such that the read data signals and the one or more DQS signal are transmitted at time t_9 by the data buffer to the MCH 101 via the respective group of data/strobe signal lines 130, with the time interval between t_9 and t_5 being consistent with a read latency R.L. associated with the system 100.

[0082] The time interval between t_4 and t_3 , i.e., the enable to write data delay EWD, is determined by the delay control circuit 650 in the DB Control circuit 310, as shown in FIG. 6. According to one embodiment, as shown in FIG. 13, the dynamic delay control circuit 650 includes a preamble detector 1310 to detect a write preamble in the DQS, a flip-flop circuit 1320 having an enable input EN receiving one of the data buffer control signals and a clock input CK receiving the buffered module clock signal CK0, and a counter circuit 1330 having a